

Data Communication Network Basics



Foreword

- From the very origins and subsequent development of human society, communication has remained a constant companion. Since the 1970s and 1980s, human society has entered the information age. For those of us living in this era, the necessity of communication is beyond question.
- In this course, communication refers to the transmission of information through a data communication network.
- This course introduces the concepts of communication and data communication networks, network types and typical topologies, network devices and their functions, and the process of information transmission. It also provides a brief introduction to network engineering and the role of network engineers.

Objectives

- Upon completion of this course, you will be able to:
 - Understand the concepts related to network communication and data communication networks.
 - Understand different types of networks and topologies.
 - Differentiate network devices of different types and understand their basic functions.
 - Describe the information transfer process.
 - Understand the concepts related to network engineering and network engineers.

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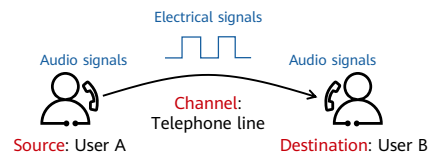
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2. Data Communication Network Devices
3. Data Communication Process
4. Network Engineering and Network Engineers

Communication

- The essence of communication lies in the transmission and exchange of information between two or more points—that is, the conveyance and interaction of information among people, between people and objects, and among objects, facilitated through certain media and actions.

Three elements of a communication system

- **Source:** device that sends information, such as a user's computer or mobile phone.
- **Channel:** medium through which information is transmitted, which can be a physical medium such as cables or optical fibers, or a wireless medium such as radio waves.
- **Destination:** device that receives information, that is, the receive device.

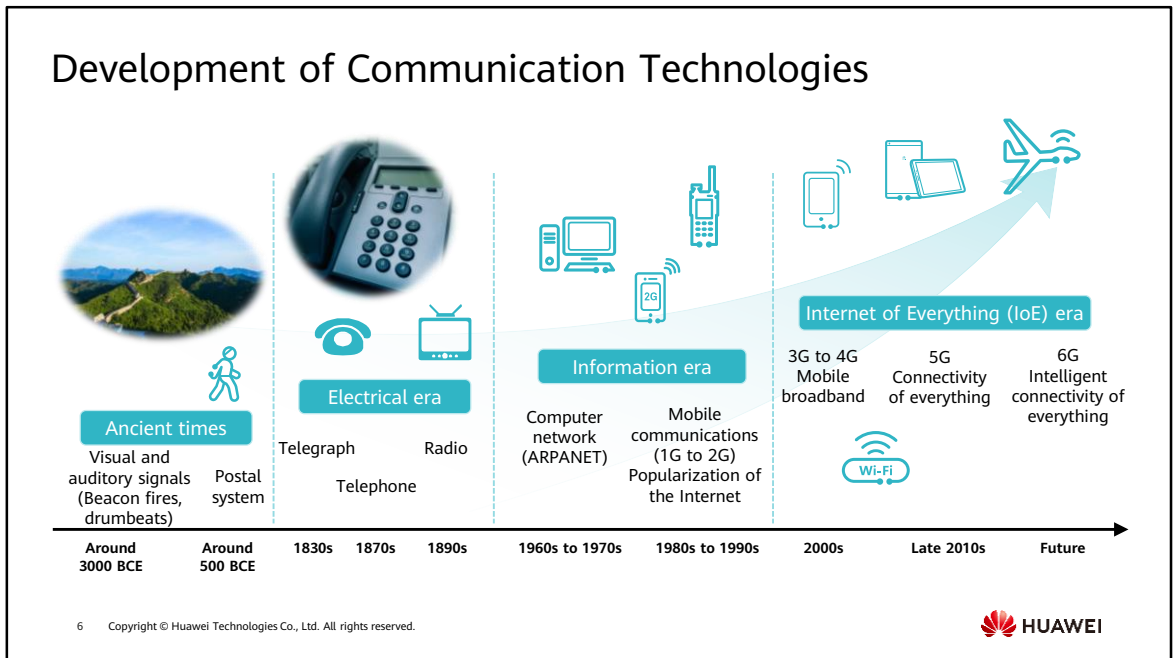


Audio transmission between users

Audio signals are converted into electrical signals at one end, transmitted through a telephone line, and restored to audio signals at the peer end.

- In addition to the communication system, communication has the following key elements:
 - **Information:** content to be transmitted, which can be in any form, such as voice, text, graphics, and data.
 - **Communication protocol:** rules (such as language, encoding mode, and communication standard) agreed by both parties to ensure that information can be correctly encoded, transmitted, and decoded.

Development of Communication Technologies



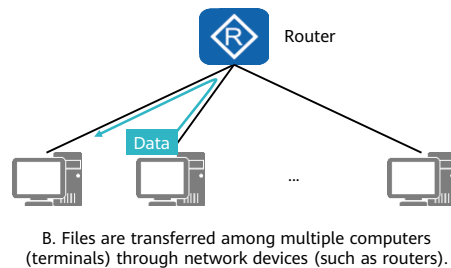
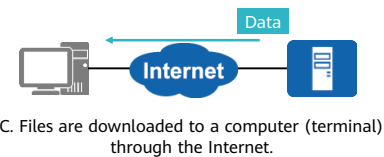
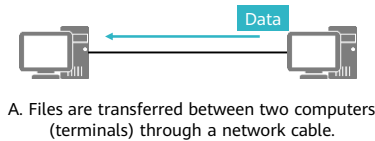
- **Ancient times:** In this era, communication relied entirely on the most basic natural media, human power, and animal power.
 - **Visual and auditory communication:** such as beacon towers, smoke signals, drum beats, and horns. These methods conveyed extremely limited information (usually prearranged signals), covered short distances, and were easily affected by weather and terrain.
 - **Postal communication:** Post stations and postal systems were established, and letters were transported by people and horses. This allowed for the transmission of larger amounts of information, but the speed was extremely slow, often taking days or even months.
- **Electrical era:** From the mid-19th century to the early 20th century, the invention and application of electricity completely transformed communication, allowing information to travel at the speed of light for the first time.
 - **Telegraph (1830s–1840s):** For the first time, people were able to communicate instantaneously over long distances (across continents or oceans). Information was transmitted through electrical wires.
 - **Telephone (1870s):** Real-time, two-way voice communication was realized, making communication more natural and direct.
 - **Radio (1890s):** Radio freed communication from the constraints of physical lines, laying the foundation for mobile communication and broadcasting. The subsequent development of AM/FM radio and television became the dominant mass media of the 20th century.
- **Information era:** From the mid-20th century to the present, the invention of

transistors, integrated circuits, and computer technologies has propelled communication into a new era of digitalization and networking.

- Computers and the Internet (1960s–1990s): The birth of the Advanced Research Projects Agency Network (ARPANET) and the adoption of Transmission Control Protocol/Internet Protocol (TCP/IP) marked the beginning of the Internet era. Communication expanded beyond human-to-human, evolving into machine-to-machine and network-to-network interactions.
- Mobile communication (1980s–present):
 - 1st generation (1G): analog voice communication, enabling people to make calls while moving.
 - 2nd generation (2G): digital voice communication, introducing Short Message Service (SMS).
 - 3rd generation (3G): mobile Internet access, enabling users to browse web pages and send and receive emails.
 - 4th generation (4G): the true era of mobile broadband, with explosive growth in applications such as smooth HD video, online gaming, and mobile payment.
 - 5th generation (5G): ultra-high speed, ultra-low latency, and massive connectivity. Beyond human communication, it aims to empower vertical industries such as the Internet of Things (IoT), autonomous driving, telemedicine, and industrial automation.
 - 1G/2G/3G/4G/5G: 1st/2nd/3rd/4th/5th-Generation Mobile Communication Technology
- The era of Internet of Everything (IoE): Communication technologies continue to accelerate in development.
 - Deepening of 5G and exploration of 6G: Enhanced versions such as 5G-A/5.5G will further improve performance. 6G is expected to be commercially deployed around 2030, aiming to support holographic communication, sensory interconnection, and digital twins, realizing "intelligent connectivity of everything."
 - Integration of IoT and Artificial Intelligence (AI): Communication will expand from humans to all things in the world. AI will empower communication networks with intelligent resource scheduling, fault prediction, and security protection, making networks more efficient and reliable.

Network Communication

- Network communication refers to the transmission of information between terminals through a data communication network.
- Examples of network communication:



- Examples of network communication:
 - A. When two computers are connected by a single network cable, they form the simplest data communication network.
 - B. A small data communication network can be built with one network device (such as a router or switch) and multiple computers. On such a network, files can be freely transferred between any two computers through the relay of the network device.
 - C. When a computer wants to obtain a file from a website, it must first connect to the Internet before it can download.
- The Internet is currently the largest computer network (that is, data communication network) in the world. Its predecessor, ARPANET, was born in 1969. The wide popularization and application of the Internet is one of the signs of entering the information age.

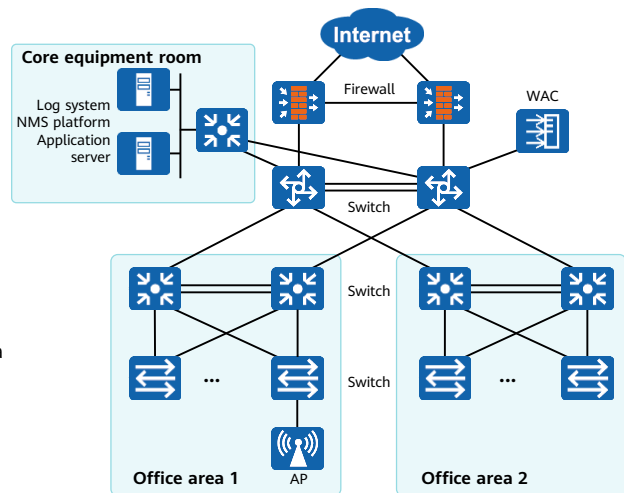
Data Communication Network

- Data communication network:

A data communication network consists of routers, switches, firewalls, wireless access controllers (WACs), wireless access points (APs), personal computers (PCs), network printers, servers, and the like.

- Functions:

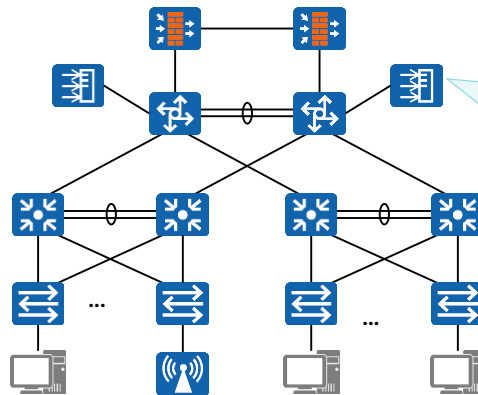
Its basic function is to implement data communication.



- Data communication network will be referred to as datacom network in this course.

Network Topology

- A network topology is a structured layout of how various network devices (such as computers, routers, and switches) are connected through transmission media (such as twisted pairs and optical fibers).

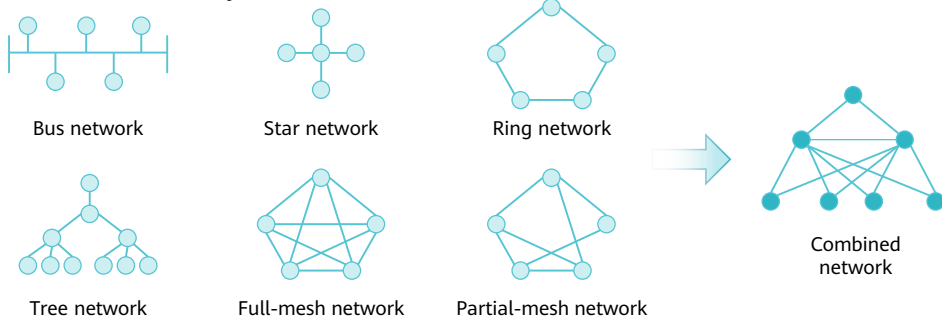


A network topology diagram describes the physical or logical structure of a network, which is essential in network engineering.

- Drawing network topologies: It is very important to master professional skills for drawing network topology diagrams, which requires a lot of practice.

Network Topology Types

- Networks can be classified into bus, star, ring, tree, full-mesh, and partial-mesh networks based on the network topology.
- In real-world networks, different topologies are often combined to strike a trade-off among cost, performance, and reliability needs.



- Bus network: All nodes are connected through a bus (such as a coaxial cable).
 - Advantage: Installation is easy with minimal cables required. Generally, a fault on a node does not affect the communication of the entire network.
 - Disadvantage: The failure of the bus affects the communication of the entire network. Information sent by a node can be received by all other nodes, which is insecure.
- Star network: All nodes are connected through a central node.
 - Advantage: New nodes can be easily added to the network. Communication data must be forwarded through the central node, facilitating network monitoring.
 - Disadvantage: The failure of the central node affects the communication of the entire network.
- Ring network: All nodes are connected to form a closed ring.
 - Advantage: Fewer cables are required.
 - Disadvantage: Adding a new node is inconvenient. You need to break the original ring before inserting the new node to form a new ring.
- Tree network: actually a hierarchical star topology.
 - Advantage: Multiple star networks can be quickly connected, facilitating network expansion.
 - Disadvantage: The higher the node, the more severe network issues caused by the node's failure.

- Full-mesh network: All nodes are interconnected through cables.
 - Advantage: It features high reliability and communication efficiency.
 - Disadvantage: Each node requires a large number of physical ports and cables for network connections. It is costly and difficult to scale.
- Partial-mesh network: Only key nodes are interconnected.
 - Advantage: The cost is lower than that of a full-mesh topology.
 - Disadvantage: The reliability is lower than that of a full-mesh topology.

Network Types

- A datacom network can be classified into different types based on different classification modes.
- There are two common classification modes.

By geographical coverage

- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)

By operation entity

- Enterprise network
- Carrier network

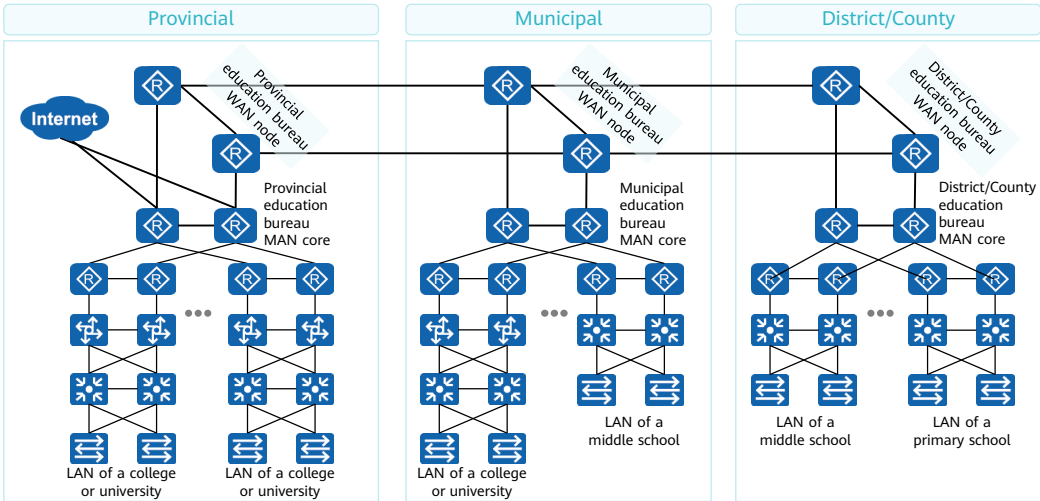
LAN, MAN, and WAN

- Datacom networks can be classified into LANs, MANs, and WANs based on geographical coverage.

LAN	MAN	WAN
• A LAN is a network that consists of computers, servers, and network devices in a geographic area. The coverage of a LAN is generally within several kilometers. • Its primary function is to connect several terminal computers that are distributed relatively close together, such as within a single household, one or more buildings, or a campus. • Technologies used: Ethernet, Wi-Fi, and so on. • Typical scenarios: office network of a company, network of an Internet cafe, and home network.	• A MAN can be regarded as a large-scale LAN, but requires higher costs and provides higher transmission rates than a LAN. It improves the transmission media used in a LAN and expands its access scope, which can encompass a university campus or a city. • Its main function is to connect hosts, databases, and LANs at different locations within the same city. • Technologies used: similar to LAN technologies, such as Ethernet (10 Gbps/100 Gbps) and WiMAX. • Typical scenarios: broadband MAN, education MAN, and municipal or provincial e-government private network.	• A WAN generally covers a large geographical area ranging from tens of to thousands of kilometers. It can connect networks of multiple cities or even networks of countries (as an international large-scale network) and provide long-distance communication. • It is mainly used to connect several LANs or MANs that are distributed far away (for example, across cities or countries). • Technologies used: PPP, MPLS, SRv6, and so on. • Typical scenario: Internet.

- WiMAX: Worldwide Interoperability for Microwave Access
- PPP: Point-to-Point Protocol
- MPLS: Multiprotocol Label Switching
- SRv6: Segment Routing over IPv6

LAN, MAN, and WAN in the Education Industry



Enterprise Network and Carrier Network

- Datacom networks can be classified into enterprise networks and carrier networks by operation entity.

Enterprise network

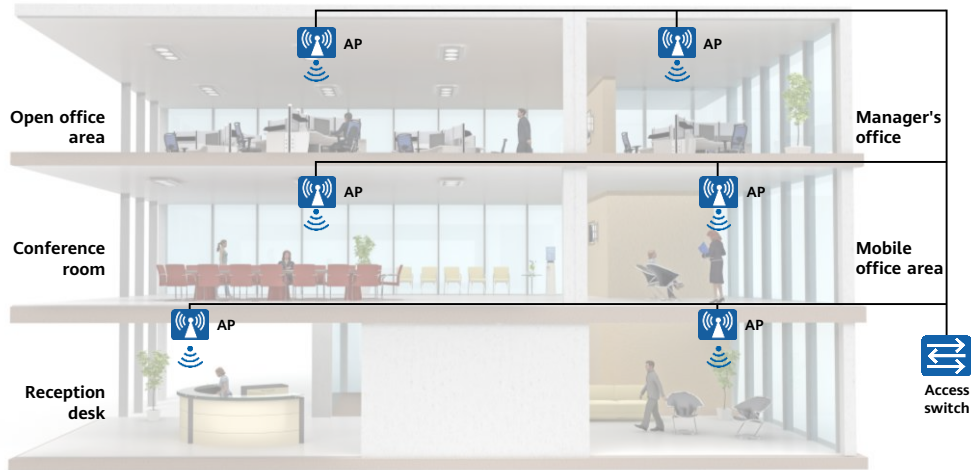
- An enterprise network is a private network constructed and maintained by a single organization (such as a company, government agency, or school). It connects employees, devices, data centers, and branch offices to support daily office operations, production, and management needs.
- Technologies used: Ethernet-related technologies (such as VLAN and VPN), Wi-Fi, firewall technologies, and so on.
- Typical scenarios: campus networks (such as office buildings, factories, and school campuses), data center networks, and enterprise-built backbone networks (WANs).

Carrier network

- A carrier network is a public network infrastructure constructed and operated by a telecom service provider, aiming to offer internet access, voice communication, and data transfer services for external customers (individual users and enterprises).
- Technologies used: BGP, MPLS, 5G core network, and so on.
- Typical scenarios: Internet backbone networks (cross-region and cross-country connections), mobile communication networks (4G/5G base stations and core networks), and enterprise private line services (MPLS VPN and SD-WAN).

- Note: Unless otherwise specified, the network described in this course refers to the enterprise network.
- VLAN: Virtual Local Area Network
- VPN: Virtual Private Network
- BGP: Border Gateway Protocol
- SD-WAN: Software-Defined Wide Area Network

Enterprise Network in Office Scenarios



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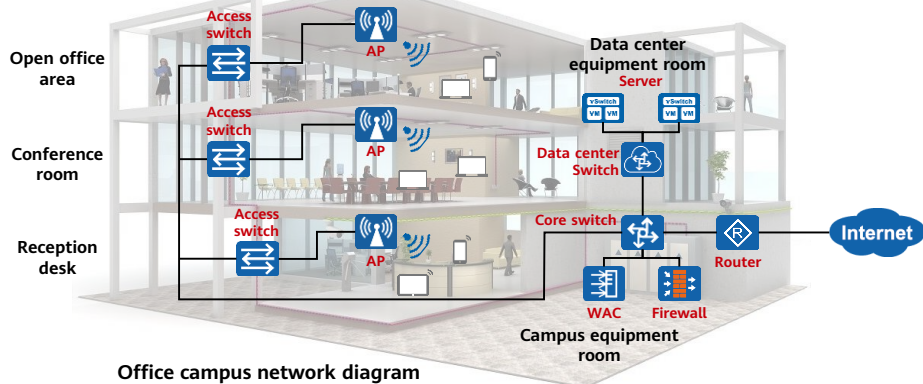
- An office network is a typical scenario of an enterprise network. It connects various network devices and terminals within the enterprise, enabling resource sharing and information exchange. It also provides a unified entry for employees to access internal resources and external network services.

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Network Devices

- A data communication network consists of a series of network infrastructure and terminals. The network infrastructure mainly includes network devices such as switches, routers, and firewalls. These devices collaborate with each other on the data communication network to receive and forward data.



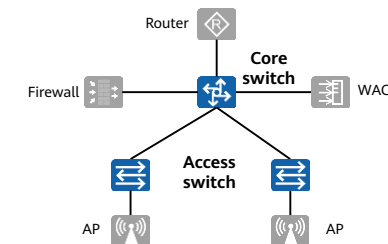
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- The figure shows a typical enterprise office campus network.
 - In the campus equipment room, egress devices such as routers and firewalls are deployed to enable secure communication with external networks. At the same time, core campus devices such as core switches and WACs are also placed there to implement internal data interconnection across the entire campus network.
 - Midsize and large enterprises often have independent data centers. Data center devices such as switches and servers are usually located in an independent data center equipment room and physically connected to the campus core, providing various application services and data storage for internal or external users.
 - On each office floor, an access switch can be installed to connect APs or wired terminals within that floor's office area for office network access. The APs then provide wireless signals to allow wireless terminals in the office area to access the network.

Switch

- A switch is a core device on a datacom network. It forwards data frames efficiently on a LAN to the destination device by identifying the destination MAC address in the data frames.
- Switches can be classified into fixed and modular switches based on their functional modules and composition. In the figure on the left, fixed switches are used as access switches on a campus network to connect user data to the network; a modular switch is used as the core switch on a campus network to aggregate traffic from all modules on the campus network for centralized forwarding.



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Fixed switch



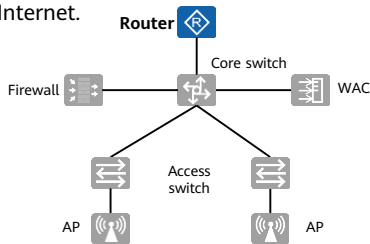
Modular switch



- Depending on the application scenario, switches can be classified into campus switches and data center switches. Data center switches connect to devices such as computing and storage devices in a data center to ensure high-speed, reliable, and accurate data forwarding. Data center switches can also be classified into fixed switches and modular switches.
- MAC: Medium Access Control

Router

- A router can forward data packets in various network scenarios. It selects a proper path based on the destination IP address of the received data packet and forwards the data packet to the next router or destination.
- Routers can be classified into backbone routers, metro routers, and access routers, depending on the application scenario. The figure on the left is a typical application scenario of an access router. The access router is used as the egress device of a campus network to connect campus data to the WAN or Internet.



Access router

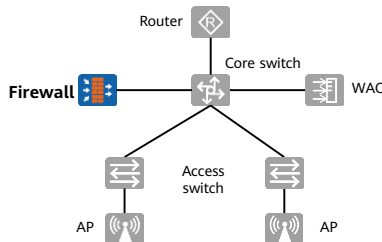


Metro router

- A router provides the following functions: routing table and routing information maintenance, route discovery and path selection, data forwarding, WAN access, network address translation, and specific security functions.

Firewall

- A firewall is a network security device used to ensure secure communication between networks. It monitors, restricts, and modifies data flows passing through it to shield information, structure, and running status of the internal network from external network users, ensuring network security.
- The figure on the left is a typical firewall application scenario. Generally, the firewall is deployed at the network egress to isolate the internal network from the external network, implementing secure Internet access for internal users and secure access for external users.

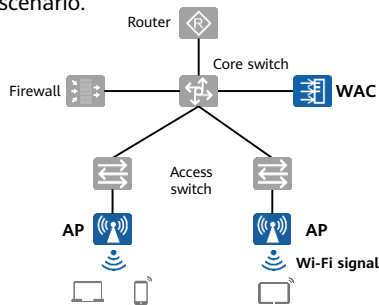


Firewall

- The main functions of a firewall include isolating networks of different security levels, implementing access control between networks of different security levels, authenticating user identities, enabling remote access, providing data encryption and virtual private network (VPN) services, performing network address translation, and other security features.

Wireless Device

- A wireless LAN (WLAN) is established using wireless technologies. It allows terminals to access the network using wireless signals for data transmission.
- Common wireless devices include Fat APs, Fit APs, and WACs, which can implement functions such as user access, authentication, and service forwarding. The figure on the left shows a typical Fit AP + WAC scenario.



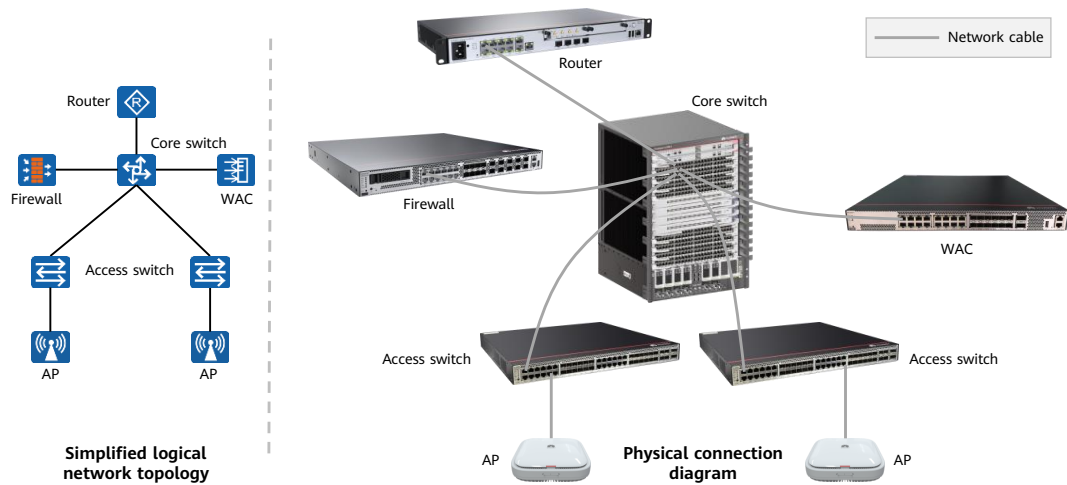
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- In a broad sense, a WLAN is a network that uses wireless signals such as radio waves, lasers, and infrared rays to replace the traditional media used for transmission on a wired LAN. The commonly known Wi-Fi refers to WLAN technology based on the IEEE 802.11 standard.
- Common WLAN devices include APs and WACs.
 - AP:
 - Supports multiple working modes, such as Fat AP, Fit AP, and Cloud AP. You can flexibly switch between these modes based on the network plan.
 - Fat AP: works independently and needs to be configured separately. It provides only basic functions and is cost-effective. Fat APs are suitable for home use.
 - Fit AP: must be used together with a WAC, which manages and configures the Fit APs in a unified manner. Fit APs provide rich functions and are applicable to large and midsize enterprises.
 - Cloud AP: must be used together with a cloud management platform, which manages and configures the Cloud APs in a unified manner. Cloud APs provide rich functions and are plug-and-play. They are applicable to small and midsize enterprises.
 - WAC:
 - Typically located at the aggregation or core layer of a network, provides high-speed, secure, and reliable WLAN services.

- Provides a large capacity and high performance. It is highly reliable, easy to install and maintain, and features such advantages as flexible networking and energy conservation.
- Note: The Institute of Electrical and Electronics Engineers (IEEE) is an organization specialized in engineering and electronics.

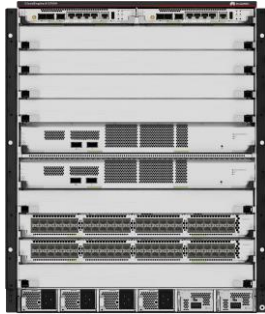
Physical Connection Diagram



- Modern network devices are no longer limited to a single function. For example, a switch integrates switching and routing functions, and a router integrates routing, security, and wireless functions. Therefore, when considering device selection, network construction, and data forwarding, it is necessary to understand the functions and features each type of device can provide, and to plan and deploy the network with a holistic view of service requirements.

Composition of Network Devices

- Switches, routers, and firewalls have similar structures and can be classified into modular and fixed devices based on functional modules. This course uses the modular switch CloudEngine S12700H-8 and fixed switch CloudEngine S5755-H24T4Y2CZ as examples to describe the composition of typical network devices.



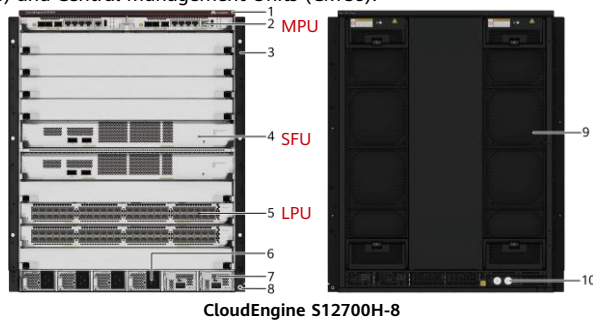
Modular switch
CloudEngine S12700H-8



Fixed switch
CloudEngine S5755-H24T4Y2CZ

Chassis Structure of Modular Devices

- Modular devices use a modular architecture and provide different functions through different boards. They are typically deployed at the core layer of a network and have high reliability.
- Based on their functions, modular devices support Main Processing Units (MPUs), Switch Fabric Units (SFUs), and Line Processing Units (LPUs). Different device models can be equipped with other functional boards, such as Service Processing Units (SPUs) and Central Management Units (CMUs).



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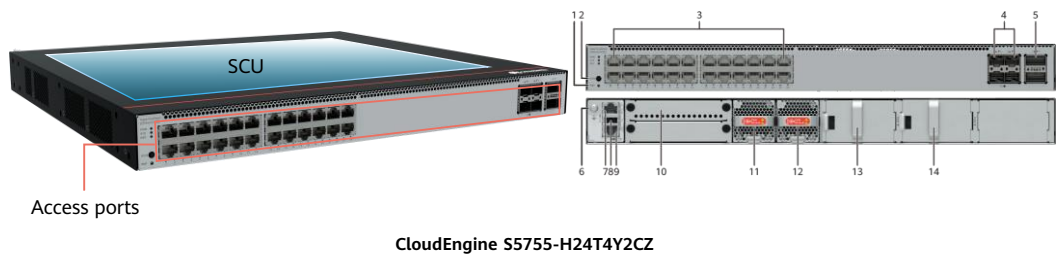


- Board functions:
 - MPU: a mandatory component that controls, manages, and monitors the system.
 - SFU: a mandatory component that implements system control, monitoring, and data exchange.
 - LPU: an optional component that processes data packets and manages traffic.
 - SPU: an optional component that performs value-added service processing on the required data packets.
 - CMU: an optional component that monitors the environment status and allows users to configure the ambient temperature threshold for monitoring.
- CloudEngine S12700H-8 chassis structure:
 1. Chassis header
 2. MPU
 3. Cable manager: used to separate cables in different slots and facilitate cable routing.
 4. SFU
 5. LPU
 6. Four power modules (PWR)

7. CMU
8. Electrostatic Discharge (ESD) jack on the front side: a jack for the ESD wrist strap. The ESD function is available only when the device is properly grounded.
9. Two fan modules
10. Ground point for a two-hole OT terminal, with a yellow grounding label attached

Chassis Structure of Fixed Devices

- Unlike modular devices, fixed devices use a fixed architecture. Their service modules are not independent boards, but are integrated into a fixed chassis. They are typically deployed at the access layer of a network or on small enterprise networks, and have a fixed number of access ports.
- For example, S series Ethernet switches use a centralized hardware platform, which consists of the chassis, power modules, fan modules, and Switch Control Unit (SCU).



- CloudEngine S5755-H24T4Y2CZ chassis structure:

1. One PNP button:

- Holding down the button for at least 6 seconds: The device restores factory settings and then is reset.
- Pressing the button: The device is reset.
- Resetting the device will interrupt services. Exercise caution when performing this operation.

2. One MODE button

3. 24 × 10/100/1000BASE-T ports

4. 4 × GE/10GE/25GE SFP28 ports

5. 2 × 40GE/100GE QSFP28 ports

6. Ground screw: used with a ground cable.

7. One console port

8. One USB port

9. One ETH management port

10. Card slot

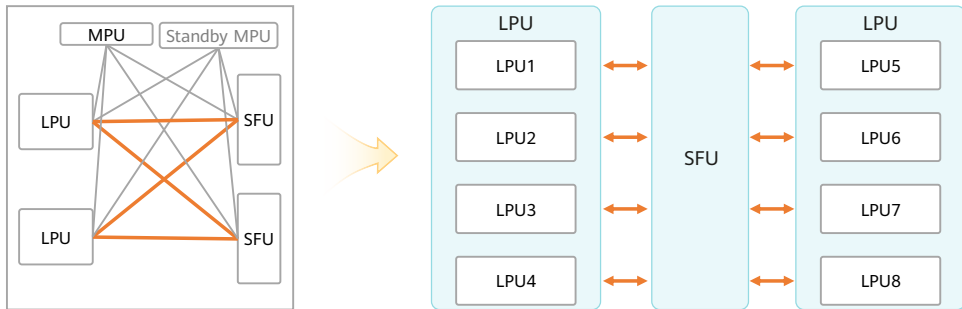
11. Fan module slot 1

12. Fan module slot 2

13. Power module slot 1

14. Power module slot 2

Logical Diagram of Module Connections



- The modules of a modular device are divided into different boards, which communicate with each other through internal connections of the modular device.
- LPUs are connected to each other through SFUs. The communication between LPUs is uniformly forwarded through SFUs.

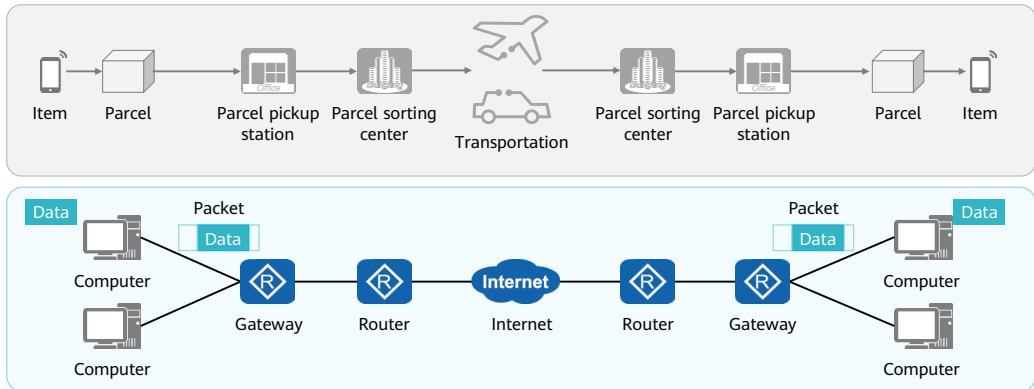
- The fixed device integrates these functional modules, and the modules communicate with each other through internal connections.

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Data Communication Process

- The process of virtual information transfer shares many similarities with the transfer of real physical objects.



- Packing and labeling (sender):
 - Sender: The item is packaged into a parcel, and a shipping label containing the sender's and recipient's addresses is attached.
 - Data sender: The application encapsulates the original "data payload" by adding a "header" and a "trailer" to form a packet. The key information in the packet is the sender's and receiver's addresses, that is, the "source address" and the "destination address."
- Local dispatch (first transfer station):
 - Parcel pickup station: The parcel is received by a courier and placed in the local parcel pickup station. The station determines where to send the parcel based on the recipient's address and notifies the next courier to collect it.
 - Gateway: The packet travels through the network cable to the "gateway." The gateway decapsulates the packet, reads the destination address, and then re-encapsulates it. Depending on the destination address, the packet leaves the local network and is sent to different "routers."
- Network sorting and transmission (intermediate node):
 - Parcel sorting center and transportation: The city's parcel sorting center forwards parcels to the destination city's sorting center using different transportation methods, based on the recipient's address.
 - Routers and the Internet backbone: The intermediate routers continuously decapsulate and encapsulate the packet, and then send the packet to the Internet backbone for transmission. Finally, the packet is sent to the

destination network.

- Final delivery and reception (receiver)
 - Recipient: The sorting center in the destination city checks the recipient's address, dispatches the parcel to the appropriate parcel pickup station, and finally the courier delivers it to the recipient. The recipient opens the parcel, confirms the item is intact, and accepts it. The entire delivery process is complete.
 - Data receiver: The packet arrives at the gateway of the destination computer's network, where it is decapsulated and encapsulated again, then forwarded to the appropriate computer based on the destination address. The computer verifies the packet, and if correct, accepts it and passes the data payload to the corresponding application for processing. A complete network communication process is thus finished.

Common Terms

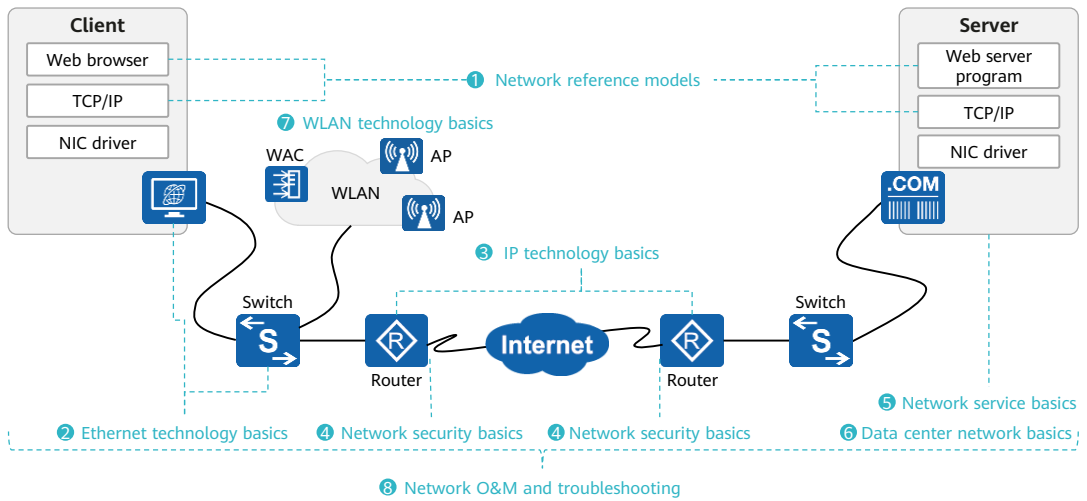
Term	Description
Data payload	Actual information intended to be transmitted
Packet	Data unit exchanged and transmitted on a network.
Header	Segment of information added before the data payload.
Trailer	Segment of information added after the data payload.
Encapsulation	Process of adding a header and a trailer to a data payload to form a new packet.
Decapsulation	Process of removing the header and trailer from a packet to obtain the data payload.
Gateway	A device that connects two different networks.
Terminal	End device of a data communication system, serving as a sender or receiver of data.

- **Data payload:** It can be understood as the actual information intended to be transmitted. In fact, within a layered communication structure, the data unit (packet) passed from an upper-layer protocol to a lower-layer protocol can be regarded as the data payload of the lower-layer protocol.
- **Packet:** The unit of data exchanged and transmitted on a network, with a certain internal format. It usually has the basic structure of "header + data payload + trailer." During transmission, the format and content of the packet may change.
- **Header:** To enable better information delivery, when assembling a packet, the segment of information added before the data payload is collectively referred to as the header.
- **Trailer:** To enable better information delivery, when assembling a packet, the segment of information added after the data payload is collectively referred to as the trailer. Note that many packets do not have a trailer.
- **Encapsulation:** A technique used in layered protocols. When a lower-layer protocol receives a message from an upper-layer protocol, it attaches that message to the data payload portion of the lower-layer message.
- **Decapsulation:** The reverse process of encapsulation, which is to remove the header and trailer of a packet to obtain the data payload.
- **Gateway:** A network device used to enable communication between networks that adopt different architectures or protocols. It provides functions such as protocol conversion, routing selection, and data exchange. A gateway is a term named according to its deployment location and function, rather than a specific

device type.

- Terminal: An end device in a data communication system, acting as either the sender or receiver of data. It provides the necessary functions for user access and protocol operations, and can be a computer, server, mobile phone, or the like.

Journey of This Course

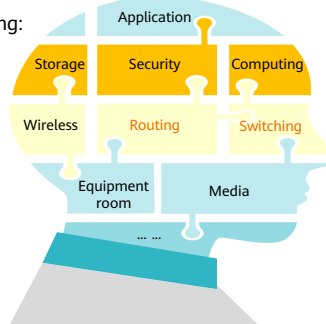


Contents

1. Data Communication Network
2. Data Communication Network Devices
3. Data Communication Process
- 4. Network Engineering and Network Engineers**

Network Engineering

- Network engineering:
 - Under the guidance of information system engineering methodologies and a well-established organizational structure, and based on the requirements of network applications, a feasible plan is designed in accordance with the standards, specifications, and technologies of computer network systems. This plan systematically integrates computer network hardware, software, and technologies to build a network system that meets user needs and offers high cost-effectiveness.
- Technical modules covered by network engineering:



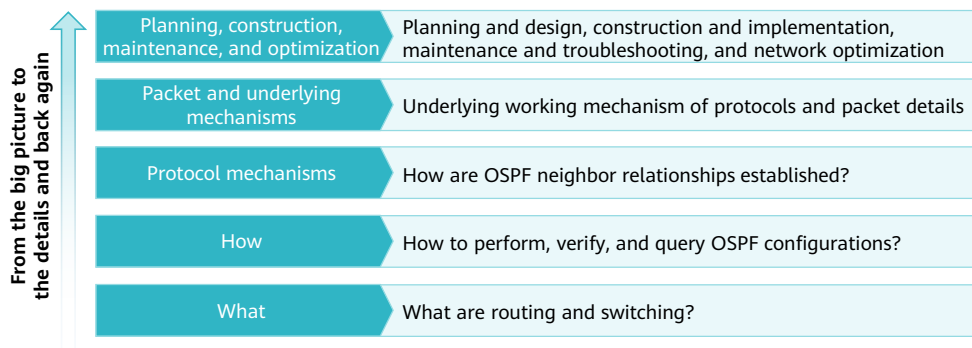
- Network engineering refers to a series of activities centered around networks, including network planning, design, implementation, testing, and troubleshooting.
- The knowledge domain of network engineering design is extensive, and routing and switching are the basics of computer networks.

Network Engineer

- Network engineer:
 - A professional in the field of network engineering who possesses specialized network technologies, professional skills, and professional qualities. With certain project implementation experience, a network engineer is able to communicate effectively onsite with customers or other project stakeholders, develop implementation plans and project schedules according to customer requirements and environmental factors (and gain stakeholder approval), mobilize resources to ensure the project is delivered on time with high quality, and, after project completion, provide training to stakeholders and deliver engineering documentation.
- Comprehensive capability models for network engineers:

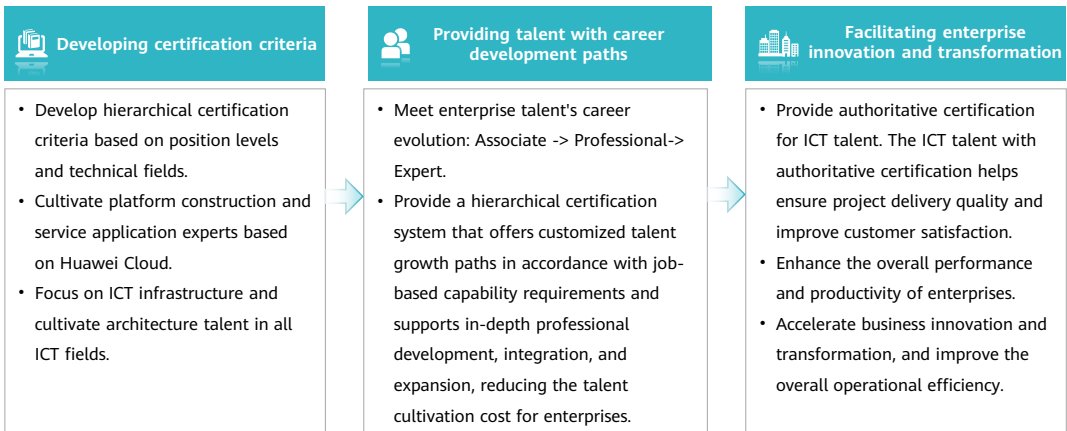
Process specifications	Business etiquette	Team collaboration
Industry knowledge	Values	Service management
Engineering knowledge	Service awareness	Presentation capability
Product expertise	Information collection	Solution
Technical knowledge	Learning competency	Communication capability
Professional knowledge	Basic qualification	Professional skills

Technical Growth Path of a Network Engineer



- OSPF: Open Shortest Path First

Huawei Certification Injects Vitality into Talent Development for Enterprises

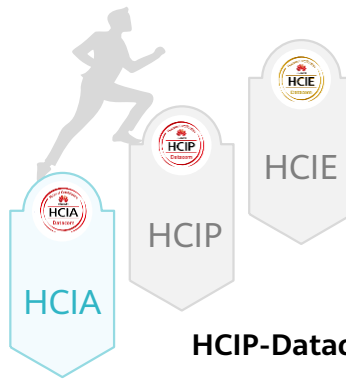


- ICT: Information & Communications Technology

Huawei Datacom Certification System

HCIA-Datacom

Aiming to cultivate network engineers with basic datacom theories and deployment skills



HCIP-Datacom

Aiming to cultivate senior network engineers with advanced knowledge and deployment skills in the datacom field

HCIE-Datacom

Aiming to cultivate network experts with solid theoretical foundation and deployment capabilities for cross-field solutions

Quiz

1. (Multiple-answer question) Which of the following communication types belong to network communications? ()
- A. Drum signaling
 - B. Pigeon post
 - C. Sending an email
 - D. Transferring files through network cables

1. CD

Summary

- This course describes the concepts of network communication and data communication networks. The basic function of data communication networks is to implement network communication.
- To understand data communication networks, it is necessary to grasp their fundamental concepts and key components, including network topologies, network types, and the various network devices and their functions.
- Finally, this course also introduces network engineering and network engineers. In addition to fundamental professional qualities, a network engineer also needs to acquire relevant skills and knowledge. Huawei datacom certification provides a specialized learning path to achieve this.

Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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Network Reference Models



Foreword

- In the digital era, information in the form of data permeates every aspect of our lives. What is data? How is it transmitted?
- To better transmit data, standards organizations and device vendors have introduced various network protocols. How do these protocols work together to effectively transmit data on data communication networks?
- In this course, we will use the network reference models to gain a simple understanding of the "life" of data.

Objectives

- Upon completion of this course, you will be able to:
 - Understand the definition of data and its transmission process.
 - Understand the concepts and advantages of network reference models.
 - Gain knowledge of common standard protocols.
 - Master the processes of data encapsulation and decapsulation.

Contents

- 1. Applications and Data**
2. Network Reference Models and Standard Protocols
3. Data Communication Process

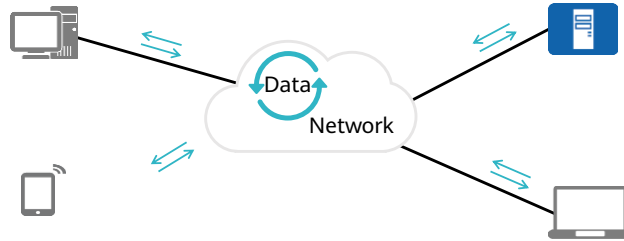
Origin of the Story: Applications

- Applications exist to meet various human needs, such as web browsing, online gaming, and video streaming.
- Along with applications, information is generated—for example, text, images, and videos are different forms of information presentation.



Application Implementation: Data

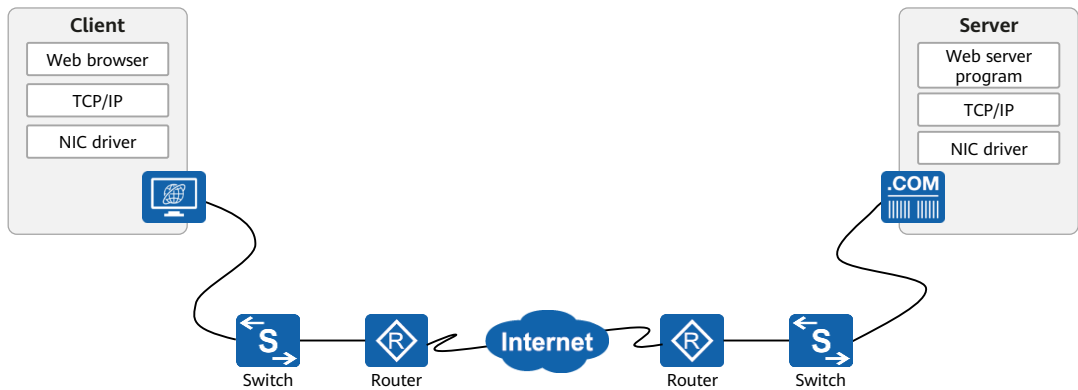
- Data generation: In the computer field, data serves as the carrier of various types of information.
- Data transmission: The data generated by most applications needs to be transmitted between devices.



- Computers can only recognize digital data composed of 0s and 1s. They do not have the ability to directly interpret various forms of information, so information must be translated into data according to certain rules.
- Humans, however, do not have the ability to read digital data. Therefore, data needs to be converted into a form that humans can understand.
- Network engineers need to pay attention to the end-to-end data transmission process.

Data Transmission Process

- A web client sending data to a web server is a typical example of a data transmission process.



- This course will help you understand the reference models and various network protocols used in data transmission.

Contents

1. Applications and Data
- 2. Network Reference Models and Standard Protocols**
3. Data Communication Process

Network Protocols and Standards Organizations

- In network communication, a network protocol refers to a series of pre-defined rules and conventions that a network device must comply with to implement communication.
- Network protocols can be classified into two types: open protocols defined by dedicated standards organizations and proprietary protocols defined by network device vendors.
- A standards organization is specifically responsible for organizing, researching, formulating, and publishing open standard protocols.
- Well-known network standards organizations:

Standards Organization	Description
Internet Engineering Task Force (IETF)	IETF is a voluntary organization responsible for developing and promoting Internet protocols (especially protocols that constitute the TCP/IP protocol suite), and releasing new or replacing old protocol standards through RFCs.
Institute of Electrical and Electronics Engineers (IEEE)	IEEE has established approximately 30% of the global standards in the fields of electronics, electrical engineering, and computer science. Those standards include well-known IEEE 802.3 (Ethernet) and IEEE 802.11 (Wi-Fi).
International Organization for Standardization (ISO)	ISO is an international organization that plays a significant role in developing computer network standards. For example, ISO defines the OSI reference model, which is defined in ISO/IEC 7498-1.

- Requirement for Comments (RFC): a publication of a standard, a protocol, or other information related to Internet operations.
- International Electrotechnical Commission (IEC): an international non-governmental standards organization that deals with electrical and electronic standards. It is responsible for international standardization in the electrical and electronic fields.

Network Reference Model: OSI Reference Model

- In the 1980s, ISO proposed the well-known Open Systems Interconnection Reference Model (OSI-RM). The emergence of the OSI reference model greatly promotes the development of network technologies.
- The OSI reference model is a seven-layer protocol model.

7. Application layer	Provides network service interfaces for applications.
6. Presentation layer	Converts data formats.
5. Session layer	Manages communication sessions.
4. Transport layer	Provides end-to-end reliable or efficient data transmission services.
3. Network layer	Performs logical addressing and path selection.
2. Data link layer	Transmits data frames reliably within the same LAN.
1. Physical layer	Transmits raw bit streams over physical media.

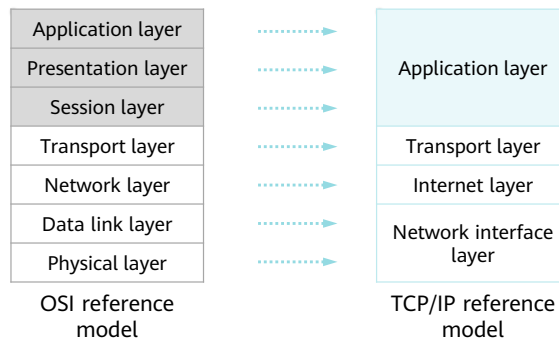
- Various functions of a network device are implemented by running various network protocols. Establishing a layered protocol model facilitates independent and free development of protocols, reduces association between protocols at different layers, and facilitates learning and understanding.
- The OSI reference model is also called the seven-layer model. The seven layers from top to bottom are as follows:
 - Application layer: provides network services for applications and is closest to users.
 - Presentation layer: provides data encoding and conversion functions so that data sent by the application layer of one system can be identified by the application layer of another system.
 - Session layer: establishes, manages, and terminates sessions between entities at the presentation layer. Communication at this layer is implemented through service requests and responses transmitted between applications on different devices.
 - Transport layer: establishes, maintains, and cancels an end-to-end data transmission process, controls transmission speeds, and adjusts data sequencing.
 - Network layer: defines logical addresses for Layer 3 devices such as routers to determine paths and transmits data from the source network to the destination network.
 - Data link layer: combines bits into bytes and then combines bytes into data

frames, implements point-to-point or point-to-multipoint direct communication of data on the data link, and performs error detection.

- Physical layer: transmits bit streams between devices and defines physical specifications such as electrical levels, speeds, and cable pins.

Network Reference Model: TCP/IP Reference Model

- The OSI reference model is complex, and the Transmission Control Protocol (TCP) and Internet Protocol (IP) are widely used in the industry. As such, the TCP/IP reference model becomes the mainstream reference model of the Internet.



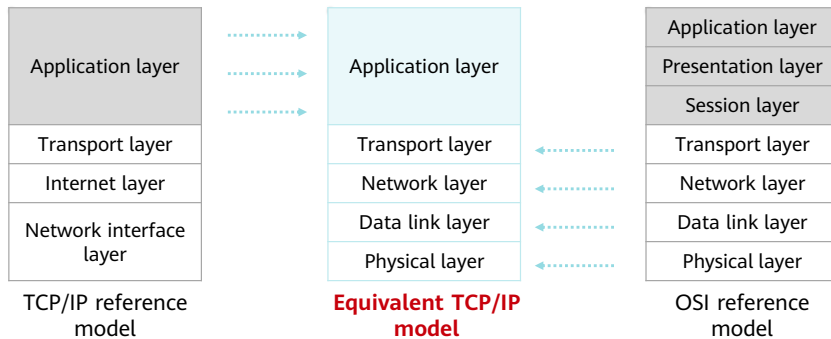
- TCP/IP reference model:
 - The TCP/IP reference model originated from the design and implementation of ARPANET. It was later enriched and refined by the IETF.
 - Unlike the theoretical OSI model, the TCP/IP reference model is practice-driven. It is a layered model summarized and abstracted from numerous RFC descriptions of specific protocols.
 - In the early 1980s, RFC 793 and RFC 791 were proposed, defining TCP (the core protocol of the transport layer) and IP (the core protocol of the network layer). These early protocol documents implied the hierarchical relationship of the network layer below and the transport layer above.
 - In the late 1980s, RFC 1122 and RFC 1123 were proposed, defining the host implementation standards and application-layer protocols of the four-layer TCP/IP model.
- The TCP/IP reference model is similar to the OSI reference model but adopts a four-layer architecture, making it simpler and easier to implement and deploy.
 - Application layer: defines user protocols (such as FTP and Telnet) that directly provide services for users and protocols (such as DNS) that support general system functions.
 - Transport layer: provides end-to-end communication services for the application layer. There are two main protocols at the transport layer: TCP and UDP.
 - Internet layer: implements host-to-host data forwarding across networks.

The main Internet protocol is IP.

- Network interface layer: is responsible for transmitting raw data frames between adjacent network nodes on a single physical network.
- In practice, the TCP/IP reference model can be referred to as the TCP/IP model, TCP/IP functional model, TCP/IP protocol model, TCP/IP protocol suite, or TCP/IP protocol stack.

Network Reference Model: Equivalent TCP/IP Model

- To meet the requirements of teaching and engineering practices, the equivalent TCP/IP model that integrates the TCP/IP reference model and the OSI reference model is proposed.



Note: This course introduces related content based on the equivalent TCP/IP model.

- Equivalent TCP/IP model:
 - The equivalent TCP/IP model is not an official model defined by standards organizations. It is a protocol framework that evolves from the practices of academia and engineering, and is used as a teaching model.
 - The equivalent TCP/IP model splits the network interface layer in the four-layer standard model into the physical layer and data-link layer, making it closer to real-world network implementations and more helpful for theoretical study and troubleshooting.
 - The equivalent TCP/IP model can help engineers better understand the details of the layered architecture by comparing it with the TCP/IP reference model and OSI reference model.
- For network engineers, the focus is on layers 1 to 4 in the model, namely, the physical layer, data link layer, network layer, and transport layer. For IT engineers, the focus is on layer 5 in the model, namely, the application layer.

Common TCP/IP Protocols

- The TCP/IP protocol stack defines a series of standard protocols.

Application layer	Telnet	FTP	TFTP	SNMP
	HTTP	SMTP	DNS	DHCP
Transport layer	TCP		UDP	
Network layer	ICMP		IGMP	
	IP			
Data link layer	PPPoE			
	Ethernet		PPP	
Physical layer	...			

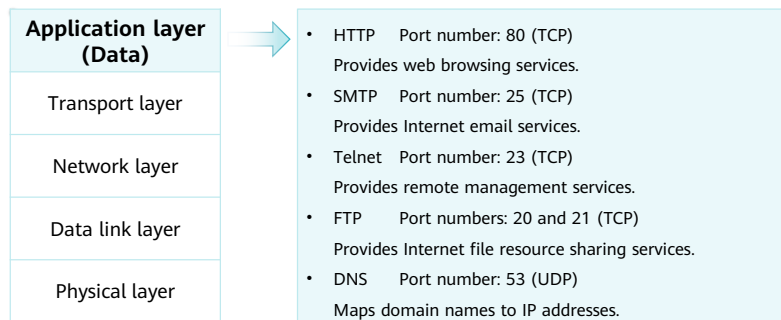
- Each layer of the TCP/IP model enables data to be transmitted over the network. These layers exchange information with each other using protocol data units (PDUs) to ensure that network devices can communicate with each other.
- The PDU at each layer contains different information, and therefore is given a different name.

- Application layer
 - Telnet: a remote login protocol used to remotely connect to a terminal. It allows users to log in to remote systems and use resources as if they were connected to a local system.
 - File Transfer Protocol (FTP): provides a method for transferring files. It allows data to be transferred from one host to another.
 - Trivial File Transfer Protocol (TFTP): a simple alternative to FTP, used in applications where complex client-server interactions are not required, limiting its service to basic file transfers without verification.
 - Simple Network Management Protocol (SNMP): allows remote users to view and modify the management information about NEs. It ensures that the management information is transmitted between any two points.
 - Hypertext Transfer Protocol (HTTP): used to access various pages on the web server.
 - Simple Mail Transfer Protocol (SMTP): used to transfer emails between computers.
 - Domain Name Service (DNS): translates host domain names into IP addresses.
 - Dynamic Host Configuration Protocol (DHCP): A DHCP server provides configuration parameters for DHCP clients upon request. These parameters typically include information necessary for clients to connect to the Internet. DHCP also provides the function of assigning IP addresses to hosts.

- Transport layer
 - Transmission Control Protocol (TCP): provides reliable, connection-oriented communication services for applications. Currently, TCP is used by many popular applications.
 - User Datagram Protocol (UDP): provides connectionless communication, without guaranteeing the reliability of transmitted data packets.
- Network layer
 - Internet Protocol (IP): encapsulates transport-layer data into data packets and forwards packets from source sites to destination sites. IP provides a connectionless and unreliable service.
 - Internet Control Message Protocol (ICMP): sends control messages based on the IP protocol and provides information about various problems that may exist in the communication environment. Such information helps administrators diagnose problems and take proper measures to resolve the problems.
 - Internet Group Management Protocol (IGMP): manages multicast group memberships. Specifically, IGMP sets up and maintains memberships between IP hosts and their directly connected multicast routers.
- Data link layer
 - Ethernet: a broadcast multi-access data link layer protocol, which is the most widely used LAN technology.
 - Point-to-Point Protocol (PPP): a data link layer protocol used in point-to-point mode, commonly employed in WANs.
 - Point-to-Point Protocol over Ethernet (PPPoE): enables multiple hosts on a network to connect to a remote access concentrator through a simple bridging access device. Common applications include dial-up Internet access in home broadband.

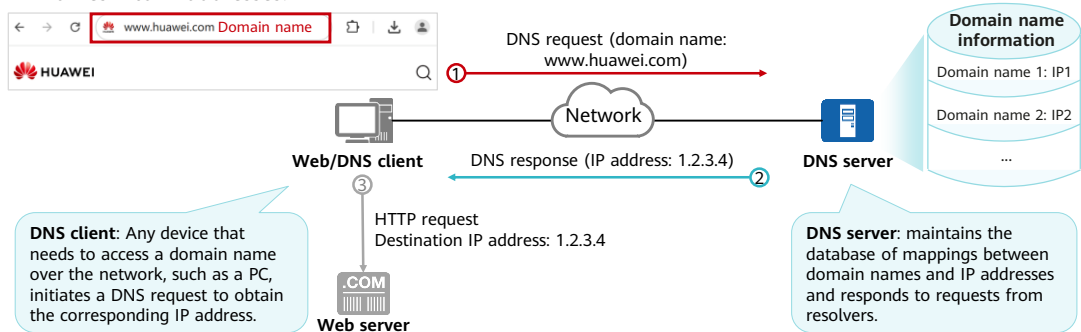
Application Layer

- The application layer provides interfaces for application software so that applications can use network services. The application layer protocols designate transport layer protocols and ports.
- PDUs transmitted at the application layer are called **data**.



Common Application Layer Protocol: DNS

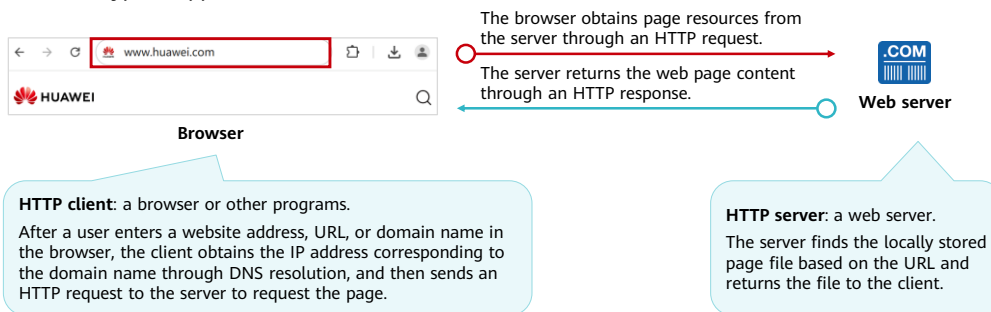
- When you enter a domain name in the browser to access a website, the domain name is resolved into an IP address, and the browser actually communicates with the IP address.
- The Domain Name System (DNS) is a typical client/server (C/S) application that resolves domain names into IP addresses.



- Each node on a network has its own unique IP address, and nodes can communicate with each other through these IP addresses. However, if communication with all nodes relies solely on IP addresses, it becomes difficult for people to remember so many of them. To solve this problem, DNS was introduced, which maps hard-to-remember IP addresses to character-based domain names.
- Domain name: a character-based identifier of a host. In most cases, the website address entered in the browser is the domain name of the website.
- Note:
 - C/S: client/server model
 - The domain names and IP addresses in the figure are only examples.

Common Application Layer Protocol: HTTP

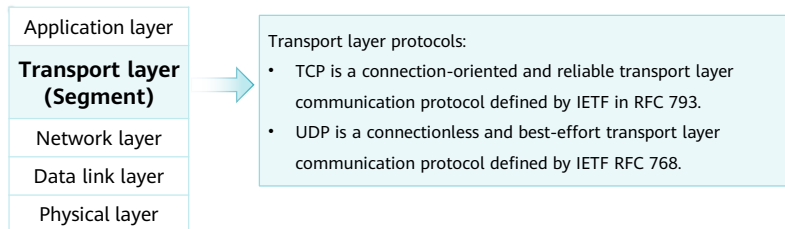
- Hypertext Transfer Protocol (HTTP) is the most widely used network protocol on the Internet. It is an application-layer communication protocol between a client browser or other programs and a web server.
- HTTP is a typical application of the C/S architecture.



- A URL uniquely identifies the address of a web page or other types of resource on the Internet.
- Common types of URLs are as follows:
 - URLs for accessing web pages, for example, `http://www.huawei.com`, which contains the protocol information **http://** and domain name information **www.huawei.com**.
 - FTP server addresses for file transfer, for example, `ftp://example.com/file.zip`, which contains the protocol information **ftp://**, FTP server domain name **example.com**, and file name **file.zip**.

Transport Layer

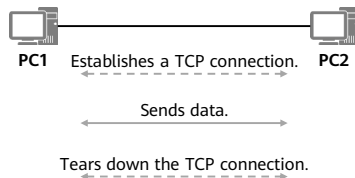
- A transport layer protocol receives data from an application layer protocol, encapsulates the data with the corresponding transport layer protocol header, and helps establish an end-to-end (port-to-port) connection.
- PDUs transmitted at the transport layer are called **segments**.



Transport Layer Protocols: TCP and UDP

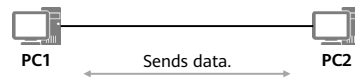
Transmission Control Protocol (TCP)

- Characteristics: A connection is established before communication. During the transmission, TCP ensures no data loss, no duplication, no errors, and in-order delivery.
- Application scenarios: web browsing, file transfer, email, and other data transmission scenarios demanding high data integrity.



User Datagram Protocol (UDP)

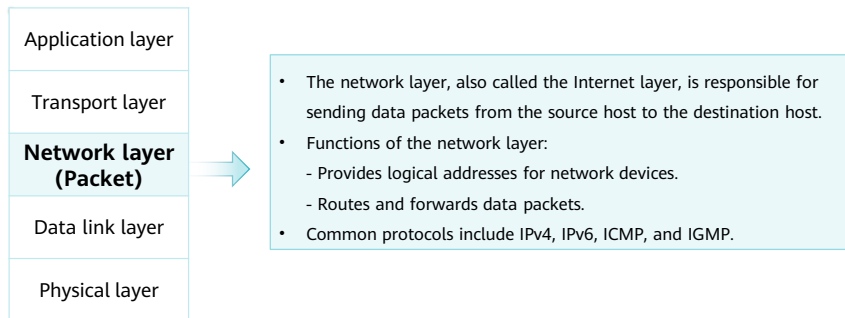
- Characteristics: Data packets are directly sent without establishing a connection. UDP does not ensure reliability but has extremely low overhead and latency.
- Application scenarios: video calls, online gaming, DNS queries, and other data transmission scenarios that demand low latency while tolerating a small amount of packet loss.



- TCP and UDP use the source port number and destination port number to identify the application that sends data and the application that receives data.

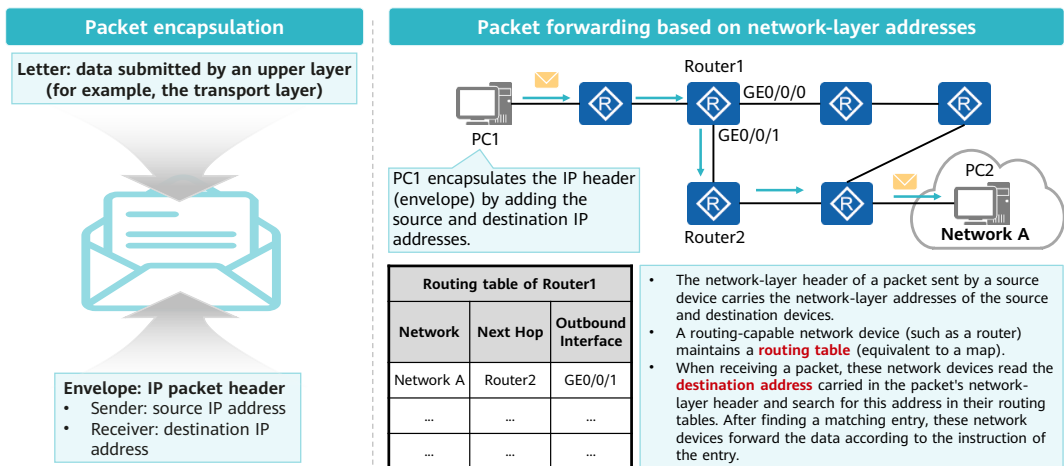
Network Layer

- The network layer transmits data from a host to another host.
- PDUs transmitted at the network layer are called **packets**.



- IPv4: Internet Protocol version 4 (IPv4) is the most widely used network layer protocol.
- IPv6: Internet Protocol version 6.

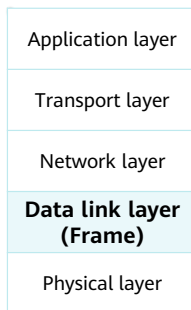
Working Process of a Network Layer Protocol



- When IP is used as the network layer protocol, both communication parties are assigned a unique IP address to identify themselves. An IP address can be written as a 32-bit binary integer. To facilitate reading and analysis, an IP address is usually represented in dotted decimal notation. In this format, the four bytes are separated and represented in decimal, with dots in between, for example, 192.168.1.1.
- Encapsulation and forwarding of IP data packets:
 - When receiving data from an upper layer (such as the transport layer), the network layer encapsulates an IP packet header and adds the source and destination IP addresses to the header.
 - Each intermediate network device (such as a router) maintains a routing table that guides IP packet forwarding like a map. After receiving a packet, the intermediate network device reads the destination address of the packet, searches the local routing table for a matching entry, and forwards the IP packet according to the instruction of the matching entry.
 - When the IP packet reaches the destination host, the destination host determines whether to accept the packet based on the destination IP address and then processes the packet accordingly.
- When the IP protocol works, routing protocols such as OSPF and BGP are used to help routers establish routing tables, and ICMP is used to control the network and diagnose the network status.

Data Link Layer

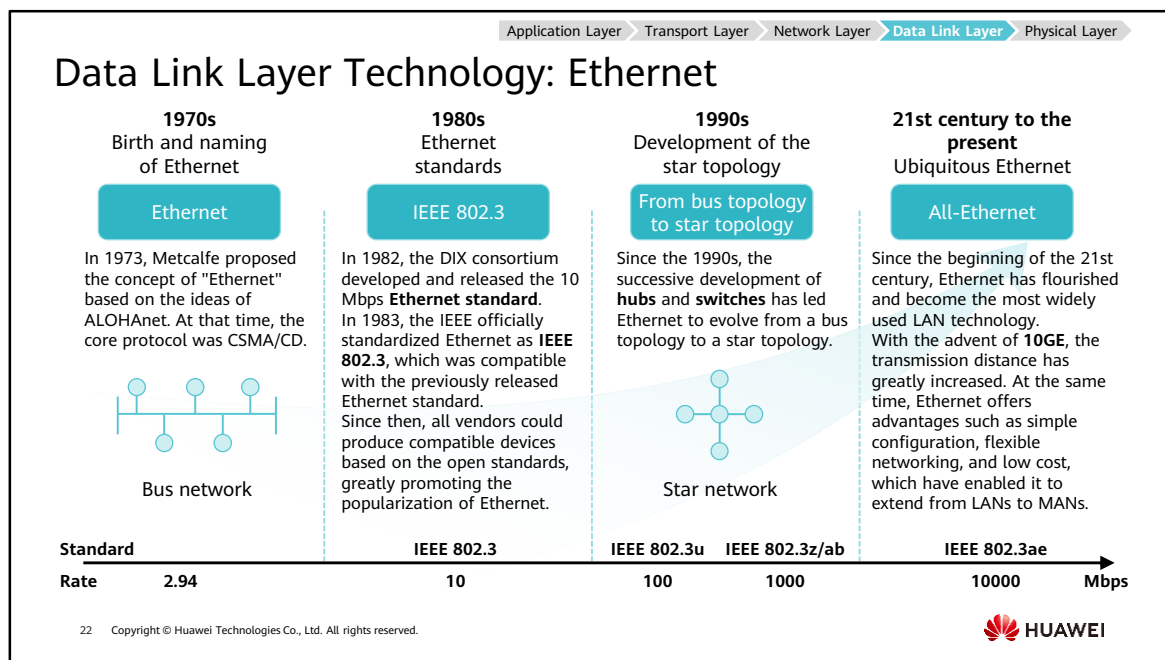
- The data link layer is located between the network layer and the physical layer and provides services for protocols such as IPv4 and IPv6 at the network layer. Ethernet is the most common data link layer protocol.
- PDUs transmitted at the data link layer are called a **frames**.



The data link layer is located between the network layer and the physical layer.

- The data link layer provides intra-segment communication for the network layer.
- The functions of the data link layer include framing, physical addressing, and error detection.
- Common data link layer protocols include Ethernet and PPP.

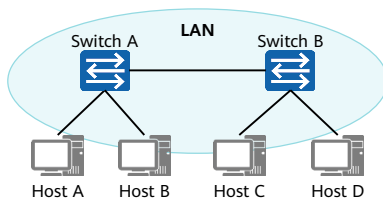
- Ethernet is the most common data link layer protocol used on LANs.
 - A LAN is a broadcast and multi-access environment. On a LAN, a terminal usually needs to access multiple services, or a server needs to provide services for multiple terminal users.
 - Ethernet is designed for such broadcast/multi-access bus or star topologies, supporting interconnection and data transmission between multiple devices.
- PPP is the most common data link layer protocol used on WANs.
 - A WAN covers a large area, spanning cities, countries, or even continents, and is used to connect LANs in different geographic locations. The core requirements of WAN links are security, reliability, and the ability to traverse third-party networks (such as carrier networks), while providing point-to-point connections between two specific endpoints, such as between an enterprise egress router and a carrier router.
 - PPP establishes a connection only between two devices, which is similar to a dedicated line. On this line, only you and the other party exist, so there is no need to use addresses to distinguish identities. The communication destination is unique. In addition, PPP has a built-in identity authentication method to ensure secure data transmission.



- Core idea of ALOHAnet: Any node can send data whenever it wants. If a collision occurs (that is, two nodes send data at the same time, causing signal confusion), the node will wait for a random period of time and then retry.
- Ethernet: The term "Ether" originates from a 19th-century physics hypothesis of an all-pervading medium filling the universe and enabling electromagnetic wave propagation. Metcalfe hoped that this new network would, like "Ether", become a ubiquitous foundational medium connecting all computers.
- Carrier Sense Multiple Access/Collision Detection (CSMA/CD): According to the original design objective of Ethernet, computers and other digital devices are connected through a shared physical line. Devices connected in this way have to adopt a collision detection and avoidance mechanism to prevent multiple devices from attempting to use the line at the same time. This mechanism is CSMA/CD.

Ethernet and MAC Address

Definition of Ethernet



- Ethernet is a set of communication protocol standards for LANs. It is a broadcast data link layer protocol and supports multi-point access. For example, the network interface of a PC complies with the Ethernet standard.
- Ethernet is mainly used to connect computers and high-speed network devices to implement fast data transmission.

Media Access Control (MAC) address

I have a unique MAC address when I leave the factory.



Host A

Name: Host A

ID number: MAC address/Ethernet address/Physical address

48 A4 72 1C 8F 4F

- A MAC address is a data link layer address, also known as an Ethernet address, physical address, or hardware address. It is written into the network card by the manufacturer during production and is used to define the location of a network device. For example, the network interface of a PC has a unique MAC address.
- Data link layer devices, such as Ethernet switches, maintain a MAC address table that guides the forwarding of data frames based on MAC addresses.

- Ethernet specifies the data frame format and media access control methods at the data link layer; it also defines the transmission media, electrical signals, and interfaces at the physical layer.
- A MAC address is 48 bits (6 bytes) long and consists of 12 hexadecimal digits, for example: 48-A4-72-1C-8F-4F.

Data Link Layer Technology: PPP

- The Point-to-Point Protocol (PPP) is a commonly used WAN data link layer protocol that provides point-to-point data encapsulation and transmission over full-duplex links.
- Application scenario: PPP is mainly used to connect enterprise branches to the headquarters (HQ) through PPP links.

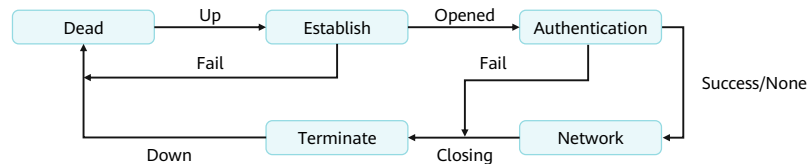


- As shown in the figure, the enterprise branch's egress router R1 is connected to the WAN-side interface of the enterprise HQ's egress router R2 through a PPP link. R1 obtains an IP address through PPP negotiation and establishes a connection with the WAN.

- Advantages of PPP:
 - PPP supports both synchronous and asynchronous links.
 - PPP features high extensibility. For example, PPP can be extended as Point-to-Point Protocol over Ethernet (PPPoE) when PPP packets need to be transmitted over an Ethernet.
 - PPP provides the Link Control Protocol (LCP) to negotiate various link layer parameters.
 - PPP provides authentication protocols Challenge-Handshake Authentication Protocol (CHAP) and Password Authentication Protocol (PAP) to enhance network security.
 - PPP provides the Network Control Protocol (NCP) to negotiate network layer parameters, thereby better supporting network layer protocols.
 - PPP has no retransmission mechanism, reducing the network cost and accelerating packet transmission.

PPP Link Establishment Process

- PPP link establishment involves link layer negotiation, authentication negotiation (optional), and network layer negotiation.
 - Link layer negotiation: LCP packets are used to negotiate link parameters and establish link layer connections.
 - (Optional) Authentication negotiation: The authentication mode (CHAP or PAP) negotiated during link establishment is used for link authentication.
 - Network layer negotiation: NCP negotiation is used to select and configure a network layer protocol and negotiate network layer parameters.

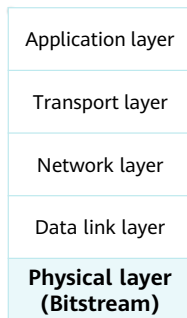


- The PPP link establishment process is as follows:
 1. Two communicating devices enter the Establish phase when starting to set up a PPP link.
 2. In the Establish phase, the two devices perform an LCP negotiation to negotiate parameters including the working mode, maximum receive unit (MRU), authentication mode, and magic number. The working mode can be either Single-link PPP (SP) or Multilink PPP (MP). If the LCP negotiation succeeds, LCP turns Opened, which indicates that a lower-layer link has been established.
 3. If authentication is configured, the two devices enter the Authenticate phase and perform CHAP or PAP authentication. If no authentication is configured, the two devices enter the Network phase.
 4. In the Authenticate phase, if authentication fails, the devices enter the Terminate phase. The link is removed and LCP turns Down. If authentication succeeds, the devices enter the Network phase and LCP remains Opened.
 5. In the Network phase, the two devices perform an NCP negotiation to select and configure a network layer protocol and to negotiate network layer parameters. After the two devices succeed in negotiating a network layer protocol, packets can be sent over this PPP link using the network layer protocol. NCP negotiation includes IP Control Protocol (IPCP) negotiation and MPLS Control Protocol (MPLSCP) negotiation. IPCP mainly negotiates the IP addresses of the two devices.

6. After NCP negotiation succeeds, packets can be sent over the PPP link. During the PPP operation, the PPP connection can be terminated at any time. A physical link disconnection, authentication failure, timeout timer expiry, or connection close by administrators through configuration can cause the two devices to enter the Terminate phase.
7. In the Terminate phase, the two devices enter the Dead phase after all resources are released. The two devices remain in the Dead phase until they start to establish a new PPP connection.

Physical Layer

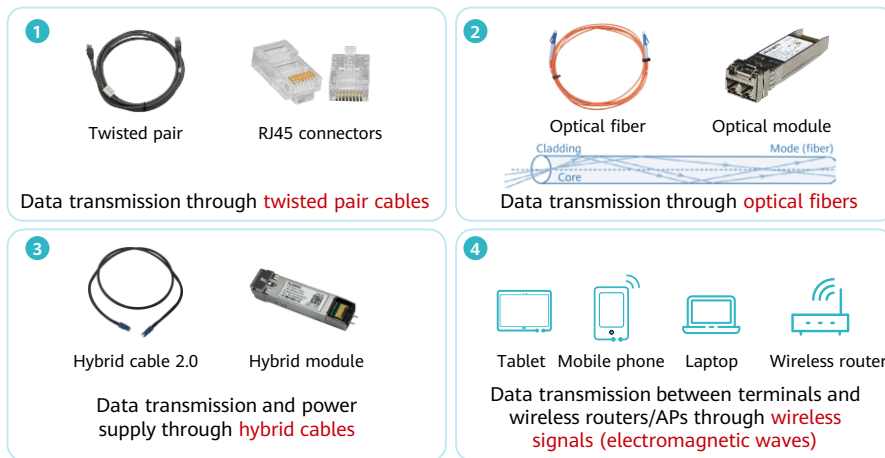
- After data reaches the physical layer, the physical layer converts digital signals into electrical signals, optical signals, or electromagnetic wave signals, depending on the physical medium.
- PDUs transmitted at the physical layer are called **bitstreams**.



The physical layer is at the bottom of the model.

- This layer transmits bitstreams on media.
- It standardizes physical features such as cables, pins, voltages, and interfaces.
- Common transmission media include twisted pairs, optical fibers, hybrid cables, and electromagnetic waves.

Common Transmission Media



- Twisted pair cable: Commonly known as network cable, it is the most widely used transmission medium on today's Ethernet. It connects and maintains the network interfaces of terminals and devices for local or remote maintenance.
 - Depending on frequency and signal-to-noise ratio (SNR), common cables include Cat 5, Cat 5e, and Cat 6 cables. These are twisted pair cables that use RJ45 connectors, with a maximum transmission distance of 100 m.
 - Depending on whether the shield layer is available, network cables can be classified into shielded twisted pair (STP) and unshielded twisted pair (UTP).
 - STP cables can reduce radiation and prevent information from being intercepted and external electromagnetic interference from entering. Compared with the same type of UTP cables, STP cables boast higher transmission rate, but they are more expensive and more difficult to install.
 - UTP cables feature low costs and light weight and are easy to bend. They rarely cause great impact on common networks. Therefore, UTP cables are more widely used.
- Optical fiber: It is a medium for optical transmission, essentially a strand of glass fiber that serves as a channel to confine optical transmission.
 - Optical fibers are generally classified into single-mode fibers and multimode fibers.
 - Single-mode fiber: transmits signals in only one mode at a given

operating wavelength. It features a wide transmission bandwidth and large transmission capacity, and is therefore suitable for long-distance transmission.

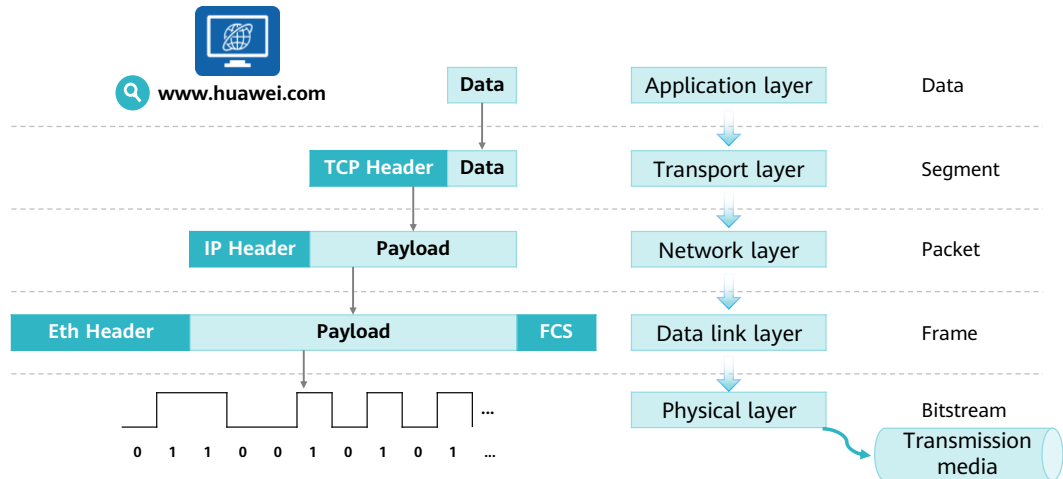
- Multimode fiber: transmits signals in multiple modes at a given operating wavelength, resulting in mode dispersion. They have a small capacity and their performance is inferior to that of single-mode fibers, making them suitable to short-distance transmission.
- Optical module: When signals are transmitted on an optical network, conversion between electrical signals and optical signals is a must. An optical module is a component that completes conversion between electrical signals and optical signals on an optical network.
- Hybrid cable: It incorporates optical fibers and copper wires within the same jacket, and can supply power to devices while transmitting data.
 - A hybrid cable incorporates optical fibers and copper wires within the same jacket. It transmits data signals through optical fibers and transmits electrical signals through copper wires, enabling long-distance power supply while ensuring high-speed data transmission.
 - Hybrid cables fall into two generations based on their connector types: hybrid cable 1.0 and hybrid cable 2.0.
 - Hybrid cable 1.0: It occupies an electrical port and an optical port when being connected to a device. To elaborate, copper wires in such a cable need to be terminated by an RJ45 connector or Phoenix connector before being inserted into the electrical port, while optical fibers in the cable need to be terminated by a common LC connector before being inserted into a common commercial optical module on the optical port. The optical port transmits data, whereas the electrical port is responsible for PoE supply.
 - Hybrid cable 2.0: It occupies just one hybrid optical-electrical port when being connected to a device. That is, the cable is fusion spliced with a pigtail or jumper with a PDLC connector before being inserted into a hybrid module on the hybrid optical-electrical port. The hybrid optical-electrical port is responsible for both data transmission and PoE power supply.
 - Hybrid module: supports optical and electrical transmission and allows for PoE power supply. It can connect to a hybrid cable 2.0 or a common optical cable.

- Electromagnetic wave: Wireless signals can be transmitted through electromagnetic waves. For example, a wireless router modulates data and sends the data through electromagnetic waves, and a wireless network interface card of a mobile terminal demodulates the electromagnetic waves to obtain the original data. Data transmission from the wireless router to mobile terminals is then complete.

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1. Applications and Data
2. Network Reference Models and Standard Protocols
- 3. Data Communication Process**

Data Encapsulation on the Sender



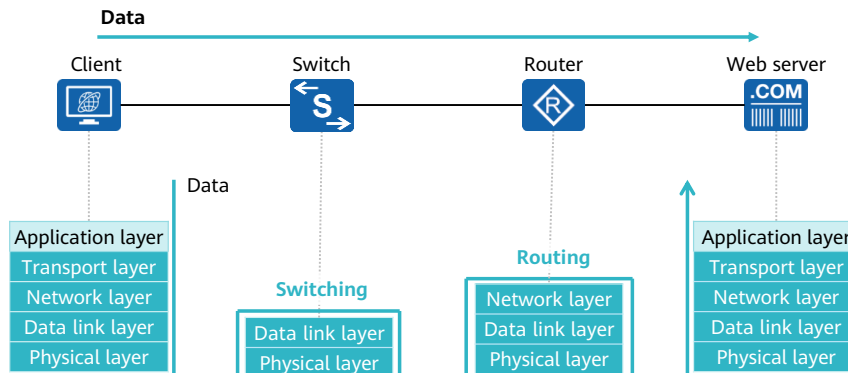
- Assume that you are using a web browser to access Huawei's official website. After you enter the website address and press Enter, the following events occur on your computer:
 1. The browser (application) invokes the HTTP (application layer protocol) to encapsulate the application layer data. (Data in the figure should also include the HTTP header, which is not shown here.)
 2. HTTP uses TCP to ensure reliable data transmission and transmits the encapsulated data to the TCP module.
 3. The TCP module adds the corresponding TCP header information (such as the source and destination port numbers) to the data received from the application layer. The PDU is called a segment. On an IPv4 network, the TCP module sends the encapsulated segment to the IPv4 module at the network layer. (On an IPv6 network, the segment is sent to the IPv6 module for processing.)
 4. After receiving the segment from the TCP module, the IPv4 module encapsulates the IPv4 header. Here, the PDU is called a packet. Ethernet is used as the data link layer protocol. Therefore, after the IPv4 module completes encapsulation, it sends the packet to the Ethernet module (such as the Ethernet NIC) at the data link layer for processing.
 5. After receiving the packet from the IPv4 module, the Ethernet module adds the corresponding Ethernet header and Frame Check Sequence (FCS) frame trailer to the packet. Now, the PDU is called a frame. After the Ethernet module completes encapsulation, it sends the data to the physical

layer.

6. Depending on the physical medium, the physical layer converts digital signals into electrical signals, optical signals, or electromagnetic wave (wireless) signals. The converted signals are then transmitted over the transmission medium.

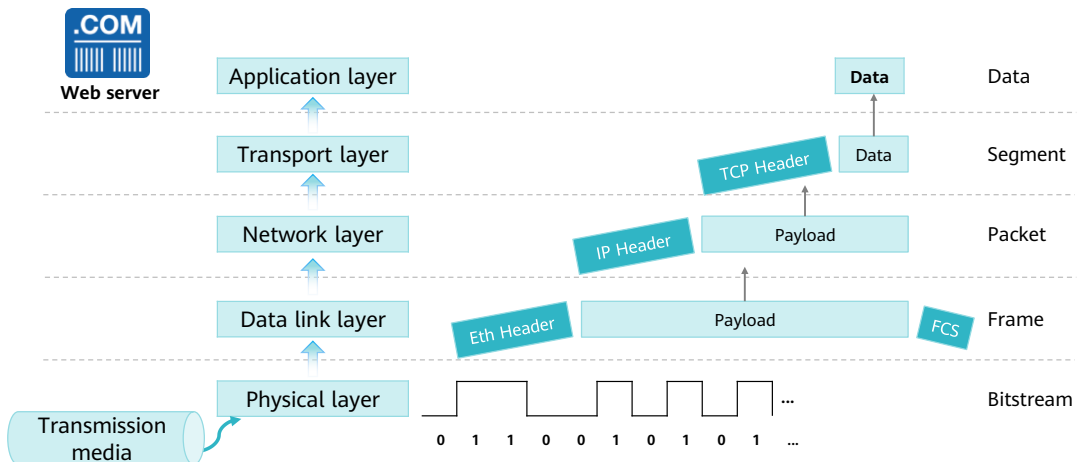
Data Transmission on an Intermediate Network

- The fully encapsulated data is transmitted on the network.



- In most cases:
 - A Layer 2 device (such as a switch) only decapsulates the Layer 2 header of the data and performs the corresponding "switching" operation based on the Layer 2 header information.
 - A Layer 3 device (such as a router) only decapsulates the Layer 3 header and performs the corresponding "routing" operation based on the Layer 3 header information.
- For details about switching and routing, see the courses Ethernet Switching Basics and IP Routing Basics.

Data Decapsulation on the Receiver



- After being transmitted over the intermediate network, the data finally reaches the destination server. Based on the information in different protocol headers, the data is decapsulated layer by layer, processed, transmitted, and finally sent to the application on the web server for processing.

Quiz

1. (Short-answer question) What are the advantages of a layered model?
2. (Short-answer question) What are the common protocols at the application layer, transport layer, network layer, and data link layer?

1. Advantages:

- Clear division of functions and boundaries between layers facilitates the development, design, and troubleshooting of each component.
- Defines the functions implemented at each layer of the model to encourage industry standardization.
- Interfaces can be provided to enable communication between hardware and software on various networks, improving compatibility.

2. Common protocols:

- Application layer: HTTP, DNS, FTP, and Telnet
- Transport layer: UDP and TCP
- Network layer: IP and ICMP
- Data link layer: Ethernet and PPP

Summary

- Both the OSI reference model and the TCP/IP reference model adopt the layered design concept.
 - Clear division of functions and boundaries between layers facilitates the development, design, and troubleshooting of each component.
 - The functions of each layer can be defined to impel industry standardization.
 - Interfaces can be provided to enable communication between hardware and software on various networks, improving compatibility.
- Data generation and transmission require collaboration between modules. Meanwhile, each module must fulfill its own responsibilities.

Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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Network Device Operating Systems



Foreword

- A network device operating system is a software system used to manage and control network devices. It provides functions such as communication, resource allocation, and security.
- Stable running and efficient management of network devices, such as switches, routers, and firewalls, depend on operating system support. Therefore, network engineers must master the methods of configuring operating systems.
- Upon completion of this course, you will be able to independently complete the initial configuration and basic O&M of network devices, laying a foundation for subsequent in-depth learning of network service configuration and management.

Objectives

- Upon completion of this course, you will be able to:
 - Understand the basic concepts of operating systems.
 - Master the login modes of network devices.
 - Perform basic CLI operations.
 - Manage file systems and configuration files.

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- 1. Operating System Overview**
2. Logging In to Network Devices
3. Configuring Network Devices

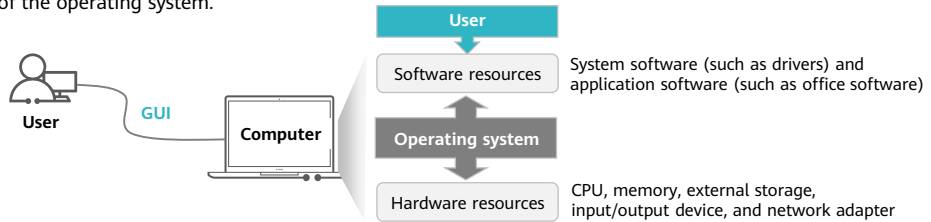
What Is an Operating System?

- An operating system is a system software that manages and controls hardware and software resources. It is like the brain of a person, which can coordinate the parts of the human body to complete various tasks.
 - Multi-task processing: Reply to messages on your phone while eating and listening to music.
 - Resource allocation: Prioritize eating before sleeping when you are both hungry and sleepy.
 - Hardware control: Control hands and feet to perform various physical movements.
 - User interface: Communicate intended actions to the brain through speech or gestures.
 - ...



Computer Operating System

- A computer operating system is a special computer program that controls a computer, acts as the bridge between the computer and user, and coordinates as well as manages the software and hardware resources of the computer.
- Users can invoke applications and access corresponding resources by using simple operation instructions through the GUI of the operating system.

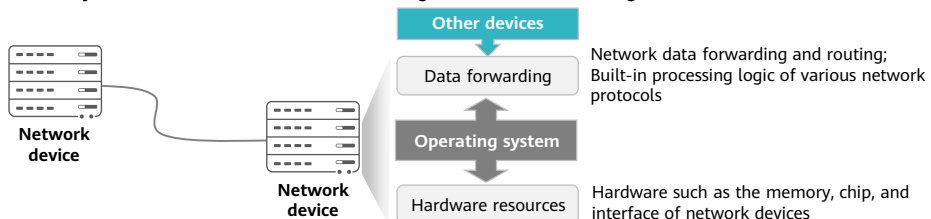


- Mainstream computer operating systems
 - By application domain: Operating systems can be classified into desktop operating systems, server operating systems, embedded operating systems, and distributed operating systems.
 - By the openness of source code: Operating systems can be classified into open-source operating systems (Linux and Unix) and closed-source operating systems (Windows and Mac OS).

- GUI: Graphical User Interface
- CPU: Central Processing Unit

Network Device Operating System

- The network device operating system is a "management brain" installed on network devices such as routers, switches, and firewalls. It controls the hardware running of network devices, processes network data forwarding, and implements communication and collaboration between devices.
- Its functions include hardware resource management, network data forwarding and routing, network protocol support, security and access control, and device configuration and monitoring.

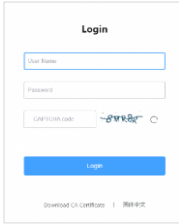
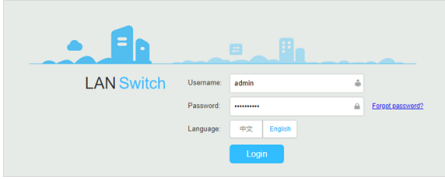


- Common network operating systems
 - Huawei YunShan OS or VRP system: is the operating system of Huawei network devices and tailored for Huawei routers, switches, and firewalls.
 - Other vendors' operating systems: Each vendor has its own operating system for managing network devices such as routers and switches.

- Versatile Routing Platform (VRP)

Huawei YunShan OS and VRP

- Huawei network device operating systems (YunShan OS or VRP) provide a command-line interface (CLI) or web UI, allowing users to log in to, configure, and manage network devices.

YunShan OS	VRP
- CLI<pre><HUAWEI> display version Huawei YunShan OS Version 1.24.0.1 (S5700 V600R024C00)</pre> - Web UI	- CLI<pre><HUAWEI> display version Huawei Versatile Routing Platform Software VRP (R) software, Version 5.170 (S5730 V200R024C00)</pre> - Web UI

- The following functions are provided regardless of which operating system is used:
 - Provides a unified user interface and management interface.
 - Implements the functions of the control plane and defines the interface specifications of the forwarding plane.
 - Achieves communication between the device forwarding plane and control plane.
- Note: This course uses the CloudEngine S5755-H switch running YunShan OS as an example to describe the login modes and basic command knowledge of network devices.

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Network Device Management

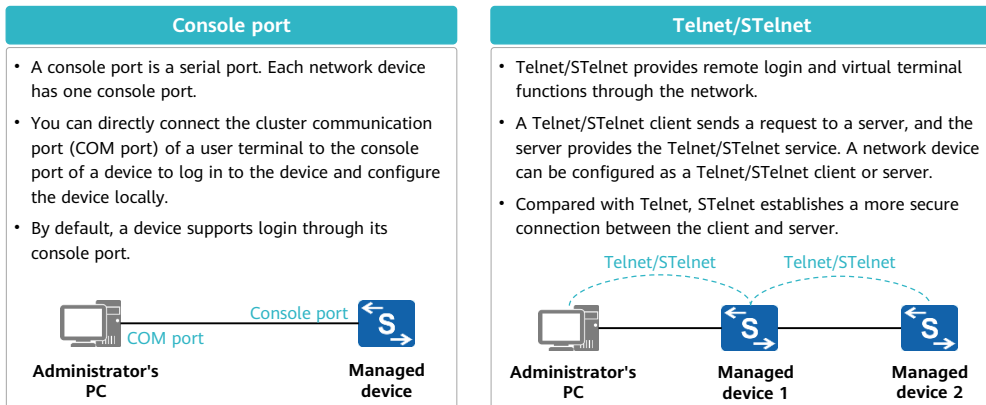
- You can typically manage network devices **using the CLI or through the web system**.
- You can manage a network device only after logging in it through the corresponding mode.

CLI	Web system
• The CLI mode requires users to use the commands provided by a device to manage and maintain the device. This mode implements fine-grained device management but requires users to be familiar with the commands. • Users can log in to the CLI of a network device through its console port or using Telnet or STelnet.	• The web system mode provides a GUI for easy and intuitive device management and maintenance. This mode, however, can be used to manage and maintain only some device functions. • Users can log in to the web system of a device through HTTPS.

- Note:
 - By default, you can log in to the CLI of a device through its console port and log in to the web system of the device through HTTPS.
 - HyperText Transfer Protocol Secure (HTTPS) is a combination of HTTP and Secure Sockets Layer (SSL) or Transport Layer Security (TLS). It provides encrypted communication and secure identification of web servers.

Logging In to the CLI of a Device

- You can log in to a network device through its console port, Telnet, or STelnet. In addition, a network device can log in to another device through Telnet or STelnet.



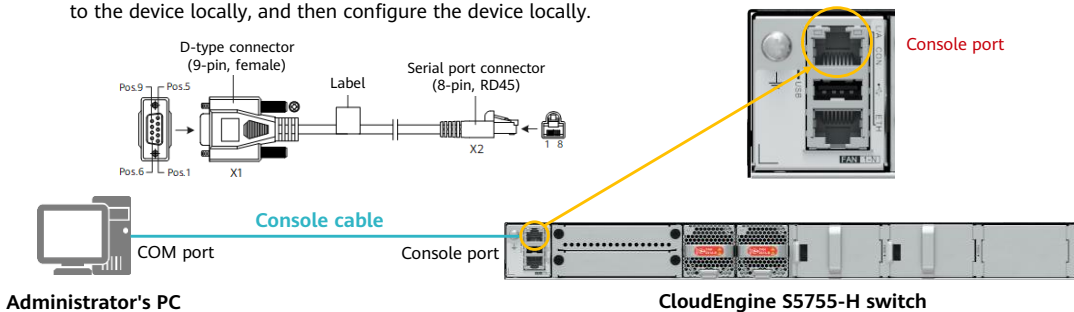
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- To configure and manage a device either locally or remotely, you need to configure the user interface, user management information, and terminal services for device login. The user interface provides a login portal, user management information ensures login security, and terminal services offer support for login protocols.
- User interface: When a user logs in to the CLI of a device, the system allocates a user interface to manage and monitor the session between the device and user. The system supports console and virtual type terminal (VTY) user interfaces.
 - Console user interface: manages and controls the users who log in through the console port.
 - VTY user interface: manages and controls the users who log in through VTY user interfaces. A VTY channel is established after a user's client establishes a Telnet or STelnet connection with the device. A maximum of 21 users (the value varies according to the device model and version) can access the device through VTY user interfaces at the same time.
- User management information: is used to authenticate and authorize users, and defines the identity credentials and permissions of login users. For details, see the course "AAA Fundamentals and Configuration".
- Terminal services: are network protocols used to establish and manage sessions. They provide underlying communication support for user interfaces, especially VTY user interfaces. Common terminal services include Telnet and STelnet. For details, see the course "Network Services and Applications".

Logging In to a Device Through Its Console Port (1)

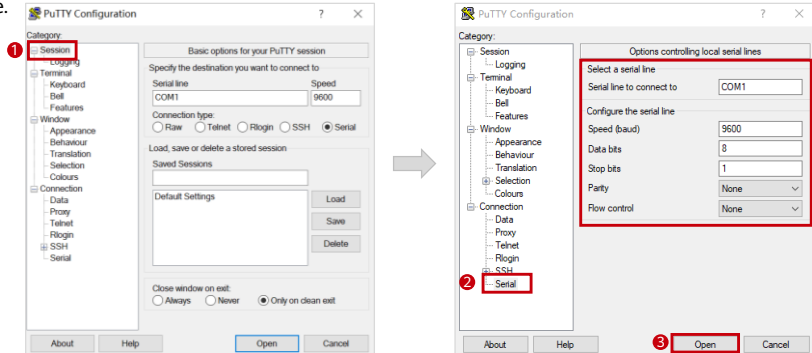
- Logging in to the CLI of a device through the console port is also called local login.
 - When configuring a device that is powered on for the first time, you can log in to the device locally through its console port.
 - You can directly connect your terminal's COM port to the device's console port, use a third-party tool to log in to the device locally, and then configure the device locally.



- You can connect the console port of a network device to the COM port of a computer using a console cable to locally commission and maintain the network device on the computer using a third-party tool.
- Connect the D-type connector of the console cable to the COM port of the computer, and connect the serial port connector to the console port of the device. If there is no serial port on the PC, connect the USB port to the PC through a serial-to-USB cable.

Logging In to a Device Through Its Console Port (2)

- You can use the third-party PuTTY tool to log in to a device. PuTTY is a software that provides Telnet, SSH, and serial port connections.
- During local login, a terminal connects to the console port of a device through a serial port. Therefore, select the serial connection type.



- Start the terminal emulation software PuTTY on the PC. Create a connection and set the port and communication parameters.
 1. Click **Session** to create a connection.
 2. Click **Serial**, select the port to be connected, and set communication parameters.
 - Select the port based on actual needs. For example, on Windows, you can open **Device Manager** to view port information and select the port to be connected.
 - Set communication parameters. Ensure that the communication parameter settings of the terminal emulation software are consistent with the default parameter settings of the device: 9600 bit/s transmission rate, 8 data bits, 1 stop bit, no parity check, and no flow control.
 3. Click **Open**.

Logging In to a Device Through Its Console Port (3)

- After a PuTTY session is opened successfully, the system displays the following information, prompting you to set a login password before logging in to the device.

```
User interface con0 is available

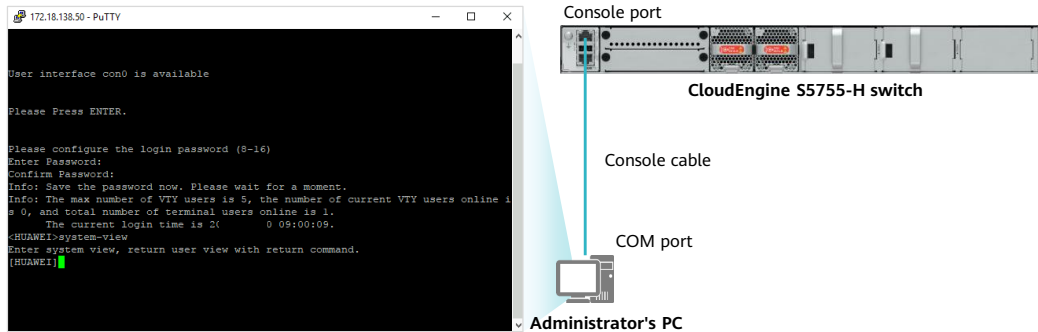
Please Press ENTER.

Please configure the login password (8-16)
Enter Password: Huawei@123
Confirm Password: Huawei@123
Info: Save the password now. Please wait for a moment.
Info: The max number of VTY users is 21, the number of current VTY users online is 0, and total number
of terminal users online is 1.
The current login time is 20XX-XX-XX 18:15:10+08:00
<HUAWEI>
```

- First login: You need to set a login password before logging in to the device.
- Non-first login: During password authentication, you need to enter the password. During AAA authentication, you need to enter the user name and password.

Performing Basic Configurations After the First Login

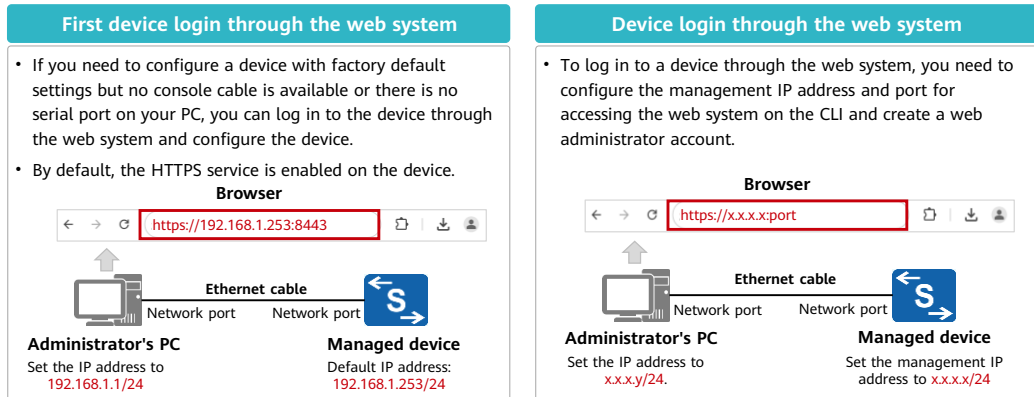
- Upon the first login, you need to set basic system parameters for the newly manufactured device. This is a prerequisite for using the device.
- For example, you need to configure the system time, device name, and management IP address of the device to provide a basic environment for subsequent service configurations.



- Note: For details about basic configuration commands after the first login, see "CLI Overview" in this course.

Logging In to the Web System of a Device

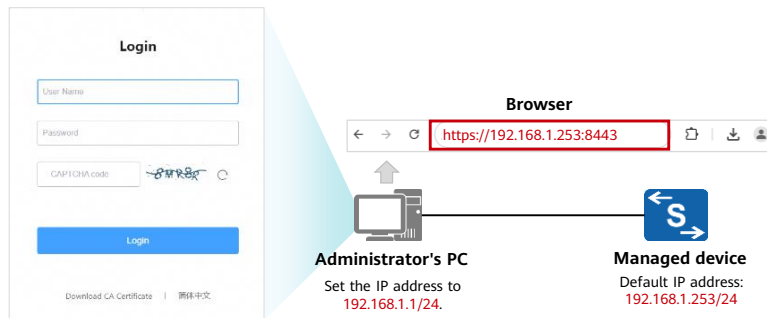
- The built-in web server function of a device provides a GUI, through which users can log in to the device from a terminal using HTTPS for device management and maintenance.



- Note:
 - IP address for web-based login: The default management IP address of the switch described in this course is 192.168.1.253/24. The default management IP address varies according to the device model and version. For details, see the product documentation. You can also configure a management IP address for the device to access the web UI.
 - Port number for web-based login: The default port number of the switch described in this course is 8443, which provides the HTTPS service. You can configure an HTTPS service port number.

Logging In to a Device Through the Web System (1)

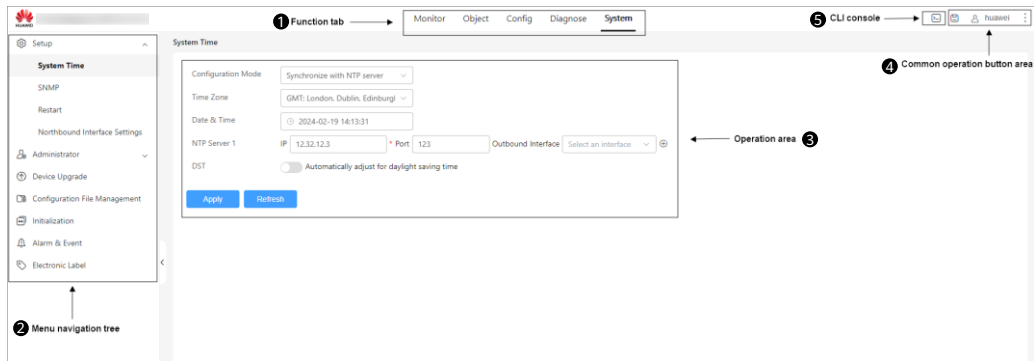
- First login: If you need to configure a device that is powered on for the first time, you can log in to the device using the default IP address.
- Non-first login: You can set the management IP address of the device through the CLI and then log in to the device through the web system.



- Procedure for the first login through the web system:
 1. Connect the network port of the administrator's PC to the management port of the device using an Ethernet cable. If the device does not have a management port, connect the network port of the PC to the first Ethernet port of the device using an Ethernet cable.
 2. Press and hold the **MODE** button for 6 seconds. When the MST indicator is steady green, the default IP address 192.168.1.253/24 is configured for the device by default.
 3. Set the IP address of the PC to be on the same network segment as the default IP address.
 4. Enter https://192.168.1.253:8443 in the address bar of the browser. If the browser displays a message indicating that the certificate is insecure, choose to continue anyway.
 5. On the page displayed for creating an administrator account upon the first login, enter the user name and password, confirm the password, and click **Create**.
 6. Click **OK** in the dialog box that is displayed after the account is successfully created. Access the login page, enter the created user name and password, and click **Login**.
 7. When you log in to the device using a new account for the first time, the page for changing the initial password is displayed. Enter the old password and new password, confirm the new password, and click **OK**.
 8. Access the login page again, enter the user name and new password and click **Login**.

Logging In to a Device Through the Web System (2)

- After logging in to the web system of the device, you can configure and manage the device on the following pages.



- Function tabs: Based on function types, the web configuration UI consists of the following five tabs:
 - Monitor:** allows you to view information such as the panel, device information, device status, interface bandwidth utilization, logs, alarms, interface statistics, and optical modules to monitor the system running status.
 - Object:** allows you to configure common elements such as addresses and services that are referenced by various service policies, simplifying service configuration.
 - Configure:** allows you to configure services such as VLAN, MAC address, interface, Address Resolution Protocol (ARP), DNS, static route, stack, and WLAN, providing detailed service configuration.
 - Diagnose:** allows you to view log information, collect device information, and diagnose device faults, providing comprehensive O&M methods.
 - System:** allows you to configure device management functions, such as administrator, clock, SNMP, and system upgrade, providing a basis for normal system running.
- Menu navigation tree: After you select a tab, menus corresponding to the tab are displayed in the navigation tree on the left. You can click menus to configure desired functions.
- Operation area: After you select a menu, the configuration page corresponding to the menu is displayed in the operation area on the right.

4. Common operation button area: This area is located in the upper right corner of the web system and provides shortcut buttons for common operations, such as logout, password change, and configuration saving.
5. CLI console: This area is located in the upper right corner of the web system and used to switch from the web system to the CLI console.

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 - Configuration File Management

Basic Command Structure

- Huawei CLI commands follow a unified structure. After a command is entered on the CLI, the CLI parses the command and executes it to implement the function of the command, such as query, configuration, or management of network devices.



- **Command word:** specifies the operation to be executed in a command, such as **display** (device status query) or **reboot** (device restart).
- **Keyword:** a special character string that is used to further restrict a command. Keywords are added to extend the command or express the command logic.
- **Parameter list:** further restricts command functions. It can contain one or more pairs of parameter names and values.

Example 1: Display interface information.
display ip interface GE1/0/1

Command word: display
Keyword: ip
Parameter name: interface
Parameter value: GE1/0/1

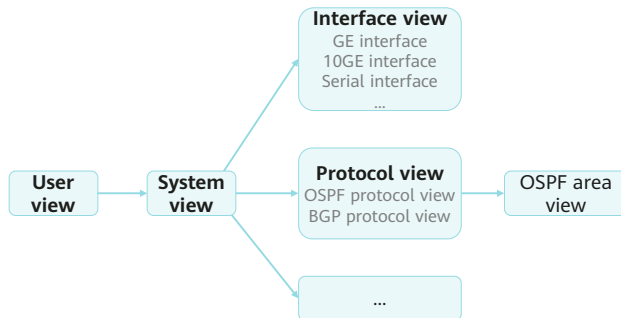
Example 2: Restart a device.
reboot

Command word: reboot

- A command consists of a command word, keywords, and parameters. A parameter consists of a parameter name and a parameter value.
- The command word, keywords, parameter names, and parameter values in a command are separated by spaces.

Command Views (1)

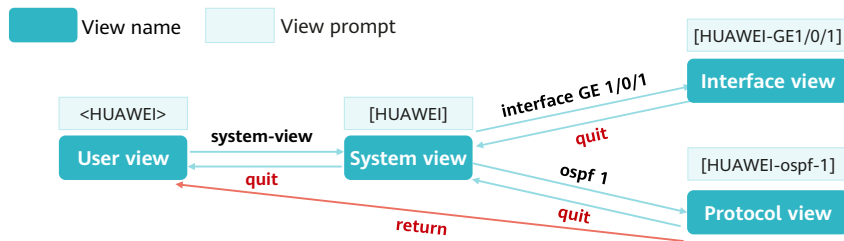
- Devices provide rich functions and a variety of configuration and query commands for device configuration, management, and maintenance. The commands are available in different command views based on their functions for ease of use. To configure a function, you need to enter the required command view and run the required commands.



- User view: allows you to check the running status and statistics of a device.
- System view: allows you to set system parameters and enter other function configuration views.
- Interface view: allows you to set interface parameters, including physical attributes, link-layer attributes, and IP addresses.
- Protocol view: allows you to configure protocol parameters. For example, in the OSPF protocol view, you can configure most OSPF parameters.

- The user view is the first view displayed after you log in to a device. Only query and tool commands are provided in the user view.
- In the user view, you can only access the system view. The system view provides global configuration commands.
- If a lower-level configuration view exists, the system view must provide a command for entering the lower-level configuration view.

Command Views (2)



Command example:

```
<HUAWEI> system-view
[HUAWEI] interface GE 1/0/1
[HUAWEI-GE1/0/1] ip address 192.168.1.1 24
[HUAWEI-GE1/0/1] quit
[HUAWEI] ospf 1
[HUAWEI-ospf-1] area 0
[HUAWEI-ospf-1-area-0.0.0.0] return
<HUAWEI>
```

```
# Access the system view from the user view.
# Access the interface view from the system view.
# Configure an IP address in the interface view.
# Return to the previous view.
# Enter the protocol view from the system view.
# Enter the OSPF area view from the protocol view.
# Return to the user view.
```

- After you log in to the device CLI, the user view is displayed first. This view provides only display commands and tool commands, such as ping and telnet. It does not provide any configuration commands.
- You can run the **system-view** command in the user view to enter the system view. Some simple global configuration functions are available in the system view.
- Some complex configuration functions, such as configuring an Ethernet interface, require many parameters. Therefore, the system provides the configuration view of an Ethernet interface. In the system view, run the **interface GE X/Y/Z** command (*X/Y/Z* indicates the number of the interface to be configured, which corresponds to the slot ID/subcard ID/interface number.). After you enter the GE interface configuration view, all commands take effect only for the GE interface.

Setting Command Privilege Levels

- The system manages commands based on command privilege levels. Each command to be run in a command view has its privilege level. A device manages users by privilege level, and maintains the relationship between user privilege levels and command privilege levels to limit user access permissions.

User Privilege Level	Command Privilege Level	Description
0	Visit level (0)	Commands at this privilege level include network diagnosis tool commands (such as ping and tracert), commands for accessing external devices from the local device, and some display commands.
1	Visit level (0) and monitoring level (1)	Commands at this privilege level are used for system maintenance, including display commands.
2	Visit level (0), monitoring level (1), and configuration level (2)	Commands at this privilege level are used for service configurations to provide network services, including routing commands and commands at each network layer.
3	Visit level (0), monitoring level (1), configuration level (2), and management level (3)	Commands at this privilege level are used for basic system operations, including file system, FTP, TFTP download, command privilege level configuration, and debugging.

Command example: Setting the command privilege level in a specified view

```
<HUAWEI> system-view
[HUAWEI] command-privilege level level/view view-name command-key
```

- After a user logs in to a device, the user can only use commands at the user's privilege level or lower. By default, the command privilege level ranges from 0 to 3, and the user privilege level ranges from 0 to 3. The table shows the mapping between user privilege levels and command privilege levels.
- The device administrator can change the command privilege level as required, enabling lower-level users to run some high-level commands. The device administrator can also increase the command privilege level to improve device security.
- Note:
 - To prevent security risks to devices, you are advised not to change the default command privilege level.
 - Not all display commands are at the monitoring level. For example, the **display current-configuration** and **display saved-configuration** commands are at privilege level 3.
 - The definition of user privilege levels varies according to the device model and version. For details, see the product documentation of the corresponding device model and version.

Editing Command Lines (1)

- You can edit commands in the device CLI.

1. Function keys

Function Key	Function
Common key	Inserts a character at the current location of the cursor and moves the cursor forward if the editing buffer is not full. Otherwise, an alarm is generated.
Backspace	Deletes the character before the cursor and moves the cursor back one character. When the cursor reaches the beginning of the command, an alarm is generated.
Left arrow key ← or <Ctrl+B>	Moves the cursor back one character. When the cursor reaches the beginning of the command, an alarm is generated.
Right arrow key → or <Ctrl+F>	Moves the cursor forward one character. When the cursor reaches the end of the command, an alarm is generated.

Editing Command Lines (2)

2. Incomplete keyword input

You do not need to enter complete keywords if the entered characters can match a unique keyword in the current view. If the current input string matches multiple commands, you need to continue typing until it matches a unique command. Then the command can be successfully delivered.

Complete Command	Incomplete Keyword	Whether a Unique Keyword Can Be Matched
display current-configuration	d cu, di cu, dis cu	Yes
	d c, dis c	No. The commands starting with d c and dis c are not unique.

Command example:

```
<HUAWEI> d cu
<HUAWEI> di cu
<HUAWEI> dis cu
<HUAWEI> d c
      ^
Error: Ambiguous command found at '^' position.
<HUAWEI> dis c
      ^
Error: Ambiguous command found at '^' position.
```



- Note: Different from the keyword in the basic command structure, the keyword descried here can be any character string except a parameter value in a command.

Editing Command Lines (3)

3. Use of the **Tab** key

After you enter an incomplete keyword and press **Tab**, the system automatically completes the keyword.

a. When the input matches a unique keyword, the system enters the unique keyword on a new line and adds a space following the keyword.

```
[HUAWEI] info-          # Enter an incomplete keyword and press Tab.
[HUAWEI] info-center    # The system replaces the entered keyword with the complete
keyword and displays the complete keyword on a new line.
```

b. When the input matches more than one keyword, press Tab repeatedly. The system then circularly displays the keywords beginning with the entered character string to help you find the desired keyword. In this case, the cursor immediately follows the end of the keyword.

```
[HUAWEI] info-center log      # Enter an incorrect keyword and press Tab.
[HUAWEI] info-center log-severity  # The system displays the prefixes of all matched keywords. In this
example, the prefix is log.
[HUAWEI] info-center logbuffer   # Press Tab to switch from one matched keyword to another.
[HUAWEI] info-center logfile     #
[HUAWEI] info-center logghost    # Stop pressing Tab when the desired keyword is displayed.
```

c. When a keyword that matches no command is entered, press Tab. The keyword is then displayed in a new line without being changed.

```
[HUAWEI] info-center loglog    # Enter an incorrect keyword and press Tab.
[HUAWEI] info-center loglog    # The system displays information in a new line, but the keyword
loglog remains unchanged, indicating that this keyword does not exist.
```

Using Command Online Help (1)

- You can use command online help to obtain real-time assistance, avoiding the need to memorize a large number of complex commands.
- When entering commands, you can enter a question mark (?) at any time to obtain online help. The online help can be classified into full help and partial help.
 1. Full help: When you enter a command, you can use the complete help of the command to obtain the prompts of all keywords and parameters.

a. Enter a question mark (?) in any command view to obtain all the commands and their simple descriptions in this command view.

```
<HUAWEI> ?  
Current view commands:  
activate      Activate locked user  
cd            Change current directory  
clear        Clear operation  
clock        Clock status and configuration information  
copy         Copy from one file to another  
...
```

- The displayed command help information is for reference only and varies according to devices.

Using Command Online Help (2)

b. When you enter some keywords of a command and a question mark (?) separated by a space, all keywords associated with this command, as well as simple descriptions, are displayed.

```
<HUAWEI> system-view
[HUAWEI] user-interface vty 0 4
[HUAWEI-ui-vty0-4] authentication-mode ?
aaa          AAA authentication
password     Authentication through the password of a user terminal interface
```

In the preceding command output, **aaa** and **password** are keywords, which are described by **AAA authentication** and **Authentication through the password of a user terminal interface**, respectively.

c. Enter some keywords of a command and a question mark (?) separated by a space to display all the parameters associated with this keyword, as well as simple descriptions.

```
<HUAWEI> system-view
[HUAWEI] ssh server timeout ?
INTEGER<1-35791> Set the authentication timeout, the default value is 60 seconds
[HUAWEI] ssh server timeout 35 ?
<cr>
[HUAWEI] ssh server timeout 35
```

INTEGER<1-35791> describes the value range of the parameter. **Set the authentication timeout, the default value is 60** briefly describes the function of this parameter. **<cr>** indicates that there is no keyword or parameter in this position. You can press **Enter** to run this command.

Using Command Online Help (3)

2. Partial help: When entering a command, if you only remember one or more of the first few characters of the command keyword, you can use the partial help function to obtain all the keywords beginning with this character or character string.

a. Enter a character string followed directly by a question mark (?) to display all the keywords that begin with this character string.

```
<HUAWEI> d?  
debugging      delete  
dir            display
```

b. Enter a command and a string followed directly by a question mark (?) to display all the keywords that begin with this character string.

```
<HUAWEI> display s?  
sysname        system
```


Interpreting the Error Messages of Commands

- The system executes only the commands that pass the syntax check. If a command fails the syntax check, the system displays an error message.

Error Message	Meaning
Error: Unrecognized command found at '^' position.	No command is found.
	No keyword is found.
Error: Wrong parameter found at '^' position.	The parameter type is incorrect.
	The parameter value exceeds the limit.
Error: Incomplete command found at '^' position.	The entered command is incomplete.
Error: Too many parameters found at '^' position.	Too many parameters are entered.
Error: Ambiguous command found at '^' position.	The entered command is not specific.

Command example:

```
[Huawei] sysname
      ^
Error: Incomplete command found at '^' position. # A supplement needs to be made at the position pointed by the arrow.

[Huawei] router if 1.1.1.1
      ^
Error: Unrecognized command found at '^' position. # An identification failure occurs at the position pointed by the arrow.
Check whether the command is correct.
```

Using an Undo Command (1)

- If a command begins with the keyword **undo**, it is an **undo** command. An **undo** command is typically used to restore a default configuration, disable a function, or delete a configuration. Almost every configuration command has its corresponding **undo** format.

1. Run an **undo** command to restore a default configuration.

```
<HUAWEI> system-view
[HUAWEI] sysname Server
[Server] undo sysname
[HUAWEI]
```

The **undo sysname** command restores the default device host name.

2. Run an **undo** command to disable a function.

```
<HUAWEI> system-view
[HUAWEI] sftp server enable
Info: Succeeded in starting the SFTP server.
[HUAWEI] undo sftp server enable
Warning: The operation will stop the SFTP server. Continue? [Y/N]:Y
Info: Succeeded in closing the SFTP server.
```

The **undo sftp server enable** command disables the SFTP server function on the device.

Using an Undo Command (2)

3. Run an **undo** command to delete a configuration.

```
<HUAWEI> system-view  
[HUAWEI] interface GE 1/0/1  
[HUAWEI-GE1/0/1] ip address 192.168.1.1 24  
[HUAWEI-GE1/0/1] undo ip address
```

The **undo ip address** command deletes an IP address from an interface.

Displaying Historical Commands

- The device automatically saves the historical commands entered by users. If you want to enter a command that has already been executed, you can use the function of displaying historical commands to find the desired one.

Operation	Command or Function Key	Result
Display historical commands.	display history-command [all-users]	The historical commands entered by the current user are displayed when all-users is not specified. The historical commands entered by all users are displayed when all-users is specified.
Display the previous historical command.	Up arrow key ↑ or Ctrl+P	The previous historical command is displayed. If the current command is the first historical command, an alarm is generated when you attempt to display the previous historical command.
Display the next historical command.	Down arrow key ↓ or Ctrl+N	The next historical command is displayed. If the current command is the latest historical command, no output is displayed and an alarm is generated when you attempt to display the next historical command.

- Historical commands entered by the current user can be deleted using the **reset history-command** command in the user view, and those entered by all users can be deleted using the **reset history-command all-users** command in the user view. Once deleted, historical commands can no longer be displayed or accessed.

- By default, the device saves the last 10 commands for each user. You can run the **history-command max-size size-value** command in a user interface view to set the number of historical commands that can be saved for the corresponding user. The maximum number is 256. The time spent in finding the desired command among all the historical commands saved on the device is related to the value specified in the **history-command max-size size-value** command. To ensure an efficient search, set an appropriate value.
- Note the following when displaying historical commands:
 - The device saves commands in the same way as how users enter them. For example, if a user enters an incomplete command, the saved command will also be incomplete.
 - If a user runs the same command several times, only the most recently entered command is saved. If a command is entered in different formats, the command in each of these formats is considered different. For example, if the **display current-configuration** command is run several times, this command last executed is saved. If the **display current-configuration** command and the **dis curr** command are run, both of them are saved.

Using Command Shortcut Keys (1)

- A device provides command shortcut keys to speed up and simplify command input.
- Command shortcut keys are classified into user-defined shortcut keys and system default shortcut keys.

1. User-defined shortcut keys

There are four shortcut keys: Ctrl+G, Ctrl+L, Ctrl+O, and Ctrl+U. A user-defined shortcut key can be associated with any command. After you press a shortcut key, the system will automatically run the command associated with the shortcut key.

```
<HUAWEI> system-view  
[HUAWEI] hotkey { CTRL_G | CTRL_L | CTRL_O | CTRL_U } command-text
```

The system supports four user-defined shortcut keys and the default values are as follows:

- Ctrl+G: **display current-configuration**
- Ctrl+L: **display ip routing-table**
- Ctrl+O: **undo debugging all**
- Ctrl+U: Null

- User-defined shortcut keys: If a user frequently uses a specific command or selection of commands, the user can use shortcut keys to define them, which improves efficiency. Only management-level users have the rights to define shortcut keys.

Using Command Shortcut Keys (2)

2. System shortcut keys

These shortcut keys are fixed in the system and cannot be defined. The table lists the common system shortcut keys.

Function Key	Function
<Ctrl+A>	Moves the cursor to the beginning of the current line.
<Ctrl+B>	Moves the cursor back one character.
<Ctrl+C>	Stops performing current functions.
<Ctrl+D>	Deletes the character where the cursor is located.
<Ctrl+E>	Moves the cursor to the end of the current line.
<Ctrl+F>	Moves the cursor forward one character.
<Ctrl+H>	Deletes the character to the left of the cursor.
<Ctrl_I>	Provides the same function as the Tab key.
<Ctrl_J> and <Ctrl_M>	Functions the same as the Enter key.
<Ctrl+X>	Deletes all characters to the left of the cursor.
<Ctrl+Y>	Deletes the character at the cursor and all characters to the right of the cursor.
<Ctrl+Z>	Returns to the user view.
<Ctrl+]>	Stops or redirects incoming connections.

Basic Configuration Commands: Configuring Login Prompts

- To provide some prompts or warnings to users, you can configure such information as titles on the device. When a user logs in to the device, the configured titles will be displayed.

- When a terminal connection is activated and a user attempts to log in, the terminal displays the title configured using the **header login** command.

Set the prompt upon a login attempt.

```
[HUAWEI] header login { information text | file file-name }
```

- If the user successfully logs in, the terminal displays the title configured using the **header shell** command.

Set the prompt upon successful login.

```
[HUAWEI] header shell { information text | file file-name }
```

- If you do not want to display login prompts, run commands to disable the display of login prompts.

Disable the function of displaying login information after a successful login.

```
[HUAWEI] system login information disable
```

- After users successfully log in to the device, login prompts are displayed, including the number of online users, current login time, first login information, and last login information.
- The commands for configuring login prompts are described as follows:
 - The **header login information** command sets the header information displayed when the terminal connection is activated during user login.
 - The **header login file** command sets the file content displayed when the terminal connection is activated during user login.
 - The **header shell information** command sets the header information displayed when a user successfully logs in to the device and establishes a session.
 - The **header shell file** command sets the file content displayed when a user successfully logs in to the device and establishes a session.

Basic Configuration Commands: Configuring the Device Time and Name

- The system time on a device is randomly set upon device delivery, but not the actual time. When the device is connected to a network, you must set the system time to the actual local time to ensure that the time in logs and alarms generated by the device is correct.

- Configure the time zone where the device is located.

```
<HUAWEI> clock timezone time-zone-name { add | minus } offset
```

- Set the current date and time.

```
<HUAWEI> clock datetime [ utc ] time date
```

The value of *time* must be in the format HH:MM:SS, which indicates the current hour, minute, and second on the device.
The value of *date* must be in the format YYYY-MM-DD, which indicates the current year, month, and day on the device.

- To differentiate devices on the network, you can set a unique device name for each device.

- Set the device name.

```
[HUAWEI] sysname host-name
```

By default, the host name of a device is HUAWEI. You can run the **undo sysname** command to restore the default host name.

- **clock timezone** *time-zone-name* { **add** | **minus** } *offset*
 - By default, a device uses the Universal Time Coordinated (UTC) time zone. The default time zone name is DefaultZoneName.
 - **add**: adds the specified time zone offset to the UTC time. The sum of the default UTC time zone and offset is the time in the time zone specified by *time-zone-name*.
 - **minus**: subtracts the specified time zone offset from the UTC time. The remainder obtained by subtracting the time zone offset (*offset*) from the default system time (UTC time) is the time in the time zone specified by *time-zone-name*.
 - After a time zone is configured, the device adds timestamps to the local log in the format of *original system time* ± *offset*. An example is Apr 27 20XX 22:36:09+08:00.

Basic Configuration Commands: Configuring a Management Address

- Each device on a network must have a globally unique management address, enabling O&M personnel to easily locate and log in to the device.
- You can configure a management address using a management interface or common interface.

- Configure a management IP address using a management interface.

<HUAWEI> system-view	# Enter the system view.
[HUAWEI] interface meth 0/0/0	# Enter the management interface view.
[HUAWEI-Meth0/0/0] ip address <i>ip-address</i> { <i>mask</i> <i>mask-length</i> }	# Configure an IP address and a mask for the management interface.
[HUAWEI-Meth0/0/0] quit	# Return to the system view.

- Configure a management IP address using a common interface.

<HUAWEI> system-view	# Enter the system view.
[HUAWEI] interface GE 1/0/1	# Enter the view of the service interface
used for management.	
[HUAWEI-GE1/0/1] undo portswitch	# Switch the working mode of the interface from Layer 2 to Layer 3.
[HUAWEI-GE1/0/1] ip address <i>ip-address</i> { <i>mask</i> <i>mask-length</i> }	# Configure an IP address and a mask for the service interface.
[HUAWEI-GE1/0/1] quit	# Return to the system view.

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File System

- The file system manages files and directories in storage media. It allows users to create, delete, and modify files and directories, as well as view the contents of files.
- Mastering the basic operations of the file system is crucial for network engineers to efficiently manage the configuration files and system files of devices.

System software

The system software is essential for device startup and operation, providing support, management, and service functionalities for the device.
The common file name extension is .cc.

Patch file

A patch is a kind of software compatible with the system software. It is used to remove a few issues in the software that need to be solved immediately.
The common file name extension is .pat.

Configuration file

A configuration file stores configuration commands, enabling a device to start with the configurations in the file.
The common file name extensions are .cfg, zip, and .dat.

Common file types

Storage Media

- Huawei devices support the following types of storage media: SDRAM, NVRAM, flash memory, SD card, and USB flash drive.

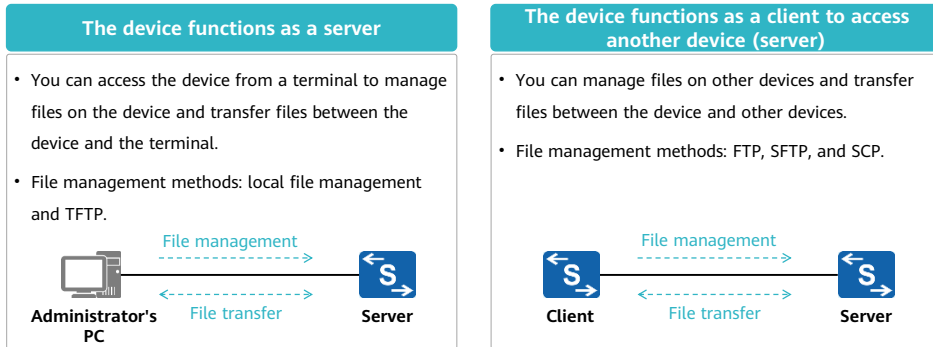
Memory	Description
Synchronous Dynamic Random Access Memory (SDRAM)	SDRAM stores the system running information and parameters. It is equivalent to a computer's memory.
Nonvolatile Random Access Memory (NVRAM)	NVRAM stores cached log files. When the timer expires or the cache is full, the s cached log files are written into the flash memory.
Flash	The flash memory is nonvolatile and can avoid data loss in case of power-off. It is used to store system software, configuration files, and so on. Patch files are uploaded by maintenance personnel and typically stored in the flash memory or SD card.
SD card	SD cards are nonvolatile and can avoid data loss in case of power-off. They are external memory media used for memory expansion. SD cards have a large storage capacity and is typically installed on a main control board. It is used to store system files, configuration files, log files, and other files.
USB	A USB flash drive is considered as an interface. It is used to connect to a large-capacity storage medium for device upgrade and data transmission.



- Huawei CloudEngine S5755-H series switches support the flash memory and USB flash drive.

Managing File Systems

- The method of managing file systems depends on the role of the device. That is, the device can function as a server or client during file management.
- File management methods include local file management, TFTP, FTP, SFTP, and SCP.



- **Local file management:** This mode applies to the scenario where users log in to the device through the console port, Telnet, or STelnet to manage directories and files in the storage media.
- **Trivial File Transfer Protocol (TFTP):** In a lab LAN with good network conditions, you can use TFTP to load and upgrade versions online. This mode applies to the environment without complex interactions between a client and a server.
- **File Transfer Protocol (FTP)** applies to the scenario that does not require high network security in file transfer, and is widely used in version upgrade.
- **Secure File Transfer Protocol (SFTP):** SFTP applies to the scenario that requires high network security and is widely used in services such as log download and configuration file backup.
- **Secure Copy Protocol (SCP):** SCP applies to the scenario that requires high network security and high file upload and download efficiency.
- **Note:** This course focuses on the commands related to local file management.

Managing Files Locally (1)

- Local file management refers to the process of managing files on a device after device login from a terminal.
1. Manage directories.

Operation	Command	Description
Display the current directory.	pwd	-
Change the current directory.	cd [<i>directory</i>]	-
Display files and subdirectories in a specified directory.	dir [/all] [<i>filename</i> /all-file systems]	-
Display all files in the backup partition.	display backup-file	-
Create a directory.	mkdir <i>directory</i>	-
Delete a directory.	rmdir <i>directory</i>	- The directory to be deleted must be empty. - A deleted directory and its files cannot be restored from the recycle bin.

Managing Files Locally (2)

2. Manage files.

Operation	Command	Description
Display the file content.	more <i>file-name</i> [<i>offset</i>] [hex]	You can also run the tail <i>file-name</i> [<i>line</i>] command to view the last specified lines in the specified file.
Copy files.	copy <i>source-filename destination-filename</i> [all]	• Before copying a file, ensure that the storage space is sufficient for the file. • If the destination file has the same name as an existing file, the system prompts you whether to overwrite the existing file.
Copy the files in the backup partition to the specified path.	copy backup-file file-name scrFile path desFile copy backup-file all path desFile	-
Move files.	move <i>source-filename destination-filename</i>	If the destination file name is the same as the name of an existing file, the system prompts you whether to overwrite the existing file.
Rename a file.	rename <i>old-name new-name</i>	-

Managing Files Locally (3)

Operation	Command	Description
Compress a file or directory.	zip <i>source-filename destination-filename</i> [password [<i>password</i>]]	-
Decompress a file.	unzip <i>source-filename destination-filename</i> [password [<i>password</i>]]	-
Delete a file.	delete [/unreserved] [/quiet] <i>filename</i> [all]	This command cannot be used to delete a directory. If the /unreserved parameter is specified, the deleted file cannot be restored.
Restore a deleted file.	undelete <i>filename</i>	After you run the delete command without specifying the /unreserved parameter to delete a file, the file is moved to the recycle bin. You can run the undelete <i>filename</i> command to restore the deleted file from the recycle bin.
Delete a file from the recycle bin.	reset recycle-bin [/f <i>filename</i>]	To permanently delete a file from the recycle bin, run this command.

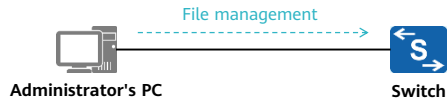
Configuration Example: Managing Files Locally

- A user logs in to the device through the console port and manages directories or files in the storage media.
- Before managing files locally, ensure that:
 - There are reachable routes between the terminal and device.
 - You have logged in to the device through the terminal.

Configuration Roadmap

The configuration roadmap is as follows:

1. View files and subdirectories in the current directory.
2. Create a directory named **test**. Copy the **vrpcfg.zip** file to the **test** directory and rename the file to **backup.zip**.
3. View files in the **test** directory.



Configuration Procedure (1)

- 1. View files and subdirectories in the current directory.

```
<HUAWEI> dir
Directory of flash:/

Idx   Attr   Size(Byte) Date           Time           FileName
0      -rw-    889         Mar 01 20XX    14:41:56       private-data.txt
1      -rw-   2,393       Mar 06 20XX    17:20:10       vrpcfg.zip
2      -rw-    812       Nov 12 20XX    15:43:10       hostkey
3      drw-     -         Mar 01 20XX    14:41:46       compatible
4      -rw-    540       Nov 12 20XX    15:43:12       serverkey
...
670,092 KB total (569,904 KB free)
```



Configuration Procedure (2)

2. Create a directory named **test**, copy the **vrpcfg.zip** file to the **test** directory, and rename the file to **backup.zip**.

Create a directory named **test**.

```
<HUAWEI> mkdir test  
Info: Create directory flash:/test/.....Done.
```

Copy the **vrpcfg.zip** file to the **test** directory and rename the file to **backup.zip**.

```
<HUAWEI> copy vrpcfg.zip flash:/test/backup.zip  
Info: Are you sure to copy flash:/vrpcfg.zip to flash:/test/backup.zip?[Y/N]: y  
100% complete  
Info: Copying file flash:/vrpcfg.zip to flash:/test/backup.zip...Done.
```

Configuration Procedure (3)

3. Verify the configuration.

Access the **test** directory.

```
<HUAWEI> cd test
```

View the current directory.

```
<HUAWEI> pwd
flash:/test/
```

View files in the **test** directory.

```
<HUAWEI> dir
Directory of flash:/test/

  Idx      Attr      Size(Byte) Date           Time           FileName
   0      -rw-        2,399   Mar 12 20XX   11:16:44    backup.zip

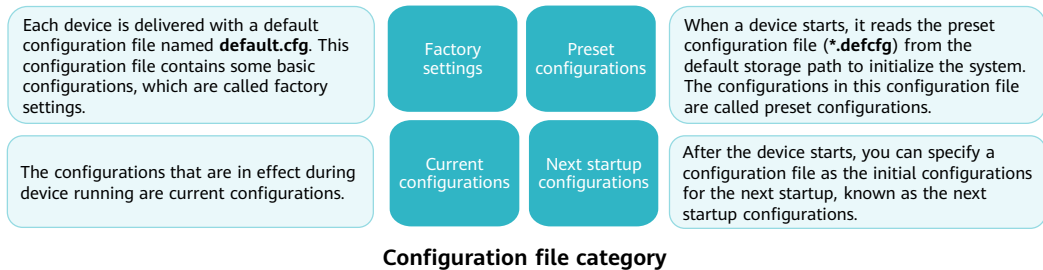
670,092 KB total (569,900 KB free)
```

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Configuration Files

- A configuration file is a collection of commands that can run on a device.
- Configuration file management allows you to view, save, compare, back up, restore, and compress configuration files, as well as delete and roll back configurations in the files. You can also specify the configuration file to be loaded at the next device startup. This ensures correct configurations on the device, prevents configuration loss, and facilitates configuration migration.



- Configuration file format
 - The configuration file can contain only configuration commands, view switching commands, # (used to switch to the system view), and **quit** commands. When other types of commands (such as **display** commands, **reset/save/ping** commands, **return** commands, and upgrade-compatible commands) are executed, the device reports an error and continues to load subsequent commands.
 - A configuration file cannot contain repeated commands.
 - Each view is displayed with an indentation of one character.
 - If commands in a configuration file need to be executed in a view, the configuration file must also contain the command for entering the view.
 - The configuration sequence and dependency must be correct.
 - Interactive commands in a configuration file support only Y and N choices. Y is the default choice, which is used automatically during configuration restoration.
 - A configuration file must be saved to the root directory of the storage media, with a file name extension as .zip, .cfg, .dat, or .defcfg.
 - A .cfg file is a text file, whose contents can be viewed directly. After a .cfg file is specified as the configuration file, the system restores the commands in the file one by one during device startup.
 - A .zip file is compressed from a .cfg file and occupies less space. After a .zip file is specified as the configuration file, the system

decompresses the file into a .cfg file, and then restores the commands in the .cfg file one by one during device startup.

- A .dat file is a compressed file, which consists of a binary database file, a text file (*.cfg), and a system hardware configuration file (**device.sys**). Only a .dat file exported from a Huawei device is supported, and the file cannot be modified manually. Otherwise, the file fails to be loaded.
- The *.defcfg file is a text file and is the preset configuration file. It must be saved in the root directory of the storage media in advance. You can run the **more *.defcfg** command to view the file content. After the **startup default-configuration** command is run, the content in the preset configuration file in the root directory is imported to the fixed directory of the device, and the preset configuration file in the root directory can be deleted.

Checking Configuration Files

Procedure	Command	Description
Check the configuration files in the storage media.	dir	-
Check the configuration files for the current and next startup.	display startup	-
Check the configurations in a specified configuration file.	display configuration <i>configuration-file</i>	-
Check the configurations in the configuration file for the next startup.	display saved-configuration	-
Check the system configurations saved last time.	display saved-configuration last	-
Check the time when the configurations are saved last time.	display saved-configuration time	-
Check all the configurations that take effect on the device.	display current-configuration [include-default]	If include-default is not specified, only configured information is displayed. If include-default is specified, both configured information and default configurations are displayed.
Check the configurations that take effect in the current view.	display this [include-default]	This command displays the configurations in the view where this command is executed. If include-default is not specified, only configured information in the current view is displayed. If include-default is specified, both configured information and default configurations in the current view are displayed.

Saving Configurations

- You can run commands to modify current device configurations, but these modified configurations will be lost once the device restarts unless you save them to the configuration file in the storage media before restarting the device.
- Use either of the following methods to save the configuration file:

- Automatically saving configurations: With the auto-save function enabled, a device automatically saves configurations to the next-startup configuration file if the current configurations are different from those in the next-startup configuration file and the configured interval or delay arrives, regardless of whether the save command is run.

Configure the system to save configurations at a scheduled time.

```
[HUAWEI] configuration file auto-save [ interval interval | delay delay-interval | cpu-limit cpu-usage ] *
```

- Manually saving configurations: The configurations are saved only after the save command is run. After the command is delivered, the current configurations are saved to the configuration file regardless of whether the current configurations are changed.

Save all configurations to the configuration file.

```
<HUAWEI> save [ configuration-file ]
```

- **configuration file auto-save [interval interval | delay delay-interval | cpu-limit cpu-usage] ***
 - **interval interval:** specifies the interval at which the system saves configurations. The value is an integer ranging from 30 to 43200, in minutes. The default value is 30.
 - **delay delay-interval:** specifies a delay after which the device automatically backs up configurations after configuration changes. The value is an integer ranging from 1 to 60, in minutes. The default value is 5.
 - **cpu-limit cpu-usage:** specifies the upper limit of the CPU usage. The value is an integer ranging from 1 to 100. The default value is 50, indicating that the upper limit of the CPU usage is 50%.
 - **default:** restores the default values of the parameters related to the auto-save function.
- **save [configuration-file]**
 - **configuration-file:** specifies the name of a configuration file. The name is a string of 5 to 64 characters in the format of *.zip, *.cfg, or *.dat.

Specifying the Configuration File for Next Startup

- When the system restarts, it uses the specified configuration file to restore configurations.

- Configure the configuration file for the next startup.

```
<HUAWEI> startup saved-configuration configuration-file
```

- Check the configuration file to be loaded for the next startup.

```
<HUAWEI> display startup
```

- Restart the device for the configuration file to take effect.

Restart the device. Before the device is restarted, you are prompted whether to save configurations.

```
<HUAWEI> reboot
```

Restart the device quickly. Before the device is restarted, you are not prompted whether to save configurations.

```
<HUAWEI> reboot fast
```

- If no configuration file is specified for the next startup, the default configuration file **default.cfg** or preset configuration file will be used for the next startup.
- If no configuration file for the next startup is available, the device starts with empty configuration.

Clearing Configurations

- A configuration file needs to be deleted in the following scenarios:
 - The software and configuration file do not match after the device software is upgraded.
 - The configuration file is damaged, or an incorrect configuration file is loaded.
- A single command is available for deleting all configurations on a device, reducing the maintenance workload and simplifying the deletion operation.

- Clear configurations for the next startup.

Clear the configuration file loaded for next startup and restart the device for the configuration to take effect.

```
<HUAWEI> reset saved-configuration  
<HUAWEI> reboot fast
```

- Restore the device to factory settings.

Restore the device to factory settings after device restart.

```
<HUAWEI> reset factory-configuration
```

After this command is executed, the system asks whether you want to restart the device. After you enter **y**, the device is restarted, and the service configurations and data files of the device are cleared.

- About the **reset saved-configuration** command:
 - Function: It clears the configurations in the startup configuration file of a device.
 - Application scenario:
 - After the device software is upgraded, the configuration file in the storage media may not match the new software version. In this case, you can clear the old configuration file.
 - If a device has been used in an application environment and the original configuration file cannot meet the requirements of the new environment, you can clear the original configuration file and reconfigure the device.
 - Impact scope:
 - Only service configurations are cleared, and configurations such as stack configurations are not cleared.
 - The BIOS password and stack configurations are not cleared.
- About the **reset factory-configuration** command:
 - Function: It restores a device to factory settings after device restart.
 - Application scenario:
 - When function exceptions occur on a device due to incorrect configurations, you need to restore the device to factory settings.
 - When a device is removed from the network or migrated, you need to

restore the device to factory settings.

- Impact scope:
 - The service configurations and data files on the device are cleared, including stack configuration.
 - The following configurations and files are cleared: next-startup configuration file, configured BIOS password, stack configuration, NETCONF database file, and data files in the file system.
 - The system software package, patch, module, and license file for next startup are not cleared.
- Command comparison
 - **reset saved-configuration** clears only service configurations.
 - **reset factory-configuration** restores the device to factory settings, which clears more configurations, including stack configuration.

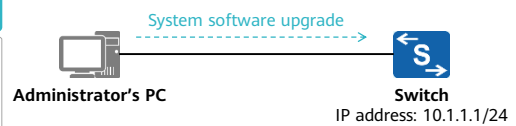
Configuration Example: Upgrading the System Software Version

- The current system software version cannot meet user needs. The device must have the new system software version with more features loaded.
- The system software of the device needs to be upgraded remotely.

Configuration Roadmap

The configuration roadmap is as follows:

1. Upload the new system software and patch file to the root directory of the device.
2. Save the current configuration so that it remains effective after upgrade.
3. Specify the system software to be loaded for next startup.
4. Specify the configuration file to be loaded for next startup.
5. Specify the patch file to be loaded for next startup.
6. Restart the device to complete upgrade.



Configuration Procedure (1)

1. Upload the new system software and patch file to the root directory of the device.

Before performing configurations, run the **display startup** command to view the configurations in the configuration file for next startup.

```
<HUAWEI> display startup
MainBoard:
Configured startup system software:    flash:/basicsoft.cc
Startup system software:               flash:/basicsoft.cc
Next startup system software:       flash:/basicsoft.cc
Startup saved-configuration file:      flash:/vrpcfg.zip
Next startup saved-configuration file: flash:/vrpcfg.zip
Startup paf file:                     default
Next startup paf file:                 default
Startup patch package:             NULL
Next startup patch package:            NULL
Startup feature software:              NULL
Next startup feature software:         NULL
Startup extended-system software:      NULL
Next startup extended-system software: NULL
```

Configuration Procedure (2)

Configure the device as an SFTP server and upload the system software **newbasicsoft.cc** and patch file **newpatch.PAT** to the device. (The procedure is not described here.)

After the upload is complete, run the **dir** command on the device to view the uploaded system software and patch file.

```
<HUAWEI> dir
Directory of flash:/

Idx  Attr  Size(Byte)      Date           Time           FileName
0   drw-  -              Apr 16 20XX    13:19:58       logfile
1   -rw-  85,925,409      Apr 16 20XX    13:18:02       basicsoft.cc
2   -rw-  4              Oct 27 20XX    17:25:22       snmpnotilog.txt
3   -rw-  6,033          Jul 16 20XX    16:40:02       private-data.txt
4   -rw-  3,275          Jul 14 20XX    14:18:08       vrpcfg.zip
5   drw-  -              Nov 14 20XX    19:14:26       sysdrv
6   -rw-  88,239,759      Jul 16 20XX    19:14:26       newbasicsoft.cc
7   -rw-  3,973,953      Jul 16 20XX    19:14:26       newpatch.PAT
...

670,092 KB total (569,904 KB free)
```

- Upload the new system software and patch file to the device. This example uses SFTP to transfer the system software and patch file. Configure the device as an SFTP server and upload the system software and patch file to the device from the SFTP client. Ensure that there is enough space in the storage device before uploading files. If the space is insufficient, delete unnecessary files to free up space in the storage device.
- For details about the SFTP configuration, see the course "Network Services and Applications".

Configuration Procedure (3)

2. Save the current configuration.

```
<HUAWEI> save
Warning: The current configuration will be written to the device. Continue? [Y/N]: y
Now saving the current configuration to the slot 6 ..
Info: Save the configuration successfully.
```

The system displays a message indicating that the current configuration will be saved to the device and asks you whether to continue. After you enter **y**, the configuration will be saved to the device.

3. Specify the system software to be loaded for next startup.

```
<HUAWEI> startup system-software newbasicsft.cc
```

4. Specify the configuration file to be loaded for next startup.

```
<HUAWEI> startup saved-configuration vrpcfg.zip
```

5. Specify the patch file to be loaded for next startup.

```
<HUAWEI> startup patch newpatch.PAT all
```


Configuration Procedure (4)

6. Verify the configuration.

Run the following command to view the files to be loaded for next startup.

```
<HUAWEI> display startup
MainBoard:
Configured startup system software:    flash:/basicsoft.cc
Startup system software:               flash:/basicsoft.cc
Next startup system software:          flash:/newbasicsoft.cc
Startup saved-configuration file:      flash:/vrpcfg.zip
Next startup saved-configuration file: flash:/vrpcfg.zip
Startup paf file:                      default
Next startup paf file:                 default
Startup patch package:                 NULL
Next startup patch package:            flash:/newpatch.PAT
Startup feature software:              NULL
Next startup feature software:         NULL
Startup extended-system software:      NULL
Next startup extended-system software: NULL
```

7. Restart the device.

```
<HUAWEI> reboot fast
```

- Since the configuration file has been saved, run the **reboot fast** command to restart the device quickly.
- Wait for several minutes until the device restart is complete. Then run the **display version** command to check the current system software version. If the version is new, the upgrade has succeeded.

Quiz

1. (Short-answer question) What is the maximum number of users who are allowed to log in to a Huawei network device concurrently through the console port?
2. (Short-answer question) How do I specify the configuration file for next startup if a device has multiple configuration files?

1. A Huawei network device allows only one user to log in through the console port at a time. Therefore, the console user ID is fixed at 0.
2. To specify a configuration file for next startup, run the **startup saved-configuration *configuration-file*** command. The value of configuration-file should contain both the file name and file name extension.

Summary

- YunShan OS and VRP are network operating systems running on Huawei datacom products. They are similar to the Windows system for computers, providing network resource allocation and device management and interconnection.
- Huawei network operating systems provide a consistent network GUI, user GUI, and management GUI, and can run on a variety of hardware platforms. In addition, they provide the CLI and web UI for users to log in to devices and configure as well as manage devices.
- Therefore, it is essential to be familiar with the CLI and master the related configurations. In addition, you need to understand some common commands and shortcut keys to lay a foundation for efficient network deployment and O&M.

Thank you.

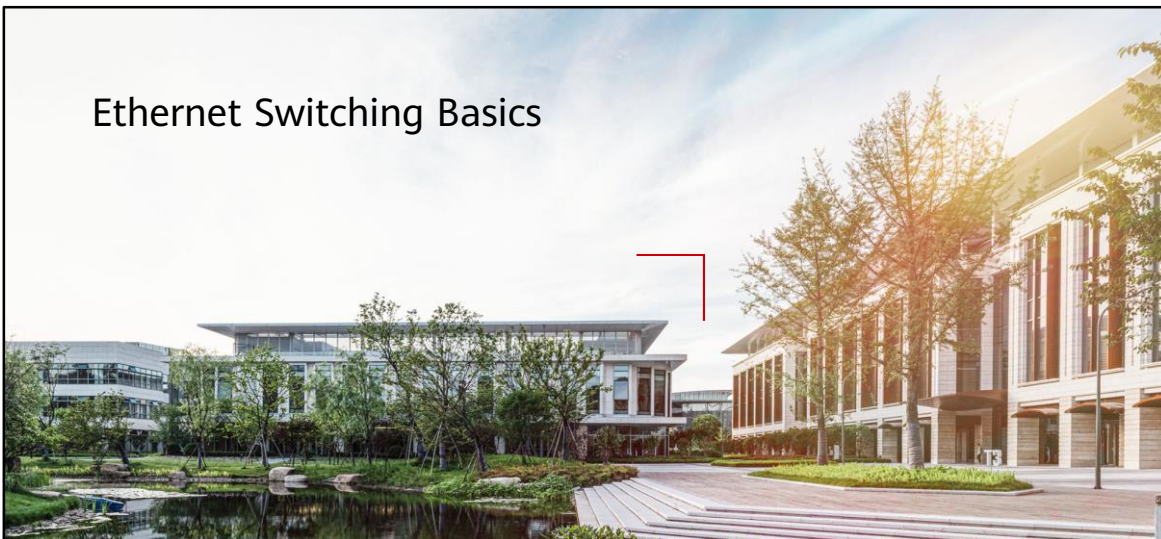
把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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Ethernet Switching Basics



Foreword

- Network communication is a complex layered encapsulation process. When upper-layer protocol data arrives at the data link layer, the data link layer adds necessary headers and trailers to the data and encapsulates the data into frames that can be transmitted over links.
- Ethernet is the dominant encapsulation technology at the data link layer. Frames encapsulated using this technology are called Ethernet frames.
- Ethernet switches are core devices that implement communication at the data link layer. They forward data based on information in headers of received Ethernet frames.
- This course describes the concepts related to the Ethernet protocol, MAC address types, and working principles of Layer 2 switches.

Objectives

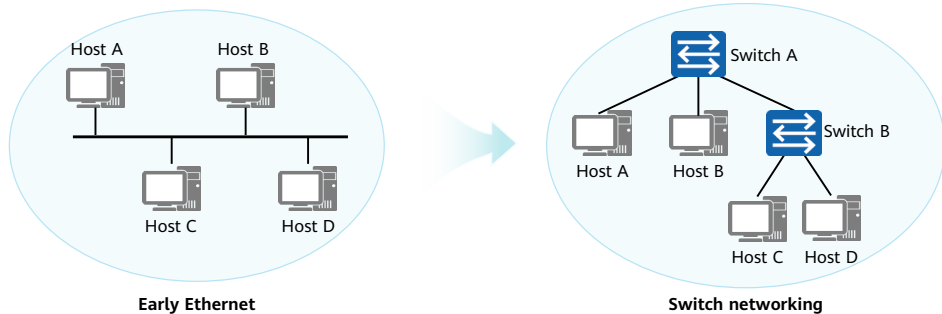
- Upon completion of this course, you will be able to:
 - Describe basic Ethernet concepts.
 - Distinguish MAC address types.
 - Describe the process of creating a MAC address entry.
 - Describe the working principles of switches.

Contents

- 1. Overview of the Ethernet Protocol**
2. Overview of Ethernet Frames
3. Working Principles of Ethernet Switches

Ethernet Protocol

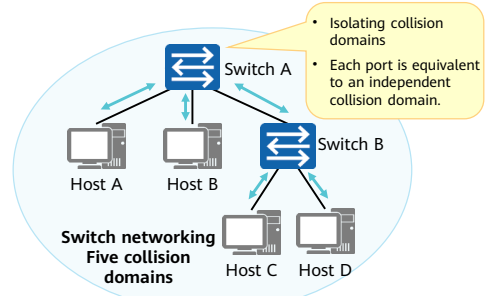
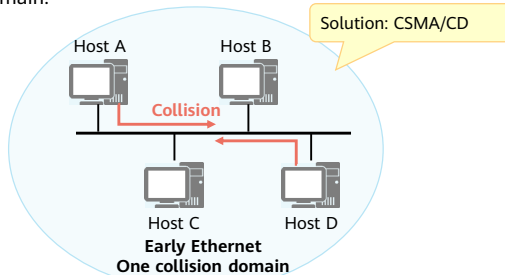
- An Ethernet network is a broadcast network built based on the Carrier Sense Multiple Access with Collision Detection (CSMA/CD) mechanism.



- Early Ethernet:
 - An Ethernet network is a broadcast network built based on the CSMA/CD mechanism. Collisions are a major factor that restricts Ethernet network performance. Early Ethernet devices such as hubs are physical-layer devices and cannot prevent collisions from spreading, which restricts the improvement of network performance.
- Switch networking:
 - Switches are Layer 2 network devices that can isolate collision domains, greatly improving Ethernet network performance. They replace hubs to become mainstream Ethernet devices. However, switches do not restrict broadcast traffic on the network, which affects network performance.

Collision Domain

- A collision domain is a set of all nodes connected to the same shared medium. All nodes in a collision domain compete for the same bandwidth. Packets sent by a node can be received by all other nodes in the same collision domain.

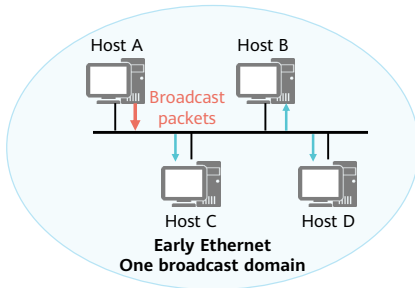


- On a traditional Ethernet network, multiple nodes on the same medium share the link bandwidth and compete for the right to use the link. As a result, a collision occurs.
- There is a higher probability that a collision occurs when more nodes are deployed on the same medium.
- Ports on a switch send and receive data independently and belong to different collision domains. Collision domains are isolated at the physical layer so that collisions will not occur between hosts (or networks) connected through this switch due to uneven traffic rates on these hosts (or networks).

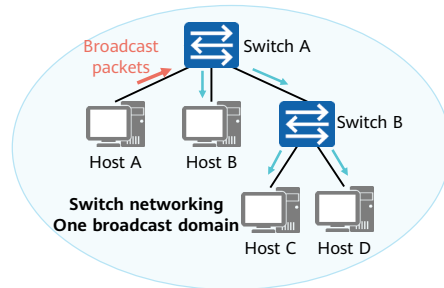
- On a shared network, CSMA/CD technology of Ethernet is used to avoid collisions. The basic working process of CSMA/CD is as follows:
 - A terminal device continuously detects the status of the shared line.
 - If the line is idle, the device sends data.
 - If the line is in use, the device waits until the line becomes idle.
 - If two devices send data at the same time, a collision occurs on the line, and signals on the line become unstable.
 - After detecting the instability, the device immediately stops sending data.
 - The device sends a series of disturbing pulses. After a period of time, the device resumes the data transmission. The device sends disturbing pulses to inform other devices, especially the device that sends data at the same time, that a collision occurred on the line.
- The working principle of CSMA/CD can be summarized as follows: listen before sending, listen while sending, stop sending upon collision, and retransmit after random delay.

Broadcast Domain

- The entire access scope of broadcast packets is called a Layer 2 broadcast domain, which is also called a broadcast domain. All hosts in the same broadcast domain can receive broadcast packets.



- On a traditional Ethernet network, multiple nodes on the same medium share a link. The broadcast packets sent by a device can be received by all the other devices.

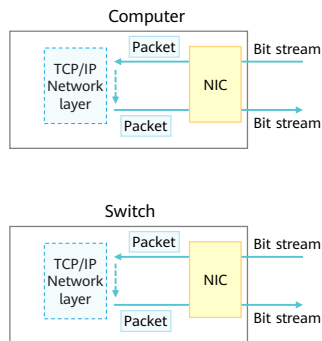


- A switch forwards broadcast packets through all ports except the port that receives the packets.
- The nodes connected to all ports of a switch belong to the same broadcast domain.

- The all-1 MAC address FF-FF-FF-FF-FF-FF is a broadcast address, and all nodes process a data frame whose destination address is the broadcast address. The entire access scope of the data frame is called a Layer 2 broadcast domain, which is also called a broadcast domain.
- Note: A Media Access Control (MAC) address uniquely identifies a network interface card (NIC) on a network. Each NIC requires and has a unique MAC address.

Ethernet NIC

- A network interface card (NIC) is a key component that connects a network device such as a computer, switch, or router to an external network.



- **Network interface**

- A network interface is also called a network port, an interface, or a port.

- **NIC**

- Each network interface corresponds to a NIC.
- A computer or switch forwards data through a NIC.

- There are many types of NICs. All the NICs mentioned in this course are Ethernet NICs, which are called Ethernet network cards or Ethernet cards for short.
- All the switches mentioned in this course are Ethernet switches. The NICs used by network interfaces that forward data on the switches are Ethernet NICs.

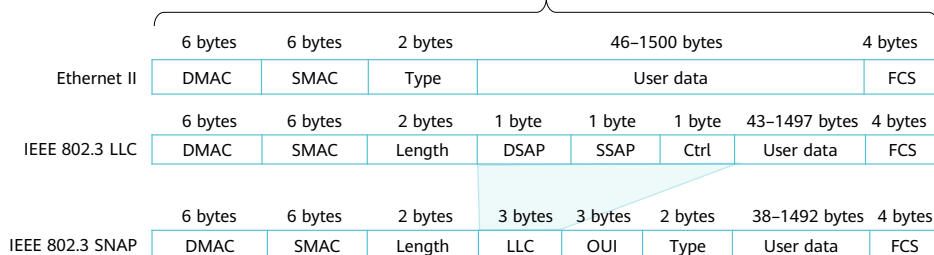
Contents

1. Overview of the Ethernet Protocol
- 2. Overview of Ethernet Frames**
3. Working Principles of Ethernet Switches

Ethernet Frame Format

- There are two standards for Ethernet frame formats: Ethernet II and IEEE 802.3.
- Ethernet II is a de facto Ethernet standard. Ethernet II frames, also known as Ethernet V2 frames, are the most common Ethernet frames.
- IEEE 802.3 LLC Ethernet frames are defined in the IEEE 802.3 standard and are evolved from Ethernet II frames. IEEE 802.3 SNAP Ethernet frames are developed by IEEE to support more upper-layer protocols and IP protocols on the basis of 802.3 LLC frames.

Total length of a data frame: 64-1518 bytes



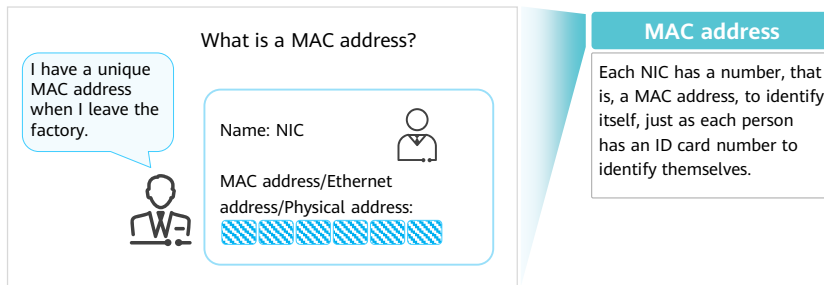
- Ethernet II frame:
 - DMAC: destination MAC address. This field identifies the receiver of the frame. The length is 6 bytes.
 - SMAC: source MAC address. This field identifies the sender of the frame. The length is 6 bytes.
 - Type: protocol type. The length is 2 bytes. Common values:
 - 0x0800: Internet Protocol Version 4 (IPv4)
 - 0x0806: Address Resolution Protocol (ARP)
- IEEE 802.3 LLC Ethernet frame:
 - Logical Link Control (LLC): This field consists of the Destination Service Access Point (DSAP), Source Service Access Point (SSAP), and Control fields.
 - DSAP: destination service access point. The length is 1 byte. If the upper-layer protocol is IP, the value is set to 0x06. The function of a service access point is similar to that of the Type field in an Ethernet II frame or the port number in TCP/UDP.
 - SSAP: source service access point. The length is 1 byte. If the upper-layer protocol is IP, the value is set to 0x06.
 - Ctrl: The length is 1 byte. The value of this field is usually set to 0x03.
- IEEE 802.3 Subnetwork Access Protocol (SNAP) Ethernet frame:
 - Organizationally unique identifier (OUI): The length is 3 bytes. The value is usually equal to the first three bytes of a MAC address, that is, the network

adapter vendor code.

- The meaning of the Type field is the same as that of the Type field in Ethernet II frames.
- The total length of a data frame ranges from 64 bytes to 1518 bytes. Why is this the case?
 - On an Ethernet network, the minimum frame length is 64 bytes, which is determined by the maximum transmission distance and the CSMA/CD mechanism.
 - The minimum frame length is specified to avoid the following situation: Site A has sent the last bit of a data packet, but the first bit of the data packet has not arrived at site B that is far away. Site B considers that the line is idle and sends data. As a result, a collision occurs.
 - An upper-layer protocol must ensure that the Data field contains at least 46 bytes. In this way, the 14-byte Ethernet frame header and 4-byte check code at the frame trailer can meet the minimum frame length of 64 bytes. If the actual data length is less than 46 bytes, the upper-layer protocol must fill in certain data units.
 - To strike a balance between transmission efficiency and transmission reliability, the maximum length of an Ethernet frame is 1518 bytes, and the corresponding IP data packet length is 1500 bytes.
 - A large frame length improves the data transmission efficiency. However, if the data frame is too long, the transmission of the frame over the shared link takes a long time, which greatly affects delay-sensitive applications.
 - As such, a compromise data frame length of 1518 bytes is used, which corresponds to an IP data packet length of 1500 bytes. This is the origin of the maximum transmission unit (MTU).

What Is a MAC Address?

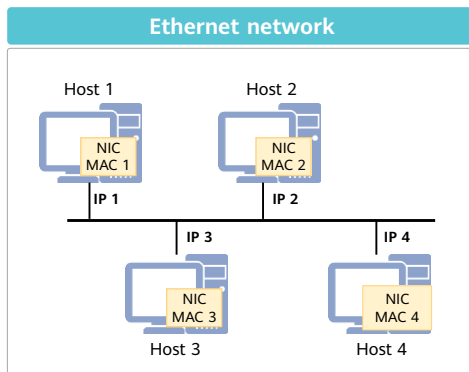
- A MAC address uniquely identifies a NIC on a network. Each NIC requires and has a unique MAC address.
- The MAC address of a NIC is globally unique.



- A MAC address, as defined and standardized in the IEEE 802 standard, identifies the position of a network device. All Ethernet NICs that comply with the IEEE 802 standard must have a MAC address. Different NICs have different MAC addresses.

IP Address vs. MAC Address

- Each Ethernet device is assigned a unique MAC address before delivery. Why is an IP address assigned to the device when it accesses a network?



Characteristics of IP addresses:

- IP addresses are unique.
- IP addresses are **changeable**.
- IP addresses are assigned based on the **network topology**.

Characteristics of MAC addresses:

- MAC addresses are unique.
- MAC addresses are **unchangeable**.
- MAC addresses are assigned based on **manufacturers**.

Is it acceptable for a network to have only either MAC or IP addresses?

- Each Ethernet device is assigned a unique MAC address before delivery. Why is an IP address still required by each host? In other words, if each host is assigned a unique IP address, why does a unique MAC address need to be embedded in a network device (such as a NIC) during production?
- The main reasons are as follows:
 - IP addresses are assigned based on network topologies, whereas MAC addresses are assigned based on manufacturers. If route selection is based on device manufacturers, this solution is not feasible.
 - When two-level addressing is used, devices are more flexible and easy to move and maintain.
 - For example, if an Ethernet NIC is faulty, you can replace it without changing its IP address. If an IP host is moved from one network to another, you can assign a new IP address to it without replacing its NIC.
- Summary:
 - An IP address uniquely identifies a node on a network, enabling data access across different network segments through IP addresses.
 - A MAC address uniquely identifies a NIC, enabling data access within the same network segment through MAC address.

MAC Address Representation

- A MAC address has 48 bits.
- A MAC address is usually represented by hexadecimal digits and hyphens (-).

For example, 00-1E-10-DD-DD-02 or 001E-10DD-DD02

Hexadecimal	00	1E	10	DD	DD	02	6 bytes
Binary	0000 0000	0001 1110	0001 0000	1101 1101	1101 1101	0000 0010	48 bits

Conversion between hexadecimal and binary digits

Power	2^3	2^2	2^1	2^0
	8	4	2	1
Bit	0	0	0	1

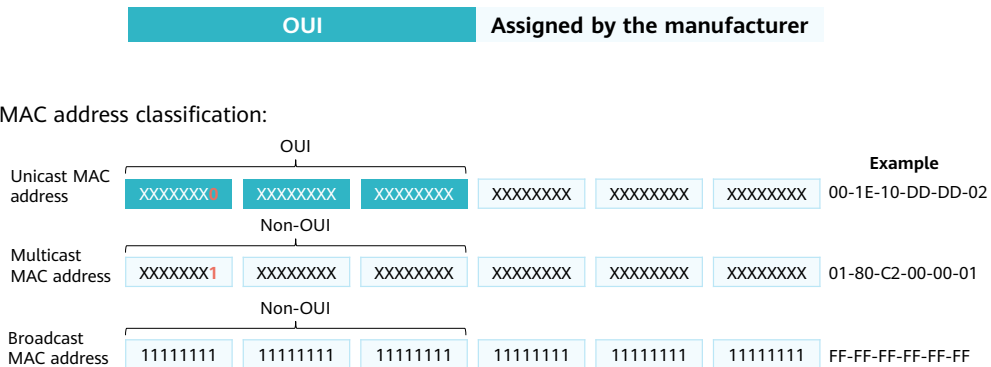
$= 0 + 0 + 0 + 1 = 1$

Power	2^3	2^2	2^1	2^0
	8	4	2	1
Bit	1	1	1	0

$= 8 + 4 + 2 + 0 = 14 = E$

MAC Address Structure and Classification

- The organizationally unique identifier (OUI) is allocated by IEEE and has a length of 24 bits.



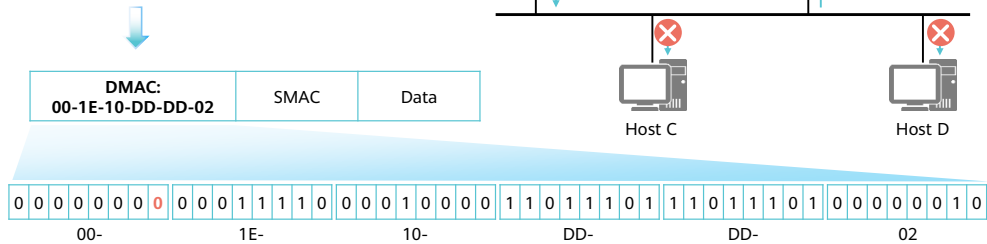
- Before manufacturing NICs, a manufacturer must register with the IEEE to obtain a 24-bit vendor code, which is also called an OUI.
- The last 24 bits of a MAC address are assigned by the manufacturer and are unique among the MAC addresses of all NICs manufactured by the manufacturer.
- MAC addresses fall into three types:
 - Unicast MAC address: also called physical MAC address. A unicast MAC address uniquely identifies a terminal on an Ethernet network and is a globally unique hardware address.
 - A unicast MAC address identifies a single node on a link.
 - A frame whose destination MAC address is a unicast MAC address is sent to a single node.
 - A unicast MAC address can be used as the source or destination address.
 - Note: Unicast MAC addresses are globally unique. If two terminals with the same MAC address are connected to a Layer 2 network (for example, due to misoperations), communication failures occur (for example, the two terminals cannot communicate with each other), and other devices cannot communicate with the two terminals.
 - Broadcast MAC address: an all-1 MAC address (FF-FF-FF-FF-FF-FF), which indicates all terminals on a LAN.
 - A broadcast MAC address can be considered as a special multicast

MAC address.

- The format of a broadcast MAC address is FFFF-FFFF-FFFF.
- A frame whose destination MAC address is a broadcast MAC address is sent to all nodes on a link.
- Multicast MAC address: indicates a group of terminals on a LAN. Except for broadcast MAC address, all the MAC addresses whose least significant bit of the first byte is 1 (for example, 01-00-00-00-00-00) are multicast MAC addresses.
 - A multicast MAC address identifies a group of nodes on a link.
 - A frame whose destination MAC address is a multicast MAC address is sent to a group of nodes.
 - A multicast MAC address can only be used as the destination address but not the source address.

Unicast Ethernet Frame

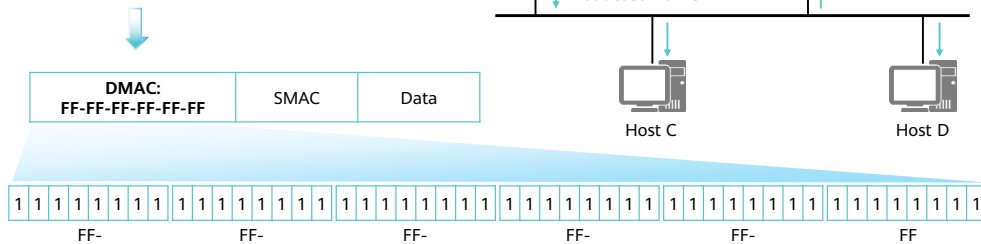
- Unicast frame for short
- The destination MAC address of a unicast frame is a unicast MAC address.



- Frames on a LAN can be sent in three modes.
- The first mode is unicast mode, in which frames are sent from a single source to a single destination.

Broadcast Ethernet Frame

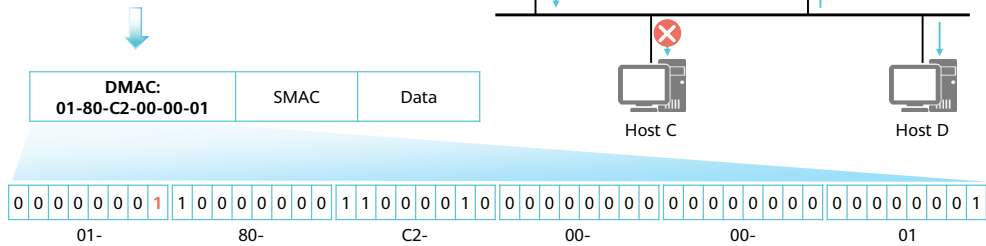
- Broadcast frame for short
- The destination MAC address of a broadcast frame is a broadcast MAC address.



- The second mode is broadcast mode, in which frames are sent from a single source to all hosts on a shared Ethernet network.
 - The destination MAC address of a broadcast frame is FF-FF-FF-FF-FF-FF. All hosts that receive the broadcast frame accept and process the frame.
 - In broadcast mode, a large amount of traffic is generated, which reduces the bandwidth utilization and affects the performance of the entire network.
 - The broadcast mode is usually used when all hosts on a network need to receive and process the same information.

Multicast Ethernet Frame

- Multicast frame for short
- The destination MAC address of a multicast frame is a multicast MAC address.

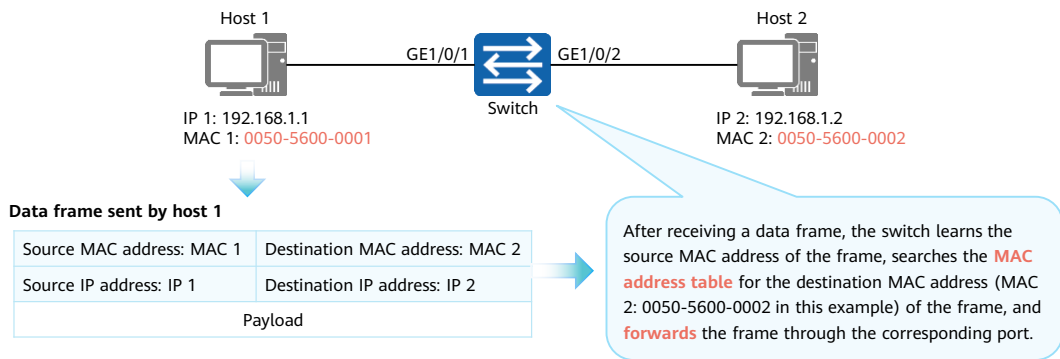


- The third mode is multicast mode.
 - Multicast forwarding can be considered as selective broadcast forwarding. Specifically, a host listens on a specific multicast address, and receives and processes frames whose destination MAC address is the multicast MAC address.
 - Generally, the multicast mode is used when a group of hosts on a network need to receive the same information, and other hosts are not affected.

Contents

1. Overview of the Ethernet Protocol
2. Overview of Ethernet Frames
- 3. Working Principles of Ethernet Switches**

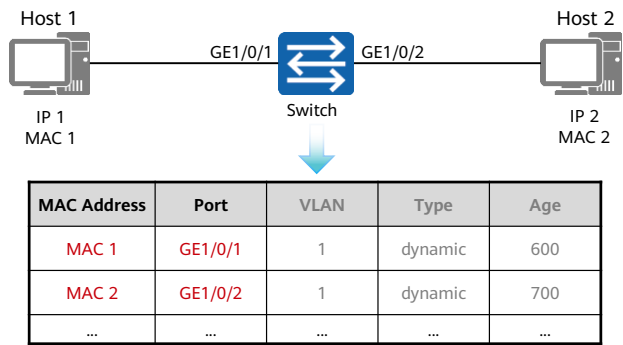
Working Principles of a Switch: Forwarding Data Frames Based on the MAC Address Table



- A Layer 2 switch works at the data link layer and forwards data frames based on MAC addresses. Different ports on a switch send and receive data independently, and each port belongs to a different collision domain. This effectively isolates collision domains on the network.
- A Layer 2 switch learns source MAC addresses of Ethernet frames to maintain the mappings between MAC addresses and ports in the MAC address table. When receiving a frame, the switch searches for the destination MAC address of the frame in the MAC address table to determine the port through which the frame is forwarded.

MAC Address Table

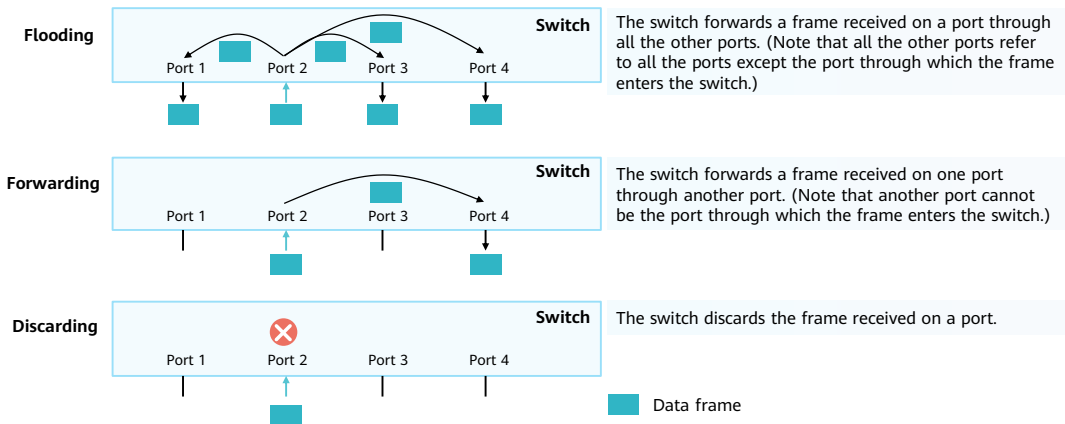
- A MAC address table on a device records the mappings between other devices' MAC addresses learned by the device, ports, and the VLANs to which the ports belong.
- This course focuses on MAC addresses and ports in the MAC address table.



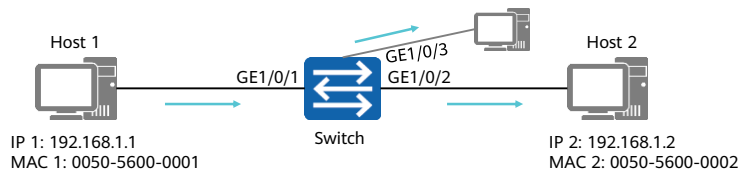
- A MAC address table records the mappings between other devices' MAC addresses learned by a switch and switch ports. When forwarding a data frame, the switch searches the MAC address table based on the destination unicast MAC address of the frame. If the MAC address table contains an entry matching the destination unicast MAC address of the frame, the switch forwards the frame through the outbound port in the entry. If no matching entry is found in the MAC address table, the switch floods the frame through all ports except the port that receives the frame.
- The MAC address table needs to be updated continuously to adapt to network changes. The dynamic MAC address entries automatically generated in the MAC address table are not always valid. Each entry has a lifetime. If an entry is not updated when its lifetime expires, the entry is deleted. The lifetime of an entry is called the aging time. If an entry is updated before the lifetime expires, the aging time of the entry is recalculated.

Three Data Frame Processing Behaviors of a Switch

- A switch processes a frame entering a port over a transmission medium in three ways:



Flooding



1 Host 1 sends three types of data frames: unicast frames, broadcast frames, and multicast frames.

2 For a unicast frame, the switch searches the MAC address table. If no matching entry is found, the switch considers the data frame as an unknown unicast frame. For broadcast and multicast frames, the switch does not need to search the MAC address table.

3 For an unknown unicast frame, a broadcast frame, or a multicast frame, the switch sends the frame through all the other ports.

Data frame sent by host 1

Unicast frame

Source MAC address: MAC 1

Destination MAC address: MAC 2

Broadcast frame

Source MAC address: MAC 1

Destination MAC address: FF-FF-FF-FF-FF-FF

Multicast frame

Source MAC address: MAC 1

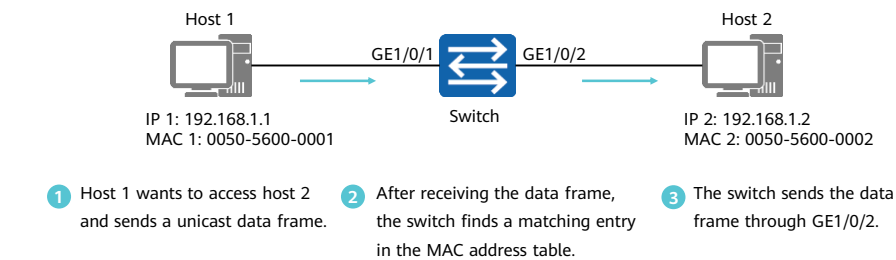
Destination MAC address: 01-00-5E-00-00-05

MAC address table of the switch

MAC Address	Port
MAC 1	GE1/0/1

- A unicast data frame whose destination MAC address does not exist in the MAC address table is called an unknown unicast frame.

Forwarding



Data frame sent by host 1

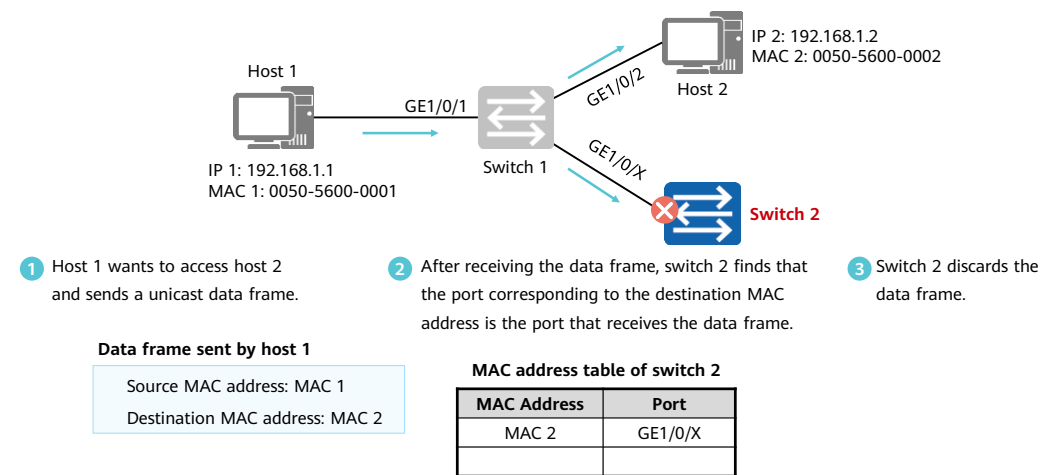
Source MAC address: MAC 1
Destination MAC address: MAC 2

MAC address table of the switch

MAC Address	Port
MAC 1	GE1/0/1
MAC 2	GE1/0/2

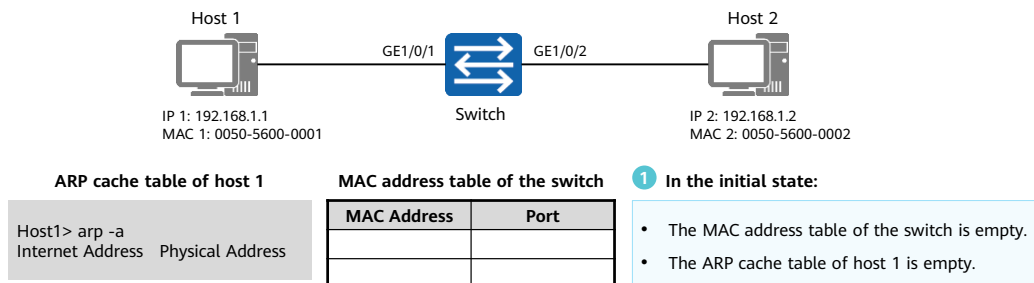
- If the received frame is a known unicast frame, the switch finds the destination MAC address of the frame in the MAC address table, and the corresponding port in the table is not the port through which the frame enters the switch over a transmission medium, the switch forwards the unicast frame.

Discarding



- If the received frame is a known unicast frame, the switch finds the destination MAC address of the frame in the MAC address table, and the corresponding port in the table is the port through which the frame enters the switch over a transmission medium, the switch discards the unicast frame.

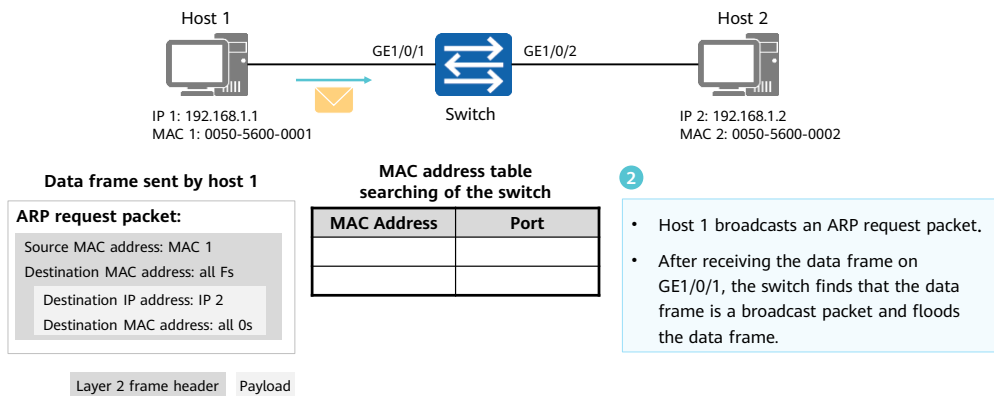
MAC Address Learning of a Switch (1)



- In the initial state:
 - The switch does not know the MAC address of the connected host, and therefore its MAC address table is empty.
 - Host 1 does not know the MAC address of host 2, and its Address Resolution Protocol (ARP) cache table is empty.
- ARP resolves a known IP address to obtain the corresponding MAC address.
- ARP provides the following functions:
 - Resolves IP addresses into MAC addresses.
 - Maintains and caches the mappings between IP addresses and MAC addresses in ARP entries.
 - Detects duplicate IP addresses on a network segment.
- Working layer of ARP:
 - ARP is a TCP/IP protocol defined in RFC 826 by the IETF. It is used to obtain the data link layer address based on the IP address. ARP is usually considered as a protocol between the network layer and the data link layer.
 - The core of ARP is to resolve IP addresses into MAC addresses. MAC addresses are the core concept at the data link layer. ARP implementation depends on the frame structure at the data link layer and does not involve protocols at the network layer or upper layers. Therefore, ARP is usually classified as a data link layer protocol.
- ARP cache table:

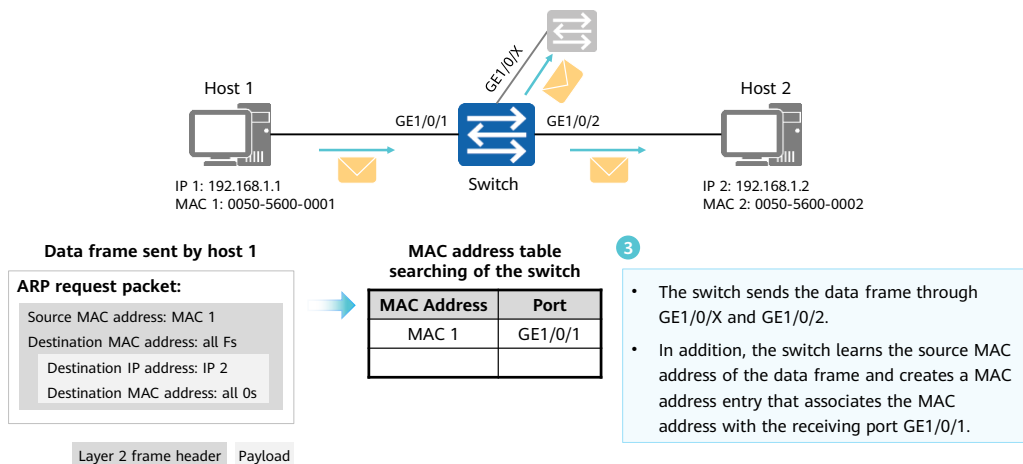
- A terminal device usually has an ARP cache table to store the mappings between IP addresses and MAC addresses.
- The learned mappings between IP addresses and MAC addresses are stored in the ARP cache table for a period of time. Within the validity period, the device can directly search the table for the destination MAC address for data encapsulation, without performing ARP query. After the validity period expires, ARP entries are automatically deleted.
- If the destination device is located on another network, the source device searches the ARP cache table for the MAC address of the gateway, sends the data to the gateway, and the gateway then forwards the data to the destination device.

MAC Address Learning of a Switch (2)



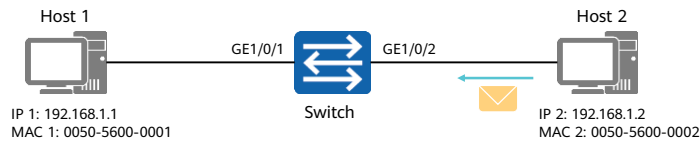
- When host 1 sends an ARP request packet, the destination MAC address of the Ethernet data frame is 0xFFFF-FFFF-FFFF. In the payload of the ARP request packet, the destination IP address is the IP address of the destination host, and the destination MAC address is all 0s.

MAC Address Learning of a Switch (3)



- The switch records the source MAC address of the received data frame and the corresponding port in the MAC address table.
- Note: The dynamically learned entries in the MAC address table are not always valid. Each entry has a lifetime. If an entry is not updated when its lifetime expires, the entry is deleted. The lifetime of an entry is called the aging time. For example, the default aging time of entries on Huawei S series switches is 300 seconds.

MAC Address Learning of a Switch (4)



4

- The hosts connected to other ports of the switch also receive the data frame but discard it.
- Host 2 receives and processes the data frame, and sends a reply data frame destined for host 1 to the switch.

Data frame sent by host 2

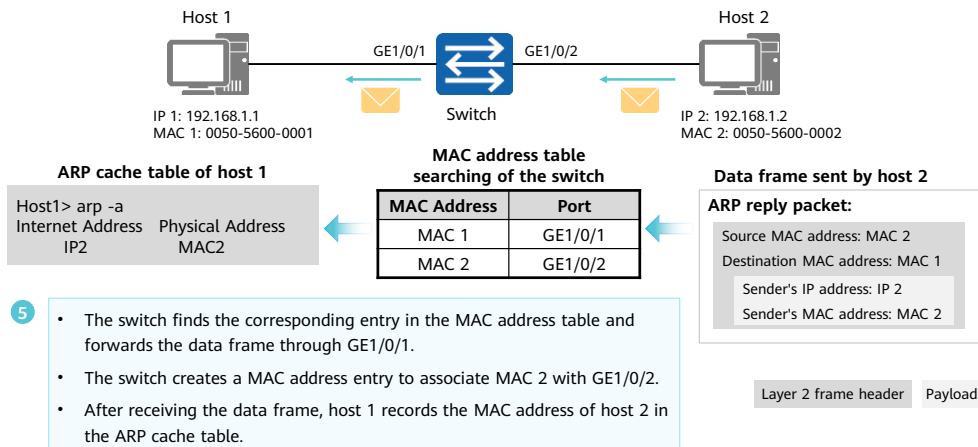
ARP reply packet:

Source MAC address: MAC 2
Destination MAC address: MAC 1
Sender's IP address: IP 2
Sender's MAC address: MAC 2

Layer 2 frame header Payload

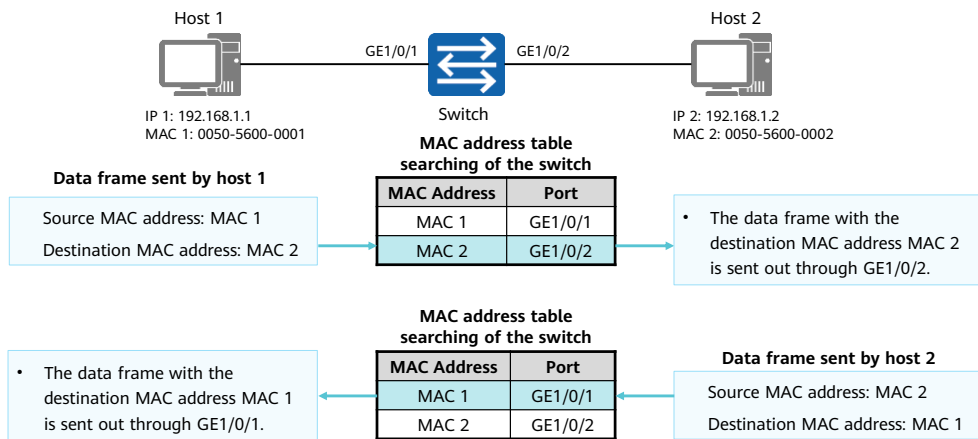
- All hosts on the broadcast network receive the data frame, but only host 2 processes the data frame because the data frame requests the MAC address corresponding to IP 2.
- Host 2 sends a reply data frame destined for host 1, and the destination MAC address of the data frame is MAC 1. In the payload of the ARP reply packet, the sender's IP address is the local IP address, and the sender's MAC address is the local MAC address.
- The ARP reply packet is a unicast packet.

MAC Address Learning of a Switch (5)



- After receiving the unicast data frame, the switch searches its MAC address table. If the switch finds a matching entry, it forwards the data frame through the corresponding port.
- In addition, the switch records the source MAC address of the received data frame and the corresponding port in the MAC address table.
- After receiving the ARP reply packet, host 1 records the mapping between IP 2 and MAC 2 in the ARP cache table.

Subsequent Data Frame Forwarding of the Switch Based on the MAC Address Table



- After creating MAC address entries, the switch directly forwards subsequently received packets sent by host 1 and host 2 to each other based on the destination MAC addresses.

Quiz

1. (Single-answer question) Based on which field of the packets received by a port does an Ethernet switch generate a MAC address entry? ()
 - A. Source MAC address
 - B. Destination MAC address
 - C. Source IP address
 - D. Destination IP address
2. (True or false) A switch has eight ports. A unicast frame enters the switch through one of the ports. However, the switch cannot find an entry matching the destination MAC address of the frame in the MAC address table. In this case, the switch discards the frame. ()
 - A. True
 - B. False

1. A

2. B

Summary

- This course provides an overview of the Ethernet protocol and describes the Ethernet frame format and MAC address.
- This course also describes the working principles of switches. After receiving a data frame, the switch learns the source MAC address of the frame, searches the MAC address table for the destination MAC address of the frame, and forwards the frame through the corresponding port.

Thank you.

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每个组织，构建万物互联的智能世界。
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organization for a fully connected,
intelligent world.

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VLAN Fundamentals and Configuration



Foreword

- Ethernet technology implements data communication over shared media based on Carrier Sense Multiple Access with Collision Detection (CSMA/CD). When there are a large number of hosts on an Ethernet network, security risks and broadcast storms may occur, degrading network performance and even causing a network breakdown.
- To solve the preceding problems, Virtual Local Area Network (VLAN) technology is developed.
- This course describes VLAN concepts, working principles of different Layer 2 interfaces, VLAN applications, data forwarding principles, and related configurations.

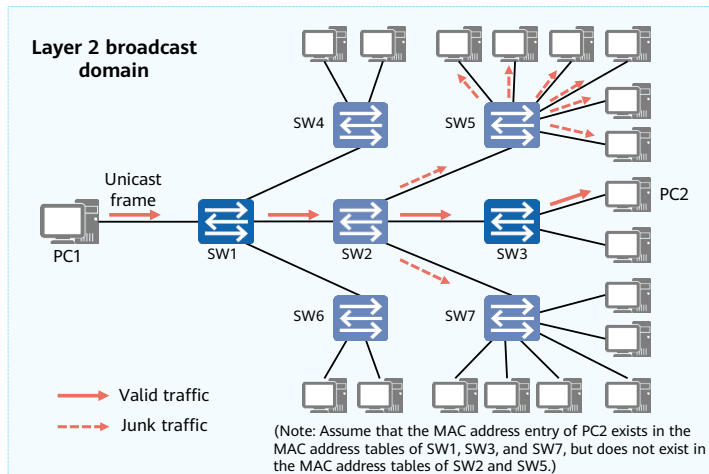
Objectives

- Upon completion of this course, you will be able to:
 - Understand the background of VLAN technology.
 - Identify the VLAN to which data belongs.
 - Master different VLAN assignment methods.
 - Describe the forwarding process of VLAN frames on a network.
 - Master basic VLAN configurations.

Contents

- 1. What Is a VLAN?**
2. Basic Concepts of VLANs
3. VLAN Application
4. VLAN Configuration Examples

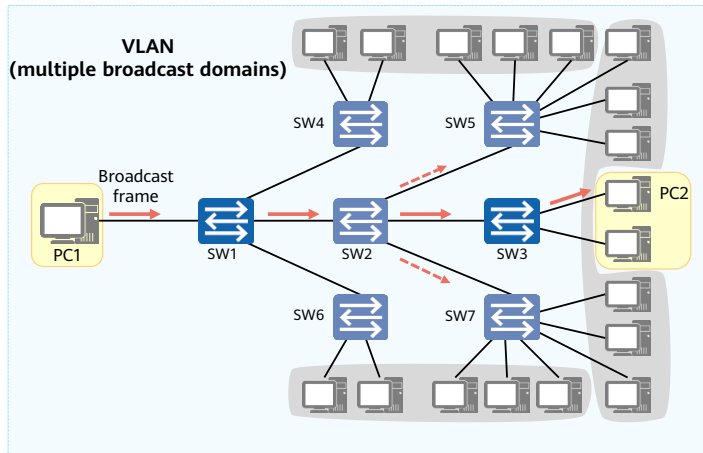
Problems Facing a Traditional Ethernet Network



- On a typical switched network, when a host sends a broadcast frame or an unknown unicast frame, the frame is flooded or even transmitted to the entire broadcast domain.
- The larger the broadcast domain is, the more serious network security and junk traffic problems are.

- As shown in the figure, PC1 sends a unicast frame to PC2. The MAC address entry of PC2 exists in the MAC address tables of SW1, SW3, and SW7, but does not exist in the MAC address tables of SW2 and SW5. In this case, SW1 and SW3 forward the unicast frame in point-to-point mode, SW7 discards the unicast frame, and SW2 and SW5 flood the unicast frame. As a result, although PC2 receives the unicast frame, many non-destination hosts on the network also receive the frame that should not be received.
- Obviously, the larger the broadcast domain is, the more serious the network security and junk traffic problems are.

Virtual Local Area Network (VLAN)



- VLAN technology isolates broadcast domains.
- Characteristics:
 - It is not restricted by geographical locations.
 - Only devices in the same VLAN can directly communicate with each other at Layer 2.

- VLAN technology is introduced to solve the problems caused by broadcast domains.
 - VLANs can be deployed on switches to logically divide a large broadcast domain into multiple smaller broadcast domains, which effectively improves network security, reduces junk traffic, and saves network resources.
- VLAN characteristics:
 - A VLAN is a broadcast domain. PCs in the same VLAN can directly communicate with each other at Layer 2. PCs in different VLANs cannot directly communicate with each other at Layer 2 and can only communicate with each other at Layer 3 to transmit information. In this way, broadcast packets are confined to a VLAN.
 - VLAN assignment is not restricted by geographical locations.
- Advantages of VLAN technology:
 - Flexible virtual workgroup establishment: VLAN technology allows hosts to be divided into different groups, and hosts in different geographical locations can be grouped together, simplifying network construction and maintenance.
 - Restricted broadcast domains: A broadcast domain is confined to a single

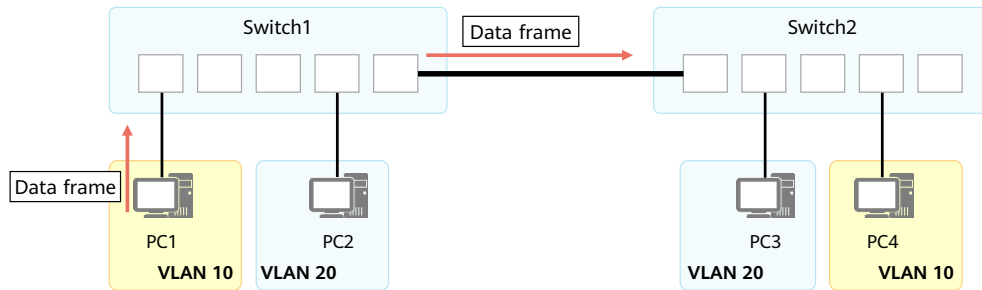
VLAN, conserving bandwidth and improving network efficiency.

- Enhanced LAN security. Packets from different VLANs are transmitted separately, so that hosts in a VLAN cannot directly communicate with hosts in another VLAN.
 - Improved network robustness: A fault in one VLAN does not affect hosts in another VLAN.
- Note: Layer 2 refers to the data link layer.

Contents

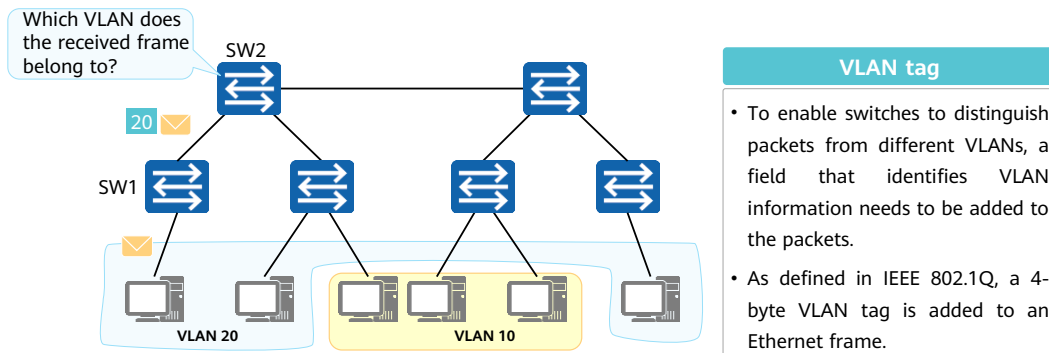
1. What Is a VLAN?
- 2. Basic Concepts of VLANs**
3. VLAN Application
4. VLAN Configuration Examples

How Do I Identify a VLAN?



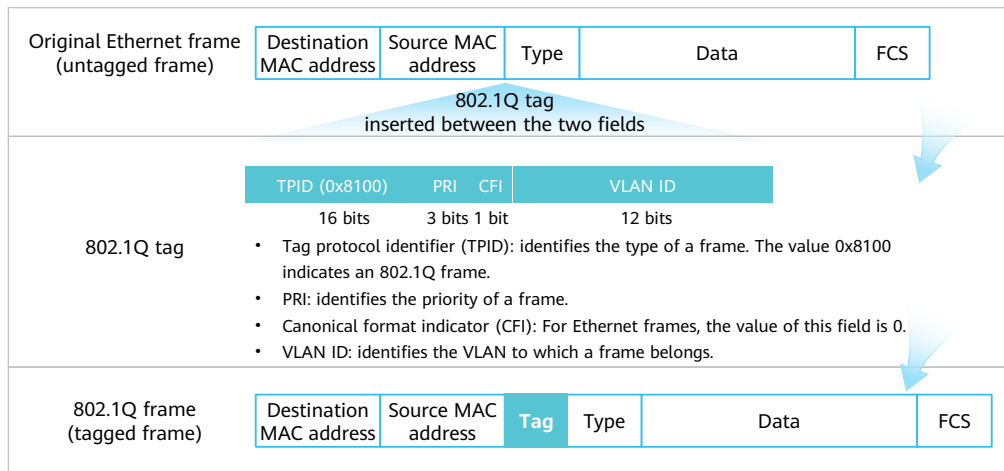
- Switch1 and Switch2 belong to the same enterprise, and the enterprise plans VLANs on the network. VLAN 10 is used for department A and VLAN 20 is used for department B. Employees in departments A and B are connected to both Switch1 and Switch2.
- A data frame sent by PC1 reaches Switch2 through the link between Switch1 and Switch2. If no processing is performed, Switch2 can neither identify the VLAN to which the data frame belongs nor determine the local VLAN to which the data frame should be sent.

VLAN Tag



- As shown in the figure, after identifying the VLAN to which a frame belongs, SW1 adds a tag to the frame at a specific position. The tag indicates the VLAN to which the frame belongs. After receiving the tagged data frame, other switches (such as SW2) can easily identify the VLAN to which the frame belongs based on the tag information.
- IEEE 802.1Q defines the format of such tagged data frames. The data frames in this format are called IEEE 802.1Q frames or VLAN frames.

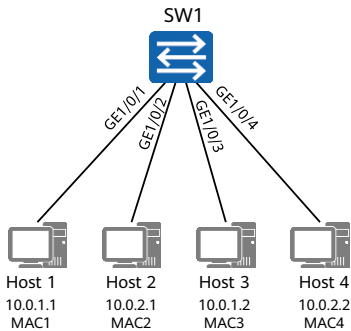
VLAN Frame



- Main fields in a VLAN frame:
 - Tag protocol identifier (TPID): indicates the data frame type. The length is 2 bytes.
 - The value 0x8100 indicates an IEEE 802.1Q VLAN frame. If a device that does not support 802.1Q receives such a frame, the device discards the frame.
 - Device vendors can define the value of this field. When the TPID value of a neighbor device is set to a value other than 0x8100, the TPID value of the local device must be changed to be the same as that of the neighbor device so that the local device can identify such frames and communicate with the neighbor device.
 - PRI: indicates the priority of a data frame. The length is 3 bits.
 - The value ranges from 0 to 7. A larger value indicates a higher priority. If congestion occurs on a network, a switch preferentially sends data frames with higher priorities.
 - Canonical format indicator (CFI): indicates whether a MAC address is encapsulated in canonical format over different transmission media. This field is used to ensure compatibility between Ethernet and token ring networks. The length is 1 bit.

- The value 0 indicates that the MAC address is encapsulated in canonical format, and the value 1 indicates that the MAC address is encapsulated in non-canonical format.
- The CFI field has a fixed value of 0 on Ethernet networks.
- VID: indicates the ID of the VLAN to which a data frame belongs. It is also called a VLAN ID. The length is 12 bits.
 - The VLAN ID ranges from 0 to 4095. The values 0 and 4095 are reserved, and therefore available VLAN IDs are in the range from 1 to 4094.
 - A switch uses the VID contained in the VLAN tag to identify the VLAN to which a data frame belongs. Broadcast frames are forwarded only in the local VLAN.
- Method of identifying frames with VLAN tags:
 - Value of the Length/Type field of frames = 0x8100
- Note: PCs cannot identify tagged frames and therefore can send or process only untagged frames. By contrast, all frames processed by switches are tagged ones, improving processing efficiency.

VLAN Assignment Methods



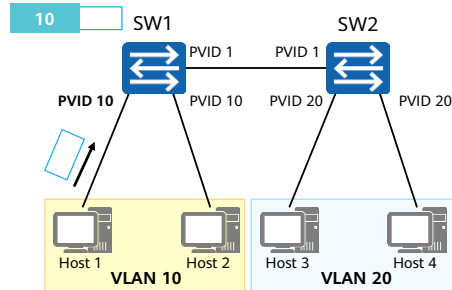
VLAN Assignment Method	VLAN 10	VLAN 20
Interface-based assignment	GE1/0/1, GE1/0/3	GE1/0/2, GE1/0/4
MAC address-based assignment	MAC1, MAC3	MAC2, MAC4
IP subnet-based assignment	10.0.1.*	10.0.2.*
Protocol-based assignment	IPv4	IPv6
Policy-based assignment	10.0.1.* + GE1/0/1 + MAC1	10.0.2.* + GE1/0/2 + MAC2

- PCs send only untagged frames. After receiving such an untagged frame, a switch that supports the VLAN feature must assign the frame to a specific VLAN based on certain rules.
- There are the following VLAN assignment methods:
 - Interface-based assignment: VLANs are assigned based on switch interfaces.
 - A network administrator preconfigures a different PVID for each interface of a switch. When an untagged frame arrives at an interface of the switch, the switch adds a tag carrying the PVID of the interface to the frame. The frame is then transmitted in the specified VLAN.
 - MAC address-based VLAN assignment: VLANs are assigned based on the source MAC addresses of frames.
 - A network administrator preconfigures the mappings between MAC addresses and VLAN IDs. When receiving an untagged frame, a switch adds a specified VLAN tag to the frame based on the mappings. The frame is then transmitted in the specified VLAN.
 - IP subnet-based VLAN assignment: VLANs are assigned based on the source IP addresses and subnet masks of frames.
 - A network administrator preconfigures the mappings between IP addresses and VLAN IDs. When receiving an untagged frame, a switch

adds a specified VLAN tag to the frame based on the mappings. The frame is then transmitted in the specified VLAN.

- Protocol-based assignment: VLANs are assigned based on the protocol (suite) types and encapsulation formats of frames.
 - A network administrator preconfigures the mappings between protocol types and VLAN IDs. When receiving an untagged frame, a switch adds a specified VLAN tag to the frame based on the mappings. The frame is then transmitted in the specified VLAN.
- Policy-based assignment: VLANs are assigned based on policies such as combinations of interfaces, MAC addresses, and IP addresses.
 - A network administrator preconfigures a policy. When receiving an untagged frame that matches the policy, a switch adds a specified VLAN tag to the frame. The frame is then transmitted in the specified VLAN.

Interface-based VLAN Assignment



The network administrator **needs to** reconfigure VLANs when hosts are moved.

Interface-based VLAN assignment

• Fundamentals

- VLANs are assigned based on switch interfaces.
- A network administrator preconfigures a different **PVID** for each interface of a switch to add the interface to the VLAN corresponding to the PVID.
- When an interface of a switch receives an untagged frame, the switch adds a tag carrying the PVID of the interface to the frame. The frame is then transmitted in the specified VLAN.

• Default VLAN (PVID)

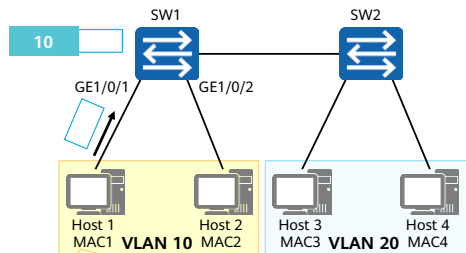
- A port VLAN ID (PVID) is the default VLAN ID of an interface.
- The value ranges from 1 to 4094.

- Assignment principle:
 - VLAN IDs are configured on physical interfaces of a switch. All PC-sent untagged frames arriving at a physical interface are assigned to the VLAN corresponding to the PVID configured on the interface.
- Characteristics:
 - This VLAN assignment method is simple, intuitive, and easy to implement. Currently, it is the most widely used VLAN assignment method.
 - If the switch interface to which a PC is connected changes, the VLAN to which the frames sent by the PC belong may also change.
- Default VLAN ID: port VLAN ID (PVID)
 - Each switch interface must be configured with a PVID. All untagged frames arriving at a switch interface are assigned to the VLAN corresponding to the PVID configured on the interface.
 - The default PVID is 1.

MAC Address-based VLAN Assignment

Mappings between MAC addresses and VLAN IDs on SW1

MAC Address	VLAN ID
MAC1	10
MAC2	10
...	...



The network administrator **does not need to** reconfigure VLANs when hosts are moved.

MAC address-based VLAN assignment

• Fundamentals

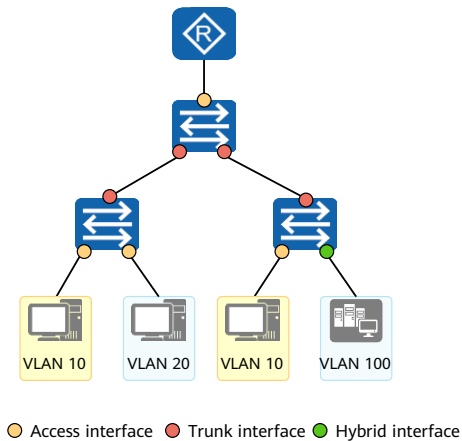
- VLANs are assigned based on source MAC addresses of frames.
- A network administrator preconfigures the mappings between MAC addresses and VLAN IDs.
- When receiving an untagged frame, a switch adds a specified VLAN tag to the frame based on the mappings. The frame is then transmitted in the specified VLAN.

• Mapping table

- The table records the mappings between MAC addresses and VLAN IDs.

- Assignment principle:
 - A switch maintains a table recording the mappings between MAC addresses and VLAN IDs. When receiving an untagged frame from a PC, the switch analyzes the source MAC address of the frame, searches the mapping table for the VLAN ID mapping the MAC address, and assigns the frame to the corresponding VLAN based on the mapping.
- Characteristics:
 - This assignment method is more complex, but more flexible.
 - If the switch interface to which a PC is connected changes, the VLAN to which the frames sent by the PC belong does not change because the MAC address of the PC remains unchanged.
 - However, as malicious PCs can easily forge MAC addresses, this assignment method is prone to security risks.

Types of Layer 2 Ethernet Interfaces



Interface type

- **Access interface**

An access interface is usually used to connect a switch to a terminal, such as a PC or server. In general, the NICs on such a terminal send and receive only untagged frames. An access interface can be added to only one VLAN.

- **Trunk interface**

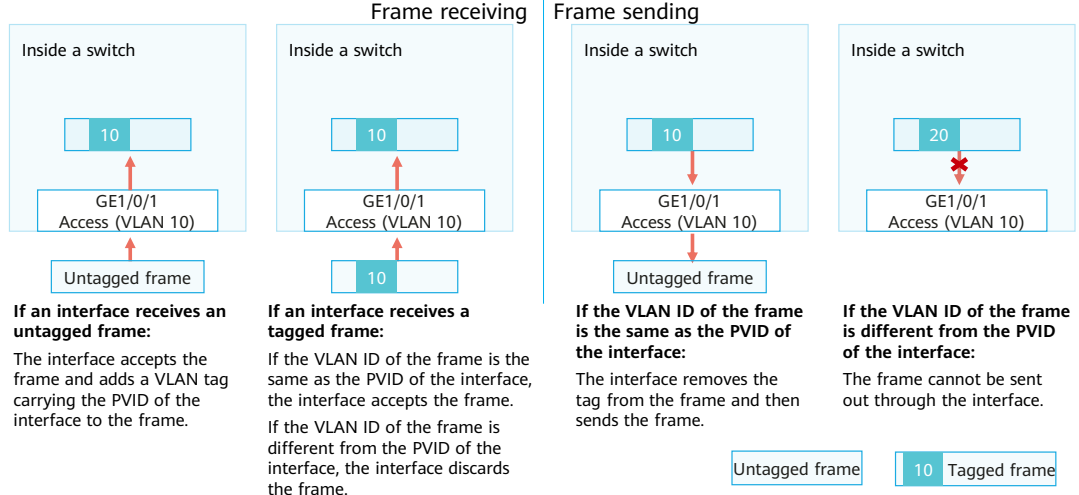
A trunk interface allows frames from multiple VLANs to pass through. These data frames are differentiated by 802.1Q tags. A trunk interface is usually used to connect a switch to another switch or a sub-interface on a device, such as a router or firewall.

- **Hybrid interface**

Similar to a trunk interface, a hybrid interface also allows frames from multiple VLANs to pass through. These frames are differentiated by 802.1Q tags. You can determine whether a hybrid interface adds tags to sent data frames of one or more VLANs.

- The interface-based VLAN assignment method varies according to the switch interface type.
- Access interface
 - An access interface often connects to a terminal (such as a PC or server) that cannot identify VLAN tags, or is used when VLANs do not need to be differentiated.
- Trunk interface
 - A trunk interface often connects to a switch, router, AP, or voice terminal that can send and receive both tagged and untagged frames.
- Hybrid interface
 - A hybrid interface can connect to a terminal (such as a PC or server) that cannot identify VLAN tags or to a switch, router, AP, or voice terminal that can send and receive both tagged and untagged frames.

Access Interface

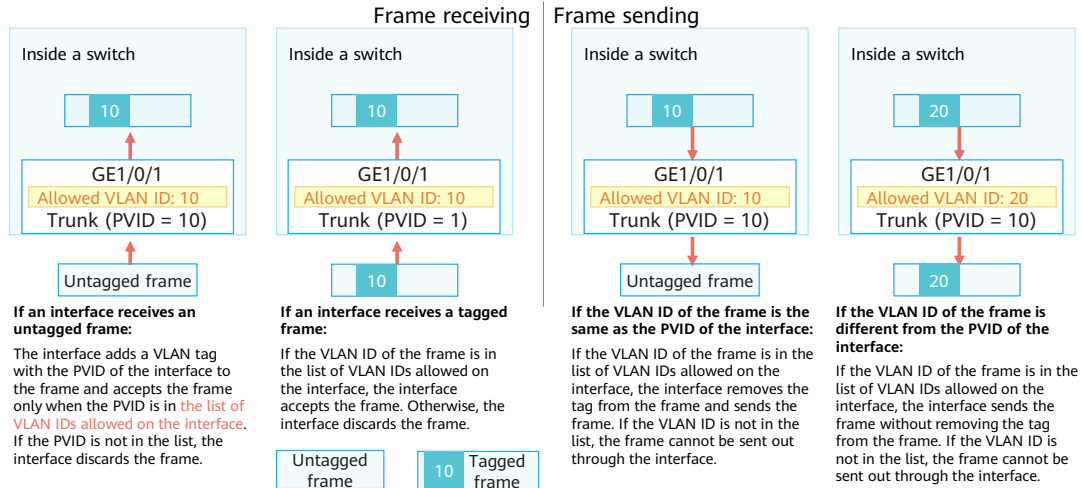


- The preceding sections describe how a switch identifies the VLAN to which a data frame belongs and VLAN assignment methods. How does a switch process untagged and tagged frames?
- Characteristics of an access interface:
 - It allows only frames whose VLAN ID is the same as the PVID of the interface to pass through.
- Frame receiving through an access interface:
 - After an access interface receives an untagged frame, the switch adds a tag with the VID being the PVID of the interface to the frame and then floods, forwards, or discards the tagged frame.
 - After an access interface receives a tagged frame, the switch checks whether the VID in the tag of the frame is the same as the PVID of the interface. If they are the same, the switch forwards the tagged frame. If they are different, the switch directly discards the tagged frame.
- Frame sending through an access interface:
 - After a tagged frame arrives at an access interface from another interface on the switch, the switch checks whether the VID in the tag of the frame is the same as the PVID of the access interface.
 - If they are the same, the switch removes the tag from the tagged

frame and sends the untagged frame.

- If they are different, the switch directly discards the tagged frame.

Trunk Interface



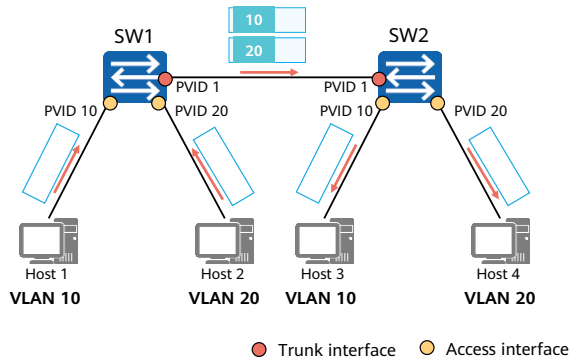
- For a trunk interface, you need to configure not only a PVID but also a list of VLAN IDs allowed on the interface. By default, VLAN 1 exists in the list.
- Characteristics of a trunk interface:
 - A trunk interface allows only frames whose VLAN IDs are in the list of VLAN IDs allowed on the interface to pass through.
 - A trunk interface allows tagged frames from multiple VLANs but untagged frames from only one VLAN to pass through.
- Frame receiving through a trunk interface:
 - After a trunk interface receives an untagged frame, the switch adds a tag with the VID being the PVID of the interface to the frame and then checks whether the PVID is in the list of VLAN IDs allowed on the interface. If the PVID is in the list, the switch forwards the tagged frame. If the PVID is not in the list, the switch directly discards the tagged frame.
 - After a trunk interface receives a tagged frame, the switch checks whether the VID in the tag of the frame is in the list of VLAN IDs allowed on the interface. If the VID is in the list, the switch forwards the tagged frame. If the VID is not in the list, the switch discards the tagged frame.
- Frame sending through a trunk interface:
 - After a tagged frame arrives at a trunk interface from another interface on

the switch, the switch checks whether the VID in the tag of the frame is in the list of VLAN IDs allowed on the trunk interface. If the VID is not in the list, the interface directly discards the tagged frame.

- After a tagged frame arrives at a trunk interface from another interface on the switch, the switch checks whether the VID in the tag of the frame is in the list of VLAN IDs allowed on the trunk interface. If the VID is in the list, the switch checks whether the VID in the tag is the same as the PVID of the trunk interface.
 - If they are the same, the switch removes the tag from the tagged frame and sends the untagged frame.
 - If they are different, the switch directly sends the tagged frame without removing the tag from the frame.

Example of Access and Trunk Interfaces

- Describe the entire process of data forwarding between hosts.

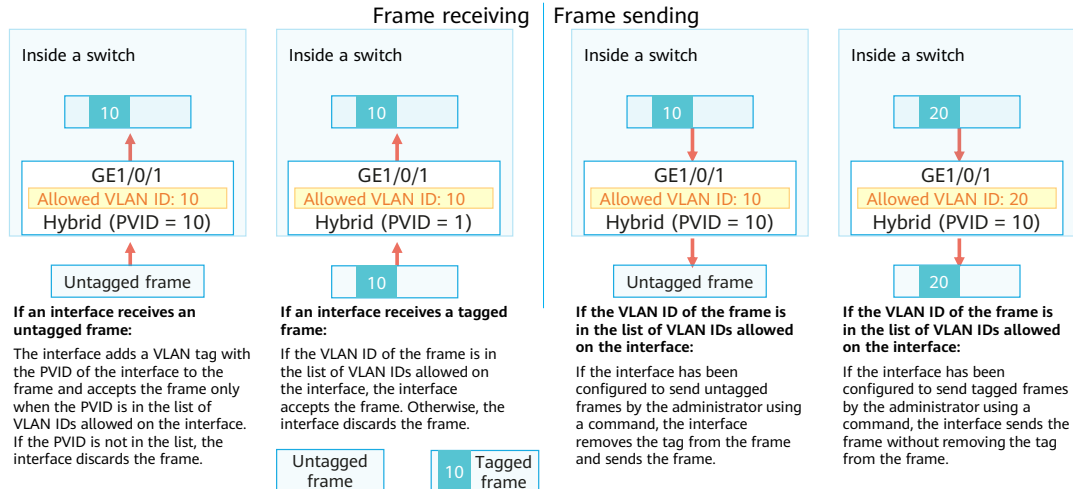


Trunk interfaces on SW1 and SW2

List of Allowed VLAN IDs	
VLAN ID	1
	10
	20

- In this example, SW1 and SW2 connect to hosts through access interfaces. PVIDs of the interfaces are shown in the figure. SW1 and SW2 connect to each other through trunk interfaces whose PVIDs are 1. The table lists the VLAN IDs allowed on the trunk interfaces.
- Describe the entire process of data forwarding between hosts.

Hybrid Interface

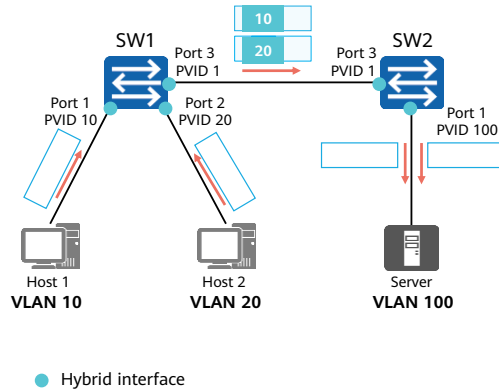


- For a hybrid interface, you need to configure not only a PVID but also two lists of VLAN IDs allowed on the interface: one untagged VLAN ID list and one tagged VLAN ID list. By default, VLAN 1 is in the untagged VLAN ID list. Frames from all the VLANs in the two lists are allowed to pass through the hybrid interface.
- Characteristics of a hybrid interface:
 - A hybrid interface allows only frames whose VLAN IDs are in the lists of VLAN IDs allowed on the interface to pass through.
 - A hybrid interface allows tagged frames from multiple VLANs to pass through. Frames sent from a hybrid interface can be either tagged or untagged, depending on the interface configuration.
 - The main difference between a hybrid interface and a trunk interface is that a hybrid interface allows untagged frames from multiple VLANs to pass through.
- Frame receiving through a hybrid interface:
 - After a hybrid interface receives an untagged frame, the switch adds a tag with the VID being the PVID of the interface to the frame and then checks whether the PVID is in the untagged or tagged VLAN ID list. If the PVID is in either list, the switch forwards the tagged frame. If the PVID is not in either list, the switch directly discards the tagged frame.

- After a hybrid interface receives a tagged frame, the switch checks whether the VID in the tag of the frame is in the untagged or tagged VLAN ID list. If the VID is in either list, the switch forwards the tagged frame. If the VID is not in either list, the switch discards the tagged frame.
- Frame sending through a hybrid interface:
 - After a tagged frame arrives at a hybrid interface from another interface on the switch, if the VID in the tag of the frame is neither in the untagged VLAN ID list nor in the tagged VLAN ID list allowed on the hybrid interface, the switch discards the tagged frame.
 - After a tagged frame arrives at a hybrid interface from another interface on the switch, if the VID in the tag of the frame is in the untagged VLAN ID list allowed on the hybrid interface, the switch removes the tag from the tagged frame and sends the untagged frame.
 - After a tagged frame arrives at a hybrid interface from another interface on the switch, if the VID in the tag of the frame is in the tagged VLAN ID list allowed on the hybrid interface, the switch directly sends the tagged frame without removing the tag from the frame.

Example of Hybrid Interfaces

- Describe the entire process for hosts to access a server.



Lists of Allowed VLAN IDs on SW1

Port 1		Port 2		Port 3	
Untagged		Untagged		Tagged	
VLAN ID	1	VLAN ID	1	VLAN ID	10
	10		20		20
	100		100		100

Lists of Allowed VLAN IDs on SW2

Port 1		Port 3	
Untagged		Tagged	
VLAN ID	1	VLAN ID	10
	10		20
	20		100
	100		

- In this example, SW1 and SW2 connect to hosts and each other through hybrid interfaces. PVIDs of the interfaces are shown in the figure. The tables list the VLAN IDs allowed on the hybrid interfaces.
- Describe the entire process for the two hosts to access the server.

Summary

Access interface	Trunk interface	Hybrid interface
Frame receiving - Untagged frame: The interface adds a tag with the PVID to the frame and accepts the frame. - Tagged frame: The interface compares the VLAN ID of the frame with the PVID. If they are the same, the interface accepts the frame. If they are different, the interface discards the frame.	Frame receiving - Untagged frame: The interface adds a tag with the PVID to the frame. If the VLAN ID is in the list of VLAN IDs allowed on the interface, the interface accepts the frame. If the VLAN ID is not in the list, the interface discards the frame. - Tagged frame: The interface checks whether the VLAN ID of the frame is in the list of VLAN IDs allowed on the interface. If the VLAN ID is in the list, the interface accepts the frame. If the VLAN ID is not in the list, the interface discards the frame.	Frame receiving - Untagged frame: The interface adds a tag with the PVID to the frame. If the VLAN ID is in the list of VLAN IDs allowed on the interface, the interface accepts the frame. If the VLAN ID is not in the list, the interface discards the frame. - Tagged frame: The interface checks whether the VLAN ID of the frame is in the list of VLAN IDs allowed on the interface. If the VLAN ID is in the list, the interface accepts the frame. If the VLAN ID is not in the list, the interface discards the frame.
Frame sending The interface compares the VLAN ID of the frame with the PVID. If they are the same, the interface removes the tag from the frame and sends the frame. If they are different, the interface discards the frame.	Frame sending - If the VLAN ID of the frame is in the list of VLAN IDs allowed on the interface and is the same as the PVID, the interface removes the tag from the frame and sends the frame. - If the VLAN ID of the frame is in the list of VLAN IDs allowed on the interface but is different from the PVID, the interface directly sends the frame without removing the tag from the frame. - If the VLAN ID of the frame is not in the list of VLAN IDs allowed on the interface, the interface directly discards the frame.	Frame sending - If the VLAN ID of the frame is not in the list of VLAN IDs allowed on the interface, the interface directly discards the frame. - If the VLAN ID of the frame is in the untagged VLAN ID list, the interface removes the tag from the frame and sends the frame. - If the VLAN ID of the frame is in the tagged VLAN ID list, the interface directly sends the frame without removing the tag from the frame.

- The processes of adding and removing VLAN tags on interfaces are as follows:
 - Frame receiving:
 - After receiving an untagged frame, access, trunk, and hybrid interfaces all add a VLAN tag to the frame. Then, trunk and hybrid interfaces determine whether to accept the frame based on whether the VID of the frame is an allowed VLAN ID, whereas an access interface accepts the frame unconditionally.
 - After receiving a tagged frame, access, trunk, and hybrid interfaces all determine whether to accept the frame based on whether the VID of the frame is an allowed VLAN ID (the allowed VLAN ID of an access interface is the default VLAN ID).
 - Frame sending:
 - An access interface directly removes the VLAN tag from a frame.
 - A trunk interface removes the VLAN tag from a frame only when the VID of the frame is the same as the PVID of the interface.
 - A hybrid interface determines whether to remove the VLAN tag from a frame based on the interface configuration.
- Therefore, frames sent by an access interface are all untagged; frames sent by a trunk interface are tagged except for frames with the PVID of the interface;

frames sent by a hybrid interface can be tagged or untagged based on the interface configuration.

Contents

1. What Is a VLAN?
2. Basic Concepts of VLANs
- 3. VLAN Application**
4. VLAN Configuration Examples

VLAN Planning

- **VLAN assignment principles**

- By service: for example, VLANs for voice, video, and document services.
- By department: for example, VLANs for engineering, marketing, and finance departments.
- By application: for example, VLANs for servers, offices, and classrooms.

- **VLAN planning example:**

- Assume that there are three buildings in a campus: administrative building with offices, finance department, and classrooms, teaching building with offices and classrooms, and office building with offices and finance department. Each building has one access switch, and the core switch is deployed in the administrative building.
- The following table describes the VLAN plan.

VLAN	IP Address Segment	Description
1	X.16.10.0/24	VLAN to which office users belong
2	X.16.20.0/24	VLAN to which finance department users belong
3	X.16.30.0/24	VLAN to which classroom users belong
100	Y.16.100.0/24	VLAN used for device management

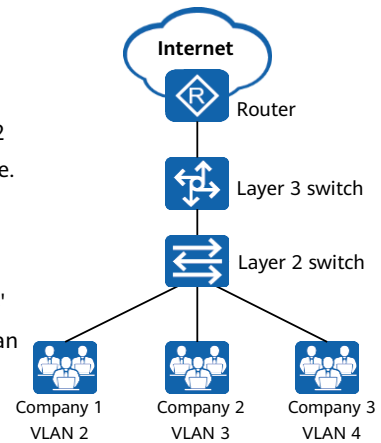
- **Tips for VLAN assignment**

VLAN IDs can be randomly assigned within the supported range. To improve VLAN ID continuity, you can associate VLAN IDs with subnets during VLAN assignment.

- You are advised to assign consecutive VLAN IDs to ensure proper use of VLAN resources. Interface-based VLAN assignment is the most commonly used method.

Interface-based VLAN Assignment

- Application scenario:
 - There are multiple companies in a building. These companies share network resources to reduce costs. Networks of the companies connect to different interfaces of the same Layer 2 switch and access the Internet through the same egress device.
- VLAN assignment:
 - To isolate the services of different companies and ensure service security, assign interfaces connected to the companies' networks to different VLANs. In this way, each company has an independent network and each VLAN is a virtual workgroup.



Contents

1. What Is a VLAN?
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Basic VLAN Configuration Commands

1. Create VLANs.

[Huawei] **vlan** *vlan-id*

This command creates a VLAN and displays the VLAN view, or displays the view of an existing VLAN.

- The value of *vlan-id* is an integer that ranges from 1 to 4094.

[Huawei] **vlan batch** { *vlan-id1* [**to** *vlan-id2*] }

This command creates VLANs in batches.

- **batch**: indicates that VLANs are created in batches.
- *vlan-id1*: specifies the ID of the first VLAN.
- *vlan-id2*: specifies the ID of the last VLAN.

- The **vlan** command creates a VLAN and displays the VLAN view, or displays the view of an existing VLAN.
- The **undo vlan** command deletes a specified VLAN.
- By default, all interfaces are added to the default VLAN, that is, VLAN 1.
 - **vlan** *vlan-id*
 - *vlan-id*: specifies a VLAN ID. The value is an integer that ranges from 1 to 4094.
 - **vlan batch** { *vlan-id1* [**to** *vlan-id2*] }
 - **batch**: indicates that VLANs are created in batches.
 - *vlan-id1* **to** *vlan-id2*: specifies the IDs of VLANs to be created in batches.
 - *vlan-id1* specifies the ID of the first VLAN.
 - *vlan-id2* specifies the ID of the last VLAN. The value of *vlan-id2* must be greater than or equal to that of *vlan-id1*. *vlan-id1* and *vlan-id2* specify a VLAN range.
 - If **to** *vlan-id2* is not specified, only the VLAN specified by *vlan-id1* is created.
 - The values of *vlan-id1* and *vlan-id2* are integers that range from 1 to

4094.

Basic Configuration Commands for Access Interfaces

1. Set the interface type.

```
[Huawei-GE1/0/1] port link-type access
```

In the interface view, set the link type of the interface to access.

2. Configure the default VLAN of the access interface.

```
[Huawei-GE1/0/1] port default vlan vlan-id
```

In the interface view, configure the default VLAN of the interface and add the interface to the VLAN.

- *vlan-id* specifies the default VLAN ID. The value is an integer that ranges from 1 to 4094.

Basic Configuration Commands for Trunk Interfaces

1. Set the interface type.

```
[Huawei-GE1/0/1] port link-type trunk
```

In the interface view, set the link type of the interface to trunk.

2. Add the trunk interface to specified VLANs.

```
[Huawei-GE1/0/1] port trunk allow-pass vlan { { vlan-id1 [ to vlan-id2 ] } | all }
```

In the interface view, add the trunk interface to specified VLANs.

3. (Optional) Configure the default VLAN of the trunk interface.

```
[Huawei-GE1/0/1] port trunk pvid vlan vlan-id
```

In the interface view, configure the default VLAN of the trunk interface.

- Command: **port trunk allow-pass vlan** { { *vlan-id1* [*to* *vlan-id2*] } | **all** }
 - *vlan-id1* [*to* *vlan-id2*]: specifies the VLANs to which a trunk interface is added.
 - *vlan-id1* specifies the ID of the first VLAN.
 - **to** *vlan-id2* specifies the ID of the last VLAN. The value of *vlan-id2* must be greater than or equal to that of *vlan-id1*.
 - The values of *vlan-id1* and *vlan-id2* are integers that range from 1 to 4094.
 - **all**: indicates that a trunk interface is added to all VLANs.
- Command: **port trunk pvid vlan** *vlan-id*, which sets the default VLAN of a trunk interface
 - *vlan-id*: specifies the default VLAN ID of a trunk interface. The value is an integer that ranges from 1 to 4094.

Basic Configuration Commands for Hybrid Interfaces

1. Set the interface type.

```
[Huawei-GE1/0/1] port link-type hybrid
```

In the interface view, set the link type of the interface to hybrid.

2. Add the hybrid interface to specified VLANs.

```
[Huawei-GE1/0/1] port hybrid untagged vlan { { vlan-id1 [ to vlan-id2 ] } | all }
```

In the interface view, add the hybrid interface to specified VLANs. Frames of these VLANs then pass through the hybrid interface in untagged mode.

```
[Huawei-GE1/0/1] port hybrid tagged vlan { { vlan-id1 [ to vlan-id2 ] } | all }
```

In the interface view, add the hybrid interface to specified VLANs. Frames of these VLANs then pass through the hybrid interface in tagged mode.

3. (Optional) Configure the default VLAN of the hybrid interface.

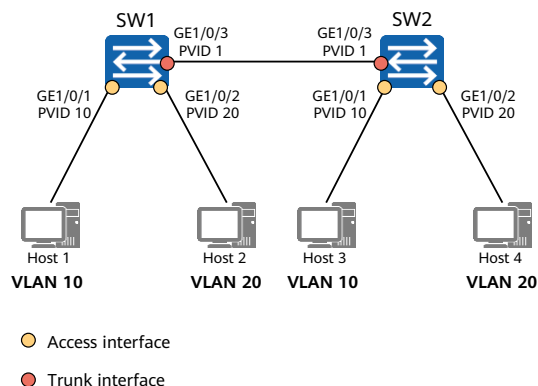
```
[Huawei-GE1/0/1] port hybrid pvid vlan vlan-id
```

In the interface view, configure the default VLAN of the hybrid interface.

- Command: **port hybrid untagged vlan { { vlan-id1 [to vlan-id2] } | all }**
 - **vlan-id1 [to vlan-id2]**: specifies the VLANs to which a hybrid interface is added.
 - **vlan-id1** specifies the ID of the first VLAN.
 - **to vlan-id2** specifies the ID of the last VLAN. The value of **vlan-id2** must be greater than or equal to that of **vlan-id1**.
 - The values of **vlan-id1** and **vlan-id2** are integers that range from 1 to 4094.
 - **all**: indicates that a hybrid interface is added to all VLANs.
- Command: **port hybrid tagged vlan { { vlan-id1 [to vlan-id2] } | all }**
 - **vlan-id1 [to vlan-id2]**: specifies the VLANs to which a hybrid interface is added.
 - **vlan-id1** specifies the ID of the first VLAN.
 - **to vlan-id2** specifies the ID of the last VLAN. The value of **vlan-id2** must be greater than or equal to that of **vlan-id1**.
 - The values of **vlan-id1** and **vlan-id2** are integers that range from 1 to 4094.
 - **all**: indicates that a hybrid interface is added to all VLANs.

- Command: **port hybrid pvid vlan** *vlan-id*, which sets the default VLAN of a hybrid interface
 - *vlan-id*: specifies the default VLAN ID of a hybrid interface. The value is an integer that ranges from 1 to 4094.

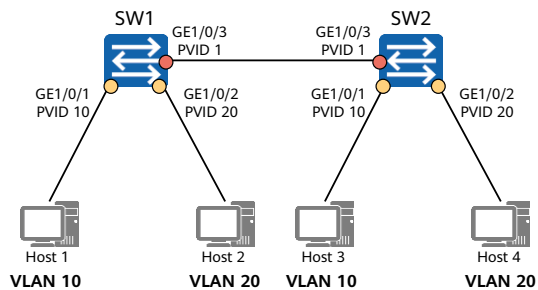
Using Trunk and Access Interfaces to Isolate Services of Different Departments



- Networking requirements:
 - Switches of an enterprise connect to many departments, and each department has multiple users. To ensure communication security, the enterprise requires that users in the same department can communicate with each other, but users in different departments cannot directly communicate with each other.
 - You can configure interface-based VLAN assignment on the switches and add the interfaces connected to users in the same department to the same VLAN. In this way, users in different VLANs cannot directly communicate at Layer 2, and users in the same VLAN can directly communicate with each other.

- Configuration roadmap:
 - Create VLANs and add interfaces connected to users to the VLANs to isolate Layer 2 traffic between users with different services.
 - Configure interface types and specify allowed VLANs for interfaces on SW1 and SW2 to allow users with the same service to communicate with each other through SW1 and SW2.

Creating VLANs

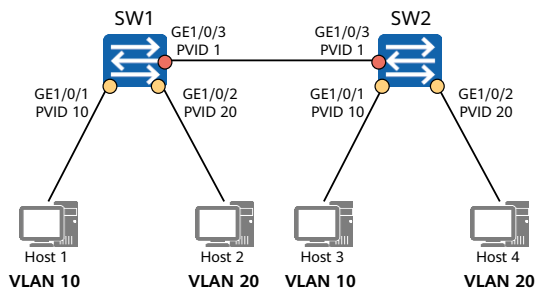


Create VLANs.

```
[SW1] vlan 10
[SW1-vlan10] quit
[SW1] vlan 20
[SW1-vlan20] quit
```

```
[SW2] vlan batch 10 20
```

Configuring Access and Trunk Interfaces



Configure access interfaces and add the interfaces to corresponding VLANs.

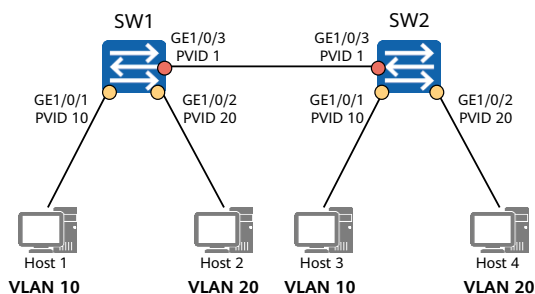
```
[SW1] interface GE 1/0/1
[SW1-GE1/0/1] port link-type access
[SW1-GE1/0/1] port default vlan 10
[SW1-GE1/0/1] quit
[SW1] interface GE 1/0/2
[SW1-GE1/0/2] port link-type access
[SW1-GE1/0/2] port default vlan 20
```

Configure a trunk interface and configure allowed VLANs for the interface.

```
[SW1] interface GE 1/0/3
[SW1-GE1/0/3] port link-type trunk
[SW1-GE1/0/3] port trunk pvid vlan 1
[SW1-GE1/0/3] port trunk allow-pass vlan 10 20
```

Note: The configuration of SW2 is similar to that of SW1.

Verifying the Configuration



[SW1] display vlan

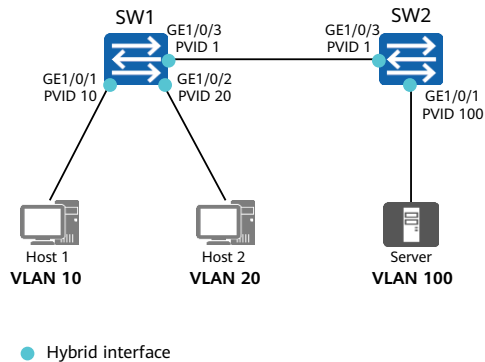
The total number of vlans is : 3

U: Up; D: Down; TG: Tagged; UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;

VID	Type	Ports
1	common	UT:GE1/0/3(U)
10	common	UT:GE1/0/1(U) TG:GE1/0/3(U)
20	common	UT:GE1/0/2(U) TG:GE1/0/3(U)
.....		

- Command: The **display vlan** command displays information about VLANs.
- Command output:
 - **Tagged/Untagged Port:** interfaces that are manually added to a VLAN in tagged or untagged mode.
 - **VID or VLAN ID:** VLAN ID.
 - **Type or VLAN Type:** VLAN type. The value **common** indicates a common VLAN.
 - **Ports:** interfaces added to a VLAN.

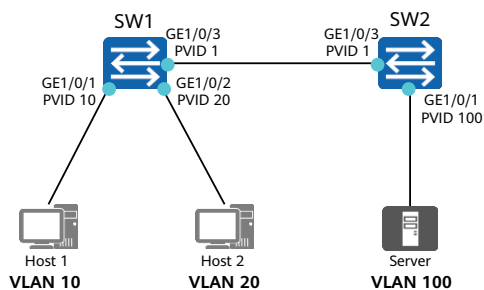
Using Hybrid Interfaces to Allow Multiple Departments to Access a Server



- Networking requirements:
 - Switches of an enterprise connect to many users, and users in different departments need to access the enterprise server. To ensure communication security, the enterprise requires that users in different departments cannot directly communicate with each other.
 - You can configure interface-based VLAN assignment and hybrid interfaces on the switches so that users in different departments cannot directly communicate with each other at Layer 2 but can directly access the enterprise server.

- Configuration roadmap:
 - Create VLANs and add interfaces connected to users to the VLANs to isolate Layer 2 traffic between users with different services.
 - Configure interface types and specify allowed VLANs for interfaces on SW1 and SW2 to allow hosts to communicate with the server through SW1 and SW2.

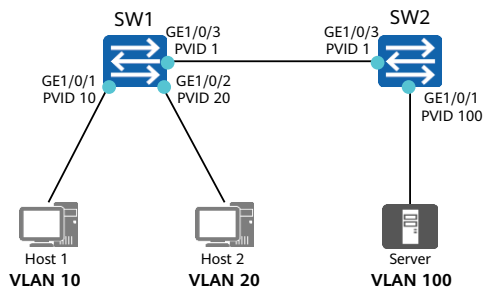
Basic Hybrid Interface Configuration (1)



The configuration of SW1 is as follows:

```
[SW1] vlan batch 10 20 100
[SW1] interface GE 1/0/1
[SW1-GE1/0/1] port link-type hybrid
[SW1-GE1/0/1] port hybrid pvid vlan 10
[SW1-GE1/0/1] port hybrid untagged vlan 10 100
[SW1-GE1/0/1] interface GE 1/0/2
[SW1-GE1/0/2] port link-type hybrid
[SW1-GE1/0/2] port hybrid pvid vlan 20
[SW1-GE1/0/2] port hybrid untagged vlan 20 100
[SW1-GE1/0/2] interface GE 1/0/3
[SW1-GE1/0/3] port link-type hybrid
[SW1-GE1/0/3] port hybrid tagged vlan 10 20 100
```

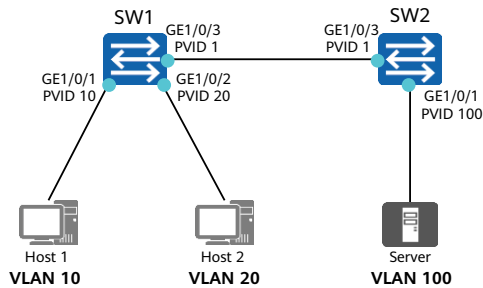
Basic Hybrid Interface Configuration (2)



The configuration of SW2 is as follows:

```
[SW2] vlan batch 10 20 100
[SW2] interface GE 1/0/1
[SW2-GE1/0/1] port link-type hybrid
[SW2-GE1/0/1] port hybrid pvid vlan 100
[SW2-GE1/0/1] port hybrid untagged vlan 10 20 100
[SW2-GE1/0/1] interface GE 1/0/3
[SW2-GE1/0/3] port link-type hybrid
[SW2-GE1/0/3] port hybrid tagged vlan 10 20 100
```

Verifying the Configuration



[SW1] display vlan

The total number of vlans is : 4

U: Up; D: Down; TG: Tagged; UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;

VID	Type	Ports
1	common	UT:GE1/0/1(U) GE1/0/2(U) GE1/0/3(U)
10	common	UT:GE1/0/1(U) TG:GE1/0/3(U)
20	common	UT:GE1/0/2(U) TG:GE1/0/3(U)
100	common	UT:GE1/0/1(U) GE1/0/2(U) TG:GE1/0/3(U)
.....		

Basic VLAN Configuration Commands

1. Associate a MAC address with a VLAN.

```
[Huawei-vlan10] mac-vlan mac-address mac-address [ mac-address-mask | mac-address-mask-length ]
```

This command associates a MAC address with a VLAN.

- *mac-address*: specifies the MAC address to be associated with a VLAN. The value is in the format of H-H-H. Each H is a hexadecimal number that contains one to four digits, such as 00e0 and fc01. If an H contains fewer than four digits, the leftmost digits are padded with zeros. For example, e0 is displayed as 00e0. The MAC address cannot be set to 0000-0000-0000, FFFF-FFFF-FFFF, or a multicast address.
- *mac-address-mask*: specifies the MAC address mask. The value is in the format of H-H-H. Each H is a hexadecimal number that contains one to four digits.
- *mac-address-mask-length*: specifies the MAC address mask length. The value is an integer that ranges from 1 to 48.

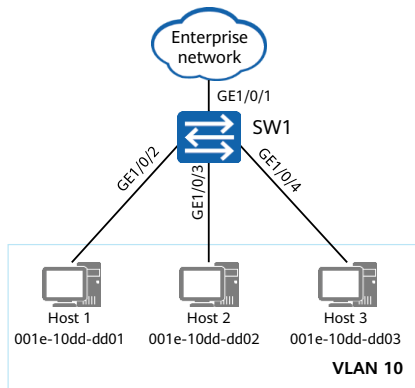
2. Enable MAC address-based VLAN assignment.

```
[Huawei-GE1/0/1] mac-vlan enable
```

This command enables MAC address-based VLAN assignment on an interface.

- Command: **mac-vlan mac-address** *mac-address* [*mac-address-mask* | *mac-address-mask-length*]
 - *mac-address*: specifies the MAC address to be associated with a VLAN.
 - The value is in the format of H-H-H. Each H is a hexadecimal number that contains one to four digits, such as 00e0 and fc01. If an H contains fewer than four digits, the leftmost digits are padded with zeros. For example, e0 is displayed as 00e0.
 - The MAC address cannot be set to 0000-0000-0000, FFFF-FFFF-FFFF, or a multicast address.
 - *mac-address-mask*: specifies the MAC address mask.
 - The value is in the format of H-H-H. Each H is a hexadecimal number that contains one to four digits.
 - *mac-address-mask-length*: specifies the MAC address mask length.
 - The value is an integer that ranges from 1 to 48.
- Command: **mac-vlan enable**, which enables MAC address-based VLAN assignment on an interface

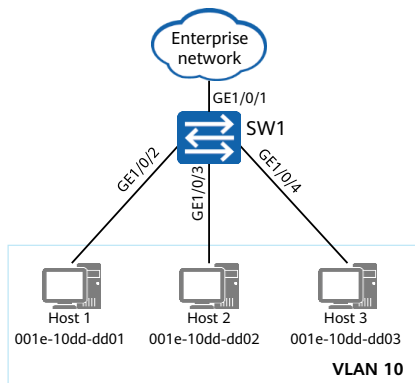
Example for Configuring MAC Address-based VLAN Assignment



- Networking requirements:
 - On the network of a company, host 1, host 2, and host 3 are moved frequently.
 - The customer requires that host 1, host 2, and host 3 belong to VLAN 10 regardless of the interfaces through which they are connected to the switch.
 - To meet this requirement, you can configure MAC address-based VLAN assignment to bind MAC addresses of hosts to a VLAN.

- Configuration roadmap:
 - Create a VLAN.
 - Add Ethernet interfaces to the VLAN in the correct mode.
 - Associate the MAC addresses of host 1, host 2, and host 3 with the VLAN so that the VLAN can be determined based on the source MAC addresses of packets.

Creating a VLAN and Associating MAC Addresses with the VLAN



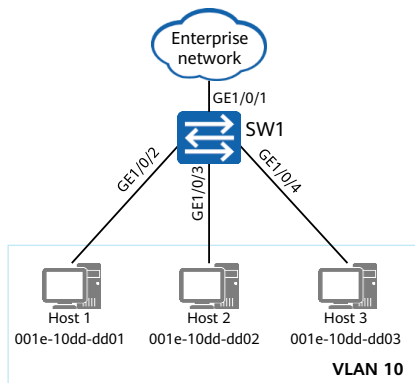
Create a VLAN.

```
[SW1] vlan 10
[SW1-vlan10] quit
```

Associate MAC addresses with the VLAN.

```
[SW1] vlan 10
[SW1-vlan10] mac-vlan mac-address 001e-10dd-dd01
[SW1-vlan10] mac-vlan mac-address 001e-10dd-dd02
[SW1-vlan10] mac-vlan mac-address 001e-10dd-dd03
[SW1-vlan10] quit
```


Adding Interfaces to the VLAN and Enabling MAC Address-based VLAN Assignment



Add interfaces to the VLAN.

```
[SW1] interface GE 1/0/1
[SW1-GE1/0/1] port link-type hybrid
[SW1-GE1/0/1] port hybrid tagged vlan 10
[SW1-GE1/0/1] quit
[SW1] interface GE 1/0/2
[SW1-GE1/0/2] port link-type hybrid
[SW1-GE1/0/2] port hybrid untagged vlan 10
```

Enable MAC address-based VLAN assignment on an interface.

```
[SW1] interface GE 1/0/2
[SW1-GE1/0/2] mac-vlan enable
[SW1-GE1/0/2] quit
```

Note: The configurations of GE1/0/3 and GE1/0/4 are similar to the configuration of GE1/0/2.

- Configure MAC address-based VLAN assignment on hybrid interfaces. This function can be used on access and trunk interfaces only when a VLAN ID assigned based on a MAC address is the same as the PVID. Therefore, you are advised to configure MAC address-based VLAN assignment on hybrid interfaces.

Verifying the Configuration

```
[SW1] display vlan
The total number of vlans is : 2
```

```
U: Up;           D: Down;           TG: Tagged; UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;
```

VID	Type	Ports
1	common	UT:GE1/0/1(U) GE1/0/2(U) GE1/0/3(U)
10	common	UT:GE1/0/2(U) GE1/0/3(U) GE1/0/4(U) TG:GE1/0/1(U)

```
[SW1] display mac-vlan mac-address all
```

MAC Address	MASK	VLAN	Priority
001e-10dd-dd01	ffff-ffff-ffff	10	0
001e-10dd-dd02	ffff-ffff-ffff	10	0
001e-10dd-dd03	ffff-ffff-ffff	10	0

```
Total MAC VLAN address count: 3
```

- Command: **display mac-vlan { mac-address { all | *mac-address* [*mac-address-mask* | *mac-address-mask-length*] } | vlan *vlan-id* }**, which displays the configuration of MAC address-based VLAN assignment
 - **all**: displays the VLAN assignment information of all MAC addresses.
 - **mac-address *mac-address***: displays the VLAN assignment information of a specified MAC address.
 - The value is in the format of H-H-H. Each H is a hexadecimal number that contains one to four digits.
 - ***mac-address-mask***: specifies the MAC address mask.
 - The value is in the format of H-H-H. Each H is a hexadecimal number that contains one to four digits.
 - ***mac-address-mask-length***: specifies the MAC address mask length.
 - The value is an integer that ranges from 1 to 48.
 - **vlan *vlan-id***: displays the MAC address-based VLAN assignment configuration of a specified VLAN.
 - The value is an integer that ranges from 1 to 4094.
- Command output:
 - **MAC Address**: MAC address.

- **MASK:** mask of a MAC address
- **VLAN:** VLAN assigned based on a MAC address.
- **Priority:** 802.1p priority of the VLAN associated with a MAC address.

Quiz

1. (Multiple-answer question) Which of the following statements about VLANs are false? ()
 - A. A trunk interface can only be connected to a switch, but cannot be connected to a terminal.
 - B. VLAN technology can isolate a large Layer 2 broadcast domain into several small Layer 2 broadcast domains.
 - C. PCs in different VLANs cannot communicate with each other.
 - D. PCs in the same VLAN can communicate with each other at Layer 2.
2. (Short-answer question) If the PVID of a trunk interface is 5 and the **port trunk allow-pass vlan 2 3** command is configured on the interface, which VLANs' frames can be transmitted through the trunk interface?

1. AC
2. Answer: After the **port trunk allow-pass vlan 2 3** command is run, frames of VLAN 5 cannot be transmitted through the trunk interface. By default, frames of VLAN 1 can be transmitted through the trunk interface. Therefore, frames of VLAN 1, VLAN 2, and VLAN 3 can be transmitted through the trunk interface.

Summary

- This course describes VLAN technology, including VLAN functions, VLAN identification and assignment, VLAN data exchange, VLAN planning and application, and basic VLAN configurations.
- VLAN technology can divide a physical LAN into multiple broadcast domains so that network devices in the same VLAN can directly communicate with each other at Layer 2, while devices in different VLANs cannot directly communicate with each other at Layer 2.

Thank you.

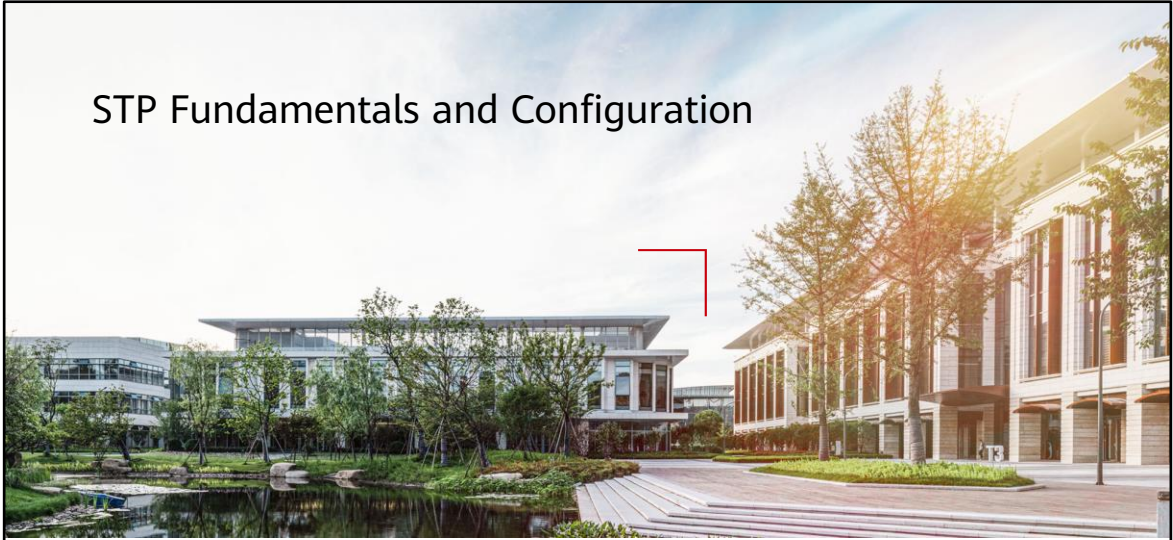
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每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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STP Fundamentals and Configuration



Foreword

- On an Ethernet switched network, redundant links are typically used to implement link backup and improve network reliability. The use of redundant links, however, may cause loops on the switched network, leading to broadcast storms and unstable MAC address entries. As a result, communication quality deteriorates and communication may even be interrupted. To prevent loops on switched networks, the Spanning Tree Protocol (STP) is introduced.
- Devices running STP discover loops on the network by exchanging information with each other and trim the ring topology into a loop-free tree topology by blocking a certain port. In this manner, looping of packets is prevented, and deterioration of device packet processing performance is avoided.
- The Rapid Spanning Tree Protocol (RSTP) is based on STP and introduces more refined improvements and enhancements to STP to implement fast network topology convergence.

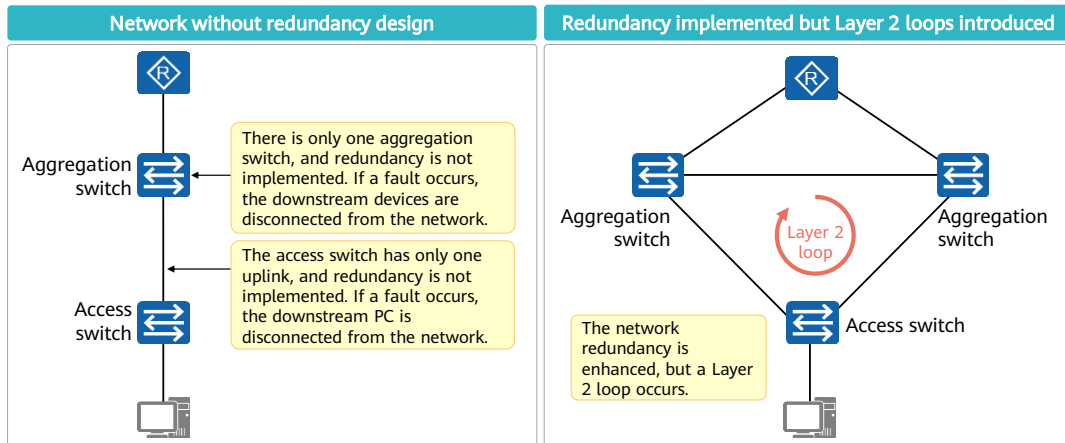
Objectives

- Upon completion of this lesson, you will be able to:
 - Describe the causes and problems of Layer 2 loops on a campus switched network.
 - Describe the basic concepts and working principles of STP.
 - Distinguish STP and RSTP and describe the improvements of RSTP over STP.
 - Complete basic STP configurations.

Contents

- 1. Overview of STP**
2. Basic Concepts of STP
3. Working Principles of STP
4. Basic STP Configuration
5. Advanced Spanning Tree Protocols

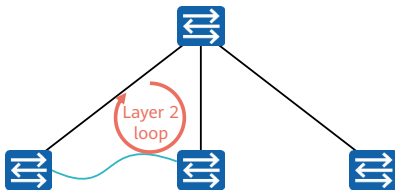
Technical Background: Redundancy and Loops on a Layer 2 Switched Network



- As the LAN scale expands, more switches are used to connect hosts. On the network shown in the figure, the access switch is connected to the upstream device through a single link. If the single link fails, users connected to the switch are disconnected from the network. Another problem is a single point of failure. That is, if the switch breaks down, users connected to the switch are disconnected from the network.
- To solve these problems, redundant links are usually used to connect switches to implement backup. Although redundant links improve network reliability, they also cause loops, which bring a series of problems, such as communication quality deterioration and communication service interruption.

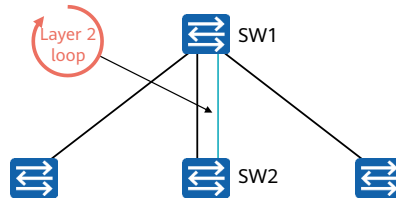
Technical Background: Layer 2 Loops Caused by Human Errors

Case 1: Layer 2 loop caused by human errors



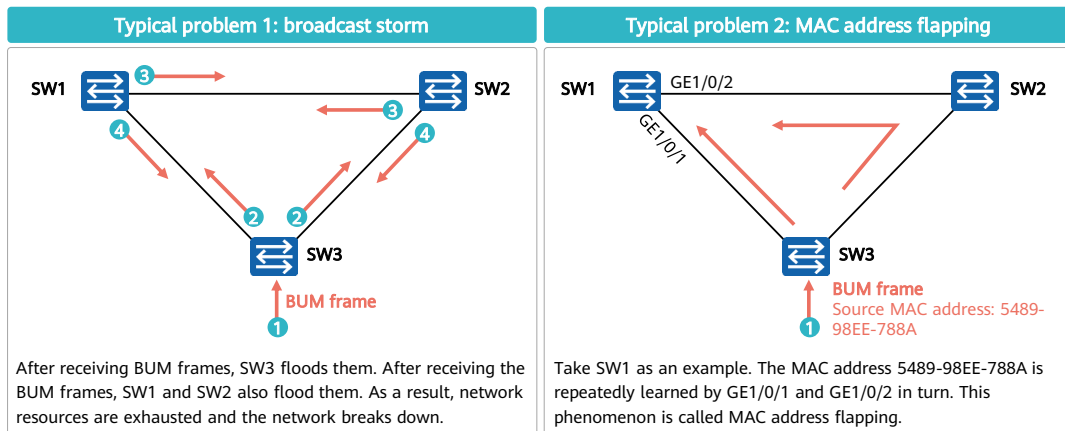
Some Layer 2 loops may be caused by negligence, for example, incorrect cable connections between devices.

Case 2: Layer 2 loop caused by human errors



Other Layer 2 loops may be caused by incorrect configurations. In this example, the network administrator does not bundle the links between SW1 and SW2 into a logical link (Eth-Trunk). As a result, a Layer 2 loop occurs.

Problems Caused by Layer 2 Loops

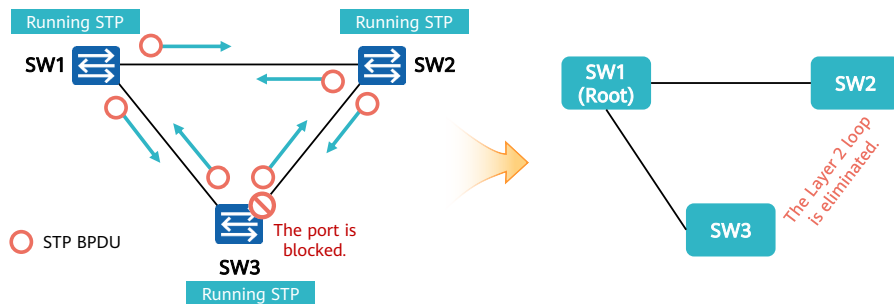


BUM frames refer to broadcast, unknown unicast, and multicast frames.

- Problem 1: broadcast storm
 - According to the forwarding principle of switches, if a switch receives a broadcast frame or a unicast frame with an unknown destination MAC address through a port, the switch forwards the frame through all ports except the port that receives the frame. If a loop occurs on the switched network, the frame is forwarded infinitely. In this case, a broadcast storm occurs, and the network is flooded with duplicate frames.
 - In this example, SW3 receives a broadcast frame and floods it. SW1 and SW2 also forward the frame through all ports except the port that receives the frame. As a result, the frame is forwarded to SW3 again. This cycle continues, causing a broadcast storm. As a result, the switch performance deteriorates rapidly and services are interrupted.
- Problem 2: MAC address flapping
 - A switch generates MAC address entries based on source addresses of received data frames and ports that receive the frames.
 - In this example, SW3 receives a broadcast frame and floods it. After receiving the broadcast frame through GE1/0/1, SW1 learns the MAC address and floods the broadcast frame, generating a mapping between the MAC address 5489-98EE-788A and GE1/0/1. After receiving the broadcast frame, SW2 learns the MAC address and floods the broadcast

frame. After receiving the broadcast frame with the source MAC address 5489-98EE-788A through GE1/0/2, SW1 learns the MAC address. The MAC address 5489-98EE-788A is repeatedly learned by GE1/0/1 and GE1/0/2 in turn. This phenomenon is called MAC address flapping.

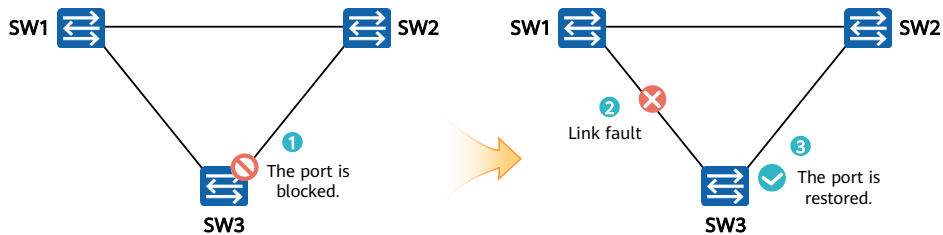
Getting to Know STP



After STP is deployed on a network, switches exchange STP bridge protocol data units (BPDUs) and calculate a loop-free topology. Finally, one or more ports on the network are blocked to eliminate loops.

- On an Ethernet network, Layer 2 loops cause problems such as broadcast storms, MAC address flapping, and duplicate data frames. To prevent loops on a switched network, STP is introduced.
- STP constructs a tree to eliminate loops on a switched network.
- STP detects loops on a network and blocks redundant links to trim the network into a loop-free tree network, preventing circular propagation of data frames on the network.

STP Dynamically Responding to Network Topology Changes and Adjusting the Blocked Port

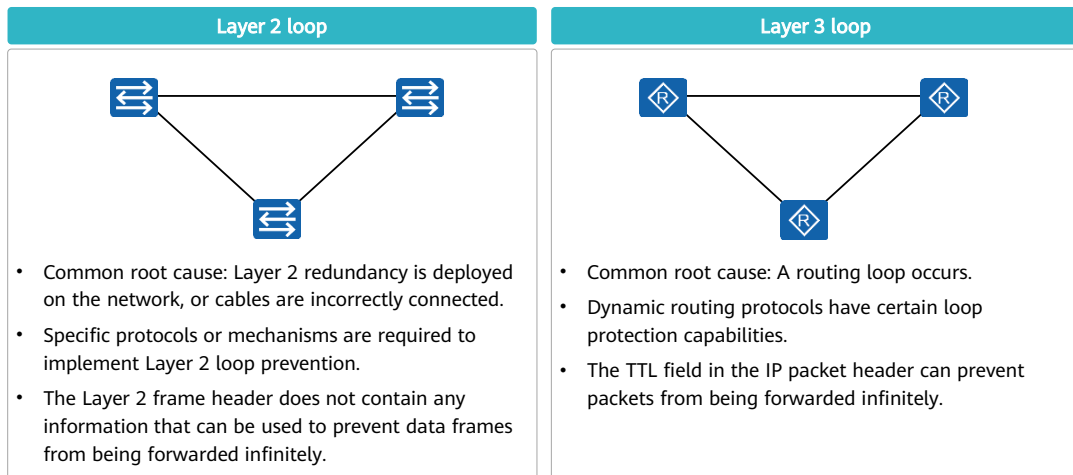


STP running on a switch continuously monitors the network topology. When the network topology changes, STP can detect the change and automatically make adjustments.

Therefore, STP can solve the problem of Layer 2 loops and provide a solution for network redundancy.

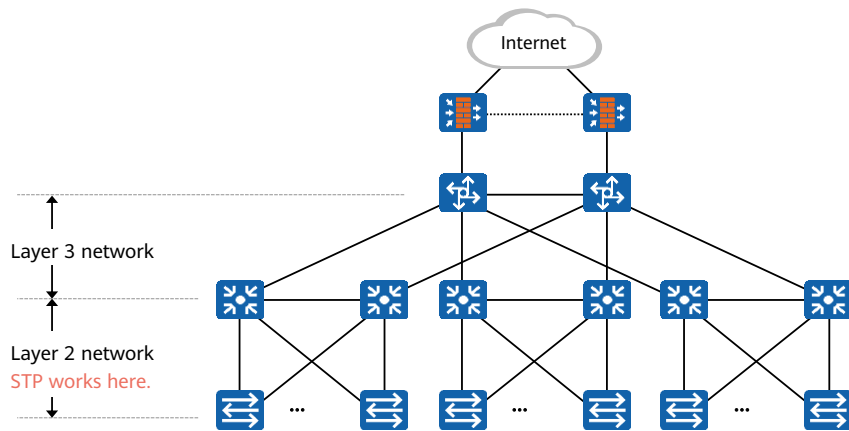
- On the network shown in the figure, switches run STP and monitor the network topology through packets. In normal cases, a port on SW3 is blocked to eliminate the loop. When the link between SW1 and SW3 is faulty, the blocked port is restored and enters the Forwarding state.

Comparison Between Layer 2 and Layer 3 Loops



- Common loops are classified into Layer 2 and Layer 3 loops.
 - Layer 2 loops are mainly caused by Layer 2 redundancy or incorrect cable connections. You can use specific protocols or mechanisms to prevent Layer 2 loops.
 - Layer 3 loops are mainly caused by routing loops. You can use dynamic routing protocols to prevent loops and the TTL field in the IP packet header to prevent packets from being forwarded infinitely.

Application Location of STP on a Campus Network

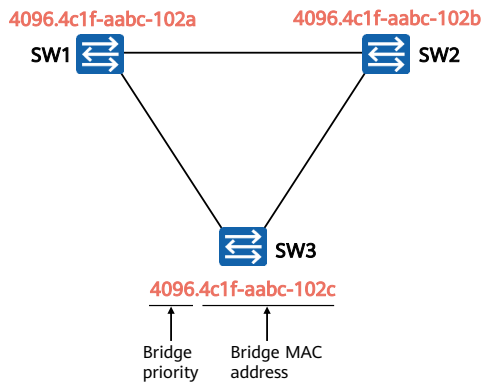


- STP is used on the Layer 2 network of a campus network to implement link backup and eliminate loops.

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Bridge ID



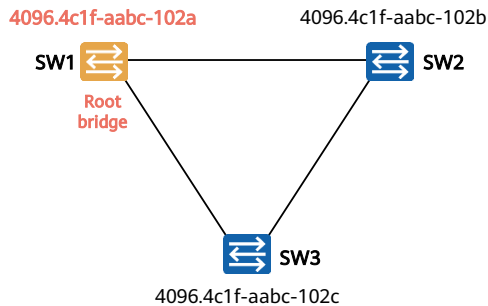
Bridge ID (BID)

- As defined in IEEE 802.1D, a BID consists of a 16-bit bridge priority and a bridge MAC address.
- Each switch running STP has a unique BID.
- The bridge priority occupies the most significant 16 bits of a BID, and the bridge MAC address occupies the least significant 48 bits.
- On an STP network, the device with the smallest BID is elected as the root bridge.

Note: Here, a bridge is a switch.

- In STP, each switch has an identifier called a bridge ID (BID). A BID consists of a 16-bit bridge priority and a 48-bit MAC address. On an STP network, the bridge priority can be configured. The value ranges from 0 to 65535. The default value is 32768. The value can be changed, but must be a multiple of 4096.

Root Bridge

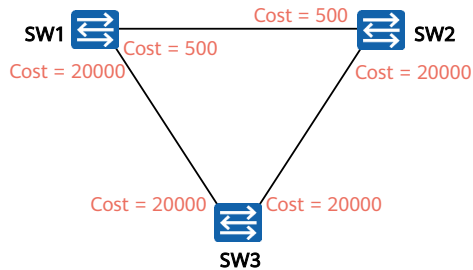


Root bridge

- One of the main functions of STP is to calculate a loop-free tree (STP tree) across a switched network.
- After STP starts working, a root bridge is elected on the switched network. The root bridge is the root of the loop-free topology calculated by STP and is an important reference point for topology calculation.
- On an STP network, the device with the smallest BID is elected as the root bridge.
- During the comparison of BIDs, the bridge priorities are compared first. The switch with the smallest priority value (a smaller priority value indicates a higher priority) is elected as the root bridge. If the priorities are the same, the MAC addresses are compared. The switch with the smallest MAC address is elected as the root bridge.

- A tree network must have a root. Therefore, the root bridge concept is introduced to STP.
- On an STP network, there is only one root bridge, which is the logical center of the entire network but not necessarily the physical center. The root bridge changes dynamically according to the network topology.
- When the root bridge is elected on the network shown in the figure, the bridge priorities of the three switches are compared first. As their bridge priority values are all 4096, their MAC addresses are compared. The switch with the smallest MAC address is elected as the root bridge. In this example, SW1 is elected as the root bridge.

Cost



Cost

- Each STP-enabled port maintains a cost. The cost of a port is used to calculate the root path cost, that is, the cost of the path to the root.
- The default cost of a port is related to the port rate, working mode, and STP cost calculation method used by the switch.
- A higher port bandwidth indicates a smaller cost.
- You can run a command to adjust the cost of a port as required.

- In STP, each port on a switch has a cost. By default, the cost of a port is related to its bandwidth. A higher bandwidth indicates a smaller cost.
- Huawei switches support multiple STP path cost calculation standards, providing the maximum compatibility in scenarios where devices from multiple vendors are used. By default, Huawei switches use the IEEE 802.1t standard to calculate path costs.

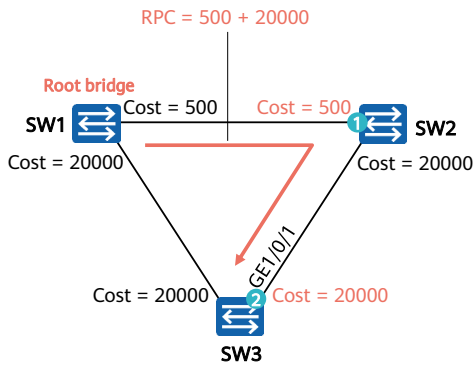
Cost Calculation Methods

Port Rate	Port Mode	STP Cost (Recommended)		
		IEEE 802.1D-1998 Standard	IEEE 802.1t Standard	Huawei Calculation Method
100 Mbit/s	Half-Duplex	19	200,000	200
	Full-Duplex	18	199,999	199
	Aggregated Link 2 Ports	15	100,000	180
1000 Mbit/s	Full-Duplex	4	20,000	20
	Aggregated Link 2 Ports	3	10,000	18
10 Gbit/s	Full-Duplex	2	2000	2
	Aggregated Link 2 Ports	1	1000	1
40 Gbit/s	Full-Duplex	1	500	1
	Aggregated Link 2 Ports	1	250	1
100 Gbit/s	Full-Duplex	1	200	1
	Aggregated Link 2 Ports	1	100	1
...				

The port cost is a cost maintained by an STP-enabled port. The cost has a default value and is related to the port rate. When the device uses different algorithms, the same port rate corresponds to different cost values.

- **Half-Duplex:** In half-duplex mode, data can be transmitted only in one direction at a time, which is similar to a walkie-talkie.
- **Full-Duplex:** In full-duplex mode, data can be sent and received in both directions at the same time, which is similar to making a phone call.
- **Aggregated Link 2 Ports:** Two ports are aggregated for link aggregation. The bandwidth of the aggregated port is the sum of the bandwidth values of the two ports.

RPC

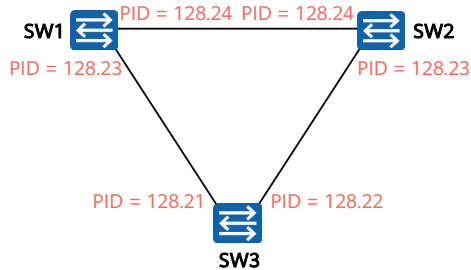


Root path cost (RPC)

- During STP topology calculation, an important step is to measure the cost from a port of a switch to the root bridge, that is, the RPC.
- The RPC from a port of a device to the root bridge is the sum of the costs of all inbound ports along the path from the root bridge to the device.
- In this example, the RPC from GE1/0/1 of SW3 to the root bridge is equal to the cost of port 1 plus the cost of port 2.

- There may be multiple paths from a non-root bridge to the root bridge. Each path has a cost, which is the sum of the costs of all the inbound ports receiving BPDUs on the path. A non-root bridge compares the costs of available paths to select the path with the smallest cost to the root bridge. The cost of the selected path is known as the RPC. A loop-free tree network is generated based on the selected path. The RPC of the root bridge is 0.

Port ID

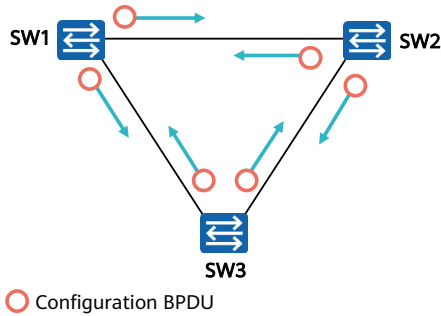


Port ID (PID)

- A switch running STP uses a PID to identify each port. PIDs are mainly used to elect designated ports in specific scenarios.
- The PID consists of a port priority (most significant 4 bits) and a port number (least significant 12 bits).
- An STP-enabled port maintains a default port priority. On Huawei switches, the default port priority is 128. You can run a command to change the port priority as required.

- Each port on a switch running STP has a PID, which consists of a port priority and a port number. The port priority value ranges from 0 to 240, in increments of 16. That is, the value must be an integer multiple of 16. By default, the port priority is 128. The PID can be used to determine the port role.

BPDUs



Bridge Protocol Data Unit (BPDU)

- BPDUs are the essential for STP to work properly. BPDUs are protocol packets of STP.
- STP-enabled switches exchange BPDUs that carry key information, enabling STP to work properly.
- BPDUs are classified into the following types:
 - Configuration BPDU
 - Topology Change Notification (TCN) BPDU
- Configuration BPDUs are the key to STP topology calculation. TCN BPDUs are triggered only when the network topology changes.

- To calculate a spanning tree, switches need to exchange related information and parameters, which are encapsulated in BPDUs.
- BPDUs are classified into configuration BPDU and TCN BPDU.
- A configuration BPDU contains parameters such as the bridge ID, path cost, and port ID. STP enables switches to transmit configuration BPDUs to elect the root switch and determine the role and state of each port on a switch. During initialization, each bridge proactively sends configuration BPDUs. After the network topology becomes stable, only the root bridge proactively sends configuration BPDUs. Other switches send their own configuration BPDUs only after receiving configuration BPDUs from upstream devices.
- TCN BPDUs are sent by downstream switches to notify upstream switches of topology changes.

Format of a Configuration BPDU

	PID	PVI	BPDU Type	Flags	Root ID	RPC	Bridge ID	Port ID	Message Age	Max Age	Hello Time	Forward Delay
Byte	Field		Description									
2	PID		Protocol ID. For STP, the value of this field is always 0.									
1	PVI		Protocol version ID. For STP, the value of this field is always 0.									
1	BPDU Type		BPDU type. If the value is 0x00, the BPDU is a configuration BPDU. If the value is 0x80, the BPDU is a TCN BPDU.									
1	Flags		Flags. STP uses only the most significant bit and the least significant bit of this field. The least significant bit is the Topology Change (TC) flag, and the most significant bit is the Topology Change Acknowledgment (TCA) flag.									
8	Root ID		Bridge ID of the root bridge.									
4	RPC		Root path cost, which is the STP cost of the path to the root bridge.									
8	Bridge ID		ID of the bridge that sends the BPDU.									
2	Port ID		Port ID (priority + port number) of the bridge that sends the BPDU.									
2	Message Age		Message lifetime, which is the number of seconds that has elapsed after the BPDU is sent from the root bridge. The value of this field is incremented by 1 each time the BPDU passes through a bridge. Therefore, the value of this field is the number of hops to the root bridge.									
2	Max Age		Maximum lifetime. If no BPDU is received before the maximum lifetime expires, the bridge considers that the link connected to the port is faulty. The default value is 20s.									
2	Hello Time		Interval at which the root bridge sends BPDUs consecutively. The default value is 2s.									
2	Forward Delay		Forwarding delay, which is the period during which a port stays in the Listening or Learning state. The default value is 15s.									

Rules for Comparing Configuration BPDUs

Field
PID
PVI
BPDUs Type
Flags
Root ID
RPC
Bridge ID
Port ID
Message Age
Max Age
Hello Time
Forward Delay

The most important task of STP is to calculate a loop-free topology on a switched network.

A major task in topology calculation is to compare configuration BPDUs. In a configuration BPDU, four fields are critical: root ID, RPC, bridge ID, and port ID.

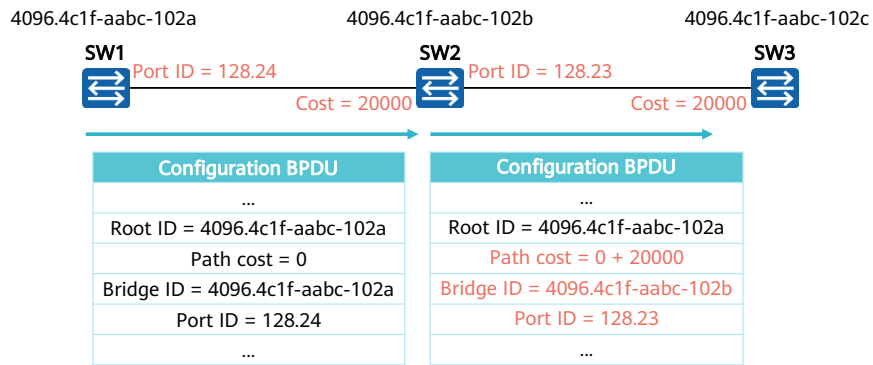
STP selects the superior configuration BPDU based on the following rules in sequence:

1. Smallest root ID
2. Smallest RPC
3. Smallest bridge ID
4. Smallest port ID

Among these four rules (each rule corresponds to a field in a configuration BPDU), the first rule is used to elect the root bridge on the network, and the other rules are used to elect the root port and designated port.

- STP operations:
 - Elect a root bridge.
 - Elect a root port on each non-root switch.
 - Elect a designated port on each link.
 - Block non-root and non-designated ports.
- STP defines three port roles: designated port, root port, and alternate port.
 - A designated port is a port through which a switch sends configuration BPDUs to the connected link. Each link has only one designated port.
 - The root port is the port with the optimal path from a non-root switch to the root bridge. A switch running STP has only one root port, but the root bridge does not have a root port.
 - If a port is neither a designated port nor a root port, the port is an alternate port. An alternate port is blocked.

Forwarding Process of Configuration BPDU

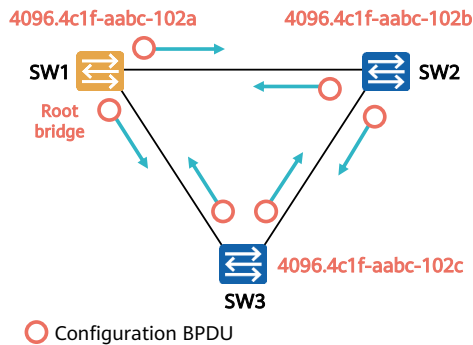


- When a switch starts, it considers itself as the root bridge and sends configuration BPDUs to other switches for STP calculation.

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STP Calculation Process (1)

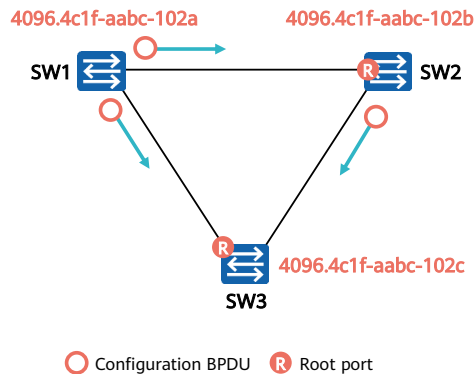


Electing a root bridge on a switched network

- After STP starts to work on a switched network, each switch sends configuration BPDUs to the network. The configuration BPDUs sent by a switch contain its own bridge ID.
- The switch with the smallest bridge ID becomes the root bridge on the network.
- There is only one root bridge on a continuous STP switched network.
- The role of the root bridge can be preempted.
- To ensure the stability of the switched network, you are advised to plan the STP networking in advance and set the bridge priority of the switch that is planned as the root bridge to the smallest value 0.

- Election process:
 - After initial startup, an STP-enabled switch considers itself as the root bridge and announces this in the BPDUs sent to other switches. Therefore, the root bridge ID in the BPDUs sent by a device is the bridge ID of the device.
 - After a switch receives a BPDU from another device on the network, the switch compares the root bridge ID in the BPDU with its own BID.
 - Switches exchange BPDUs and compare BIDs. The switch with the smallest BID is elected as the root bridge, and the other switches work as non-root bridges.
 - On the network shown in the figure, the priorities are compared first during root bridge election. If SW1, SW2, and SW3 have the same priority, their MAC addresses are compared to elect the one with the smallest MAC address. SW1 has the smallest BID. Therefore, SW1 is elected as the root bridge, and SW2 and SW3 are non-root bridges.
- Note:
 - The role of the root bridge can be preempted. When a switch with a smaller BID joins the network, STP recalculation is performed on the network to elect a new root bridge.

STP Calculation Process (2)



Electing a root port on each non-root bridge

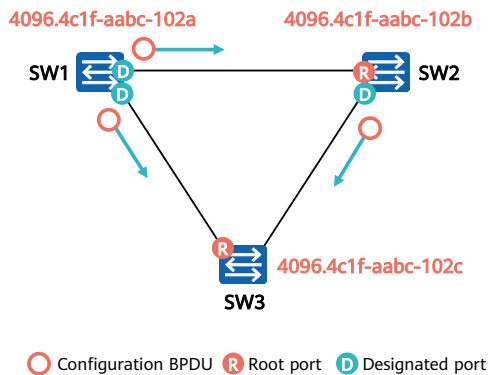
- A non-root bridge has only one root port.
- When a non-root bridge has multiple ports connected to the network, the root port is the port that receives the superior configuration BPDU.
- You can consider the root port on a non-root bridge as the port that points toward the root bridge.

- What is a root port?
 - A non-root bridge has multiple ports connected to the network. To ensure that the working path from the non-root bridge to the root bridge is optimal and unique, a port called the root port must be determined from ports of the non-root bridge. The non-root bridge uses the root port to exchange packets with the root bridge.
 - After the root bridge is elected, it continuously sends BPDUs, and non-root bridges continuously receive BPDUs from the root bridge. Therefore, a port closest to the root bridge must be elected as the root port on all non-root bridges. After network convergence is completed, the root port continuously receives BPDUs from the root bridge.
 - In this way, the root port ensures the uniqueness and optimality of the working path between a switch and the root bridge.
- Election process:
 - A switch has multiple ports connected to the network. Each port receives BPDUs that contain key fields such as Root ID, RPC, BID, and PID. The values of these fields are compared for root port election.
 - The RPCs are compared first. STP uses RPCs as an important basis for determining the root port. A smaller RPC value indicates a higher priority.

Therefore, the switch elects the port with the smallest RPC as the root port.

- If all ports have the same RPC, the switch compares the BIDs in BPDUs received from upstream switches. A smaller value indicates a higher priority. Therefore, the switch elects the port receiving the smallest BID as the root port.
- If the upstream switches have the same BID, the local switch compares the PIDs in BPDUs received from the upstream switches. A smaller value indicates a higher priority. Therefore, the switch elects the port receiving the smallest PID as the root port.
- If the upstream switches have the same PID, the local switch compares the PIDs of its own ports. A smaller value indicates a higher priority. Therefore, the switch elects the port with the smallest PID as the root port.

STP Calculation Process (3)

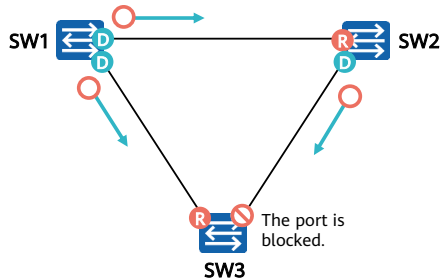


Electing a designated port on each link

- After the root port is elected, a non-root bridge calculates a configuration BPDU based on the superior BPDU received on the port, and then compares the calculated configuration BPDU with the configuration BPDUs received on all other ports except the root port.
 - If the former is superior, the port is a designated port.
 - If the latter is superior, the port is a non-designated port.
- Generally, all ports on the root bridge are designated ports.

- What is a designated port?
 - The working path between each link on the network and the root bridge must be unique and optimal. When a link has two or more paths to the root bridge (the link connects to multiple switches or multiple ports on the same switch), the switches (possibly more than one) connected to the link must determine a unique designated port.
 - Therefore, a designated port is elected for each link and is used to send BPDUs to the link.
- Election process:
 - The designated port is also determined by comparing RPCs. The port with the smallest RPC is elected as the designated port.
 - If the RPCs are the same, the BIDs of the switches at both ends of the link are compared. A smaller value indicates a higher priority. Therefore, the port on the switch with the smallest BID is elected as the designated port.
 - If the BIDs are the same, the PIDs of the ports at both ends of the link are compared. A smaller value indicates a higher priority. Therefore, the port with the smallest PID is elected as the designated port.

STP Calculation Process (4)



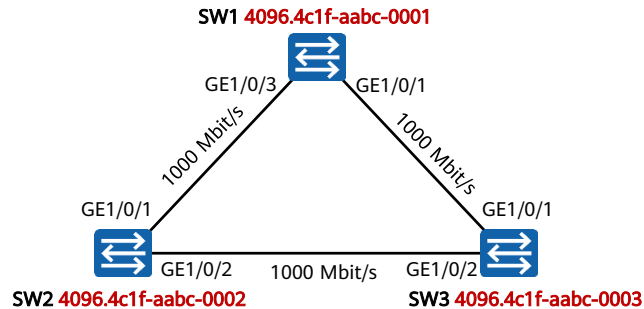
○ Configuration BPDU ○ Root port ○ Designated port

Blocking a non-designated port

- On a switch, a port that is neither a root port nor a designated port is called a non-designated port.
- The last step of STP is to block non-designated ports on the network. After this step is complete, a Layer 2 loop on the network is eliminated.

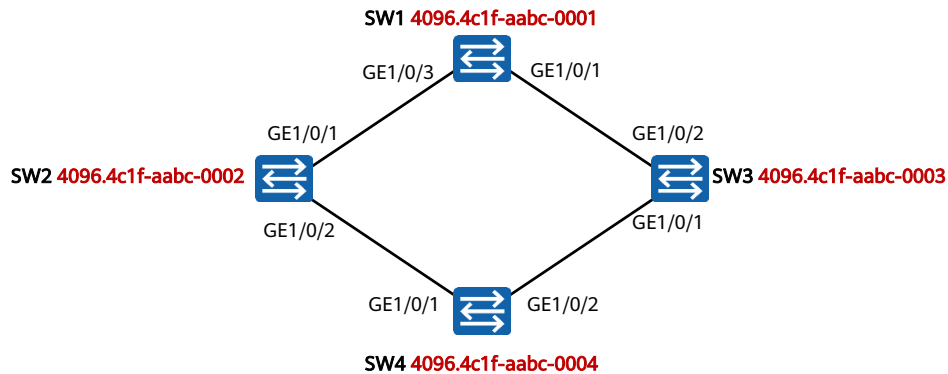
- After the root port and designated port are determined, all the other ports on a switch are called alternate ports.
- Blocking non-designated ports
 - STP logically blocks non-designated ports to prevent them from forwarding user data frames generated and sent by terminal computers.
 - Once non-designated ports are logically blocked, an STP tree (loop-free working topology) is generated.
- Note:
 - Non-designated ports can receive and process BPDUs.
 - Root ports and designated ports can send and receive BPDUs, and forward user data frames.

Question 1: Identifying the Root Bridge and Port Roles in the Following Topology



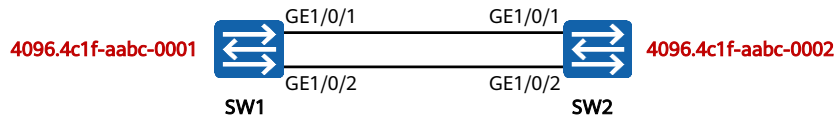
- On the network shown in the figure, the root bridge is elected first. The three switches have the same bridge priority. Therefore, their bridge MAC addresses are compared, and the switch with the smallest bridge MAC address is elected as the root bridge. In this example, SW1 is elected as the root bridge.
- Then, root ports are elected. GE1/0/1 on SW2 is the closest to the root bridge and has the smallest RPC. Therefore, GE1/0/1 on SW2 is the root port. Similarly, GE1/0/1 on SW3 is also the root port.
- Next, designated ports are elected. SW1 is the root bridge, so GE1/0/3 and GE1/0/1 on SW1 are designated ports. GE1/0/2 on SW2 receives configuration BPDUs from SW3. BIDs are compared. The BID of SW2 is better than that of SW3. Therefore, GE1/0/2 on SW2 is the designated port.
- Finally, GE1/0/2 on SW3, which is neither a root port nor a designated port, becomes an alternate port.

Question 2: Identifying the Root Bridge and Port Roles in the Following Topology



- On the network shown in the figure, the root bridge is elected first. The four switches have the same bridge priority. Therefore, their bridge MAC addresses are compared, and the switch with the smallest bridge MAC address is elected as the root bridge. In this example, SW1 is elected as the root bridge.
- Next, root ports are elected. GE1/0/1 on SW2 is the closest to the root bridge and has the smallest RPC. Therefore, GE1/0/1 on SW2 is the root port. Similarly, GE1/0/2 on SW3 is also the root port. The two ports on SW4 have the same RPC. Then, the BID of SW2 connected to GE1/0/1 on SW4 and the BID of SW3 connected to GE1/0/2 on SW4 are compared, and the port with a smaller BID is elected as the root port. In this example, GE1/0/1 on SW4 is elected as the root port.
- Next, designated ports are elected. SW1 is the root bridge, so GE1/0/3 and GE1/0/1 on SW1 are designated ports. GE1/0/2 on SW2 receives configuration BPDUs from SW4. RPCs are compared. The RPC of SW2 is smaller than that of SW4. Therefore, GE1/0/2 on SW2 is the designated port. Similarly, GE1/0/1 on SW3 is the designated port.
- Finally, GE1/0/2 on SW4, which is neither a root port nor a designated port, becomes an alternate port.

Question 3: Identifying the Root Bridge and Port Roles in the Following Topology

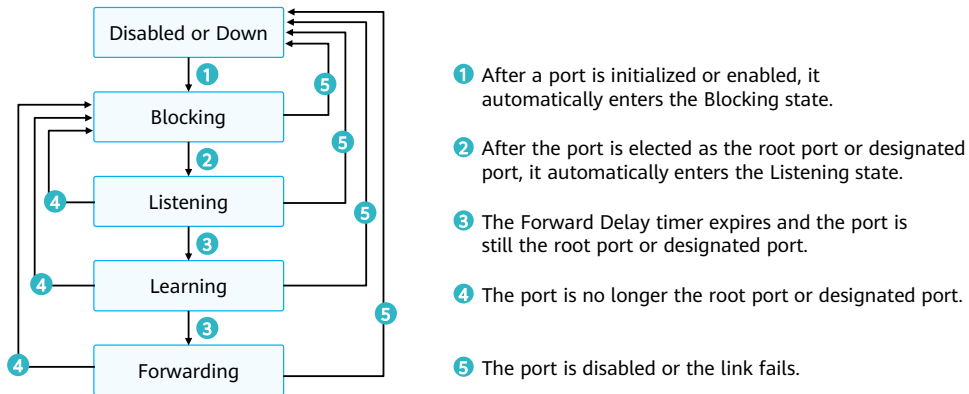


- On the network shown in the figure, the root bridge is elected first. The two switches have the same bridge priority. Therefore, their bridge MAC addresses are compared, and the switch with the smaller bridge MAC address is elected as the root bridge. In this example, SW1 is elected as the root bridge.
- Next, the root port is elected. The two ports on SW2 have the same RPC and the same remote BID. Then, the remote PIDs of the two ports are compared. The remote PID of GE1/0/1 on SW2 is 128.1, and the remote PID of GE1/0/2 on SW2 is 128.2. A smaller PID indicates a higher priority. Therefore, GE1/0/1 on SW2 is elected as the root port.
- Then, designated ports are elected. SW1 is the root bridge, so GE1/0/1 and GE1/0/2 on SW1 are designated ports.
- Finally, GE1/0/2 on SW2, which is neither a root port nor a designated port, becomes an alternate port.

STP Port States

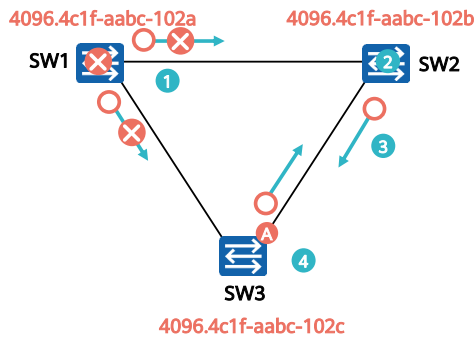
State Name	State Description
Disabled	The port cannot send or receive BPDUs or service data frames. For example, the port is Down.
Blocking	The port is blocked by STP. A port in the Blocking state cannot send BPDUs, but can continuously listen to BPDUs. The port cannot receive or send service data frames, and does not learn MAC addresses.
Listening	STP preliminarily considers a port in this state as a root port or designated port, but the port state is not determined, meaning that the STP calculation process is not completed yet. In this case, the port can send and receive BPDUs, but cannot send or receive service data frames or learn MAC addresses.
Learning	A port in this state listens to service data frames, but cannot forward service data frames. It learns MAC addresses after receiving service data frames.
Forwarding	A port in this state can send and receive service data frames and process BPDUs. A port can enter the Forwarding state only when it is a root port or designated port.

STP Port State Transition



- The figure shows the port state transition mechanism of STP. A device running STP has five port states:
 - Forwarding: A port in this state can forward both user traffic and BPDUs. Only the root port or designated port can enter the Forwarding state.
 - Learning: A port in this state can create MAC address entries based on the received user traffic, but does not forward user traffic. The Learning state is added to prevent temporary loops.
 - Listening: A port in this state can forward BPDUs, but cannot forward user traffic.
 - Blocking: A port in this state can only receive and process BPDUs, but cannot forward BPDUs or user traffic. This state is the final state of an alternate port.
 - Disabled: A port in this state does not process or forward BPDUs, and does not forward user traffic.

Topology Change: Root Bridge Fault

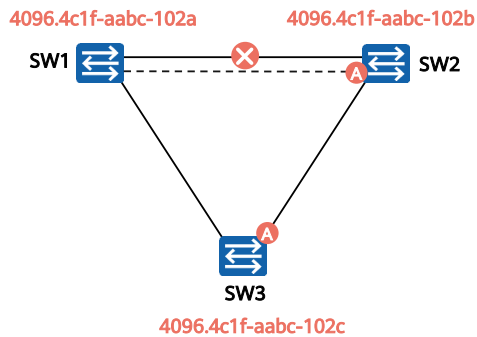


Recovery process from a root bridge fault

1. SW1 (root bridge) fails and stops sending BPDUs.
2. SW2 waits for the Max Age timer (20s) to expire, during which received BPDUs become invalid, and SW2 cannot receive new BPDUs from the root bridge. This indicates that the upstream device is faulty.
3. Non-root bridges send configuration BPDUs to each other to re-elect a new root bridge.
4. After the re-election, port A of SW3 restores to the Forwarding state after two Forward Delay intervals (15s for each interval).
 - Non-root bridges start to re-elect a root bridge after BPDUs age.
 - If the root bridge fails, the recovery process takes about 50s.

- Root bridge fault:
 - On a stable STP network, non-root bridges periodically receive BPDUs from the root bridge.
 - If the root bridge fails and stops sending BPDUs, downstream switches cannot receive BPDUs from the root bridge.
 - If downstream switches do not receive any BPDUs, the Max Age timer (default value: 20s) expires, causing the previously received BPDUs to become invalid. In this case, non-root bridges send configuration BPDUs to each other to re-elect a new root bridge.
- Port state:
 - The alternate port of SW3 enters the Listening state from the Blocking state after 20s and then enters the Learning state. Finally, the port enters the Forwarding state to forward user traffic.
- Convergence time:
 - If the root bridge is faulty, the recovery process takes about 50s, which is equal to the value of the Max Age timer plus twice the value of the Forward Delay timer.

Topology Change: Physical Link Fault



Recovery process from a direct link fault

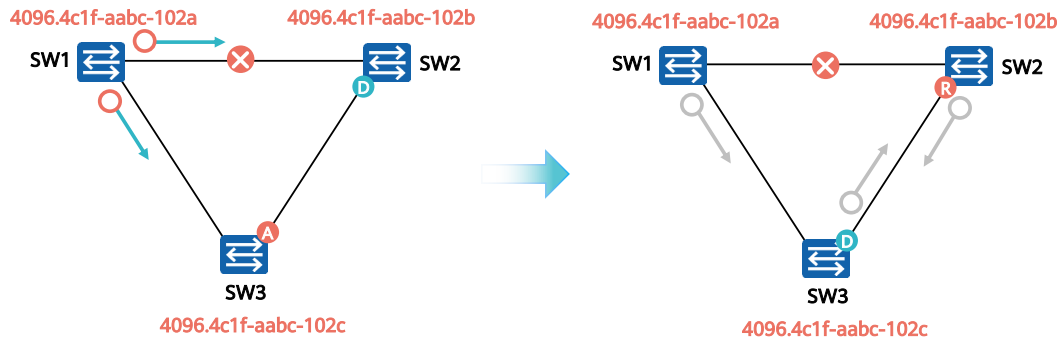
When the network is stable, if SW2 detects that the link of the root port fails, the alternate port enters the Forwarding state after twice the value of the Forward Delay timer (15s).

- After SW2 detects a physical fault on the direct link, the alternate port becomes the root port.
- If a direct link fails, the alternate port restores to the Forwarding state after 30s.

- Physical link fault:
 - When two switches are connected through two links, one is the primary link and the other is the backup link.
 - When the network is stable, SW2 detects that the link of the root port fails (the port status changes to Down). In this case, the alternate port will enter the Forwarding state to forward user traffic.
- The port state transition process is as follows:
 - The alternate port changes from the Blocking state to the Listening state, and then to the Learning state, and finally to the Forwarding state.
 - Convergence time: If a direct link fails, the alternate port restores to the Forwarding state after 30s.

Topology Change: Non-Physical Link Fault

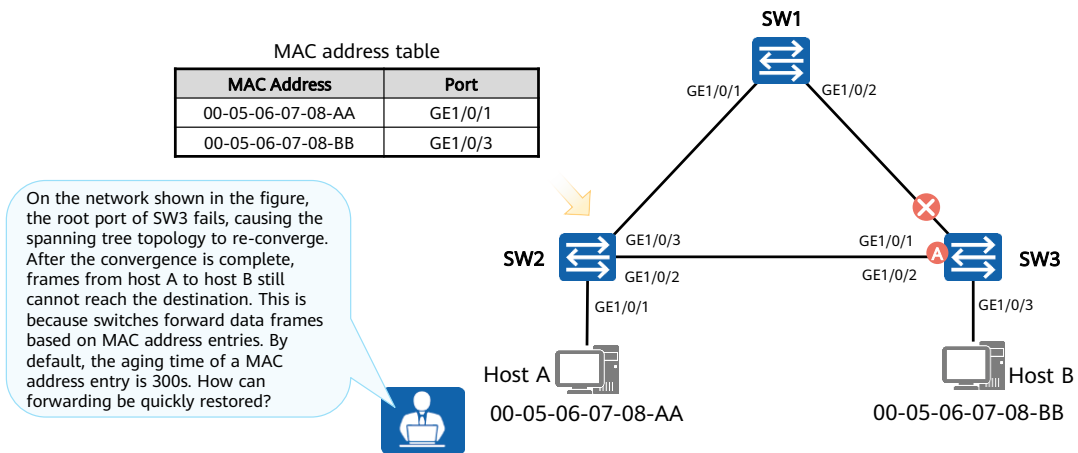
- After a non-physical link fault occurs, the alternate port of SW3 restores to the Forwarding state after about 50s.



- Non-physical link fault:
 - On a stable STP network, non-root bridges periodically receive BPDUs from the root bridge.
 - If a fault (non-physical fault) occurs on the link between SW1 and SW2, SW2 cannot receive BPDUs from the root bridge SW1, and the Max Age timer (20s by default) expires. As a result, the previously received BPDUs become invalid.
 - In this case, the non-root bridge SW2 considers the root bridge faulty and itself as the new root bridge. It then sends its own configuration BPDU to SW3 to notify SW3 that it is the new root bridge.
 - After receiving the non-superior BPDU from SW2, SW3 forwards the superior BPDU received from SW1 to SW2.
 - SW2 finds that the BPDU received from SW3 is superior and therefore no longer declares itself as the root bridge. It then re-determines its port role.
- Port state:
 - The alternate port of SW3 enters the Listening state from the Blocking state after 20s and then enters the Learning state. Finally, the port enters the Forwarding state to forward user traffic.
- Convergence time:

- If a non-physical link fault occurs, the recovery process takes about 50s, which is equal to the value of the Max Age timer plus twice the value of the Forward Delay timer.

Incorrect MAC Address Entries Due to a Topology Change (1)



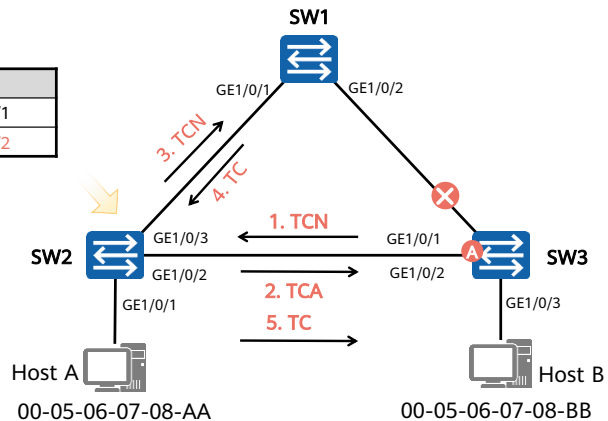
- On a switched network, switches rely on MAC address entries to forward data frames. By default, the aging time of a MAC address entry is 300s. If the spanning tree topology changes, the data forwarding path of a switch also changes. If the MAC address entries that have not aged out in the MAC address table are used, data forwarding errors occur. This requires MAC address entries to be updated in a timely manner after the topology changes.
- In this example, the MAC address entries on SW2 indicate that frames can reach host A through GE1/0/1 and reach host B through GE1/0/3. The root port of SW3 fails, causing the spanning tree topology to re-converge. After the convergence is complete, frames from host A to host B still cannot reach the destination. This is because the aging time of a MAC address entry is 300s. After the frames from host A to host B reach SW2, SW2 still forwards the data frames through GE1/0/3.

Incorrect MAC Address Entries Due to a Topology Change (2)

MAC address table

MAC Address	Port
00-05-06-07-08-AA	GE1/0/1
00-05-06-07-08-BB	GE1/0/2

- TCN BPDUs are generated when the network topology changes.
- Packet format: protocol ID, version ID, and type.
- Topology change: The TCA and TC bits in the Flags field of a configuration BPDU are used.



- During the topology change, the root bridge detects the fault in the spanning tree topology through TCN BPDUs. The root bridge generates TC BPDUs to instruct other switches to accelerate aging of existing MAC address entries.
- The following describes how MAC address entry update is implemented when the topology changes:
 - After detecting that the network topology changes, SW3 continuously sends TCN BPDUs to SW2.
 - After receiving the TCN BPDUs from SW3, SW2 sets the TCA bit in the Flags field of the configuration BPDUs to 1 and sends the configuration BPDUs to SW3, instructing SW3 to stop sending TCN BPDUs.
 - SW2 forwards the TCN BPDUs to the root bridge.
 - SW1 sets the TC bit in the Flags field of the configuration BPDUs to 1 and sends the configuration BPDUs to instruct downstream devices to change the aging time of MAC address entries from the default 300s to the Forward Delay time (15s by default).
 - After a maximum of 15s, the incorrect MAC address entries on SW2 are automatically deleted. Then, SW2 re-learns MAC address entries and forward packets.

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1. Overview of STP
2. Basic Concepts of STP
3. Working Principles of STP
- 4. Basic STP Configuration**
5. Advanced Spanning Tree Protocols

Basic STP Configuration Commands (1)

1. Configure the spanning tree working mode.

```
[HUAWEI] stp mode { stp | rstp | mstp }
```

A switch supports three spanning tree working modes: STP, RSTP, and Multiple Spanning Tree Protocol (MSTP). By default, the switch works in MSTP mode.

2. (Optional) Configure the root bridge.

```
[HUAWEI] stp root primary
```

Configure the current device as the root bridge. By default, a switch does not function as the root bridge of any spanning tree. After this command is configured, the priority of the device is automatically set to 0 and cannot be changed.

3. (Optional) Configure the secondary root bridge.

```
[HUAWEI] stp root secondary
```

Configure the current switch as the secondary root bridge. By default, a switch does not function as the secondary root bridge of any spanning tree. After this command is configured, the priority of the device is 4096 and cannot be changed.

Basic STP Configuration Commands (2)

1. (Optional) Configure the STP priority of a switch.

```
[HUAWEI] stp priority priority
```

By default, the priority of a switch is 32768.

2. (Optional) Configure the path cost of a port.

```
[HUAWEI] stp pathcost-standard { dot1d-1998 | dot1t | legacy }
```

Configure the path cost calculation method. By default, the IEEE 802.1t (dot1t) standard method is used to calculate the path cost.

All switches on the same network must use the same path cost calculation method.

```
[HUAWEI-GE1/0/1] stp cost cost
```

Set the path cost of the current port.

Basic STP Configuration Commands (3)

1. (Optional) Configure the port priority.

```
[HUAWEI-GE1/0/1] stp priority priority
```

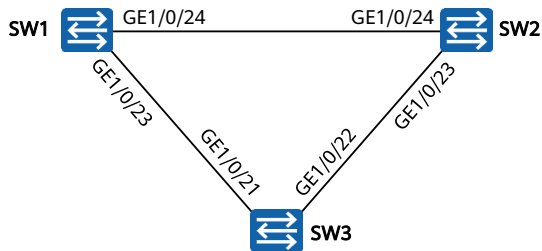
Configure the port priority. By default, the priority of a switch port is 128.

2. Enable STP, RSTP, or MSTP.

```
[HUAWEI] stp enable
```

Enable STP, RSTP, or MSTP on a switch. By default, STP, RSTP, or MSTP is enabled on a device.

Case: Basic STP Configuration



- Deploy STP on the three switches to eliminate Layer 2 loops on the network.
- Perform configurations to specify SW1 as the root bridge and ensure that GE1/0/22 on SW3 is blocked by STP.

The configuration of SW1 is as follows:

```
[SW1] stp mode stp
[SW1] stp enable
[SW1] stp priority 0
```

The configuration of SW2 is as follows:

```
[SW2] stp mode stp
[SW2] stp enable
[SW2] stp priority 4096
```

The configuration of SW3 is as follows:

```
[SW3] stp mode stp
[SW3] stp enable
```

Case: Checking Brief STP State Information

Check brief STP port state information on SW3.

```
<SW3> display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GE1/0/21	ROOT	FORWARDING	NONE
0	GE1/0/22	ALTE	DISCARDING	NONE

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Disadvantages of STP

- Although STP can prevent network loops, its low network convergence performance affects users' communication quality. If the network topology changes frequently, connectivity will be repeatedly unavailable, leading to frequent communication interruptions, which is intolerable to users.
- STP does not clearly distinguish between port states and port roles, which makes it difficult for beginners to learn and deploy.
- The effectiveness of a network protocol often depends on how well it differentiates between various scenarios.
 - From the perspective of users, the Listening, Learning, and Blocking states are no different, as ports in none of them forward user traffic.
 - From the perspective of port use and configuration, the key difference between ports lies not in their states but in the roles they play.
 - The root and designated ports can be both in the Listening or Forwarding state.
- The STP algorithm is a passive algorithm and relies on timers to detect topology changes, resulting in slow network convergence.
- The STP algorithm requires that the root bridge send configuration BPDUs after the network topology becomes stable and other devices process and spread the configuration BPDUs to the entire network. This is one of the main reasons for slow topology convergence.

Overview of RSTP

- RSTP, defined in IEEE 802.1w, is an enhancement of STP. RSTP optimizes STP in many aspects while remaining compatible with STP and achieving faster convergence.
- RSTP introduces new port roles. An alternate port enables a switch to immediately obtain a new path to the root bridge when the root port fails. A backup port serves as a backup for a designated port, enabling the bridge to quickly obtain a backup path to the root bridge.
- RSTP reduces the five port states to three based on whether a port forwards user traffic and learns MAC addresses.
- In addition, RSTP introduces the concept of edge port, which enables a switch port connected to a terminal to enter the Forwarding state immediately after initialization, improving working efficiency.

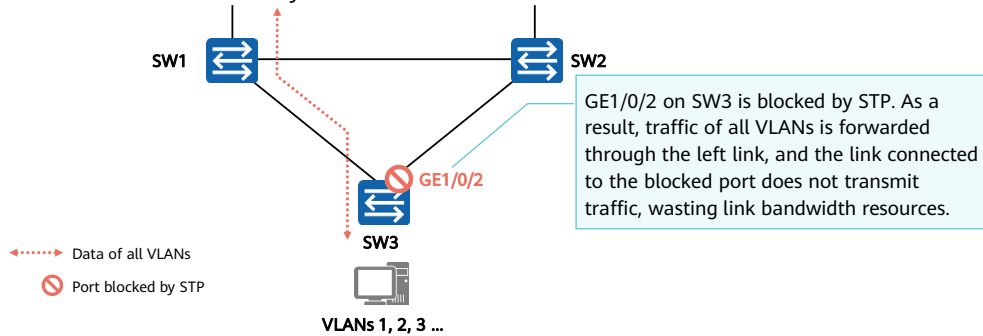
- IEEE 802.1w, released in 2001, defines RSTP. RSTP is an enhancement of STP and implements fast network topology convergence.
- RSTP is evolved from STP and has the same basic idea as STP. When the topology of a switched network changes, RSTP can use the Proposal/Agreement mechanism to restore network connectivity more quickly.
- To address the limitations of STP, RSTP deletes three port states, adds two port roles, and decouples port states from port roles. In addition, RSTP adds some enhanced features and protection measures to ensure network stability and implement fast convergence.
- RSTP is backward compatible with STP. However, you are advised not to use RSTP and STP together because using them together will cancel the advantage of fast convergence in RSTP and highlight the drawback of slow convergence in STP.
- Other improvements of RSTP over STP:
 - RSTP processes configuration BPDUs differently from STP:
 - After the topology becomes stable, the sending mode of configuration BPDUs is optimized.
 - A shorter BPDU timeout period is used.
 - The method of processing inferior BPDUs is optimized.
 - RSTP changes the configuration BPDU format and makes full use of the

Flags field in STP to describe port roles.

- Compared with STP, RSTP optimizes the processing of topology changes, accelerating the response to topology changes.

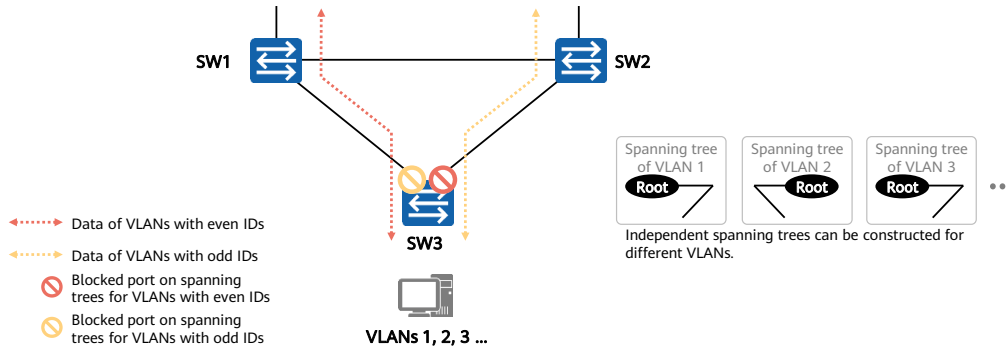
Defect of STP/RSTP: All VLANs Sharing One Spanning Tree

- RSTP makes improvements over STP to implement fast network topology convergence.
- However, STP and RSTP both have a defect: all VLANs on a LAN share one spanning tree, so inter-VLAN load balancing cannot be implemented. Once a link is blocked, it no longer transmits traffic, and packets from some VLANs may fail to be forwarded.



VBST

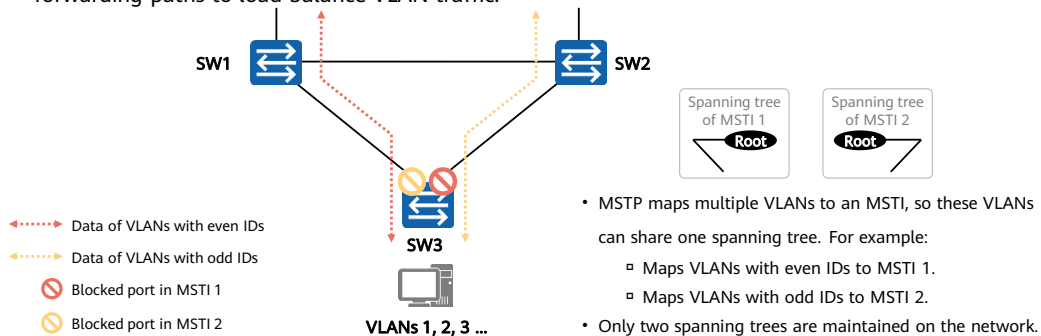
- Huawei proposes the VLAN-based Spanning Tree (VBST) solution. In this solution, a spanning tree is constructed in each VLAN so that traffic from different VLANs is forwarded through different spanning trees, thereby implementing load balancing.



- VBST offers the following benefits to an enterprise network:
 - Eliminates possible communication loops on the network.
 - Implements reuse of links and load balancing of traffic, thereby improving link bandwidth utilization.
 - Reduces configuration and maintenance costs.
- If there are a large number of VLANs on a network, spanning tree calculation for each VLAN consumes a lot of switch resources.

MSTP

- To fix the defect of STP and RSTP, IEEE released the 802.1s standard that defines the Multiple Spanning Tree Protocol (MSTP) in 2002.
- Compatible with STP and RSTP, MSTP implements fast convergence while providing multiple forwarding paths to load balance VLAN traffic.



- MSTP divides a switched network into multiple regions, each of which has multiple spanning trees that are independent of each other.
- Each spanning tree is called a Multiple Spanning Tree Instance (MSTI).
- An MSTI is a spanning tree corresponding to a set of VLANs.
- Binding multiple VLANs to a single MSTI reduces communication costs and resource usage.
- The topology of each MSTI is calculated independently, and traffic can be load balanced among MSTIs.
- Multiple VLANs that have the same topology can be mapped to one MSTI. The state of a port in the MSTI determines whether the port forwards packets from these VLANs.

Quiz

1. (Single-answer question) Which of the following statements is false about STP port states? ()
- A. A blocked port does not listen or send BPDUs.
 - B. A port in the Learning state learns MAC addresses but does not forward data.
 - C. A port in the Listening state continuously listens to BPDUs.
 - D. If a blocked port does not receive BPDUs within a certain period, it automatically switches to the Listening state.

1. A

Summary

- STP is a protocol used to eliminate loops on a LAN. Devices running STP exchange information with one another to discover loops on the network and then block certain ports to eliminate loops. As the scale of LANs continues to grow, STP has become one of the most important protocols used on LANs.
- After STP is deployed on a switched Ethernet network, if a loop occurs on the network, STP calculates the topology to implement the following functions:
 - Loop prevention: STP blocks redundant links to prevent potential loops on the network.
 - Link backup: If an active link fails and there is a redundant link, STP activates the redundant link to restore network connectivity.
- As a long-standing protocol, STP cannot meet the requirements of modern campus networks. However, understanding the working principles of STP helps you further understand and master the principles and deployment of RSTP and MSTP.

Thank you.

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每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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Ethernet Link Aggregation



Foreword

- As networks grow in scale, users require backbone links to provide higher bandwidth and reliability. In conventional technologies, the usual way to increase the bandwidth is to use high-speed cards or devices that support high-speed cards, which is costly and inflexible.
- Eth-Trunk (also called Ethernet link aggregation) uses link aggregation technology to bundle multiple physical interfaces into a logical interface, thereby increasing the link bandwidth without upgrading the hardware. In addition to increasing the bandwidth, Eth-Trunk provides the link backup mechanism to improve the reliability of links between devices.
- This course describes how to use link aggregation technology to improve network bandwidth and ensure high reliability.

Objectives

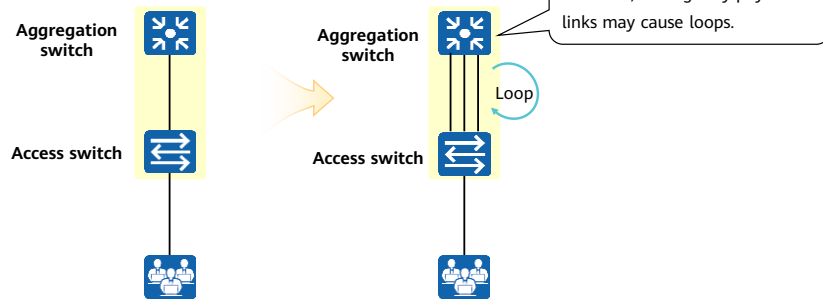
- Upon completion of this course, you will be able to:
 - Describe the functions of link aggregation.
 - Describe the types of link aggregation.
 - Understand the negotiation process of an Eth-Trunk in LACP mode.
 - Describe common application scenarios of link aggregation.

Contents

- 1. Basic Concepts of Link Aggregation**
2. Link Aggregation in Manual Mode
3. Link Aggregation in LACP Mode
4. Link Aggregation Configuration Examples

Background of Link Aggregation

- As networks grow in scale, users require backbone links to provide higher bandwidth and reliability.
- The simplest way to meet these requirements is to add more links between network devices. This can increase the link bandwidth and improve network reliability.

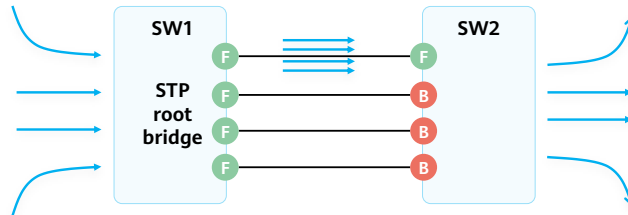


Using STP to Prevent Loops Causes New Problems

- When there are multiple links between devices, only one link forwards traffic due to STP. As a result, the bandwidth of the link between devices cannot be increased.

F Port that forwards traffic

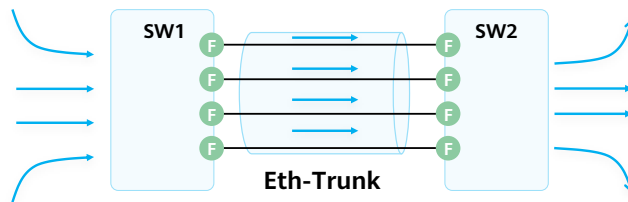
B Port that is blocked by STP and does not forward traffic



Ethernet Link Aggregation

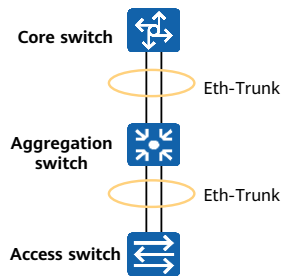
- Ethernet link aggregation, also known as Eth-Trunk, bundles multiple physical interfaces into a logical interface to increase the link bandwidth, without the need to upgrade the hardware.

F Port that forwards traffic



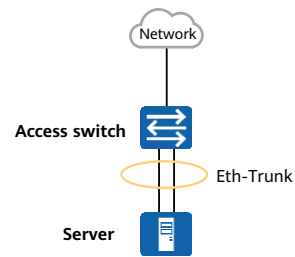
Application Scenarios of Link Aggregation

Between switches



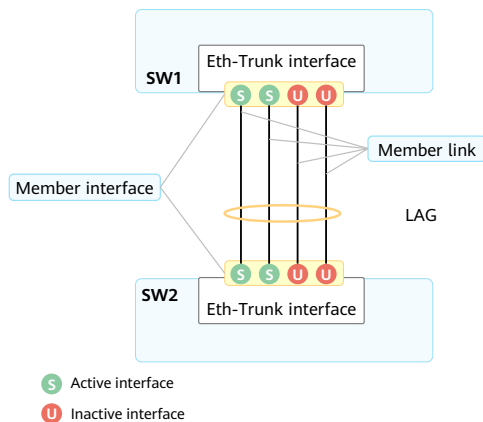
To ensure the link bandwidth and reliability between switches, you can deploy multiple physical links between switches and use Eth-Trunk.

Between a switch and a server



To improve the access bandwidth and reliability of the server, you can bind two or more physical Network Interface Cards (NICs) into a NIC group and establish an Eth-Trunk between the server and the switch.

Basic Concepts of Link Aggregation



- Link Aggregation Group (LAG): A LAG is a logical link formed by bundling multiple links. Each LAG has one logical interface, known as a LAG interface or an Eth-Trunk interface.
- Member interface and member link: The physical interfaces that constitute an Eth-Trunk interface are known as member interfaces. The link corresponding to a member interface is known as a member link.
- Active interface and active link: An active interface, also called a selected interface, is a member interface that participates in data forwarding. The link corresponding to an active interface is called an active link.
- Inactive interface and inactive link: An inactive interface, also called an unselected interface, is a member interface that does not participate in data forwarding. The link corresponding to an inactive interface is called an inactive link.
- Aggregation mode: Link aggregation can be classified into link aggregation in **manual mode** and link aggregation in **Link Aggregation Control Protocol (LACP) mode**, depending on whether LACP is enabled.

- An Eth-Trunk interface can be treated as a physical Ethernet interface. The only difference between an Eth-Trunk interface and a physical Ethernet interface is that an Eth-Trunk interface must select one or more member interfaces to forward traffic.
- The restrictions on link aggregation vary according to network device models. The following uses Huawei S3700, S5700, and S6700 series switches as an example.
 - When adding an interface to an Eth-Trunk interface, ensure that the interface is of the default type. In addition, the member interface cannot be configured with certain services, such as a static MAC address. Because the services supported by interfaces are different, the types of services that cannot be configured on an interface added to an Eth-Trunk interface cannot be listed. For details, see the error message displayed on the device.
 - The numbers of physical interfaces at both ends of an Eth-Trunk must be the same and the physical interfaces must have the same rate, duplex mode, and flow control configuration.
 - It is recommended that the same jumbo frame length be configured for physical interfaces at both ends of an Eth-Trunk.
 - If an interface of the local device is added to an Eth-Trunk interface, the directly connected interface on the peer device must also be added to the Eth-Trunk interface so that the two devices can communicate with each other.
 - Devices at both ends of an Eth-Trunk must use the same link aggregation mode.
 - Interfaces operating at different rates and in different duplex modes can be added to the same Eth-Trunk interface. In addition, optical and electrical interfaces can be added to the same Eth-Trunk interface.
 - A member interface of an Eth-Trunk interface cannot be an Eth-Trunk

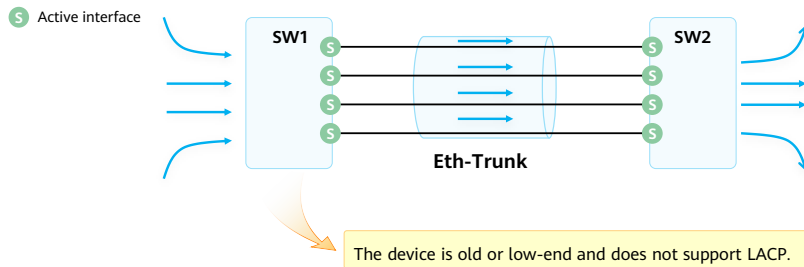
interface.

- Interfaces operating at different rates, in different duplex modes, and on different cards can be added to the same Eth-Trunk interface in LACP mode. However, member interfaces working at different rates cannot be in the forwarding state at the same time, and member interfaces working in half-duplex mode cannot forward traffic. Before the configuration, check the cards where member interfaces reside, interface rates, and duplex modes.

Contents

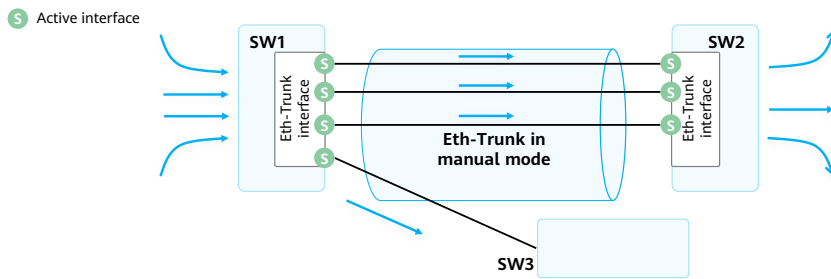
1. Basic Concepts of Link Aggregation
- 2. Link Aggregation in Manual Mode**
3. Link Aggregation in LACP Mode
4. Link Aggregation Configuration Examples

Manual Mode



- Manual mode: Eth-Trunk interfaces and their member interfaces are manually configured, and LACP is not used for negotiation between the two ends.
- In normal cases, all links are active links. In this mode, all active links participate in data forwarding and evenly load balance traffic. If an active link fails, the remaining active links automatically load balance the traffic evenly.
- If one of the two devices involved in link aggregation does not support LACP, the manual mode can be used.

Defects of the Manual Mode (1)



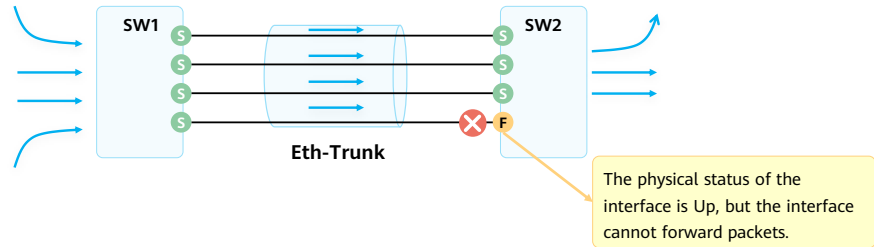
- To ensure that an Eth-Trunk interface works properly, ensure that the peer interfaces of all member interfaces in the local Eth-Trunk interface meet the following requirements:
 - They are on the same device.
 - They are added to the same Eth-Trunk interface.
- In manual mode, devices do not exchange packets. Therefore, the administrator must manually check whether the requirements are met.

- In the preceding figure, four interfaces on SW1 are added to the same Eth-Trunk interface. However, the peer end of one of these interfaces is SW3 instead of SW2. As a result, some traffic is load balanced to SW3, causing communication exceptions.

Defects of the Manual Mode (2)

S Active interface

F Faulty interface

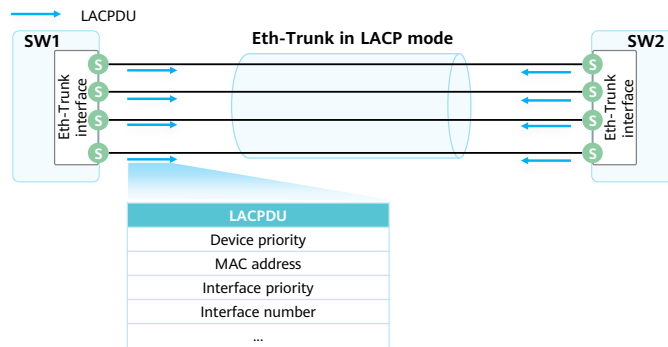


- In manual mode, a device can determine whether the peer interface is working properly only based on the physical status of the interface.

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LACP Mode

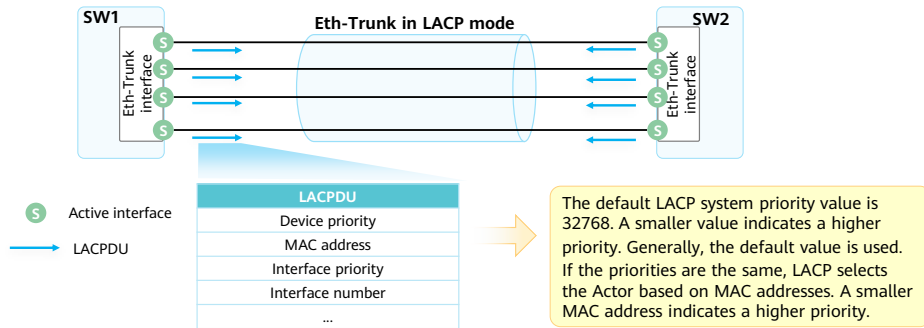


- LACP mode: is a link aggregation mode using LACP. Devices exchange Link Aggregation Control Protocol Data Units (LACPDU) for negotiation to ensure that the peer interfaces are member interfaces of the same Eth-Trunk interface on the same device.
- An LACPDU contains information such as the device priority, MAC address, interface priority, and interface number.

- LACP modes are classified into static LACP mode and dynamic LACP mode.
- In dynamic LACP mode, LACPDU are exchanged in the same way as in static LACP mode. Where they differ is how Eth-Trunks in two modes respond to an LACP negotiation failure:
 - In static LACP mode, an Eth-Trunk goes Down and cannot forward data after an LACP negotiation failure.
 - In dynamic LACP mode, an Eth-Trunk goes Down after an LACP negotiation failure, but its member interfaces inherit VLAN attributes of the Eth-Trunk and can independently forward data at Layer 2.
- Eth-Trunks in dynamic LACP mode are used only when Huawei devices connect to servers. For other scenarios, configuring Eth-Trunks in static LACP mode is recommended because Eth-Trunks in dynamic LACP mode may incur a risk of loops.
- Note: Some switches support only static LACP mode.

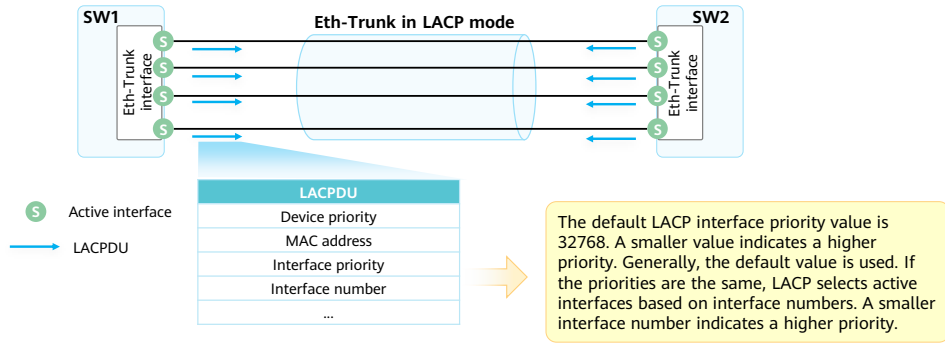
System Priority

- In LACP mode, the number of active interfaces selected by the devices at both ends must be the same; otherwise, the LAG cannot be set up. Devices exchange LACPDU to determine the Actor, and the Partner selects active interfaces according to the Actor. LACP system priorities are used to determine the Actor. A smaller value indicates a higher priority.



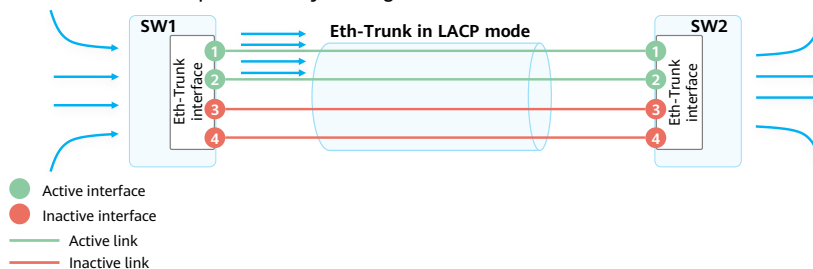
Interface Priority

- After the Actor is elected, both ends select active interfaces based on the interface priorities of the Actor. An interface with a higher priority is preferentially selected as an active interface. A smaller LACP interface priority value indicates a higher LACP interface priority.



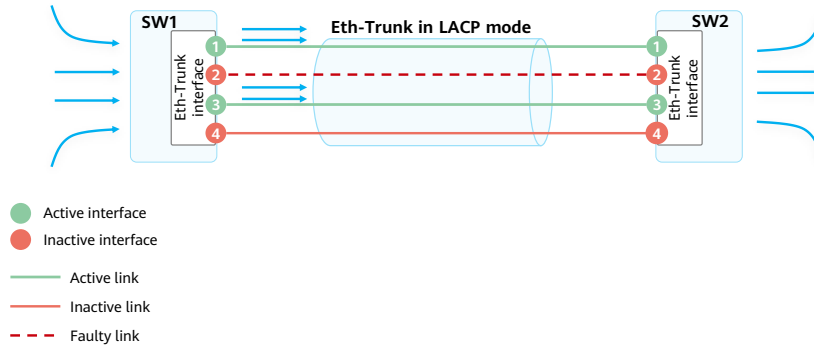
Maximum Number of Active Interfaces (1)

- In LACP mode, the maximum number of active interfaces can be configured. When the number of member interfaces exceeds the maximum number of active interfaces, the interfaces with higher priorities and smaller interface numbers are selected as active interfaces, and the other interfaces function as backup interfaces (inactive interfaces). In addition, the links corresponding to active interfaces become active links, and the links corresponding to inactive interfaces become inactive links. A switch sends and receives packets only through active interfaces.

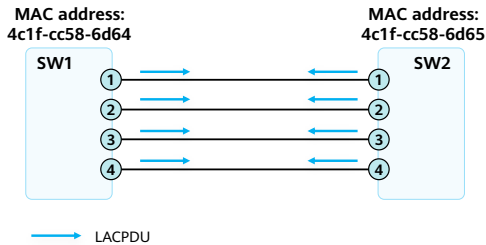


Maximum Number of Active Interfaces (2)

- If an active link fails, an inactive link with the highest priority (determined based on the interface priority and interface number) can be selected to replace the faulty link. The selected link becomes active. In this way, the overall bandwidth remains unchanged and services are not interrupted.



Active Link Election (1)



- Link aggregation in LACP mode is configured on SW1 and SW2. The maximum number of active interfaces is set to 2 at both ends.
- The two devices exchange LACPDU to elect the Actor. SW1 has a higher priority and is elected as the Actor in LACP negotiation.

- Link aggregation in LACP mode is configured on SW1 and SW2, and four interfaces numbered 1, 2, 3, and 4 are added to the Eth-Trunk on both devices. The maximum number of active interfaces in the Eth-Trunk is set to 2 on SW1 and SW2. The default settings are retained for other parameters (system priority and interface priority).
- SW1 and SW2 send LACPDU through the member interfaces 1, 2, 3, and 4.
- SW1 and SW2 receive LACPDU from each other, compare the system priorities (both are the default value 32768), and then compare the MAC addresses. The MAC address of SW1 is 4c1f-cc58-6d64, and that of SW2 is 4c1f-cc58-6d65. SW1 has a smaller MAC address and is elected as the Actor.

Active Link Election (2)

MAC address:
4c1f-cc58-6d64

SW1



MAC address:
4c1f-cc58-6d65

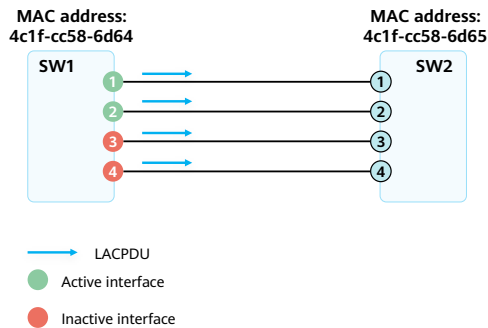
SW2



- SW1 elects active interfaces by comparing interface priorities and interface numbers. The interfaces have the same interface priority. Therefore, interfaces 1 and 2 with smaller interface numbers become active interfaces.

● Active interface
● Inactive interface

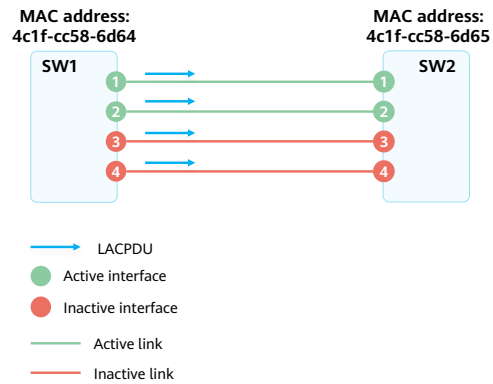
Active Link Election (3)



- SW1 sends LACPDU to notify the peer end of the active interface election result.

- LACP uses three flags in an LACPDU to identify the state of an interface. If an interface is active, the values of the following three flags are 1:
 - Synchronization
 - Collecting
 - Distributing
- If an interface is inactive, the values of the three flags are 0.

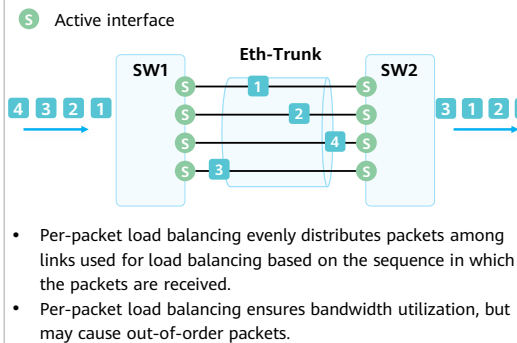
Active Link Election (4)



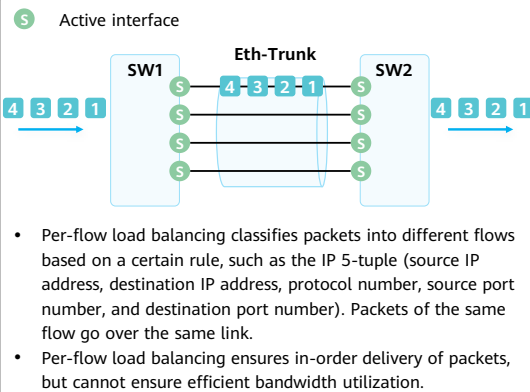
- SW2 determines the local active interfaces based on the election result of SW1, and the corresponding links become active links.
- The active link election of the Eth-Trunk is complete.

Load Balancing Classification

Per-packet load balancing



Per-flow load balancing

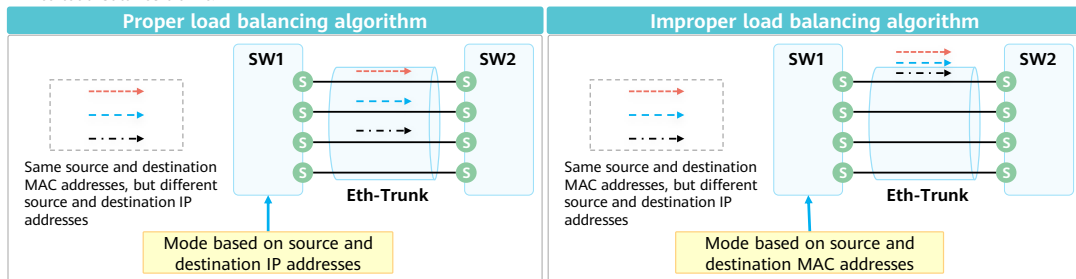


To prevent out-of-order packets, Eth-Trunk uses the per-flow load balancing mechanism.

- Load balancing allows a network node to distribute traffic among multiple links for forwarding. Load balancing can be classified into route load balancing, tunnel load balancing, and Eth-Trunk interface load balancing.
 - Route load balancing enables traffic to be balanced among multiple forwarding paths to the same destination. If there are multiple routes with the same destination address and mask but different next hops, outbound interfaces, or tunnel IDs, route load balancing can be implemented,
 - Tunnel load balancing, when used on a VPN, enables traffic to be balanced among the ingress PE's multiple tunnels that are destined for the same destination PE.
 - Eth-Trunk interface load balancing enables traffic to be balanced among multiple member links of an Eth-Trunk after multiple physical interfaces with the same link layer protocol are bundled into the Eth-Trunk interface.
- This course uses only Eth-Trunk interface load balancing as an example to describe load balancing classification.

Specifying Rules for Distinguishing Data Flows

- Eth-Trunk supports load balancing based on IP addresses or MAC addresses of packets. You can configure different load balancing modes to load balance data flows among different member interfaces. A load balancing mode is valid only locally for outgoing packets.
- Common load balancing modes can be based on the source IP address, source MAC address, destination IP address, destination MAC address, source and destination IP addresses, and source and destination MAC addresses.
- In real-world scenarios, you need to configure a proper load balancing mode based on service traffic characteristics. If a service traffic parameter changes frequently, using a load balancing mode based on this frequently-changing parameter will make it easier to load balance traffic.



- If the IP addresses of packets change frequently, using load balancing based on the source IP address, destination IP address, or both can better balance load among physical links.
- If the MAC addresses of packets change frequently but the IP addresses are fixed, using load balancing based on the source MAC address, destination MAC address, or both can better balance load among physical links.
- If a load balancing mode does not match the actual service characteristics, traffic may be unevenly load balanced. As a result, some member links may be heavily loaded, while other member links are idle. For example, if the source and destination IP addresses of packets change frequently but the source and destination MAC addresses are fixed, using load balancing based on source and destination MAC addresses may cause all traffic to be carried over only one member link.

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Configuration Commands (1)

1. Create a LAG.

```
[HUAWEI] interface eth-trunk trunk-id
```

Create an Eth-Trunk interface and enter the Eth-Trunk interface view.

2. Configure a link aggregation mode.

```
[HUAWEI-Eth-Trunk1] mode { lacp | manual load-balance }
```

The **mode lacp** command sets the link aggregation mode to LACP, and the **mode manual load-balance** command sets the link aggregation mode to manual.

Note: The link aggregation modes at both ends must be the same.

3. Add an interface to the LAG (Ethernet interface view).

```
[HUAWEI-GE1/0/1] eth-trunk trunk-id
```

In the interface view, add the interface to the Eth-Trunk.

Configuration Commands (2)

4. Add an interface to the LAG (Eth-Trunk interface view).

```
[HUAWEI-Eth-Trunk1] trunkport interface-type interface-number
```

In the Eth-Trunk interface view, add an interface to the LAG. An interface can be added to a LAG through operations performed in either step 3 or step 4.

5. Enable the function of adding interfaces working at different rates to the same Eth-Trunk interface.

```
[HUAWEI-Eth-Trunk1] lACP mixed-rate link enable
```

By default, after interfaces working at different rates are added to an Eth-Trunk interface in static or dynamic LACP mode, traffic is forwarded at only one of the rates through the corresponding member interfaces. This operation cannot implement load balancing on the Eth-Trunk interface.

6. Configure the LACP system priority.

```
[HUAWEI] lACP priority priority
```

A smaller priority value indicates a higher priority. By default, the LACP system priority value is 32768.

Configuration Commands (3)

7. Configure the LACP priority of an interface.

[HUAWEI-GE1/0/1] **lACP priority** *priority*

In the interface view, configure the LACP priority of the interface. By default, the LACP priority of an interface is 32768. A smaller value indicates a higher LACP priority.

This command can be configured for an interface only after the interface is added to a LAG.

8. Configure the maximum number of active interfaces.

[HUAWEI-Eth-Trunk1] **lACP max active-linknumber** *link-number*

When configuring the command, ensure that the maximum number of active interfaces on the local end is the same as that on the peer end. The maximum number of active interfaces can be configured only in LACP mode.

9. Configure the minimum number of active interfaces.

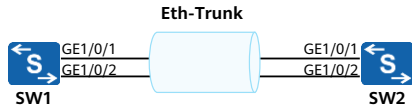
[HUAWEI-Eth-Trunk1] **least active-linknumber** *number*

The lower threshold for the number of active interfaces on the local end can be different from that on the peer end. The minimum number of active interfaces can be configured in both manual mode and LACP mode.

The minimum number of active interfaces is configured to ensure the minimum bandwidth. When the number of active links falls below the lower threshold, the Eth-Trunk interface goes Down.

- The maximum number of active interfaces that can be configured varies according to switch models. For details, see the product documentation.
- The minimum number of active interfaces is configured to ensure the minimum bandwidth. If the bandwidth is too low, some services that require high link bandwidth are abnormal. To prevent this issue, the Eth-Trunk interface goes Down, and services are switched to other paths, ensuring normal running of the services.

Example for Configuring Link Aggregation in Manual Mode



- Requirements:

- Both SW1 and SW2 are connected to VLAN 10 and VLAN 20.
- SW1 and SW2 are connected through two Ethernet links. To provide link redundancy and ensure transmission reliability, configure link aggregation in manual mode on SW1 and SW2.

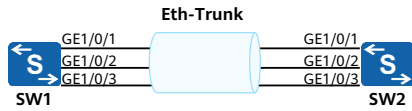
The configuration of SW1 is as follows:

```
[SW1] interface eth-trunk 1
[SW1-Eth-Trunk1] trunkport GE 1/0/1 to 1/0/2
[SW1-Eth-Trunk1] port link-type trunk
[SW1-Eth-Trunk1] port trunk allow-pass vlan 10 20
```

The configuration of SW2 is as follows:

```
[SW2] interface eth-trunk 1
[SW2-Eth-Trunk1] trunkport GE 1/0/1 to 1/0/2
[SW2-Eth-Trunk1] port link-type trunk
[SW2-Eth-Trunk1] port trunk allow-pass vlan 10 20
```

Example for Configuring Link Aggregation in LACP Mode (1)



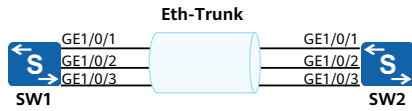
- Requirements:

- Both SW1 and SW2 are connected to VLAN 10 and VLAN 20.
- SW1 and SW2 are connected through three Ethernet links. To provide link redundancy and ensure transmission reliability, configure link aggregation in LACP mode on SW1 and SW2, manually adjust the priorities so that SW1 becomes the Actor, and set the maximum number of active interfaces to 2. One link functions as the backup link.

The configuration of SW1 is as follows:

```
[SW1] interface eth-trunk 1
[SW1-Eth-Trunk1] mode lacp
[SW1-Eth-Trunk1] lacp max active-linknumber 2
[SW1-Eth-Trunk1] trunkport GE 1/0/1 to 1/0/3
[SW1-Eth-Trunk1] port link-type trunk
[SW1-Eth-Trunk1] port trunk allow-pass vlan 10 20
[SW1-Eth-Trunk1] quit
[SW1] lacp priority 30000
```

Example for Configuring Link Aggregation in LACP Mode (2)



- Requirements:
 - Both SW1 and SW2 are connected to VLAN 10 and VLAN 20.
 - SW1 and SW2 are connected through three Ethernet links. To provide link redundancy and ensure transmission reliability, configure link aggregation in LACP mode on SW1 and SW2, manually adjust the priorities so that SW1 becomes the Actor, and set the maximum number of active interfaces to 2. One link functions as the backup link.

The configuration of SW2 is as follows:

```
[SW2] interface eth-trunk 1
[SW2-Eth-Trunk1] mode lacp
[SW2-Eth-Trunk1] lacp max active-linknumber 2
[SW2-Eth-Trunk1] trunkport GE 1/0/1 to 1/0/3
[SW2-Eth-Trunk1] port link-type trunk
[SW2-Eth-Trunk1] port trunk allow-pass vlan 10 20
[SW2-Eth-Trunk1] quit
```

Quiz

1. (Short-answer question) What are the differences between per-packet load balancing and per-flow load balancing?
2. (Short-answer question) How is the Actor elected in link aggregation in LACP mode?

1. Per-packet load balancing: Each data packet is transmitted over a different link, which may cause out-of-order packets. Per-flow load balancing: Packets of the same flow are forwarded through the same link, which prevents out-of-order packets. However, a single flow cannot use the logical bandwidth of the entire Eth-Trunk interface.
2. System priorities are compared. A smaller value indicates a higher priority. If system priorities are the same, MAC addresses are compared. A smaller MAC address indicates a higher priority. The device with a higher priority becomes the Actor.

Summary

- Link aggregation technology can be used to improve link reliability, link utilization, and link bandwidth. Link aggregation can be classified into link aggregation in manual mode and link aggregation in LACP mode.
- In LACP mode, devices exchange LACPDU for negotiation to implement link backup. If an active link fails, the backup link is elected as the active link to forward traffic.
- To ensure the order in which packets arrive, link aggregation uses per-flow load balancing.

Thank you.

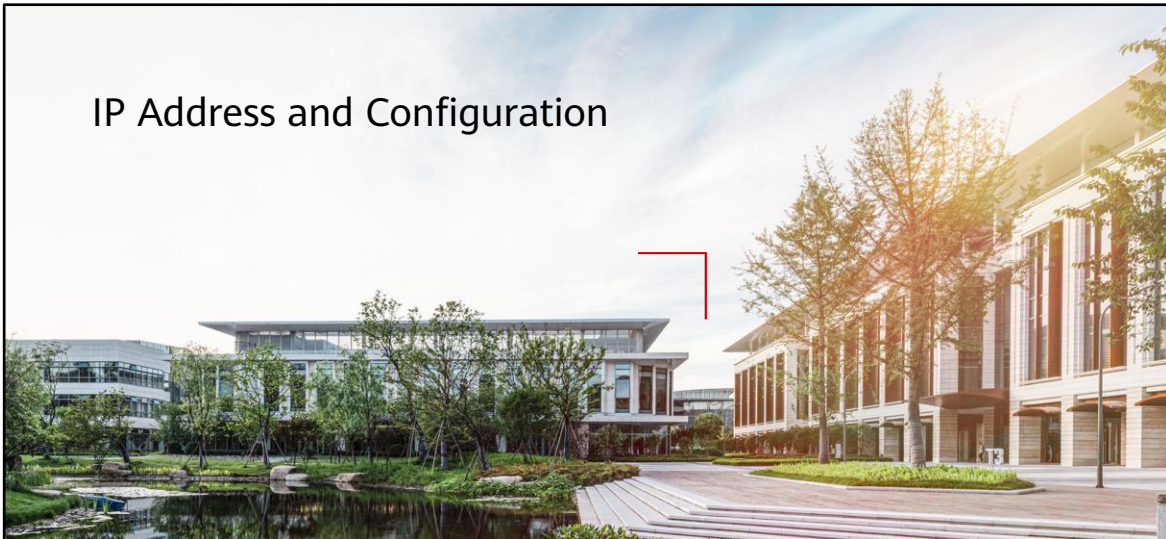
把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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IP Address and Configuration



Foreword

- Internet Protocol version 4 (IPv4) is the core protocol of the TCP/IP protocol suite. It works at the network layer of the TCP/IP protocol stack.
- The network layer provides the connectionless data transmission service. That is, no connection needs to be established before data packets are sent, and each IP data packet is sent independently.
- The Dynamic Host Configuration Protocol (DHCP) dynamically manages and configures user IP addresses in a centralized manner. It simplifies network deployment and scale-out, even for small networks.
- This course describes the basic concepts of IPv4 addresses, how to divide subnets, and the fundamentals of IPv4 address autoconfiguration.

Objectives

- Upon completion of this course, you will be able to:
 - Describe the main protocols of the network layer.
 - Describe the concept and classification of IPv4 addresses and special IP addresses.
 - Subnet the network based on service requirements.
 - Describe the fundamentals of DHCP.

Contents

- 1. Network Layer Protocols**
2. Introduction to IPv4 Addresses
3. Subnetting
4. ICMP
5. Fundamentals and Configuration of DHCP

Network Layer Protocols

- The network layer is often referred to as the IP layer. However, network layer protocols are not limited to IP, but also include other protocols such as the Internet Control Message Protocol (ICMP) and Internet Group Management Protocol (IGMP).



TCP/IP reference model

- For more information about the Internet Protocol, see RFC 791 at <https://www.rfc-editor.org/rfc/rfc791.html>.
- IGMP will be described in the HCIP-Datacom courses.

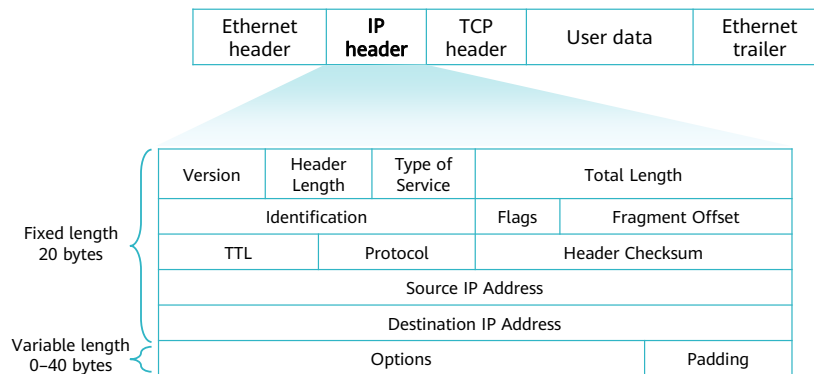
IP

- IP is short for Internet Protocol. The Internet Protocol itself is the name of a protocol document, which contains minimal content and primarily defines and describes the format of IP packets.
- IP usually refers to IP-related information rather than the Internet Protocol itself.

Function	Version
• Provides logical addresses for devices at the network layer. • Routes and forwards data packets.	• IPv4 • IPv6

- IP has two versions: IPv4 and IPv6. Currently, most IP services on the Internet are IPv4 services, but are gradually transitioning to IPv6 services. Unless otherwise specified, IP mentioned in this course refers to IPv4.
- Internet Protocol version 6 (IPv6) is the second-generation standard network layer protocol. It is a set of specifications designed by the Internet Engineering Task Force (IETF) and is an upgraded version of IPv4.

IPv4 Packet Format



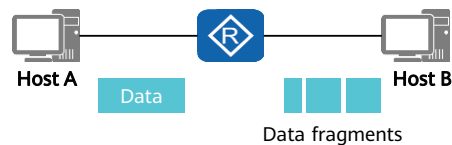
- The IP packet header contains the following fields:
 - Version: 4 bits. The value 4 indicates IPv4.
 - Header Length: 4 bits. It indicates the length of the header. If the Options field is not carried, the length is 20 bytes. If the Options field is carried, the maximum length is 60 bytes.
 - Type of Service: 8 bits. It indicates the service type.
 - Total Length: 16 bits. It indicates the total length of the IP packet.
 - Identification: 16 bits. This field is used during fragment reassembly.
 - Flags: 3 bits. It indicates flags.
 - Fragment Offset: 13 bits. It is the fragment offset. This field is used during fragment reassembly.
 - Time to Live: 8 bits. It indicates the time to live.
 - Protocol: 8 bits. It indicates the next-layer protocol. This field specifies the protocol used by the data carried in the packet, so that the IP layer of the destination host can determine the process to which the data is to be submitted.
 - Common values:
 - 1: ICMP
 - 2: IGMP
 - 6: TCP
 - 17: UDP
 - Header Checksum: 16 bits. It indicates the checksum of the header.
 - Source IP Address: 32 bits. It indicates the source IP address.
 - Destination IP Address: 32 bits. It indicates the destination IP address.
 - Options: variable length. It is the options field.

- Padding: variable length. It is the padding field, which is filled with all 0s.

Data Packet Fragmentation

- IP packets forwarded on a network can have different lengths. If the length of a packet exceeds the maximum length supported by a data link, the packet needs to be divided into several smaller fragments before being transmitted on the link.
- The process of dividing a packet into multiple fragments is called fragmentation.

Version	Header Length	Type of Service	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol	Header Checksum		
Source IP Address				
Destination IP Address				
Options				Padding

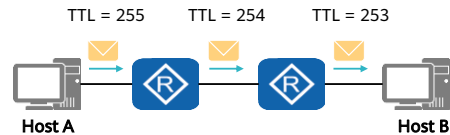


- Identification: 16 bits. It indicates the identifier assigned by the sender. This field is used during fragment reassembly.
- Flags: 3 bits. It indicates flags.
 - Reserved bit: 0 (reserved)
 - Don't Fragment bit: The value 1 indicates that the packet cannot be fragmented, and the value 0 indicates that the packet can be fragmented.
 - More Fragments bit: The value 1 indicates that there are more fragments, and the value 0 indicates the last fragment.
- Fragment Offset: 13 bits. It is the fragment offset. This field is used during fragment reassembly. It indicates the relative position of the fragment in the original packet after the packet is fragmented. This field is used together with the More Fragments bit to help the receiver reassemble the fragments.

TTL

- The Time to Live (TTL) field specifies the number of routers that a data packet can pass through.
- The TTL value is decremented by 1 each time a data packet passes through a router. When the TTL value becomes 0, the data packet is discarded.

Version	Header Length	Type of Service	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol	Header Checksum		
Source IP Address				
Destination IP Address				
Options				Padding

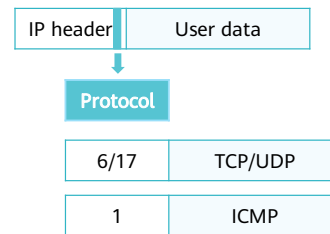


- When packets are forwarded between network segments, loops may occur if routes are not properly planned on network devices. As a result, packets are looped infinitely on the network and cannot reach the destination. After a loop occurs, all packets sent to the destination are forwarded infinitely. As the number of such packets increases, congestion occurs on the network.
- To prevent network congestion caused by loops, the IP header contains the TTL field. The TTL value is decremented by 1 each time the packet passes through a Layer 3 device. The initial TTL value is set by the source device. When the TTL value is decremented to 0, the packet is discarded. In addition, the device that discards the packet sends an ICMP error message to the source device based on the source IP address in the packet header.
- Note: You can also configure the network device not to send ICMP error messages to the source device.

Protocol

- The Protocol field in the IP header of a packet identifies the protocol that will continue to process the packet.
- That is, the protocol used by the data carried in the packet is specified, so that the IP layer of the destination host can report the data to the corresponding process for processing.

Version	Header Length	Type of Service	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol	Header Checksum		
Source IP Address				
Destination IP Address				
Options				Padding



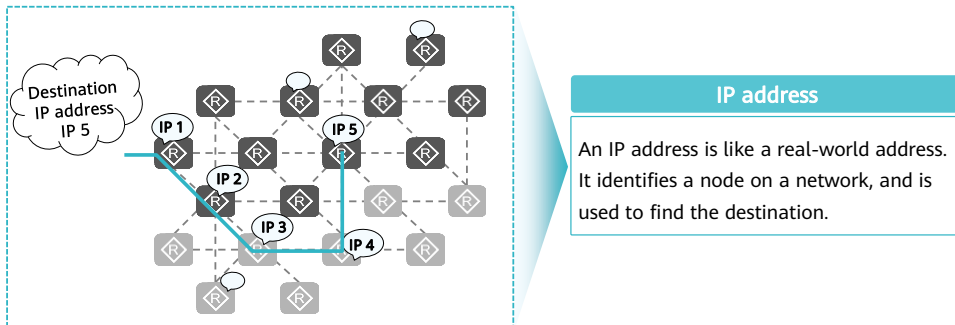
- After receiving and processing a packet, the network layer of the destination device needs to determine the next step for processing the packet. The Protocol field in the IP header of the packet identifies the protocol that will continue to process the packet.
- This field can identify a network layer protocol, such as the Internet Control Message Protocol (ICMP) (the corresponding value is 0x01), or identify an upper-layer protocol, such as the Transmission Control Protocol (TCP) (the corresponding value is 0x06) or User Datagram Protocol (UDP) (the corresponding value is 0x11).

Contents

1. Network Layer Protocols
- 2. Introduction to IPv4 Addresses**
3. Subnetting
4. ICMP
5. Fundamentals and Configuration of DHCP

What Is an IP Address?

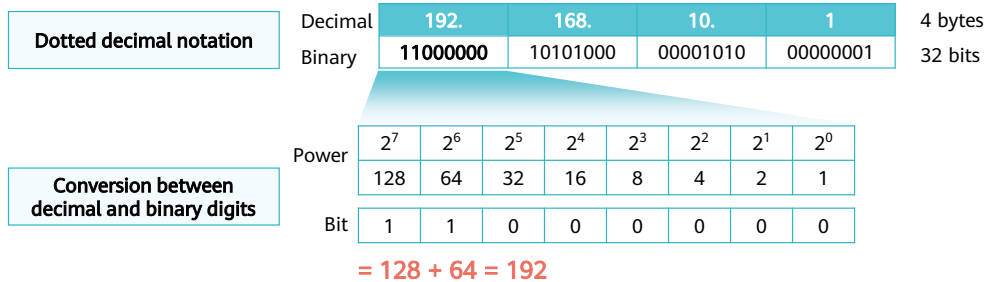
- An IP address identifies a node (or an interface of a network device) on a network.
- IP addresses are used for addressing of IP packets on the network.



- On an IP network, if a user wants to connect a computer to the Internet, the user needs to apply for an IP address for the computer.
- An IP address is an attribute of an interface on a network device. To assign an IP address to a device is to assign an IP address to an interface of the device actually. If a device has multiple interfaces, each interface usually requires at least one IP address.
- Note: Interfaces that require IP addresses are usually interfaces of routers and computers.

IP Address Format

- An IPv4 address has 32 bits.
- An IPv4 address is usually expressed in dotted decimal notation.



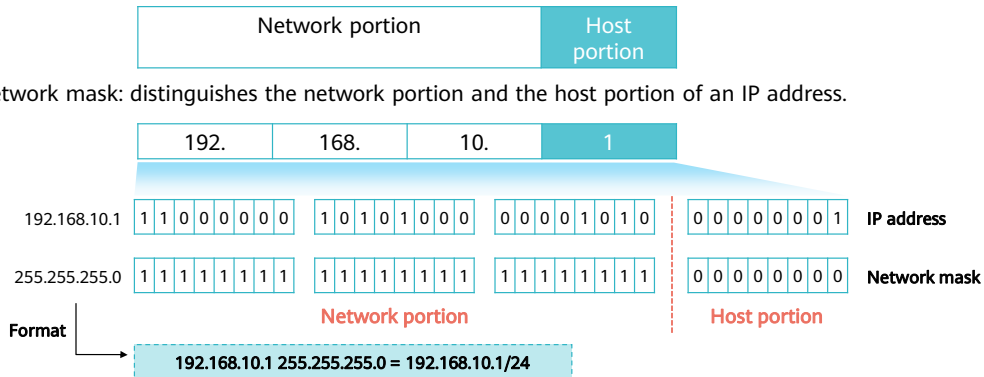
- IPv4 address range: 0.0.0.0 to 255.255.255.255

- An IP address is 32 bits long and consists of four bytes. For ease of reading and writing, an IP address is usually expressed in dotted decimal notation.
- IPv4 address range
 - 00000000.00000000.00000000.00000000 to 11111111.11111111.11111111.11111111, that is, 0.0.0.0 to 255.255.255.255

IP Address Structure

- Network portion: identifies a network.
- Host portion: identifies a host on a network.

- Network mask: distinguishes the network portion and the host portion of an IP address.

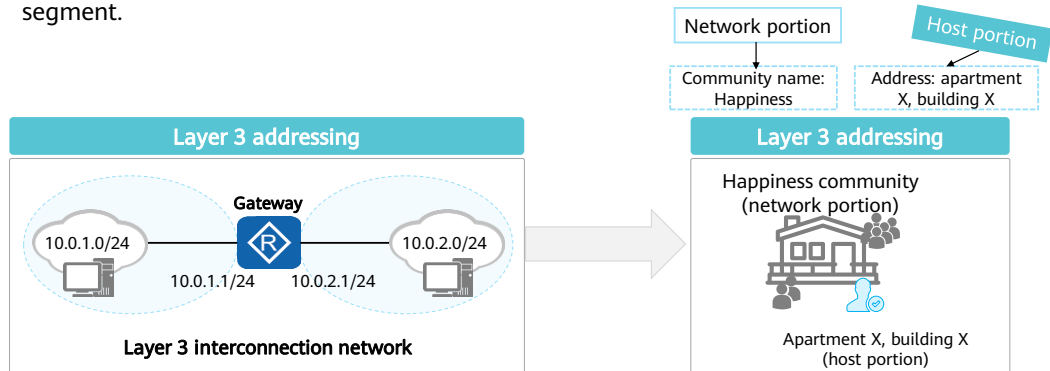


- An IPv4 address consists of two portions:
 - Network portion (network ID): identifies a network.
 - An IP address does not contain any geographical information about a host. The network to which the host belongs can be determined only based on the network ID.
 - Network devices with the same network ID are located on the same network, regardless of their physical locations.
 - Host portion (host ID): identifies a host on a network.
- Network mask (also called subnet mask):
 - A network mask contains 32 bits, which is the same as the number of bits in an IP address. Network masks are also usually expressed in dotted decimal notation.
 - A network mask is not an IP address. In binary notation, it is a string of consecutive 1s followed by a string of consecutive 0s.
 - Generally, the number of 1s in a network mask is called the length of the network mask. For example, the length of the network mask 0.0.0.0 is 0, and the length of the network mask 252.0.0.0 is 6.
 - A network mask is usually used together with an IP address. The bits with the value of 1 correspond to the network portion in an IP address, and the bits with the value of 0 correspond to the host portion. It helps identify the network ID and host ID in an IP address. That is, the number of 1s in a network mask is the number of bits in the network ID of an IP address, and the number of 0s in a network mask is the number of bits in the host ID of

an IP address.

IP Addressing

- Network portion: identifies a network to which an IP address belongs.
- Host portion: identifies a host on a network. It can uniquely identify a device on a network segment.



- A network ID identifies the network where a host resides, similar to the function of identifying a residential community in a specific district of a city in a province.
- A host ID identifies a specific host interface on the network defined by a network ID, similar to the function of identifying a specific apartment in a building.
- Network addressing:
 - Layer 2 network addressing: A host interface can be directly located based on the IP address.
 - Layer 3 network addressing: A gateway is used to forward data packets between different network segments.
- Gateway:
 - During packet forwarding, the forwarding path and the interface to the destination network segment need to be determined first. If the destination host and source host are on different network segments, packets need to be forwarded to the gateway first, and then the gateway forwards the packets to the destination network segment.
 - A gateway is a device that receives and processes packets sent by hosts on the local network segment and forwards the packets to the destination network segment. To implement this function, the gateway must know the route to the destination network segment. The address of the gateway interface connected to the local network segment is the gateway address of

the network segment.

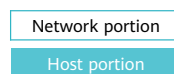
IP Address Classification

- IP addresses are classified into five classes to facilitate IP address management and networking:

Class A	0NNNNNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	0.0.0.0–127.255.255.255	Assigned to hosts
Class B	10NNNNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	128.0.0.0–191.255.255.255	
Class C	110NNNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	192.0.0.0–223.255.255.255	
Class D	1110NNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	224.0.0.0–239.255.255.255	For multicast
Class E	1111NNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	240.0.0.0–255.255.255.255	For research

- Default network masks of Classes A, B, and C addresses

- Class A: 8 bits, 0.0.0.0/8 to 127.255.255.255/8
- Class B: 16 bits, 128.0.0.0/16 to 191.255.255.255/16
- Class C: 24 bits, 192.0.0.0/24 to 223.255.255.255/24



- IP addresses are classified into five classes to facilitate IP address management and networking:
 - The easiest way to determine the class of an IP address is to check the first bits in its network ID. The class fields of Class A, Class B, Class C, Class D, and Class E are binary numbers 0, 10, 110, 1110, and 1111, respectively.
 - Classes A, B, and C addresses are unicast IP addresses (except some special addresses), and only these three classes of addresses can be assigned to hosts.
 - Class D addresses are multicast IP addresses.
 - Class E addresses are used for special experimental purposes.
 - This course focuses only on Classes A, B, and C addresses.
- Comparison of Classes A, B, and C addresses:
 - A network that uses Class A addresses is called a Class A network. A network that uses Class B addresses is called a Class B network. A network that uses Class C addresses is called a Class C network.
 - A Class A network has an 8-bit network ID, which contains a few bits. However, a Class A network allows a large number of host interfaces. The first bit of the network ID is always 0, and the address range is from 0.0.0.0 to 127.255.255.255.
 - A Class B network has a 16-bit network ID. The first two bits of the network ID are always 10, and the address range is from 128.0.0.0 to 191.255.255.255.
 - A Class C network has a 24-bit network ID, which contains many bits. However, a Class C network allows only a small number of host interfaces. The first three bits of the network ID are always 110, and the address range is from 192.0.0.0 to 223.255.255.255.
- Note:
 - A host generally refers to a router or computer. The IP address of an interface on a host is generally called the host IP address.

- Multicast address: Multicast can implement one-to-many message transmission.

IP Address Types

- Generally, the network range defined by a network ID is called a network segment.

- Network address:** identifies a network.

Example: 192.168.10.0/24

192.	168.	10.	00000000
------	------	-----	----------

- Broadcast address:** is a special address used to send data to all hosts on the network.

Example: 192.168.10.255/24

192.	168.	10.	11111111
------	------	-----	----------

- Available address:** identifies an address that can be assigned to a node or network device interface on the network.

Example: 192.168.10.1/24

192.	168.	10.	00000001
------	------	-----	----------

Note

- The network address and broadcast address cannot be directly used by nodes or network devices.
- The number of available addresses on a network segment is $2^n - 2$, where n is the number of bits in the host portion.

- Network address
 - The network ID is X , and each bit of the host ID is 0.
 - It cannot be assigned to a specific host interface.
- Broadcast address
 - The network ID is X , and each bit of the host ID is 1.
 - It cannot be assigned to a specific host interface.
- Available address
 - It is also called a host address, and can be assigned to a specific host interface.
- The number of available addresses on a network segment is calculated as follows:
 - If the host ID of a network segment contains n bits, the number of IP addresses is 2^n , and the number of available IP addresses is $2^n - 2$ (the network address and broadcast address are excluded).

IP Address Calculation

- Calculate the network address, broadcast address, and number of available addresses of the Class B address 172.16.10.1/16.

	172.	16.	00001010.	00000001
--	------	-----	-----------	----------

IP address	1 0 1 0 1 1 0 0	0 0 0 1 0 0 0 0	0 0 0 0 1 0 1 0	0 0 0 0 0 0 0 1
Network mask	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0
Network address	1 0 1 0 1 1 0 0	0 0 0 1 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0
Broadcast address	1 0 1 0 1 1 0 0	0 0 0 1 0 0 0 0	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1

Change all host bits to 0 to obtain the network address.
172.16.0.0

Change all host bits to 1 to obtain the broadcast address.
172.16.255.255

Number of IP addresses: $2^{16} = 65536$

Number of available IP addresses: $2^{16} - 2 = 65534$

Available IP address range: 172.16.0.1 to 172.16.255.254

Exercise

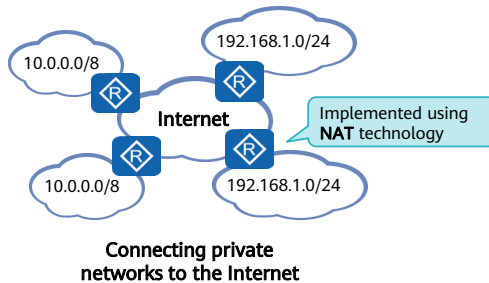
Calculate the network address, broadcast address, and number of available addresses of the Class A address 10.128.20.10/8.

- Network address: Change all host bits of an IP address to 0. The obtained address is the network address of the network to which the IP address belongs.
- Broadcast address: Change all host bits of an IP address to 1. The obtained address is the broadcast address of the network to which the IP address belongs.
- Answer to the exercise:
 - Network address: 10.0.0.0
 - Broadcast address: 10.255.255.255
 - Number of IP addresses: 2^{24}
 - Number of available IP addresses: $2^{24} - 2$
 - Available IP address range: 10.0.0.1 to 10.255.255.254

Public and Private IP Addresses

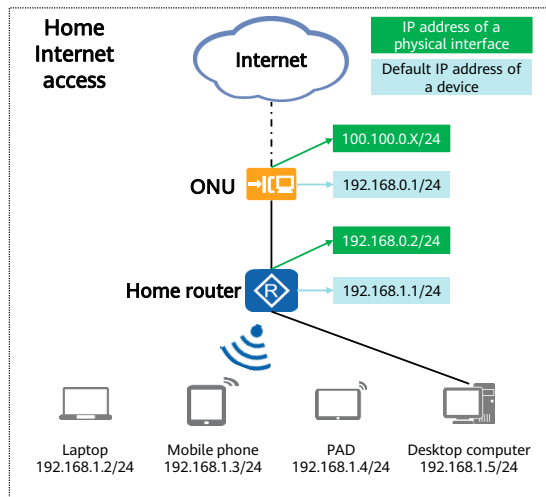
- Public IP addresses are assigned by the Internet Assigned Numbers Authority (IANA) to ensure that each public IP address is unique on the Internet.
- Private IP addresses are used on internal networks. These addresses are not directly used on the Internet. Instead, they are translated into public IP addresses using Network Address Translation (NAT) technology for external communication. In the IP address space, some Classes A, B, and C addresses are reserved.

- Class A: 10.0.0.0 to 10.255.255.255
- Class B: 172.16.0.0 to 172.31.255.255
- Class C: 192.168.0.0 to 192.168.255.255



- Private addresses can be used only on an internal network and cannot be used on a public network.
 - Public IP address: A network device connected to the Internet must have a public IP address.
 - Private IP address: The use of private IP addresses allows networks to be expanded more freely because the same private IP address can be reused on different private networks.
- Connecting a private network to the Internet: A private network is not allowed to connect to the Internet because it uses private IP addresses. Driven by actual requirements, many private networks want to connect to the Internet to communicate with the Internet and communicate with other private networks over the Internet. The interconnection between a private network and the Internet must be implemented using Network Address Translation (NAT) technology.
- Note:
 - NAT: stands for Network Address Translation. It is used to translate private IP addresses into public IP addresses.
 - IANA: stands for Internet Assigned Numbers Authority.

Application Example of Private and Public IP Addresses



- In a home Internet access scenario, both private and public IP addresses are usually used.
 - Private IP addresses: 192.168.1.X/24 and 192.168.0.X/24. Private IP addresses are assigned by users. By default, they are Class C IP addresses.
 - Public IP address: 100.100.0.X/24. Only the uplink interface of the optical network unit (ONU) uses a public IP address assigned by a network service provider.
- When various types of terminals access the Internet, address translation is performed twice. In the first translation, an IP address is translated into 192.168.0.2 (IP address of the interface connecting the home router to the ONU). In the second translation, an IP address is translated into 100.100.0.X (public IP address obtained by the ONU).

- Optical network unit (ONU) is a network device provided by a network service provider for optical-to-electrical information conversion.
- The default IP address of a device is usually the built-in IP address of the device before delivery. The default IP address varies according to vendors, but it is generally a Class C address. This address is used for device management and terminal access.

Special IP Addresses

- In the IP address space, some IP addresses have special meanings and functions.

Special IP Address	Address Range	Function
Limited broadcast address	255.255.255.255	It can be used as the destination address. Packets with this address as the destination address are sent to all hosts on the network segment of the sender.
Any address	0.0.0.0	It can be used as the source address and represents all networks in the routing table.
Loopback address	127.0.0.0/8	It is used to test the software system of a device.
Link-local address	169.254.0.0/16	If a host fails to automatically obtain an address, the host can use an address on this network segment for temporary communication.

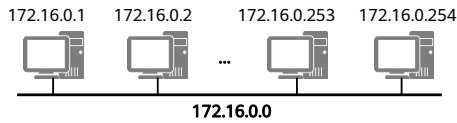
- 255.255.255.255
 - This address is called a limited broadcast address and can be used as the destination IP address of an IP packet.
 - After receiving an IP packet whose destination IP address is a limited broadcast address, a router stops forwarding the IP packet.
- 0.0.0.0
 - When a host interface does not obtain an IP address during startup, it can send a DHCP request packet with the destination IP address 255.255.255.255 and source IP address 0.0.0.0 to the network. The host expects that the DHCP server will assign an available IP address to it after receiving the request packet.
 - In the routing table, 0.0.0.0/0 represents the default route.
- 127.0.0.0/8
 - This address is a loopback address and can be used as the destination IP address of an IP packet. It is used to test the software system of a device.
 - An IP packet whose destination IP address is a loopback address cannot leave the device that sends the packet.
- 169.254.0.0/16
 - If a network device is configured to automatically obtain an IP address but does not find an available DHCP server on the network, the device uses an IP address on the network segment 169.254.0.0/16 for temporary

communication.

Contents

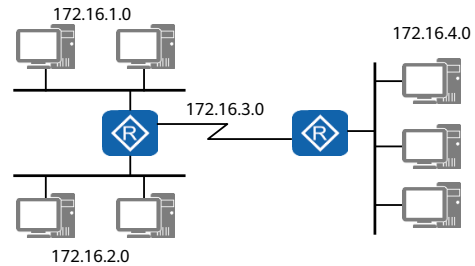
1. Network Layer Protocols
2. Introduction to IPv4 Addresses
- 3. Subnetting**
4. ICMP
5. Fundamentals and Configuration of DHCP

Why Is Subnetting Required?



$2^{16} = 65536$ IP addresses

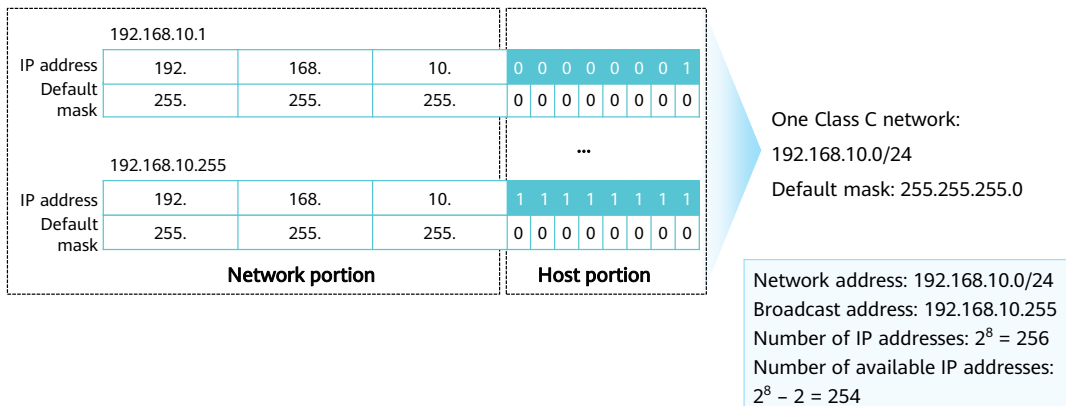
- A Class B address is used for a broadcast domain, wasting addresses.
- The broadcast domain is too large. Once a broadcast storm occurs, the internal network is overloaded.



- The network can be divided into multiple subnets, each of which belongs to a different broadcast domain.
- In this way, broadcast domains are smaller and network planning is more reasonable.
- IP addresses are properly used.

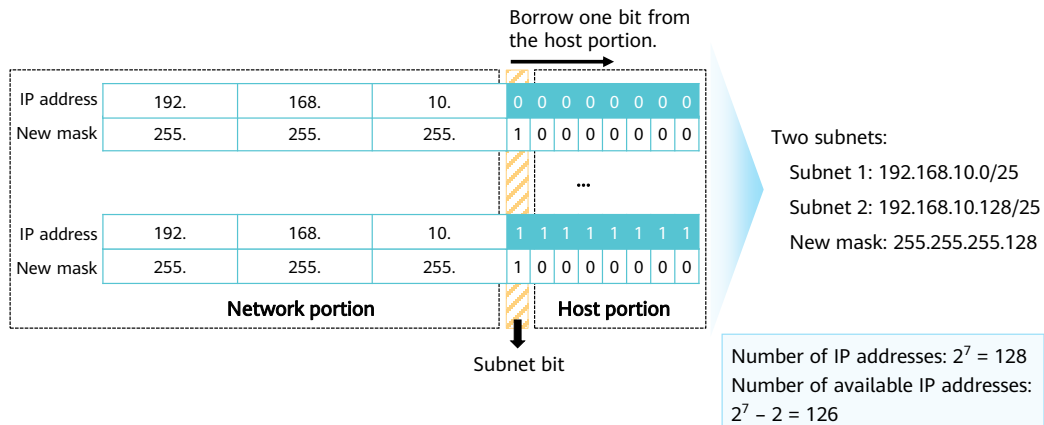
- Classful addressing is too rigid and coarse-grained. As a result, a large number of host IDs cannot be fully utilized, wasting a lot of IP address resources.
- Therefore, subnetting, that is, Variable Length Subnet Mask (VLSM), can be used to reduce address waste. A large classful network can be divided into several small subnets, making the use of IP addresses more appropriate.

Subnetting Process: Analysis of the Original Network Segment



- Assume that there is a Class C network segment address 192.168.10.0/24. This network segment has 256 IP addresses, among which 254 IP addresses are available.

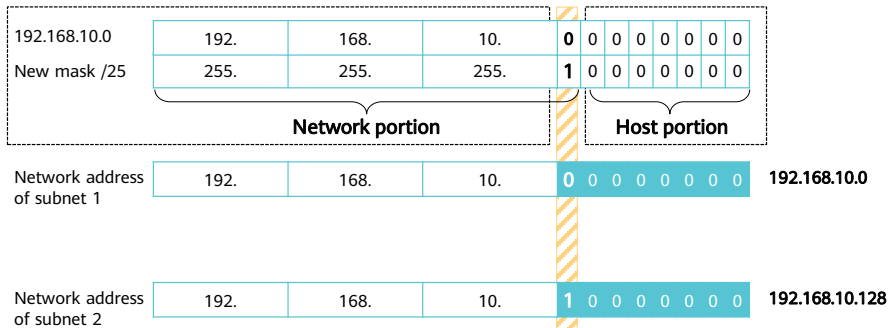
Subnetting Process: Borrowing a Bit from the Host Portion



- Now, one bit is borrowed from the host ID to the 24-bit network ID. In this way, the network ID is extended to 25 bits, and the host ID is reduced to 7 bits. The borrowed bit is the subnet bit. In this case, the network mask is changed to 25 bits, that is, 255.255.255.128 or /25.
- Subnet bit: The value can be 0 or 1. In this way, two new subnets are obtained.
- According to the calculation, there are 128 IP addresses on the network.

Subnetting Process: Calculating Subnet Network Addresses

- The host bits are all 0s. Calculate subnet network addresses.



- Calculate network addresses. The host bits are all 0s.
 - If the subnet bit is 0, the network address is 192.168.10.0.
 - If the subnet bit is 1, the network address is 192.168.10.128.

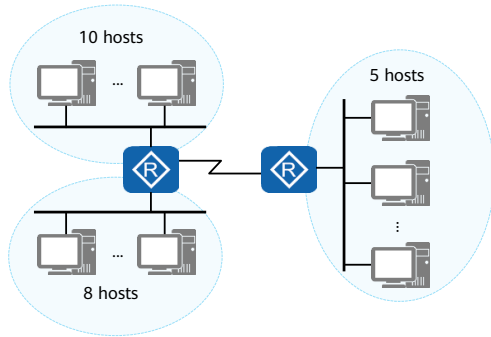
Subnetting Process: Calculating Subnet Broadcast Addresses

- The host bits are all 1s. Calculate subnet broadcast addresses.

192.168.10.0	192.	168.	10.	0	0	0	0	0	0	0	0
New mask /25	255.	255.	255.	1	0	0	0	0	0	0	0
Network portion				Host portion							
Network address of subnet 1	192.	168.	10.	0	0	0	0	0	0	0	0
Broadcast address of subnet 1	192.	168.	10.	0	1	1	1	1	1	1	1
Network address of subnet 2	192.	168.	10.	1	0	0	0	0	0	0	0
Broadcast address of subnet 2	192.	168.	10.	1	1	1	1	1	1	1	1

- Calculate broadcast addresses. The host bits are all 1s.
 - If the subnet bit is 0, the broadcast address is 192.168.10.127.
 - If the subnet bit is 1, the broadcast address is 192.168.10.255.

Exercise: Calculating Subnets (1)



- There is a Class C network segment 192.168.1.0/24. Use variable-length subnet masks to allocate IP addresses to three subnets.

- Calculation:** (The following uses 10 hosts as an example.)

Step 1: Calculate the number of required host bits.

$$2^n - 2 \geq 10$$

$$n \geq 4 \text{ bits, host bits}$$

Step 2: Borrow bits from the host portion.

Borrow 4 bits from the host portion.

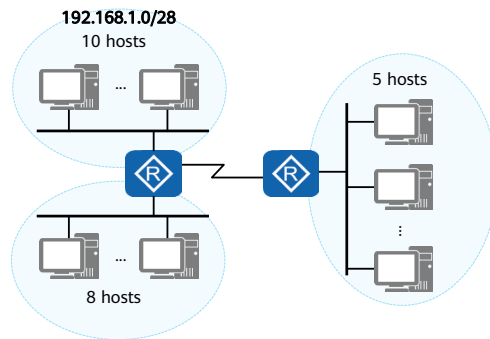
IP address	192.	168.	1.	0	0	0	0	0	0	0	0
Subnet mask	255.	255.	255.	1	1	1	1	0	0	0	0

Subnet bits

Number of subnets:
 $2^4 = 16$ subnets.

- In actual network planning, the subnet with more hosts is planned first.

Exercise: Calculating Subnets (2)



Step 3: Calculate subnet network addresses.

IP address	192.	168.	1.	0	0	0	0	0	0	0	0	
New mask	255.	255.	255.	1	1	1	1	0	0	0	0	
Subnet 1	192.	168.	1.	0	0	0	0	0	0	0	0	192.168.1.0/28
Subnet 2	192.	168.	1.	0	0	0	1	0	0	0	0	192.168.1.16/28
Subnet 3	192.	168.	1.	0	0	1	0	0	0	0	0	192.168.1.32/28
Subnet 16	192.	168.	1.	1	1	1	1	0	0	0	0	192.168.1.240/28

- The network addresses of the subnets are as follows:

- 192.168.1.0
- 192.168.1.16
- 192.168.1.32
- 192.168.1.48
- 192.168.1.64
- 192.168.1.80
- 192.168.1.96
- 192.168.1.112
- 192.168.1.128
- 192.168.1.144
- 192.168.1.160
- 192.168.1.176
- 192.168.1.192
- 192.168.1.208
- 192.168.1.224

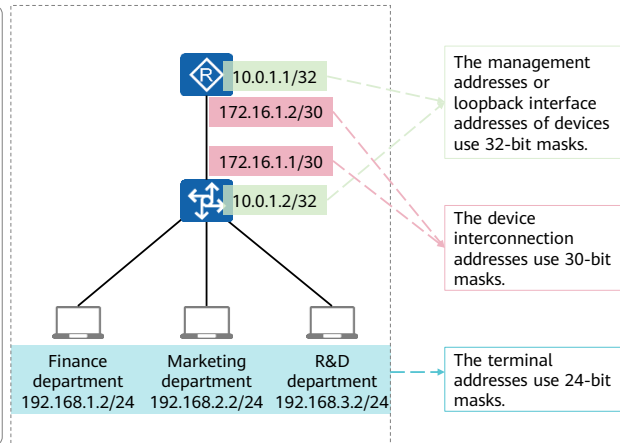
▫ 192.168.1.240

Suggestions on Subnet Mask and IPv4 Address Planning

Planning suggestions

- **Hierarchy:** To implement hierarchical division of a network, you need to consider both geographical and service factors and use the top-down method for division, thereby effectively managing the network and simplifying routing tables.
- **Contiguity:** Contiguous addresses facilitate route summarization on a hierarchical network, greatly reducing the number of routing entries and improving the route lookup efficiency.
- **Scalability:** When addresses are allocated, sufficient addresses must be reserved at each layer. When the network scale is expanded, the continuity of address allocation can be ensured to implement long-term network planning.
- **Efficiency:** When dividing a network into subnets, make full use of address resources to ensure that the number of subnets meets the requirement on the number of hosts.

Address planning practice

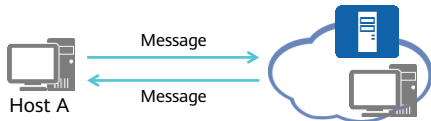


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- 4. ICMP**
5. Fundamentals and Configuration of DHCP

ICMP

- ICMP is used to transmit various error and control information between network devices.
- ICMP plays an important role in collecting various network information and diagnosing and troubleshooting network faults.



Ethernet header	IP header	ICMP message	Ethernet trailer
-----------------	-----------	--------------	------------------

Type	Code	Checksum
Message Body (variable length)		

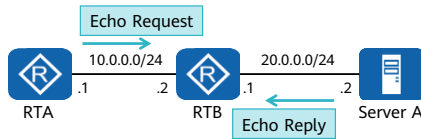
Type	Code	Description
0	0	Echo Reply
3	0	Net Unreachable
3	1	Host Unreachable
3	2	Protocol Unreachable
3	3	Port Unreachable
5	0	Redirect Datagram for the Network
8	0	Echo Request

- ICMP works at the network layer to forward IP packets more efficiently and increase the chance of successful packet exchange. ICMP allows hosts or devices to report errors during packet transmission.
- ICMP messages are encapsulated in IP packets. If the Protocol value in the IP header is 1, the used protocol is ICMP.
- The format of an ICMP message depends on the Type and Code fields. The Type field indicates the message type, and the Code field contains specific parameters of the message type.
- The Checksum field is used to check whether the message is complete.
- The Message Body field is 32 bits long and contains variable parameters. Generally, this field is not used and is set to 0.
 - In an ICMP Redirect message, this field specifies a gateway IP address. A host redirects packets to the specified gateway based on this address.
 - In an Echo Request message, this field contains the identifier and sequence number. The source end associates a received Echo Reply message with an Echo Request message it sends based on the two parameters. Especially when the source end sends multiple Echo Request messages to the destination end, the Echo Request and Echo Reply messages must be matched based on the identifiers and sequence numbers.
- For more information about ICMP, see RFC 792 at <https://www.rfc->

editor.org/rfc/rfc792.html.

ICMP Error Check

- ICMP Echo messages are used to diagnose the connectivity between the source and destination, and provide other information such as the round-trip time of packets.



Function: ping

The **ping** command is a command used on network devices and the Windows, Unix, and Linux platforms. It is a small and practical application based on ICMP. Ping is often used to test network reachability to a destination node.

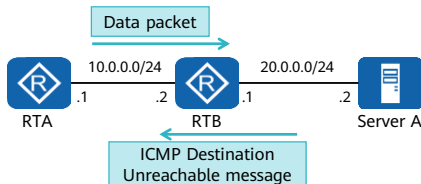
```
[RTA] ping 20.0.0.2
PING 20.0.0.2: 56 data bytes, press CTRL_C to break
Reply from 20.0.0.2: bytes=56 Sequence=1 ttl=254 time=70 ms
Reply from 20.0.0.2: bytes=56 Sequence=2 ttl=254 time=30 ms
Reply from 20.0.0.2: bytes=56 Sequence=3 ttl=254 time=30 ms
Reply from 20.0.0.2: bytes=56 Sequence=4 ttl=254 time=40 ms
Reply from 20.0.0.2: bytes=56 Sequence=5 ttl=254 time=30 ms
```

```
--- 20.0.0.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 30/40/70 ms
```

- A typical application of ICMP is ping. Ping is a common tool for checking network connectivity and collecting related information. In the **ping** command, you can specify different parameters, such as the ICMP message length, number of ICMP messages to be sent, and timeout period for waiting for a reply. The device constructs ICMP messages based on the parameters and sends them to perform a ping test.

ICMP Error Report

- ICMP defines various error messages for diagnosing network connectivity problems. Based on the error messages, the source device can determine the cause of a data transmission failure. For example, when a network device cannot access a target network, it automatically sends an ICMP Destination Unreachable message to the sender.



Function: tracer

Tracer traces the forwarding path of a packet hop by hop based on the TTL value in the packet header. Tracer is an effective method to check the packet loss and delay and to help administrators find routing loops on a network.

```
[RTA] tracer 20.0.0.2
```

```
traceroute to 20.0.0.2(20.0.0.2), max hops: 30 ,packet length: 40,press CTRL_C to break
```

1	10.0.0.2	80 ms	10 ms	10 ms
2	20.0.0.2	30 ms	30 ms	20 ms

- ICMP defines various error messages for diagnosing network connectivity problems. Based on the error messages, the source device can determine the cause of a data transmission failure.
 - If a loop occurs on a network, packets are looped on the network and the TTL times out. In this case, the network device sends a TTL timeout message to the sender.
 - If the destination is unreachable, the intermediate network device sends a Destination Unreachable message to the sender. Destination unreachable involves many scenarios. If the network device cannot find the destination network, it sends a destination network unreachable message. If the network device cannot find the destination host on the destination network, it sends a destination host unreachable message.
- Tracer is another typical application of ICMP. Tracer traces the forwarding path of a packet hop by hop based on the TTL value in the packet header. To trace the path to a specific destination address, the source end first sets the TTL value of the packet to 1. After the packet reaches the first node, the TTL times out. This node then sends a TTL timeout message carrying the timestamp to the source end. Then, the source end sets the TTL value of the packet to 2. After the packet reaches the second node, the TTL times out. This node also returns a TTL timeout message. The process repeats until the packet reaches the destination. In this way, the source end can trace each node that the packet passes through based on the information in the returned messages, and calculate the round-trip time

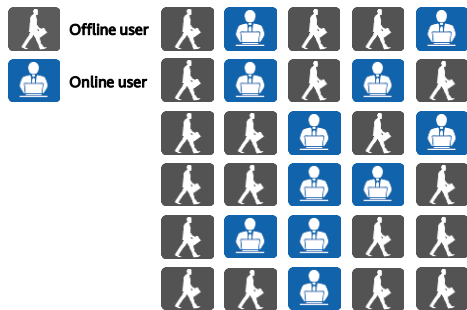
based on the timestamp information.

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 - Fundamentals of DHCP
 - Basic DHCP Configuration

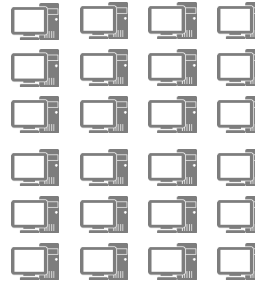
Defects of Static IP Address Configuration

Low utilization



- On an enterprise network, each user uses a fixed IP address, which results in low IP address utilization. Some IP addresses may remain unused for a long time.

Heavy workload

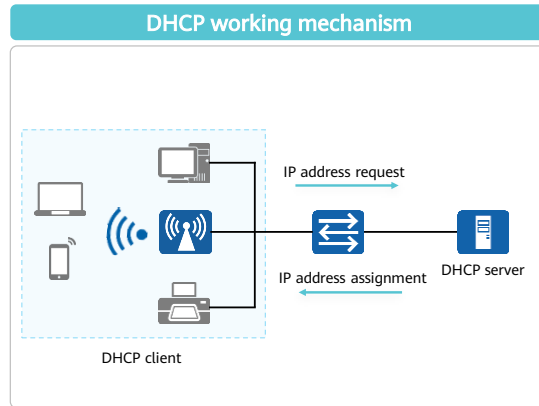


☐ Monday: Add 10 terminals.
☐ Tuesday: Add 20 terminals.
☐ ...

Daily work of the network administrator

- The network administrator needs to plan IP addresses in advance and assign them to individuals.
- The network administrator needs to perform configurations in a unified manner. The workload is heavy and the work is repetitive.

Basic Concepts of DHCP



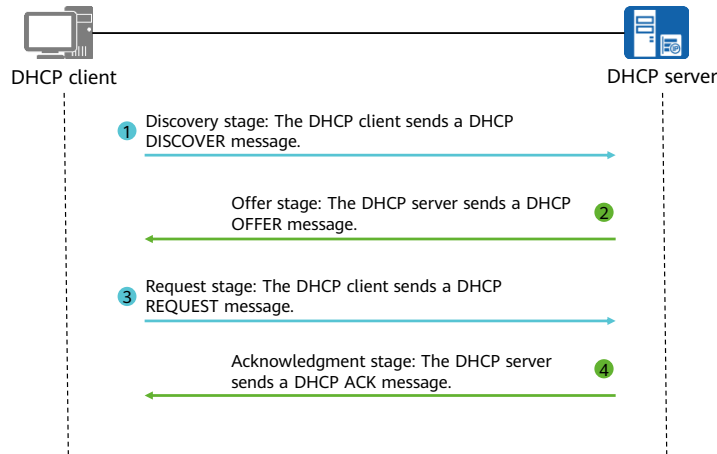
- To address the defects of static IP address configuration, the Dynamic Host Configuration Protocol (DHCP) is developed to dynamically and properly assign IP addresses.
- DHCP uses the client/server architecture.
- DHCP has the following advantages:
 - Reduced client configuration and maintenance costs: DHCP is easy to configure and deploy. For non-technical users, DHCP minimizes configuration-related operations on clients and reduces remote deployment and maintenance costs.
 - Centralized management: A DHCP server can manage the configurations of multiple network segments. When the configuration of a network segment changes, the administrator only needs to update the related configuration on the DHCP server.

DHCPv4 Message Types

Message Name	Description
DISCOVER	This message is the first message sent by a DHCP client to discover a DHCP server when the client accesses the network for the first time.
OFFER	A DHCP server sends this message in response to a DHCP DISCOVER message. A DHCP OFFER message carries various configuration information.
REQUEST	- After the client starts, it broadcasts a DHCP REQUEST message to respond to a DHCP OFFER message sent by a DHCP server. - After the client restarts, it broadcasts a DHCP REQUEST message to confirm the configuration information (including the assigned IP address). - After the client obtains an IP address, it unicasts or broadcasts a DHCP REQUEST message to renew the IP address lease.
ACK	A DHCP server sends this message to acknowledge a DHCP REQUEST message sent from a DHCP client. After receiving a DHCP ACK message, the DHCP client obtains the configuration information (including an IP address).
NAK	A DHCP server sends this message to reject a DHCP REQUEST message from a DHCP client. For example, if a DHCP server cannot find a matching lease record after receiving a DHCP REQUEST message, it sends a DHCP NAK message to notify the DHCP client that no IP address is available.
DECLINE	A DHCP client sends this message to notify a DHCP server that the assigned IP address conflicts with another IP address. The DHCP client then applies to the DHCP server for another IP address.
RELEASE	A DHCP client sends this message to release its IP address assigned by a DHCP server. After receiving a DHCP RELEASE message, the DHCP server can assign this IP address to another DHCP client.
INFORM	A DHCP client sends this message to a DHCP server to obtain other network configuration parameters, such as the gateway address and DNS server address, after it has obtained an IP address.

- A DHCP server and a DHCP client communicate with each other through DHCP messages. DHCP messages are transmitted using the User Datagram Protocol (UDP). A DHCP client uses source port number 68, and a DHCP server uses source port number 67.
- A DHCP message contains the Options field, which is used to store the control information and parameters allocated to a DHCP client.

Process for a DHCP Client to Obtain an IP Address for the First Time

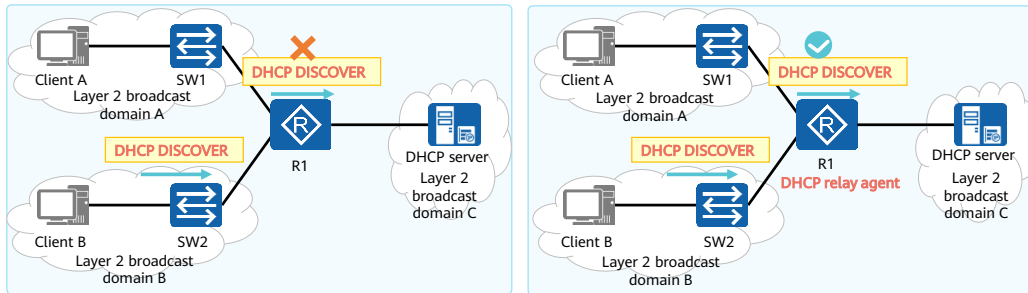


- Step 1: Discovery stage
 - A DHCP client that accesses the network for the first time does not know the IP address of the DHCP server. To obtain the IP address of the DHCP server, the DHCP client broadcasts a DHCP DISCOVER message with the destination IP address 255.255.255.255 to all devices on the same network segment.
- Step 2: Offer stage
 - All DHCP servers on the same network segment as the DHCP client receive the DHCP DISCOVER message. Each DHCP server selects an address pool on the same network segment as the IP address of the interface that receives the DHCP DISCOVER message, selects an available IP address from the address pool, and sends a DHCP OFFER message to the DHCP client.
 - In most cases, the leases of IP addresses are specified in an address pool. If the DHCP DISCOVER message sent by the DHCP client carries an expected lease, the DHCP server compares the expected lease with the specified lease and assigns the IP address with a shorter lease to the DHCP client.
- Step 3: Request stage
 - If multiple DHCP servers send DHCP OFFER messages to the DHCP client, the DHCP client usually accepts the first received DHCP OFFER message and broadcasts a DHCP REQUEST message, which contains the identifier of the DHCP server to be selected and the IP address of the client.
- Step 4: Acknowledgment stage
 - After receiving the DHCP REQUEST message from the DHCP client, the DHCP server sends a DHCP ACK message to the client, indicating that the IP

address requested in the DHCP REQUEST message is assigned to the client.

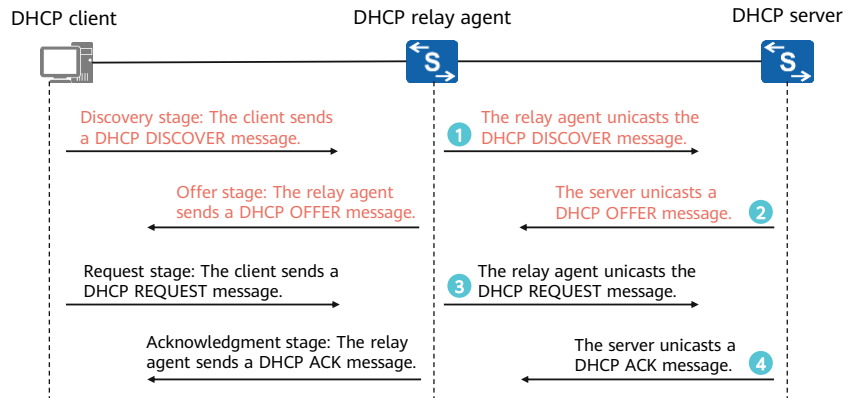
What Is a DHCP Relay Agent?

- A DHCP relay agent is used to solve the problem that the DHCP server and client are not in the same broadcast domain. It provides the relay function for DHCP broadcast packets. It can transparently transmit broadcast packets from a DHCP client to a DHCP server in another broadcast domain and transparently transmit response packets from a DHCP server to a DHCP client in another broadcast domain.



- As the network scale expands and more network devices are deployed, different users in an enterprise may be located on different network segments. In normal cases, a DHCP server cannot assign addresses to users on multiple network segments. If the DHCP server is still required to assign IP addresses to the users, DHCP messages need to be sent across network segments.

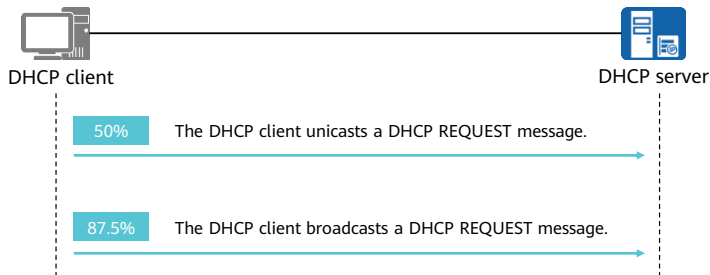
Working Principles of a DHCP Relay Agent



- The working principles of a DHCP client and a DHCP server when the DHCP client accesses a network for the first time in a scenario where a DHCP relay agent is deployed are the same as those in a scenario where no DHCP relay agent is deployed. The main difference is that the DHCP relay agent forwards DHCP messages between the DHCP server and DHCP client to ensure that the DHCP server and DHCP client can communicate with each other properly.
- The DHCP relay agent and DHCP server send DHCP messages to each other in unicast mode.

DHCP Lease Renewal

- IP addresses that are dynamically assigned by a DHCP server to DHCP clients have leases.
- When the lease of an IP address expires or the client goes offline and releases the IP address, the server reclaims the IP address, which can then be assigned to another client. This mechanism improves IP address utilization and prevents IP addresses from being occupied after clients go offline.
- If a DHCP client wants to continue using an IP address, the client needs to renew the lease of the IP address.



- When the lease reaches 50% of its validity period, the DHCP client unicasts a DHCP REQUEST message to the DHCP server to request lease renewal. If the DHCP client receives a DHCP ACK message, the IP address lease is successfully renewed (counted from 0). If the DHCP client receives a DHCP NAK message, the DHCP client must send a DHCP DISCOVER message to apply for a new IP address.
- If no response is received from the DHCP server when the lease reaches 87.5% of its validity period, the DHCP client broadcasts a DHCP REQUEST message to request lease renewal. If the DHCP client receives a DHCP ACK message, the IP address lease is successfully renewed (counted from 0). If the DHCP client receives a DHCP NAK message, the DHCP client must send a DHCP DISCOVER message to apply for a new IP address.
- If no response is received when the lease expires, the DHCP client stops using the current IP address and sends a DHCP DISCOVER message to apply for a new one.

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 - Fundamentals of DHCP
 - Basic DHCP Configuration

Basic DHCP Configuration (1)

1. Enable the DHCP function.

```
[HUAWEI] dhcp enable
```

The **dhcp enable** command is the prerequisite for configuring DHCP-related functions including the DHCP relay and DHCP server functions. These functions take effect only after the DHCP function is enabled using the **dhcp enable** command.

2. Enable the DHCP server function based on an interface.

```
[HUAWEI-GE0/0/1] dhcp select interface
```

3. Enable the DHCP server function globally.

```
[HUAWEI-GE0/0/1] dhcp select global
```

4. Create an address pool.

```
[HUAWEI] ip pool ip-pool-name
```

```
[HUAWEI-ip-pool-ip-pool-name] network ip-address [ mask { mask | mask-length } ]
```

Configure the range of IP addresses that can be dynamically allocated in the global address pool.

- An interface address pool is created by configuring IPv4 addresses on an interface connecting a DHCPv4 server to a DHCPv4 client. IPv4 addresses in the interface address pool are on the same network segment as the interface's address, and can be allocated only to clients connected to the interface. An interface address pool applies only to the scenario where a DHCPv4 server and client are on the same network segment.
- A global address pool is an address pool created in the system view for a specified network segment. Addresses in the global address pool can be assigned to clients connected to all interfaces of the device. If a DHCPv4 server and client are located on different network segments, a DHCPv4 relay agent needs to be deployed.

Basic DHCP Configuration (2)

5. Configure an address lease for the address pool.

```
[HUAWEI-ip-pool-ip-pool-name] lease { day day [ hour hour [ minute minute ] ] | unlimited }
```

By default, the lease of IP addresses is one day.

6. Configure the egress gateway address for DHCP clients.

```
[HUAWEI-ip-pool-ip-pool-name] gateway-list ip-address &<1-8>
```

7. Assign a fixed IP address to a DHCP client.

```
[HUAWEI-ip-pool-ip-pool-name] static-bind ip-address ip-address mac-address mac-address [ option-template template-name | description description ]
```

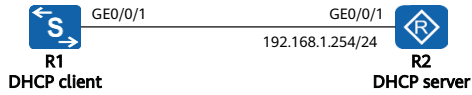
8. Enable the DHCP relay function.

```
[HUAWEI-GE0/0/1] dhcp select relay
```

```
[HUAWEI-GE0/0/1] dhcp relay server-ip ip-address
```

Configure the IP address of the DHCP server on the DHCP relay agent.

Configuration Example



- A network engineer requires that R1 obtain an IP address through DHCP.
- The interface-based DHCP server function is enabled on R2.

The configuration of R2 is as follows:

```
[R2] dhcp enable
[R2] interface GE 0/0/1
[R2-GE0/0/1] undo portswitch
[R2-GE0/0/1] ip address 192.168.1.254 24
[R2-GE0/0/1] dhcp select interface
```

The configuration of R1 is as follows:

```
[R1] dhcp enable
[R1] interface GE 0/0/1
[R1-GE0/0/1] undo portswitch
[R1-GE0/0/1] ip address dhcp-alloc
```

Check the IP address obtained by R1.

```
[R1] display ip interface GE 0/0/1
.....
Internet Address is allocated by DHCP, 192.168.1.105/24
.....
```

Quiz

1. (Single-answer question) Which of the following is the class of the IP address 201.222.5.64? ()
 - A. Class A
 - B. Class B
 - C. Class C
 - D. Class D
2. (Multiple-answer question) A company has been assigned a Class C network address segment 192.168.20.0/24. A department in the company has 40 hosts. Which of the following subnets can be allocated to the department? ()
 - A. 192.168.20.64/26
 - B. 192.168.20.64/27
 - C. 192.168.20.128/26
 - D. 192.168.20.184/29

1. C

2. AC

Quiz

3. (Single-answer question) Which of the following is the correct sequence of interactions when a DHCP client obtains an IP address for the first time? ()
- A. DISCOVER - REQUEST - OFFER - ACK
 - B. DISCOVER - OFFER - REQUEST - ACK
 - C. REQUEST - DISCOVER - ACK - OFFER
 - D. OFFER - DISCOVER - ACK - REQUEST
4. (Multiple-answer question) In which of the following situations does a DHCP client send a DHCP REQUEST message? ()
- A. Confirming the selected server when applying for an IP address for the first time
 - B. Requesting lease renewal when the lease reaches 50% of the validity period
 - C. Proactively releasing the assigned IP address
 - D. Requesting lease renewal when the lease reaches 87.5% of the validity period

3. B

4. ABD

Summary

- An IPv4 address is a 32-bit binary number, expressed in dotted decimal notation, which helps you memorize and identify it.
- An IPv4 packet consists of a header and a data field. The first part of the header has a fixed length of 20 bytes and is mandatory for all IPv4 packets. The Options field following the fixed part of the header has a variable length.
- Private addresses are internal network or host addresses that can be used only on an internal network and cannot be used on the Internet.
- Dividing the network portion of an IP address into several subnets solves the problem of low utilization of the IP address space.
- Compared with manual IP address configuration, DHCP prevents manual configuration errors and helps network administrators perform unified maintenance and management.

Thank you.

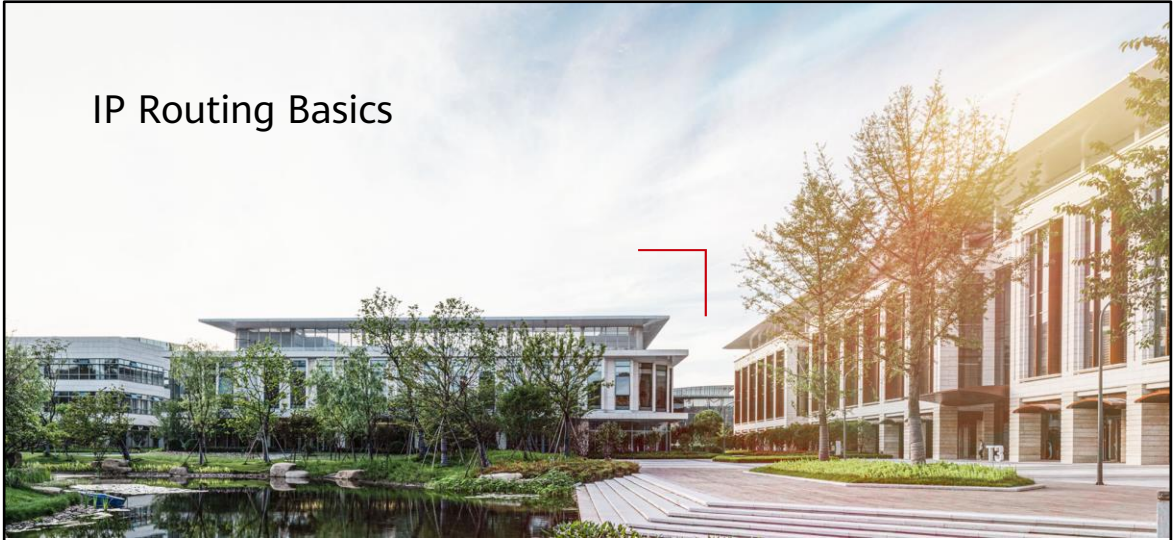
把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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IP Routing Basics



Foreword

- Data communication networks typically contain multiple IP subnets. Layer 3 devices are required to enable communication between these subnets. With routing capabilities, these devices can forward data across subnets.
- Routing is a fundamental function of data communication networks. Routing information defines the path used to forward packets, and the routing process itself is the packet forwarding process.
- With advances in hardware and software, switches have also gained routing capabilities.
- This course introduces the fundamental concepts and forwarding principles of routing, along with methods for implementing inter-VLAN communication.

Objectives

- Upon completion of this course, you will be able to:
 - Describe the fundamental working principles of routers.
 - Explain how routers select optimal routes.
 - Identify and interpret the contents of a routing table.
 - Describe the process of route forwarding.
 - Understand and apply inter-VLAN communication methods.

Contents

1. Overview of IP Routing

- **Basic Routing Concepts**

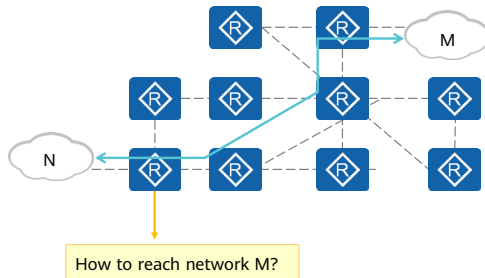
- Routing Table Details
- Route Forwarding

2. Direct Routes

3. Static Routes

4. Advanced Routing Features

Background: Inter-Subnet Communication

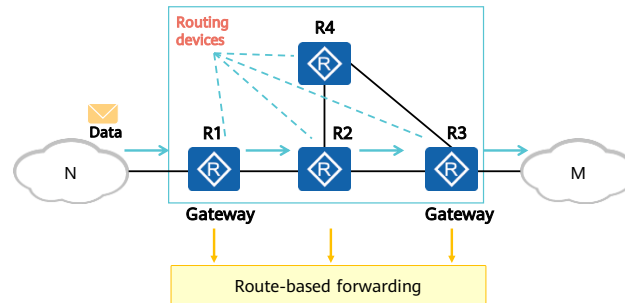


- An IP address uniquely identifies a node on a network. Each IP address belongs to a specific subnet, and different subnets may be located in different network domains.
- To ensure effective IP addressing, connectivity is required between these subnets across different domains.

- A unique network node is identified by its specific IP address. Each IP address belongs to a specific subnet. These subnets can be globally distributed, collectively forming a vast, interconnected network.
- To enable inter-subnet communication, network devices must forward IP packets from a source subnet to the target destination subnet.

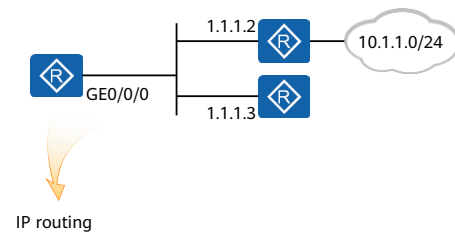
Routing

- The term "routing" in computer networking refers to path selection. It describes the process of determining and allocating paths for packet forwarding.
- A routing device is a network device that forwards packets to the destination subnet based on routing information. The most common routing device is a router.
- Routing devices maintain a routing table, which stores routing information.



- In route-based forwarding, a gateway or intermediate router selects the appropriate path based on the destination IP address of a received IP packet, and forwards the packet to the next router. The last-hop router along the path performs Layer 2 addressing, and forwards the packet to the destination host.
- Intermediate nodes rely on the routing table to determine the optimal path for packet forwarding.
- A routing entry specifies an outbound interface and next hop used to forward IP packets to the corresponding next-hop device.

Routing Information



IP routing

Destination Network/Mask	Outbound Interface	Next Hop
10.1.1.0/24	GE0/0/0	1.1.1.2

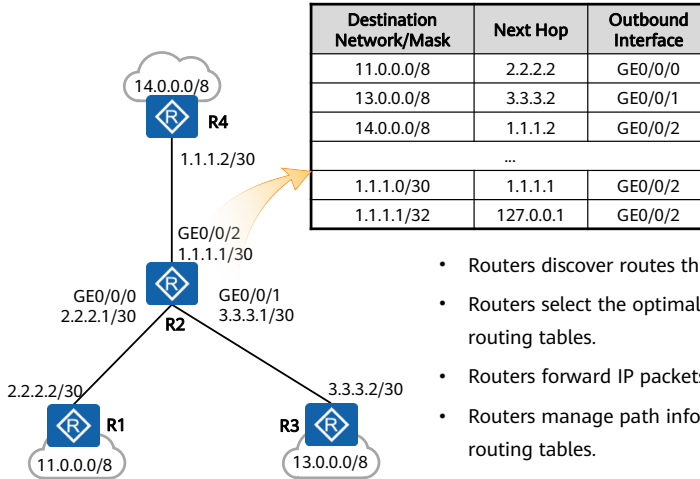
- A routing entry contains the following information:

- Destination network/Mask: identifies the destination subnet.
- Outbound interface: indicates the interface through which a data packet is sent out of the local router.
- Next hop: specifies the IP address of the interface on the next-hop router along the forwarding path.

The preceding information defines the destination subnet and the path used for IP packet forwarding.

- Using the information in a routing entry, a router forwards IP packets to the destination network along the defined path.
- The destination network and mask identify the destination address of an IP packet. Once an IP packet matches a routing entry, the router determines the forwarding path based on the outbound interface and next hop specified in this entry.
- An outbound interface alone is insufficient to identify the next-hop device for forwarding IP packets; the specific next-hop address must also be specified.

Routing Table

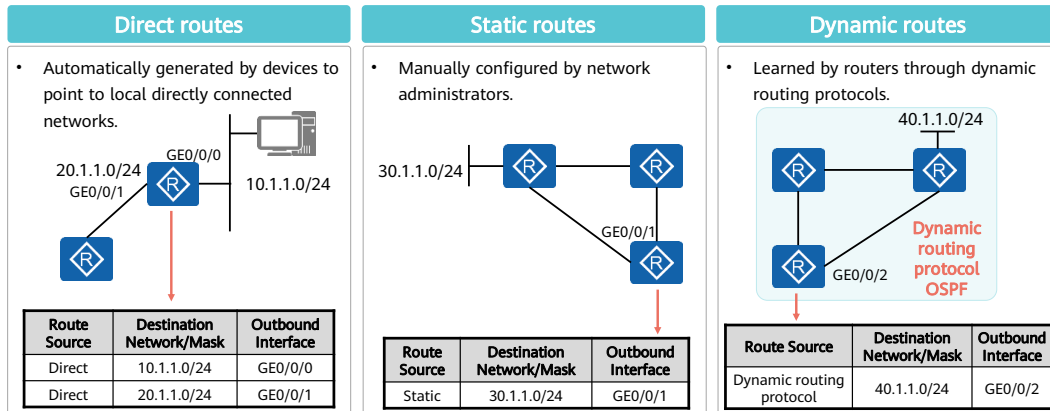


- Routers discover routes through multiple methods.
- Routers select the optimal route and install it in their routing tables.
- Routers forward IP packets based on the routing table.
- Routers manage path information by maintaining their routing tables.

- Routers forward packets based on the routing table.
- A routing table consists of detailed, individual routing entries.
- A routing table stores only the optimal routes, not all discovered routes.
- Routers manage routing information by maintaining and managing the routing entries in the routing table.

Methods of Obtaining Routing Information

- Routers forward packets according to their routing tables. To enable route forwarding, routers must first obtain routing information. The following describes the common methods of obtaining routing information.



- Direct routes are destined for subnets directly attached to router interfaces. They are automatically generated by devices.
- Static routes are manually configured by network administrators.
- Dynamic routes are learned by routers through dynamic routing protocols, such as Open Shortest Path First (OSPF).
- OSPF is a widely used dynamic routing protocol.

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1. Overview of IP Routing

- Basic Routing Concepts
- **Routing Table Details**
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4. Advanced Routing Features

Checking an IP Routing Table

```
<Huawei> display ip routing-table
Route Flags: R - relay, D - download to fib
```

Routing Tables: Public
Destinations : 6 Routes : 6

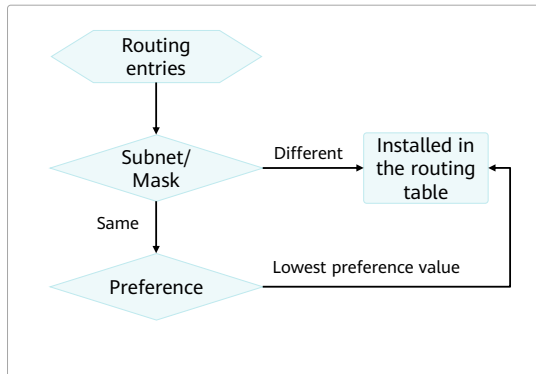
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
1.1.1.1/32	Static	60	0	D	0.0.0.0	NULL0
2.2.2.2/32	Static	60	0	D	100.0.0.2	Vlanif100
100.0.0.0/24	Direct	0	0	D	100.0.0.1	Vlanif100
100.0.0.1/32	Direct	0	0	D	127.0.0.1	Vlanif100
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Diagram labels with arrows pointing to the table columns:

- Destination network address/Mask length (points to Destination/Mask)
- Protocol type (points to Proto)
- Preference (points to Pre)
- Cost (metric) value (points to Cost)
- Flags (points to Flags)
- Next-hop address (points to NextHop)
- Outbound interface (points to Interface)

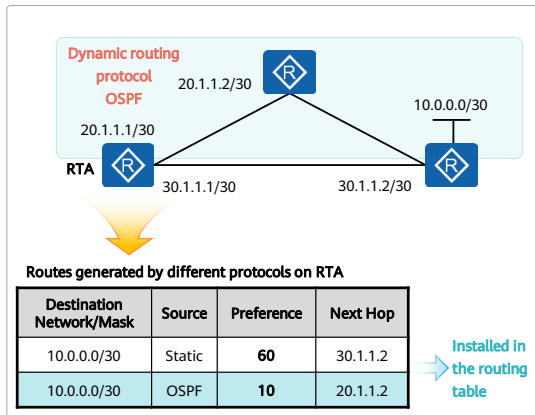
- **Destination/Mask:** specifies the destination network address and mask length of the route.
- **Proto:** specifies the protocol through which a router learns the route.
- **Pre:** specifies the preference of the route. Multiple routes to the same destination may exist, with different next hops and outbound interfaces. These routes can be discovered by different routing protocols or manually configured. The router selects the route with the highest preference (the lowest numerical value) as the optimal route.
- **Cost:** defines the cost of the route. When multiple routes to the same destination have the same preference, the route with the lowest cost is selected as the optimal route.
- Preference compares routes across different routing protocols, while Cost compares routes within the same routing protocol. Cost is also known as the metric.
- **NextHop:** specifies the IP address of the next-hop device's interface to which packets need to be forwarded to reach the destination network.
- **Interface:** specifies the local outbound interface through which the router sends packets toward the destination.

Route Preference



- When a router learns multiple routes to the same destination network (with identical destination addresses and masks) from different sources, it compares their preferences. The route with the lowest preference value—representing the highest priority—is selected and installed into the routing table.

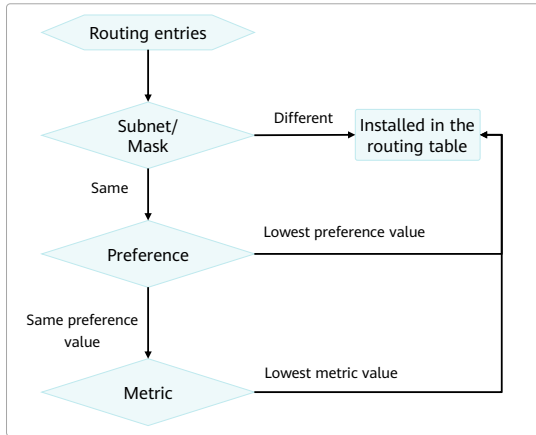
Example of Route Preference Comparison



- RTA discovers two routes to 10.0.0.0/30: a dynamically learned OSPF route and a manually configured static route. In this case, RTA compares the preferences of the two routes and selects the one with the lower preference value.
- Each routing protocol has a specific preference.
- OSPF has a higher priority than static routes, so the OSPF route is installed into the routing table.

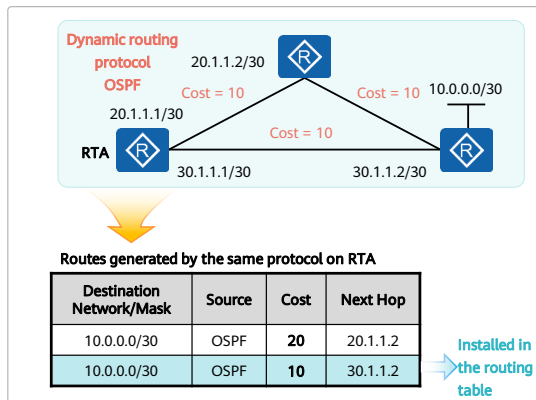
- Direct routes have a preference value of 0. OSPF internal routes have a default preference value of 10. Static routes have a default preference value of 60. OSPF external routes have a default preference value of 150.

Metric



- When a router discovers multiple routes to the same destination network through the same routing protocol (resulting in identical preferences), it selects the optimal route based on the metrics of these routes.
- The metric of a route represents the cost of reaching the destination address.
- Common metrics include the hop count, bandwidth, and delay.
- Routes with lower metrics are preferred, and the one with the lowest metric value is installed into the routing table.
- The metric is also known as the cost.

Example of Metric Comparison



- RTA learns two OSPF routes to the destination 10.0.0.0/30. Since both routes originate from the same protocol and share the same preference, RTA proceeds to compare the metrics of the two routes.
- The two routes have different metric values. The OSPF route with the next hop 30.1.1.2 has a lower metric, so it is installed into the routing table.

- When multiple routes to the same destination discovered by a single protocol have identical costs, they are considered equal-cost routes. These routes can be used to achieve load balancing.

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1. Overview of IP Routing

- Basic Routing Concepts
- Routing Table Details
- Route Forwarding

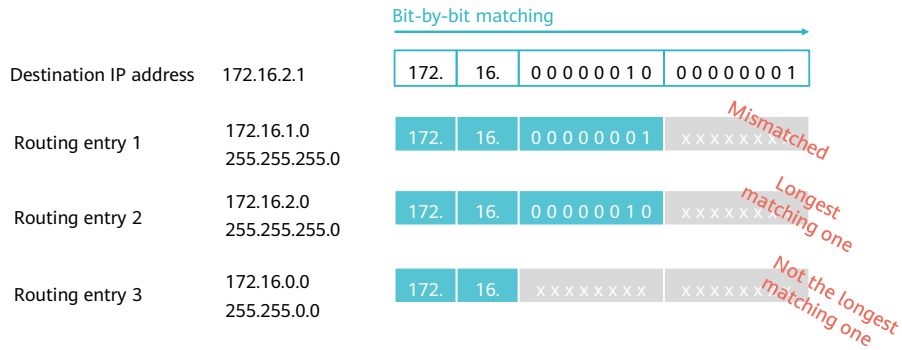
2. Direct Routes

3. Static Routes

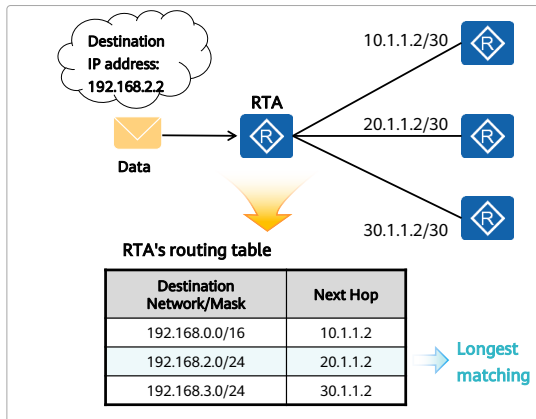
4. Advanced Routing Features

Longest Prefix Match

- When a router receives an IP packet, it compares the destination IP address against all entries in its local routing table bit-by-bit. The packet is forwarded according to the entry with the longest matching prefix. This process is known as the longest prefix match mechanism.

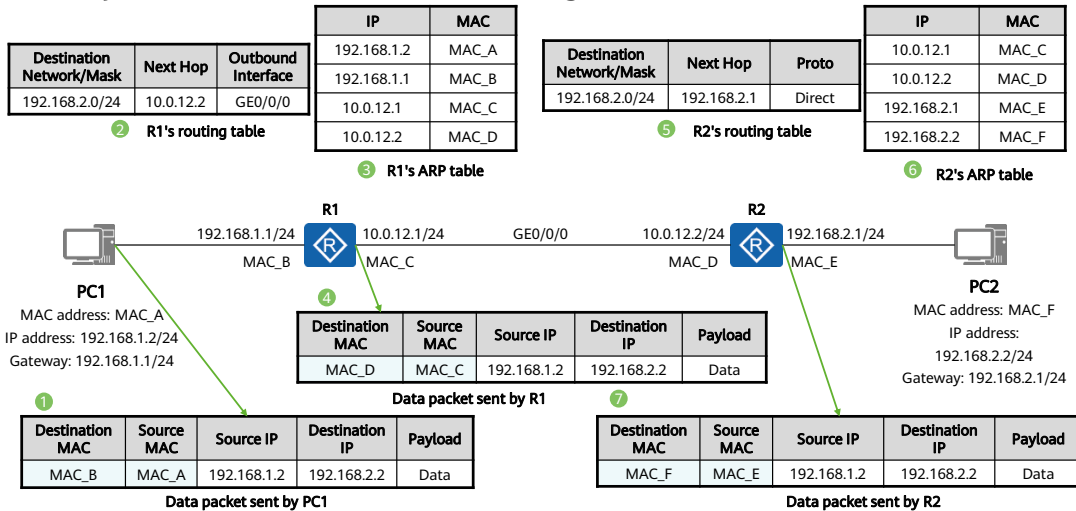


Example of Longest Prefix Match



- In RTA's routing table, two routes to 192.168.2.2 exist: one with a 16-bit mask and the other with a 24-bit mask. According to the longest prefix match rule, the route with the 24-bit mask is preferred and used to forward packets destined for 192.168.2.2.

Example of the Data Forwarding Process: PC1 -> PC2



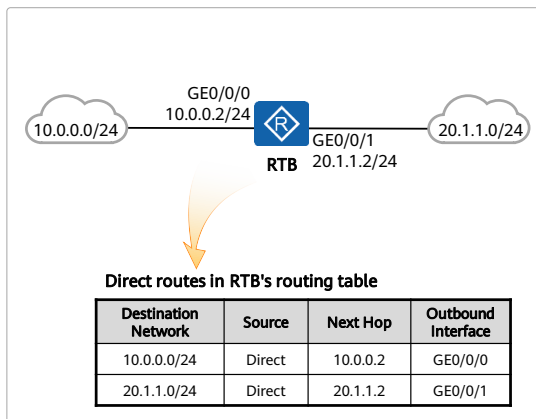
- In earlier sessions, we have explained how devices in the same subnet use ARP to establish IP-to-MAC mappings. This process will not be repeated here. Assume that each device already has an ARP table for other devices in the same subnet.
- The process of PC1 sending a packet to PC2 is as follows:
 - Since the destination IP address is on a different subnet from PC1, PC1 sends the data packet to the gateway. PC1 encapsulates the data packet with source MAC address MAC_A, destination MAC address MAC_B, source IP address 192.168.1.2, and destination IP address 192.168.2.2, and sends the packet through its network interface.
 - R1 receives the packet. After decapsulation, it finds that the destination MAC address matches its interface. R1 then decapsulates the network layer, and finds that the destination IP address is 192.168.2.2. R1 checks its routing table and identifies the next hop as 10.0.12.2.
 - R1 checks its ARP table for the next-hop address, and finds that 10.0.12.2 corresponds to MAC_D.
 - R1 re-encapsulates the data link layer header with source MAC address MAC_C and destination MAC address MAC_D, while the layers above the network layer remain unchanged.
 - R2 receives the packet. After decapsulation, it finds that the destination MAC address matches its interface. R2 then decapsulates the network layer, and finds that the destination IP address is 192.168.2.2, which is a directly connected network.

6. R2 queries its ARP table and finds that 192.168.2.2 corresponds to MAC_F.
 7. R2 re-encapsulates the data link layer header with source MAC address MAC_E and destination MAC address MAC_F, while the layers above the network layer remain unchanged.
- The above process ensures that PC1 can successfully send packets to PC2. However, PC2 may not be able to send packets back to PC1, because routing is inherently unidirectional. Successful return traffic requires all devices along the reverse path to have valid routes back to the source network.

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- 2. Direct Routes**
3. Static Routes
4. Advanced Routing Features

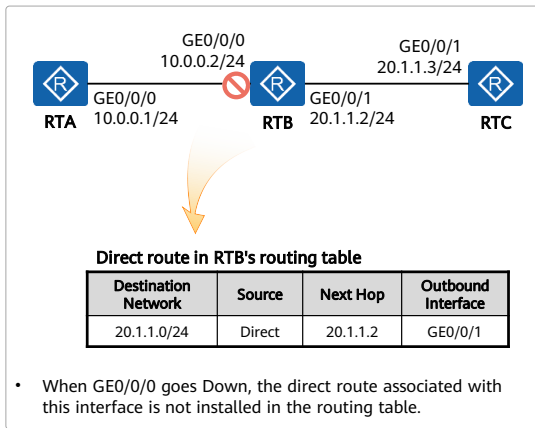
Direct Routes (1)



- A direct route is automatically generated by a device and points to a local directly connected network.
- When a router is the last-hop router, IP packets to be forwarded will match a direct route, and the router forwards the packets directly to the destination host.
- When a direct route is used for packet forwarding, the destination IP address of the packet and the IP address of the router's outbound interface are on the same subnet.

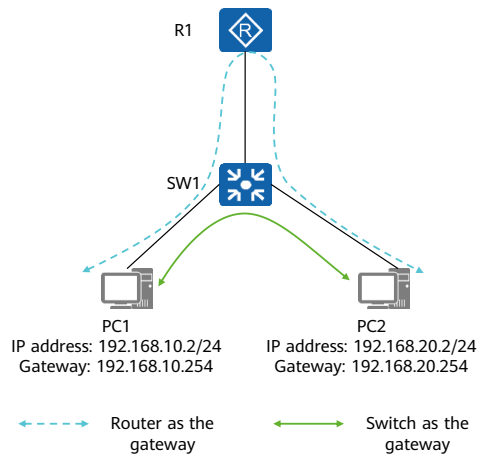
- The next-hop address of a direct route is not the interface address of another device. Because the destination subnet is directly connected to the local interface, this interface acts as the final hop in the path. Therefore, the next-hop address in the routing table is represented by the local interface's own address.
- When forwarding a packet based on a direct route, the router performs a local ARP lookup and forwards the packet directly to the destination host. In this case, the router serves as the last-hop router.

Direct Routes (2)



- Not all direct routes generated by interfaces are installed in the routing table. Only the direct routes on the interfaces whose physical status and protocol status are both Up are installed.

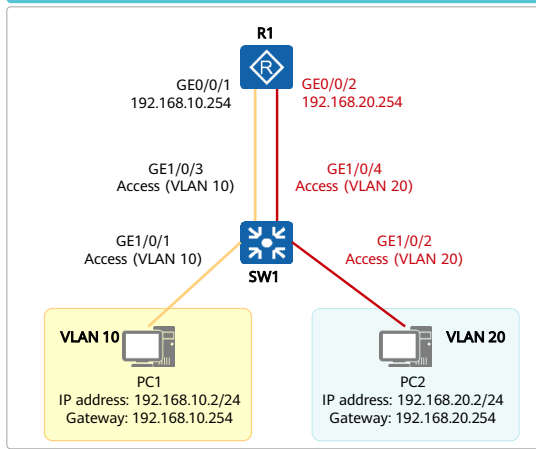
Application of Direct Routes



- As shown in the diagram, gateways for multiple subnets are typically deployed on a single device (for example, R1 or SW1) to facilitate centralized terminal management.
- When terminals in different subnets communicate with each other, the gateway only needs to query direct routes since it is directly connected to the terminal networks. This approach greatly reduces the workload of network administrators.
- When a router acts as the gateway, two methods are available for inter-subnet communication:
 - Using physical interfaces
 - Using sub-interfaces
- When a switch serves as the gateway, terminal connectivity is established through VLANIF interfaces.

Using Physical Interfaces on a Router

Physical connection diagram

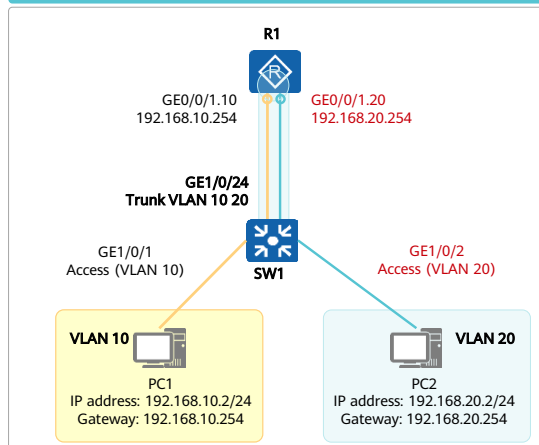


- A Layer 3 interface on a router functions as a gateway to forward traffic from the local subnet to other subnets.
- A router's Layer 3 interface cannot process data frames with VLAN tags. To enable communication, the switch interface connected to the router needs to be configured as an access interface.
- A router's physical interface serves as the gateway for a VLAN. Therefore, each VLAN requires one dedicated physical interface.
- Since routers, as Layer 3 forwarding devices, typically have a limited number of interfaces, this solution offers poor scalability.

- VLANs are configured on the Layer 2 switch, with each VLAN connected to the router through a dedicated switch interface.
- The router provides two physical interfaces as default gateways for PCs in VLAN 10 and VLAN 20, respectively. Inter-VLAN communication is enabled through these physical router interfaces.

Using Sub-interfaces on a Router

Physical connection diagram

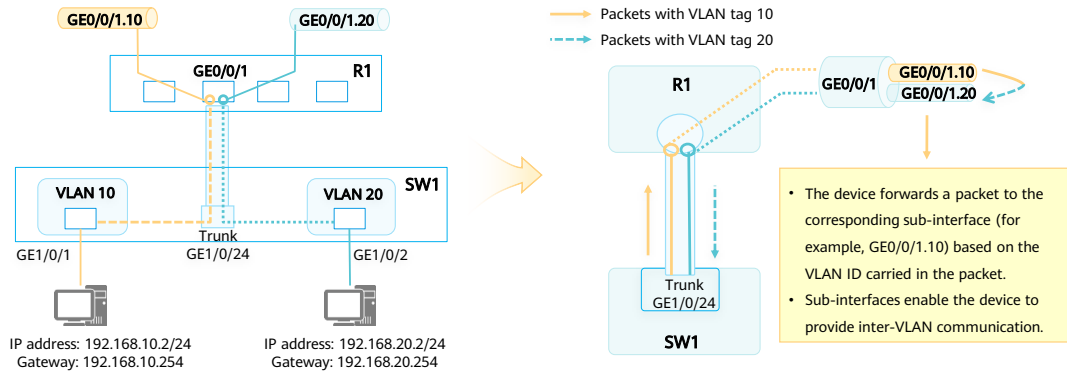


- A sub-interface is a logical interface created based on an Ethernet interface of a router, identified by the physical interface ID plus a sub-interface ID. Like a physical interface, a sub-interface supports Layer 3 forwarding.
- Unlike a physical interface, a sub-interface can terminate VLAN-tagged data frames.
- Multiple sub-interfaces can be created on a single physical interface. By connecting this physical interface to the trunk interface of a switch, the router enables Layer 3 forwarding for multiple VLANs through one physical interface.

- Connect R1 to SW1 through the physical interface GE0/0/1, and create two sub-interfaces (GE0/0/1.10 and GE0/0/1.20) on the physical interface. Configure the two sub-interfaces as the default gateways for VLAN 10 and VLAN 20, respectively.
- By default, a Layer 3 sub-interface does not accept VLAN-tagged packets and will discard them as invalid. To ensure proper communication, configure VLAN termination on sub-interfaces to remove VLAN tags.

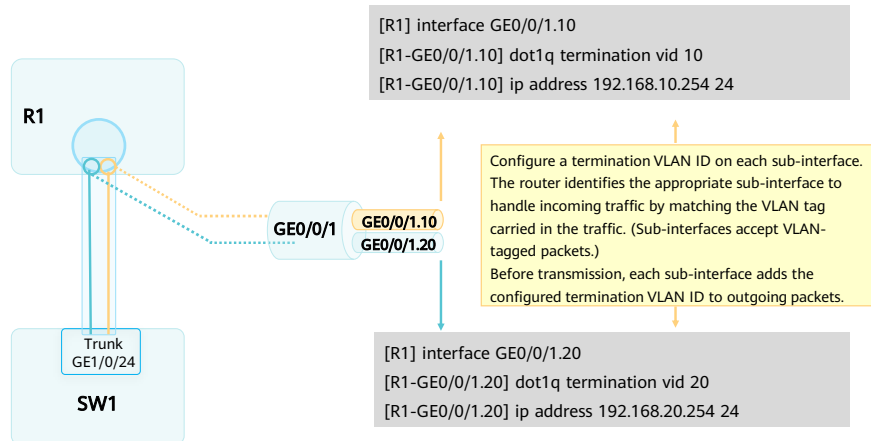
Packet Processing on a Sub-interface

- The switch interface connected to a router needs to be configured as a trunk interface. The router forwards the received packets to the corresponding sub-interface based on the VLAN tags carried in the packets.



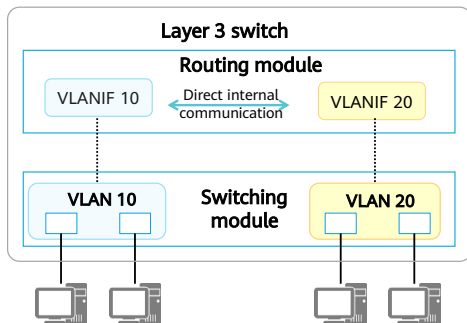
- VLAN termination on a sub-interface involves the following actions:
 - Removes VLAN tags from incoming packets for Layer 3 forwarding or other processing.
 - Adds VLAN tags to outgoing packets prior to transmission.

Sub-interface Configuration Example



- The **interface** *interface-type interface-number.sub-interface number* command creates a sub-interface. *sub-interface number* specifies the logical interface channel of a physical interface. For ease of management, a sub-interface ID is usually set to match the VLAN ID to be terminated.
- The **dot1q termination vid** command specifies the VLAN ID for single-layer dot1q termination on a sub-interface. By default, no VLAN ID is configured for single-layer dot1q termination on a sub-interface.

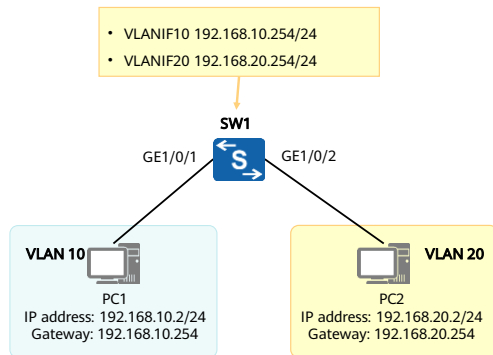
Layer 3 Switches and VLANIF Interfaces



- A Layer 2 switch supports only Layer 2 switching.
- In addition to Layer 2 switching, a Layer 3 switch supports route forwarding through Layer 3 interfaces such as VLANIF interfaces.
- A VLANIF interface is a Layer 3 logical interface that can remove and add VLAN tags in packets, enabling communication between devices across VLANs.
- A VLANIF interface number corresponds to a VLAN ID. For example, VLANIF 10 corresponds to VLAN 10.

- A Layer 2 interface cannot be configured with an IP address, whereas a Layer 3 interface can.
- Typical Layer 2 interfaces are physical interfaces on Layer 2 switches (switches with only the Layer 2 switching capability). On most Layer 3 switches (switches with both Layer 2 and Layer 3 switching capabilities), physical interfaces function as Layer 2 interfaces by default.
- Typical Layer 3 interfaces are router interfaces.
- In addition to physical Layer 3 interfaces, there are logical Layer 3 interfaces, such as VLANIF interfaces on switches or sub-interfaces on routers (for example, GE0/0/1.10).
- Layer 2 interfaces do not isolate broadcast domains. When a Layer 2 interface receives a broadcast frame, it floods the frame. Layer 3 interfaces isolate broadcast domains. When a Layer 3 interface receives a broadcast frame, it terminates the frame instead of flooding it.

VLANIF Interface Configuration Example



- Configuration requirements:
There are two PCs, one in VLAN 10 and the other in VLAN 20. Enable communication between the two PCs through a Layer 3 switch.

Basic configuration:

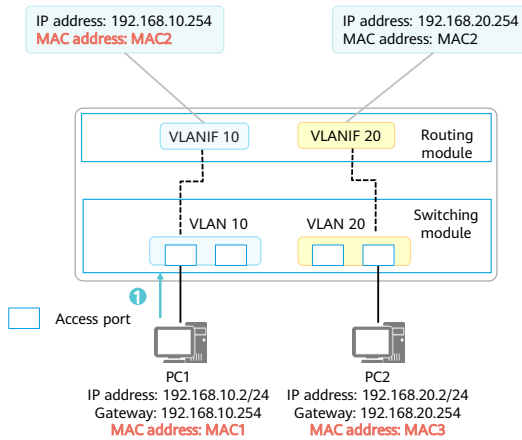
```
[SW1]vlan batch 10 20
[SW1] interface GE 1/0/1
[SW1-GE1/0/1] port link-type access
[SW1-GE1/0/1] port default vlan 10
[SW1-GE1/0/1] quit
[SW1] interface GE 1/0/2
[SW1-GE1/0/2] port link-type access
[SW1-GE1/0/2] port default vlan 20
```

VLANIF interface configuration:

```
[SW1] interface Vlanif 10
[SW1-Vlanif10] ip address 192.168.10.254 24
[SW1-Vlanif10] quit
[SW1] interface Vlanif 20
[SW1-Vlanif20] ip address 192.168.20.254 24
```

- The **interface vlanif *vlan-id*** command creates a VLANIF interface and displays its view. *vlan-id* specifies the ID of the VLAN associated with the VLANIF interface. The IP address of a VLANIF interface is used as the gateway address for the host in the corresponding VLAN. Therefore, it must be in the same subnet as the host's IP address.

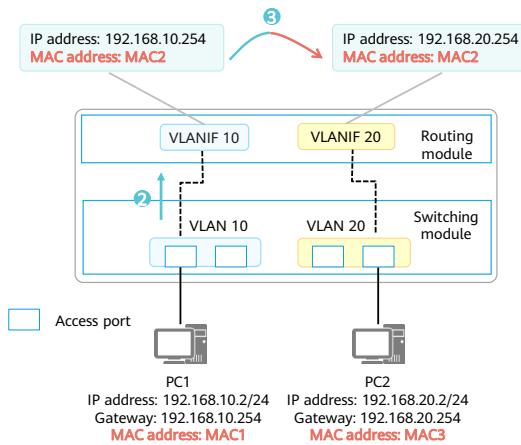
VLANIF Forwarding Process (1)



1. PC1 performs a calculation using its local IP address, local subnet mask, and the destination IP address. Upon identifying that PC2 resides on a different subnet, PC1 concludes that the communication requires Layer 3 forwarding, and sends the traffic destined for PC2 to the gateway. In the data frame sent by PC1, the source MAC address is MAC1, and the destination MAC address is MAC2.

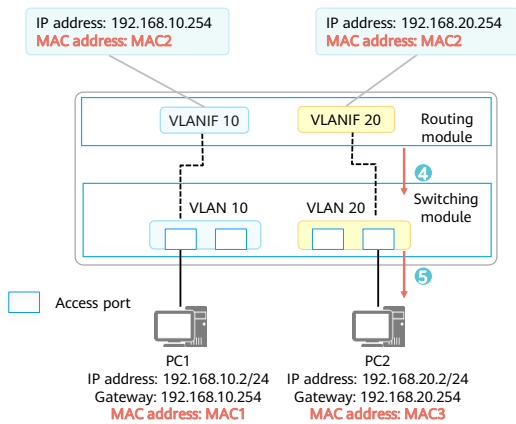
- Assume that the corresponding ARP entries already exist on the PCs and Layer 3 switch.

VLANIF Forwarding Process (2)



2. Upon receiving the packet from PC1, the switch performs decapsulation, and finds that the destination MAC address is the MAC address of VLANIF 10. It then hands over the packet to the routing module for further processing.
3. The routing module identifies the destination IP address as 192.168.20.2, requiring Layer 3 forwarding. After performing a routing table lookup, the routing module locates the direct route associated with VLANIF 20.

VLANIF Forwarding Process (3)

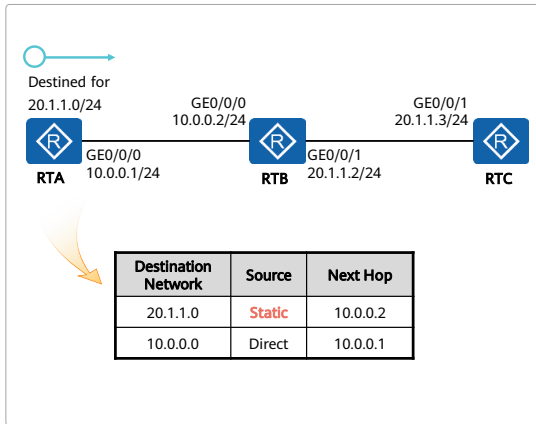


4. The matched route is a direct route, indicating that the packet has reached the last hop. Therefore, the switch looks up 192.168.20.2 in its ARP table to obtain the corresponding MAC address. This MAC address is then handed over to the switching module for data frame re-encapsulation.
5. The switching module checks the MAC address table to identify the outbound interface and determine whether a VLAN tag is required. The resulting data frame sent by the switching module carries source MAC address MAC2, destination MAC address MAC3, and no VLAN tag.

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Application Scenarios of Static Routes



- Static routes are manually configured by network administrators. They are easy to configure and impose minimal system overhead, making them ideal for small, stable network topologies.
- However, static routes cannot automatically adapt to topology changes and therefore require manual updates.
- For example, packets destined for 20.1.1.0/24 do not match any route in RTA's routing table when only direct routes are present. In this case, a static route can be manually configured on RTA to forward these packets to the next hop 10.0.0.2.

Configuring Static Routes

1. Configure a static route with the next-hop IP address specified.

```
[Huawei] ip route-static ip-address { mask | mask-length } nexthop-address
```

2. Configure a static route with the outbound interface specified.

```
[Huawei] ip route-static ip-address { mask | mask-length } interface-type interface-number
```

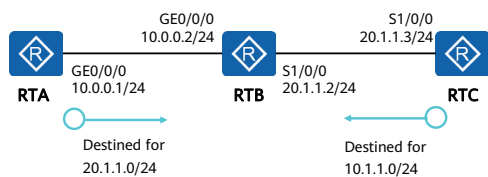
3. Configure a static route with both the outbound interface and next-hop IP address specified.

```
[Huawei] ip route-static ip-address { mask | mask-length } interface-type interface-number [ nexthop-address ]
```

When creating a static route, you can specify both the outbound interface and next-hop IP address. Alternatively, depending on the type of outbound interface, you can specify only one of them:

- Point-to-point (P2P) interfaces (for example, serial interfaces): Only the outbound interface needs to be specified.
- Broadcast interfaces (for example, Ethernet interfaces) and virtual template (VT) interfaces: The next-hop IP address must be specified.

Configuration Example



- Configure static routes on RTA and RTC to enable communication between the 10.0.0.0/24 and 20.1.1.0/24 networks.
- Because packets are forwarded hop by hop, every router along the path from the source to the destination must have a route to the destination network.
- As communication is bidirectional, static routes must be configured for both forward and return traffic.

RTA configuration:

```
[RTA] ip route-static 20.1.1.0 255.255.255.0 10.0.0.2
```

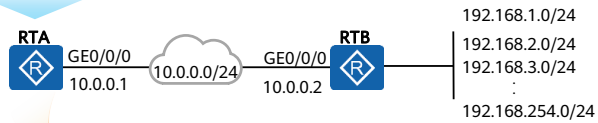
RTC configuration:

```
[RTC] ip route-static 10.0.0.0 255.255.255.0 S1/0/0
```

Default Routes

- A default route is used only when no matching entry is found in the routing table.
- A default route in the routing table is denoted by 0.0.0.0/0.

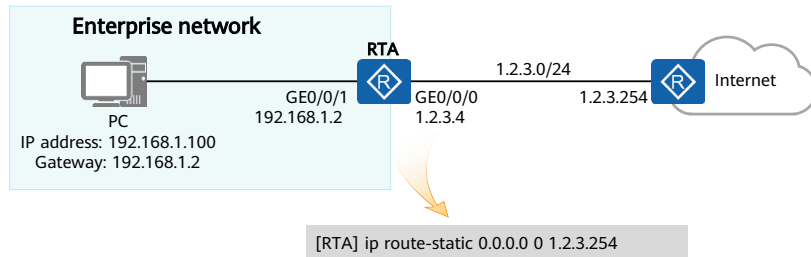
RTA forwards packets destined for non-local networks to the next-hop address 10.0.0.2.



```
[RTA] ip route-static 0.0.0.0 0 10.0.0.2
```

Application Scenarios of Default Routes

- Default routes are typically used at the egress of an enterprise network. For example, you can configure a default route on the egress device to forward IP packets destined for any address on the Internet.

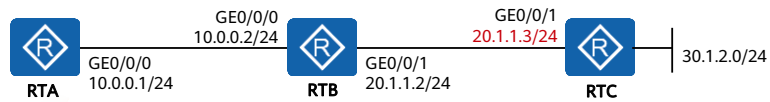


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Route Recursion (1)

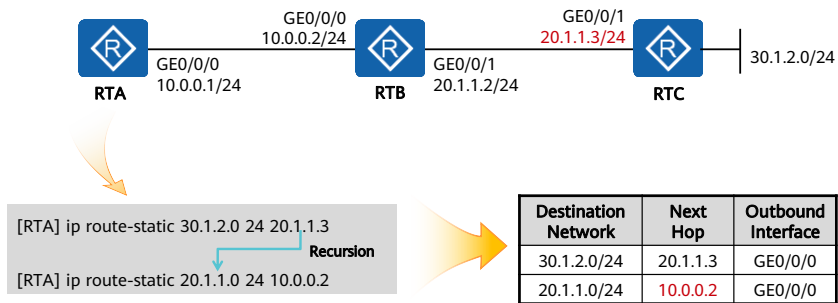
- A route can guide forwarding only when it has a directly connected next hop. However, a generated route may point to a non-local next hop. The process of resolving a directly connected next hop and its corresponding outbound interface is known as route recursion.



[RTA] ip route-static 30.1.2.0 24 20.1.1.3

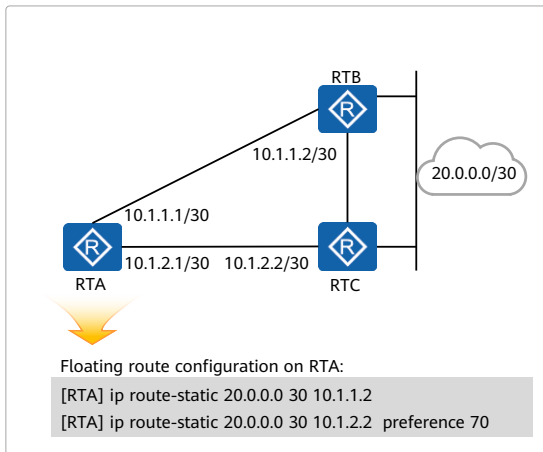
The next hop of the route to 30.1.2.0/24 is 20.1.1.3, which is not on a directly connected network. If no route to 20.1.1.3 exists in the routing table, this static route will be invalid and cannot be installed in the routing table.

Route Recursion (2)



Add a route to 20.1.1.3 with the next-hop IP address set to 10.0.0.2, which is on a directly connected network.
Through recursive lookup, the route to 30.1.2.0/24 can resolve its next hop to a directly connected next hop.

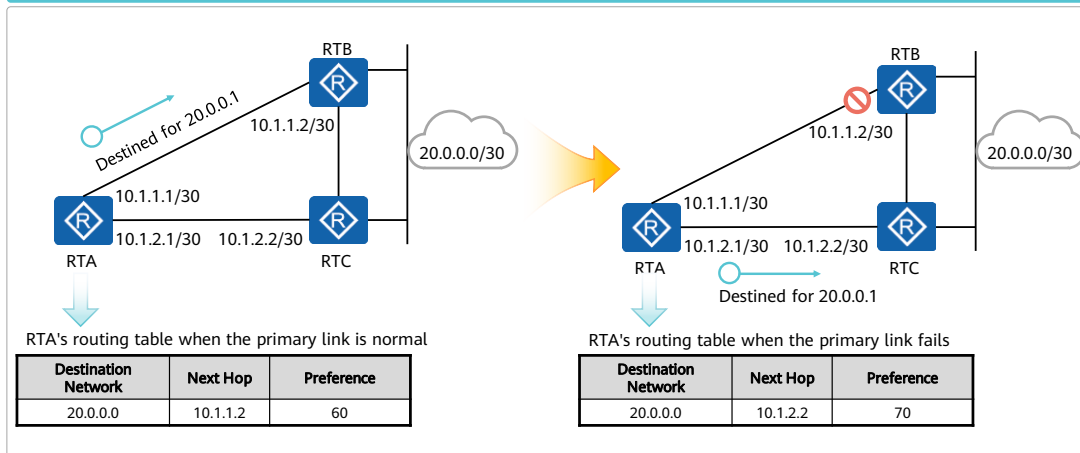
Floating Routes



- Static routes support manually configured preferences. By configuring static routes with the same destination and mask but different preferences and next hops, you can implement forwarding path backup.
- A floating static route serves as a backup to the primary route, providing an alternative path if the primary link fails. This backup route does not appear in the routing table as long as the next hop of the primary route is reachable.

Example of a Floating Route

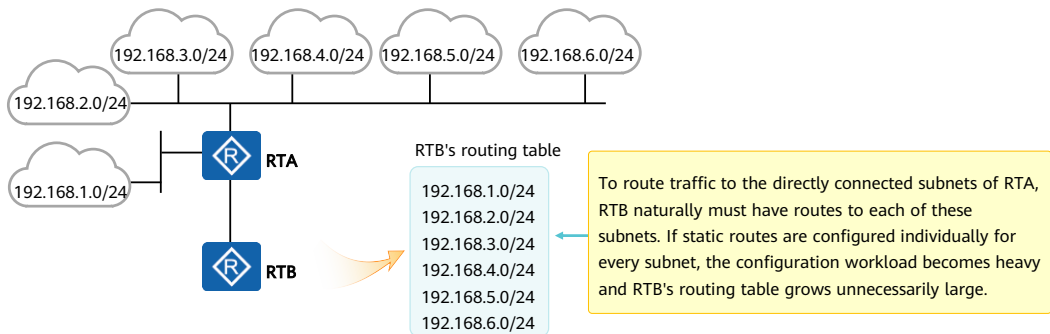
Floating route switchover



- When the link between RTA and RTB is normal, both routes to 20.0.0.0/30 are valid. RTA compares their preferences: the route with next hop 10.1.1.2 has a preference of 60, while the route with next hop 10.1.2.2 has a preference of 70. Therefore, the route with next hop 10.1.1.2 (preference 60) is installed in the routing table.
- If the link between RTA and RTB fails, the next hop 10.1.1.2 becomes unreachable, invalidating the corresponding route. In this case, only one route to 20.0.0.0/30 remains, which is then installed in the routing table. RTA forwards traffic destined for 20.0.0.1 to the next hop 10.1.2.2.

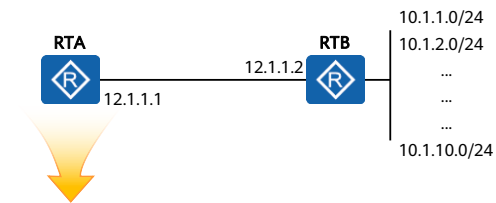
Background for Route Summarization

- Subnetting and Variable Length Subnet Mask (VLSM) help reduce address space waste, but they also increase the number of routing entries in the routing table.
- To minimize routing entries, use the route summarization function.



- On large-scale networks, routers or other routing-capable devices need to maintain a large number of routing entries. This consumes significant device resources, and as the routing table grows, the efficiency of route lookups decreases. To ensure IP reachability from routers to all subnets while minimizing resource usage, it is important to reduce the size of routing tables.
- With proper network planning, several methods can be used to reduce the routing table size. One of the most common and effective approaches is route summarization.

Overview of Route Summarization



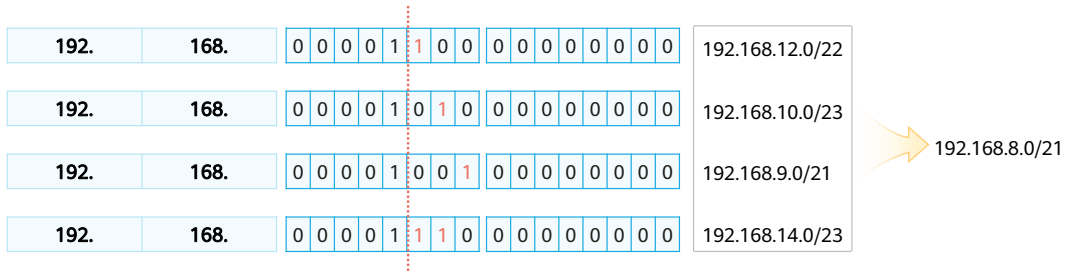
```
[RTA] ip route-static 10.1.0.0 16 12.1.1.2
```

Multiple static routes to the directly connected subnets of RTB—10.1.1.0/24, 10.1.2.0/24, ..., and 10.1.10.0/24—are configured on RTA. Since these routes share the same next hop, they can be summarized into one route.

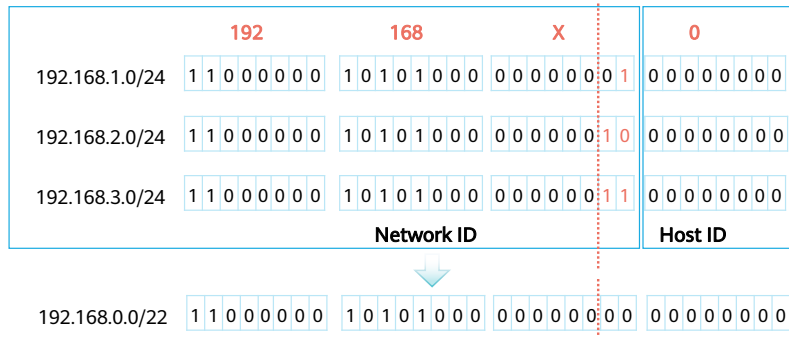
- Route summarization combines multiple specific routes with the same prefix into a single summarized route, reducing the size of the routing table and improving resource efficiency.
- Route summarization is based on Classless Inter-Domain Routing (CIDR), which aggregates addresses with the same prefix into one.
- The individual routes before summarization are called specific routes or detailed routes, while the resulting single route after summarization is known as a summary route or aggregated route.

CIDR

- CIDR identifies networks and subnets using IP addresses and mask lengths, replacing the traditional classful addressing model (such as Classes A, B, and C).
- CIDR allows for prefixes of arbitrary length, treating the IP address space as a continuous pool. This enables multiple contiguous prefixes to be aggregated into a single network, effectively reducing the number of entries in the routing table.

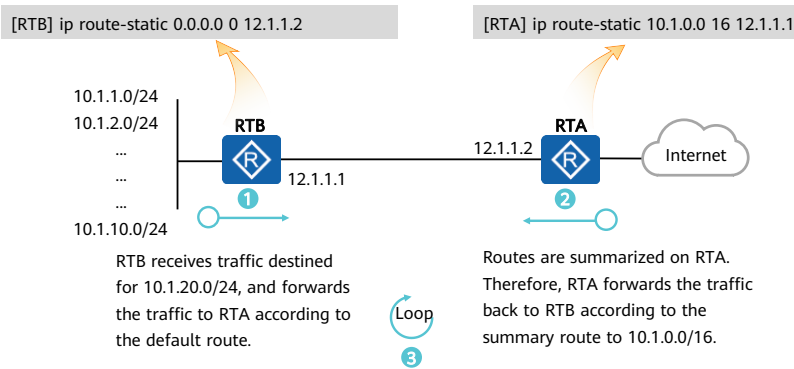


Route Summarization Calculation

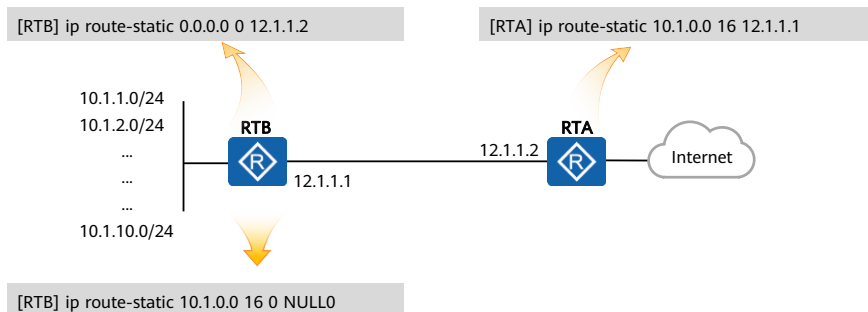


- When summarizing a set of continuous and regular IP subnets into a summary route that encompasses them, ensure that the subnet mask of the summary route is as long as possible.
- The key is to convert the destination network addresses of the specific routes into binary, align them, and then identify the identical leading bits across all addresses.

Issues Caused by Route Summarization



Addressing Issues Caused by Route Summarization



- Solution: Add a route pointing to Null0 on RTB to resolve the routing loop issue.

- Generally, every route, whether statically configured or dynamically learned, must be associated with an outbound interface.
- The outbound interface in a route can be a physical interface such as a 100M or GE interface, or a logical interface such as a VLANIF or tunnel interface.
- Among the various types of outbound interfaces, there is a unique one called the NULL (invalid) interface, which has only a single identifier—0.
- NULL0 is a system-reserved logical interface. When a network device forwards packets through a route whose outbound interface is NULL0, these packets are immediately discarded, as if dropped into a black hole. For this reason, routes pointing to NULL0 are often referred to as black hole routes.

Quiz

1. (Short-answer question) How does a router select the optimal route?
2. (Short-answer question) When inter-VLAN communication is implemented through sub-interfaces of a router, what configurations are required on the switch interface connected to the router?
3. (Short-answer question) Upon receiving a data frame, how does a Layer 3 switch determine whether to forward it based on the MAC address table or by performing a routing table lookup?

1. The router first compares the route preferences. If the preferences are identical, it then compares the route metrics. If the metrics are also the same, the routes are installed in the routing table as equal-cost routes.
2. The switch interface must be configured as a trunk or hybrid interface, allowing the specific VLANs associated with terminals to pass through.
3. A Layer 3 switch determines whether to perform Layer 2 or Layer 3 forwarding by inspecting the destination MAC address of the data frame. If the destination MAC address belongs to the switch itself, the frame is processed at Layer 3. If the destination MAC address corresponds to another device on the network, the frame is forwarded at Layer 2.

Summary

- When a router learns multiple routes to the same destination network (with identical destination addresses and masks) from different sources, it selects the optimal route and installs it into the routing table.
- Upon receiving a data packet, a router looks up the destination IP address in its routing table. If a matching entry is found, the packet is forwarded through the outbound interface and next hop specified in this entry; otherwise, the packet is discarded.
- Routing is a hop-by-hop process. Every router along the path from source to destination must have a valid route to the destination. If any router lacks such a route, packet loss occurs.
- Because data communication is typically bidirectional, round-trip routes must be established to ensure successful communication in both directions.
- Inter-VLAN communication can be implemented using physical router interfaces, router sub-interfaces, or VLANIF interfaces.

Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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OSPF Fundamentals and Configuration



Foreword

- Because static routes are manually configured by network administrators, they must be updated manually when the network topology changes. This limits the large-scale deployment of static routing on live networks.
- Dynamic routing protocols are widely adopted in modern networks due to their flexibility, reliability, and scalability. Among them, Open Shortest Path First (OSPF) is one of the most commonly used.
- Defined in RFC 2328, OSPF is a link-state routing protocol.
- This course covers OSPF fundamental concepts, working principles, and basic configuration.

Objectives

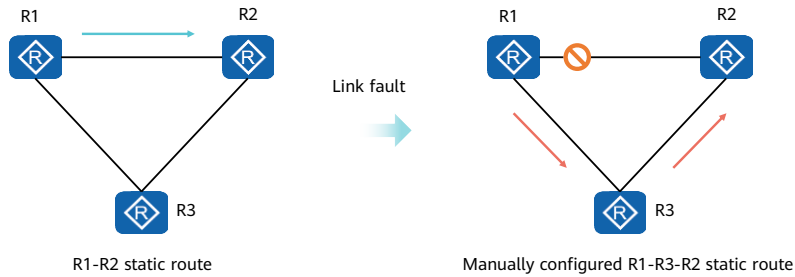
- Upon completion of this course, you will be able to:
 - Describe the advantages and classification of dynamic routing protocols.
 - Explain OSPF fundamental concepts and typical networking scenarios.
 - Explain the working principles of OSPF.
 - Perform basic OSPF configuration.

Contents

1. **Overview of OSPF**
2. Working Principles of OSPF
3. Typical OSPF Configuration

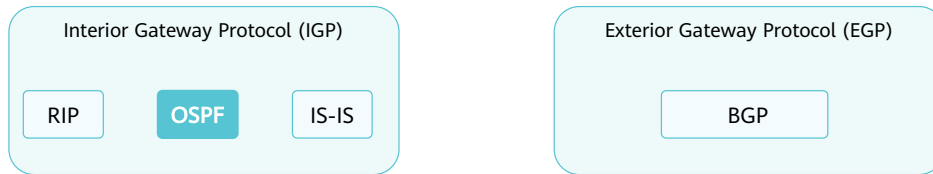
Why Dynamic Routing Protocols Are Required?

- Static routes are manually configured and maintained by engineers using simple, straightforward commands, making them suitable for small or stable networks. However, static routing has the following drawbacks:
 - Not scalable for large networks: As the number of devices grows, the configuration workload increases significantly.
 - No dynamic response to network changes: When the network changes, static routes cannot automatically converge and must be updated manually by engineers.

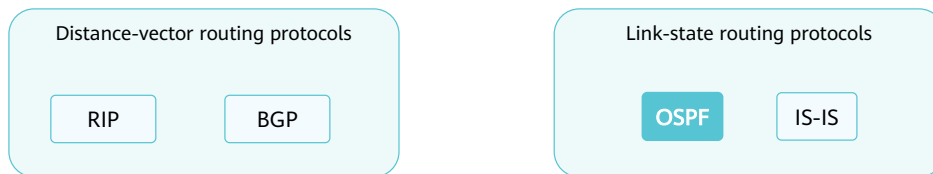


Classification of Dynamic Routing Protocols

By scope of operation



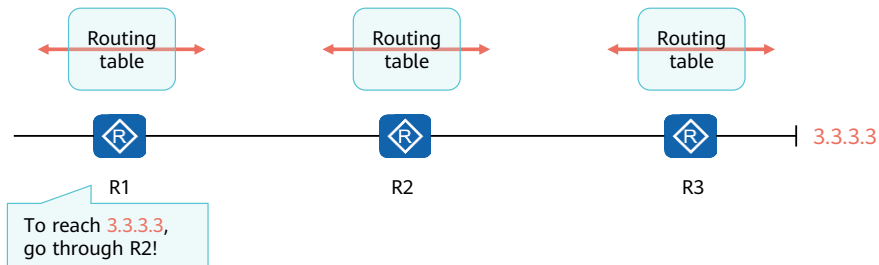
By working mechanism and algorithm



- The Routing Information Protocol (RIP) is a distance-vector routing protocol that uses hop count as its metric to measure the distance between the local device and the destination network.
- Intermediate System to Intermediate System (IS-IS) is a link-state routing protocol that calculates routes using the Shortest Path First (SPF) algorithm.
- The Border Gateway Protocol (BGP) is a distance-vector routing protocol designed to establish routing reachability between autonomous systems (ASs) and to select the optimal routes.

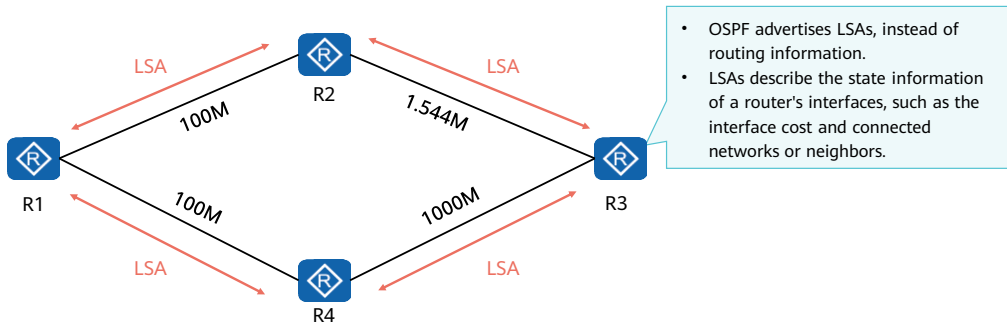
Distance-Vector Routing Protocols

- Routers running distance-vector routing protocols periodically flood their routing tables. Through route exchange, each router learns routes from its neighbors and updates its own routing table accordingly.
- On a network, routers do not have knowledge of the overall topology. Instead, they only know the direction toward a destination and the distance to reach it. This is the fundamental essence of the distance-vector algorithm.



Link-State Routing Protocols: LSA Flooding

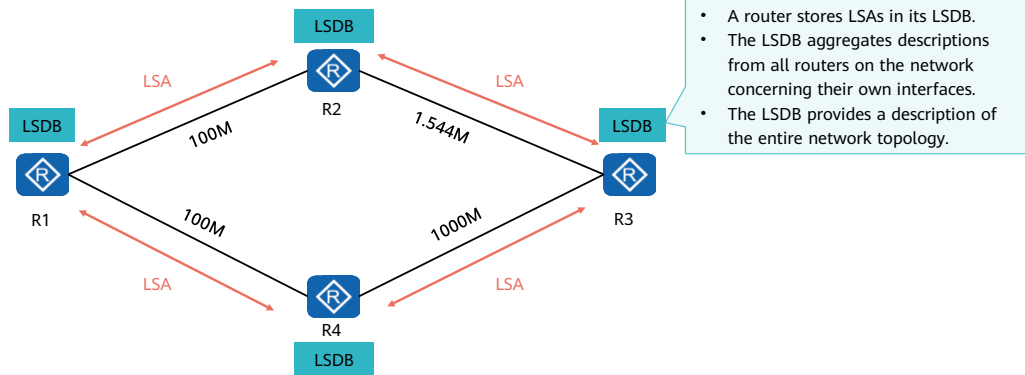
- Unlike distance-vector routing protocols, link-state routing protocols advertise link states instead of routing tables. Two routers running a link-state routing protocol first establish a neighbor relationship, and then exchange Link State Advertisements (LSAs).



- An LSA can be simply understood as a message generated by each router to describe the state of its directly connected interfaces, including information such as the interface cost and the relationships with neighboring routers.

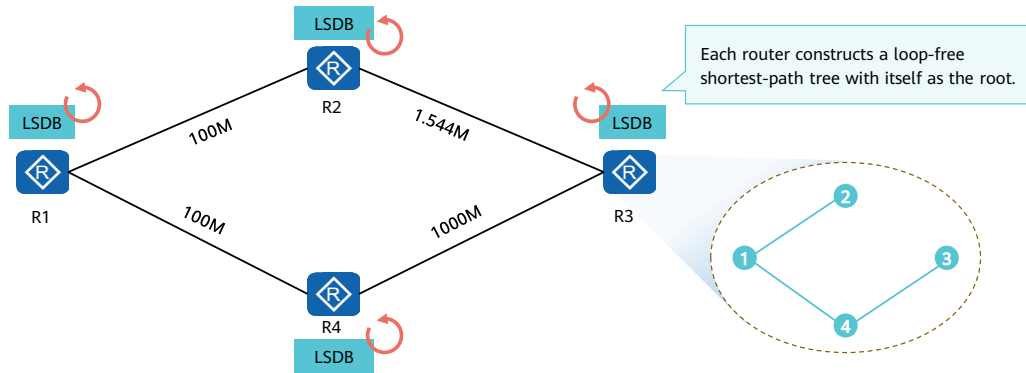
Link-State Routing Protocols: LSDB Construction

- Each router generates LSAs and stores the received LSAs in its own Link-State Database (LSDB). By building the LSDB, a router can learn the complete network topology.



Link-State Routing Protocols: SPF Calculation

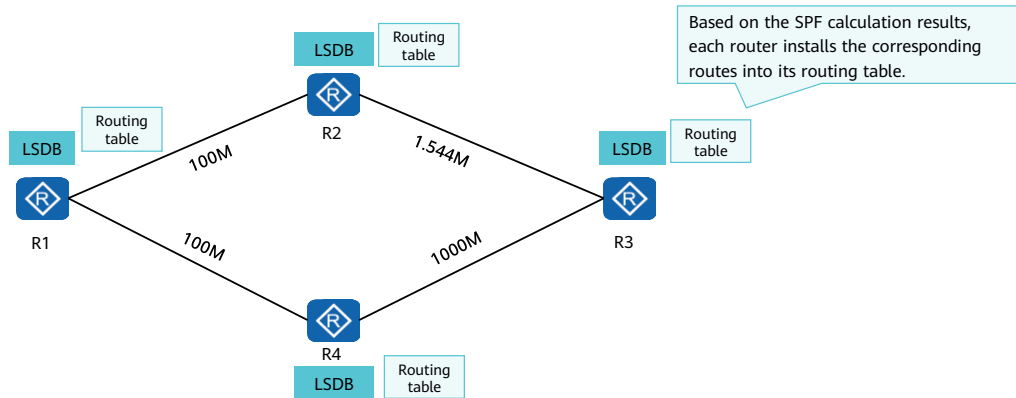
- Each router uses the Shortest Path First (SPF) algorithm to calculate routes based on the information contained in the LSDB. By doing so, each router constructs a loop-free shortest-path tree with itself as the root. This tree enables the routers to determine the optimal path to every destination on the network.



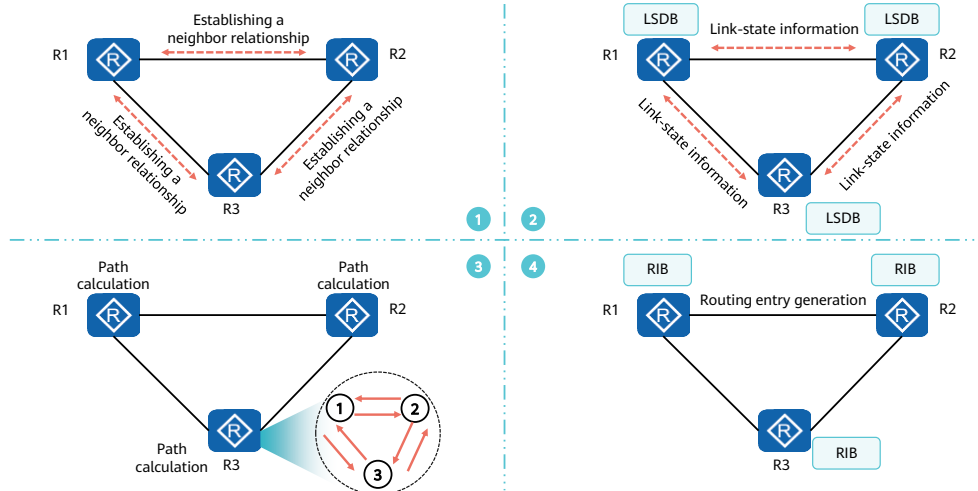
- SPF is a core algorithm of the OSPF routing protocol. It enables routers to make optimal routing decisions on complex networks.

Link-State Routing Protocols: Routing Table Generation

- Finally, each router installs the optimal paths it has calculated into its routing table.



Summary of Link-State Routing Protocols



- The operation of link-state routing protocols consists of four key steps.
 - Step 1: Establish neighbor relationships between adjacent routers.
 - Step 2: Exchange link-state information and synchronize the LSDB between neighbors.
 - Step 3: Calculate the optimal paths.
 - Step 4: Generate and install routes into the routing table based on the shortest-path tree.
- RIB refers to routing information base.

Overview of OSPF

OSPF is a typical link-state routing protocol and one of the most widely used IGPs.

OSPFv2 (RFC 2328) is used for IPv4, while OSPFv3 (RFC 2740) is used for IPv6. Unless otherwise specified, OSPF refers to OSPFv2 in this document.

OSPF routers exchange link-state information, instead of routing information. Link-state information is key to OSPF topology building and route calculation.

OSPF routers collect link-state information on the network and store it in the LSDB. With the LSDB, routers within an area gain a complete view of the network topology, allowing them to calculate loop-free paths.

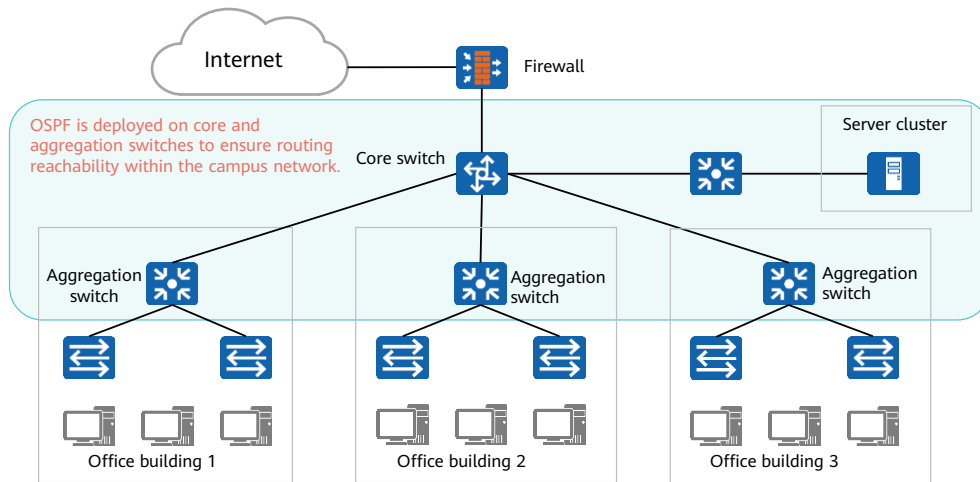
Each OSPF router uses the SPF algorithm to calculate the shortest paths to destinations. The router then generates routes based on these paths and installs the routes into the routing table.

OSPF supports Variable Length Subnet Mask (VLSM) and manual route summarization.

The multi-area design enables OSPF to support larger-scale networks.

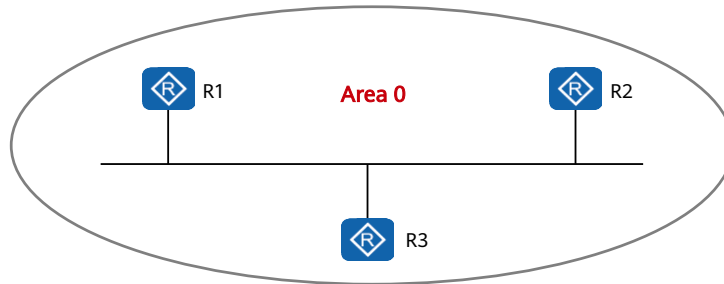
- OSPFv2 is short for OSPF version 2.
- OSPFv3 is short for OSPF version 3.

Application of OSPF on Campus Networks



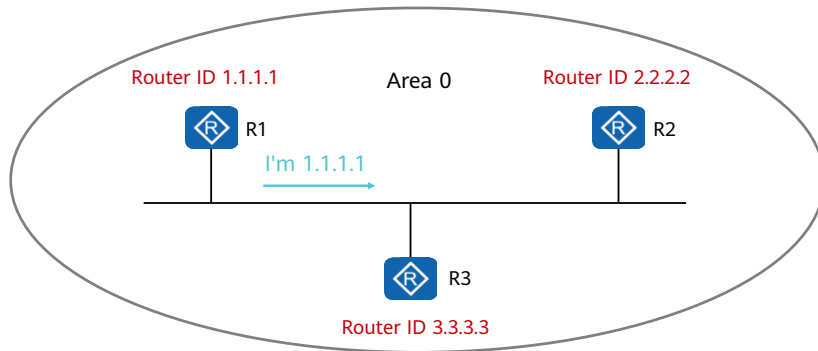
Basic OSPF Terminology: Area

- An OSPF area serves as a logical grouping of devices, distinguished by its area ID.



Basic OSPF Terminology: Router ID

- A router ID uniquely identifies a router within an OSPF domain.
- The router ID can be manually configured or automatically assigned by the system.

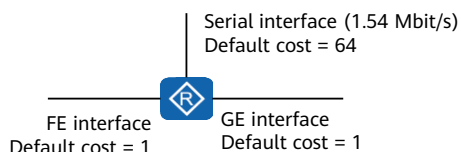


- In practical deployments, OSPF router IDs are usually configured manually. Ensure that each device within the same OSPF domain has a unique router ID. A common practice is to configure the router ID to match the IP address of a specific interface on the device, typically a loopback interface.

Basic OSPF Terminology: Metric

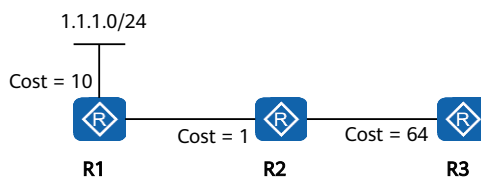
- OSPF uses cost as the routing metric. Each OSPF-enabled interface maintains an interface cost. By default, the interface cost is calculated as follows: $\frac{100 \text{ Mbit/s}}{\text{Interface bandwidth}}$. In this formula, 100 Mbit/s is the default OSPF reference bandwidth, which can be manually adjusted if required.
- The cost of an OSPF route is the cumulative cost of all outbound interfaces along the path from the local router to the destination network.

OSPF interface cost



- The OSPF cost varies across interfaces based on their respective bandwidth.

Cumulative cost of an OSPF path



- In R3's routing table, the cost of the OSPF route to 1.1.1.0/24 is 75 (10 + 1 + 64).

OSPF Packet Types

- There are five types of OSPF packets.

Packet Type	Function
Hello	Sent periodically to discover and maintain OSPF neighbor relationships.
Database Description (DD)	Provides a summary of the local LSDB and is used for database synchronization between two routers.
Link State Request (LSR)	Requests specific LSAs from a neighbor. Devices send LSR packets only after successfully exchanging DD packets.
Link State Update (LSU)	Delivers the requested LSAs to the neighbor.
Link State Acknowledgment (LSAck)	Acknowledges the receipt of LSAs.

Three OSPF Tables: Neighbor Table

- OSPF maintains three key tables: OSPF neighbor table, LSDB, and OSPF routing table. For the OSPF neighbor table, note the following:
 - An OSPF neighbor relationship is required before link-state information can be exchanged.
 - OSPF neighbor relationships are established through the exchange of Hello packets.
 - The OSPF neighbor table contains the neighbor state information, which can be queried using the **display ospf peer** command.

[R1] display ospf peer

Router ID: 1.1.1.1



R1

GE0/0/0
10.1.1.1/30

Router ID: 2.2.2.2



R2

GE0/0/0
10.1.1.2/30

```
<R1> display ospf peer
  OSPF Process 1 with Router ID 1.1.1.1
    Neighbors
Area 0.0.0.0 interface 10.1.1.1 (GE0/0/0)'s neighbors
Router ID: 2.2.2.2 Address: 10.1.1.2 GR State: Normal
State: Full Mode:Nbr is Master Priority: 1
DR: 10.1.1.1 BDR: 10.1.1.2 MTU: 0
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00:00:05
Authentication Sequence: [ 0 ]
```

- The OSPF neighbor table contains various critical information, such as the router ID and interface address of the peer device.

Three OSPF Tables: LSDB

- For the OSPF LSDB, note the following:
 - The LSDB of a device stores both the LSAs it generates and those received from neighbors. In this example, the LSDB of R1 contains three LSAs.
 - Type** indicates the type of an LSA, and **AdvRouter** identifies the originating router of the LSA.
 - The LSDB can be queried using the **display ospf lsdb** command.

[R1] display ospf lsdb

Router ID: 1.1.1.1



R1

GE0/0/0

10.1.1.1/30

Router ID: 2.2.2.2



R2

GE0/0/0

10.1.1.2/30

<R1> display ospf lsdb

OSPF Process 1 with Router ID 1.1.1.1

Link State Database

Area: 0.0.0.0

Type	LinkState ID	AdvRouter	Age	Len	Sequence	Metric
Router	2.2.2.2	2.2.2.2	98	36	8000000B	1
Router	1.1.1.1	1.1.1.1	92	36	80000005	1
Network	10.1.1.2	2.2.2.2	98	32	80000004	0

- For more information about LSAs, refer to HCIP-Datacom.

Three OSPF Tables: OSPF Routing Table

- For the OSPF routing table, note the following:
 - The OSPF routing table of a router is distinct from its IP routing table. In this example, the OSPF routing table contains three routes.
 - The OSPF routing table contains essential forwarding information, such as **Destination**, **Cost**, and **NextHop**.
 - The OSPF routing table can be queried using the **display ospf routing** command.

[R1] display ospf routing

Router ID: 1.1.1.1



R1

GE0/0/0
10.1.1.1/30

Router ID: 2.2.2.2



R2

GE0/0/0
10.1.1.2/30

```
<R1> display ospf routing
OSPF Process 1 with Router ID 1.1.1.1
Routing Tables
Routing for Network
```

Destination	Cost	Type	NextHop	AdvRouter	Area
1.1.1.1/32	0	stub	1.1.1.1	1.1.1.1	0.0.0.0
10.1.1.0/20	1	Transit	10.1.1.1	1.1.1.1	0.0.0.0
2.2.2.2/32	1	stub	10.1.1.2	2.2.2.2	0.0.0.0

```
Total Nets: 3
Intra Area: 3 Inter Area: 0 ASE: 0 NSSA: 0
```

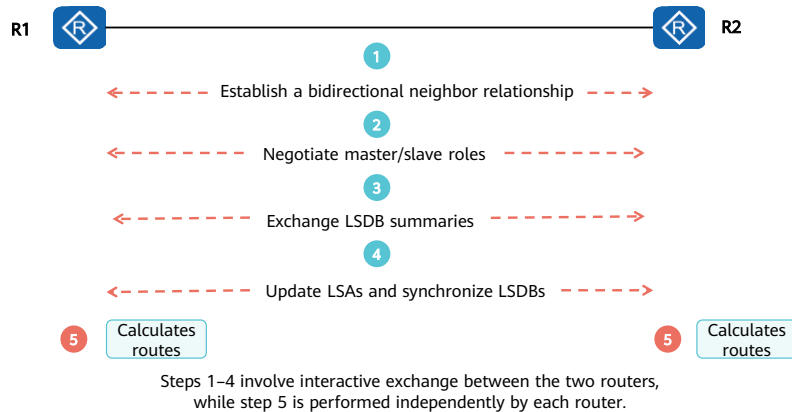
- For more information about the OSPF routing table, refer to HCIP-Datacom.

Contents

1. Overview of OSPF
- 2. Working Principles of OSPF**
3. Typical OSPF Configuration

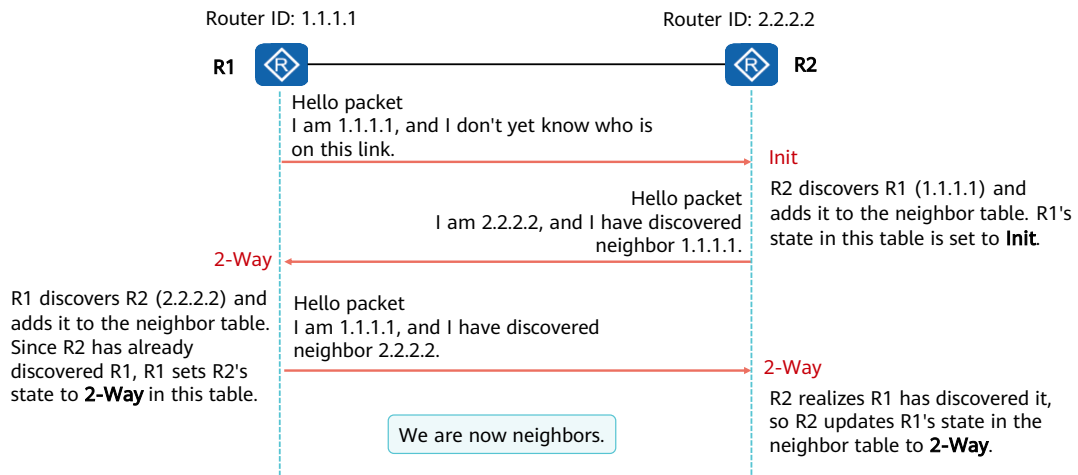
Process of Establishing an OSPF Adjacency

- OSPF establishes an adjacency through four main steps: neighbor relationship establishment, master/slave negotiation, LSDB exchange, and LSDB synchronization.



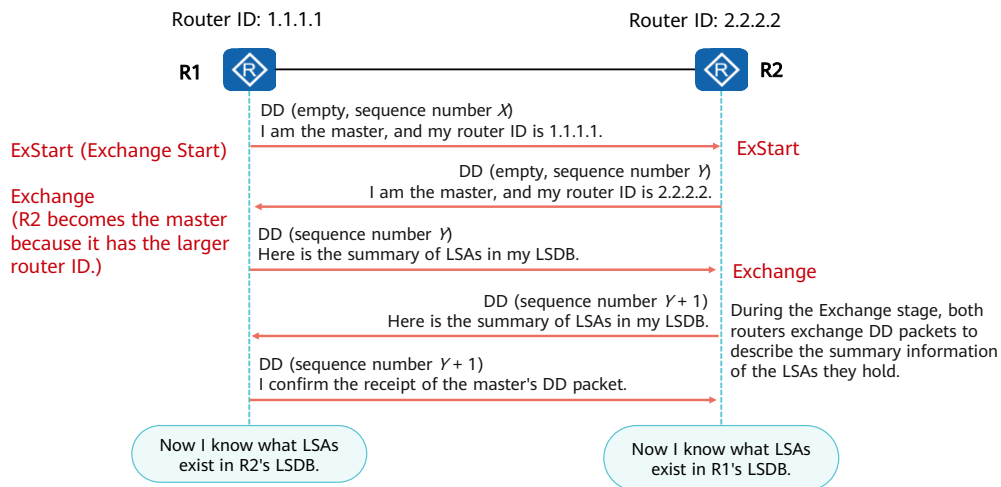
- There are two key concepts in OSPF router relationships: neighbor relationship and adjacency.
- In a simple topology where two routers are directly connected, enabling OSPF on their interconnected interfaces causes them to send and listen for Hello packets. Once they discover each other through these Hello packets, a neighbor relationship is established.
- Forming a neighbor relationship is only the initial step. The routers then exchange a series of packets, such as DD, LSR, LSU, and LSAck packets. Once their LSDBs are fully synchronized, the two routers form an adjacency, allowing each to begin independent route calculation.

Process of Establishing an OSPF Adjacency (Step 1)



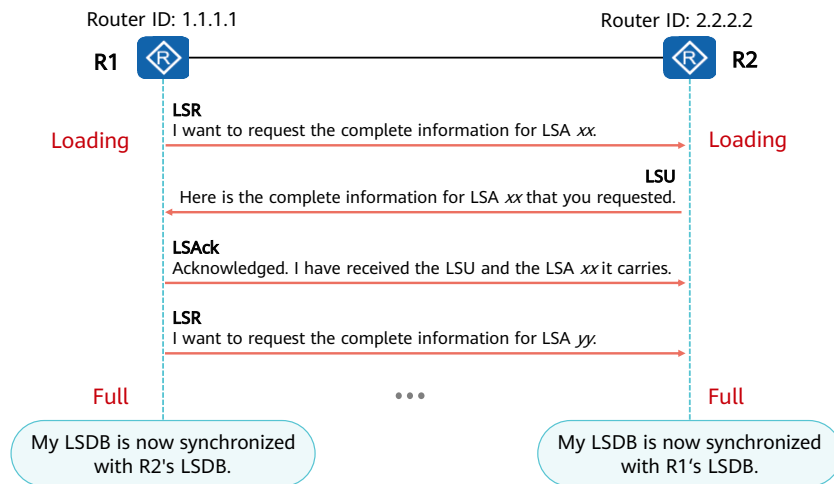
- When an OSPF router receives the first Hello packet from another router, it transitions from the initial Down state to the Init state.
- When an OSPF router receives a Hello packet where the Neighbor field contains its own router ID, it transitions from the Init state to the 2-Way state.

Process of Establishing an OSPF Adjacency (Steps 2 and 3)



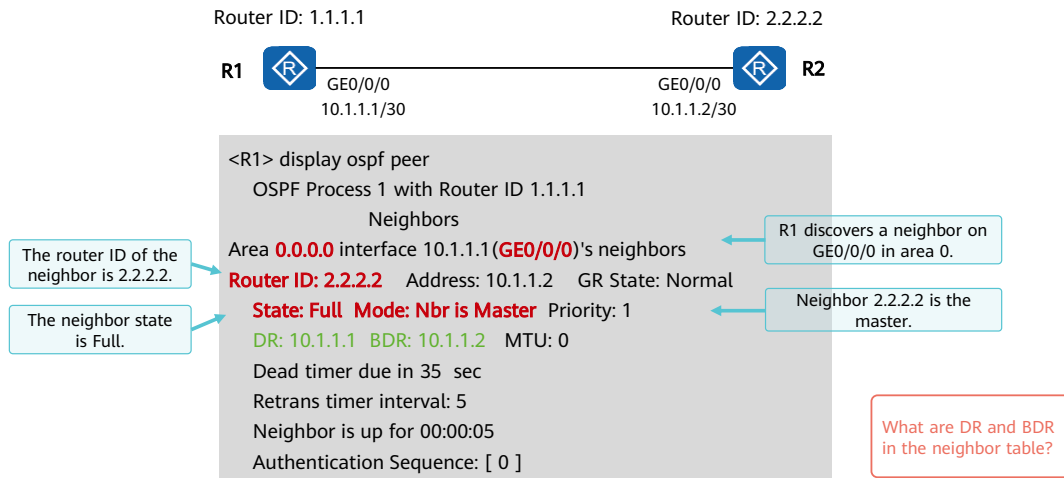
- After the neighbor state transitions from 2-Way to ExStart, the routers begin master/slave negotiation.
 - R1 sends the first empty DD packet with sequence number X to R2.
 - R2 also sends the first DD packet with sequence number Y to R1.
 - The master/slave relationship is determined by comparing router IDs. (The larger router ID wins.) Since R2's router ID is greater than R1's, R2 becomes the master. After the election, R1's state changes from ExStart to Exchange.
- Once R1 enters the Exchange state, it sends a DD packet containing a summary of its LSDB, adopting the sequence number of the master (R2). Upon receiving this packet, R2 also transitions from ExStart to Exchange.
- R2 then sends a DD packet with its LSDB summary to R1, using sequence number $Y+1$.
- As the slave, R1 acknowledges each DD packet sent by the master (R2) by responding with a DD packet that carries the same sequence number as R2's.
- After the final DD packet is sent, R1 transitions its neighbor state to Loading.

Process of Establishing an OSPF Adjacency (Step 4)



- Upon transitioning to the Loading state, R1 sends an LSR to R2, requesting the LSAs that are discovered through DD packets in the Exchange stage but are missing in its local LSDB.
- When receiving the request, R2 responds to R1 with an LSU containing details of the requested LSAs.
- When receiving the LSU, R1 sends an LSAck to R2 to confirm the receipt, ensuring reliable transmission.
- Meanwhile, R2 also requests and receives LSAs from R1. When both routers' LSDBs are fully synchronized and identical, the neighbor state transitions to Full, signifying the successful establishment of an OSPF adjacency.

OSPF Neighbor Table Review



- The parameters in the **display ospf peer** command output are described as follows:
 - OSPF Process 1 with Router ID 1.1.1.1**: indicates that the local OSPF process ID is 1 and the local OSPF router ID is 1.1.1.1.
 - Router ID**: indicates the OSPF router ID of the neighbor.
 - Address**: indicates the interface address of the neighbor.
 - GR State**: indicates the OSPF Graceful Restart (GR) status (a high-availability optimization function). The default value is **Normal**.
 - State**: indicates the neighbor state. Normally, after LSDB synchronization, the stable state is **Full**.
 - Mode**: identifies the local device's role as **Master** or **Slave** during the exchange of link-state information.
 - Priority**: indicates the priority of the neighbor router (used in DR election).
 - DR**: indicates the designated router.
 - BDR**: indicates the backup designated router.
 - MTU**: indicates the MTU value of the neighbor's interface.
 - Retrans timer interval**: indicates the interval for retransmitting LSAs, in seconds.

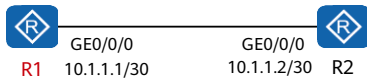
- **Authentication Sequence:** indicates the authentication sequence number.

OSPF Network Types

- Before learning the concepts of designated router (DR) and backup designated router (BDR), it is essential to first understand OSPF network types.
- The OSPF network type is a key interface variable. It determines how OSPF operates on an interface—for example, how OSPF packets are transmitted and whether DR/BDR election is required.
- The default OSPF network type of an interface depends on the data link layer encapsulation in use.
- OSPF defines four network types: broadcast, Non-Broadcast Multiple Access (NBMA), Point-to-Multipoint (P2MP), and Point-to-Point (P2P).

Router ID: 1.1.1.1

Router ID: 2.2.2.2



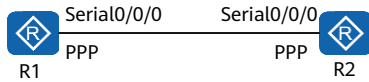
[R1-GE0/0/0] ospf network-type ?

broadcast	Specify OSPF broadcast network
nbma	Specify OSPF NBMA network
p2mp	Specify OSPF point-to-multipoint network
p2p	Specify OSPF point-to-point network

OSPF Network Types (1)

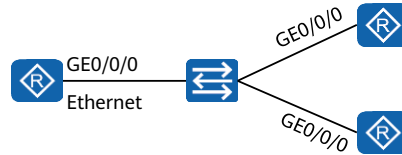
- Generally, the OSPF network types of interfaces at both ends of a link must be the same. Otherwise, the routers cannot establish an OSPF neighbor relationship.
- When necessary, you can change the OSPF network type of an interface using configuration commands. For example, you may change the network type from Broadcast Multiple Access (BMA) to P2P.

P2P



- P2P refers to a network environment where only two network devices are connected on a single link.
- A typical example is a PPP link. When an interface uses PPP encapsulation, the default OSPF network type of the interface is P2P.

BMA



- BMA, also known as a broadcast network, refers to an environment that allows the connection of multiple devices and supports broadcast communication.
- A typical example is Ethernet. When an interface uses Ethernet encapsulation, the default OSPF network type of the interface is BMA.

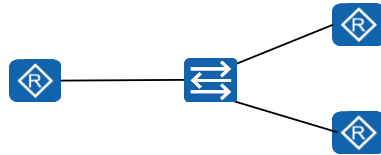
OSPF Network Types (2)

NBMA



- NBMA refers to an environment that allows the connection of multiple network devices but does not support broadcast communication.
- A typical example is a Frame Relay (FR) network.

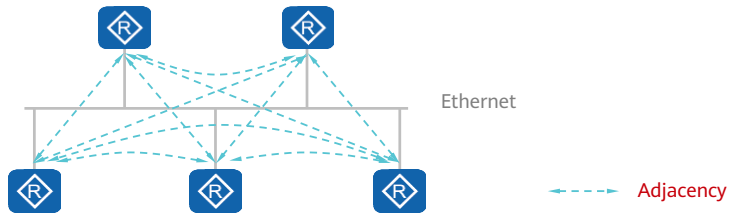
P2MP



- P2MP can be viewed as a network formed by bundling one end of multiple P2P links.
- P2MP is not the default network type of any data link layer protocol. This type must be manually configured from other network types.
- A common practice is to change a non-fully meshed NBMA network to a P2MP network.

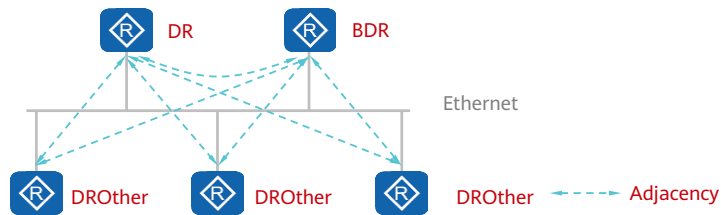
Background of DR and BDR

- Multi-access (MA) networks are classified into two types: BMA and NBMA. Ethernet is a typical broadcast MA network.
- On an MA network, if every OSPF router establishes OSPF adjacencies with all other routers, an excessive number of OSPF adjacencies will exist. This increases device workload and the volume of flooded OSPF packets.
- When the topology changes, LSA flooding can consume unnecessary bandwidth and device resources.



DR and BDR

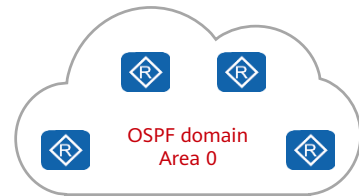
- To optimize OSPF adjacencies on an MA network, OSPF defines three router roles: DR, BDR, and DROther.
- Only the DR and BDR establish adjacencies with other OSPF routers. DROthers do not form adjacencies with each other, and they remain in the 2-Way state.
- The BDR monitors the status of the DR, and takes over its role if the current DR fails.



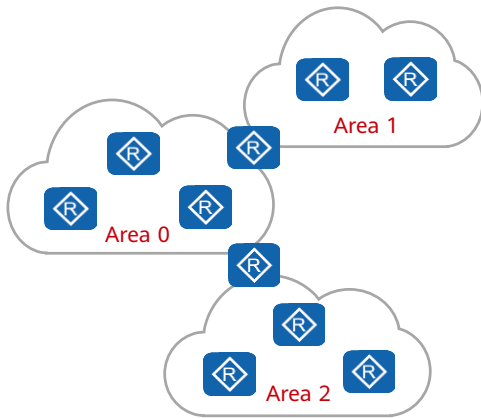
- If the DR fails, devices on the network must elect a new DR and synchronize with it. To accelerate this process, OSPF introduces the BDR. As a backup for the DR, the BDR is elected alongside the DR; it establishes adjacencies and exchanges routing information with all other devices on the local network segment. When the DR fails, the BDR immediately assumes the DR role. Because no new election is required and adjacencies are already established, this transition is fast. A new BDR is then elected; although this process takes longer, it does not affect route calculation.
- DR and BDR roles are not manually configured; instead, they are elected collectively by all devices on the local network segment. The DR priority of an interface determines its eligibility, and only devices with a DR priority greater than 0 can participate in the election. The election process relies on Hello packets, which act as "votes." Each device advertises its preferred DR in the Hello packets exchanged on the network segment. If two devices both declare themselves as DR, the one with the higher DR priority wins; if priorities are equal, the device with the larger router ID wins.
- A router with a priority of 0 will never be elected as the DR or BDR.
- According to protocol specifications, OSPF DR/BDR elections are non-preemptive. Therefore, changing a device's DR priority after the election will not alter the current DR or BDR.

OSPF Domain and Single Area

- An OSPF domain is the network scope consisting of all routers running the OSPF protocol and sharing the same routing information.
- OSPF routers flood LSAs within a single area. The LSDB must be synchronized across the area to ensure that all routers share a consistent view of the network topology.
- If an OSPF domain contains only one area, as the network grows and the number of OSPF routers increases, several issues arise:
 - The LSDB becomes very large, which also expands the OSPF routing table. This leads to increased resource consumption, degraded device performance, and a negative impact on data forwarding.
 - Route calculation based on a massive LSDB becomes increasingly complex.
 - When the network topology changes, LSA flooding across the entire domain and network-wide SPF recalculation impose a heavy burden.



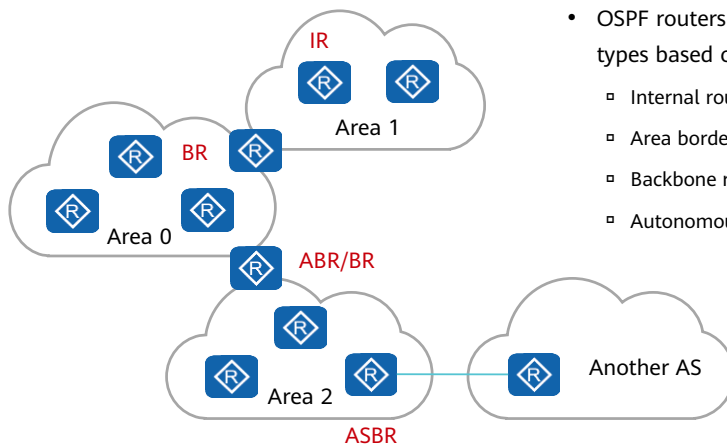
OSPF Multi-Area



- An OSPF domain can be divided into multiple areas to support larger networks.
- The OSPF multi-area design limits the scope of LSA flooding, effectively confining the impact of topology changes within a single area and achieving network optimization.
- Route summarization can be performed at area boundaries to reduce the size of the routing table.

- Area classification: OSPF areas are classified into backbone and non-backbone areas. The backbone area is area 0, and all other areas are non-backbone areas.
- Multi-area interconnection principle: To prevent inter-area routing loops, non-backbone areas cannot connect directly to each other. All non-backbone areas must connect to the backbone area.

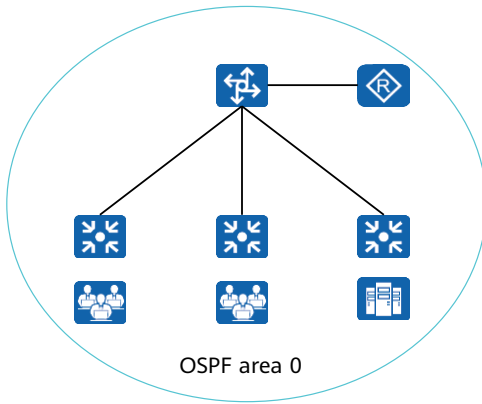
OSPF Router Types



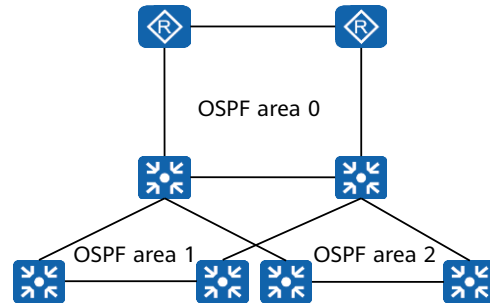
- OSPF routers are classified into the following types based on their locations or functions:
 - Internal router (IR)
 - Area border router (ABR)
 - Backbone router (BR)
 - Autonomous system boundary router (ASBR)

- Internal router (IR): All interfaces of this router belong to the same OSPF area.
- Area border router (ABR): This router has interfaces in two or more areas, with at least one interface in the backbone area.
- Backbone router (BR): This router has at least one interface in the backbone area.
- Autonomous system boundary router (ASBR): This router exchanges routing information with other ASs.
- An AS is a collection of IP networks that are controlled by a single administrative entity and use a unified routing policy. Each AS has a unique number.
- For OSPF, any router that imports external routes is considered an ASBR.

Typical OSPF Single-Area and Multi-Area Topologies



Small- and medium-sized enterprise network (single-area)



Large enterprise network (multi-area)

- For small- and medium-sized enterprise networks, where the scale is limited and the number of routing devices is small, it is feasible to place all devices within a single OSPF area.
- For large enterprise networks, which are extensive, contain many routing devices, and have a clear hierarchical structure, it is recommended to deploy OSPF using a multi-area design.

Contents

1. Overview of OSPF
2. Working Principles of OSPF
- 3. Typical OSPF Configuration**

Basic OSPF Configuration Commands (1)

1. Create and run an OSPF process.

```
[Huawei] ospf [ process-id | router-id router-id ]
```

process-id identifies an OSPF process. The default process ID is 1. OSPF supports multiple processes, meaning a single device can run several independent OSPF processes without interference. *router-id* specifies a device ID. If no router ID is specified in the command, the system automatically selects an interface IP address as the router ID.

2. Create an OSPF area and enter its view.

```
[Huawei-ospf-1] area area-id
```

The **area** command creates an OSPF area and displays its view.

The value of *area-id* can be a decimal integer or in dotted decimal notation. If the value is an integer, it ranges from 0 to 4294967295.

3. Specify the interface to run OSPF.

```
[Huawei-ospf-1-area-0.0.0.0] network network-address wildcard-mask
```

The **network** command specifies the interface running OSPF and the area to which the interface belongs. *network-address* specifies the subnet address of the interface. *wildcard-mask* specifies the inverse of the subnet mask, calculated by flipping the bits (changing 0s to 1s and 1s to 0s). For example, a wildcard mask of 0.0.0.255 represents a 24-bit mask.

Basic OSPF Configuration Commands (2)

4. Set a cost for an OSPF interface.

```
[Huawei-GE0/0/0] ospf cost cost
```

The **ospf cost** command sets a cost for running OSPF on an interface. By default, OSPF automatically calculates the cost of an interface based on the interface bandwidth. The cost value ranges from 1 to 65535.

5. Set an OSPF reference bandwidth.

```
[Huawei-ospf-1] bandwidth-reference value
```

The **bandwidth-reference** command sets the reference bandwidth used in the formula for calculating the interface cost. The value ranges from 1 to 2147483648, in Mbit/s. The default value is 100.

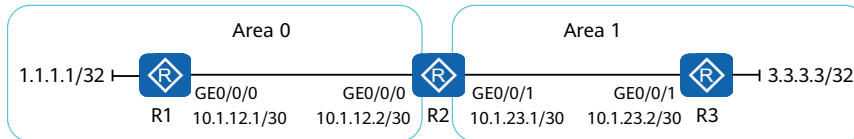
6. Set a DR priority for the interface.

```
[Huawei-GE0/0/0] ospf dr-priority priority
```

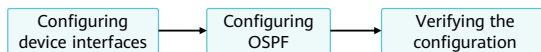
The **ospf dr-priority** command sets the priority of an interface in the DR election. A larger *priority* value increases the chance of being elected as the DR. The value ranges from 0 to 255.

OSPF Configuration Example

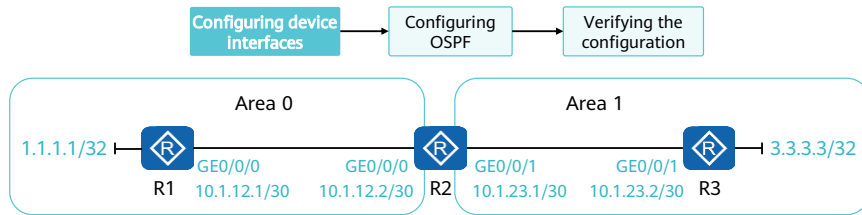
- There are three routers: R1, R2, and R3. R1 and R3 are connected to networks 1.1.1.1/32 and 3.3.3.3/32 (simulated using Loopback0), respectively. OSPF needs to be configured to enable communication between the two networks. The topology is shown below.



- Configuration process:



Configuring Interfaces



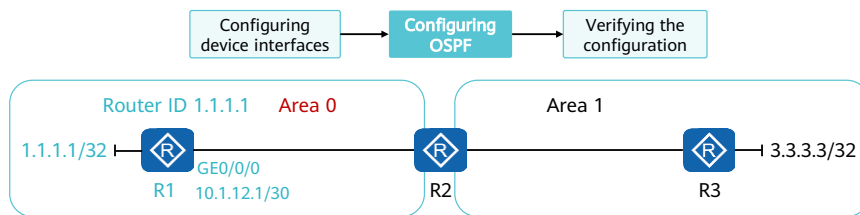
- Configure IP addresses for the interfaces on R1, R2, and R3 as planned.

```
[R1] interface LoopBack 0
[R1-LoopBack0] ip address 1.1.1.1 32
[R1-LoopBack0] quit
[R1] interface GE 0/0/0
[R1-GE0/0/0] ip address 10.1.12.1 30
```

```
[R3] interface LoopBack 0
[R3-LoopBack0] ip address 3.3.3.3 32
[R3-LoopBack0] quit
[R3] interface GE 0/0/1
[R3-GE0/0/1] ip address 10.1.23.2 30
```

```
[R2] interface GE 0/0/0
[R2-GE0/0/0] ip address 10.1.12.2 30
[R2-GE0/0/0] quit
[R2] interface GE 0/0/1
[R2-GE0/0/1] ip address 10.1.23.1 30
```

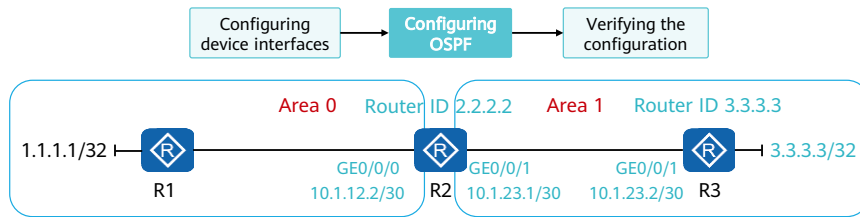
Configuring OSPF (1)



- OSPF parameter plan: The OSPF process ID is 1. The router IDs of R1, R2, and R3 are 1.1.1.1, 2.2.2.2, and 3.3.3.3, respectively.
- Procedure:
 - Create and run an OSPF process.
 - Create an OSPF area and enter its view.
 - Specify the interface to run OSPF.

```
[R1] ospf 1 router-id 1.1.1.1
[R1-ospf-1] area 0
[R1-ospf-1-area-0.0.0.0] network 1.1.1.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0] network 10.1.12.0 0.0.0.3
```

Configuring OSPF (2)

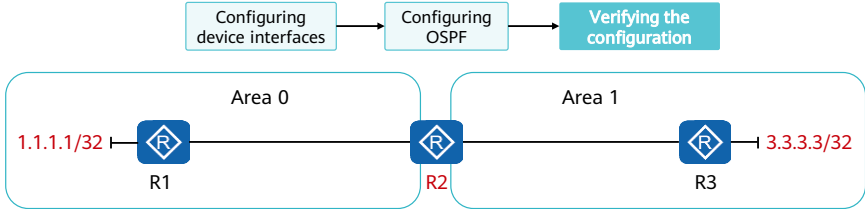


- When configuring OSPF multi-area, ensure that the corresponding subnets are advertised within their respective areas.

```
[R2] ospf 1 router-id 2.2.2.2
[R2-ospf-1] area 0
[R2-ospf-1-area-0.0.0.0] network 10.1.12.0 0.0.0.3
[R2-ospf-1-area-0.0.0.0] quit
[R2-ospf-1] area 1
[R2-ospf-1-area-0.0.0.1] network 10.1.23.0 0.0.0.3
```

```
[R3] ospf 1 router-id 3.3.3.3
[R3-ospf-1] area 1
[R3-ospf-1-area-0.0.0.1] network 3.3.3.3 0.0.0.0
[R3-ospf-1-area-0.0.0.1] network 10.1.23.0 0.0.0.3
```

Verification (1)



- Check the OSPF neighbor table on R2.

<R2> display ospf peer brief

OSPF Process 1 with Router ID 2.2.2.2

Peer Statistic Information

Area Id	Interface	Neighbor id	State
0.0.0.0	GE0/0/0	1.1.1.1	Full
0.0.0.1	GE0/0/1	3.3.3.3	Full

Area ID of a neighbor. → 0.0.0.0

→ Full

The neighbor state is **Full**, indicating that the adjacency has been established.

Verification (2)

- On R1, check the routing table and ping 3.3.3.3 from 1.1.1.1.

The route 3.3.3.3/32 is learned from OSPF.

Ping 3.3.3.3 from 1.1.1.1.

```
<R1> display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 10    Routes : 10
Destination/Mask    Proto  Pre  Cost  Flags NextHop         Interface
1.1.1.1/32          Direct  0    0      D  127.0.0.1       LoopBack0
3.3.3.3/32          OSPF  10  2    D  10.1.12.2      GE 0/0/0
10.1.12.0/30        Direct  0    0      D  10.1.12.1       GE 0/0/0
...
<R1> ping -a 1.1.1.1 3.3.3.3
PING 3.3.3.3: 56 data bytes, press CTRL_C to break
Reply from 3.3.3.3: bytes=56 Sequence=1 ttl=254 time=50 ms
...
```


Quiz

1. (Multiple-answer question) On a broadcast network, which of the following states are considered stable for the neighbor relationship between OSPF routers? ()
 - A. ExStart
 - B. 2-Way
 - C. Exchange
 - D. Full
2. (Multiple-answer question) Which of the following pairs of routers will establish an adjacency? ()
 - A. Two routers on a P2P link
 - B. DR and BDR on a broadcast network
 - C. Two DROthers on an NBMA network
 - D. BDR and DROther on a broadcast network

1. BD
2. ABD

Summary

- OSPF is one of the most widely used routing protocols on live networks. This course provides an initial overview of OSPF's fundamental concepts, working principles, and basic configuration.
- Key concepts of OSPF include router ID, area, OSPF neighbor table, LSDB, and OSPF routing table. Understanding how OSPF neighbor relationships and adjacencies are established is essential to mastering link-state routing protocols.
- Advanced OSPF concepts include LSA types, SPF calculation mechanism, and special OSPF areas, among others. For further your expertise, continue with the Huawei HCIP-Datacom certification.

Thank you.

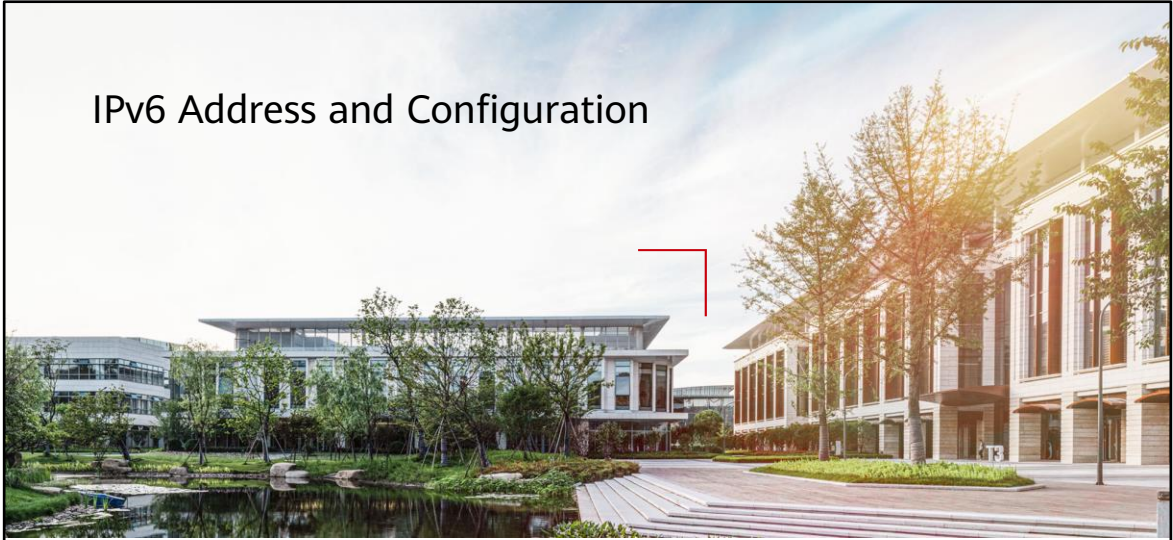
把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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IPv6 Address and Configuration



Foreword

- IPv4 is a widely used Internet protocol. When the Internet was starting up, IPv4 developed rapidly due to its simple design, easy implementation, and good interoperability. However, with the expansion of the Internet and the development of new technologies such as 5G and Internet of Things (IoT), IPv4 faces more and more challenges.
- IPv6 is an upgraded version of IPv4. It addresses many defects of IPv4, such as address scarcity, large routing tables, and insufficient support for mobile devices.
- In addition to providing similar functions as ICMP, ICMPv6 has many extensions.
- One of the prominent features of IPv6 is that it supports address autoconfiguration for network nodes, greatly simplifying the work of network administrators.
- This course describes basic IPv6 concepts, ICMPv6, and IPv6 address autoconfiguration.

Objectives

- Upon completion of this course, you will be able to:
 - Describe the advantages of IPv6 over IPv4.
 - Describe the format of an IPv6 packet header.
 - Describe IPv6 address types.
 - Describe the functions of ICMPv6.
 - Describe the fundamentals of IPv6 address autoconfiguration.

Contents

1. Overview of IPv6

- IPv6 Address Classification
 - IPv6 Packet Format

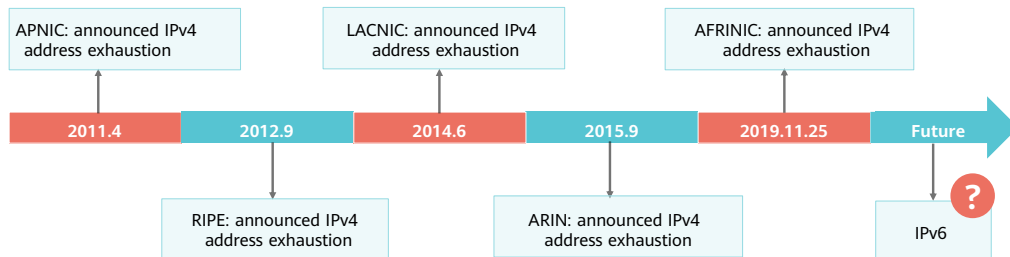
2. ICMPv6

3. IPv6 Address Autoconfiguration

4. IPv6 Address Configuration Example

IPv4 Addresses Exhaustion

- On February 3, 2011, the Internet Assigned Numbers Authority (IANA) announced even allocation of its last 4.68 million IPv4 addresses to five Regional Internet Registries (RIRs) around the world. The IANA thereafter had no available IPv4 address resources.



- The IANA is the organization responsible for assigning global Internet IP addresses. The IANA assigns some IPv4 addresses to continent-level RIRs, and then each RIR assigns addresses in its regions. The five RIRs are as follows:
 - RIPE: Réseaux IP Européens, which is a European IP address registration center and serves Europe, Middle East, and Central Asia.
 - LACNIC: Latin American and Caribbean Internet Address Registry, which is an Internet address registration center for Latin America and the Caribbean and serves the Central America, South America, and the Caribbean.
 - ARIN: American Registry for Internet Numbers, which is an Internet number registration center in the United States and serves North America and some Caribbean regions.
 - AFRINIC: Africa Network Information Centre, which serves Africa.
 - APNIC: Asia Pacific Network Information Centre, which serves Asia and the Pacific.
- IPv4 has proven to be a very successful protocol. It has survived the development of the Internet from a small number of computers to hundreds of millions of computers. However, the protocol was designed decades ago based on the network scale at that time. From today's view, the designers of IPv4 have not fully estimated and anticipated the Internet. With the expansion of the Internet and the launch of new applications, IPv4 has shown its limitations.
- The rapid expansion of the Internet scale was unforeseen at that time. Especially over the past decade, the Internet has experienced explosive growth and has been accessed by numerous households. It has become a necessity in people's daily life. In this case, IPv4 address exhaustion is becoming an urgent issue.
- In the 1990s, the IETF launched technologies such as Network Address Translation (NAT) and Classless Inter-Domain Routing (CIDR) to delay IPv4 address exhaustion. However, these transition solutions can only slow down the

speed of address exhaustion, but cannot fundamentally solve the issue.

Why Is IPv6 Used?

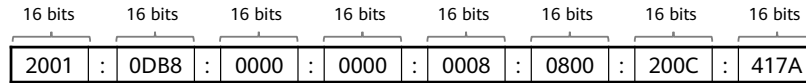
Nearly infinite address space	The 128-bit address length provides numerous addresses, meeting the requirements of emerging services such as IoT and facilitating service evolution and expansion.
Hierarchical address structure	IPv6 addresses are allocated more properly than IPv4 addresses, facilitating route summarization (reducing the size of IPv6 routing tables) and fast route query.
Plug and play	IPv6 supports stateless address autoconfiguration (SLAAC), simplifying terminal access.
Simplified packet header	The simplified packet header improves efficiency. Extension headers are used to support new applications, facilitating forwarding and processing of network devices such as routers and reducing investment costs.
Security features	Internet Protocol Security (IPsec), source address authentication, and other security features ensure end-to-end security, preventing NAT from damaging the integrity of end-to-end communication.
Mobility	The real-time communication of a mobile network is greatly enhanced, improving network performance.
Enhanced Quality of Service (QoS) features	The Flow Label field is additionally defined, and can be used by an application or a terminal. Specific resources are allocated for special services and data flows.

- **Nearly infinite address space:** This is the most obvious advantage over IPv4. An IPv6 address is 128 bits long. The number of IPv6 addresses is approximately 8×10^{28} times that of IPv4 addresses. It is claimed that each grain of sand on the earth can be assigned an IPv6 address. This makes it possible for a large number of terminals to be online at the same time and unified addressing management to be implemented, providing strong support for the Internet of Everything (IoE).
- **Hierarchical address structure:** IPv6 addresses are divided into different address segments based on application scenarios because of the nearly infinite address space. In addition, the continuity of unicast IPv6 address segments is strictly required to prevent "holes" in IPv4 address segments. This facilitates IPv6 route summarization and reduces the size of IPv6 address tables.
- **Plug and play:** Any host or terminal must have a specific IP address to obtain network resources and transmit data. Traditionally, IP addresses are assigned manually or automatically using DHCP. In addition to the preceding two methods, IPv6 supports SLAAC.
- **End-to-end network integrity:** Using NAT widely on IPv4 networks damages the integrity of end-to-end connections. After IPv6 is used, NAT devices are no longer required, and online behavior management and network monitoring become simple. In addition, complex NAT adaptation code does not need to be developed for applications.
- **Enhanced security:** IPsec was initially designed for IPv6. Therefore, IPv6-based protocol packets (such as routing protocol and neighbor discovery packets) can be encrypted in end-to-end mode. Currently, this function is not widely used. The security of IPv6 data plane packets is basically the same as that of IPv4 packets + IPsec.

- High scalability: IPv6 extension headers are not a part of the main data packet. However, if necessary, the extension headers can be inserted between the IPv6 basic header and payload to assist IPv6 in encryption, mobility, optimal path selection, and QoS, improving packet forwarding efficiency.
- Improved mobility: When a user is moved from one network segment to another, traffic can be directly forwarded through routes, reducing the cost of traffic forwarding and improving network performance and reliability.
- Further enhanced QoS: IPv6 retains all the QoS attributes of IPv4 and additionally defines a 20-bit Flow Label field, which can be used by applications or terminals. This field can be used to allocate specific resources to special services and data flows. Currently, this mechanism is not fully developed and applied.
- QoS is a network resource management technology that aims to ensure the transmission quality of key service flows through differentiated service policies.

IPv6 Address

- An IPv6 address is 128 bits long. Generally, an IPv6 address is divided into eight groups by colon, and each group contains 16 bits and is expressed in hexadecimal notation.



The letters in an IPv6 address are case insensitive. For example, **A** is equivalent to **a**.

- Similar to an IPv4 address, an IPv6 address is expressed in the format of IPv6 address/mask length.
 - Example: 2001:0DB8:2345:CD30:1230:4567:89AB:CDEF/64
IPv6 address: 2001:0DB8:2345:CD30:1230:4567:89AB:CDEF
Subnet ID: 2001:0DB8:2345:CD30::/64

IPv6 Address Abbreviation Specifications

- For convenience, IPv6 addresses can be abbreviated according to the following rules:

IPv6 address abbreviation specifications

2001	:	0DB8	:	0000	:	0000	:	0008	:	0800	:	200C	:	417A
------	---	------	---	------	---	------	---	------	---	------	---	------	---	------

The leading 0s in each 16-bit group can be omitted. However, if all bits in a 16-bit group are 0s, at least one 0 must be reserved. The trailing 0s cannot be omitted.

2001	:	DB8	:	0	:	0	:	8	:	800	:	200C	:	417A
------	---	-----	---	---	---	---	---	---	---	-----	---	------	---	------

If one or more consecutive 16-bit groups contain only 0s, they can be replaced using a double colon (::). However, this substitution can only be applied once in an IPv6 address.

2001	:	DB8	:	::	:	8	:	800	:	200C	:	417A
------	---	-----	---	----	---	---	---	-----	---	------	---	------

If an abbreviated IPv6 address contains two double colons (::), the IPv6 address cannot be restored to the original one.

IPv6 address abbreviation examples

Before abbreviation:

0000:0000:0000:0000:0000:0000:0000:0001

After abbreviation: ::1

Before abbreviation:

2001:0DB8:0000:0000:FB00:1400:5000:45FF

After abbreviation: 2001:DB8:FB00:1400:5000:45FF

Before abbreviation:

2001:0DB8:0000:0000:0000:2A2A:0000:0001

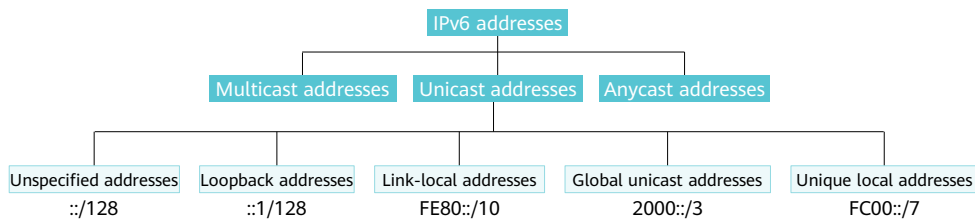
After abbreviation: 2001:DB8::2A2A:0:1

Before abbreviation:

2001:0DB8:0000:1234:FB00:0000:5000:45FF

After abbreviation: 2001:DB8::1234:FB00:0:5000:45FF
or 2001:DB8:0:1234:FB00::5000:45FF

IPv6 Address Types



- Unicast address: identifies an interface. A packet destined for a unicast address is sent to the interface having that unicast address. In IPv6, it is common that an interface has multiple IPv6 addresses.
- Multicast address: identifies multiple interfaces. A packet destined for a multicast address is sent to all the interfaces having that multicast address. Only the interfaces that join a multicast group listen to the packets destined for the corresponding multicast address.
- Anycast address: identifies a group of network interfaces (usually on different nodes). A packet sent to an anycast address is routed to the nearest interface having that address, according to routing results.
- No broadcast addresses are defined in IPv6.

- Unspecified address: 0:0:0:0:0:0:0:0/128 or ::/128. The address is used as the source address of some packets, for example, Neighbor Solicitation (NS) messages sent during duplicate address detection or request packets sent by a client during DHCPv6 initialization.
- Loopback address: 0:0:0:0:0:0:0:1/128 or ::1/128, which is used for local loopback (same function as 127.0.0.1 in IPv4). The data packets sent to ::1 are actually sent to the local end and can be used for loopback tests of local protocol stacks.

IPv6 Unicast Address Structure

- An IPv6 unicast address is composed of two parts:
 - Network prefix: equivalent to the network ID of an IPv4 address. The length of the network prefix is variable (n bits).
 - Interface ID: equivalent to the host ID of an IPv4 address. The length of the interface ID is 128 bits minus the length of the network prefix.
- It is required that both the network prefix and interface ID be 64 bits for common IPv6 unicast addresses, such as global unicast addresses and link-local addresses.



- Global unicast addresses that start with the binary value 000 can use a non-64-bit network prefix. Such addresses are not covered in this course.

Interface ID of an IPv6 Address

- An interface ID can be manually configured, automatically generated by the system, or generated based on the IEEE EUI-64 specification.
- The method of generating an interface ID based on the IEEE EUI-64 specification is most commonly used. In this method, the MAC address of an interface is converted into the interface ID of an IPv6 address.

MAC address (hexadecimal)

08-70-5A-90-1A-01

MAC address (binary)

The seventh bit is 0, indicating that the MAC address is a global management address.

00001000 - 01110000 - 01011010 - 10010000 - 00011010 - 00000001

2 The seventh bit is set to 1.

FFFE is inserted at this position.

00001000 - 01110000 - 01011010 - 11111111 - 11111110 - 10010000 - 00011010 - 00000001

00001010 - 01110000 - 01011010 - 11111111 - 11111110 - 10010000 - 00011010 - 00000001

Interface ID generated based on the IEEE EUI-64 specification

0A - 70 - 5A - FF - FE - 90 - 1A - 01

- Currently, the interface ID of an IPv6 address can be generated in the following ways:
 - Generated based on the IEEE EUI-64 specification
 - The typical length of an interface ID is 64 bits. The IEEE EUI-64 specification defines a method of automatically generating a 64-bit interface ID based on a 48-bit MAC address.
 - The conversion algorithm is as follows: Set the seventh bit to 1 and insert two bytes (FFFE) into the middle (at the 24th bit) of the MAC address.
 - This method of generating an interface ID based on a MAC address can reduce the configuration workload. Only an IPv6 prefix needs to be obtained to form an IPv6 address with the interface ID.
 - The biggest disadvantage of this method is that attackers can deduce Layer 3 IPv6 addresses based on Layer 2 MAC addresses.
 - Randomly generated by a device
 - The device randomly generates an interface ID. Currently, the Windows operating system uses this method.
 - Manually configured

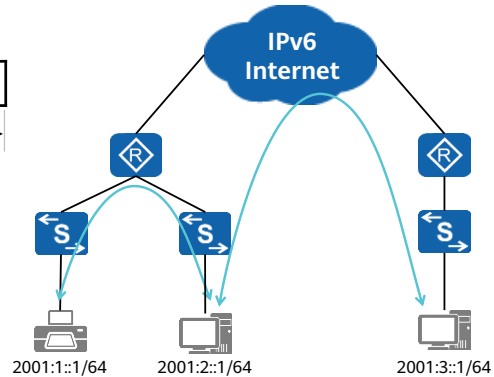
- An interface ID can be manually specified.

Global Unicast Address

- A global unicast address (GUA) is also called an aggregatable GUA. Addresses of this type are globally unique and are used by hosts that need to access the Internet. They are equivalent to public IPv4 addresses.

3 bits	45 bits	16 bits	64 bits
001	Global routing prefix	Subnet ID	Interface ID
Network portion			Host portion

- Generally, the network portion of a GUA is 64 bits long, and the interface ID is also 64 bits long.
- Global routing prefix: is assigned by a provider to an organization and is generally at least 45 bits long.
- Subnet ID: is used by an organization to perform subnetting based on network requirements.
- Interface ID: identifies a device's interface.



- You can apply for a GUA from a carrier or the local IPv6 address management organization.

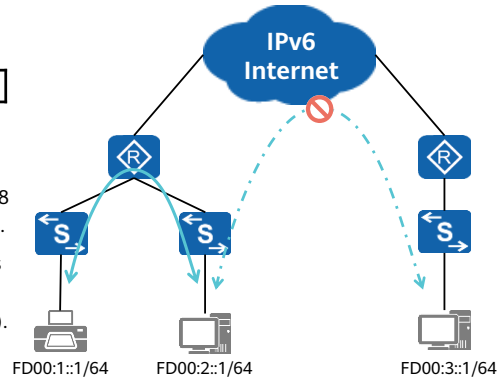
Unique Local Address

- A unique local address (ULA) is a private IPv6 address that can be used only on an intranet. Addresses of this type cannot be routed on an IPv6 public network and therefore cannot be used to directly access a public network.

8 bits	40 bits	16 bits	64 bits
1111 1101	Global ID	Subnet ID	Interface ID

Generated using a pseudo-random algorithm

- ULAs use the FC00::/7 address block in which only FD00::/8 is currently used. FC00::/8 is reserved for future expansion.
- Although a ULA is valid only in a limited scope, it also has a globally unique prefix (generated using a pseudo-random algorithm, but the conflict probability is very low).

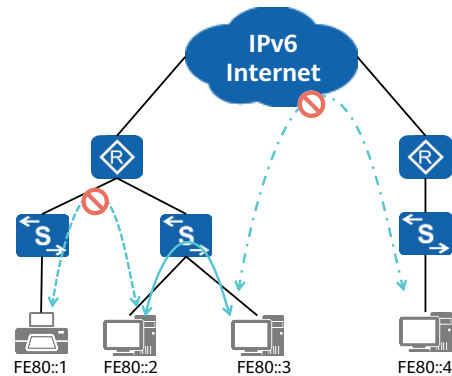


Link-Local Address

- A link-local address (LLA) is another type of IPv6 address with a limited application scope. The valid scope of an LLA is the local link, with the prefix of FE80::/10.

10 bits	54 bits	64 bits
1111 1110 10	0	Interface ID
Fixed at 0		

- An LLA is used for communication on a single link, such as during IPv6 SLAAC and IPv6 neighbor discovery.
- Data packets with the source or destination IPv6 address being an LLA are not forwarded out of the originating link. In other words, the valid scope of an LLA is the local link.
- Each IPv6 interface must have an LLA. Huawei devices support automatic generation and manual configuration of LLAs.



IPv6 Multicast Address

- An IPv6 multicast address identifies a multicast group. Packets destined for a multicast address are sent to members in the multicast group. A multicast address is composed of the prefix (FF::/8) as well as the Flags, Scope, and Group ID fields.

8 bits	4 bits	4 bits	80 bits	32 bits
11111111	Flags	Scope	Reserved (The value must be 0.)	Group ID

- Flags**

- Indicates a permanent or temporary multicast address.
 - The value 0000 indicates a permanent multicast address.
 - The value 0001 indicates a temporary multicast address.

- Scope**

- Indicates the scope of a multicast group.

- Group ID**

- Indicates the multicast group ID.

0: reserved

1: interface-local scope, which spans only a single interface on a node and is useful only for loopback transmission of multicast

2: link-local scope (for example, FF02::1)

5: site-local scope

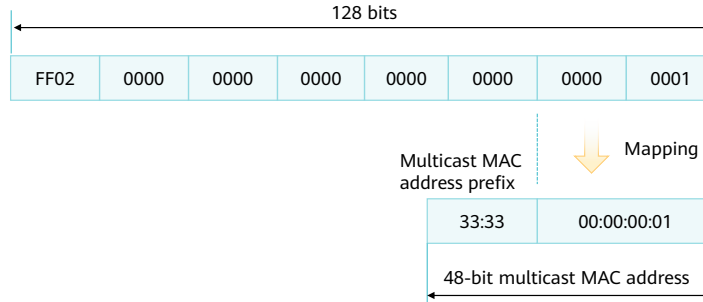
8: organization-local scope

E: global scope

F: reserved

IPv6 Multicast MAC Address

- The destination IP address of an IPv6 multicast packet is an IPv6 multicast address, and the destination MAC address is a multicast MAC address.
- The first 16 bits of a multicast MAC address are 33:33, which is the MAC address prefix reserved for IPv6 multicast. The last 32 bits are directly mapped from the last 32 bits of a multicast IPv6 address.

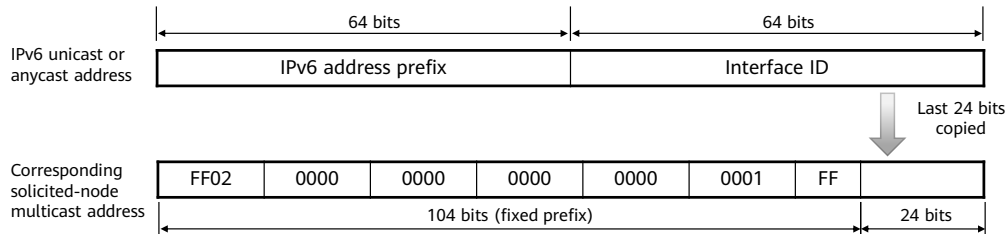


- When unicast IP packets are transmitted on an Ethernet network, the destination MAC address is the MAC address of the next hop. However, when multicast packets are transmitted, their destination is a group of unspecific members but not a specific receiver. Therefore, the destination MAC address is a multicast MAC address.
- IPv4 multicast MAC address
 - As defined by the IANA, the 24 most significant bits of an IPv4 multicast MAC address are 0x01005E, the 25th bit is 0, and the 23 least significant bits are mapped from the 23 least significant bits of an IPv4 multicast address.
 - The four most significant bits of an IPv4 multicast address are 1110, which indicates multicast. However, only 23 bits of the 28 least significant bits of the IPv4 multicast address are mapped to an IPv4 multicast MAC address. As a result, five bits of the IPv4 multicast address are lost. Therefore, 32 IPv4 multicast addresses are mapped to the same IPv4 multicast MAC address. During Layer 2 processing, a device may need to receive multicast data from multicast groups other than the local IPv4 multicast group. In this case, the redundant multicast data needs to be filtered by the upper layer.
- IPv6 multicast MAC address
 - When an IPv6 multicast packet is transmitted over an Ethernet link, the corresponding MAC address is 0x3333-*X-X-X-X*, where *X-X-X-X* is the direct

mapping of the last 32 bits of the multicast IP address.

Solicited-Node Multicast Address

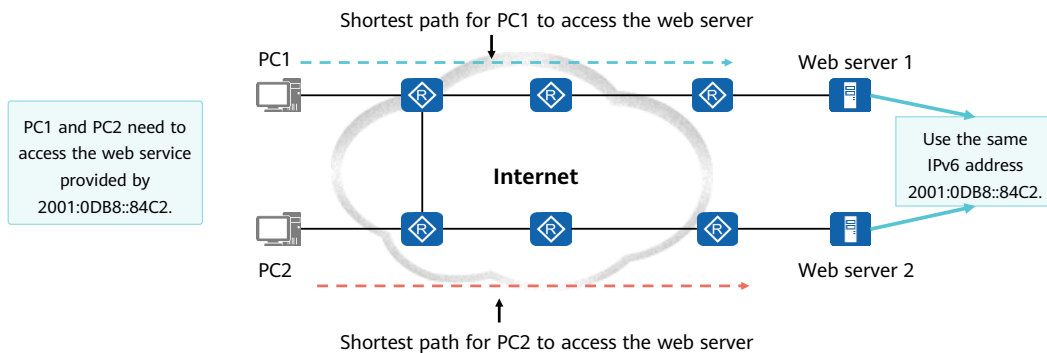
- If a node has an IPv6 unicast or anycast address, a solicited-node multicast address is generated for the node, and the node joins the corresponding multicast group. This address is mainly used for neighbor discovery and duplicate address detection. A solicited-node multicast address is valid only on the local link.



- An application scenario example of a solicited-node multicast address is as follows: In IPv6, ARP and broadcast addresses are canceled. When a device needs to request the MAC address corresponding to an IPv6 address, the device still needs to send a request packet, which is a multicast packet. The destination IPv6 address of the packet is the solicited-node multicast address corresponding to the target IPv6 unicast address. Because only the target node listens to the solicited-node multicast address, the multicast packet is received only by the target node, without affecting the network performance of other non-target nodes.

IPv6 Anycast Address

- An anycast address identifies a group of network interfaces, which usually belong to different nodes.



- The anycast process involves an anycast packet initiator and one or more responders.
 - An initiator of an anycast packet is usually a host requesting a service (for example, a web service).
 - The format of an anycast address is the same as that of a unicast address. The only difference is that a device can send packets to multiple devices with the same anycast address.
- Anycast addresses have the following advantages:
 - Provide service redundancy. For example, a user can obtain the same service (for example, a web service) from multiple servers that use the same anycast address. These servers are all responders of anycast packets. If no anycast address is used and a server fails, the user needs to obtain the address of another server to establish communication again. If an anycast address is used and a server fails, the user can automatically communicate with another server that uses the same address, to implement service redundancy.
 - Provide better services. For example, a company deploys two servers — one in province A and the other in province B — to provide the same web service. Based on the optimal route selection rule, users in province A preferentially access the server deployed in province A when accessing the web service provided by the company. This increases the access speed,

reduces the access delay, and greatly improves user experience.

Comparison Between IPv6 and IPv4 Addresses

Item	IPv4	IPv6
Address space	2^{32}	2^{128}
Expression mode	Dotted decimal notation	Colon-hexadecimal notation
Address type	Unicast, multicast, and broadcast	Unicast, multicast, and anycast
Others	Classes A, B, and C addresses and addresses of other classes	There is no such concept in IPv6.
	Multicast address (224.0.0.0/4)	IPv6 multicast address (FF00::/8)
	Broadcast address	There is no such concept in IPv6.
	Unspecified address 0.0.0.0/32	Unspecified address ::/128
	Loopback address 127.0.0.0/8	Loopback address ::1/128
	Public IP address	Global unicast address
	Private IP addresses (10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16)	Unique local address (FD00::/8)
	APIPA address (169.254.0.0/16)	Link-local address (FE80::/10)

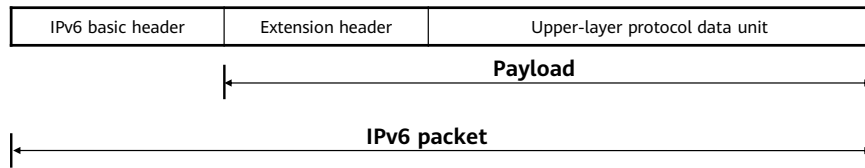
- Automatic Private IP Addressing (APIPA) is a DHCP failover mechanism. When the DHCP server fails, an IP address within the private IP address range from 169.254.0.1 to 169.254.255.254 is automatically assigned using APIPA, and the default subnet mask 255.255.0.0 is used. A client uses the ARP protocol to check and ensure that the selected address is unique on the LAN. APIPA can implement automatic configuration of the TCP/IP protocol on a single network segment without a DHCP server.

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1. Overview of IPv6

- IPv6 Address Classification
 - IPv6 Packet Format
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- 3. IPv6 Address Autoconfiguration
- 4. IPv6 Address Configuration Example

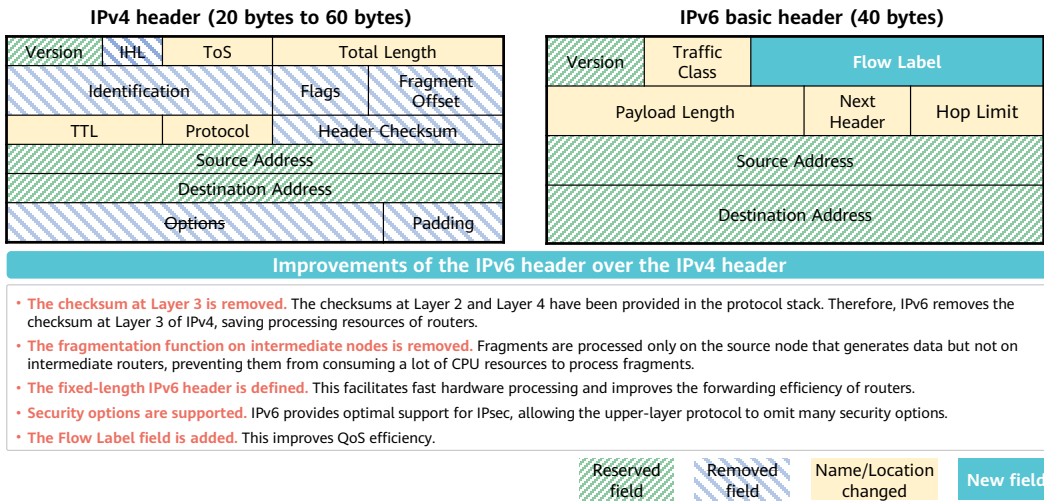
IPv6 Packet Composition



- An IPv6 packet is generally composed of the following parts:
 - IPv6 basic header: provides basic packet forwarding information. Routers use the information to forward most packets.
 - Extension headers: provide extended packet forwarding information, such as segmentation and encryption. This part is optional and does not need to be processed by each router. A sender adds one or more extension headers only when a router or destination node needs to perform special processing.
 - Upper-layer protocol data unit: is generally composed of an upper-layer protocol header and its payload. This part is similar to the upper-layer protocol data unit in an IPv4 packet.

- As shown in the figure, an IPv6 packet is composed of the following parts:
 - IPv6 basic header
 - Each IPv6 packet must contain a header with a fixed length of 40 bytes.
 - The basic header provides basic packet forwarding information and is parsed by all routers on the forwarding path.
 - Extension headers
 - IPv6 extension headers are optional headers that may follow the IPv6 basic header. An IPv6 packet can contain no extension header, or it can contain one or more extension headers with different lengths. The IPv6 basic header and extension headers replace the IPv4 header and its options. Extension headers enhance the functions of IPv6 and make IPv6 extensible. Unlike the options in an IPv4 header, the maximum length of an IPv6 extension header is not limited. Therefore, an extension header can contain all the extension data required for IPv6 communication. An extension header provides extended packet forwarding information and is not parsed by all routers on the forwarding path. Generally, it is parsed only by the destination router.
 - Upper-layer protocol data unit
 - An upper-layer protocol data unit is generally composed of the upper-layer protocol header and its payload, which can be an ICMPv6 packet, a TCP packet, or a UDP packet.

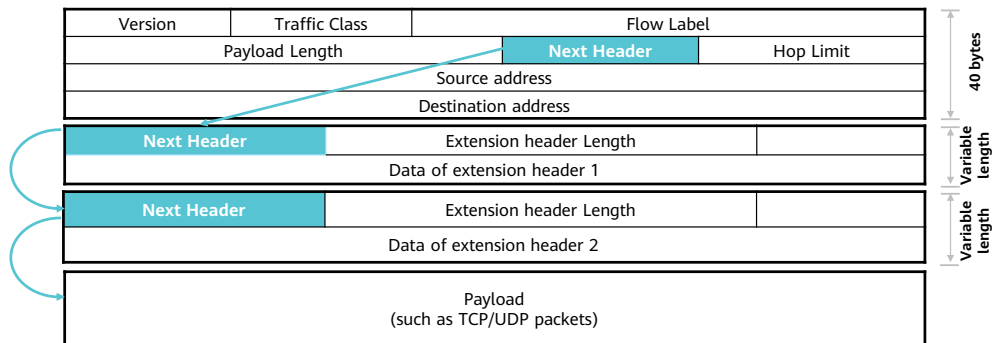
IPv6 Basic Header



- The IPv6 basic header is also called the fixed header, which contains eight fields. The total length of the fixed header is 40 bytes. The eight fields are Version, Traffic Class, Flow Label, Payload Length, Next Header, Hop Limit, Source Address, and Destination Address.
- Version
 - This field indicates the version of IP and its value is 6. The length is 4 bits.
- Traffic Class
 - This field indicates the class or priority of an IPv6 packet and its function is similar to that of the ToS field in the IPv4 header. The length is 8 bits.
- Flow Label
 - This field is a new field, which is not contained in an IPv4 header. It is used to identify a specific packet sequence between the source and destination nodes. This field requires special processing by intermediate IPv6 routers. The length of this field is 20 bits. Generally, the flow label, together with the source and destination IPv6 addresses, can determine a flow.
- Payload Length
 - This field indicates the length of the payload in an IPv6 packet. The payload refers to the extension header and upper-layer protocol data unit that follow the IPv6 basic header. This field is 16 bits long and can indicate the payload with a maximum length of 65535 bytes. If the payload length exceeds 65535 bytes, this field is set to 0, and the Jumbo Payload option in the Hop-by-Hop Options header is used to express the actual payload length.

- Next Header
 - This field defines the type of the first extension header (if any) that follows the IPv6 basic header or the protocol type in the upper-layer protocol data unit. The length of this field is 8 bits.
- Hop Limit
 - This field is similar to the Time to Live (TTL) field in the IPv4 header. It defines the maximum number of hops that an IP packet can pass through. The value is decreased by 1 each time an IP packet passes through a router. The packet is discarded if Hop Limit is decreased to zero. The length of this field is 8 bits.
- Source Address
 - This field indicates the address of the sender. The length is 128 bits.
- Destination Address
 - This field indicates the address of the receiver. The length is 128 bits.

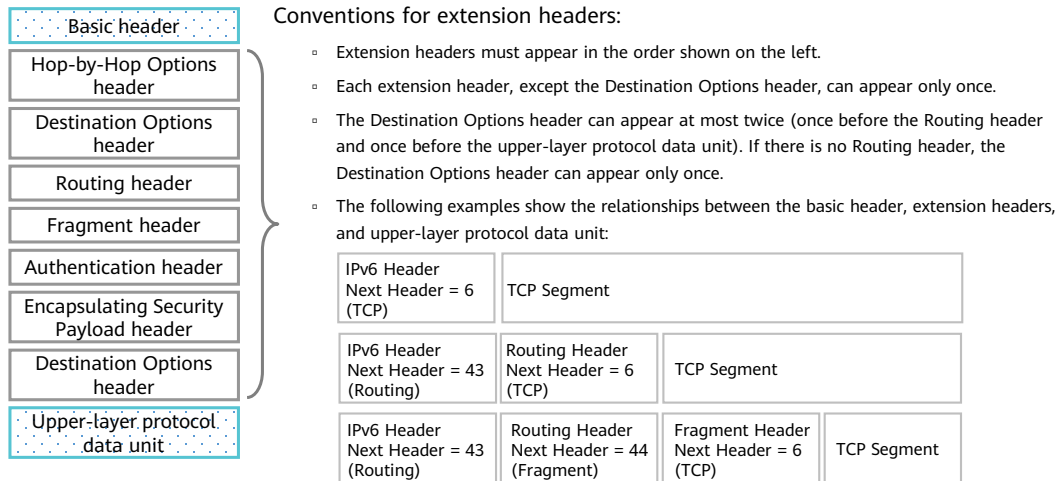
IPv6 Extension Headers (1)



- **Next Header:** The length is 8 bits. This field is similar to the Next Header field in the IPv6 basic header, indicating the type of the next extension header (if any) or upper-layer protocol.
- **Extension Header Length:** The length is 8 bits. This field indicates the extension header length excluding the length of the Next Header field.
- **Extension Header Data:** The length is variable. This field includes a series of options and the padding field.

- The Options field in an IPv4 header is placed in extension headers of an IPv6 packet. IPv6 extension headers are optional headers that may follow the IPv6 basic header. Why are extension headers designed in IPv6? Each intermediate router must check whether the options contained in the IPv4 header exist. If the options exist, the intermediate router must process them. This reduces the efficiency for routers to forward IPv4 packets. Therefore, the Options field is placed in extension headers in IPv6 to resolve this issue. In this case, the intermediate router does not need to process each possible option, accelerating packet processing and improving forwarding performance.
- A typical IPv6 packet does not contain any extension header. A sender adds one or more extension headers only when a router or destination node needs to perform special processing. Unlike IPv4, IPv6 has variable-length extension headers, which are not limited to 40 bytes, to facilitate further extension. To improve extension header processing efficiency and transport protocol performance, IPv6 requires that the extension header length be an integral multiple of 8 bytes.

IPv6 Extension Headers (2)



- Currently, RFC 2460 defines the following six IPv6 extension headers:
 - Hop-by-Hop Options header: This header is processed by each hop on the path. It can contain multiple options, such as the router alert option.
 - Destination Options header: This header is processed by the destination node. It can contain multiple options, such as the home address option of Mobile IPv6.
 - Routing header: This header is used to specify the source route, and is similar to the IPv4 source routing option. The source node of an IPv6 packet uses this header to specify the intermediate nodes that the packet must pass through to reach the destination. The destination address in the IPv6 basic header is not the final destination address of a packet but the first address listed in the Routing header.
 - Fragment header: This header contains information about IP packet fragmentation, and is processed only by the destination node.
 - Authentication header: This header is an IPsec extension header, and is processed only by the destination node.
 - Encapsulating Security Payload header: This header is an IPsec extension header, and is processed only by the destination node.
- The Hop-by-Hop Options and Destination Options headers provide option functions and support extensibility (such as mobility). Options are TLV-coded.
- If a packet does not contain any extension headers, that is, the packet contains only the basic header and the upper-layer protocol data unit, the value of the Next Header field in the basic header indicates the upper-layer protocol type. In the preceding examples, the value of the Next Header field in the basic header is

6, indicating that the upper-layer protocol is TCP. If the packet contains one extension header, the value of the Next Header field in the basic header indicates the extension header type. In the preceding examples, the value of the Next Header field in the basic header is 43, indicating that the extension header immediately following the basic header is the Routing header. The value of the Next Header field in the extension header indicates the upper-layer protocol type. Similarly, if the packet contains multiple extension headers, the value of the Next Header field in each extension header indicates the type of the extension header immediately following it, and the value of the Next Header field in the last extension header indicates the upper-layer protocol.

IPv6 Packet Processing Mechanism

IPv6 Basic Header Next Header = 0 (Hop-by-Hop Options Header)
IPv6 Hop-by-Hop Options Header Next Header = 51 (Authentication Header)
IPv6 Authentication Header Next Header = 6 (TCP)
TCP Segment

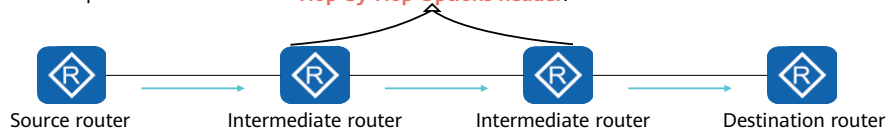
Constructs an IPv6
packet as required.

IPv6 Basic Header Next Header = 0 (Hop-by-Hop Options Header)
IPv6 Hop-by-Hop Options Header Next Header = 51 (Authentication Header)
IPv6 Authentication Header Next Header = 6 (TCP)
TCP Segment

Process the **basic header** and
Hop-by-Hop Options header.

IPv6 Basic Header Next Header = 0 (Hop-by-Hop Options Header)
IPv6 Hop-by-Hop Options Header Next Header = 51 (Authentication Header)
IPv6 Authentication Header Next Header = 6 (TCP)
TCP Segment

Processes all headers.



- The length of the basic header is fixed, **improving forwarding efficiency**.
- The extension headers **meet special requirements**.

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Overview of ICMPv6

- ICMPv6 is one of the basic IPv6 protocols.
- ICMPv6 messages are widely used in other protocols, such as NDP and path MTU discovery.
- ICMPv6 controls key IPv6 processes, such as address autoconfiguration, address resolution, address conflict detection, route selection, and error control.
- ICMPv6 messages are classified into error messages and informational messages.

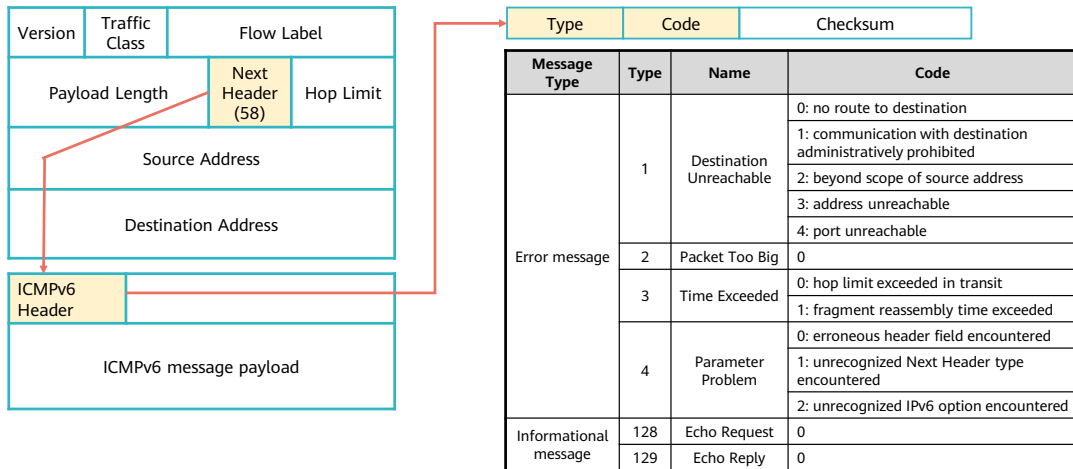
Error messages

Error messages are used to report errors that occur during IPv6 packet forwarding, such as destination unreachable and timeout.

Informational messages

Informational messages can be used to implement functions such as inter-node communication on a link and multicast member management on a subnet.

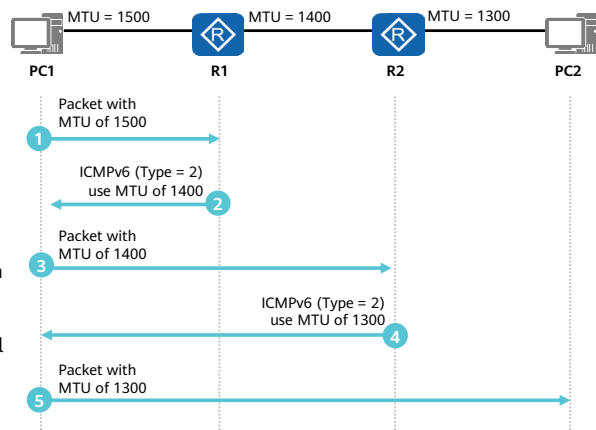
ICMPv6 Message Format



- If the value of the Next Header field in an IPv6 header is 58, the message is an ICMPv6 message.
- Type: indicates the message type. The value range [0, 127] indicates error messages, and the value range [128, 255] indicates informational messages.
- Code: provides an additional level of message granularity.
- Checksum: indicates the checksum of an ICMPv6 message.

PMTU

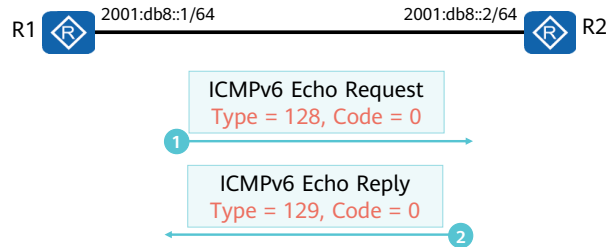
- In IPv6, an intermediate forwarding device does not fragment IPv6 packets, and packet fragmentation is performed on the source node.
- The path MTU (PMTU) is the minimum interface MTU on a path.
- The path MTU discovery (PMTUD) mechanism is used to discover the MTU on the path from the source to the destination, preventing packets from being fragmented during forwarding.
- With PMTUD, the data sender can use the optimal PMTU discovered to communicate with the destination node. This prevents packets from being fragmented by intermediate routers during transmission from the source to the destination.



1. PC1 sends an IPv6 packet with an MTU of 1500 bytes to PC2.
2. After checking that the packet is too large and the MTU of the outbound interface is 1400 bytes, R1 sends an ICMPv6 (Type = 2) message indicating that the MTU is 1400 bytes to PC1.
3. PC1 sends an IPv6 packet with an MTU of 1400 bytes.
4. After the packet reaches R2, R2 checks that the MTU of the outbound interface is 1300 bytes and sends an ICMPv6 (Type = 2) message indicating that the MTU is 1300 bytes to PC1.
5. PC1 sends an IPv6 packet with an MTU of 1300 bytes.

IPv6 Ping

- Ping is implemented based on ICMPv6 informational messages.
 - Echo Request: is sent to a destination node, which responds with an Echo Reply message. The Type and Code field values in an Echo Request message are 128 and 0, respectively.
 - Echo Reply: After receiving an Echo Request message, ICMPv6 responds with an Echo Reply message. The Type and Code field values in an Echo Reply message are 129 and 0, respectively.

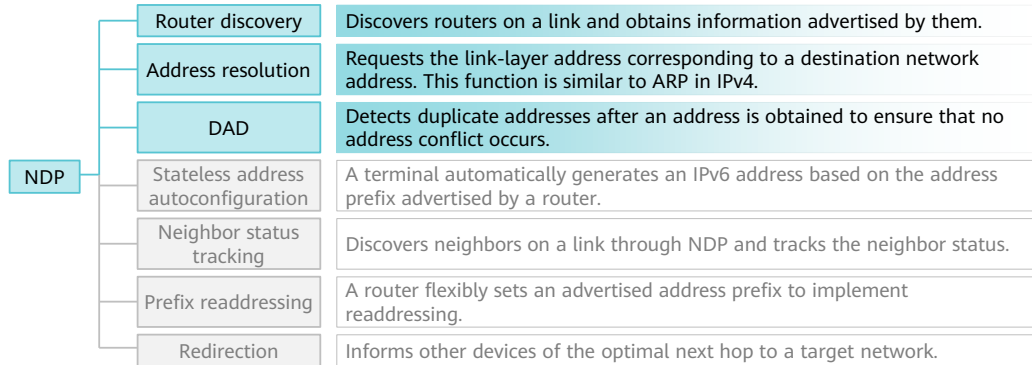


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Overview of NDP

- The Neighbor Discovery Protocol (NDP) is an important basic protocol in the IPv6 protocol system. It defines the use of ICMPv6 messages to implement functions such as router discovery, Duplicate Address Detection (DAD), and address resolution.



- This course describes only the router discovery, address resolution, and DAD functions.

NDP Message Types and Functions

- NDP uses the following types of ICMPv6 messages:

- Router Solicitation (RS)
- Router Advertisement (RA)
- Neighbor Solicitation (NS)
- Neighbor Advertisement (NA)

Function \ ICMPv6 Message	RS 133	RA 134	NS 135	NA 136	Redirect (Type = 137)
Router discovery	•	•			
Address resolution			•	•	
DAD			•	•	

Router Discovery

- Router discovery is a process in which a host discovers a router on a local link and determines its configuration.

Functions

Router discovery: is a process in which a host locates neighboring routers and selects a router as the default gateway.

Prefix discovery: is a process in which a host discovers a group of IPv6 prefixes on a local link for address autoconfiguration of the host.

Parameter discovery: is a process in which a host discovers operation parameters, such as the default limit on the number of hops allowed for outgoing packets and address configuration mode.

Messages

RS

RA

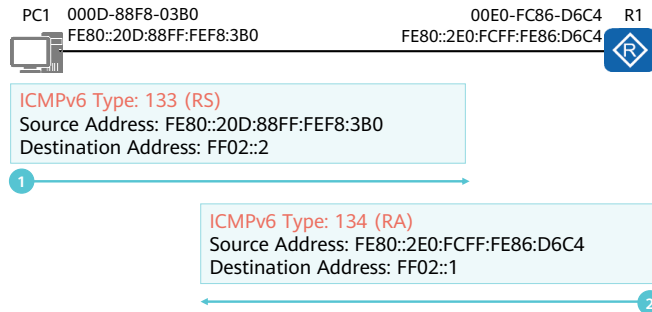
Modes

A host sends an RS message to trigger a router to respond with an RA message.

A router periodically sends RA messages.

Triggered by RS Messages Sent by a Host

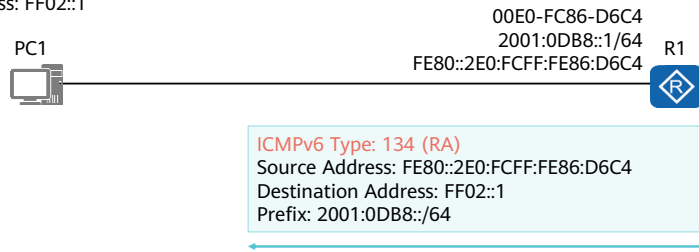
- When a host starts, it sends RS messages to all the routers on a local link, triggering them to respond with RA messages. After discovering routers on the local link, the host automatically configures a router as the default router. In addition, the host sets up the default routing table and prefix list, and sets other configuration parameters.



1. A host starts and sends RS messages to all the routers on a local link.
2. The routers respond with RA messages.
3. After the preceding steps are performed, the host generates a default route with the next-hop address being the link-local address of the default router.

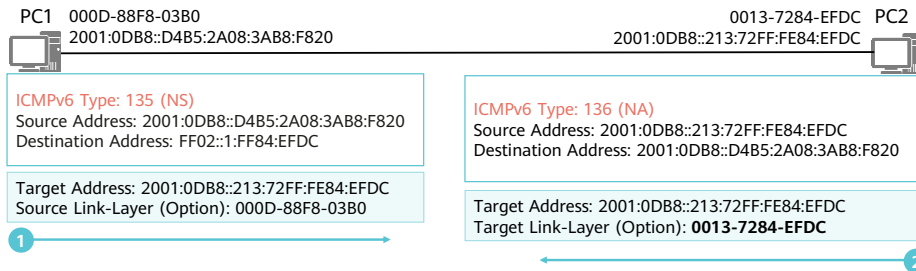
A Router Periodically Sends Messages

- A router periodically sends RA messages. The interval for sending RA messages is a random value within a specified range. The default maximum interval is 600s, and the default minimum interval is 200s.
- The addresses of these RA messages must meet the following requirements:
 - Source Address: It must be the link-local address of the outbound interface.
 - Destination Address: FF02::1



Address Resolution

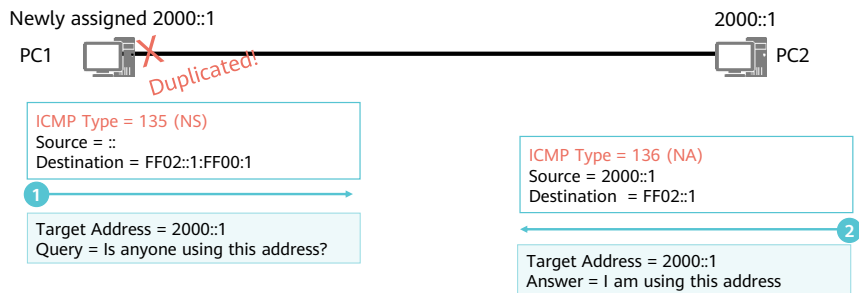
- IPv6 address resolution is implemented using ICMPv6 NS and NA messages.
- Layer 3 address resolution has the following advantages:
 - Layer 3 address resolution enables Layer 2 devices to use the same address resolution protocol.
 - Layer 3 security mechanisms can be used to prevent address resolution attacks.
 - NS messages are multicast, reducing loads on Layer 2 networks.



- In this example, if PC1 wants to send a packet to PC2 but does not know PC2's link-layer address, the following protocol exchange process needs to be performed:
 - PC1 sends an NS message to the network. The destination address of the message is the solicited-node multicast address (FF02::1:FF84:EFDC) corresponding to PC2, and the Options field contains PC1's link-layer address 000D-88F8-03B0.
 - After receiving the NS message, PC2 finds that the destination address of the message is FF02::1:FF84:EFDC, which is the multicast address of the group to which PC2 belongs. Therefore, PC2 processes the message and updates its neighbor cache entries based on the source address and Source Link-Layer address option in the NS message.
 - PC2 responds to the NS message with an NA message in which the Target Link-Layer option carries its link-layer address 0013-7284-EFDC.
 - After receiving the NA message, PC1 obtains PC2's link-layer address and creates a neighbor cache entry for the destination node.
- After the preceding process is complete, PC1 and PC2 obtain each other's link-layer addresses and create neighbor cache entries (similar to ARP entries in IPv4), so that they can communicate with each other.

DAD (1)

- Duplicate Address Detection (DAD) enables a node to detect whether an IPv6 address is used by another node before assigning it to an interface. This ensures that duplicate unicast addresses do not exist on a network.
- DAD must be performed before any unicast IPv6 address (including a link-local address) is used by an interface.



- DAD is a process in which a node determines whether the address to be used is used by another node. Before automatically configuring an IPv6 unicast address for an interface, a node must ensure that this address is unique on a local link and is not used by another node. The node sends an NS message to the local link once by default. If the node does not receive an NA message within a specified period, it considers that the temporary unicast address is unique on the local link and can be assigned to the interface. Otherwise, the temporary unicast address is duplicate with an existing one and cannot be assigned to the interface.

DAD (2)

- An address is called a tentative address before it passes DAD. An interface cannot use a tentative address for unicast communication.
- If DAD is performed for two nodes configured with the same address, the receiver does not use this address when receiving an NS message.



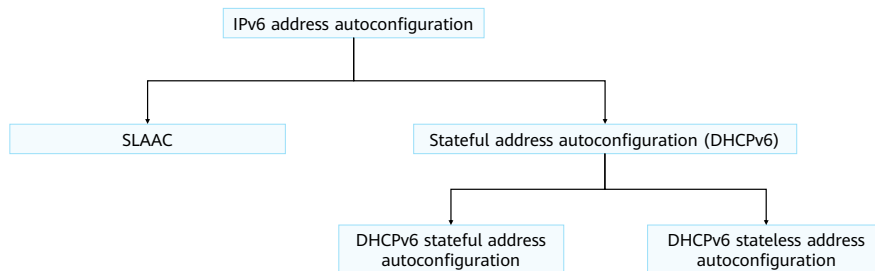
- Special scenario: Two hosts are assigned the same IP address. Assume that both PC1 and PC2 want to use the address 2000::1. If PC1 sends an NS message first, PC2 will not send an NS message (or an NA message) after receiving the NS message. Instead, PC2 does not use the address 2000::1 and wait for a new address to be generated. If both PC1 and PC2 receive an NS message, they do not use the address 2000::1.

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IPv6 Address Autoconfiguration Modes

- IPv6 address autoconfiguration can be classified into the following types:
 - Stateless address autoconfiguration (SLAAC): It is implemented based on NDP.
 - Stateful address autoconfiguration: The server that assigns IPv6 addresses must store the status information of each node and manage the stored information. Stateful address autoconfiguration is implemented based on the Dynamic Host Configuration Protocol for IPv6 (DHCPv6).



- DHCPv6 address allocation can be implemented in the following modes:
 - DHCPv6 stateful address autoconfiguration: A DHCPv6 server automatically allocates IPv6 addresses/PD prefixes and other network configuration parameters (such as DNS parameters).
 - DHCPv6 stateless address autoconfiguration: Host IPv6 addresses are automatically generated through RA messages. A DHCPv6 server allocates only configuration parameters (including DNS parameters) except IPv6 addresses.

How Does an IPv6 Client Select an Address Autoconfiguration Mode?

- The mode (DHCPv6 or SLAAC) that an IPv6 client uses to obtain an IPv6 address is determined by the flags in RA messages delivered by the gateway and the operating system of the terminal.

ICMPv6
Type: 134 (RA)
...
Flags: 0xC0
11000000 = Managed address configuration: Set
11000000 = Other configuration: Set
...

- M bit: Managed Address Configuration flag

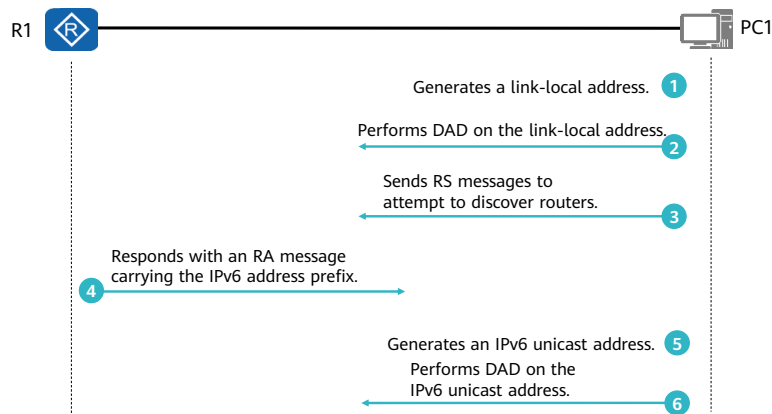
- By default, the M bit in an RA message is 0. A terminal that receives the RA message uses the IPv6 address prefix contained in the message to construct a unicast address. That is, stateless address autoconfiguration is used.
- If the M bit in an RA message is set to 1, a terminal that receives the RA message uses DHCPv6 to obtain the address and other information. When the M bit is 1, the terminal can ignore the O bit. That is, the terminal uses DHCPv6 to obtain the address and other information (such as DNS information).

- O bit: Other Configuration flag

- By default, the O bit in an RA message is 0. A terminal that receives the RA message performs stateless address autoconfiguration. That is, a routing device advertises other configuration information except an IPv6 address to the terminal through the RA message.
- If the O bit in an RA message is set to 1, a terminal that receives the RA message uses DHCPv6 to obtain other parameters except the address.

- Currently, smart terminals running the Android operating system cannot obtain IPv6 addresses through DHCPv6 stateful autoconfiguration.

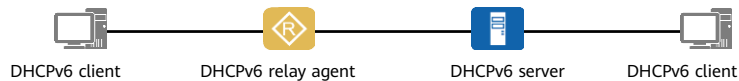
IPv6 Stateless Address Autoconfiguration Process



1. PC1 automatically generates a link-local address based on a local interface ID.
2. PC1 performs DAD on the link-local address. If the address does not conflict with another, it can be used. In this case, PC1 has the IPv6 connection capability.
3. PC1 sends RS messages to attempt to discover IPv6 routers on the link.
4. R1 sends an RA message, which carries the IPv6 address prefix that can be used for stateless address autoconfiguration. (The router can also send RA messages even if it does not receive RS messages.)
5. PC1 parses the RA message to obtain the IPv6 address prefix, and generates an IPv6 unicast address based on this prefix and the local interface ID.
6. PC1 performs DAD on the IPv6 unicast address. If no conflict is detected, PC1 uses this address.

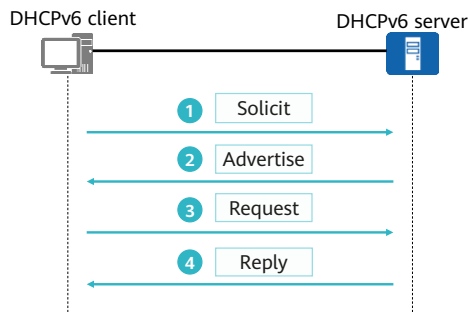
DHCPv6 Network Composition

- The basic DHCPv6 architecture involves the following roles:
 - DHCPv6 client: exchanges DHCPv6 messages with a DHCPv6 server to obtain an IPv6 address/prefix and other network configuration parameters.
 - DHCPv6 server: processes requests for address allocation, lease renewal, and address release from a DHCPv6 client or a DHCPv6 relay agent, and allocates an IPv6 address/prefix and other network configuration parameters to the client.
 - DHCPv6 relay agent: forwards DHCPv6 messages between a DHCPv6 client and a DHCPv6 server to help the client configure its address.



- A DHCPv6 server can allocate addresses and network configuration parameters to clients that access the network for the first time in two modes: DHCPv6 four-message exchange and DHCPv6 two-message exchange.
- The DHCPv6 four-message exchange is used when multiple DHCPv6 servers exist on the network.

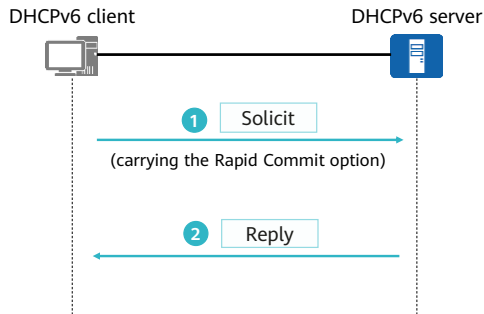
Four-Message Exchange



1. The DHCPv6 client sends a Solicit message to request the DHCPv6 server to allocate an IPv6 address/prefix and network configuration parameters to it.
2. The DHCPv6 server sends an Advertise message to the DHCPv6 client, notifying the client of the IPv6 address/prefix and network configuration parameters that can be allocated to it.
3. If the DHCPv6 client receives Advertise messages from multiple DHCPv6 servers, it selects a server based on the sequence in which the messages are received and the server priorities. The client then sends a Request message to the server, requesting an IPv6 address/prefix and network configuration parameters.
4. The DHCPv6 server responds with a Reply message to confirm the allocation of the IPv6 address/prefix and network configuration parameters to the client.

- In the four-message exchange, a DHCPv6 client and a DHCPv6 server exchange four messages to complete the process of obtaining parameters such as prefixes/addresses.

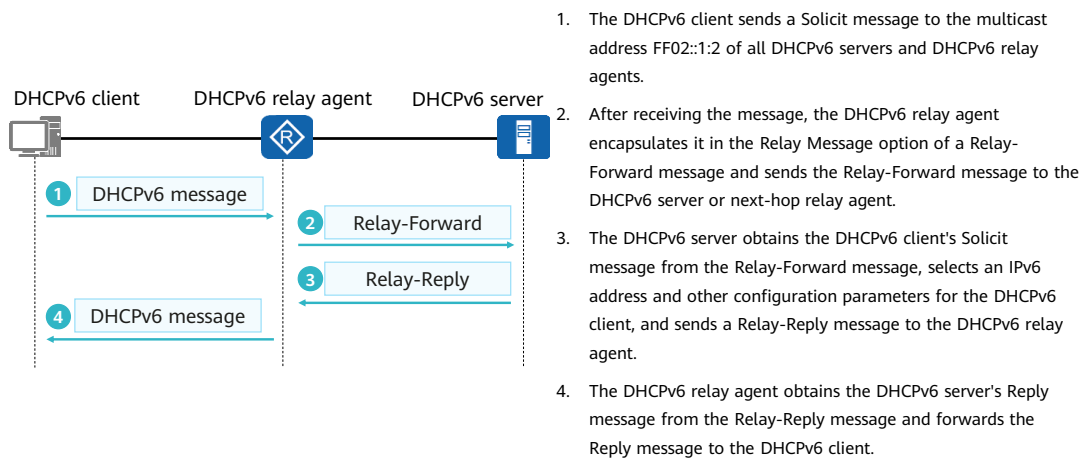
Two-Message Exchange



1. The DHCPv6 client sends a Solicit message carrying the Rapid Commit option.
2. After receiving the Solicit message, the DHCPv6 server checks whether it supports fast address allocation. If so, the server returns a Reply message to allocate an IPv6 address/prefix and other network configuration parameters to the client. If the DHCPv6 server does not support fast address allocation, the four-message exchange is used.

- The two-message exchange is usually used when there is only one DHCPv6 server on the network. A DHCPv6 client multicasts a Solicit message to locate the DHCPv6 server that can allocate an address and configuration parameters to it. After receiving the Solicit message, the DHCPv6 server responds with a Reply message carrying the address and configuration parameters allocated to the DHCPv6 client, completing the address request and allocation process.
- The two-message exchange improves the efficiency of the DHCPv6 address allocation process. On a network with multiple DHCPv6 servers, the DHCPv6 servers can allocate IPv6 addresses to a DHCPv6 client and respond with Reply messages. However, the client uses only the IPv6 address and configuration parameters allocated by one DHCPv6 server. To prevent the preceding situation, the administrator can configure only one DHCPv6 server to support the two-message exchange.

Working Process of a DHCPv6 Relay Agent



- When a DHCPv6 server and a DHCPv6 client are on different network segments, a DHCPv6 relay agent is required to obtain IPv6 addresses/prefixes and other network configuration parameters.

Address Autoconfiguration Comparison

Address Configuration	Based on DHCPv6	Based on NDP
Address management	Stateful. A server stores information about user address or prefix allocation and release.	Stateless. User address allocation information is not stored.
Deployment value	128-bit addresses and prefixes of different lengths can be allocated, providing high scalability.	Only 64-bit prefixes are supported, resulting in poor scalability.
Implementation difficulty	Complex configuration	Simple configuration
Security	Application layer protocol with high security	Weak security

Contents

1. Overview of IPv6
2. ICMPv6
3. IPv6 Address Autoconfiguration
- 4. IPv6 Address Configuration Example**

Basic IPv6 Configuration (1)

1. Enable IPv6.

```
[Huawei-GE0/0/0] ipv6 enable
```

Enable the IPv6 function on the interface.

2. Configure a link-local address for the interface.

```
[Huawei-GE0/0/0] ipv6 address ipv6-address link-local
```

```
[Huawei-GE0/0/0] ipv6 address auto link-local
```

Configure a link-local address for the interface manually or automatically in the interface view.

3. Configure a global unicast address for the interface.

```
[Huawei-GE0/0/0] ipv6 address { ipv6-address prefix-length | ipv6-address/prefix-length }
```

```
[Huawei-GE0/0/0] ipv6 address auto { global | dhcp }
```

Configure a global unicast address for the interface manually or automatically (stateful or stateless) in the interface view.

Basic IPv6 Configuration (2)

4. Configure an IPv6 static route.

```
[Huawei] ipv6 route-static dest-ipv6-address prefix-length { interface-type interface-number [ nexthop-ipv6-address ] | nexthop-ipv6-address } [ preference preference ]
```

5. Check IPv6 information about an interface.

```
[Huawei] display ipv6 interface [ interface-type interface-number | brief ]
```

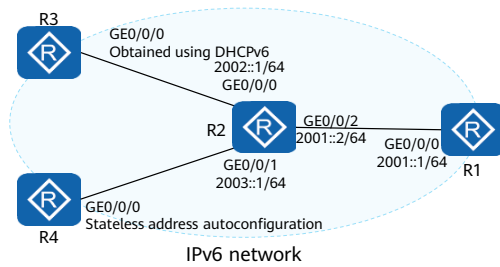
6. Enable the function of sending RA messages.

```
[Huawei-GE0/0/0] ipv6 nd ra halt disable
```

By default, a Huawei router's interface does not send ICMPv6 RA messages. As a result, other devices on the link connected to the interface cannot perform stateless address autoconfiguration.

To implement IPv6 stateless address autoconfiguration, you need to manually enable the function of sending RA messages.

Example for Configuring a Small IPv6 Network (1)



- Configuration requirements:

- Connect R1 and R2 through interfaces with static IPv6 addresses.
- Configure R2 as a DHCPv6 server to assign a global unicast address to GE0/0/0 of R3.
- Enable GE0/0/0 of R4 to perform stateless address autoconfiguration based on the RA messages sent by R2.
- Configure static routes to enable communication between devices.

1. Enable the IPv6 function on interfaces of R1, R2, R3, and R4, and enable the interfaces to automatically generate link-local addresses. (The configuration of R1 is used as an example.)

```
[R1] interface GE 0/0/0
[R1-GE0/0/0] ipv6 enable
[R1-GE0/0/0] ipv6 address auto link-local
```

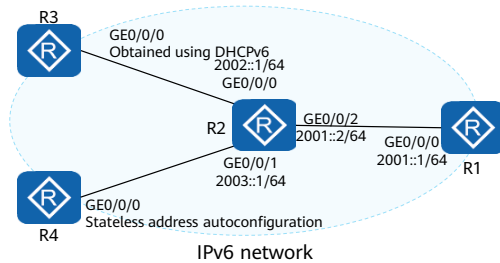
2. Configure static IPv6 global unicast addresses on interfaces of R1 and R2.

```
[R1] interface GE 0/0/0
[R1-GE0/0/0] ipv6 address 2001::1 64
```

```
[R2] interface GE 0/0/0
[R2-GE0/0/0] ipv6 address 2002::1 64
[R2-GE0/0/0] quit
[R2] interface GE 0/0/1
[R2-GE0/0/1] ipv6 address 2003::1 64
[R2-GE0/0/1] quit
[R2] interface GE 0/0/2
[R2-GE0/0/2] ipv6 address 2001::2 64
```

- Note: NetEngine AR5700, AR6700, and AR8000 series routers can function as DHCPv6 servers starting from V600R025C00.

Example for Configuring a Small IPv6 Network (2)



- Configuration requirements:

- Connect R1 and R2 through interfaces with static IPv6 addresses.
- Configure R2 as a DHCPv6 server to assign a global unicast address to GE0/0/0 of R3.
- Enable GE0/0/0 of R4 to perform stateless address autoconfiguration based on the RA messages sent by R2.
- Configure static routes to enable communication between devices.

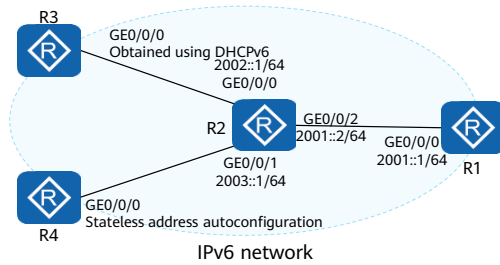
3. Configure the DHCPv6 server function on R2.
Configure an interface of R3 to obtain a global unicast address using DHCPv6 and learn the default route to the IPv6 gateway R2.

```
[R2] dhcp enable
[R2] dhcpv6 pool pool1
[R2-dhcpv6-pool-pool1] address prefix 2002::/64
[R2-dhcpv6-pool-pool1] quit
[R2] interface GE 0/0/0
[R2-GE0/0/0] dhcpv6 server pool1
[R2-GE0/0/0] ipv6 nd ra halt disable
[R2-GE0/0/0] ipv6 nd autoconfig managed-address-flag
[R2-GE0/0/0] ipv6 nd autoconfig other-flag
[R2-GE0/0/0] quit
```

```
[R3] dhcp enable
[R3] interface GE 0/0/0
[R3-GE0/0/0] ipv6 address auto dhcp
[R3-GE0/0/0] ipv6 address auto global default
```

- The **ipv6 nd autoconfig managed-address-flag** command configures the managed address configuration flag in RA messages.
- The **ipv6 nd autoconfig other-flag** command configures the other configuration flag in RA messages.

Example for Configuring a Small IPv6 Network (3)



4. Enable the function of sending RA messages on R2.
Configure R4 to obtain an address through stateless address autoconfiguration.

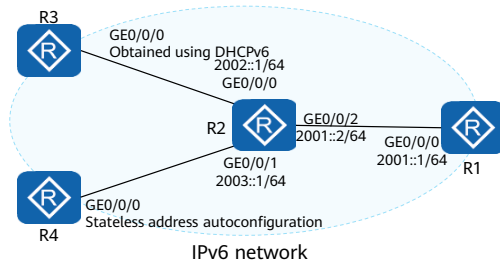
```
[R2] interface GE 0/0/1
[R2-GE0/0/1] ipv6 nd ra halt disable
```

```
[R4] interface GE 0/0/0
[R4-GE0/0/0] ipv6 address auto global
```

- Configuration requirements:

- Connect R1 and R2 through interfaces with static IPv6 addresses.
- Configure R2 as a DHCPv6 server to assign a global unicast address to GE0/0/0 of R3.
- Enable GE0/0/0 of R4 to perform stateless address autoconfiguration based on the RA messages sent by R2.
- Configure static routes to enable communication between devices.

Example for Configuring a Small IPv6 Network (4)



5. Configure static routes on R4.

```
[R4] ipv6 route-static 2001:: 64 2003::1
[R4] ipv6 route-static 2002:: 64 2003::1
```

6. Configure an aggregated static route on R1.

```
[R1] ipv6 route-static 2002:: 15 2001::2
```

- Configuration requirements:

- Connect R1 and R2 through interfaces with static IPv6 addresses.
- Configure R2 as a DHCPv6 server to assign a global unicast address to GE0/0/0 of R3.
- Enable GE0/0/0 of R4 to perform stateless address autoconfiguration based on the RA messages sent by R2.
- Configure static routes to enable communication between devices.

- Similar to IPv4 static routes, IPv6 static routes can be configured on an IPv6 network to implement network connectivity if the network scale is small.

Quiz

1. (Short-answer question) What is the most compressed format of the IPv6 address 2001:0DB8:0000:0000:032A:0000:0000:2D70?
2. (Short-answer question) What is the process of stateless address autoconfiguration for an IPv6 host?
3. (Short-answer question) What are the differences between an IPv6 header and an IPv4 header?
4. (Short-answer question) What messages are used to implement IPv6 address resolution?

1. 2001:DB8::32A:0:0:2D70 or 2001:DB8:0:0:32A::2D70
2. An IPv6 host obtains an address prefix from an RA message sent by a router's interface, and then generates an interface ID by inserting the 16-bit FFFE into the existing 48-bit MAC address of the host's interface. After generating an IPv6 address, the IPv6 host checks whether the address is unique through DAD.
3. The differences between IPv6 and IPv4 are as follows:
 - The packet format of basic header + extension header is used.
 - The checksum at Layer 3 in IPv4 is removed. The checksums at Layer 2 and Layer 4 are sufficiently robust. Therefore, IPv6 removes the checksum at Layer 3 in IPv4, saving router processing resources.
 - The fragmentation function on intermediate nodes is removed. Fragments are processed only on the source node that generates data but not on intermediate routers, preventing them from consuming a lot of CPU resources to process fragments.
 - The fixed-length IPv6 header is defined. This facilitates fast hardware processing and improves the forwarding efficiency of routers.
 - Security options are supported. IPv6 provides optimal support for IPsec, allowing the upper-layer protocol to omit many security options.
 - The Flow Label field is added. This improves QoS efficiency.

4. NS and NA messages.

Summary

- IPv6 provides a large address space. It simplifies the packet header, improving the packet forwarding efficiency of routers. IPv6 addresses are easy to plan and allocate, facilitating route summarization. In addition, IPv6 supports plug and play and enhances QoS.
- ICMPv6 is one of the basic IPv6 protocols. There are two types of ICMPv6 messages: error messages and informational messages. They are used by IPv6 nodes to report errors and information generated during packet processing. Error messages are used to report errors that occur during IPv6 packet forwarding, such as destination unreachable and timeout. Informational messages are used to implement functions such as router discovery and DAD.
- IPv6 address autoconfiguration can be classified into NDP-based stateless address autoconfiguration and DHCPv6-based stateful address autoconfiguration. Stateless address autoconfiguration supports only the allocation of 64-bit prefixes, resulting in poor scalability and weak address management and control.
- The IPv6 address autoconfiguration mode used by a client is determined by the M bit and O bit settings in RA messages sent by a router. On an actual network, you need to determine an appropriate address autoconfiguration mode based on the actual scenario.

Thank you.

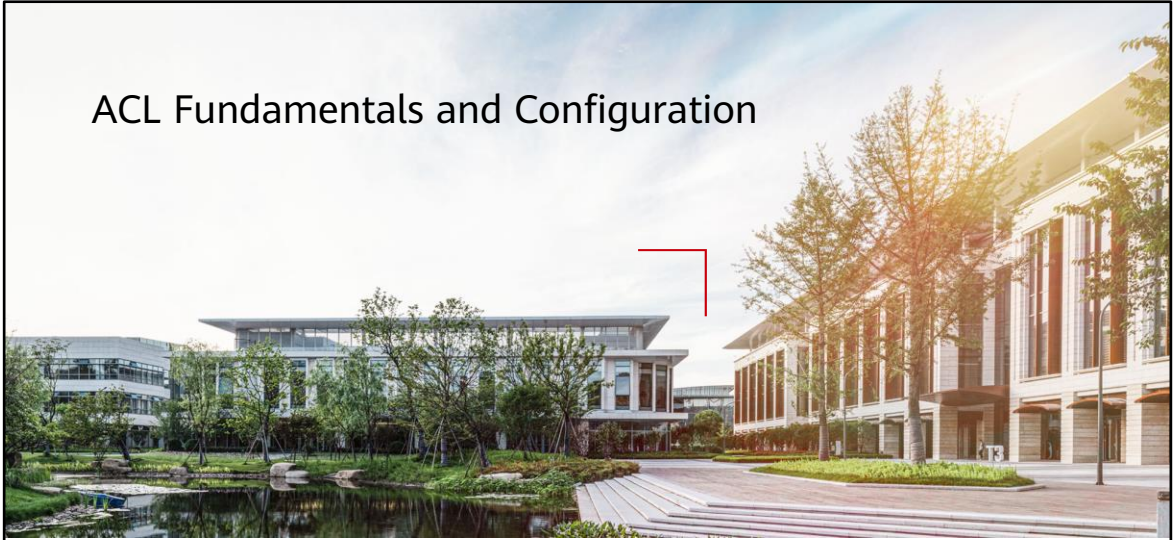
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每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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ACL Fundamentals and Configuration



Foreword

- With the rapid development of networks, the demands for network security and Quality of Service (QoS) are continuously increasing. The Access Control List (ACL) is a technology closely related to network security and QoS.
- ACLs accurately identify and control packets on a network to manage network access behaviors, prevent network attacks, and improve bandwidth utilization. In this way, ACLs ensure network security and QoS.
- This course describes the fundamentals and configurations of ACLs.

Objectives

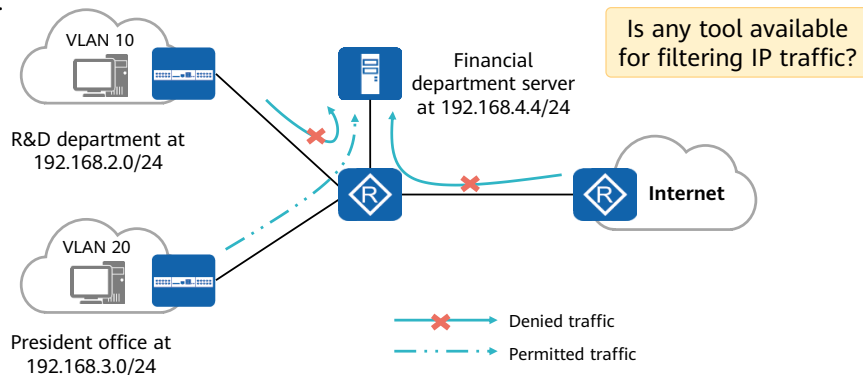
- Upon completion of this course, you will be able to:
 - Describe the fundamentals of ACLs.
 - Understand the types and characteristics of ACLs.
 - Complete basic ACL configurations.

Contents

- 1. ACL Overview**
2. Fundamentals of ACL
3. ACL Configuration Examples

Technical Background: A Tool Is Required to Filter Traffic

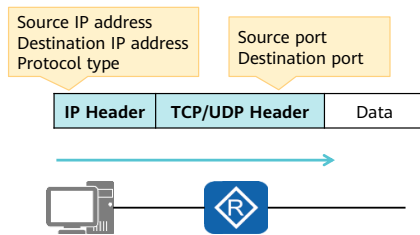
- To ensure financial data security, an enterprise prohibits the R&D department and external networks from accessing the financial department server but allows the president office to access it.



- As networks continue to grow rapidly, enterprise networks encounter the following issues:
 - Resources on the key servers in an enterprise campus network are obtained without permission, making confidential information vulnerable to leakage and posing potential security risks.
 - Viruses on the Internet spread to the campus intranet, threatening intranet security.
 - Network bandwidth is consumed indiscriminately by various services, making it impossible to guarantee sufficient bandwidth for delay-sensitive services such as voice and video, lowering user experience.
- These issues severely affect network communication, making it urgent to enhance both network security and QoS. For example, a tool is required to filter certain types of traffic.

ACL Overview

- An ACL is a collection of one or more rules. A rule refers to a judgment statement that describes a packet matching condition, which can be a source address, destination address, or port number.
- An ACL is essentially a rule-based packet filter. The device filters packets based on given ACL rules and permits or denies the matched packets based on the processing policy of the service module to which the ACL is applied.



ACL Application

- Matching IP traffic
- Invoked in traffic filtering
- Invoked in routing policies
- Invoked in QoS
- Others

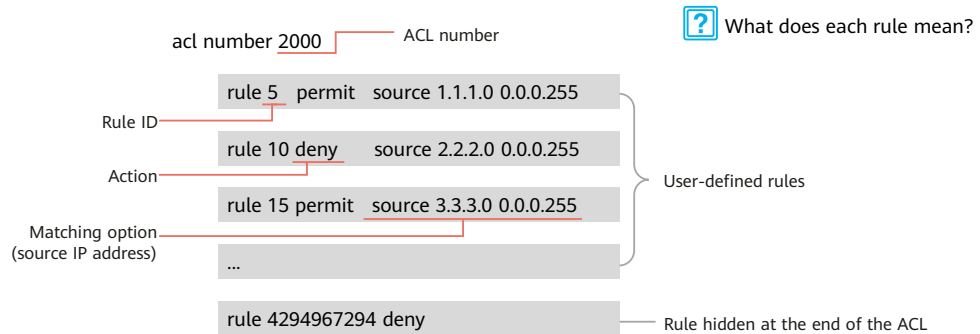
- This course uses IP packet filtering as an example to illustrate ACLs.

Contents

1. ACL Overview
- 2. Fundamentals of ACL**
3. ACL Configuration Examples

ACL Composition

- An ACL consists of several permit or deny statements. Each statement is a rule of the ACL, and the permit or deny keyword in the statement specifies the action corresponding to the rule.



- ACL composition:
 - ACL number: Each ACL needs to be allocated an ACL number, which identifies an ACL. ACL number ranges vary depending on ACL types.
 - Rule: Each statement is a rule of the ACL.
 - Rule ID: Each ACL rule has an ID that identifies the rule. Rule IDs can be manually defined or automatically allocated by the system. A rule ID ranges from 0 to 4294967294. All ACL rules are arranged in the ascending order of rule ID.
 - Action: Each rule contains a permit or deny action. ACLs are usually used together with other technologies, and the meanings of the permit and deny actions may vary depending on the scenario.
 - For example, if an ACL is used together with traffic filtering technology (that is, the ACL is invoked in a traffic filter), the permit action allows traffic to pass through and the deny action rejects traffic.
 - Matching option: ACLs support various matching options. In this example, the matching option is a source IP address. The ACL also supports other matching options, such as Layer 2 Ethernet frame header information (including source and destination MAC addresses and Ethernet frame protocol type), Layer 3 packet information (including destination address and protocol type), and

Layer 4 packet information (including TCP/UDP port number).

Rule ID

acl number 2000				
	Rule ID			Equivalent to 0.0.0.0
rule	5	deny	source 10.1.1.1	0
rule	10	deny	source 10.1.1.2	0
rule	15	permit	source 10.1.1.0	0.0.0.255
Step = 5				

 How do I add a rule?

rule 11 deny source 10.1.1.3 0

acl number 2000				
rule	5	deny	source 10.1.1.1	0
rule	10	deny	source 10.1.1.2	0
rule	11	deny	source 10.1.1.3	0
rule	15	permit	source 10.1.1.0	0.0.0.255

Rule ID and Step

- **Rule ID:**
Each rule in an ACL has an ID.
- **Step:**
A step is an increment between neighboring rule IDs automatically allocated by the system. The default step is 5. Setting a step facilitates rule insertion between existing rules of an ACL.
- **Rule ID allocation rules:**
When assigning an ID to the first ACL rule that does not have a manually specified ID, the system uses the step value (5 for example) as the ID of the rule. For each subsequent rule, the system assigns an ID that is greater than the current largest rule ID in the ACL and is the smallest integer multiple of the step value.

- Rule ID and step:
 - Rule ID: Each ACL rule has an ID that identifies the rule. Rule IDs can be manually defined or automatically allocated by the system.
 - Step: When the system automatically allocates IDs to ACL rules, the increment between neighboring rule IDs is called a step. The default step is 5. Therefore, rule IDs are 5, 10, 15, and so on.
 - If a rule is manually added to an ACL but no ID is specified, the system allocates to this rule an ID that is greater than the current largest rule ID in the ACL and is the smallest integer multiple of the step value.
 - The step value is configurable. For example, if the step is changed to 2, the system automatically rennumbers the rule IDs to 2, 4, 6, and so on.

Wildcard (1)

acl number 2000

rule			source	10.1.1.1	0
rule	5	deny	source	10.1.1.1	0
rule	10	deny	source	10.1.1.2	0
rule	15	permit	source	10.1.1.0	0.0.0.255

Wildcard

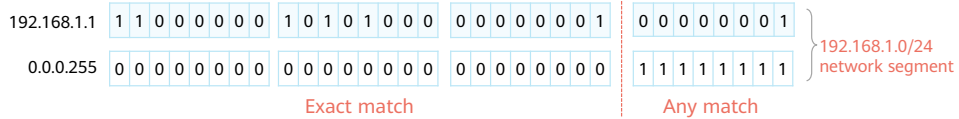
Wildcard

- A wildcard is a 32-bit number that indicates which bits in an IP address need to be strictly matched and which do not.
- A wildcard is usually expressed in dotted decimal notation, as a network mask is expressed. However, their meanings are different.

Matching rules:

The value 0 indicates exact match, and the value 1 indicates any match.

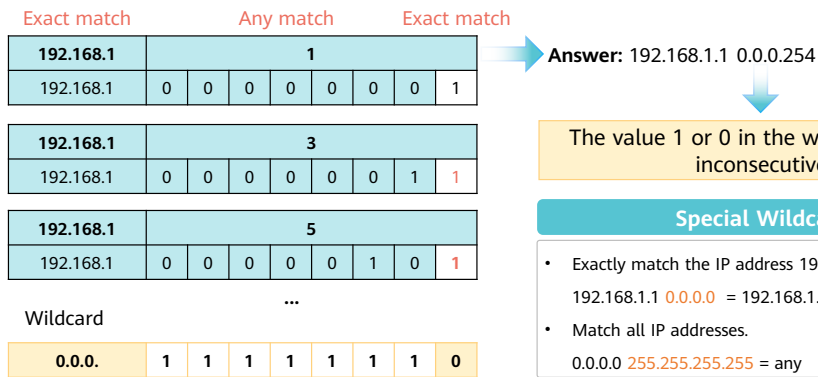
? How do I match the network segment address corresponding to 192.168.1.1/24?



- rule 5: denies packets with the source IP address 10.1.1.1.
- rule 10: denies packets with the source IP address 10.1.1.2.
- rule 15: permits packets with the source IP address on network segment 10.1.1.0/24.

Wildcard (2)

- A wildcard can be used to match odd-numbered IP addresses in the network segment 192.168.1.0/24, such as 192.168.1.1, 192.168.1.3, and 192.168.1.5.



The value 1 or 0 in the wildcard can be inconsecutive.

Special Wildcard

- Exactly match the IP address 192.168.1.1.
192.168.1.1 0.0.0.0 = 192.168.1.1 0
- Match all IP addresses.
0.0.0.0 255.255.255.255 = any

- How do I set the wildcard to match the odd-numbered IP addresses in the network segment 192.168.1.0/24?
 - First, list all odd-numbered IP addresses such as 192.168.1.1, 192.168.1.5, and 192.168.1.11.
 - After the last eight bits are converted into binary numbers, the corresponding addresses are 192.168.1.00000001, 192.168.1.00000101, and 192.168.1.00001011.
 - We can observe the common pattern: The seven most significant bits of the last octet can be any value, and the least significant bit is fixed to 1. Therefore, the answer is 192.168.1.1 0.0.0.254 (0.0.0.1111110).
- There are two special wildcards.
 - If a wildcard comprising all 0s is used to match an IP address, the address is exactly matched.
 - If a wildcard comprising all 1s is used to match 0.0.0.0, all IP addresses are matched.

ACL Classification

Category	Description	ACL Number
Basic ACL	Defines packet filtering rules based on information such as source IPv4 addresses, fragment information, and effective time ranges.	2000 to 2999
Advanced ACL	Defines packet filtering rules based on information such as source and destination IPv4 addresses, IP protocol types, source and destination TCP port numbers, source and destination UDP port numbers, fragment information, and effective time ranges.	3000 to 3999
Layer 2 ACL	Defines packet filtering rules based on the information in Ethernet frame headers, such as source and destination MAC addresses, VLAN IDs, and Layer 2 protocol types.	4000 to 4999
User-defined ACL	Defines packet filtering rules based on packet headers, offsets, character string masks, and user-defined character strings. With a user-defined ACL configured, the system performs an AND operation on the packet bytes starting at a specified offset behind the packet header and the character string mask, compares the extracted character string against the user-defined character string, and then obtains matched packets.	5000 to 5999
User ACL	Defines packet filtering rules based on information such as source IPv4 addresses, source user control list (UCL) groups, destination IPv4 addresses, destination UCL groups, IP protocol types, ICMP types, source/destination TCP port numbers, and source/destination UDP port numbers.	6000 to 9999
ARP-based ACL	Defines packet filtering rules based on the source and destination IPv4 addresses and source and destination MAC addresses of ARP packets.	23000 to 23999

Classification by Function

- ACLs are classified by creation mode as follows:
 - A named ACL is created using a name. This type of ACL is named in the format of name + number. The number can be manually specified or automatically assigned by the system. If a number is not manually specified for an ACL, the system assigns the largest number available for the ACL.
 - A numbered ACL is created using a number.

Examples of Basic and Advanced ACLs

- Basic ACL

Number range:
2000–2999

Source IP address			
IP Header	TCP/UDP Header	Data	
acl number 2000			
rule	5	deny	source 10.1.1.1 0
rule	10	deny	source 10.1.1.2 0
rule	15	permit	source 10.1.1.0 0.0.0.255

- Advanced ACL

Number range:
3000–3999

Source IP address,
destination IP address,
and protocol type

Source port
Destination port

IP Header

TCP/UDP Header

Data

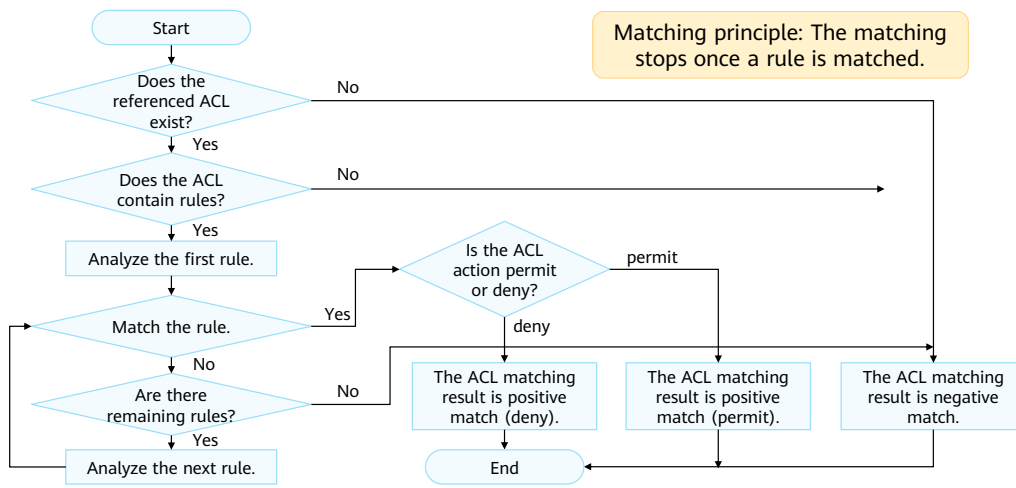
acl number 3000

rule 5 permit ip source 10.1.1.0 0.0.0.255 destination 10.1.3.0 0.0.0.255

rule 10 permit tcp source 10.1.2.0 0.0.0.255 destination 10.1.3.0 0.0.0.255 destination-port eq 21

- eq: matches packets with a value equal to the specified value.
- gt: matches packets with a value greater than the specified value.
- lt: matches packets with a value less than the specified value.

ACL Matching Mechanism



- The ACL matching mechanism is as follows:
 - After receiving a packet, the device configured with an ACL matches the packet against ACL rules one by one. If the packet does not match a rule, the device continues to match the packet against the next rule.
 - If the packet matches an ACL rule, the device takes the action defined in the rule and stops the rule matching.
- Matching process: The device checks whether an ACL is configured.
 - If no ACL is configured, the device returns the result "negative match".
 - If an ACL is configured, the device checks whether the ACL contains rules.
 - If the ACL does not contain rules, the device returns the result "negative match".
 - If the ACL contains rules, the device matches a packet against the rules in ascending order of rule ID.
 - If the packet matches a permit rule, the device stops matching and returns the result "positive match (permit)".
 - If the packet matches a deny rule, the device stops matching and returns the result "positive match (deny)".
 - If the packet does not match any rule in the ACL, the device

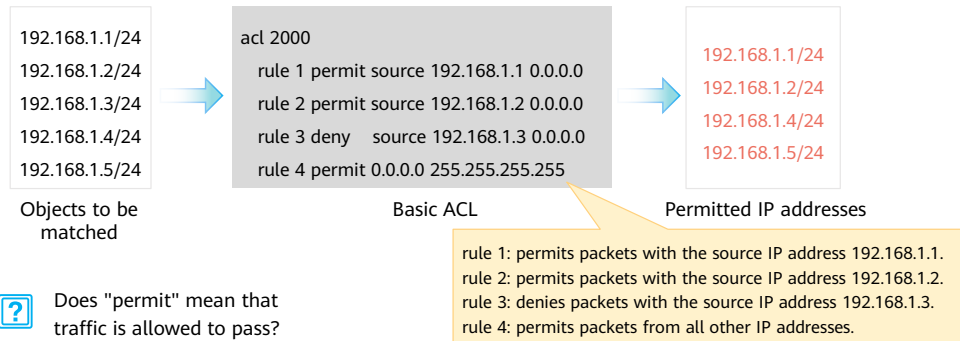
returns the result "negative match".

- The ACL matching results include "positive match" and "negative match".
 - Positive match: Packets match a rule in an ACL. The result is "positive match" regardless of whether packets match a permit or deny rule in an ACL.
 - Negative match: This refers to any of the following situations—No ACL exists, the ACL does not contain rules, or packets do not match any rule in an ACL.
- Matching principle: The matching stops once a rule is matched.

ACL Matching Order and Result

- Configuration order (config mode)

- The system matches packets against ACL rules in ascending order of rule ID.



- An ACL can consist of multiple deny or permit statements, with each defining a rule. Rules may overlap or conflict. Therefore, the order in which ACL rules are matched is critical.
- Huawei devices support two matching orders: automatic order (auto) and configuration order (config). The default matching order is config.
 - In auto order, the system arranges rules in descending order of precision ("depth first" principle), and matches packets against the rules in the same order—descending order of precision.
 - In config order, the system matches packets against ACL rules in ascending order of rule ID.
 - If a new rule is added later, it is inserted into the appropriate position based on its rule ID, and packets are still matched against ACL rules in ascending order of rule ID.
- Matching result:
 - First, understand what ACL 2000 represents:
 - rule 1: permits packets with the source IP address 192.168.1.1.
 - rule 2: permits packets with the source IP address 192.168.1.2.
 - rule 3: denies packets with the source IP address 192.168.1.3.

- rule 4: permits packets from all other IP addresses.
- When packets with the source IP address 192.168.1.3 pass through the device configured with the ACL:
 - The device matches the packets against rule 1. The matching result is "negative match."
 - The device continues to match the packets against rule 2. The matching result is still "negative match."
 - The device continues to match the packets against rule 3. The matching result is "positive match," and the action is deny.
- The permit or deny action in an ACL rule only specifies the matching action of the ACL, but does not determine whether the traffic is permitted or denied. The ultimate action for processing the traffic is determined by the service module.

Default ACL Actions and Processing Mechanisms of Different Service Modules

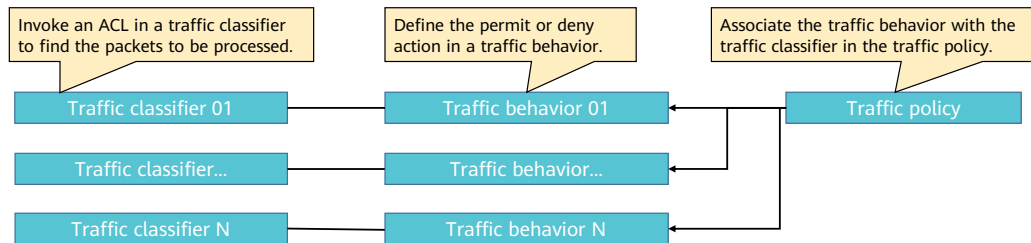
Service Module	Default ACL Action	ACL Processing Mechanism				
		Packets Match the Permit Rule	Packets Match the Deny Rule	Packets Do Not Match Any Rule in an ACL	No Rule Configured in an ACL	ACL Not Created
Telnet	deny	Allow login	Deny login	Deny login	Allow login	Allow login
STelnet	deny	Allow login	Deny login	Deny login	Allow login	Allow login
HTTP	deny	Allow login	Deny login	Deny login	Allow login	Allow login
SNMP	deny	Allow login	Deny login	Deny login	Allow login	Allow login
FTP	deny	Allow login	Deny login	Deny login	Allow login	Allow login
TFTP	deny	Allow login	Deny login	Deny login	Allow login	Allow login
SFTP	deny	Allow login	Deny login	Deny login	Allow login	Allow login
Traffic policy	permit	- If the permit action is configured in the traffic behavior, packets are allowed to pass. - If the deny action is configured in the traffic behavior, packets are discarded. - If another action is configured in the traffic behavior, the corresponding action is performed.	Discard packets	The function does not take effect and packets are forwarded based on the original forwarding mode.	The function does not take effect and packets are forwarded based on the original forwarding mode.	The function does not take effect and packets are forwarded based on the original forwarding mode.

How to invoke an ACL to filter packets?

- Service modules such as Telnet, FTP, and SFTP can directly invoke ACLs.
- For example, to associate the FTP server access permission with ACL 2001, run the following command: **[HUAWEI] ftp server acl 2001**

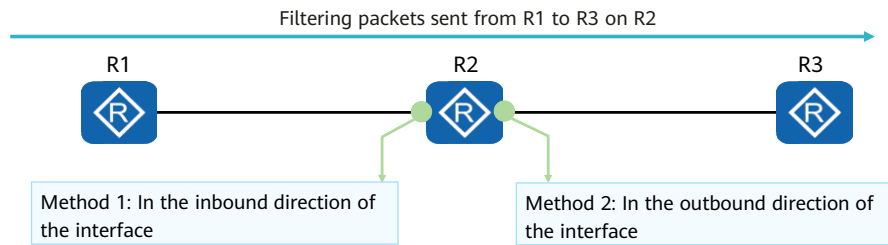
Using MQC to Invoke an ACL for Packet Filtering

- Modular QoS Command-Line Interface (MQC) classifies packets that share common characteristics into a type, allowing the device to provide the same service for packets of the same type and differentiated services for packets of different types.
- MQC involves three entities: traffic classifier, traffic behavior, and traffic policy.
- Packet filtering can be easily implemented based on MQC.



- Leveraging MQC, users can perform fine-grained classification of network traffic and specify appropriate services for traffic of different types as needed, thereby enhancing the serviceability of the network.
- A traffic classifier defines a group of traffic matching rules to classify packets.
- A traffic behavior defines an action to be taken on packets of a specified type.
- A traffic policy binds traffic classifiers to traffic behaviors, and actions defined in traffic behaviors are taken on packets of specified types. Multiple pairs of traffic classifiers and traffic behaviors can be bound to a traffic policy.

Location Where Packet Filtering Is Performed



Contents

1. ACL Overview
2. Fundamentals of ACL
- 3. ACL Configuration Examples**

Configuring a Basic ACL

1. Create a basic ACL.

```
[HUAWEI] acl [ number ] basic-acl-number
```

Create a numbered basic ACL (2000–2999) and enter its view.

```
[HUAWEI] acl name basic-acl-name basic
```

Create a named basic ACL and enter its view.

2. Configure a rule for the basic ACL.

```
[HUAWEI-acl4-basic-2000] rule [ rule-id ] [ name rule-name ] { permit | deny } [ fragment-type fragment | source { source-ip-address { source-wildcard | 0 | src-netmask } | any } | time-range time-name |
```

In the basic ACL view, you can run this command to configure a rule for the basic ACL.

Configuring an Advanced ACL

1. Create an advanced ACL.

```
[HUAWEI] acl [ number ] advance-acl-number
```

Create a numbered advanced ACL (3000–3999) and enter its view.

```
[HUAWEI] acl name advance-acl-name advance
```

Create a named advanced ACL and enter its view.

2. Configure a rule for the advanced ACL.

```
[HUAWEI-acl4-advance-3000] rule [ rule-id ] [ name rule-name ] { permit | deny } ip { destination { destination-ip-address { destination-wildcard | 0 | des-netmask }
```

In the advanced ACL view, you can run this command to configure a rule for the advanced ACL.

- Advanced ACLs support a wide range of matching rules. This slide shows only some of them. For more ACL configuration commands, see the product documentation.

Configuring MQC to Invoke ACLs

1. Create a traffic classifier, enter its view, and bind the created ACL to the traffic classifier.

```
[HUAWEI] traffic classifier classifier-name [ type { and | or } ]
```

The **type** parameter specifies the logical operator between rules in the traffic classifier. If this parameter is not specified, the logical relationship between rules is OR by default.

```
[HUAWEI-classifier-classifier-name] if-match acl { acl-number | acl-name }
```

Define matching rules in the traffic classifier.

2. Create a traffic behavior, enter its view, and specify an action for the traffic behavior.

```
[HUAWEI] traffic behavior behavior-name
```

```
[HUAWEI-behavior-behavior-name] permit | deny
```

3. Create a traffic policy, enter its view, and bind the traffic classifier and traffic behavior to the traffic policy.

```
[HUAWEI] traffic policy policy-name
```

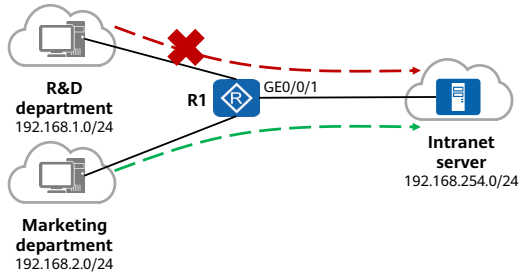
```
[HUAWEI-trafficpolicy-policy-name] classifier classifier-name behavior behavior-name
```

4. Apply the traffic policy to an interface.

```
[HUAWEI-GE0/0/1] traffic-policy policy-name { inbound | outbound }
```

- For more MQC configuration commands, see the product documentation.

Example for Configuring Traffic Filtering



- Configuration requirements:

- Block access from the R&D department to the intranet server but permit access from the marketing department.
- Configure a traffic policy in the outbound direction of GE0/0/1 on R1.

1. Configure an ACL.

```
[R1] acl 3000
[R1-acl4-advance-3000] rule deny ip source 192.168.1.0 0.0.0.255
destination 192.168.254.0 0.0.0.255
[R1-acl4-advance-3000] rule permit ip source 192.168.2.0
0.0.0.255 destination 192.168.254.0 0.0.0.255
```

2. Configure a traffic classifier, traffic behavior, and traffic policy.

```
[R1] traffic classifier test01
[R1-classifier-test01] if-match acl 3000
[R1-classifier-test01] quit
[R1] traffic behavior test01
[R1-behavior-test01] permit
[R1-behavior-test01] quit
[R1] traffic policy test01
[R1-trafficpolicy-test01] classifier test01 behavior test01
```

3. Apply the traffic policy.

```
[R1] interface GE 0/0/1
[R1-GE0/0/1] traffic-policy test01 outbound
[R1-GE0/0/1] quit
```

Quiz

1. (Single-answer question) Which of the following rules is a valid basic ACL rule?

()

- A. rule permit ip
- B. rule deny ip
- C. rule permit source any
- D. rule deny tcp source any

1. (Short-answer question) Which parameters can you use to define advanced ACL rules?

1. C
2. Advanced ACL rules can be defined using parameters such as the source/destination IP address, source/destination port number, protocol type, and TCP flag (such as SYN, ACK, or FIN).

Summary

- ACL is a widely used network technology. Their fundamental mechanism is that a device configured with an ACL matches passing packets against predefined rules and performs the specified actions on packets that match those rules. These matching rules and their corresponding actions are defined based on specific network requirements. The variety of processing actions and matching rules enables ACLs to serve a wide range of purposes.
- ACL technology is often used in conjunction with other technologies such as firewalls, routing policies, QoS, and traffic filtering.

Thank you.

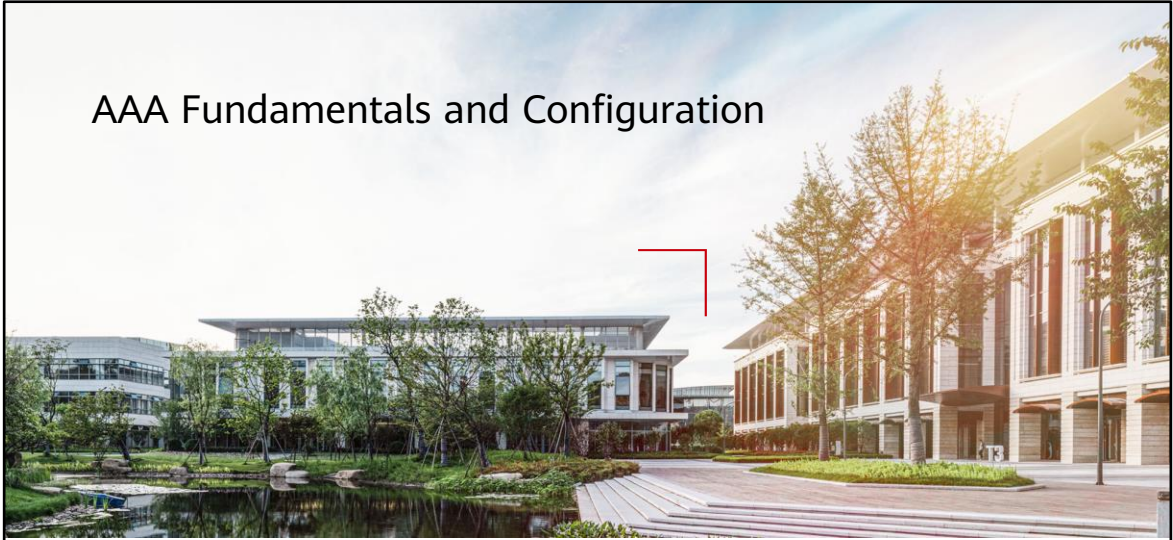
把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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AAA Fundamentals and Configuration



Foreword

- There are a large number of users on the network. As such, user management is one of the most basic security management requirements.
- AAA is short for authentication, authorization, and accounting. It is a management framework that provides a security mechanism for authorizing some users to access specified resources and recording the operations of these users. AAA is widely used because of its good scalability and easy implementation of centralized management of user information.
- AAA can be implemented using multiple protocols. In real-world applications, the Remote Authentication Dial-In User Service (RADIUS) protocol is most commonly used.
- This course describes the basic concepts, implementation, basic configurations, and typical application scenarios of AAA.

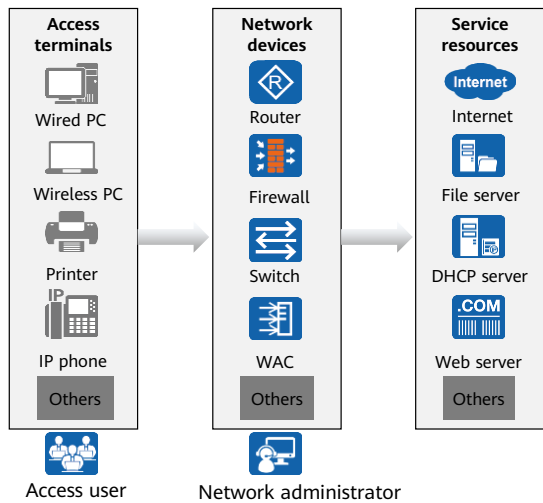
Objectives

- Upon completion of this course, you will be able to:
 - Describe the basic concepts of AAA.
 - Describe the application scenarios of AAA.
 - Complete basic AAA configurations.

Contents

- 1. AAA Overview**
2. Deploying AAA
3. Configuration Examples for AAA

Background of User-based Network Permission Control

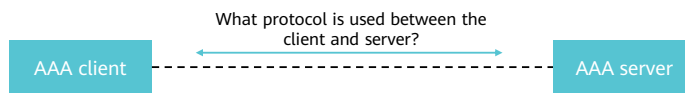


- In the figure, network participants can be classified into two types:
 - Access user: a person who uses various types of terminals to enjoy network services.
 - Network administrator: a person who manages network devices and ensures normal network running.
- To ensure secure network running, the following requirements must be met:
 - Ensure that network participants are authorized users.
 - Ensure that network participants obtain network permissions as required.
 - Ensure that network participants pay for network services.
- On a large-scale network, although all related devices on the network have IP addresses, the IP addresses are not fixed. If the preceding three requirements are managed and controlled based on IP addresses, the configuration is complex and the workload is heavy.
- User-based network security control is a common method.

- On a large-scale network, it is infeasible to implement network permission control based on MAC addresses. This is because the configuration is complex and the workload is heavy. In addition, to implement MAC address-based control, network devices and access terminals must be on the same Layer 2 network.
- This course focuses on network security control by network administrators.

Basic Architecture of AAA

- As a management mechanism of network security, AAA provides a modular way of performing the following services:
 - Authentication: confirms the identities of users accessing the network and determines whether they are authorized.
 - Authorization: grants differentiated rights to users and limits the services available to them.
 - Accounting: records all the operations of a user during the network service process, including the used service types, start time, and data traffic, to collect and record data about the network resource usage of the user for implementing time- or traffic-based accounting and network monitoring.
- AAA uses the client/server structure.
 - An AAA client is typically a network device that authenticates users and manages user access.
 - An AAA server is a collective term for authentication, authorization, and accounting servers and implements centralized user information management.



Protocols Implementing AAA

RADIUS

- RADIUS is a distributed information exchange protocol.
- RADIUS combines authentication and authorization, which cannot be separated. If authentication succeeds, authorization also succeeds.
- RADIUS uses UDP for transmission and uses ports 1812 and 1813 for authentication and accounting, respectively.

HWTACACS

- HWTACACS is a centralized information exchange protocol.
- HWTACACS provides independent authentication, authorization, and accounting, which can be implemented on different servers.
- HWTACACS uses TCP port 49 to transmit data.

LDAP

- LDAP is a TCP/IP-based directory access protocol.
- LDAP is used to store the data that is not frequently changed, such as email addresses and contact lists.
- LDAP does not support accounting.

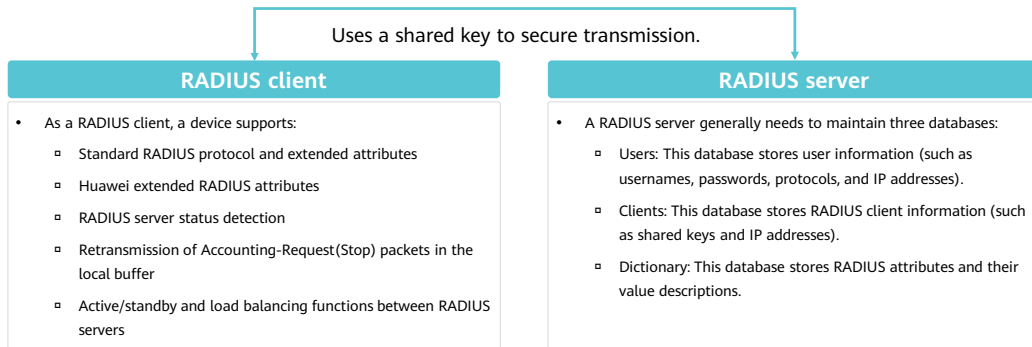
AD

- AD server is a type of LDAP server software running on the Windows operating system.
- AD stores the user information of the operating system for authentication and authorization when users log in to the operating system.
- AD does not support accounting.

In real-world applications, RADIUS is most commonly used.

- At the very beginning, RADIUS was only the AAA protocol used for dial-up users. As the user access mode diversifies, such as Ethernet access, RADIUS can also be applied to these access modes.
- LDAP is short for Lightweight Directory Access Protocol.
- HWTACACS is short for Huawei Terminal Access Controller Access Control System.
- AD is short for Active Directory.

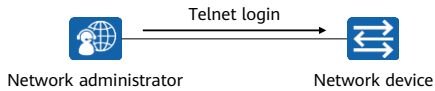
Functions of the RADIUS Client and Server



- Authentication messages between a RADIUS server and RADIUS clients are exchanged using a shared key. The shared key is a character string that is transmitted in out-of-band mode, is known to both clients and the server, and does not need to be transmitted independently on the network.

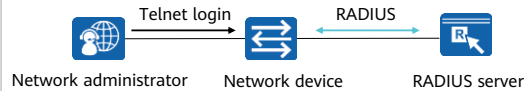
Common AAA Application Scenarios

Local authentication and authorization for administrative users



- After local AAA schemes are configured on a network device, the device compares the user name and password of a network administrator with the locally configured user name and password for authentication when the network administrator logs in to the device.
- After the authentication succeeds, the network device grants the specified rights to the network administrator.

RADIUS authentication and authorization for administrative users



- AAA schemes are configured on a network device to interconnect the device with a RADIUS server.
- After a network administrator enters the user name and password, the network device sends the information to the RADIUS server for authentication.
- After the authentication succeeds, the network device grants the specified rights to the network administrator.

- Telnet is an application layer protocol used for remote login on a computer network. It allows users to log in to a remote host from a local host and use the command line interface (CLI) to perform operations.
- In addition to Telnet, network administrators also need to use the following protocols to implement permission control through AAA:
 - File Transfer Protocol (FTP) is a standard network protocol used to transfer files on a computer network. It allows users to upload files from one host to another or download files from one host to another.
 - Secure Shell (SSH) is a security protocol that uses encryption and authentication mechanisms to implement network services (such as remote access and file transfer) securely over an insecure network.

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1. AAA Overview
- 2. Deploying AAA**
3. Configuration Examples for AAA

- This course uses Huawei S3700&S5700&S6700 series campus switches as an example to describe how to deploy AAA on network devices.

AAA Schemes Supported by Huawei Switches

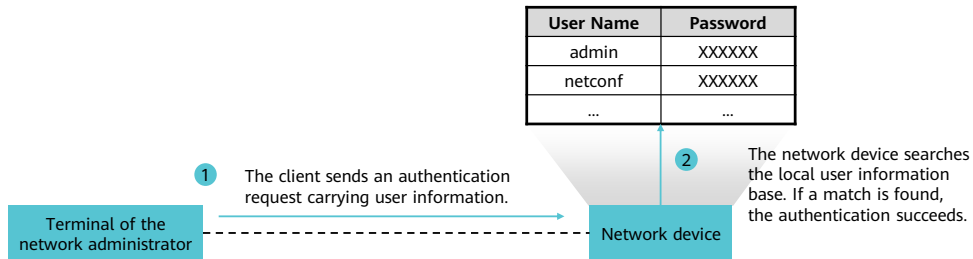
- On Huawei switches, authentication, authorization, and accounting (AAA) schemes are deployed independently.
- Huawei switches functioning as AAA servers do not support the accounting service.

Authentication scheme	Authorization scheme	Accounting scheme
• Local authentication • Third-party certification ▫ RADIUS authentication ▫ HWTACACS authentication ▫ AD authentication ▫ LDAP authentication ▫ Non-authentication	• Local authorization • Third-party authorization: ▫ RADIUS authorization ▫ HWTACACS authorization ▫ AD authorization ▫ LDAP authorization ▫ Non-authorization	• RADIUS accounting • HWTACACS accounting • Non-accounting

- Non-authentication: Users are completely trusted without validity check. This method is rarely used.
- Non-authorization: Users are not authorized.
- Third-party authentication and authorization are also called server authentication and authorization.

Local Authentication

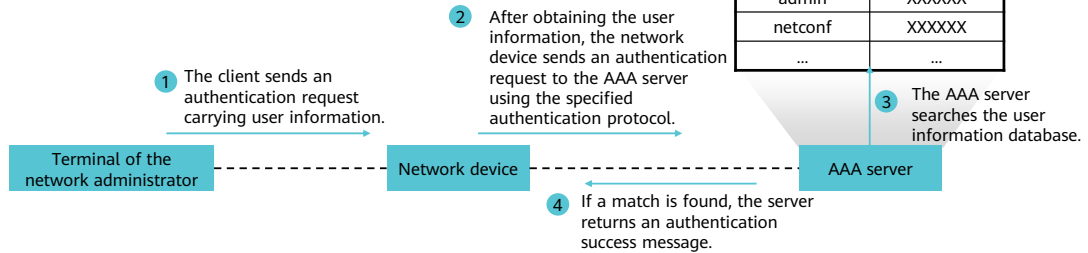
- Local authentication: The device functions as an authentication server and has user information configured. This method features fast processing and low operation costs. However, the information storage capacity is subject to the device hardware.
- Local authentication is typically used to authenticate administrators.



- In local authentication, AAA is not used between the network administrator's terminal and network device. During terminal authentication, user information is carried by applications, such as FTP, Telnet, and SSH.

Third-party Authentication

- Third-party authentication: User information is configured on the AAA server.
- Third-party authentication is typically used to authenticate access users.

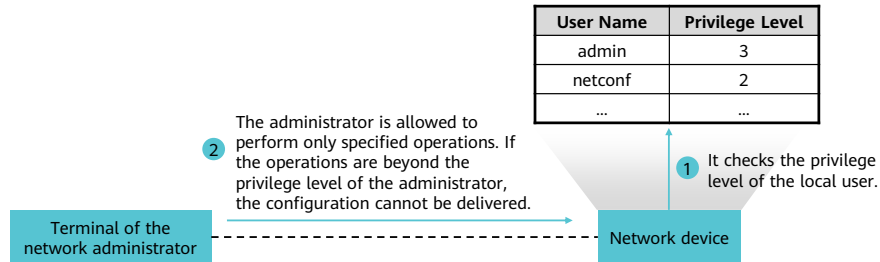


- An authentication scheme enables you to specify one or more authentication methods to be used. If multiple authentication methods are specified, the authentication methods configured later are used as backup authentication methods.
- The network device uses the first method listed in the authentication scheme to authenticate users; if that method does not respond, the device selects the next method in the authentication scheme. This process continues until there is successful communication with a listed authentication method or the authentication method list is exhausted, in which case authentication fails.

- The device attempts authentication with the next listed authentication method only when there is no response from the previous method. If authentication fails at any point in this cycle — meaning that the AAA server responds by denying the user access — the authentication process stops and no other authentication methods are attempted.

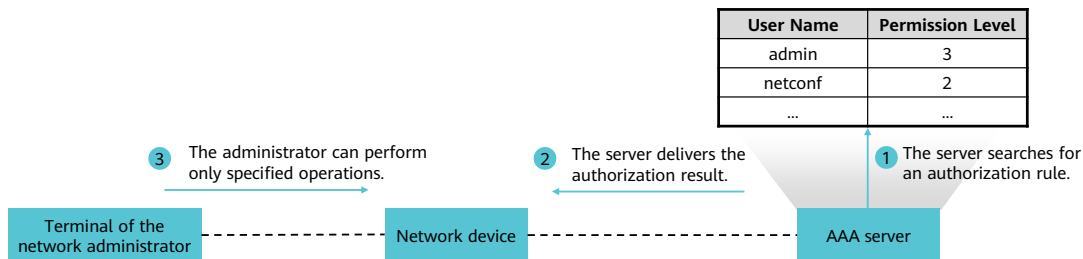
Local Authorization

- The "authentication + privilege level" method is typically used to control access of the administrators (login users) to the device, improving device operation security. Authentication restricts the administrators' access to the device and the privilege level defines the commands that the administrators can enter after logging in to the device.



- In local authorization, AAA is not used between the network administrator's terminal and network device. When the terminal accesses the network device through a specific application (such as FTP, Telnet, or SSH), the terminal can obtain only the rights corresponding to the user privilege level.

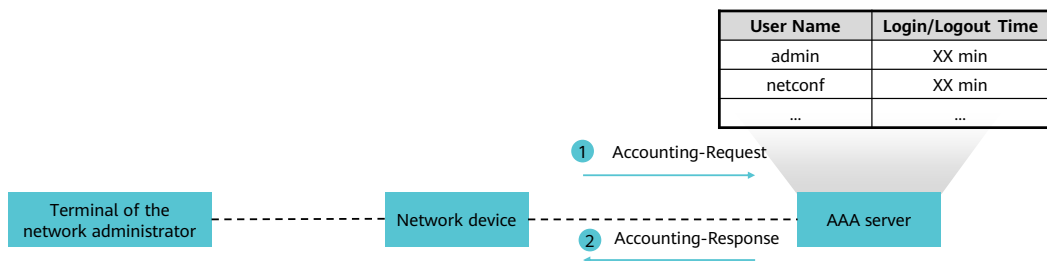
Third-party Authorization



- An authorization scheme enables you to specify one or more authorization methods to be used.
- When multiple authorization methods are specified, the configuration sequence determines the order in which each authorization method takes effect. The authorization method configured earlier takes effect preferentially. The authorization methods configured later function as backup authorization methods and are used when the previous authorization method does not respond.
- If the previous method responds with an authorization failure message, the AAA server refuses to provide services for the user. In this case, authorization ends and the next listed method is not attempted.

Accounting Scheme

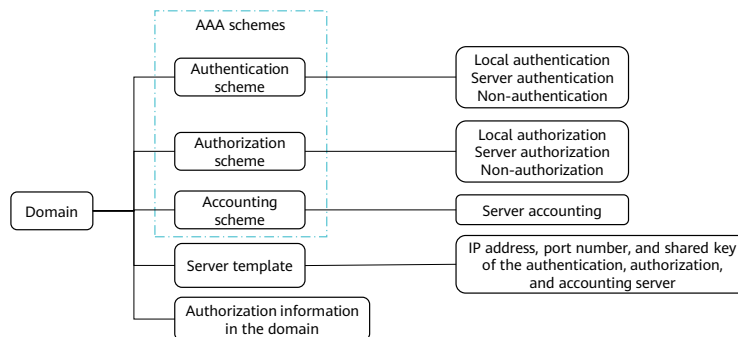
- Accounting packets are classified into Accounting-Request and Accounting-Response packets.
- After the accounting function is enabled, the device sends Accounting-Request packets recording user activities to the AAA server. The AAA server then performs user accounting and auditing based on information in the packets.



- Only one accounting method can be specified at a time in an accounting scheme.

Domain-based User Management

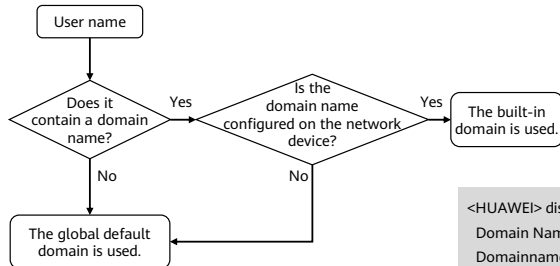
- To facilitate management, network devices manage users based on domains. Each user belongs to a domain.
- A user uses the AAA configuration in the domain to which the user belongs.



- Configurations including AAA schemes, a server template, and authorization information are bound to a domain for unified management.
 - AAA schemes: includes an authentication scheme, an authorization scheme, and an accounting scheme. These schemes define the authentication, authorization, and accounting methods and the sequence in which the methods take effect. If server authentication, authorization, and accounting are used, you need to configure a server template. If local authentication and authorization are used, you need to configure a local user.
 - Server template: is used to configure the server used for authentication, authorization, or accounting. If server authorization is configured, users obtain authorization information from the server and domain.
 - Authorization information in a domain: Authorization information can be configured in a domain.
- If local authorization is used, users obtain authorization information from the domain.
- If server authorization is used, users obtain authorization information from the server or domain. Authorization information configured in a domain has lower priority than that delivered by a server. If the two types of authorization information conflicts, authorization information delivered by the server takes effect. If no conflict occurs, the two types of authorization information take effect simultaneously. In this manner, you can increase authorization flexibility by means of domain-based management, regardless of the authorization attributes provided by the server.

Name of the Domain to Which Users Belong

- The domain to which a user belongs is determined by the user name used by the user to log in to a network device. If the user name does not contain a domain name or the domain name contained in the user name is not configured on the device, the device cannot determine the domain to which the user belongs. In this case, the device adds the user to the global default domain based on the user type.



User	Supported User Access Mode	Global Default Domain
Administrator	Is also called a login user and refers to the user who can log in to the device. Includes users who log in to the device through FTP, SSH, Telnet, the console port, and HTTP.	default_admin

<HUAWEI> display aaa configuration

```
Domain Name Delimiter: @                #Domain name delimiter
Domainname parse direction: Left to right #Domain name parsing direction
Domainname location: After-delimiter      #Domain name location
Administrator user default domain: default_admin #Global default administrative domain
Normal user default domain : default      #Global default common domain
```

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 - Configuring Local Authentication for Administrators
 - Configuring RADIUS Authentication for Administrators

Common Commands (1)

1. Enter the AAA view.

```
[HUAWEI] aaa
```

Exit the system view and enter the AAA view.

2. Create a local administrator and configure a password for the local administrator.

```
[HUAWEI-aaa] local-user user-name password irreversible-cipher ir-password
```

Parameter	Description
<i>user-name</i>	Name of a local administrator. If the user name contains a domain name delimiter, for example, an at sign (@), the character string before @ is the user name and that following @ is the domain name. If the user name does not contain @, the entire character string represents the user name and the domain name is the default one.
irreversible-cipher <i>ir-password</i>	Login password of a user. irreversible-cipher : indicates that a user password is encrypted using an irreversible encryption algorithm. When this parameter is set, unauthorized users cannot obtain the plain text by using a decryption algorithm, ensuring user security.

Common Commands (2)

3. Configure an access type for the local administrator.

```
[HUAWEI-aaa] local-user user-name service-type { none | { http | ftp | ssh | telnet | terminal }* }
```

Parameter	Description
ftp	Indicates an FTP user.
terminal	Indicates a terminal user, which is typically a user connected using a console port.
telnet	Indicates a Telnet user, which is typically a network administrator.
ssh	Indicates an SSH user.
http	Indicates an HTTP user.
none	Indicates that no access type is specified.

Common Commands (3)

4. Set a privilege level for the local administrator.

```
[HUAWEI-aaa] local-user user-name privilege level level
```

level: specifies the privilege level of a local administrator. After logging in to the device, a user can run only the commands of the same privilege level or lower privilege levels. The value ranges from 0 to 3. A larger value indicates a higher privilege level.

- For details about how to configure other functions, see the product documentation. For example, configure the validity period, access time range, and password policy for the local administrator account.

Configuration Example for AAA



- An enterprise requires that only users who pass local AAA authentication can operate network devices.
- Configuration roadmap:
 - Configure the user name and password for a local administrator.
 - Configure an access type for the local administrator.
 - Set a privilege level for the local administrator.

1. Create a local AAA user and set an access type and a privilege level for the user.

```
[SW1] aaa
[SW1-aaa] local-user huawei password irreversible-cipher Huawei@123
[SW1-aaa] local-user huawei service-type telnet
[SW1-aaa] local-user huawei privilege level 3
[SW1] quit
```

2. Check information about the created user.

```
[SW1] display local-user username huawei
The contents of local user(s):
Password          : *****
State             : active
Service-type-mask : T           #The access type is Telnet.
Privilege level : 3           #The privilege level is 3.
Ftp-directory      : -
Access-limit       : Yes
Access-limit-max   : 4294967295
...
```

- Note:
 - Only the configuration for creating a local administrator is provided here. Other configurations are required for authenticating the local administrator.
 - The alarm information automatically displayed on the device is not provided in this example.

Contents

1. AAA Overview
2. Deploying AAA
3. Configuration Examples for AAA
 - Configuring Local Authentication for Administrators
 - Configuring RADIUS Authentication for Administrators

Configure an Authentication Scheme

1. Enter the AAA view.

```
[HUAWEI] aaa
```

Exit the system view and enter the AAA view.

2. Create an authentication scheme and enter the authentication scheme view.

```
[HUAWEI-aaa] authentication-scheme authentication-scheme-name
```

```
[HUAWEI-aaa-authentication-scheme-name] authentication-mode { hwtacacs | { local | local-case } | radius |  
ldap | ad | haca } * [ none ]
```

local: authenticates users locally and sets local usernames to case-insensitive.

local-case: authenticates users locally and sets local usernames to case-sensitive.

radius: authenticates users using a RADIUS server. If RADIUS authentication is required, you must configure an authentication server in the RADIUS server template.

- For details about the parameters that are not described here, see the product documentation.

Configure an Accounting Scheme

1. Enter the AAA view.

```
[HUAWEI] aaa
```

Exit the system view and enter the AAA view.

2. Create an accounting scheme.

```
[HUAWEI-aaa] accounting-scheme accounting-scheme-name
```

Create an accounting scheme and enter the accounting scheme view.

```
[HUAWEI-aaa-accounting-scheme-name] accounting-mode { hwtacacs | none | radius | haca }
```

radius: indicates that accounting is performed by a RADIUS server.

none: indicates that accounting is not performed.

- For details about the parameters that are not described here, see the product documentation.

Configuring a RADIUS Server Template

1. Create a RADIUS server template and enter its view.

```
[HUAWEI] radius-server template template-name
```

2. Configure an IPv4 RADIUS authentication server.

```
[HUAWEI-radius-template-name] radius-server authentication ip-address port
```

Configure an IP address and a port number for the RADIUS authentication server.

3. Configure an IPv4 RADIUS accounting server.

```
[HUAWEI-radius-template-name] radius-server accounting ip-address port
```

Configure an IP address and a port number for the RADIUS accounting server.

4. Configure a shared key.

```
[HUAWEI-radius-template-name] radius-server shared-key cipher key-string
```

For security purposes, you are advised to periodically change the shared key.

Configuring a Domain

1. Enter the AAA view.

```
[HUAWEI] aaa
```

Exit the system view and enter the AAA view.

2. Create a domain and enter the domain view.

```
[HUAWEI-aaa] domain domain-name
```

3. Apply AAA schemes to the domain.

```
[HUAWEI-aaa-domain-name] authentication-scheme authentication-scheme-name
```

```
[HUAWEI-aaa-domain-name] accounting-scheme accounting-scheme-name
```

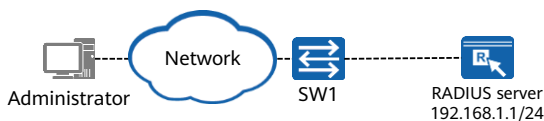
Specify the authentication and accounting schemes to be applied to the domain.

4. Apply a RADIUS server template to the domain.

```
[HUAWEI-aaa-domain-name] radius-server template-name
```

Specify the RADIUS server template to be applied to the specified domain.

Configuration Example for AAA



- An enterprise requires that only users who pass RADIUS authentication can operate network devices.
- Configuration roadmap:
 - Configure a RADIUS server template.
 - Configure AAA schemes.
 - Apply the AAA schemes and RADIUS server template to a domain.

1. Configure a RADIUS server template.

```
[SW1] radius-server template 1
[SW1-radius-1] radius-server authentication 192.168.1.1 1812
[SW1-radius-1] radius-server accounting 192.168.1.1 1813
[SW1-radius-1] radius-server shared-key cipher Huawei@123
[SW1-radius-1] quit
```

2. Configure AAA schemes.

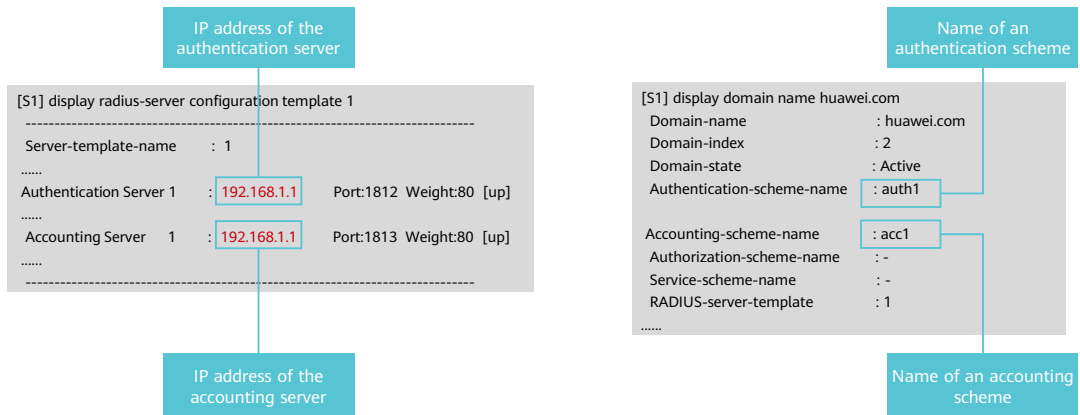
```
[SW1] aaa
[SW1-aaa] authentication-scheme auth1
[SW1-aaa-authen-auth1] authentication-mode radius
[SW1-aaa-authen-auth1] quit
[SW1-aaa] accounting-scheme acc1
[SW1-aaa-accounting-acc1] accounting-mode radius
[SW1-aaa-accounting-acc1] quit
```

3. Apply the AAA schemes and RADIUS server template to a domain.

```
[SW1-aaa] domain huawei.com
[SW1-aaa-domain-huawei.com] authentication-scheme auth1
[SW1-aaa-domain-huawei.com] accounting-scheme acc1
[SW1-aaa-domain-huawei.com] radius-server 1
[SW1-aaa-domain-huawei.com] quit
[SW1-aaa] quit
```

- Note:
 - Configuring a RADIUS server includes adding a device, adding a user, and setting the user privilege level to 3. These configurations are not provided here.
 - The alarm information automatically displayed on the device is not provided in this example.

Verifying the Configuration



Quiz

1. (Short-answer question) What authentication, authorization, and accounting modes are supported by AAA?
2. (Short-answer question) When a new common user is configured with local authentication but is not associated with a user-defined domain, which domain does the user belong to?

1. AAA supports the following authentication modes: non-authentication, local authentication, and server authentication. AAA supports the following authorization modes: non-authorization, local authorization, and server authorization. AAA supports two accounting modes: non-accounting and server accounting.
2. If no domain is specified for a user during the creation of the user, the user is associated with the default_admin domain.

Summary

- AAA improves the security of enterprise networks.
 - Authentication: verifies whether users can obtain access rights and determines which users can access the network.
 - Authorization: authorizes users to access specific services.
 - Accounting: records the network resource usage of users.
- AAA can be implemented either locally or through a remote server.
- Of the protocols that are used to implement AAA, RADIUS is the most commonly used.

Thank you.

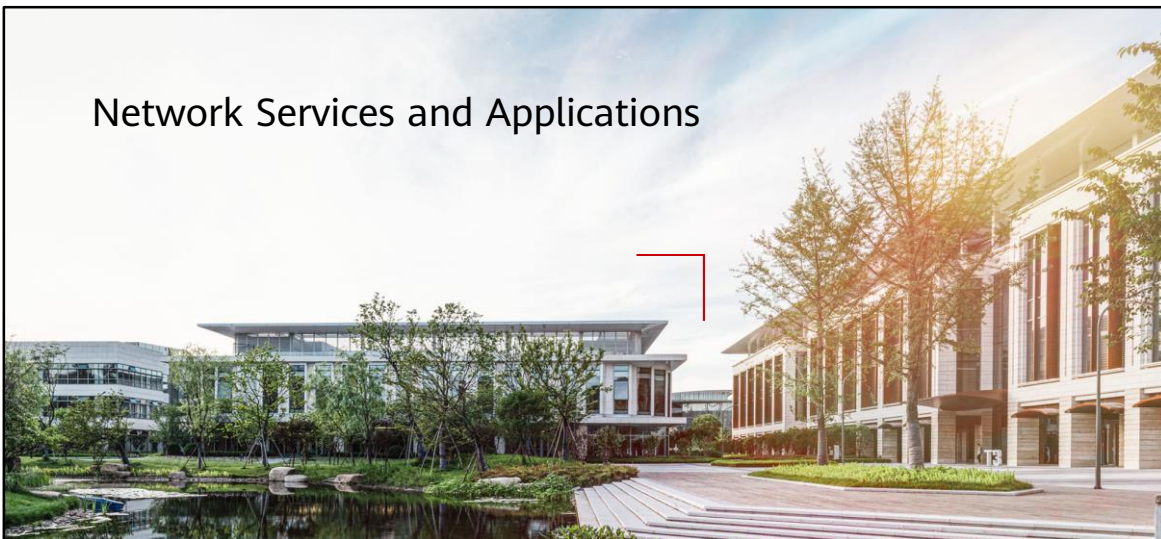
把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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Network Services and Applications



Foreword

- Networks have become an integral part of everyday life. Activities such as file transfers, email communication, online video streaming, web browsing, and online gaming all rely heavily on network connectivity. Due to the hierarchical network model, ordinary users can enjoy various application-layer services without the need to understand the underlying communication principles or technical details.
- In previous courses, we explored technologies related to the data link layer and the network layer. In this course, we will move on to the transport-layer protocols and examine common network services and applications such as FTP, Telnet, and SSH.

Objectives

- Upon completion of this course, you will be able to:
 - Explain the relationship between transport-layer protocols and application-layer protocols.
 - Describe the differences in application scenarios of TCP and UDP.
 - Understand the fundamentals of FTP.
 - Understand the fundamentals of Telnet.
 - Understand the fundamentals and configuration method of SSH.

Contents

1. Overview of Network Services

- 2. FTP for File Transfer
- 3. Telnet for Remote Login
- 4. SSH

Overview of Network Services

- Application-layer services are built on top of transport-layer protocols, leveraging their end-to-end communication capabilities to implement specific functions.
- Transport-layer protocols, such as Transmission Control Protocol (TCP) and User Datagram Protocol (UDP), provide reliable and efficient data transmission channels for the application layer:
 - TCP ensures accurate and ordered data delivery, making it suitable for services such as web browsing and email.
 - UDP offers lightweight transmission, ideal for real-time scenarios such as video and voice services.
- Application-layer protocols, such as HTTP and DNS, define the rules for data exchange based on transport-layer services, transforming underlying network connections into applications that users can directly access. The transport and application layers work together to support a wide range of network services and applications.

Introduction to TCP and UDP

TCP

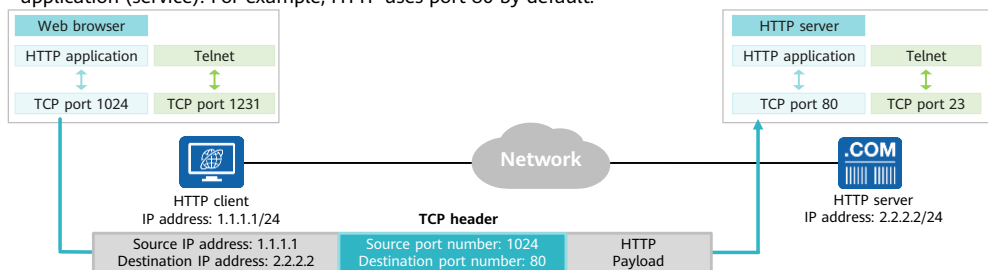
- **Connectivity:** TCP is a connection-oriented transport-layer protocol.
- **Reliability:** TCP provides highly reliable data transmission. Through mechanisms such as acknowledgments, retransmission, sequencing, and congestion control, TCP ensures that data is transmitted accurately, without loss or duplication, and in the correct order.
- **Transmission efficiency:** TCP has low transmission efficiency but ensures fairness across the network.

UDP

- **Connectivity:** UDP is a connectionless transport-layer protocol.
- **Reliability:** UDP provides best-effort data transmission. It does not guarantee reliability, meaning that data may be lost, duplicated, or transmitted out of order. These issues must be handled by the application layer if needed.
- **Transmission efficiency:** UDP offers low latency, high speed, and high efficiency.

TCP vs. UDP: Port Number

- TCP and UDP use the **source port number** and **destination port number** to identify the application that sends data and the application that receives data.
 - The source port number used by a client is typically assigned dynamically. It is chosen from the range of unused ports on the system, usually greater than 1023.
 - The destination port number is determined by the application on the server. It is the port listened on by the application (service). For example, HTTP uses port 80 by default.



- The value port number range for both TCP and UDP is 0–65535 ($2^{16}-1=65535$).
- Source ports are typically chosen from unused ports greater than 1023 because ports 0–1023 are well-known ports reserved for standard network services. For example, HTTP uses TCP port 80, Telnet uses TCP port 23, and DNS uses UDP port 53.

TCP vs. UDP: Segment Structure

TCP segment structure

Source port (16)			Destination port (16)	
Sequence number (32)				
Acknowledgement number (32)				
Header length (4)	Reserved (4)	Control bits (8)	Window (16)	
Checksum (16)			Urgent (16)	
Options (0–40)				
Data (varies)				

TCP header
20 bytes

UDP segment structure

Source port (16)	Destination port (16)	} UDP header 8 bytes
Length (16)	Checksum (16)	
Data (if any)		

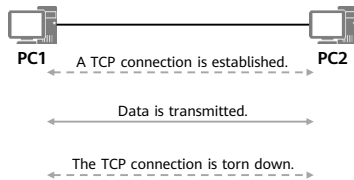
- TCP header:
 - Source Port: identifies the application that sends data. This field is 16 bits long.
 - Destination Port: identifies the application that receives the data. This field is 16 bits long.
 - Sequence Number (Seq): Each byte in a TCP stream is assigned a sequence number. This field specifies the sequence number of the first byte of data in this segment. This field is 32 bits long.
 - Acknowledgment Number (Ack): indicates the sequence number of the first octet expected in the next segment from the peer. The value is the sequence number of the last octet in the segment received last time plus 1. This field is valid only when the ACK flag is set to 1. This field is 32 bits long.
 - Header Length: indicates the length of the TCP header, measured in units of 32-bit (4-byte) words. If no Options field is present, the value is 5, indicating a header length of 20 bytes. This field is 4 bits long.
 - Reserved: is a field reserved for future use, and must be set to 0. This field is 4 bits long.
 - Control bits: include flags such as URG, ACK, SYN, and FIN, which represent different states of a TCP segment. This field is 8 bits long.
 - URG: If this flag is set to 1, the urgent pointer field is valid.

- ACK: If this flag is set to 1, the acknowledgment number field is valid.
- SYN: If this flag is set to 1, sequence numbers are synchronized.
- FIN: If this flag is set to 1, the sender has no more data to transmit.
- Window: indicates the TCP flow control window. It specifies the maximum number of octets that the receiver is currently prepared to receive. The maximum window size is 65,535 bytes. This field is 16 bits long.
- Checksum: holds the checksum for error control. It is mandatory in TCP. The checksum is calculated and stored by the sender and verified by the receiver. The checksum calculation includes the TCP header, TCP data, and a 12-byte pseudo header added before the TCP segment. This field is 16 bits long.
- Urgent: valid only when the URG flag is set to 1. The urgent mode in TCP allows the sender to indicate that certain data should be processed urgently by the receiver. This urgent pointer indicates how many bytes of urgent data are present in this segment (urgent data appears at the beginning of the data in this segment). This field is 16 bits long.
- Options: an optional field used to extend TCP functionality. This field is 0 to 40 bytes in length.
- UDP header:
 - Source Port: identifies the application that sends data. This field is 16 bits long.
 - Destination Port: identifies the application that receives the data. This field is 16 bits long.
 - Length: specifies the total length of the UDP header and data. The minimum value is 8 bytes (the size of the UDP header). Because this field is 16 bits, the maximum UDP datagram size is 65,535 bytes ($2^{16}-1=65535$). This field is 16 bits long.
 - Checksum: field calculated over the UDP header and UDP payload. This field is 16 bits long.

TCP vs. UDP: Working Mechanism

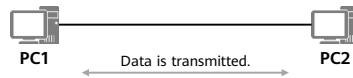
TCP: Connection-Oriented Reliable Transmission

- A reliable connection must be established through a three-way handshake before data transmission.
- During transmission, TCP ensures no data loss, no duplication, no errors, and ordered transmission.
- When communication is completed, the connection is terminated through a four-way handshake.



UDP: Connectionless Best-Effort Transmission

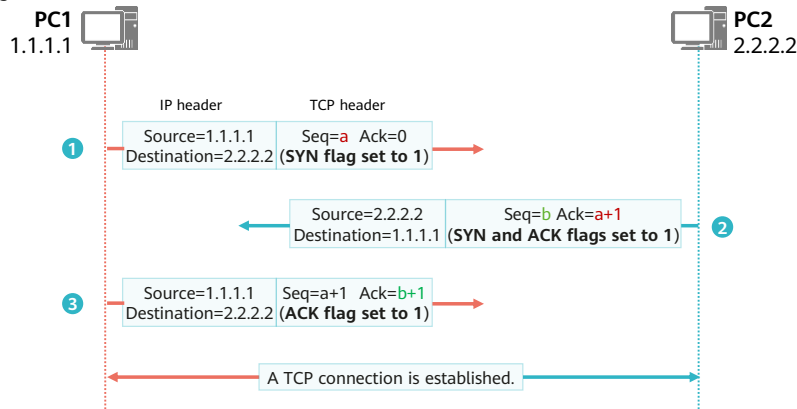
- UDP transmits data directly without the need to establish a connection or maintain connection states.
- UDP does not guarantee reliability, but offers lower overhead and latency.



- Note: UDP operates without establishing a connection. This course focuses only on the mechanisms for establishing and terminating a TCP connection.

TCP Working Mechanism: Connection Establishment

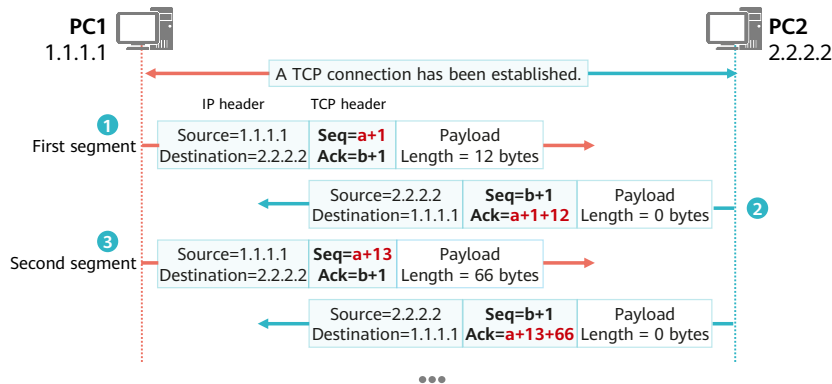
- Any application carried over TCP must establish a connection through a three-way handshake before transmitting data.



- The process of establishing a TCP connection is as follows:
 - The initiator of the TCP connection (PC1) sends the first TCP segment with the SYN flag set to 1. In this segment, the initial sequence number **a** is a randomly generated value, and the acknowledgment number is **0** since PC1 has received no segment from PC2.
 - Upon receiving the segment with the SYN flag set to 1 from PC1, the receiver (PC2) replies with a TCP segment with both SYN and ACK flags set to 1. In this segment, the initial sequence number **b** is also a randomly generated value. As this segment is a response to PC1, its acknowledgment number is **a + 1**.
 - After receiving the TCP segment with both SYN and ACK flags set to 1 from PC2, PC1 replies with a TCP segment with the ACK flag set to 1, sequence number set to **a + 1**, and acknowledgment number set to **b + 1** (acknowledging that it has received the segment from PC2). Once PC2 receives the segment, a bidirectional TCP connection is established.

TCP Working Mechanism: Data Transmission

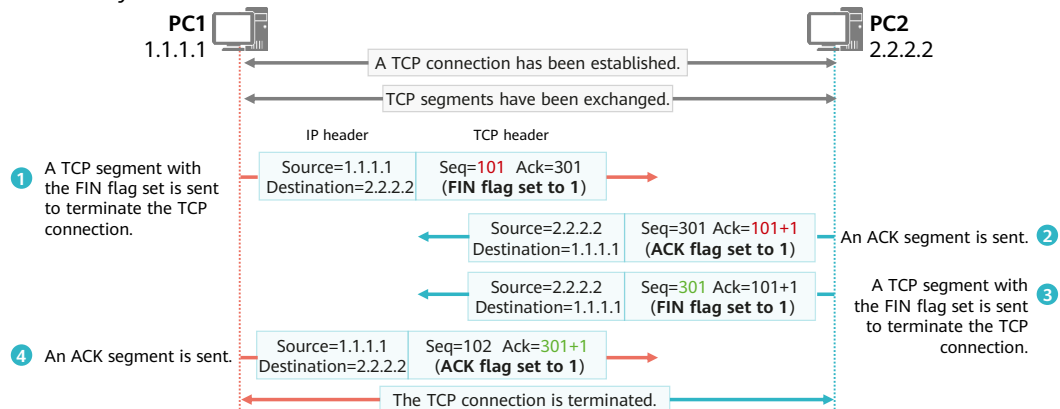
- After a TCP connection is established, TCP uses the Sequence Number and Acknowledgment Number fields to ensure reliable and ordered data transmission.



- Assume that PC1 needs to send data to PC2, and PC2 does not send any data in return. The transmission process is as follows:
 - PC1 numbers all bytes of the data segment to be transmitted sequentially. Suppose the first byte is numbered $a+1$, the second byte $a+2$, and so on. For each segment, PC1 uses the sequence number of the first byte in that segment as the Sequence Number field (for example, the first segment sent by PC1 has Seq = $a+1$). Each segment carries a payload of 12 bytes.
 - After receiving a TCP segment from PC1, PC2 must acknowledge it and request the next segment. The Acknowledgment Number in the returned segment is calculated as follows: Ack = received sequence number ($a+1$) + payload length (12) = $a+1+12$
 - When PC1 receives the acknowledgment from PC2 and sees that Ack = $a+1+12$, it knows that the bytes corresponding to sequence numbers $a+1$ through $a+12$ have been successfully received. PC1 then sends the next segment starting from the byte whose sequence number is $a+13$.
- To improve transmission efficiency, TCP allows multiple segments to be sent at a time and acknowledged cumulatively by the receiver. This course does not cover these details. For more information, see RFC 793.

TCP Working Mechanism: Connection Termination

- When data transmission is complete, TCP terminates the connection through a four-way handshake to release system resources.



- TCP supports full-duplex data transmission, meaning that data can be sent in both directions simultaneously. Before data transmission begins, TCP establishes a bidirectional connection through a three-way handshake. Therefore, when the transmission is complete, both directions of the connection must be closed.
- As shown in the figure:
 - PC1 sends a TCP segment with the FIN flag set and no payload.
 - Upon receiving the TCP segment from PC1, PC2 replies with a TCP segment with the ACK flag set.
 - If PC2 has no more data to send, it immediately sends a TCP segment with the FIN flag set. If PC2 still has data to transmit, it sends a TCP segment with the FIN flag set to terminate the connection only after all remaining data has been sent.
 - When PC1 receives the TCP segment with the FIN flag set from PC2, it replies with a TCP segment with the ACK flag set. At this point, the bidirectional TCP connection is fully terminated.

Application Scenarios of TCP and UDP

TCP

- Suitable for scenarios where data reliability is more important than speed.
- Examples: web browsing (HTTP/HTTPS), file transfer (FTP), email (SMTP), and remote login (Telnet/SSH)

UDP

- Suitable for scenarios that require low latency and high real-time performance, and can tolerate a certain degree of data loss.
- Examples: video and voice communication, online gaming, DNS queries, and broadcast services

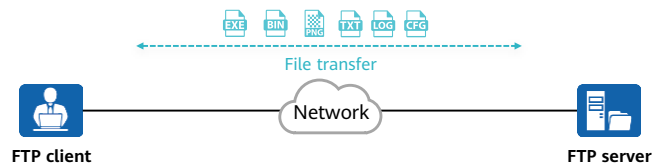
- Because TCP provides higher reliability for data transmission, this course focuses only on application-layer services that run over TCP, including FTP, Telnet, and SSH.

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1. Overview of Network Services
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FTP

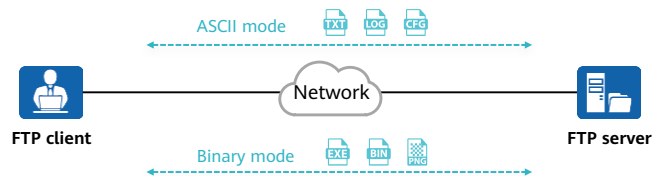
- File Transfer Protocol (FTP) is a standard network protocol used to transfer files between hosts on a computer network. It allows users to upload files from one host or download files to another host.
- FTP adopts the client/server model.
 - FTP client: transfers files and manages files and directories on the server.
 - FTP server: stores files and runs continuously, waiting for connection requests from clients.
- Application scenarios: FTP is widely used in file-related operations such as version upgrades.



- Common network devices, such as routers and switches, can function as FTP clients and servers.

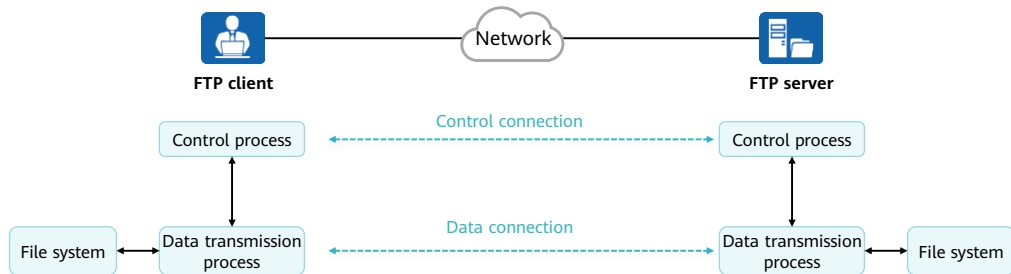
FTP Transfer Mode

- FTP supports two transfer modes depending on the types of files to be transferred.
 - ASCII mode: used to transfer text files (such as TXT, LOG, and CFG). The sender converts the characters into ASCII format before transmission, and the receiver converts them back into characters after receiving the data. This improves transmission efficiency, and is recommended for transferring configuration and log files of network devices.
 - Binary mode: used to transfer non-text files (such as CC, BIN, EXE, and PNG), such as images and executable programs. The sender transmits the raw file content directly in binary form without any format conversion. This mode is recommended for transferring software files of network devices.



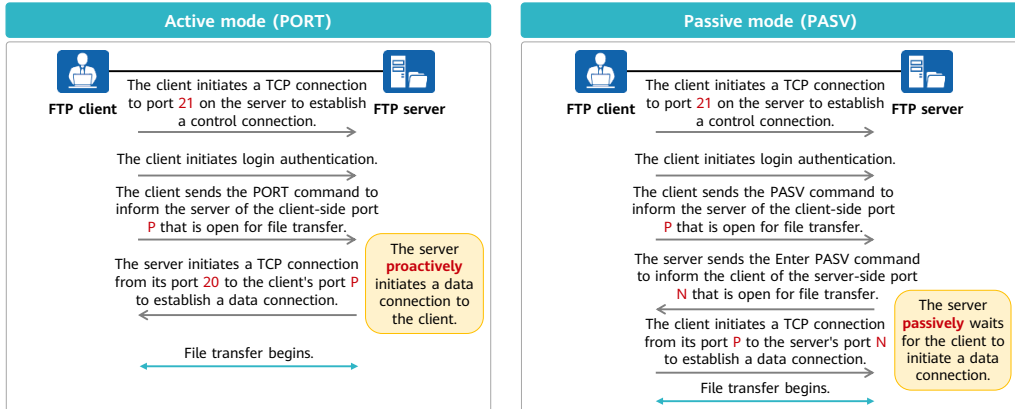
FTP Connection

- After the FTP client and server establish a TCP connection, files can be uploaded or downloaded.
 - Control connection (TCP port 21): used to transfer FTP control commands. The control connection remains open throughout the FTP session.
 - Data connection (typically TCP port 20): used to transfer actual file content or directory lists. The data connection is closed after each file transfer is completed.



FTP Working Mode

- FTP supports two working modes based on how a data connection is established between the server and client: active mode and passive mode.



- Active mode:
 - In active mode, the FTP client uses a random port (typically greater than 1024) to send a connection request to port 21 on the FTP server. After the FTP server accepts the request, a control connection is established for transmitting control messages.
 - At the same time, the FTP client starts to listen on another random port P (also typically greater than 1024) and informs the server of this port using the **PORT** command. When data needs to be transmitted, the FTP server initiates a connection from its port 20 to the client's port P to establish a data connection.
 - In active mode, a control connection is established through TCP port 21, and a data connection is established through TCP port 20.
- Passive mode:
 - In passive mode, the FTP client uses a random port (typically greater than 1024) to send a connection request to port 21 on the FTP server. The server accepts the request and establishes a control connection for transmitting control messages.
 - At the same time, the FTP client starts to listen on another random port P (also typically greater than 1024) and informs the server of this port using the **PASV** command. After receiving the **PASV** command, the FTP server enables a random port N (typically greater than 1024) and informs the client of this port using the **Enter PASV** command. When data needs to be transmitted, the FTP client initiates a connection from its port P to the

server's port N to establish a data connection.

- In passive mode, a control connection is established through TCP port 21, and a data connection is established through a random TCP port.

FTP Configuration Roadmap

FTP Server Configuration

- Configure the FTP server function and parameters.
- Configure a local FTP user.
- (Optional) Configure FTP access control.

FTP Client Configuration

- Log in to another device using FTP to perform file operations.
- Disconnect from the FTP server.

Configuring a Device as an FTP Server (1)

1. Enable the FTP server function.

```
[HUAWEI] ftp [ ipv6 ] server enable
```

By default, the FTP server function is disabled.

2. Specify a source interface or source address for the FTP server.

```
[HUAWEI] ftp server source { -a ip-address | -i { interface-type interface-number | interface-name } }
```

```
[HUAWEI] ftp ipv6 server source -a ipv6-address [ -vpn-instance vpn-instance-name ]
```

```
[HUAWEI] ftp [ ipv6 ] server source all-interface
```

Specifying a source interface or source address for the FTP server allows the device to filter incoming and outgoing packets to enhance security.

After specifying a source address for the server, you can log in to the server only through this configured address.

Run one of the preceding commands based on the actual situation.

- Note: This course uses the Huawei NetEngine AR5700 router as an example to describe how to configure FTP.

Configuring a Device as an FTP Server (2)

3. Configure a local FTP user.

Configure a local user, with a user name and password specified.

```
[HUAWEI] aaa  
[HUAWEI-aaa] local-user user-name password irreversible-cipher password
```

Configure the local user privilege level.

```
[HUAWEI-aaa] local-user user-name privilege level level
```

You must set the user privilege level to the management level (level 3). Otherwise, the FTP connection cannot be established.

Set the service type of the local user to FTP.

```
[HUAWEI-aaa] local-user user-name service-type ftp
```

Configure an authorized directory for the FTP user.

```
[HUAWEI-aaa] local-user user-name ftp-directory directory
```

- Configuring a local FTP user: When a network administrator manages files on network devices through FTP, AAA can be used to control their permissions.

Configuring a Device as an FTP Client (1)

1. Connect to another device using FTP.

```
<HUAWEI> ftp [ -a source-ip-address | -i { interface-type interface-number | interface-name } ] host-ip [ port-number ] [ vpn-instance vpn-instance-name | public-net ]
```

2. Perform file operations using FTP.

After logging in to the FTP server, you can run FTP commands to perform operations on files, including managing directories, managing files, configuring the file transfer mode, and viewing online help of FTP commands.

Common operation commands:

ascii	Set the file transfer type to ASCII (default mode).
binary	Set the file transfer type to support binary images.
ls	List the contents of the current or remote directory.
passive	Enable the passive mode (enabled by default).
get	Download a remote file to the local host.
put	Upload a local file to the remote host.

- In the FTP client view, you can run the following FTP commands to perform operations on files:
 - **ascii**: switches to the ASCII transfer mode.
 - **binary**: switches to the binary transfer mode.
 - **ls**: displays the file list on the FTP server. You can also run the **dir** command.
 - **passive**: configures the passive mode. The **undo passive** command configures the active mode.
 - **get**: downloads files from the FTP server to the local host.
 - **put**: uploads local files to the FTP server.

Configuring a Device as an FTP Client (2)

3. Disconnect from the FTP server.

Disconnect from the FTP server and return to the user view.

```
[ftp] bye
```

```
[ftp] quit
```

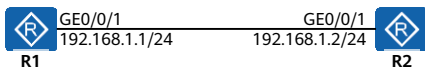
Disconnect from the FTP server and remain in the FTP client view.

```
[ftp] close
```

```
[ftp] disconnect
```

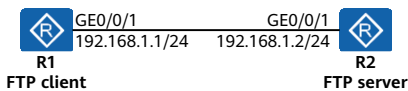
FTP Configuration Example

- As shown in the figure, R1 and R2 are directly connected. To upgrade system software of R2, it is required that R1 act as the FTP client and R2 as the FTP server. The system software needs to be uploaded from R1 to R2 and the current configuration file of R2 needs to be saved to R1 for backup.
- In addition, an ACL policy needs to be configured to allow access to R2 only from R1.



- Configuration roadmap:
 - Configure the FTP server function on R2, including the source address for the server. In addition, configure an FTP user on R2, including the user name, password, user privilege level, service type, and authorized directory.
 - Configure access control for the FTP server (R2).
 - Save the current configuration file of R2.
 - Log in to R2 from R1 using FTP.
 - Upload system software from R1 to R2 and back up R2's configuration file to R1.

Configuration Procedure (1)



1. Configure the IP address of the FTP server.

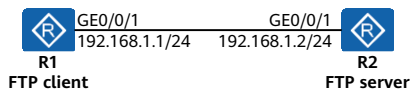
```
<HUAWEI> system-view
[HUAWEI] sysname R2
[R2] interface GE 0/0/1
[R2-GE0/0/1] undo portswitch
[R2-GE0/0/1] ip address 192.168.1.2 24
[R2-GE0/0/1] quit
```

2. Configure the FTP function. In addition, configure an FTP user, with the user name **admin1** and password **Huawei@123**.

```
[R2] ftp server enable
[R2] ftp server source -i GE 0/0/1
[R2] aaa
[R2-aaa] local-user admin1 password
Please configure the login password (8-128)
It is recommended that the password consist of four types of
characters, including lowercase letters, uppercase letters,
numerals and special characters.
Please enter password: Huawei@123
Please confirm password: Huawei@123
[R2-aaa] local-user admin1 privilege level 3
[R2-aaa] local-user admin1 service-type ftp
Warning: If the service type of local users is configured multiple
times, the new configuration overwrites the previous one. The
new configuration may cause local user authentication to fail.
Continue? [Y/N]: y
[R2-aaa] local-user admin1 ftp-directory flash:/
[R2-aaa] quit
```

- Weak Security Algorithm or Protocol Feature Package WEAKEA:
 - Some protocols or algorithms inherently have lower security. Using these protocols or algorithms may pose security risks. For security purposes, you are advised not to use insecure protocols or algorithms. If their use is required, the weak security algorithm or protocol feature package WEAKEA must be installed.
 - Weak security protocols include Telnet, FTP, TFTP, SSHv1, SNMPv1, and SNMPv2c.
- When configuring an FTP server, you can install the weak security algorithm or protocol feature package WEAKEA using the **install feature-software WEAKEA** command in the user view.
- Note: The actual command output varies according to the device. The command output in this configuration example is for reference only.

Configuration Procedure (2)



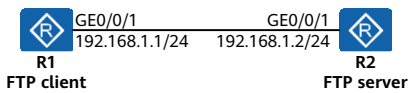
3. Configure access control for the FTP server.

```
[R2] acl number 2001
[R2-acl4-basic-2001] rule 5 permit source 192.168.1.1 0
[R2-acl4-basic-2001] rule 10 deny
[R2-acl4-basic-2001] quit
[R2] ftp server acl 2001
[R2] quit
```

4. Save the device's current configuration file.

```
<R2> save
Warning: The current configuration will be written to the
device. Continue? [Y/N]: y
```

Configuration Procedure (3)



5. Configure an IP address for the FTP client.

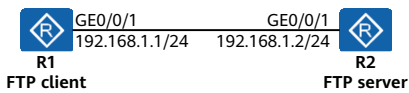
```
<HUAWEI> system-view
[HUAWEI] sysname R1
[R1] interface GE 0/0/1
[R1-GE0/0/1] undo portswitch
[R1-GE0/0/1] ip address 192.168.1.1 24
[R1-GE0/0/1] return
```

6. Connect to R2 from R1 using FTP. Enter the user name **admin1** and password **Huawei@123** and configure the binary mode for file transfer.

```
<R1> ftp 192.168.1.2
Warning: FTP is not secure. Using SFTP is recommended.
Trying 192.168.1.2 ...
Press CTRL + K to abort
Connected to 192.168.1.2.
220 FTP service ready.
User(192.168.1.2:(none)): admin1
331 Password required for admin.
Enter password: Huawei@123
230 User logged in.
[ftp]
[ftp] binary
200 Type set to I.
```

- When configuring an FTP client, you can install the weak security algorithm or protocol feature package WEAKEA using the **install feature-software WEAKEA** command in the user view.

Configuration Procedure (4)



7. Upload the system software package from R1 to R2.

```
[ftp] put devicesoft.cc
200 Port command okay.
150 Opening BINARY mode data connection for /devicesoft.cc
/ 100% [*****]
226 Transfer complete.

FTP: 5027 byte(s) send in 0.000 second(s)
```

8. Back up the configuration file of R2.

```
[ftp] get vrpcfg.zip
200 Port command okay.
150 Opening BINARY mode data connection for /vrpcfg.zip.
/ 100% [*****]
226 Transfer complete.

FTP: 4599 byte(s) received in 0.010 second(s)
449.100Kbyte(s)/sec.
[ftp] quit
221 Server closing.
```

Verifying the Configuration

1. Run the **dir** command on R2 to check whether the system software package has been uploaded to it.

```
<R2> dir
Directory of flash:/

Idx Attr  Size(Byte) Date          Time          FileName
0 -rw- 14        Mar 13 20XX 14:13:38 back_time_a
1 drw- -        Mar 11 20XX 00:58:54 logfile
2 -rw- 4        Nov 17 20XX 09:33:58 snmpnotilog.txt
3 -rw- 11,238 Mar 12 20XX 21:15:56 private-data.txt
4 -rw- 1,257 Mar 12 20XX 21:15:54 vrpcfg.zip
5 -rw- 14        Mar 13 20XX 14:13:38 back_time_b
6 -rw- 107,973,953 Mar 13 20XX 14:24:24 devicesoft.cc
7 drw- -        Oct 31 20XX 10:20:28 sysdrv
8 drw- -        Feb 21 20XX 17:16:36 compatible
9 drw- -        Feb 09 20XX 14:20:10 selftest
10 -rw- 19,174 Feb 20 20XX 18:55:32 backup.cfg
11 -rw- 23,496 Oct 15 20XX 20:59:36 20XX1015.zip
12 -rw- 588    Nov 04 20XX 13:54:04 servercert.der
13 -rw- 320    Nov 04 20XX 13:54:26 serverkey.der
14 drw- -        Nov 04 20XX 13:58:36 security
...
670,092 KB total (569,904 KB free)
```

2. Run the **dir** command on R1 to check whether R2's configuration file has been downloaded to R1.

```
<R1> dir
Directory of flash:/

Idx Attr  Size(Byte) Date          Time          FileName
0 -rw- 14        Mar 13 20XX 14:13:38 back_time_a
1 drw- -        Mar 11 20XX 00:58:54 logfile
2 -rw- 4        Nov 17 20XX 09:33:58 snmpnotilog.txt
3 -rw- 11,238 Mar 12 20XX 21:15:56 private-data.txt
4 -rw- 1,257 Mar 12 20XX 21:15:54 vrpcfg.zip
5 -rw- 14        Mar 13 20XX 14:13:38 back_time_b
6 -rw- 107,973,953 Mar 13 20XX 14:24:24 devicesoft.cc
7 drw- -        Oct 31 20XX 10:20:28 sysdrv
8 drw- -        Feb 21 20XX 17:16:36 compatible
9 drw- -        Feb 09 20XX 14:20:10 selftest
10 -rw- 19,174 Feb 20 20XX 18:55:32 backup.cfg
11 -rw- 23,496 Oct 15 20XX 20:59:36 20XX1015.zip
12 -rw- 588    Nov 04 20XX 13:54:04 servercert.der
13 -rw-        Nov 04 20XX 13:54:26 serverkey.der
14 drw- -        Nov 04 20XX 13:58:36 security
...
670,092 KB total (569,904 KB free)
```

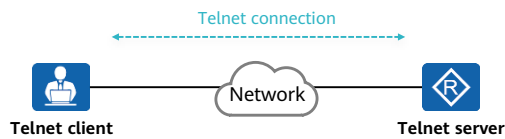


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2. FTP for File Transfer
- 3. Telnet for Remote Login**
4. SSH

Remote Login Protocol

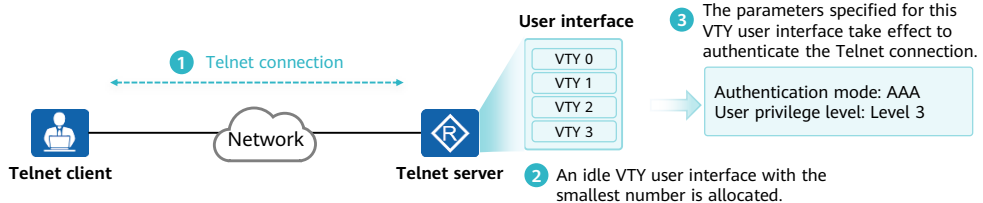
- Telnet is an application-layer protocol used for remote login over a computer network. It allows users to log in from one host to another remote host and operate as if they were using a local CLI.
- Unlike managing a device through a console port, Telnet does not require a dedicated cable connected directly to the console port. As long as the Telnet client has a reachable IP address and can communicate with the TCP port 23 of the device, Telnet access is possible.
- Telnet uses the client/server architecture. A device that can be managed using Telnet is called a Telnet server, while the terminal initiating a connection to the server is referred to as a Telnet client.



- Common network devices, such as routers and switches, can function as Telnet servers and clients.

Virtual User Interface

- When a user logs in to a device through the console port, Telnet, or other methods, the system allocates a user interface to manage and monitor the current session between the user and device. Each user interface view can be configured with a set of parameters to define the user's authentication mode and privilege level after login. When a user logs in, these parameters take effect and restrict the user accordingly.
- For Telnet access, the corresponding user interface type is Virtual Type Terminal (VTY).



- When a user sets up a Telnet or SSH connection with the device from a terminal, a VTY is created to provide the user with access to the device. Up to 21 users (the exact number varies depending on the device model and version) can access the device through VTYS simultaneously.
- User interface numbering:
 - When you log in to a device, the device automatically assigns the lowest-numbered available user interface of the corresponding type to you based on your login method.
 - User interfaces are identified using a relative numbering format that consists of the user interface type followed by its number. For VTY interfaces, the available numbers range from 0 to 20.
- Authentication modes for user interfaces:
 - After authentication modes are configured for user interfaces, the device verifies a user's identity upon their logins.
 - The following user authentication modes are supported:
 - Password authentication: requires only a password.
 - AAA authentication: requires both a user name and password. This mode is typically configured for Telnet users.
- User privilege levels for user interfaces:
 - A device supports hierarchical user privilege levels to control what commands a logged-in user can execute.

- If password authentication is configured, the levels of commands available to a user are determined by the privilege level assigned to the user interface they log in through.
- If AAA authentication is configured, the levels of commands available to a user are determined by the privilege level assigned to that user in the AAA configuration.

Telnet Configuration Roadmap

Configuring Telnet Login

- Configure the Telnet server function and related parameters.
- Configure a user interface for Telnet login.
- Configure a local Telnet user.
- (Optional) Configure Telnet access control.

Configuring a Device to Access Another Device as a Telnet Client

- Log in to another device using Telnet.

Configuring the Telnet Server Function and Related Parameters

1. Enable the Telnet server function.

```
[HUAWEI] telnet [ ipv6 ] server enable
```

By default, the Telnet server function is disabled.

2. Specify a source interface for the Telnet server.

```
[HUAWEI] telnet server source -i { interface-type interface-number | interface-name }
```

```
[HUAWEI] telnet ipv6 server source -a ipv6-address [ -vpn-instance vpn-instance-name ]
```

```
[HUAWEI] telnet [ ipv6 ] server source all-interface
```

By default, no source interface is specified for the Telnet server. Run one of the preceding commands based on the actual situation.

- Specifying a source interface for the Telnet server: If a loopback interface is designated as the source interface, ensure that this loopback interface has been created. Otherwise, the configuration fails.
- Note: This course uses the Huawei NetEngine AR5700 router as an example to describe how to configure Telnet.

Configuring a User Interface for Telnet Login

1. Enter the VTY user interface view.

```
[HUAWEI] user-interface vty first-ui-number [ last-ui-number ]
```

2. Configure a user privilege level and other attributes for the user interface. This example uses VTY 0.

Configure a user privilege level for the user interface.

```
[HUAWEI] user-interface vty 0  
[HUAWEI-ui-vty0] user privilege level level
```

By default, the user privilege level of a VTY user interface is 0.

Configure a user authentication mode.

```
[HUAWEI-ui-vty0] authentication-mode aaa
```

Configure the VTY user interface to support the Telnet protocol.

```
[HUAWEI-ui-vty0] protocol inbound { all | ssh | telnet }
```

- By default, a VTY user interface supports all protocols, including Telnet.

Configuring a local Telnet User

1. Configure a local user with a user name and password specified.

```
[HUAWEI] aaa  
[HUAWEI-aaa] local-user user-name password irreversible-cipher password
```

2. Set the service type of the local user to Telnet.

```
[HUAWEI-aaa] local-user user-name service-type telnet
```

3. Configure the local user privilege level.

```
[HUAWEI-aaa] local-user user-name privilege level level
```

After logging in, a user can run only the commands at or below their assigned user privilege level, which helps maintain device security.

- Configuring a local Telnet user: When a network administrator logs in to network devices through Telnet for configuration and commissioning, AAA can be used to control the administrator's permissions.
- If the user privilege level configured for a user interface conflicts with that defined for a local Telnet user, the latter takes effect.
- Note: This slide presents the commands for configuring local users in AAA authentication mode.

Configuring a Device as a Telnet Client

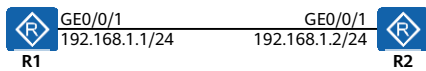
1. Log in to another device using Telnet.

```
<HUAWEI> telnet [ -i { interface-type interface-number | interface-name } ] [ vpn-instance vpn-instance-name ] [ -a source-ip-address ] host-ip-address [ port-number ]
```

A Telnet client can connect to the server successfully without an explicit port number only when the server listens on port 23. If the server listens on a port other than 23, a port number must be specified for the Telnet client to log in.

Telnet Configuration Example

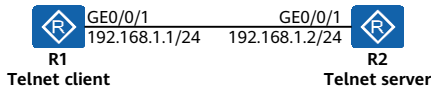
- As shown in the figure, R1 and R2 are directly connected. It is required that R1 act as a Telnet client to manage and maintain R2 that functions as a Telnet server.
- A Telnet user needs to be configured on the server, and AAA authentication needs to be configured for that user. In addition, an ACL needs to be configured to allow Telnet access to R2 only from R1.



- Configuration roadmap:

- Configure Telnet login. Specifically, enable the Telnet server function to remotely maintain network devices.
- Configure the user name and password for an administrator, and configure an AAA authentication policy to ensure that only users who pass the authentication can log in to the device.
- Set VTY user interface parameters, including configuring an ACL to ensure that only users matching the ACL can log in to the device.

Configuration Procedure (1)



1. Configure an IP address for the server.

```
<HUAWEI> system-view
[HUAWEI] sysname R2
[R2] interface GE 0/0/1
[R2-GE0/0/1] undo portswitch
[R2-GE0/0/1] ip address 192.168.1.2 24
[R2-GE0/0/1] quit
```

2. Enable the Telnet server function.

```
[R2] telnet server enable
[R2] telnet server-source all-interface
```

3. Configure VTY interfaces for Telnet on the server.

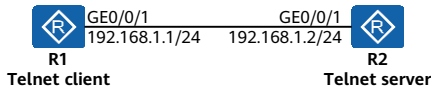
```
[R2] user-interface vty 0 4
[R2-ui-vty0-4] authentication-mode aaa
[R2-ui-vty0-4] user privilege level 3
[R2-ui-vty0-4] protocol inbound telnet
[R2-ui-vty0-4] quit
```

4. Configure a local user, with the user name **admin1** and password **Huawei@123**.

```
[R2] aaa
[R2-aaa] local-user admin1 password
Please configure the login password (8-128)
It is recommended that the password consist of four types of
characters, including lowercase letters, uppercase letters,
numerals and special characters.
Please enter password: Huawei@123
Please confirm password: Huawei@123
[R2-aaa] local-user admin1 service-type telnet
[R2-aaa] local-user admin1 privilege level 3
[R2-aaa] quit
```

- When configuring a Telnet server, you can install the weak security algorithm or protocol feature package WEAKEA using the **install feature-software WEAKEA** command in the user view.
- Note: The actual command output varies according to the device. The command output in this configuration example is for reference only.

Configuration Procedure (2)



5. Configure an ACL on R2 to allow access only from R1.

```
[R2] acl 2000
[R2-acl4-basic-2000] rule 5 permit source 192.168.1.1 0
[R2-acl4-basic-2000] rule 10 deny
[R2-acl4-basic-2000] quit
[R2] user-interface vty 0 4
[R2-ui-vty0-4] acl 2000 inbound
[R2-ui-vty0-4] quit
```

6. Configure an IP address for the client.

```
<HUAWEI> system-view
[HUAWEI] sysname R1
[R1] interface GE 0/0/1
[R1-GE0/0/1] undo portswitch
[R1-GE0/0/1] ip address 192.168.1.1 24
[R1-GE0/0/1] return
```

7. Log in to R2 from R1 using Telnet.

```
<R1> telnet 192.168.1.2
Trying 192.168.1.2...
Press CTRL+K to abort
Connected to 192.168.1.2...
Warning: Telnet is not a secure protocol, and it is recommended
to use Stelnet.
```

```
Username: admin1
Password: Huawei@123
Warning: The initial password poses security risks.The password
needs to be changed, Continue? [Y/N]: y
Please enter old password: Huawei@123
Please enter new password: Huawei@1234
Please confirm new password: Huawei@1234
The password has been changed successfully.
Info: The max number of VTY users is 21, the number of current
VTY users online is 4, and total number of terminal users online
is 4.
The current login time is 20XX-12-15 14:23:00.
<R2>
```

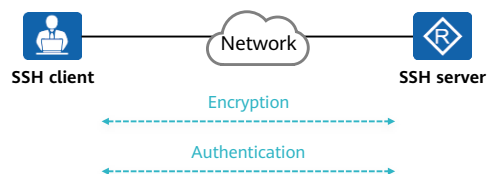
- When configuring a Telnet client, you can install the weak security algorithm or protocol feature package WEAKEA using the **install feature-software WEAKEA** command in the user view.
- By default, a local administrator needs to change the login password upon the first login. This is to prevent potential risks caused by the initial password, thereby enhancing device security.

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- 4. SSH**

SSH

- Secure Shell (SSH) is a network security protocol that provides secure access and file transfer services over untrusted networks through encryption and authentication.
 - Encryption: All data exchanged between the client and server is encrypted.
 - Authentication: Multiple authentication modes are supported to verify the identities of both ends of the connection, ensuring that the communication is established with the intended party.
- SSH adopts the client/server architecture and sets up a secure channel over TCP for data transmission. It typically relies on TCP port 22, and can be configured to use other ports to reduce exposure to attacks.



- SSH is available in two versions: earlier SSH 1.x series and SSH 2.0. Compared with SSH 1.x, SSH 2.0 introduces an expanded architecture that supports a wider range of authentication methods and key exchange algorithms, while also enhancing service capabilities such as SFTP.
- Common network devices, such as routers and switches, can function as SSH servers and clients.

Components of SSH

- SSH consists of three components: transport layer protocol, user authentication protocol, and connection protocol.

Transport layer protocol	User authentication protocol	Connection Protocol
- The SSH transport layer protocol handles version negotiation, algorithm negotiation, and key exchange. - It typically runs over a TCP connection. - During negotiation, it determines the key exchange algorithms, public key algorithms, symmetric encryption algorithms, and message authentication algorithms to be used.	- The SSH user authentication protocol provides client authentication to the server. - It operates on top of the transport layer protocol. - Two authentication methods are commonly used: - Password authentication: The client provides a user name and password to authenticate with the server. - Public key authentication: The server authenticates the client by verifying the client's digital signature using a public key.	- The SSH connection protocol establishes session connections and provides multiple channels for different purposes. - It runs above both the transport layer protocol and the user authentication protocol. - It multiplexes the encrypted session into several logical channels, enabling features such as interactive shell sessions, remote command execution, and TCP/IP forwarding. All channels share a session connection.

- Algorithms:
 - Diffie-Hellman (DH) algorithm: used to generate session keys. Common algorithms include diffie-hellman-group14-sha1 and diffie-hellman-group1-sha1.
 - Public key algorithm: used for digital signatures and user authentication. Common algorithms include ssh-rsa and ssh-dss.
 - Symmetric encryption algorithm: used to encrypt sessions. Common algorithms include aes128-ctr and 3des-cbc.
 - Message authentication algorithm: used to ensure data integrity. Common algorithms include hmac-sha1 and hmac-md5.
- Public key authentication:
 - An asymmetric key pair is used: a private key stored securely on the client and a public key placed on the server. During authentication, the client uses its private key to sign a random challenge (that is, generate a digital signature) and sends it to the server. The server verifies the signature using the public key stored locally. If the verification succeeds, the client is authenticated.
- TCP/IP connection forwarding:
 - SSH can forward traffic from other TCP ports through its encrypted channel. This allows insecure protocols, such as FTP and Telnet, to be transmitted securely, preventing user names, passwords, and sensitive data from being exposed in plaintext.

SSH Fundamentals

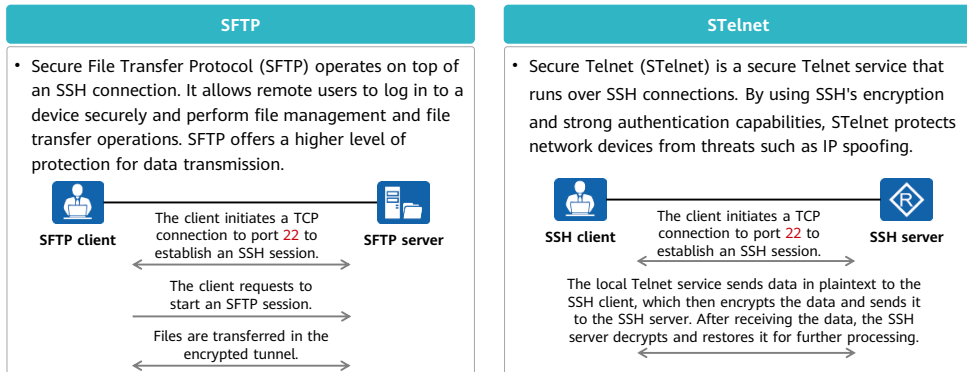
- To establish a secure SSH connection, the client and the server go through the following stages during the communication process:

Stage	Description	SSH Component
Connection setup	The SSH server listens for incoming client requests on TCP port 22. When a client initiates a connection, the two sides establish a TCP session.	/
Version negotiation	The client and server determine the SSH protocol version to use.	Transport layer protocol
Algorithm negotiation	SSH supports multiple algorithms. Based on their supported algorithms, the server and client negotiate the following algorithms: Diffie-Hellman (DH) algorithm for generating session keys, encryption algorithm for encrypting data, public key algorithm for digital signature and authentication, and hash-based message authentication code (HMAC) algorithm for data integrity protection.	
Key exchange	The server and client dynamically generate a session key to protect data transmission and a session ID to identify the SSH connection through DH exchanges. The client also authenticates the server during this stage.	
User authentication	The client sends an authentication request to the server, and the server authenticates the client.	User authentication protocol
Session request	After the authentication succeeds, the client sends a session request to the server, requesting the server to provide a certain type of service (SFTP or STelnet). That is, the client requests to establish a session with the server.	Connection protocol
Session interaction	After a session is established, the server and client exchange data over the session.	

- HMAC: Hash-based Message Authentication Code

Application Scenarios of SSH

- FTP and Telnet lack secure authentication mechanisms and transmit data in plaintext over TCP, creating significant security risks that no longer meet modern network security requirements. SSH addresses these issues by providing an encrypted communication channel that can secure services like FTP and Telnet.



- SFTP:**
 - SFTP is a secure protocol based on SSH 2.0. It is a completely different protocol from FTP and operates as a subsystem of SSH. Unlike FTP, SFTP uses a single TCP connection to carry both commands and data. It encrypts data and supports SSH authentication, making it well-suited for secure transmission.
 - When an authorized user logs in, they provide a valid user name along with the corresponding password and private key. After the server verifies the credentials, the user can remotely manage and transfer files just as they would with FTP. All data exchanged during the session is encrypted using the negotiated session key.
 - An attacker without the correct private key and password cannot pass authentication. In addition, attackers cannot obtain the session keys used between legitimate clients and the server, meaning only the intended client and server can decrypt the transmitted data. This ensures the security of data transmission.
- STelnet:**
 - STelnet is an application that runs on top of SSH 2.0. It is not a new protocol; instead, it uses an SSH-encrypted tunnel to carry insecure Telnet traffic. Compared with Telnet, STelnet encrypts all communication, preventing eavesdropping and tampering, and supports SSH-based authentication. It is therefore suitable for secure remote login to network devices.
 - After the SSH client and server negotiate and establish a secure connection, the client can log in to the server just as they would with Telnet.

SSH Configuration Roadmap

SSH server configuration

- Configure the SSH server function and related parameters.
- Configure VTY user interfaces to support SSH.
- Configure an SSH user.
- Configure the device as an STelnet or SFTP server.

SSH client configuration

- Configure the mode for connecting a device to the SSH server for the first time.
- Set SSH client parameters.
- Configure the device as an STelnet or SFTP client.

Configure the SSH Server Function and Parameters (1)

1. (Optional) Generate a local key pair.

Generate a local RSA key pair.

```
[HUAWEI] rsa local-key-pair create
```

Generate a local DSA key pair.

```
[HUAWEI] dsa local-key-pair create
```

Generate a local ECC key pair.

```
[HUAWEI] ecc local-key-pair create
```

2. Enable a public key algorithm for the SSH server.

```
[HUAWEI] ssh server publickey { dsa | ecc | rsa | x509v3-ssh-rsa | rsa_sha2_256 | rsa_sha2_512 | sm2 |  
x509v3-rsa2048-sha256 | x509v3-ecdsa-sha2 | sm2-sm3 } *
```

By default, the ECC, RSA_SHA2_256, and RSA_SHA2_512 public key algorithms are enabled.

- Note: This course uses the Huawei NetEngine AR5700 router as an example to describe how to configure SSH.
- Generate a local key pair.
 - This step is optional and is used when the client needs to authenticate the server.
 - If no local key pair is manually generated, the server automatically creates one based on the negotiated public key algorithm when the client initiates a login request. This allows the client to authenticate the server.
 - For details about how to generate a key pair, see the corresponding product documentation.
- An asterisk (*) at the end of a command indicates that multiple algorithms can be configured simultaneously. These algorithms form the list of supported options on the server. During communication, the server selects an appropriate algorithm from this list based on what the client supports.

Configure the SSH Server Function and Parameters (2)

3. Enable the SSH server function.

Configure the device as an STelnet server.

```
[HUAWEI] stelnet [ ipv4 | ipv6 ] server enable
```

By default, the STelnet service is disabled.

Configure the device as an SFTP server.

```
[HUAWEI] sftp [ ipv4 | ipv6 ] server enable
```

By default, the SFTP service is disabled.

4. Configure the port that the SSH server listens on.

```
[HUAWEI] ssh [ ipv4 | ipv6 ] server port port-number
```

By default, the SSH server listens on port 22.

If a new port number is configured, the SSH server first terminates all existing SSH connections and then begins listening on the new port. This is more secure because it prevents attackers from connecting to the server straight through the standard SSH port.

Configure the SSH Server Function and Parameters (3)

5. (Optional) Configure extended attributes for the SSH server.

Configure a DH algorithm list for the SSH server.

```
[HUAWEI] ssh server key-exchange { dh_group_exchange_sha256 | dh_group_exchange_sha1 |  
dh_group1_sha1 | ecdh_sha2_nistp256 | ecdh_sha2_nistp384 | ecdh_sha2_nistp521 | sm2_kep |  
dh_group14_sha1 | dh_group16_sha512 | curve25519_sha256 | sm2_sm3 } *
```

By default, the SSH server uses the dh_group_exchange_sha256, dh_group16_sha512, and curve25519_sha256 key exchange algorithms.

Set the minimum key length supported during DH group exchanges between the SSH server and client.

```
[HUAWEI] ssh server dh-exchange min-len min-len
```

By default, the minimum key length supported during DH group exchanges between the SSH server and client is 3072 bits.

Configure the SSH Server Function and Parameters (4)

Configure an encryption algorithm list for the SSH server.

```
[HUAWEI] ssh server cipher { des_cbc | 3des_cbc | aes128_cbc | aes256_cbc | aes128_ctr | aes256_ctr |  
arcfour128 | arcfour256 | aes192_cbc | aes192_ctr | aes128_gcm | aes256_gcm | blowfish_cbc | sm4_cbc |  
sm4_gcm | sm4_ctr } *
```

By default, the SSH server uses the AES256_GCM, AES128_GCM, AES256_CTR, AES192_CTR, and AES128_CTR encryption algorithms.

Configure an HMAC authentication algorithm list for the SSH server.

```
[HUAWEI] ssh server hmac { md5 | md5_96 | sha1 | sha1_96 | sha2_256 | sha2_256_96 | sha2_512 | sm3 |  
sha2_256_etm | sha2_512_etm } *
```

By default, the SSH server uses the SHA2_512_ETM, SHA2_256_ETM, SHA2_512, and SHA2_256 HMAC authentication algorithms.

- Upon startup with factory defaults, a device uses SHA2_512 and SHA2_256 as the HMAC authentication algorithms when operating as an SSH server.
- Upon startup with factory defaults, RSA_SHA2_256 and RSA_SHA2_512 public key algorithms are enabled.

Configure the SSH Server Function and Parameters (5)

6. Specify a source interface or source address for the SSH server.

Configure the SSH server to use a specified interface as the source interface.

```
[HUAWEI] ssh server-source -i interface-type interface-number
```

Configure the SSH server to use all valid interfaces as source interfaces.

```
[HUAWEI] ssh server-source all-interface
```

Configure a source IPv4 address for the SSH server.

```
[HUAWEI] ssh server-source -a ip-address [ -vpn-instance vpnName ]
```

Configure the SSH server to use all valid interfaces using IPv6 addresses as source interfaces.

```
[HUAWEI] ssh ipv6 server-source all-interface
```

Configure a source IPv6 address for the SSH server.

```
[HUAWEI] ssh server-source -a ipv6-address [ -vpn-instance vpn-instance-name ]
```

- By default, no source interface or source IPv6 address is specified for an SSH server.

Configuring VTY User Interfaces to Support SSH

1. Enter the VTY user interface view.

```
[HUAWEI] user-interface vty first-ui-number [ last-ui-number ]
```

2. Configure a user privilege level and other attributes for the user interface. This example uses VTY 0.

Configure a user authentication mode.

```
[HUAWEI] user-interface vty 0  
[HUAWEI-ui-vty0] authentication-mode aaa
```

By default, no authentication mode is configured for a VTY user interface.

To configure the VTY user interface to support SSH, configure AAA authentication for the VTY user interface. If AAA authentication is not set, the **protocol inbound ssh** command does not take effect.

Configure the VTY user interface to support SSH.

```
[HUAWEI-ui-vty0] protocol inbound { all | ssh | telnet }
```

- By default, a VTY user interface supports all protocols, including SSH.

Configuring an SSH User

1. Create an SSH user.

```
[HUAWEI] ssh user user-name
```

2. Configure an authentication mode for the SSH user.

```
[HUAWEI] ssh user user-name authentication-type { password | rsa | password-rsa | all | dsa | password-dsa | ecc | password-ecc | sm2 | password-sm2 | password-x509v3-rsa | x509v3-rsa | password-x509v3-ecdsa-sha2 | x509v3-ecdsa-sha2 | sm2-sm3 | password-sm2-sm3 }
```

By default, no authentication mode is configured for an SSH user.

If no SSH user is configured using the **ssh user user-name** command, run the **ssh authentication-type default password** command to configure password authentication as the default authentication mode. In this case, you only need to configure AAA users. This helps to simplify the configuration if there are a large number of users.

3. Configure a service type for the SSH user.

```
[HUAWEI] ssh user user-name service-type { all | { sftp | stelnet | snetconf }* }
```

By default, no service type is configured for an SSH user.

- Authentication mode for an SSH user:
 - Password authentication relies on AAA. When a user logs in using any of the following authentication methods—password, password-rsa, password-x509v3-rsa, password-dsa, password-ecc, password-x509v3-ecdsa-sha2, password-sm2-sm3, and password-sm2—a local user with the same user name must be created in the AAA view.
 - When RSA, DSA, SM2, or ECC authentication is used, both the server and client must generate their respective RSA, DSA, SM2, or ECC key pairs, and each side must configure the peer's public key locally.

Configuring the SSH Client (1)

1. (Optional) Generate a local key pair.

Generate a local RSA key pair.

```
[HUAWEI] rsa local-key-pair create
```

Generate a local DSA key pair.

```
[HUAWEI] dsa local-key-pair create
```

Generate a local ECC key pair.

```
[HUAWEI] ecc local-key-pair create
```

2. (Optional) Assign the RSA, DSA, or ECC public key to the SSH client.

```
[HUAWEI] ssh client peer server-name assign { rsa-key | dsa-key | ecc-key | sm2-key } key-name
```

By default, no public key is assigned to an SSH client.

- Generating a local key pair:
 - This step is optional and is used when the server needs to authenticate the client.
 - This configuration is required when RSA, DSA, or ECC authentication is used to log in to the SSH server, and can be skipped if password authentication is used.

Configuring the SSH Client (2)

3. Configure the mode used by the device when establishing its initial connection with an SSH server.

Enable the first login function for the SSH client.

```
[HUAWEI] ssh client first-time enable
```

4. Enable a public key algorithm for the SSH client.

```
[HUAWEI] ssh client publickey { dsa | ecc | rsa | sm2 | rsa_sha2_256 | rsa_sha2_512 | x509v3-ssh-rsa |  
x509v3-rsa2048-sha256 | x509v3-ecdsa-sha2 | sm2-sm3 } *
```

Configuring the SSH Client (3)

5. (Optional) Configure SSH client parameters.

Configure an encryption algorithm list for the SSH client.

```
[HUAWEI] ssh client cipher { des_cbc | 3des_cbc | aes128_cbc | aes256_cbc | aes128_ctr | aes256_ctr |  
arcfour128 | arcfour256 | aes192_cbc | aes192_ctr | aes128_gcm | aes256_gcm | sm4_cbc | sm4_gcm |  
sm4_ctr } *
```

Configure an HMAC authentication algorithm list for the SSH client.

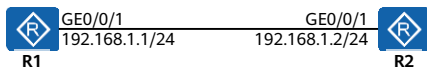
```
[HUAWEI] ssh client hmac { md5 | md5_96 | sha1 | sha1_96 | sha2_256 | sha2_256_96 | sha2_512 | sm3 |  
sha2_256_etm | sha2_512_etm } *
```

Configure a key exchange algorithm list for the SSH client.

```
[HUAWEI] ssh client key-exchange { dh_group_exchange_sha256 | dh_group_exchange_sha1 |  
dh_group1_sha1 | ecdh_sha2_nistp256 | ecdh_sha2_nistp384 | ecdh_sha2_nistp521 | sm2_kep |  
dh_group14_sha1 | dh_group16_sha512 | curve25519_sha256 | sm2_sm3 } *
```

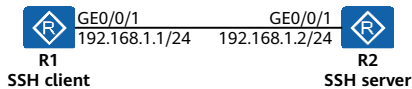
STelnet Configuration Example

- As shown in the figure, R2 operates as an SSH server and has the STelnet server function enabled. R1, acting as an SSH client, can log in to the server using various authentication modes. The following example describes how to use password authentication to log in to the server through STelnet.



- Configuration roadmap:
 - Configure an SSH user named **admin1** on the SSH server and specify the password authentication mode for this user.
 - Enable the STelnet server function on the SSH server.
 - Set the service type of the SSH user **admin1** to STelnet on the SSH server.
 - Specify the port number that the SSH server listens on. This can prevent attackers from connecting to the server straight through the standard SSH port.
 - Configure the SSH user **admin1** to log in to the SSH server through STelnet.

Configuration Procedure (1)



1. Configure an IP address for the server.

```
<HUAWEI> system-view
[HUAWEI] sysname R2
[R2] interface GE 0/0/1
[R2-GE0/0/1] undo portswitch
[R2-GE0/0/1] ip address 192.168.1.2 24
[R2-GE0/0/1] quit
```

2. Configure a VTY user interface on the SSH server.

```
[R2] user-interface vty 0 4
[R2-ui-vty0-4] authentication-mode aaa
[R2-ui-vty0-4] protocol inbound ssh
[R2-ui-vty0-4] user privilege level 3
[R2-ui-vty0-4] quit
```

3. Create an SSH user with the user name **admin1** and password **Huawei@123**, and configure password authentication for this user.

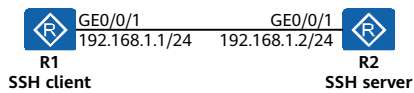
```
[R2] aaa
[R2-aaa] local-user admin1 password
Please configure the login password (8-128)
It is recommended that the password consist of four types of
characters, including lowercase letters, uppercase letters,
numerals and special characters.
Please enter password: Huawei@123
Please confirm password: Huawei@123
[R2-aaa] local-user admin1 service-type ssh
[R2-aaa] local-user admin1 privilege level 3
[R2-aaa] quit
[R2] ssh user admin1
[R2] ssh user admin1 authentication-type password
```

4. Enable the STelnet function, and set the service type of the created SSH user to STelnet.

```
[R2] stelnet server enable
[R2] ssh server-source all-interface
[R2] ssh user admin1 service-type stelnet
[R2] ssh server port 1025
```

- Note: The actual command output varies according to the device. The command output in this configuration example is for reference only.

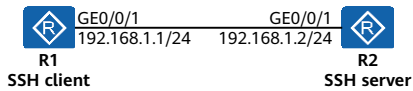
Configuration Procedure (2)



5. Configure the lists of supported encryption algorithms, HMAC authentication algorithms, DH algorithms, and public key algorithms for the SSH server, and specify the minimum key length.

```
[R2] ssh server cipher aes128_ctr aes256_ctr aes192_ctr aes128_gcm aes256_gcm
[R2] ssh server hmac sha2_512_etm sha2_256_etm sha2_512 sha2_256
[R2] ssh server key-exchange dh_group_exchange_sha256 dh_group16_sha512 curve25519_sha256
[R2] ssh server publickey ecc rsa_sha2_256 rsa_sha2_512
[R2] ssh server dh-exchange min-len 3072
```

Configuration Procedure (3)



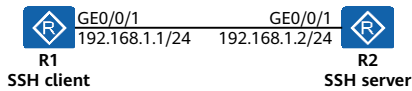
6. Configure an IP address for the client.

```
<HUAWEI> system-view
[HUAWEI] sysname R1
[R1] interface GE 0/0/1
[R1-GE0/0/1] undo portswitch
[R1-GE0/0/1] ip address 192.168.1.1 24
[R1-GE0/0/1] quit
```

7. Configure the lists of supported encryption algorithms, HMAC authentication algorithms, key exchange algorithms, and public key algorithms on the client.

```
[R1] ssh client cipher aes128_ctr aes256_ctr aes192_ctr aes128_gcm aes256_gcm
[R1] ssh client hmac sha2_512_etm sha2_256_etm sha2_512 sha2_256
[R1] ssh client key-exchange dh_group_exchange_sha256 dh_group16_sha512 curve25519_sha256
[R1] ssh client publickey ecc_rsa_sha2_256 rsa_sha2_512
```

Configuration Procedure (4)



8. Connect the STelnet client to the SSH server.

Enable the first login function for the SSH client.

```
[R1] ssh client first-time enable
[R1] quit
```

Connect the STelnet client R1 to the SSH server in password authentication mode by entering the configured user name and password.

After a successful login, the user view of the SSH server is displayed.

```
<R1> stelnet 192.168.1.2 1025
Trying 192.168.1.2 ...
Press CTRL+K to abort
Connected to 192.168.1.2 ...
The server's key fingerprint is ssh-eccdsa 521
pJUvdbNGyjaBwLSCnJa573J0hN+1yYa0ZkapY6e6BRU.
The server is not authenticated. Continue to access it?[Y/N]: y
Save the server's public key? [Y/N]: y
The server's public key will be saved with the name 120.18.65.14.
Please wait...

Please input the username: admin1
Enter password: Huawei@123
Warning: The initial password poses security risks.The password
needs to be changed, Continue? [Y/N]: y
Please enter old password: Huawei@123
Please enter new password: Huawei@1234
Please confirm new password: Huawei@1234
The password has been changed successfully.
Info: The max number of VTY users is 21, the number of current
VTY users online is 4, and total number of terminal users online
is 4.
The current login time is 20XX-12-31 11:22:06.
<R2>
```

Verifying the Configuration

1. Check the SSH status.

```
[R2] display ssh server status
SSH Version : 2.0
SSH authentication timeout (Seconds) : 60
SSH authentication retries (Times) : 3
SSH server key generating interval (Hours) : 0
SSH version 1.x compatibility : Enable
SSH server keepalive : Disable
SFTP IPv4 server : Disable
SFTP IPv6 server : Disable
STELNET IPv4 server : Enable
STELNET IPv6 server : Enable
NETCONF IPv4 server : Enable
NETCONF IPv6 server : Enable
NETCONF IPv4 server port(830) : Disable
NETCONF IPv6 server port(830) : Disable
SCP IPv4 server : Enable
SCP IPv6 server : Enable
SSH port forwarding : Disable
SSH IPv4 server port : 1025
SSH IPv6 server port : 1025
ACL name :
ACL number :
ACL6 name :
ACL6 number :
SSH server ip-block : Enable
```

2. Check SSH user information.

```
[R2] display ssh user-information
-----
User Name : admin1
Authentication type : password
User public key name : --
User public key type : --
Sftp directory : flash:
Service type : stelnet
-----
Total 1, 1 printed
```

Quiz

1. (Short-answer question) Which FTP mode is recommended for transferring log files and configuration files on network devices?
2. (Single-answer question) Which TCP port does an STelnet client use by default to establish a connection with the SSH server for remote login? ()
 - A. 20
 - B. 21
 - C. 22
 - D. 23

1. ASCII mode. The binary mode is more suitable for transferring non-text files that cannot undergo character encoding conversion, such as EXE, BIN, and CC files.
2. C

Summary

- Application-layer network services are built on top of transport-layer protocols (such as TCP and UDP), leveraging their end-to-end communication capabilities to implement specific functions.
- FTP is a protocol used to transfer files on a computer network, and Telnet is a protocol used for remote login. Both protocols establish connections on top of TCP to provide reliable data transmission. However, because they transmit data in plaintext and lack user authentication, they no longer meet current network security requirements.
- SSH is a protocol designed to provide secure remote login and other secure network services over untrusted networks. It creates an encrypted tunnel that allows a client to securely connect to a server and perform operations such as executing commands and managing files. SFTP is a secure file transfer protocol implemented over SSH, while STelnet is an SSH-based implementation of Telnet that enables secure remote system management and CLI operations.

Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
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organization for a fully connected,
intelligent world.

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NAT Fundamentals and Configuration



Foreword

- As the Internet continues to evolve and network applications grow rapidly, the limited pool of public IPv4 addresses has become a constraint on network development. To address this constraint, Network Address Translation (NAT) was introduced.
- NAT enables intranet hosts to access the Internet. It not only alleviates the shortage of public IPv4 addresses, but also prevents extranet hosts from directly reaching intranet hosts that use private addresses, thereby enhancing intranet security.
- In this course, you will learn the background of NAT, examine how different NAT types operate, and understand their application scenarios.

Objectives

- Upon completion of this course, you will be able to:
 - Understand the technical background of NAT.
 - Understand the classification and implementation of NAT.
 - Understand the scenarios in which each type of NAT is applied.

Contents

- 1. Overview of NAT**
2. Source NAT
3. Destination NAT
4. Bidirectional NAT

IPv4 Address Exhaustion

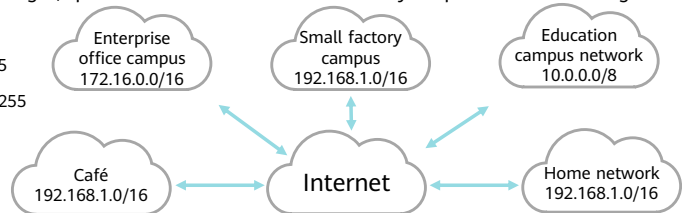
- As the number of Internet users continues to grow, public IPv4 addresses have become increasingly scarce.
- In addition, the allocation of public IPv4 addresses is uneven across regions, leaving some areas with a severe shortage of available public IPv4 addresses.
- To mitigate this problem, it is essential to adopt transition technologies.



Popularization of Private IP Networks

- As the demand for building local area networks (LANs) in enterprises and households surged, an RFC introduced dedicated private IP address spaces. These addresses can be reused across different LANs, effectively meeting the networking needs of both organizations and home users.
- Private IP addresses:
 - These addresses can be freely used by organizations and individuals, but they cannot communicate directly over the Internet and are limited to intranet use.
 - Within Class A, B, and C address ranges, specific blocks are reserved exclusively for private IP addressing:

- Class A: 10.0.0.0–10.255.255.255
- Class B: 172.16.0.0–172.31.255.255
- Class C: 192.168.0.0–192.168.255.255



- Public IP addresses are managed and allocated by designated authorities and can communicate directly over the Internet.
- Private IP addresses cannot be routed across the Internet. This is because, under normal circumstances, Internet service provider (ISP) routers do not maintain routes for destinations within private IP address ranges.

Emergence of NAT

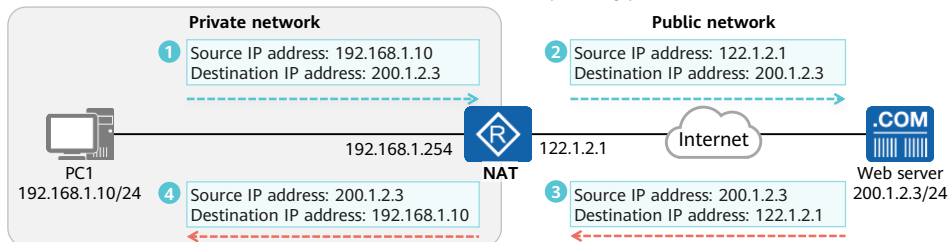
- NAT was introduced to mitigate the challenges caused by IPv4 address exhaustion.
- NAT enables the reuse of limited public IPv4 addresses through address translation, thereby alleviating IPv4 address shortages for an extended period.
- NAT offers the following benefits:
 - By reusing addresses, NAT allows all devices within a private network to share a single public IP address, significantly reducing the demand for public IPv4 addresses.
 - A NAT device acts as a gateway between public and private networks, enabling communication between them.
 - NAT hides the intranet topology, preventing private addresses from being exposed to external threats.
 - By keeping the public-facing IP address unchanged, NAT allows flexible intranet design.



- Limitations of NAT:
 - Because NAT alters IP header information, the two endpoints of a connection can no longer see each other's real IP addresses, which complicates troubleshooting. For example, IP addresses recorded in logs are NAT-translated, so tracking the activity of specific intranet devices requires checking NAT mapping entries.
 - NAT introduces additional latency and impacts device performance, since translating every packet consumes CPU and memory resources on routers.
 - Some applications do not work properly with NAT due to protocol compatibility issues. Some application protocols, such as FTP, require NAT devices to support Application Level Gateway (ALG) to function properly.

Fundamentals of NAT

- NAT is widely deployed to translate IP addresses in IP packets. It is typically implemented on egress devices, such as routers or firewalls.
- A typical application scenario for NAT is in private networks—such as enterprise campuses or home networks—where intranet hosts use private IP addresses and the network's egress device performs NAT. When traffic flows from the private network to the Internet, the NAT device translates the source IP address of each packet into a designated public IP address. Conversely, for traffic coming from the Internet to the private network, the NAT device translates the destination IP address back to the corresponding private IP address.



- If a host using a private IP address needs to access the Internet, NAT must be configured on the network's egress device to translate the private source address of outgoing packets into a public one. This translation allows response packets from Internet servers to be routed to the public address and ultimately forwarded to the NAT device at the network egress.
- Note: This course uses the Huawei NetEngine AR5700 router as an example to explain how NAT works and how to configure NAT.

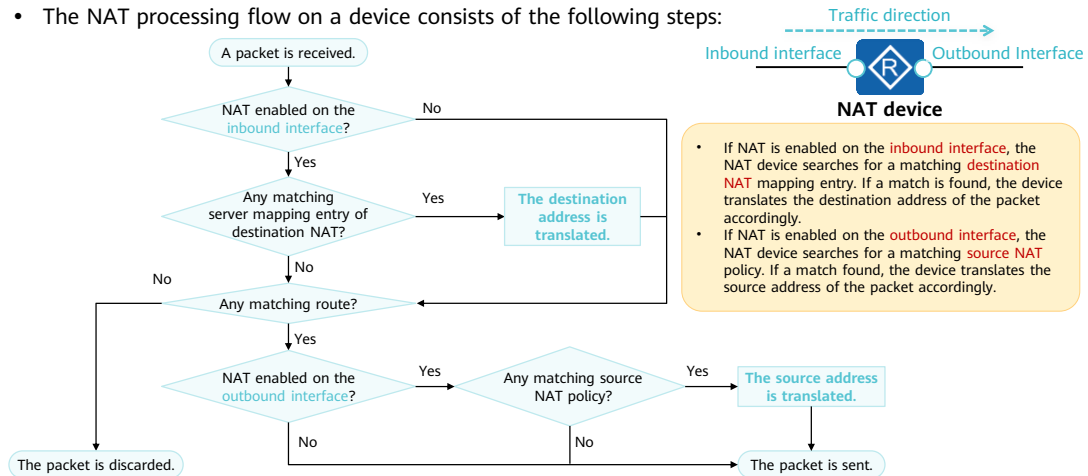
NAT Classification

- NAT can be classified based on whether it translates the source IP address, destination IP address, or both. Accordingly, NAT is categorized into source NAT, destination NAT, and bidirectional NAT.

Type		Translated Address	Port Translation
Source NAT	NAT No-PAT	Source IP address	No
	NAPT	Source IP address	Yes
	Easy IP	Source IP address	Yes
Destination NAT	NAT Server	Destination IP address	Optional
Bidirectional NAT	Source NAT + NAT Server	Source IP address + destination IP address	Optional

NAT Processing Flow

- The NAT processing flow on a device consists of the following steps:



- NAT processing flow:

- When a NAT device receives a packet, it first checks whether NAT is enabled on the inbound interface. If NAT is enabled, the device proceeds to Step 2. If not, it moves to Step 3.
- The device then checks the server mapping table generated by NAT Server. If the packet matches an entry, the device translates the destination address accordingly and proceeds to Step 3. If no match is found, the device proceeds to Step 3 with no action performed.
- The device searches for a matching route for the packet. If a match is found, the device proceeds to step 4. If there is no matching route, the device discards the packet.
- The device then checks whether NAT is enabled on the outbound interface. If NAT is enabled, the device proceeds to Step 5. If not, the device proceeds to Step 6.
- The device checks for a matching source NAT policy. If a match is found, the device translates the source address of the packet and proceeds to Step 6. If no match is found, the device directly proceeds to Step 6.
- The device sends the packet along the matching route.

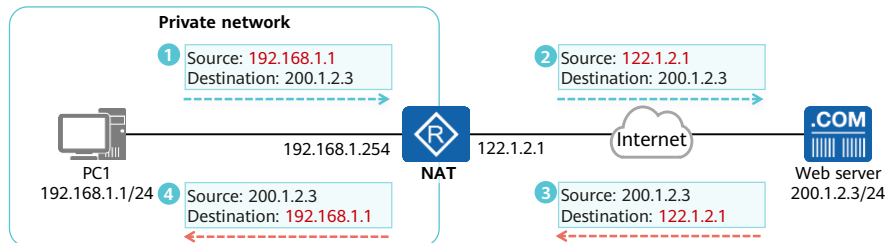
- A NAT device performs address translation on packets passing through an interface only when NAT is enabled on that interface.

Contents

1. Overview of NAT
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Introduction to Source NAT

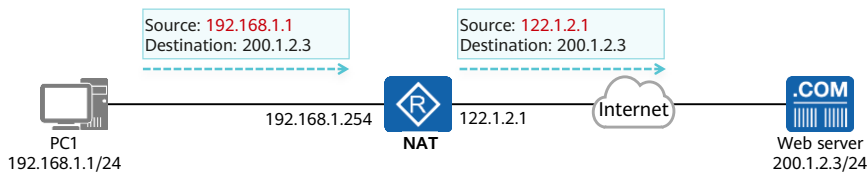
- Source NAT translates the source IP address in a packet header.
- By translating private IP addresses to public ones, source NAT enables hosts on private networks to access the Internet using public addresses.



- As shown in the figure, when the intranet host PC1 attempts to access the web server on the Internet, the NAT device processes packets as follows:
 - When receiving an Internet-bound packet from the intranet host, the NAT device translates the private source IP address of the packet into a public IP address.
 - When receiving a return packet from the web server, the NAT device translates the public destination IP address back into the private IP address of the host.

NAT Policy

- A device implements source NAT by applying NAT policy rules.



- A NAT policy rule consists of a matching condition, post-NAT address (an address from the address pool or the outbound interface address), and the action to take.
- If multiple NAT policy rules are configured, the device matches the rules from top to bottom in the order they appear in the rule list.

NAT Policy Rule

Matching Condition: Source Address	Post-NAT Address	Action
192.168.1.1	122.1.2.1	Source NAT

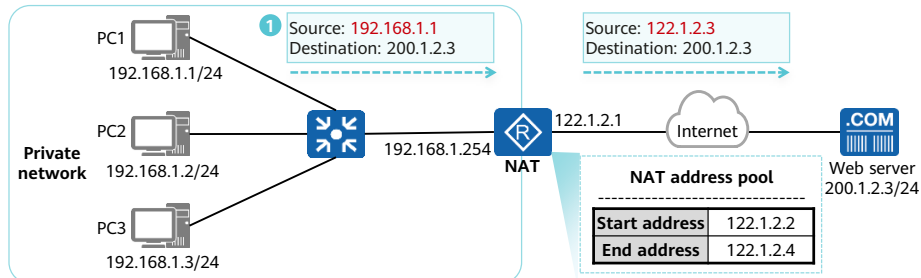
NAT policy rule 1
NAT policy rule 2
NAT policy rule 3
...

Matching order

- A NAT policy rule consists of the following elements:
 - Matching condition: defines which traffic the rule applies to. Typically, a matching condition can include the source address, destination address, outbound interface, service, and time range. You can configure matching conditions as needed so that the device performs NAT only on the intended traffic.
 - Post-NAT address: specifies the address to which the source address of matching traffic will be translated. Such addresses can come from a pre-configured address pool, or be taken from the outbound interface.
 - Action: defines the type of source NAT to perform, such as NAPT or Easy IP.
- Matching order of NAT policy rules:
 - NAT policy rules are matched sequentially in the order as they appear in the rule list, from top to bottom. Once a match is found for a packet, no further rules are checked for that packet.
 - Newly added rules and rules whose NAT actions are modified are automatically placed at the end of the rule list.
 - The **rule move** command can be used to reorder NAT policy rules, allowing administrators to adjust rule priorities and the matching order.

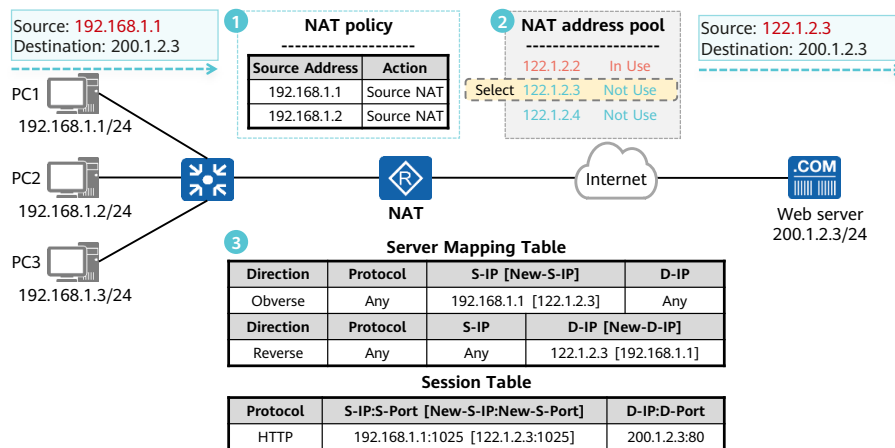
Fundamentals of NAT No-PAT

- No-Port Address Translation (NAT No-PAT) is a type of NAT that **translates only addresses**. It translates each private address to a dedicated public address, while keeping port numbers unchanged.
- Application scenario: typically used in networks where public IP resources are plentiful and only a limited number of intranet hosts need Internet access. In this case, the number of required public addresses equals the number of intranet users accessing the Internet simultaneously.



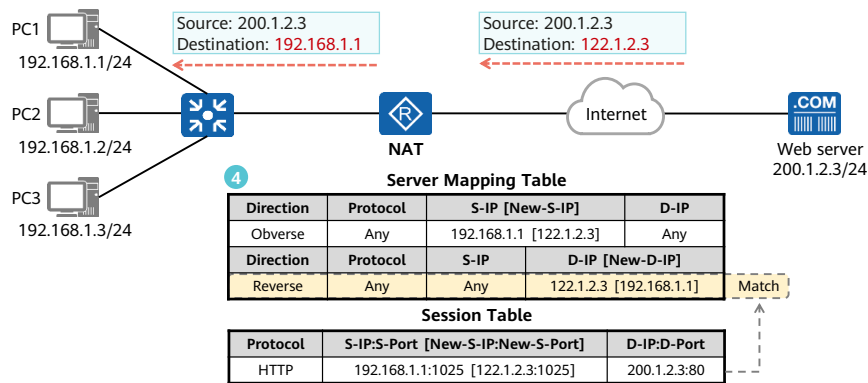
- With NAT No-PAT configured, the NAT device translates each private address to a dedicated public address from the pre-configured NAT address pool. When all addresses in the pool are in use, NAT is not performed for subsequent intranet hosts attempting to access the Internet, until a public address becomes available in the pool.
- Note: The implementation and feature support of NAT No-PAT varies across software versions of routers. For details, see the product documentation of the corresponding version.

NAT No-PAT Process (1)



- When PC1 initiates access to the web server, the NAT device processes packets as follows:
 - After receiving a packet from PC1, the NAT device evaluates the configured NAT policies and determines that source NAT is required for this packet.
 - The NAT device selects an available public IP address from the NAT address pool using the round-robin algorithm and replaces the packet's source IP address with the selected public address.
 - The NAT device creates the corresponding server mapping entry and session table entry, and then forwards the packet to the Internet.

NAT No-PAT Process (2)



4. When the NAT device receives the response packet from the web server, it looks up the session table and searches for a matching server mapping entry. If a match is found, the device replaces the packet's destination IP address with PC1's private IP address and forwards the packet to the private network.

Server Mapping Table and Session Table

- Server mapping table:
 - The server mapping table stores mappings between private and public IP addresses.
 - When NAT No-PAT or NAT Server is configured, the NAT device automatically generates server mapping entries to maintain the pre-defined mappings of addresses, and, where applicable, port numbers.
- Session table:
 - The session table maintains dynamic session information for each translated flow. Each entry corresponds to a single flow identified by the 5-tuple (source and destination IP addresses, source and destination ports, and protocol), along with the current session state.
 - Session entries are created when packets undergo NAT processing and are automatically aged out when their corresponding session ends. When NAPT or Easy IP is configured, the NAT device uses the session table to maintain dynamic mappings of addresses and port numbers.

Server Mapping Table

Direction	Protocol	S-IP [New-S-IP]	D-IP
Obverse	Any	192.168.1.1 [122.1.2.3]	Any
Direction	Protocol	S-IP	D-IP [New-D-IP]
Reverse	Any	Any	122.1.2.3 [192.168.1.1]

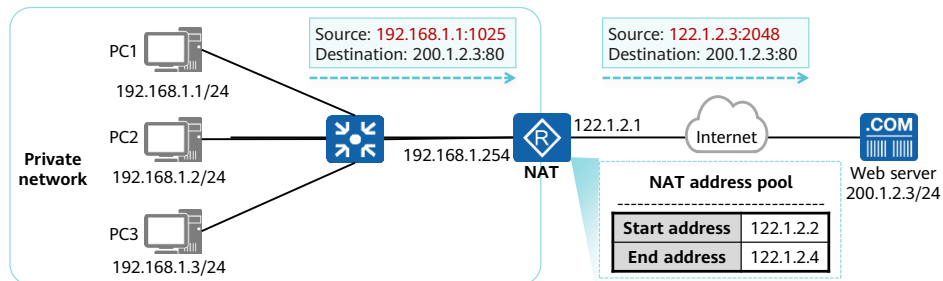
Session Table

Protocol	S-IP:S-Port [New-S-IP:New-S-Port]	D-IP:D-Port
HTTP	192.168.1.1:1025 [122.1.2.3:1025]	200.1.2.3:80

- The server mapping table consists of obverse and reverse server mapping entries, each serving a distinct role in NAT processing:
 - Obverse server mapping entries accelerate NAT processing for Internet-bound packets originating from intranet hosts. This allows the device to perform translations more efficiently.
 - Reverse server mapping entries apply to NAT processing for packets initiated by Internet users toward intranet hosts. In NAT No-PAT scenarios, reverse access is only possible after an intranet host has first initiated Internet-bound traffic and a corresponding server mapping entry has been created.

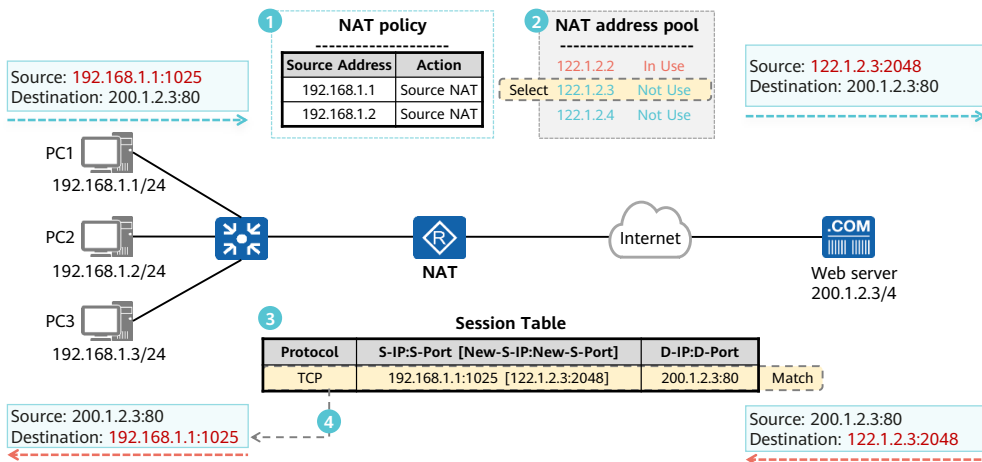
Fundamentals of NAPT

- Network Address and Port Translation (NAPT) is a form of NAT that **translates both IP addresses and port numbers**. By mapping private addresses and assigning translated port numbers from the range 2048–65535, NAPT enables multiple intranet hosts to share **one or more public addresses**.
- NAPT is well suited for scenarios where public addresses are limited but many intranet hosts need Internet access.



- In this mechanism, when selecting a public IP address from the address pool for translation, the NAT device translates not only the IP address but also the port number. This enables a one-to-many mapping between public and private addresses, significantly improving the utilization of public IP addresses. With NAPT, the NAT device distinguishes users by assigning distinct translated port numbers, enabling support for a larger number of concurrent Internet access sessions from intranet hosts.

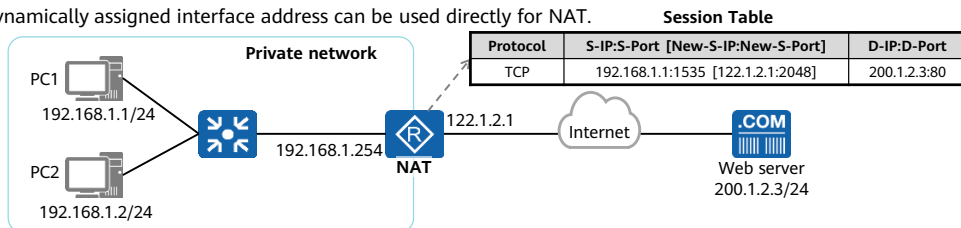
NAPT Processing



- When PC1 initiates access to the web server, the NAT device processes packets as follows:
 - After receiving a packet from PC1, the NAT device evaluates the configured NAT policies and determines that source NAT is required for this packet.
 - The NAT device selects a public IP address from the NAT address pool using a round-robin algorithm. It replaces the packet's source address with the selected public address and assigns a new translated source port.
 - The NAT device creates a corresponding session entry and forwards the translated packet to the Internet.
 - When the NAT device receives the response packet from the web server, it looks up the session table to locate the matching entry. It then restores the destination address to PC1's private address and replaces the destination port with the original port before forwarding the packet to the private network.

Fundamentals of Easy IP

- Easy IP is a type of NAT that uses **the public IP address of the outbound interface** (Internet-facing WAN interface) as the translated source address, while **translating both IP addresses and ports**, with translated ports assigned from the range 2048–65535.
- Like NAT, Easy IP translates both IP addresses and ports. The difference is that Easy IP does not rely on an address pool; instead, it uses an interface address as the post-NAT public address.
- Easy IP is suitable for networks where a fixed public IP address is not available. A typical example is a private network where the egress device obtains a public IP address dynamically through DHCP or PPPoE. In such cases, the dynamically assigned interface address can be used directly for NAT.



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- When PC1 initiates access to the web server, the NAT device processes packets as follows:
 1. After receiving a packet from PC1, the NAT device checks it against the configured NAT policies and determines that source NAT is required for this packet.
 2. The NAT device replaces the packet's source IP address with the public IP address of the Internet-facing WAN interface and assigns a public source port number. It then creates a corresponding session entry and forwards the translated packet to the Internet.
 3. When the NAT device receives the response packet from the web server, it looks up the session table to locate the matching entry. It then restores the destination address to PC1's private address and replaces the destination port with the original port before forwarding the packet to the private network.
- In this mechanism, because both the IP address and port are translated, multiple intranet hosts can share a single public IP address. With Easy IP, the NAT device distinguishes users by assigning distinct translated port numbers, enabling support for a larger number of concurrent Internet access sessions from intranet hosts.

Source NAT Configuration Procedure: Creating a Source NAT Address Pool (1)

1. Create a source NAT address pool and enter the address pool view.

```
[HUAWEI] nat address-group group-name [ group-number]
```

2. Specify the range of IP addresses in the address pool.

Method 1: Configure one or more address ranges.

```
[HUAWEI-address-group-addressgroup1] section [ id ] start-ipv4 [ end-ipv4]
```

Method 2: Specify a large address range and exclude specific addresses or ports from the range.

```
[HUAWEI-address-group-addressgroup1] section [ id ] start-ipv4 [ end-ipv4 ]  
[HUAWEI-address-group-addressgroup1] exclude-ip { ip-address1 [ to ip-address2 ] | ip-address1 mask {  
mask-value | mask-length } }  
[HUAWEI-address-group-addressgroup1] exclude-port port1 [ to port2 ]
```

- Note: This course uses the Huawei NetEngine AR5700 running V600R024C00 as an example to describe how to configure source NAT.

Source NAT Configuration Procedure: Creating a Source NAT Address Pool (2)

- Set the NAT mode of the address pool to NAPT.

```
[HUAWEI-address-group-addressgroup1] mode nat
```

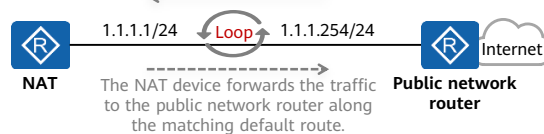
This type of NAT enables many private addresses to share one or more public addresses, and translates both source addresses and port numbers.

- Configure the device to generate open programming routes (OPRs), which function similarly to blackhole routes, to prevent routing loops.

```
[HUAWEI-address-group-addressgroup1] route enable
```

The device is enabled to generate OPRs destined for addresses in the NAT address pool.

The Internet router matches traffic against service routes and sends it back to the NAT device.



- Open Programming Route (OPR):
 - As shown in the figure, when an Internet user initiates traffic toward an address in the NAT address pool, the packet is forwarded to the NAT device. Because no matching session entry exists, the NAT device forwards the packet to the Internet router along a default route. The Internet router then matches the packet against its service routing table and forwards it back to the NAT device. As a result, the packet loops repeatedly between the NAT device and Internet router, creating a routing loop.
 - After the **route enable** command is configured, the NAT device generates an OPR route for each address in the NAT address pool. Such routes function similarly to blackhole routes: any traffic destined for an address in the NAT pool is discarded by the NAT device, preventing routing loops.
 - Generating OPR routes for addresses in the NAT address pool is mandatory when the NAT address pool resides in a different network segment from the Internet-facing WAN interface, recommended when the NAT address pool is in the same segment as the WAN interface, and unnecessary when the pool contains the WAN interface's address.

Source NAT Configuration Procedure: Configuring a Source NAT Policy (1)

1. Enable the NAT function in the interface view.

```
[HUAWEI-10GE0/0/1] nat enable
```

2. Create a NAT policy and enter the NAT policy view.

```
[HUAWEI] nat-policy
```

3. Create a NAT policy rule and enter the NAT policy rule view.

```
[HUAWEI-policy-nat] rule name rule-name
```

Source NAT Configuration Procedure: Configuring a Source NAT Policy (2)

4. Configure matching conditions for the NAT policy rule.

Matching Condition	Command
Source IP address	source-address { <i>ipv4-address</i> { <i>ipv4-mask-length</i> mask <i>mask-address</i> wildcard <i>wildcard</i> } range <i>ipv4-start-address</i> <i>ipv4-end-address</i> any }
	source-address-exclude { <i>ipv4-address</i> { <i>ipv4-mask-length</i> mask <i>mask-address</i> wildcard <i>wildcard</i> } range <i>ipv4-start-address</i> <i>ipv4-end-address</i> }
Destination IP address	destination-address { <i>ipv4-address</i> { <i>ipv4-mask-length</i> mask <i>mask-address</i> wildcard <i>wildcard</i> } range <i>ipv4-start-address</i> <i>ipv4-end-address</i> any }
	destination-address-exclude { <i>ipv4-address</i> { <i>ipv4-mask-length</i> mask <i>mask-address</i> wildcard <i>wildcard</i> } range <i>ipv4-start-address</i> <i>ipv4-end-address</i> }
Outbound interface	egress-interface <i>interface-type</i> <i>interface-number</i>



- Source IP address: specifies the source IP address of the traffic to be matched. You can use the **source-address-exclude** command to exclude certain addresses in the specified address range from NAT.
- Destination IP address: specifies the destination IP address of the traffic to be matched. You can use the **destination-address-exclude** command to exclude certain addresses in the specified address range from NAT.
- Outbound interface: specifies the outbound interface of the traffic to be matched.

Source NAT Configuration Procedure: Configuring a Source NAT Policy (3)

Matching Condition	Command
Service	• service protocol { udp tcp } [source-port { <i>source-port</i> <i>start-source-port</i> to <i>end-source-port</i> } &<1-64> destination-port { <i>destination-port</i> <i>start-destination-port</i> to <i>end-destination-port</i> } &<1-64>] * • service protocol sctp • service protocol icmp [icmp-type { <i>icmp-name</i> <i>icmp-type-number</i> { <i>icmp-code-number</i> [to <i>icmp-code-number</i>] } &<1-64> }] • service protocol protocol-number • service-exclude protocol { udp tcp } [source-port { <i>source-port</i> <i>start-source-port</i> to <i>end-source-port</i> } &<1-64> destination-port { <i>destination-port</i> <i>start-destination-port</i> to <i>end-destination-port</i> } &<1-64>] * • service-exclude protocol sctp • service-exclude protocol icmp [icmp-type { <i>icmp-name</i> <i>icmp-type-number</i> { <i>icmp-code-number</i> [to <i>icmp-code-number</i>] } &<1-64> }] • service-exclude protocol protocol-number • service any

- Service: specifies the service port and protocol of the traffic to be matched.

Source NAT Configuration Procedure: Configuring a Source NAT Policy (4)

5. Define the action to be performed on traffic matching the NAT policy rule.

```
[HUAWEI-policy-nat-rule-policy1] action { source-nat { address-group address-group-name | easy-ip } | no-nat }
```

- **source-nat** indicates that source NAT is performed on the matching traffic. NAT takes effect if **address-group** is specified, and NAT in Easy IP mode takes effect if **easy-ip** is specified.
- **no-nat** indicates that NAT is not performed on the matching traffic.

- The **no-nat** action is mainly used for special clients.
- For example, to perform NAT for all hosts in the 192.168.1.0/24 subnet except 192.168.1.2, you can first configure a policy to exclude traffic sourced from 192.168.1.2 from NAT and configure another policy to perform NAT on traffic sourced from 192.168.1.0/24.

```
[R1] nat-policy
```

```
[R1-policy-nat] rule name policy_nat1
```

```
[R1-policy-nat-rule-policy_nat1] source-address 192.168.1.2 24
```

```
[R1-policy-nat-rule-policy_nat1] action no-nat
```

```
[R1-policy-nat-rule-policy_nat1] quit
```

```
[R1-policy-nat] rule name policy_nat2
```

```
[R1-policy-nat-rule-policy_nat2] source-address 192.168.1.0 24
```

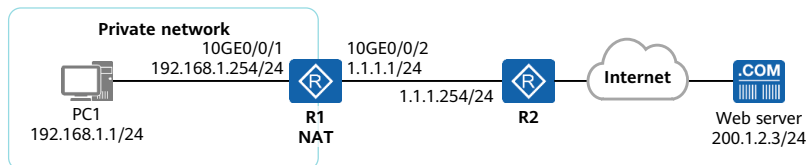
```
[R1-policy-nat-rule-policy_nat2] action source-nat address-group  
addressgroup1
```

```
[R1-policy-nat-rule-policy_nat2] quit
```

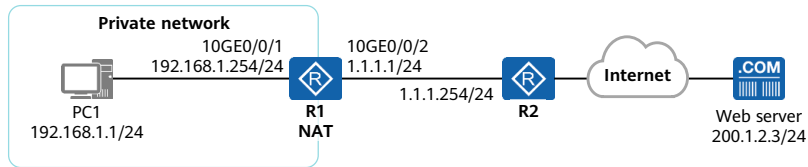
```
[R1-policy-nat] quit
```

Example for Configuring NAPT for Intranet Users to Access the Internet

- On the enterprise network shown in the figure, R1 is deployed at the egress. A source NAT policy needs to be configured on R1 so that intranet users on the 192.168.1.0/24 network segment can access the Internet.
- In addition to the public IP address of the WAN interface (1.1.1.1/24) on R1, the enterprise has obtained six public IP addresses (1.1.1.10 to 1.1.1.15) from the ISP for address translation. R2 is the access gateway provided by the ISP.



Enabling NAT



1. Configure IP addresses for interfaces and enable the NAT function on 10GE0/0/2 of R1.

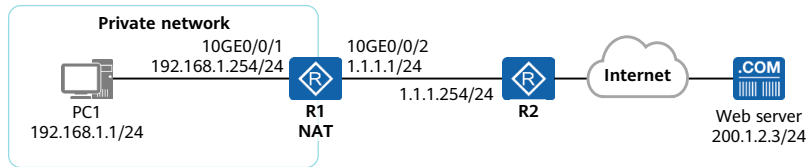
Configure an IP address for 10GE0/0/1.

```
<HUAWEI> system-view
[HUAWEI] sysname R1
[R1] interface 10GE 0/0/1
[R1-10GE0/0/1] undo portswitch
[R1-10GE0/0/1] ip address 192.168.1.254 24
[R1-10GE0/0/1] quit
```

Configure an IP address for 10GE 0/0/2 and enable NAT on this interface.

```
[R1] interface 10GE 0/0/2
[R1-10GE0/0/2] undo portswitch
[R1-10GE0/0/2] ip address 1.1.1.1 24
[R1-10GE0/0/2] nat enable
[R1-10GE0/0/2] quit
[R1]
```

Configuring a NAT Address Pool and a NAT Policy



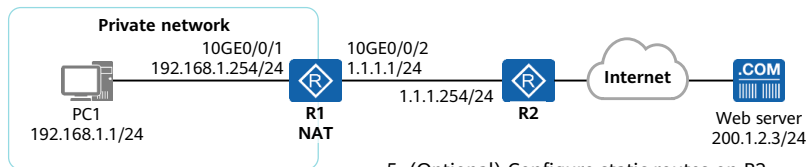
2. Configure a NAT address pool.

```
[R1] nat address-group addressgroup1
[R1-address-group-addressgroup1] mode pat
[R1-address-group-addressgroup1] section 1.1.1.10 1.1.1.15
[R1-address-group-addressgroup1] route enable
[R1-address-group-addressgroup1] quit
```

3. Configure a NAT policy to perform source NAT on Internet access traffic sourced from a specific intranet network segment.

```
[R1] nat-policy
[R1-policy-nat] rule name policy_nat1
[R1-policy-nat-rule-policy_nat1] source-address 192.168.1.0 24
[R1-policy-nat-rule-policy_nat1] action source-nat address-group addressgroup1
[R1-policy-nat-rule-policy_nat1] quit
[R1-policy-nat] quit
```

Configuring Routes



4. Configure a default route on R1, so that it can forward traffic sent from intranet users to the ISP router.

```
[R1] ip route-static 0.0.0.0 0 1.1.1.254
```

5. (Optional) Configure static routes on R2.

If the addresses in the NAT address pool are on a different network segment from 10GE0/0/2, configure static routes destined for the addresses in the NAT address pool on R2, with the next hop at 1.1.1.1. As such, R2 can forward traffic returned from the Internet to R1.

The post-NAT public addresses are not assigned to interfaces, which results in R2 not being able to use a routing protocol to discover routes to these public addresses. In most cases, you need to contact the ISP network administrator to configure static routes as needed.

Verification

1. Verify that the intranet PC can access the Internet.

```
PC1> ping 200.1.2.3

Pinging 200.1.2.3 with 32 bytes of data:
Reply from 200.1.2.3: bytes=32 time=3ms TTL=253
Reply from 200.1.2.3: bytes=32 time=3ms TTL=253
Reply from 200.1.2.3: bytes=32 time=3ms TTL=253
Reply from 200.1.2.3: bytes=32 time=6ms TTL=253

Ping statistics for 200.1.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 6ms, Average = 3ms
```

2. Check the NAT session table on R1.

```
<R1> display session all verbose
Session Table Information:
Protocol           : 1 (ICMP)
SrcAddr VPN        : 192.168.1.1
DestAddr VPN        : 200.1.2.3
Type Code IcmpId    : 8 0 4885
Time To Live        : 8 s
NAT Info
  New SrcAddr        : 1.1.1.10
  New DestAddr        : -
  New IcmpId          : 2050

Total : 1
```

Check NAT information. If the post-NAT IP address is an address in the NAT address pool, the NAT policy is successfully configured.

- Notes:

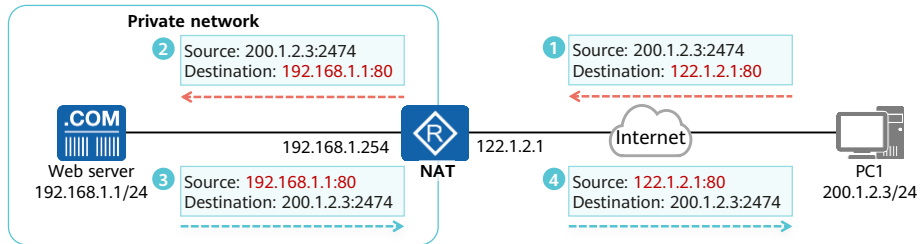
- ICMP does not use transport-layer ports. Consequently, a NAT device cannot translate port numbers for ICMP Echo Request and Echo Reply messages (messages used for ping operations) in the same way as it does for TCP or UDP traffic. Instead, the NAT device relies on the ICMP identifier (the **IcmpId** field shown in the preceding command output), which functions similarly to a port number, to distinguish ICMP sessions from different intranet hosts, or from different processes on the same host.
- To trace these ICMP sessions, the NAT device maintains a mapping table that includes the protocol, IP address, and ICMP identifier. This table allows the device to perform address translation and ensure that ICMP messages are forwarded correctly in both directions.

Contents

1. Overview of NAT
2. Source NAT
- 3. Destination NAT**
4. Bidirectional NAT

Introduction to Destination NAT

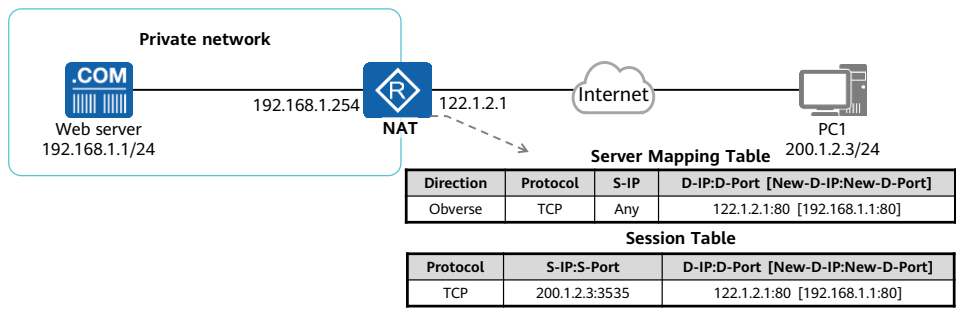
- Destination NAT translates the destination IP address in a packet header.
- By translating a public IP address into a private IP address, destination NAT enables Internet users to access intranet servers using public IP addresses.



- As shown in the figure, when the extranet host PC1 attempts to access the intranet web server, the NAT device processes packets as follows:
 - When receiving a packet destined for the intranet web server from the extranet host, the NAT device translates the public destination address of the packet into a private address.
 - When receiving a return packet from the intranet web server, the NAT device translates the private source address into the corresponding public address.

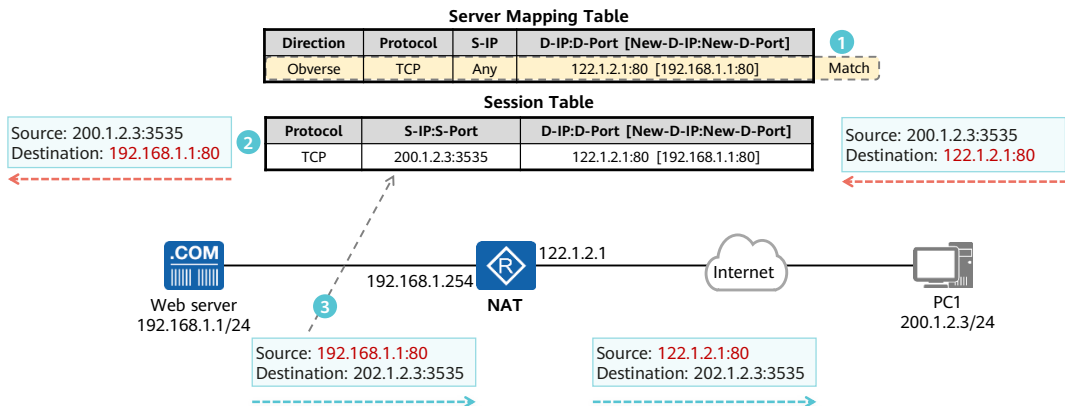
Fundamentals of NAT Server

- NAT Server is a form of **static destination** NAT. It translates a public IP address into a fixed private address.
- NAT Server is suitable for scenarios where there are fixed mappings between private IP addresses and public IP addresses, private IP addresses and public port numbers, private port numbers and public IP addresses, or private port numbers and public port numbers. This type of NAT is commonly used when an organization, such as a school or enterprise, needs to provide services for extranet users.



- NAT Server maintains a fixed one-to-one mapping between **[public address:port]** and **[private address:port]**.
- Through these mappings, extranet hosts can access an intranet server using the server's assigned public address and port.

NAT Server Processing



- When PC1 initiates access to the web server, the NAT device processes packets as follows:
 - After receiving the first packet from the Internet host destined for the public address 122.1.2.1, the NAT device searches for a matching server mapping entry. Upon a successful match, the device translates the destination IP address into 192.168.1.1.
 - The NAT device creates a session entry and forwards the translated packet to the private network.
 - After receiving the response packet from the web server to PC1, the NAT device matches the packet against the session table, translates the source address back to 122.1.2.1, and forwards the packet to the Internet.
 - For subsequent packets exchanged between PC1 and the web server, the NAT device performs address translation directly based on the existing session entry, without re-checking the server mapping table.
- To implement NAT Server, you need to define fixed mappings between private and public addresses on the NAT device. Once configured, the NAT device automatically generates server mapping entries to maintain these mappings. These entries remain persistent until the NAT Server configuration is deleted.

NAT Server Configuration Procedure

1. Enable the NAT function in the interface view.

```
[HUAWEI-10GE0/0/1] nat enable
```

2. Configure NAT Server.

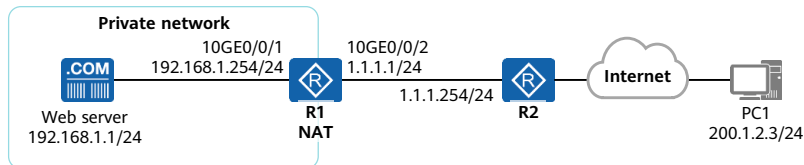
A single public IP address or Internet-facing interface is accessible to Internet users. As such, all Internet users access intranet servers through that same public address or interface.

Protocol Type Specified	Command
No protocol is specified.	• nat server <i>name</i> global interface { <i>interface-name</i> <i>interface-type interface-num</i> } inside <i>host-address</i> <i>host-port</i> [no-reverse] [vpn-instance <i>vpn-instance-name2</i>] [nat-disable] [description <i>description</i>] • nat server <i>name</i> [vpn-instance <i>vpn-instance-name1</i>] global <i>global-address</i> [<i>global-address-end</i>] inside <i>host-address</i> [<i>host-address-end</i>] [no-reverse] [vpn-instance <i>vpn-instance-name2</i>] [route] [description <i>description</i>]
A protocol is specified.	• nat server <i>name</i> protocol <i>protocol-type</i> global interface { <i>interface-name</i> <i>interface-type interface-num</i> } <i>global-port</i> [<i>global-port-end</i>] inside <i>host-address</i> <i>host-port</i> [<i>host-port-end</i>] [no-reverse] [vpn-instance <i>vpn-instance-name2</i>] [nat-disable] [description <i>description</i>] • nat server <i>name</i> [vpn-instance <i>vpn-instance-name1</i>] protocol <i>protocol-type</i> global <i>global-address</i> [<i>global-address-end</i>] <i>global-port</i> [<i>global-port-end</i>] inside <i>host-address</i> [<i>host-address-end</i>] <i>host-port</i> [<i>host-port-end</i>] [no-reverse] [vpn-instance <i>vpn-instance-name2</i>] [route] [description <i>description</i>]

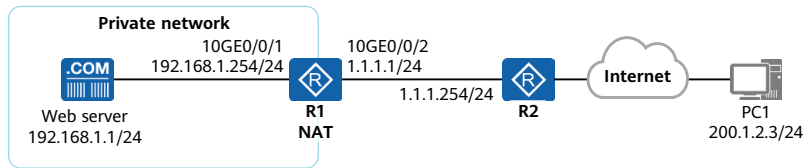
- When no protocol is specified, intranet servers advertise a single public IP address/Internet-facing WAN interface to Internet users. In this case, all Internet users access intranet servers through that same public IP address or interface.
- When a protocol is specified, only the ports associated with that protocol are translated. In this way, selected ports of intranet servers are open to Internet users, improving security. However, this may affect the forwarding of protocols that use dynamic ports. In such cases, NAT ALG must be configured to ensure proper handling of the traffic.
- Key parameters:
 - **no-reverse**: configures the device not to generate reverse server mapping entries. If this parameter is not specified, the device generates both obverse and reverse server mapping entries.
 - **route**: indicates that OPRs are delivered to prevent routing loops. By default, no OPR is delivered.
- Note: This course uses the Huawei NetEngine AR5700 running V600R024C00 as an example to describe how to configure NAT Server.

Example for Configuring NAT Server for Internet Users to Access Intranet Servers

- On the enterprise network shown in the figure, R1 is deployed at the egress. NAT Server needs to be configured on R1 to enable the intranet web server to provide services for Internet users.
- The enterprise has only one available public IP address 1.1.1.1. This IP address is assigned to the WAN interface of R1 and used by the intranet server to provide services to Internet users. R2 is the access gateway provided by the ISP.



Enabling NAT



1. Configure IP addresses for interfaces and enable the NAT function on 10GE0/0/2 of R1.

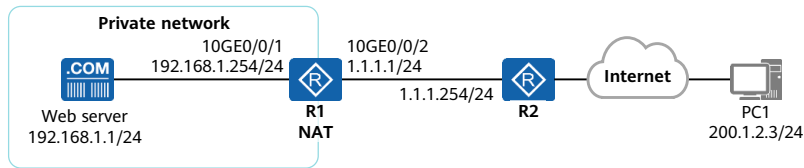
Configure an IP address for 10GE0/0/1.

```
<HUAWEI> system-view
[HUAWEI] sysname R1
[R1] interface 10GE 0/0/1
[R1-10GE0/0/1] undo portswitch
[R1-10GE0/0/1] ip address 192.168.1.254 24
[R1-10GE0/0/1] quit
```

Configure an IP address for 10GE0/0/2 and enable NAT.

```
[R1] interface 10GE 0/0/2
[R1-10GE0/0/2] undo portswitch
[R1-10GE0/0/2] ip address 1.1.1.1 24
[R1-10GE0/0/2] nat enable
[R1-10GE0/0/2] quit
[R1]
```

Configuring NAT Server and Routes



2. Configure NAT Server.

```
[R1] nat server policy_web protocol tcp global 1.1.1.1 8080 inside 192.168.1.1 www route
```

3. Configure a default route, so that traffic from the intranet server can reach the ISP router.

```
[R1] ip route-static 0.0.0.0 0.0.0.0 1.1.1.254
```

4. (Optional) Configure static routes on R2.

If the public address defined in NAT Server is on a different network segment from 10GE0/0/2, configure a static route destined for the public address used for NAT Server on R2, with the next hop at 1.1.1.1. As such, R2 can then forward traffic destined for the server to R1. In most cases, you need to contact the ISP network administrator to configure static routes as needed.

Verification

1. Check the NAT session table on R1 when an Internet user accesses the intranet server.

```
<R1> display session all verbose
Session Table Information:
Protocol      : 6(TCP)
SrcAddr Port Vpn : 200.1.2.3/24 2474
DestAddr Port Vpn : 1.1.1.1 8080
Time To Live   : 60s
NAT Info
New SrcAddr    : -
New SrcPort    : -
New DestAddr   : 192.168.1.1
New DestPort   : 80

Total : 1
```

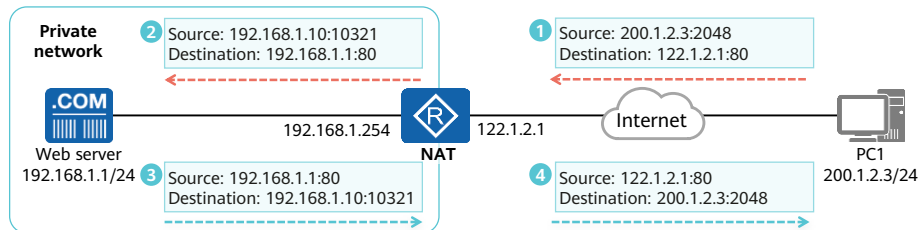
If the corresponding entry is generated and the post-NAT address in the entry is the private IP address of the intranet server, the NAT policy is successfully configured.

Contents

1. Overview of NAT
2. Source NAT
3. Destination NAT
- 4. Bidirectional NAT**

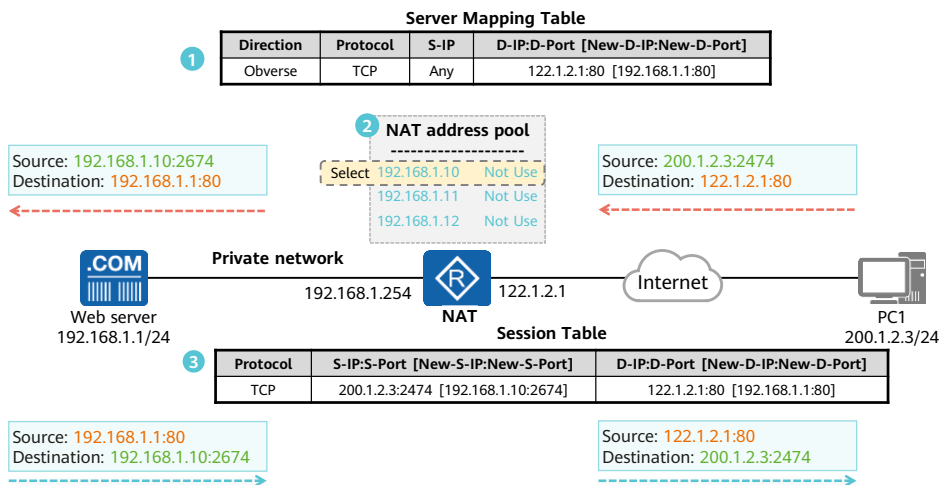
Introduction to Bidirectional NAT

- Bidirectional NAT translates both the source and destination IP addresses of a packet.
- Bidirectional NAT is not a standalone feature, but rather a combination of **source NAT** and **destination NAT**. When receiving a packet, the NAT device translates both the source and destination addresses.
- Bidirectional NAT is suitable for scenarios where both source and destination addresses require translation and the destination address has a fixed mapping. For example, an Internet user wants to access an intranet server through a combination of NAPT and NAT Server, or an intranet user wants to use a public address to access an intranet server.



- A NAT device can implement bidirectional NAT through a combination of source NAT and NAT Server.

Bidirectional NAT Processing



- The NAT device processes packets as follows:
 - When receiving a request packet, the NAT device translates the packet's destination address and port number according to the server mapping entry generated by NAT Server.
 - The NAT device replaces the source address of the packet with an available private address from the source NAT address pool, replaces the source port number with a new one, creates a session entry accordingly, and sends the resulting packet out.
 - When receiving a response packet, the NAT device searches the session table for a matching entry. If a match is found, the device translates the source and destination addresses as well as the source and destination ports according to the matching entry and then sends the resulting packet out.

Quiz

1. (Short-answer question) Which type of NAT can enable access to intranet servers from Internet users?
2. (Short-answer question) What are the advantages of NAPT compared with No-PAT?

1. NAT Server is specifically designed to allow Internet users to initiate connections to intranet servers.
2. NAPT can translate multiple private IP addresses into one public IP address, improving public IP address utilization.

Summary

- Using private IP addresses within a private network and applying NAT at the network egress can effectively reduce the number of public IPv4 addresses required and mitigate the shortage of public IPv4 addresses.
 - Source NAT, including NAT No-PAT, NAPT, and Easy IP, translates source addresses of intranet hosts when they access the Internet.
 - Destination NAT, such as NAT Server, enables intranet hosts to provide services for Internet users.
 - Bidirectional NAT combines source NAT and destination NAT to translate both source and destination addresses.

Thank you.

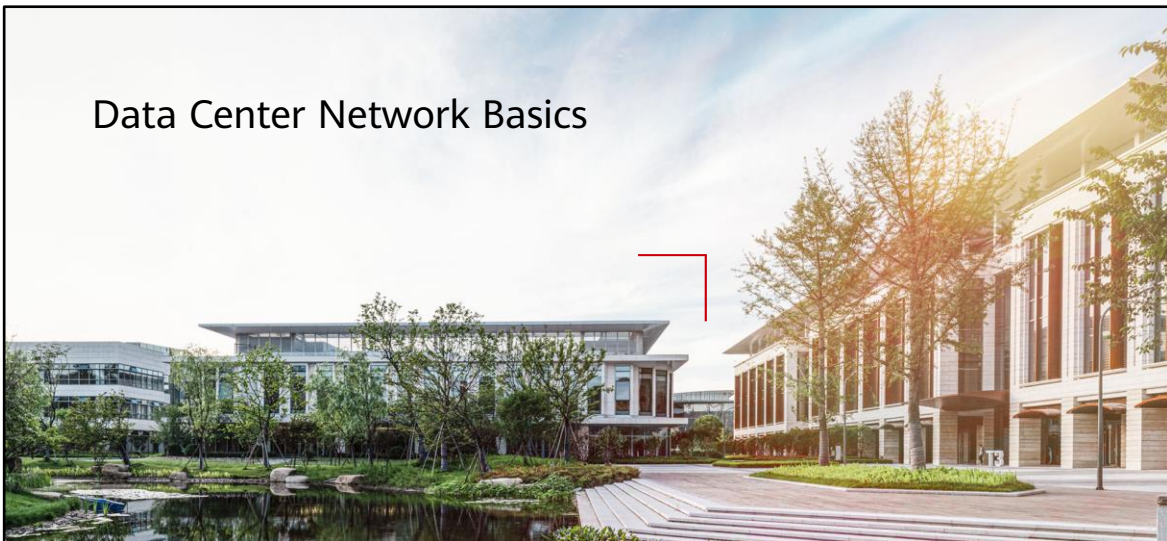
把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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Data Center Network Basics



Foreword

- A data center network (DCN) is an application scenario of enterprise networks, providing services such as data storage and application access for enterprise users. As the foundation carrying enterprise services, DCNs continue to develop and evolve.
- With the deepening of digital transformation, the application scenarios of DCNs are also divided into general-purpose computing and intelligent computing. The former handles massive services and requires stable networks with low latency. The latter is applicable to artificial intelligence (AI) scenarios and requires high-bandwidth and lossless networks for large-scale parallel collaboration. This fundamental difference drives the development of network technologies in different scenarios, and the network architecture has become the key cornerstone that determines the overall computing power efficiency and scale.
- This course describes the evolution of data centers (DCs), network requirements in general-purpose computing and intelligent computing scenarios, and related network technologies.

- AI is a new technological science that studies and develops theories, methods, technologies, and application systems for mimicking, extending, and augmenting human intelligence.

Objectives

- Upon completion of this course, you will be able to:
 - Describe the composition and development history of DCs.
 - Describe the development phases of DCNs.
 - Describe the requirements and related network technologies of DCNs in general-purpose computing scenarios.
 - Describe the requirements and related network technologies of DCNs in intelligent computing scenarios.

Contents

1. DC Evolution

- What Is a DC
 - DC Development and Evolution

2. DCN Basics

Why Do We Need DCs?

- As enterprises develop, the amount of data that enterprises need to process every day increases. The processing capability of PCs in offices is no longer sufficient to meet the requirements of enterprises. To provide a more efficient method for processing information and data, enterprises build or rent DCs to centrally process massive amounts of data, thereby meeting the needs of enterprise development.



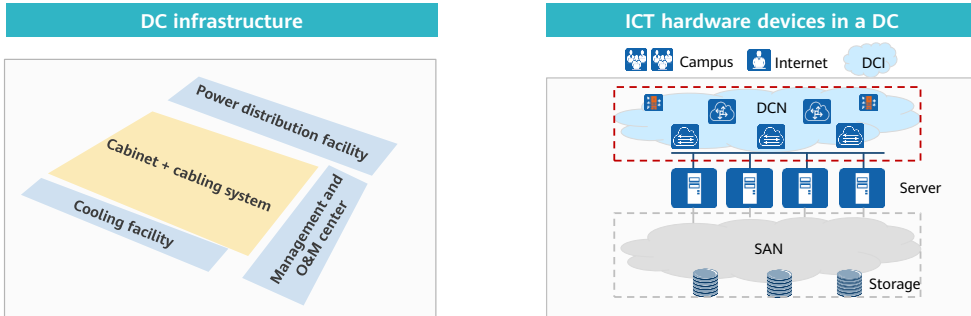
Small enterprises use PCs to process data.

With the development of enterprises, more and more data needs to be maintained.

Data is centrally processed in DCs, and large enterprises use data through DCs.

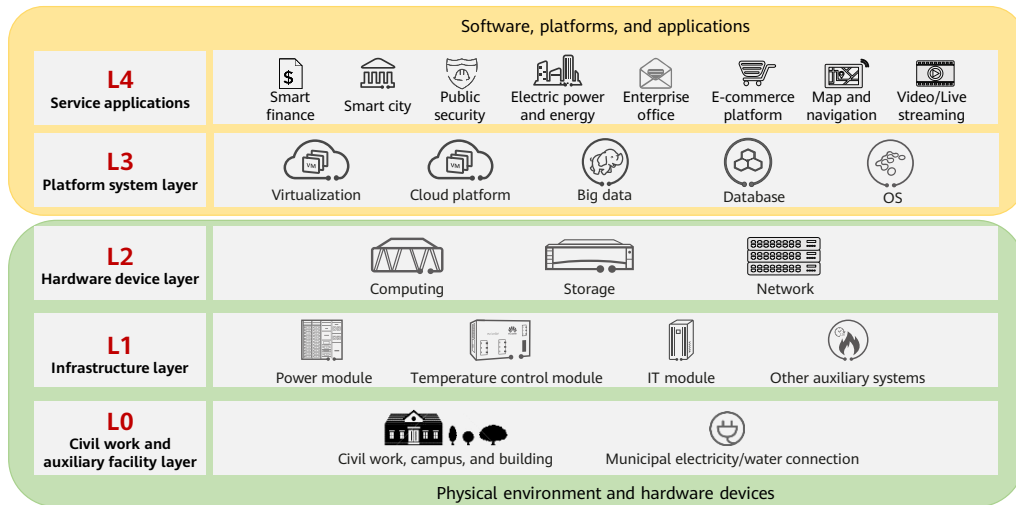
What Is a DC?

- A DC is a place where enterprises process and store massive amounts of data.
- It is a complex set of facilities including buildings. It includes not only the computer system and related systems (for example, communications and storage systems), but also data communication connections, environmental control devices, monitoring devices, and various security devices.



- Data Center Interconnect (DCI): network interconnection between two DCs for cross-DC service transmission.
- Storage area network (SAN): a high-speed, dedicated network used to connect multiple servers to centralized storage devices, enabling the servers to access storage resources at a high speed.
- Campus: indicates a campus network in the figure on the right, which is a local area network (LAN) that connects people and things within a specified area. The scale of a campus network can vary greatly, ranging from a small office home office (SOHO) to large environments such as school campuses, enterprise campuses, or shopping centers.

What Does a DC Have?



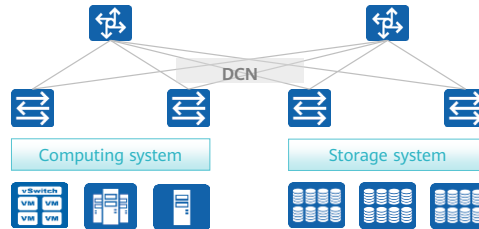
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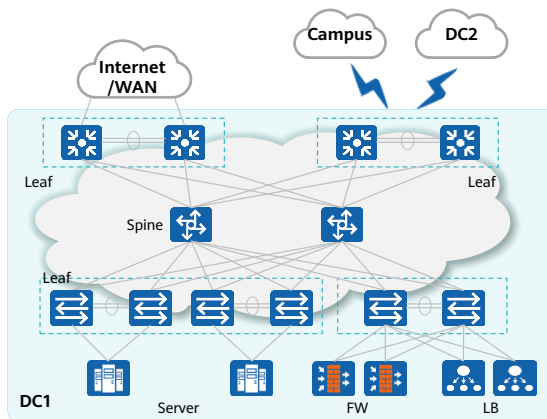
- Generally, a DC can be divided into the following levels:
 - L0: civil engineering and auxiliary facilities, including the buildings of equipment rooms.
 - L1: infrastructure layer, consisting of equipment room auxiliary facilities, such as interior construction, power supply, cooling, and fire protection.
 - L2: hardware device layer, consisting of equipment room infrastructure, such as servers, storage devices, and networks.
 - L3: platform system layer, consisting of system platforms deployed based on hardware devices, such as virtualization, cloud platform, and big data system.
 - L4: service application layer, consisting of application services of various industries.
- In this course, subsequent discussions related to DCs, such as concepts, topologies, and architectures, are based on L2 and above.

Overall Architecture of the Hardware Device Layer in a DC

- For enterprises, a DC is essentially an expanded version of a PC, which is responsible for computing, storing, and forwarding enterprise data. A modern DC consists of the following parts:
 - Computing system: Composed of a large number of servers, it is the heart of the DC, responsible for processing massive amounts of data.
 - Storage system: consists of different types of storage devices. Storage devices are used to store massive data and information.
 - DCN: consists of different types of network devices, such as switches and firewalls. It connects the computing system and storage system in the DC. All data exchanges between the computing system and storage system are implemented through the DCN.



DCN



- A DCN is the infrastructure that carries DC services and is responsible for data forwarding.
- The network architecture of a DCN varies depending on service scenarios.
- Multiple DCNs can connect to branches of enterprises or organizations across regions. DCNs can also connect to the Internet, WANs, and other DCs.

- Spine: DC backbone switch, which provides high-speed IP forwarding and connects to leaf nodes through high-speed interfaces.
- Leaf: DC leaf node, that is, access switch, which connects various network devices and servers to the DCN.
- FW: firewall
- LB: load balancer

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1. DC Evolution

- What Is a DC
- DC Development and Evolution

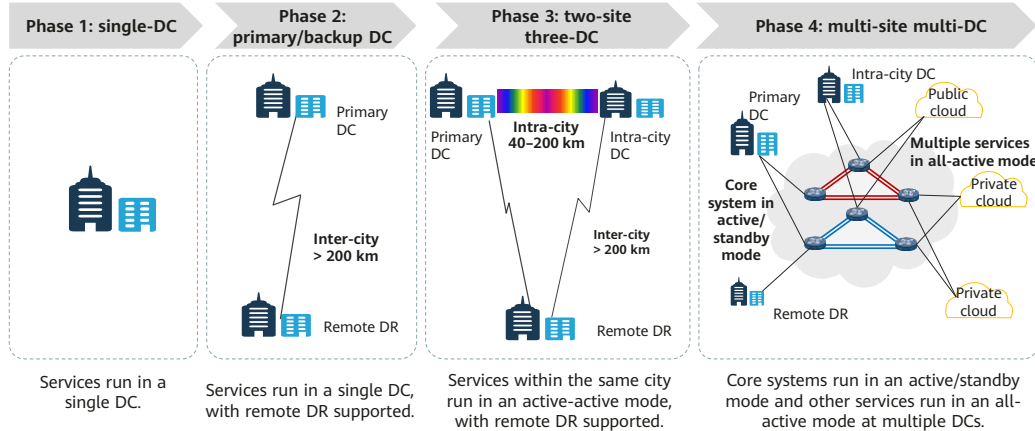
2. DCN Basics

Birth and Development of DCs

1940s	1960s	1990s	2010s
Origin: devices requiring an equipment room Supercomputer equipment room  The first supercomputer, ENIAC, had a volume of 3,000 cubic feet, occupied 170 square meters of floor space, and weighed 30 tons. <i>Server scale:</i> 1 server	Prototype: servers placed without order Server farm  As computers became smaller in size, they could be physically concentrated in a single room, without interconnection networking or management and O&M. Multiple servers	Debut: server networking in the C/S architecture DC  With the emergence of the client-server technology model, servers were placed in a separate room, with cabling used to interconnect the devices. Hundreds of servers	Boom: cloud computing/AI infrastructure Cloud data center/ Intelligent computing center  With the explosive growth of cloud computing and AI services, DCN traffic has rapidly increased, and DCs with a scale of 300,000 servers have emerged. Tens of thousands of servers

- Cloud/Cloud computing: Refers to a model that provides computing resources (including servers, storage, databases, networking, software, analytics, and intelligence) on demand through the Internet (the "cloud"). Its core concept is to treat computing as a utility, just like water, electricity, or gas, where users pay based on usage without the need to build and maintain their own power plants.

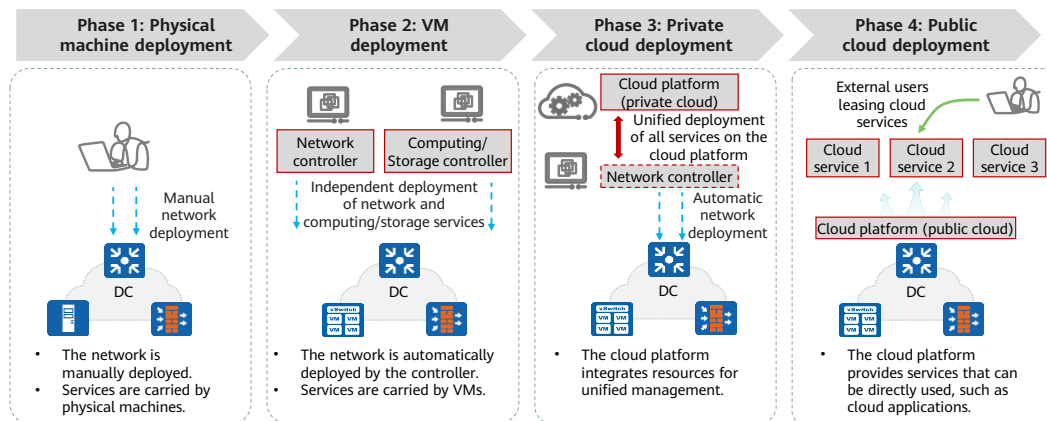
DC Evolution Process: From Single-Center to Multi-Site and Multi-Center



To ensure service reliability, DCs have evolved from a single center to multiple centers in different sites.

- **Public cloud:** A cloud model in which cloud service providers offer computing resources and services to the public or large industry groups through the Internet. All infrastructure, platforms, and software are owned and managed by cloud service providers. Users can obtain services by sharing these resources.
- **Private cloud:** A cloud model in which cloud computing resources and services are dedicated to a single organization. It can be deployed on-premises or hosted by a third-party service provider. Its core purpose is to provide the organization with a dedicated, controlled, and secure cloud environment.
- **DR: Disaster Recovery.**

DC Evolution Process: From Physical Machine Deployment to Cloud-based Deployment



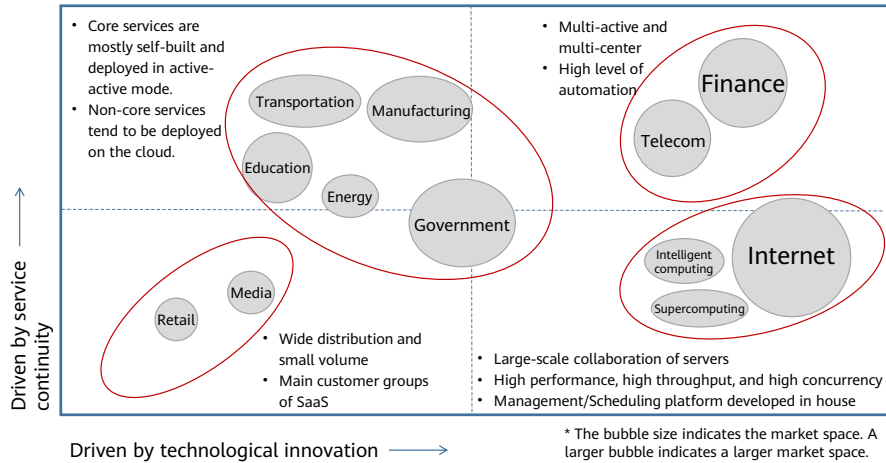
Technological innovation drives DC service innovation, and DCNs evolve accordingly.

- Phase 1: DC services (computing/storage services) are carried by physical servers, and the DCN is manually deployed by network engineers. After the DCN is successfully deployed, the DC provides external services through the DCN.
- Phase 2: The computing/storage services of the DC are delivered by the computing/storage controller, and the network services are delivered by the network controller. However, the network and computing/storage services are deployed independently. In this phase, automatic network configuration can be implemented to some extent. When a new computing service is rolled out, the network can detect the computing service and deliver the network configuration in real time, implementing fast service rollout on the entire network. In this phase, DC services are mainly carried by VMs that are virtualized from physical servers.
 - Virtualization: Virtualization technologies are used to run multiple VMs on a physical server, improving physical resource utilization and enabling fast service deployment and elastic service expansion. It can be further classified into computing virtualization, storage virtualization, and network virtualization.
- Phase 3: DC services are deployed on a private cloud. The private cloud platform integrates computing, storage, and network resources to implement unified management and service delivery. In this phase, the private cloud platform can interconnect with the network controller to automatically deliver configurations

to physical network devices.

- Phase 4: DC services are deployed on both private and public clouds to implement hierarchical security, service disaster recovery, and cost reduction. In addition, some Internet enterprises can transform into cloud service providers and build their own cloud platforms (public clouds) to provide external users or traditional enterprises with resources or services that can be directly used, such as scalable cloud computing and storage resources and cloud applications.

DCs Spanning Every Sector: the Foundation of Digital Transformation



- Software as a Service (SaaS): Common cloud applications are SaaS services.

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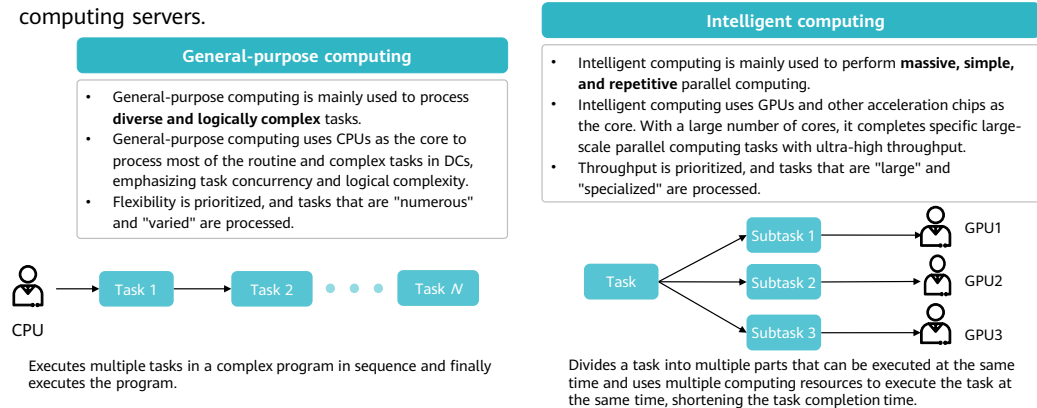
1. DC Evolution

2. DCN Basics

- General-Purpose Computing DCN
 - Intelligent Computing DCN
 - Dual-Track Architecture of DCs

Application Scenarios of DCs

- With the continuous evolution of DCs, application scenarios are gradually divided into general-purpose computing and intelligent computing due to the differences in the tasks processed by internal computing servers.

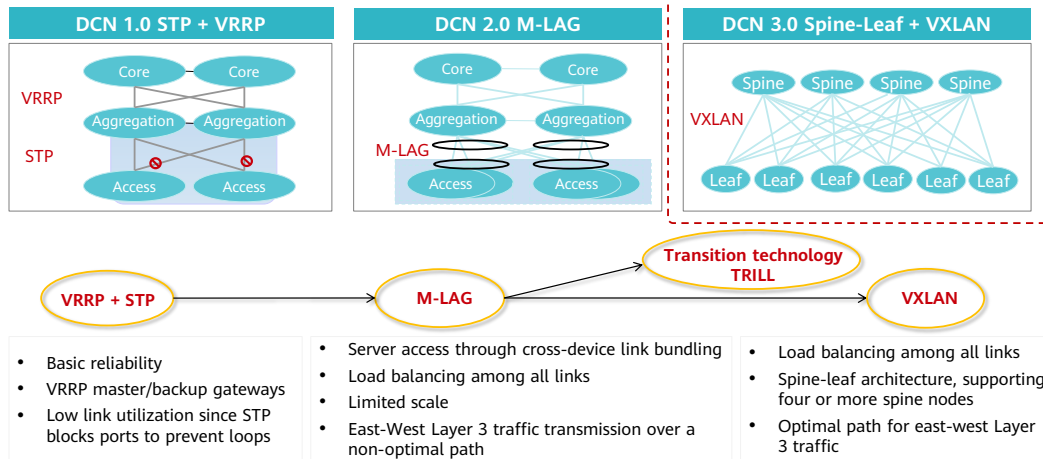


- In general-purpose computing, the CPU is like an **all-powerful top engineer**. Although it has relatively few cores, each core is very powerful and can quickly switch between different tasks. This makes it suitable for running applications that require frequent judgment, jump, and complex logic. For example, when a web page is opened, the CPU needs to take a series of tightly linked steps that are difficult to split into large-scale parallel tasks. These steps include code execution, database query, and page assembly.
 - Typical applications of general-purpose computing include web servers (accessing websites such as Amazon and YouTube) and enterprise databases (storing and querying transaction records and user information). Most of the software and services people encounter in daily life involve general-purpose computing.
- In intelligent computing, the GPU is like a **team of tens of thousands of primary school students**. The computing capability of each primary school student (a single core) is far inferior to that of the top engineer (the CPU), but their sheer numbers (tens of thousands of cores) give them an overwhelming speed when solving the same simple arithmetic problem simultaneously. The GPU's strength lies in breaking down a huge task into millions of identical, simple subtasks and having thousands of cores process them at the same time.
 - Typical applications of intelligent computing in DCs include AI model training, AI model inference, and scientific computing (such as climate

simulation).

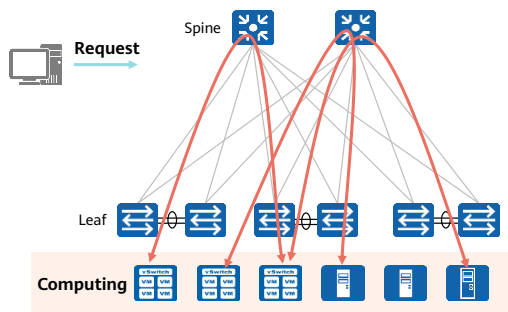
- For example, training an AI model essentially means fine-tuning a huge mathematical model with billions or even trillions of parameters. Each adjustment requires the same multiplication and addition operations on massive data. The GPU's working mode fits perfectly with the needs of AI.

Three Versions of the Evolution of General-Purpose Computing DCN Architecture



- The current general-purpose computing DCN is DCN3.0.
- Multichassis Link Aggregation Group (M-LAG): a link aggregation technology.
- Virtual Extensible LAN (VXLAN): a network virtualization technology at the physical network layer.
- Transparent Interconnection of Lots of Links (TRILL): a Layer 2 technology, which is seldom used due to its difficult deployment.
- Virtual Router Redundancy Protocol (VRRP): a network fault tolerance protocol. It is used to coordinate multiple physical routers (or Layer 3 switches) to virtualize a single, highly available virtual router, thereby providing uninterrupted default gateway services for hosts within a LAN.
- Spanning Tree Protocol (STP): a network protocol that works at the data link layer. By logically blocking specific ports, it constructs a loop-free logical tree topology within a switching network that has physical loops. This allows for link redundancy while avoiding broadcast storms, MAC address table instability, and other problems caused by loops.

Traffic Patterns in General-Purpose Computing Scenarios

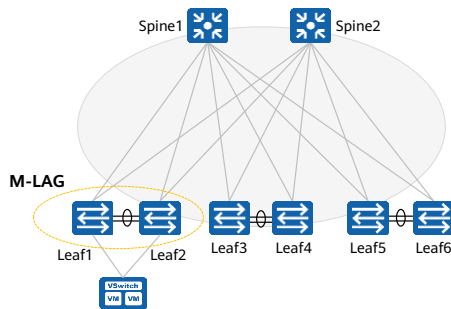


An external client initiates a request, triggering multiple calls between backend servers to exchange data and resulting in intensive east-west traffic.

East-West traffic

- General-purpose computing starts with a client request, which triggers dozens of backend service calls, resulting in intensive east-west lateral traffic. Each call involves a small amount of data, but when aggregated, the workload becomes latency-sensitive. Round-trip delays must be reduced to the microsecond level; otherwise, user experience deteriorates exponentially.

Requirements of General-Purpose Computing on DCNs (1)



The spine-leaf architecture is used to meet the requirements for elastic scalability.
M-LAG is used to meet the requirements for high aggregate bandwidth and high availability.

Elastic scalability

- In cloud DCs, VMs or containers need to be created dynamically based on customer demands. The network must flexibly adapt to these dynamic changes and ensure that newly deployed compute nodes can seamlessly connect to the network.

High aggregate bandwidth

- Although the data volume of a single request is small, massive concurrent requests can instantly converge into TB-level traffic.
- High aggregate bandwidth can effectively cope with such instantaneous traffic peaks and ensure that the network does not become congested due to excessive load.

High availability

- Core systems such as e-commerce and payment systems are intolerant of interruptions in seconds, necessitating device redundancy, multipath links, and millisecond-level failover.
- Network infrastructure must exhibit exceptional reliability, with mechanisms for fast fault detection and convergence.

Requirements of General-Purpose Computing on DCNs (2)

User isolation

- DCs need to serve an increasing number of users, and massive computing services require isolation.

A VLAN ID is 12 bits long, which can represent only 4,096 logical units.

The number of tenants supported by a large DC is much greater than 4,096.

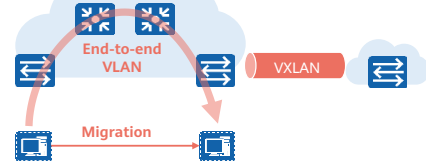
DC

The VXLAN ID is 24 bits, supporting up to 16 million users.

Expanded VM migration range

- To ensure resilience for critical services, remote backup must be implemented, driving the establishment of DCs across multiple physical locations.
- Services require VMs to be able to migrate across physical locations.

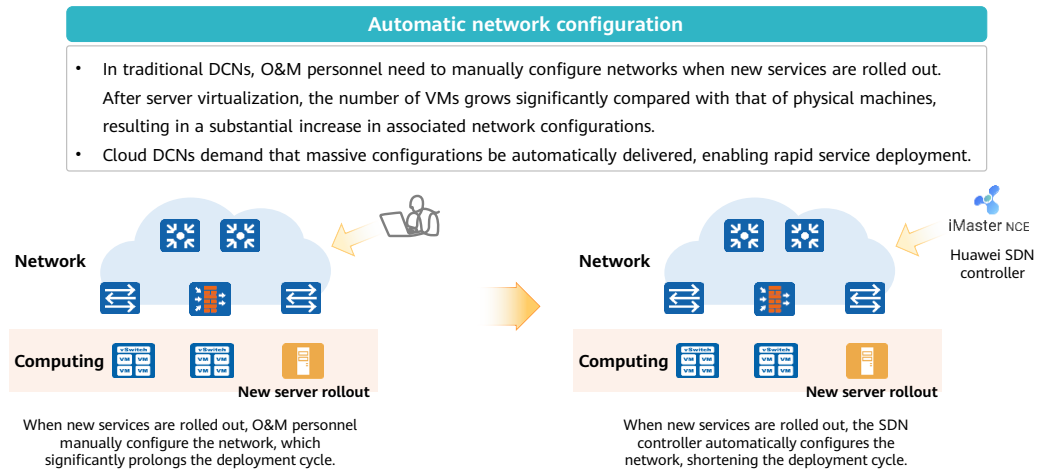
VMs can be migrated only within a VLAN, and the number of VLANs is limited.



Migration

The establishment of VXLAN tunnels expands the VM migration scope.

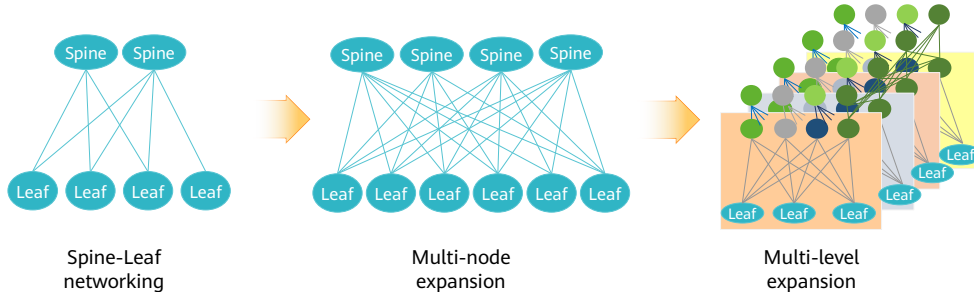
Requirements of General-Purpose Computing on DCNs (3)



- **Software-Defined Networking (SDN):** The core idea is to separate the network's control plane from the data plane, and manage and control the network in a unified and intelligent manner through an independent, centralized SDN controller.

Elastic Scalability: Spine-Leaf Architecture

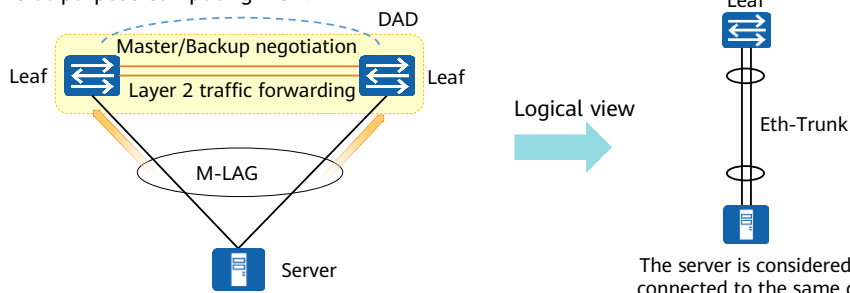
- Spine-Leaf is a new type of DCN architecture, consisting of spine nodes and leaf nodes. A spine node provides high-speed IP forwarding. A leaf node provides access to networks.
- The spine-leaf architecture provides **high scalability**, meeting the requirements of general-purpose computing for DCNs.



- In terms of hardware, both spine and leaf nodes are switches. Spine nodes can be regarded as core switches on a campus network, while leaf nodes are access switches.
- Characteristics of the spine-leaf architecture:
 - Each lower-level node (leaf) connects to all corresponding higher-level nodes (spine), forming a full-mesh topology.
 - There is no east-west link between nodes at the same level.
 - Typically, the spine-leaf architecture is similar to a modular switch, where the leaf node is like an LPU, responsible for receiving external traffic and the spine node is like an SFU, responsible for transmitting traffic between leaf nodes.
- High scalability of the spine-leaf architecture:
 - The number of spine nodes can be expanded to four or more. The maximum number of spine nodes depends on the number of uplink ports on leaf nodes.
 - Based on the two-level spine-leaf design, the architecture can be further extended to a three-level spine-leaf, supporting high-speed data exchange among a larger number of leaf nodes.

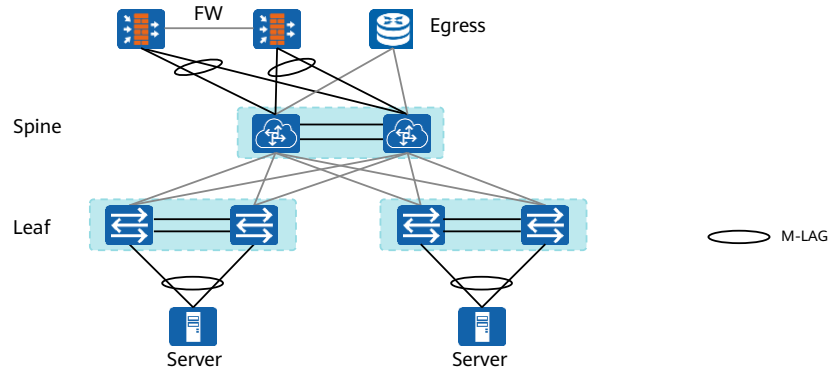
High Aggregate Bandwidth and High Availability: M-LAG

- M-LAG is a technology that implements link aggregation across devices.
- It improves link reliability from the board level to the device level. By implementing inter-device link aggregation (Eth-Trunk), M-LAG not only increases server access bandwidth but also provides backup protection, thereby meeting the requirements of high aggregate bandwidth and high availability on the general-purpose computing DCN.



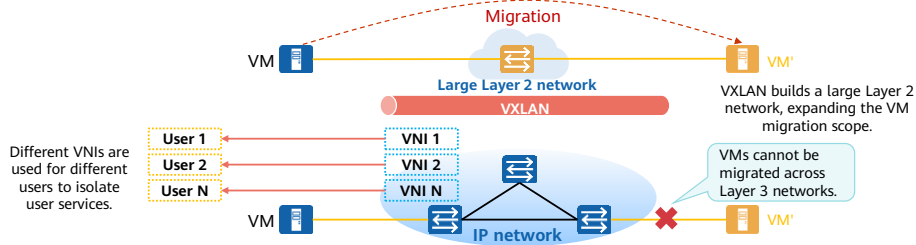
Application of the Spine-Leaf + M-LAG Architecture in DCs

- The general-purpose computing DCN adopts the spine-leaf architecture to meet scalability requirements during service expansion. To meet high availability requirements, M-LAG is recommended for connecting servers and value-added service devices (such as firewalls).



User Isolation and VM Migration Scope Expansion: VXLAN

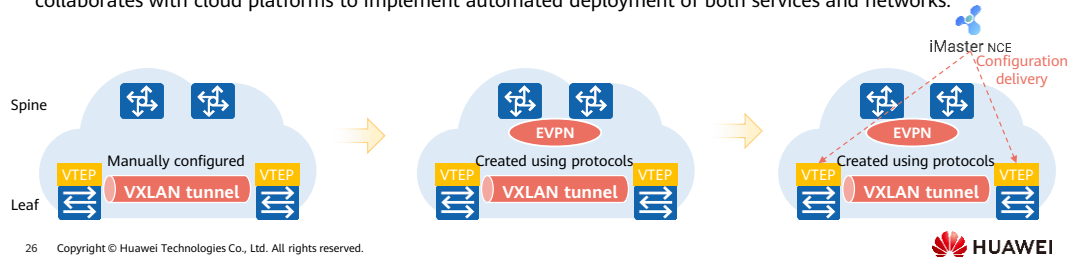
- VXLAN is a tunneling technology primarily designed to address user isolation and network scalability challenges in cloud DC environments.
 - User isolation: Similar to traditional VLAN isolation, VXLAN uses the VXLAN Network Identifier (VNI) field to distinguish between different networks, thereby isolating users. Compared with VLAN, VXLAN offers a much larger VNI range, supporting a greater number of isolated user networks.
 - Network scalability (VM migration range expansion): VXLAN can overlay a Layer 2 virtual network (large Layer 2 network) on any network with reachable routes. Terminals connected to such a large Layer 2 network can directly communicate with each other at Layer 2, meeting the VM migration requirements and expanding the migration scope.



- On a VXLAN network, a user can have one or more VNIs. Each VNI is 24 bits long, supporting up to 16 million users, thereby meeting the requirements for isolating a large number of users in large-scale cloud computing service scenarios.
- As shown in the figure, after VXLAN is deployed on an existing IP network with reachable routes, terminals can communicate with each other at Layer 2 across a Layer 3 network. That is, VMs are unaware of the IP network and consider that they are connected to a Layer 2 network. Such a network is called a large Layer 2 network.

Automatic Network Configuration

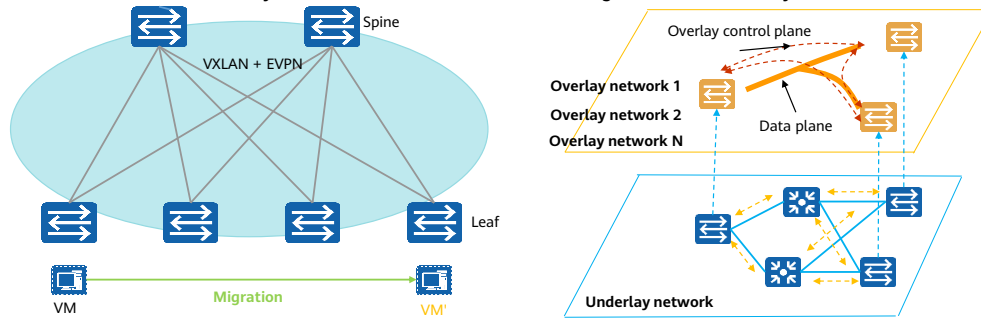
- In the early stage, VXLAN was deployed in static mode, requiring manual creation of VXLAN tunnels, which involved heavy configuration workloads. Moreover, VXLAN itself lacks a control plane, and automatic VXLAN Tunnel Endpoint (VTEP) discovery and host information collection were implemented through traffic flooding on the data plane, resulting in significant amounts of flooding traffic within DCNs. To address these issues, VXLAN is combined with Ethernet Virtual Private Network (EVPN) to enable automatic VXLAN tunnel creation, automatic VTEP discovery, and mutual advertisement of host information.
- On a large Layer 2 network, an SDN controller is introduced to facilitate control and deployment. The controller manages devices through network management protocols, automates the delivery of network configurations, and collaborates with cloud platforms to implement automated deployment of both services and networks.



- VXLAN Tunnel Endpoints (VTEP): encapsulates and decapsulates VXLAN packets.

Application of VXLAN in DCs

- VXLAN builds an overlay virtual LAN on top of the traditional IP-routed underlay network, with all DC services running on the overlay network.
- Within the DC, multiple overlay networks can be established on the underlay network. Different users utilize different overlay networks, and user isolation is implemented through the VNI field. During service running, users are unaware of the underlay network. In addition, VMs can be migrated on the overlay network.



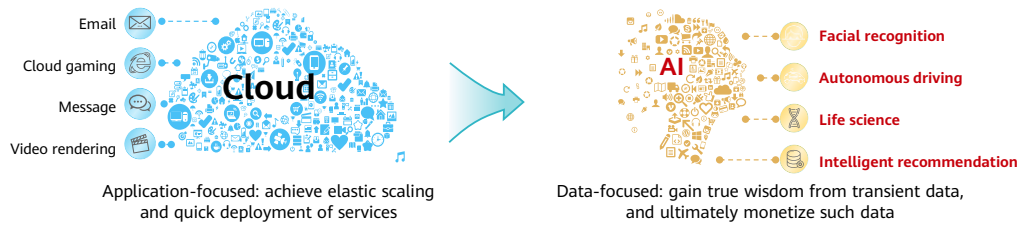
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1. DC Evolution

2. DCN Basics

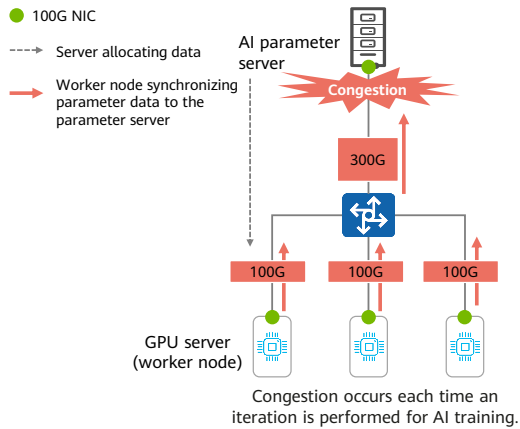
- General-Purpose Computing DCN
- Intelligent Computing DCN
- Dual-Track Architecture of DCs

The AI Era Has Arrived



- The era of AI has arrived. For example, facial recognition, autonomous driving, and intelligent recommendation are now closely related to our daily life, work, and study. These technologies cannot be implemented without the participation of AI.
- To implement these technologies, engineers need to collect and filter massive amount of data, build AI training models, and use the obtained data to train the models. With the rapid increase of AI application data, the scale of the AI algorithms continuously increases, and the training tasks of the AI models require a massive amount of computing.
- Since the growth rate of GPU server computing power still lags far behind the rapid rise in computing demands from AI applications, AI clusters have become the preferred choice today for meeting the computing requirements of AI applications.

AI Clusters Require High-Speed Networks with Zero Packet Loss

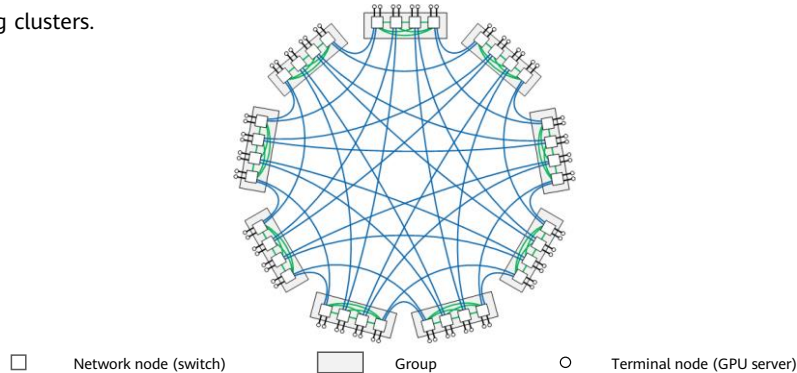


- During the iteration of AI computing, a large amount of burst traffic is allocated to worker nodes within several milliseconds. When intermediate parameters are transmitted and updated, small-scale traffic sent to the parameter server may cause incast. In the process of these flow exchanges, the network may experience packet loss, congestion, and load imbalance. As a result, the flow completion time (FCT) of some flows is extended. If some flows are delayed, storage and computing resources may not be fully utilized, which further delays the completion time of the entire application.
- To meet the requirements of AI computing, a high-speed DCN with zero packet loss, that is, an intelligent computing network, needs to be built.

- Incast refers to a many-to-one communication pattern (for example, worker nodes sending data to a parameter server, as shown in the figure). It is commonly seen in DCs where large-scale distributed storage or computing services are deployed.

Architecture of Intelligent Computing DCNs

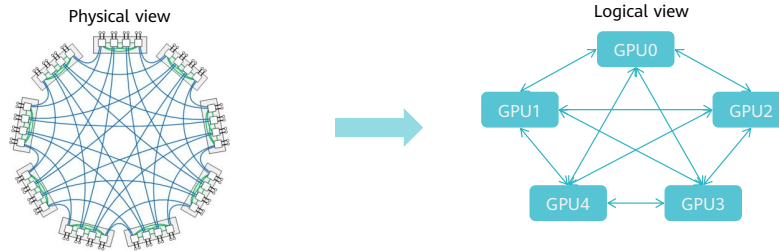
- In addition to continuing the use of the spine-leaf architecture from general-purpose computing DCNs, intelligent computing DCNs also adopt the dragonfly architecture. With its small network diameter and lower cost, the dragonfly architecture has become a viable option for ultra-large-scale intelligent computing clusters.



- In the dragonfly architecture, there are multiple groups, with full-mesh connections established between groups and within a group. This means each pair of groups are connected through one or more links; each network node in a group is directly connected to other nodes in the group and can also be connected to other groups and terminal nodes.
- The dragonfly architecture is an alternative to the spine-leaf architecture. Its core topology uses a group-based flat interconnection structure, which optimizes inter-group connections and reduces the network diameter (number of hops) to 2–3. This design enables effective control over the growth in the number of core switches during ultra-large-scale scale-out expansion, thereby offering advantages in both latency and cost. Therefore, the dragonfly architecture is primarily suited for ultra-large-scale intelligent computing centers and high-performance computing clusters that require interconnection of tens of thousands to even hundreds of thousands of compute nodes. In contrast, the traditional spine-leaf architecture is mainly applicable to intelligent computing clusters with thousands of GPUs/NPUs (thousands of compute nodes) or less.

Traffic Patterns in Intelligent Computing Scenarios

- The core of intelligent computing (especially AI training) lies in hundreds or even thousands of servers (primarily GPU servers) working together to complete a large-scale parallel computing task. The network traffic pattern is "All-to-All," meaning that at the end of each training iteration, all GPUs must synchronize their data.
- In AI training, once any single GPU finishes its computation, it must wait for data from all other GPUs to arrive (synchronization barrier). The higher the network latency, the longer GPUs remain idle, which reduces the effective utilization of the entire cluster's computing power. Therefore, in this scenario, the network performance can be directly converted into the cluster utilization.



- With the expansion of network scale, intelligent computing services of all types face increasing concurrency. This easily leads to network congestion, increased forwarding latency, and in severe cases, packet loss, which can degrade service quality or even cause service unavailability. Therefore, to support these real-time services, network congestion must be addressed. While the best way to solve congestion is to increase network bandwidth, it is economically unsustainable in terms of operations and maintenance. The most effective solution is to use a guaranteed policy to manage network traffic and maximize network utilization.

Requirements of Intelligent Computing on DCNs

Ultimate bandwidth per port

- AI training and similar services require rapid synchronization between servers, which in turn demands a sustained throughput of 400 Gbit/s or even 800 Gbit/s per single network port.
- Ultra-high network bandwidth ensures that data synchronization can be completed within a very short time, improving training efficiency.

Lossless network for AI training

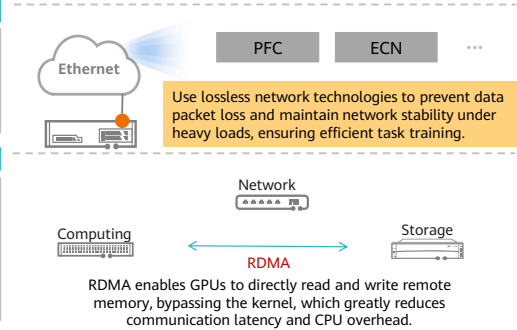
- In AI training, any packet loss triggers retransmission, which can instantly slow down the overall training process. Therefore, it is necessary to build a lossless network environment with zero packet loss to ensure non-blocking data flow.

Ultra-low latency

- The traditional TCP/IP protocol stack consumes CPU resources and introduces high latency, while intelligent computing services demand networks with much lower latency.
- RDMA enables GPUs to directly read and write remote memory, bypassing the kernel, which greatly reduces communication latency and CPU overhead.

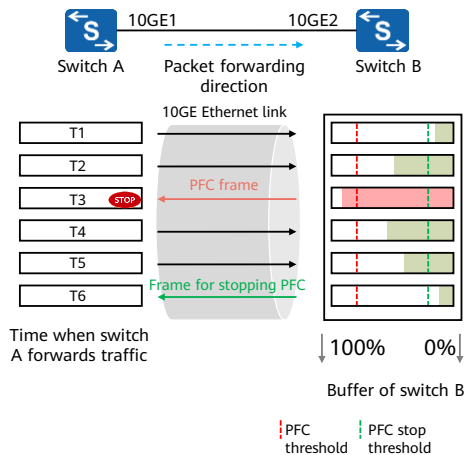


Huawei 100GE/400GE/800GE optical modules



- In intelligent computing scenarios, lossy networks cannot meet the requirements of high-performance systems. The intelligent lossless network (intelligent computing network) employs various lossless networking technologies to achieve maximum throughput and minimum latency while ensuring zero packet loss. The two most fundamental protocols on an intelligent computing network are Explicit Congestion Notification (ECN) and Priority-based Flow Control (PFC).

Intelligent Lossless Network Technology: PFC

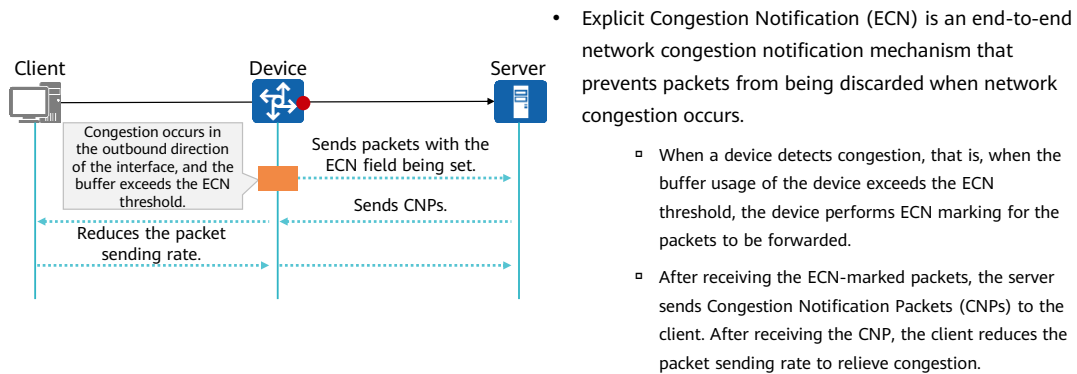


- Priority-based Flow Control (PFC): It manages the transmission rate between two nodes to prevent the transmit end from exhausting the receive end's buffer, thereby resolving the mismatch between the transmit end's transmission rate and the receive end's processing rate.

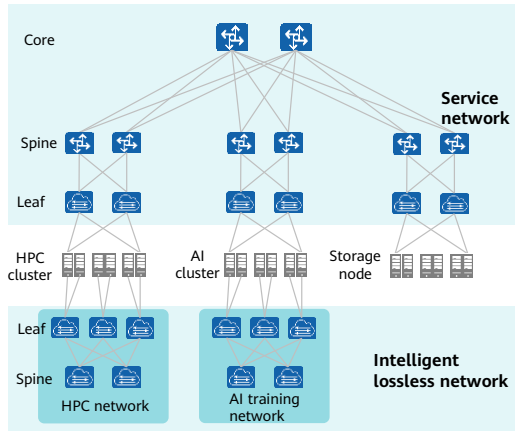
- When the buffer usage of a specific priority queue at the receive end exceeds the PFC threshold, the receive end sends a PFC frame to the transmit end.
- After receiving the PFC frame, the transmit end stops sending traffic to the receive end.
- When the buffer usage of a specific priority queue at the receive end falls below the PFC stop threshold, the receive end sends a frame for stopping PFC to the transmit end.
- After receiving the frame, the transmit end resumes sending traffic to the receive end.

- As shown in the figure, the working mechanism of PFC is as follows:
 - Switch A forwards traffic to switch B. Over time, the buffer of switch B gradually increases. When the buffer of switch B exceeds the PFC threshold, switch B sends a PFC frame to switch A.
 - After receiving the PFC frame, switch A stops sending traffic to switch B.
 - Switch B continues to process the traffic in the buffer. When the buffer of switch B falls below the PFC stop threshold, switch B sends a frame for stopping PFC to switch A.
 - After receiving the frame, switch A resumes sending traffic to switch B.

Intelligent Lossless Network Technology: ECN



Application of Intelligent Lossless Network Technologies in DCs



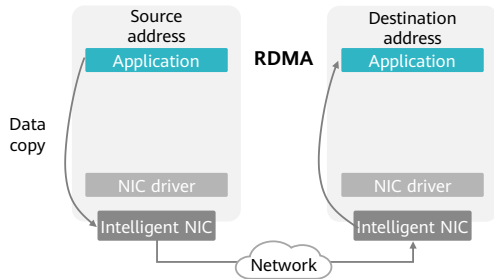
- Service networks are often deployed as TCP/IP lossy networks.
- For different industries or enterprises, the methods of area division vary.

Lossless intelligent networks can be applied to different computing scenarios in DCs:

- For example, high-performance computing (HPC) networks and AI distributed training networks are generally deployed as closed networks.
- Depending on the scale of cluster access nodes, select the two-layer or three-layer spine-leaf networking architecture. Based on the port bandwidth, select proper leaf and spine switches and deploy lossless network technologies.

RDMA Protocol Stack

- Remote Direct Memory Access (RDMA): leverages related hardware and network technologies to directly transfer data from the memory of one server to the memory of another. This significantly reduces the server data processing delay during network transmission.
- RDMA over Converged Ethernet (RoCE): a network protocol that allows RDMA over Ethernet. The outer header is an Ethernet header, which allows RDMA to be used over standard Ethernet infrastructure, provided that the server's NIC supports RoCE. Among RoCE protocols, RoCEv2 is the most widely used.



- NICs can directly transmit data with application memory without copying data between application memory and the kernel. With this, the transmission latency is significantly reduced.

- In the preceding process, applications can send commands to NICs without invoking the kernel memory. The protocol stack is offloaded to NICs, and the CPU and kernel do not need to be involved. In this way, ultra-low-latency data processing and ultra-high-throughput traffic transmission can be easily implemented.

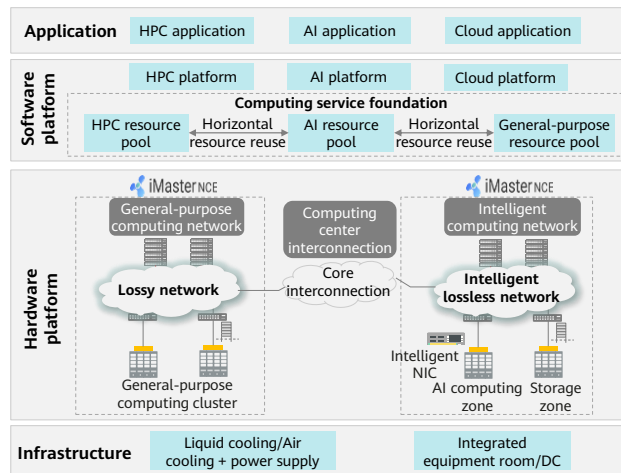
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Dual-Track Architecture of DCs



- On future DCNs, general-purpose computing clusters and intelligent computing clusters will share common infrastructure (such as power and cooling), but their network planes will remain separated. This separation of network planes is referred to as the dual-track architecture of DCs. In this architecture, the upper-layer service platform schedules the two types of tasks to their respective resource pools, achieving physical coexistence with logical isolation.
- Unified resource scheduling can effectively reduce construction and O&M costs, while maintaining clear performance boundaries to ensure that the two types of networks coexist without interference.
- The general-purpose computing network is like the arteries of a city, ensuring service continuity; the intelligent computing network is like a dedicated racetrack, unleashing the full potential of computing power. The two are not substitutes, but rather complementary.
- In actual deployment, only by understanding the differences, designing as needed, and implementing unified scheduling can DCs remain both stable and fast in the AI era. The dual-track architecture is the inevitable choice for future large-scale DCs.

Quiz

1. (Single-answer question) In the data center general-purpose computing scenario, which of the following technologies is used to isolate multiple tenants? ()
 - A. VXLAN
 - B. M-LAG
 - C. RDMA
 - D. PFC
2. (Single-answer question) On the data center intelligent computing network, when congestion occurs, simply increasing network bandwidth continuously can achieve a lossless network. ()
 - A. True
 - B. False

1. A

2. B

Summary

- Data centers are classified into general-purpose computing data centers and intelligent computing data centers based on the types of internal services, each employing different network technologies.
- General-purpose computing data center networks primarily adopt a spine-leaf architecture, combined with M-LAG and VXLAN technologies, to meet user requirements for network isolation, scalability, and availability in such scenarios.
- In addition to the spine-leaf architecture, the intelligent computing center network also supports the dragonfly architecture. It uses intelligent lossless network technologies (PFC and ECN) and the RDMA protocol stack to provide a lossless network with zero packet loss and ultra-low latency for intelligent computing services.

Thank you.

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WLAN Technology Basics



Foreword

- Wired LANs use cables or optical fibers as transmission media, which are widely used but are expensive and lack user mobility. As the demand for network portability and mobility increased, traditional wired LANs were no longer able to meet users' needs, leading to the emergence of Wireless Local Area Network (WLAN) technology.
- 802.11 was originally a wireless networking standard defined by the Institute of Electrical and Electronics Engineers (IEEE) in 1997. This standard has been continuously revised and improved since then, forming the 802.11 series standards.
- This chapter describes the definition, basic concepts and key technologies of WLAN, and 802.11 standards.

Objectives

- On completion of this lesson, you will be able to:
 - Describe the definition of WLAN.
 - Describe the basic concepts of WLAN.
 - Describe the common technical principles of WLAN.
 - Describe the differences between 802.11 protocols.
 - Describe the development of Wi-Fi generations.

Contents

- 1. WLAN Overview**
2. Basic WLAN Concepts
3. Key WLAN Technologies
4. Introduction to 802.11 Standards

Background of WLAN

- Before WLAN, computer networks relied on cables, which restricted users to fixed locations.
- WLAN emerged as a solution to these limitations, providing mobile connectivity that wired networks cannot offer. This is also the core value of WLAN.

Breaking free from cables

- With the rise of laptops, users want the flexibility to connect anywhere in the office without being tied to fixed network ports.



Overcoming cabling challenges and costs

- In many environments such as historic buildings, warehouses, or factories, deploying wired networks is impractical or prohibitively expensive.



Supporting massive IoT connectivity

- Today, smart devices such as speakers and TVs can seamlessly connect to the Internet or to each other through WLAN.



Definition of WLAN

- A WLAN is a type of local area network that connects devices and transmits data using wireless technologies (such as Wi-Fi and Bluetooth), eliminating the need for traditional wired cables.
- Core features of WLAN:

Comparison with LAN

- LAN: uses physical cables (such as twisted pairs and coaxial cables) to connect devices.
- WLAN: uses radio waves as the transmission medium instead of physical cables, offering mobility and deployment flexibility.

Coverage

- The coverage is typically limited to homes, offices, and campuses, ranging from a single room to an entire building (tens to hundreds of meters).

Technical implementation

- WLANs operate on specific unlicensed frequency bands, among which the most common ones are 2.4 GHz and 5 GHz and Wi-Fi 7 introduces the 6 GHz frequency band.
- Standardized IEEE 802.11 protocols ensure interoperability among devices from different vendors.

Relationship with Wi-Fi

- Wi-Fi is the most widely adopted WLAN technology and is a trademark owned by the Wi-Fi Alliance.
- In simple terms, all Wi-Fi is WLAN, but not all WLAN is Wi-Fi. In everyday usage, however, the two terms are often treated as synonyms.

- The WLANs in the following courses are implemented based on Wi-Fi technology (IEEE 802.11 standard).

WLANs in Daily Use

- WLANs provide the wireless infrastructure for everyday Internet access, streaming, and gaming. Home Wi-Fi, corporate wireless networks, and public hotspots in cafes are all types of WLAN.

Home WLAN

- Wireless Internet access for mobile phones, tablets, and laptops
- Connectivity for smart home devices (such as smart speakers, cameras, and robotic vacuums)
- Streaming and online gaming on smart TVs and game consoles



Enterprise and school campus WLANs

- Wireless office and video conferencing for employees
- Wireless coverage across classrooms, libraries, and dormitories
- Wi-Fi in hotels and other public venues

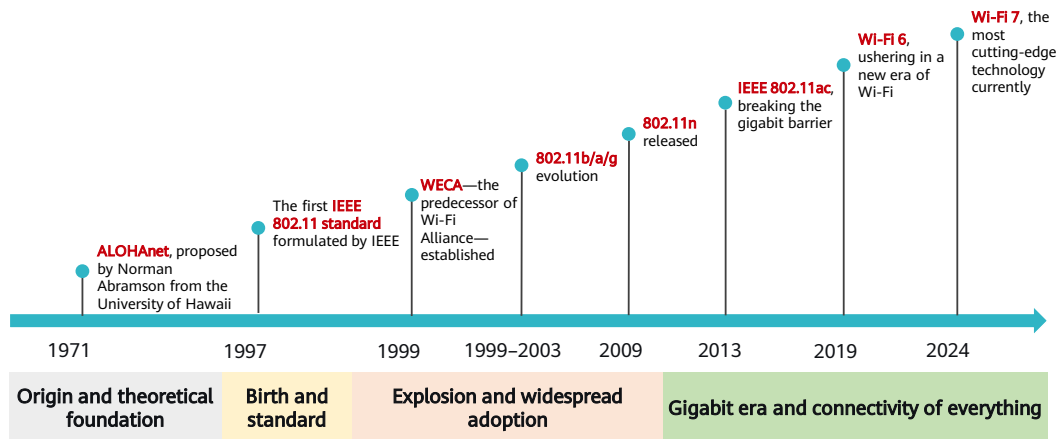


Public WLAN

- Free Wi-Fi in cafes, airports, and subway stations



Origin and Development of WLAN



- Origin and theoretical foundation
 - Wireless networking traces back to World War II, when the U.S. Army used radio signals to transmit data. They developed a set of radio transmission technologies and strong encryption that influenced later systems. In 1971, researchers at the University of Hawaii created ALOHAnet, the first packet-based radio network. The ALOHA protocol used by ALOHAnet became a foundational idea for later Wi-Fi core protocols.
- Birth and standard
 - The IEEE developed the first IEEE 802.11 standard, which defined the MAC and physical layer (PHY) specifications for WLANs. Early wireless rates were 1 Mbit/s and 2 Mbit/s, marking the beginning of wireless standardization.
- Explosion and widespread adoption
 - In 1999, the Wireless Ethernet Compatibility Alliance (WECA) was formed to promote the development of the IEEE 802.11b specifications. In October 2002, WECA was renamed the Wi-Fi Alliance. It is a commercial alliance that owns the Wi-Fi trademark. The alliance runs certification programs that ensure Wi-Fi device interoperability and develops WLAN technologies based on IEEE 802.11 standards.
 - In 1999, 802.11b was released, which became the first widely accepted Wi-

Fi standard.

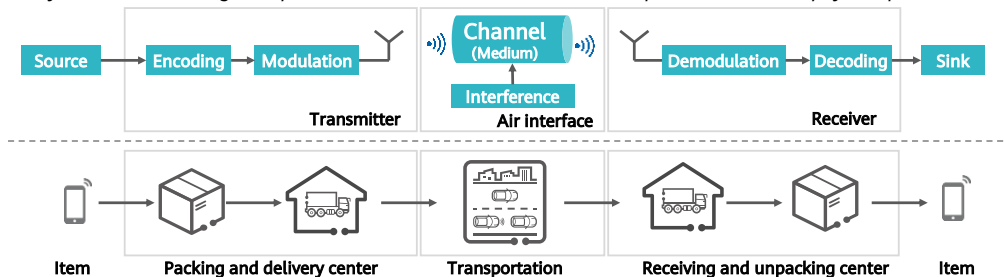
- In 1999, 802.11a was released, which operated on the 5 GHz frequency band and provided a rate of up to 54 Mbit/s, with strong anti-interference capabilities.
- In 2003, 802.11g was released, which operated on the 2.4 GHz frequency band and provided a rate of up to 54 Mbit/s, becoming the mainstream standard at that time.
- In 2009, 802.11n was released, which supported both the 2.4 GHz and 5 GHz frequency bands and increased the theoretical rate to 600 Mbit/s, achieving a major leap.
- Gigabit era and connectivity of everything
 - We have entered the era of the all-wireless office, with wireless connectivity at the center of daily work. In 2018, the next-generation Wi-Fi standard Wi-Fi 6 (802.11ax) was released, marking a milestone in the Wi-Fi development history. The core value of Wi-Fi 6 is to further improve the transmission rate, leading wireless communication into the 10 Gbps era.
 - In 2024, Wi-Fi 7 (IEEE 802.11be) was released. It is a cutting-edge technology designed to provide ultimate throughput and low latency.

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Wireless Communication System

- Wireless communication uses radio waves to carry information—data, voice, images, etc.—between locations without physical cables.
- In a wireless communication system, the transmitter converts information into a digital signal using source coding for efficient processing. It then applies channel coding and modulation to convert the digital signal into radio waves for transmission. The receiver demodulates and decodes the received radio waves to recover the original information.
- The system works like a global postal service that delivers invisible data packets instead of physical parcels.



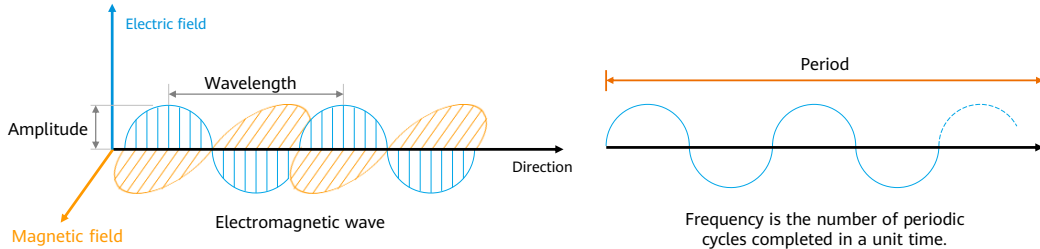
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- Basic concepts in a wireless communication system:
 - **Source:** the entity that originates information, for example, a phone, computer, or sensor.
 - **Transmitter:** the "packing and delivery center" of information. It processes and converts the source data into a form suitable for transmission on a channel.
 - **Channel:** the "transportation path" of information. In a wireless system, it is free space (air or vacuum) and is the core medium of communication.
 - **Receiver:** the "receiving and unpacking center" of information. It captures signals from the channel and reverses the transmitter's operations to recover the original information.
 - **Sink:** the final recipient of information, for example, a friend's phone, a server, or another sensor.
- In wired communication systems, the device interfaces and connecting cables are physically visible. In wireless communication systems, however, the interfaces and the connection medium are not visible; the connection occurs over the air interface (also called the space interface), which is the invisible channel between the transmitter and receiver.

Radio Wave

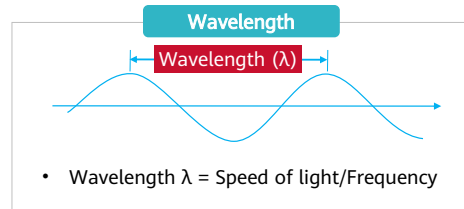
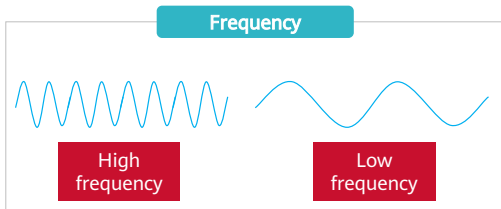
- A changing current in a conductor produces radio waves. Using modulation, information is encoded onto those radio waves. When the waves reach a receiver, the varying electromagnetic field induces currents in the conductor. Demodulation extracts the encoded information from those currents, completing the transmission.
- A radio wave is an electromagnetic wave with relatively low frequency and a long wavelength within the electromagnetic spectrum.



- Electromagnetic waves, also called electromagnetic radiation, transmit energy and momentum through space. They consist of electric and magnetic fields that oscillate in phase and are perpendicular to each other. The direction of propagation is perpendicular to the plane formed by the electric and magnetic fields.
- The key characteristics of electromagnetic waves are described by two parameters:
 - Frequency: number of times a wave oscillates in 1 second, in hertz (Hz).
 - Wavelength: distance between two adjacent wave crests.
- The electromagnetic spectrum arranges all electromagnetic waves by frequency (or wavelength). From low to high frequency, the order of the electromagnetic spectrum is: radio wave → microwave → infrared → visible light → ultraviolet → X-ray → gamma ray. Radio waves occupy the lowest-frequency end of the spectrum.

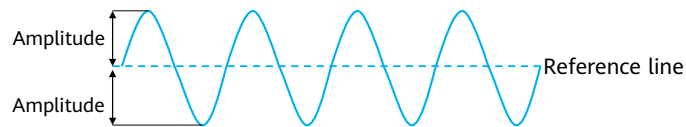
Frequency and Wavelength

- Frequency is the number of complete vibration cycles that occur in a unit time. A complete cycle is from one peak to the next peak, or from one trough to the next trough.
- A wave's frequency is its oscillation rate. One complete cycle (one complete vibration cycle) per second equals 1 Hz.
- Wavelength is the distance between corresponding points on two adjacent cycles, which is also the distance traveled during one vibration cycle. It is typically expressed as the distance between adjacent crests or troughs. Waves with higher frequency have shorter wavelengths.



Amplitude

- Amplitude refers to the distance from the equilibrium position (reference line) of an oscillating quantity to its maximum displacement.
- As shown in the figure, the dashed line represents the equilibrium position (reference line). The vertical distance from the reference line to a crest is the amplitude. Similarly, the distance from the reference line to a trough is also the amplitude.
- In wireless communication, amplitude measures the signal strength. Larger amplitude changes usually indicate a stronger signal and a longer transmission distance.

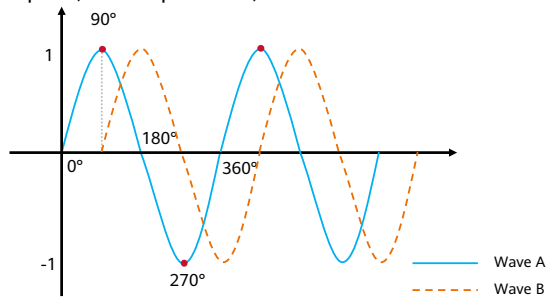


- Amplitude and frequency are independent concepts. A wave can have a large amplitude but a low frequency (like a slow, loud drumbeat), or it can have a small amplitude but a high frequency (like a faint, sharp bird call).

Phase

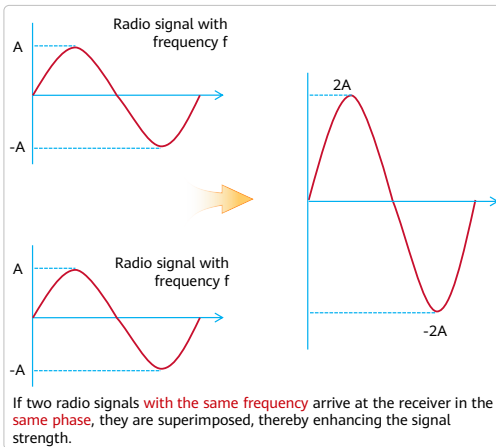
- Phase refers to the relative position of two signals in a signal period. It is a relative term describing the relationship between two waves of the same frequency. It is usually expressed in degrees or radians, and is also called phase angle.
- A signal waveform changes periodically, and one cycle of the waveform is 360° . When measuring the phase, the wavelength is generally divided into 360 parts, each equal to 1° , with 0° defined as the start time of wave propagation.

- As shown in the figure on the right, if wave A starts to propagate at 0° and wave B starts to propagate at 90° , the two waves are 90° out of phase.
- Waves with the same frequency that start at different times significantly affect electromagnetic wave transmission.

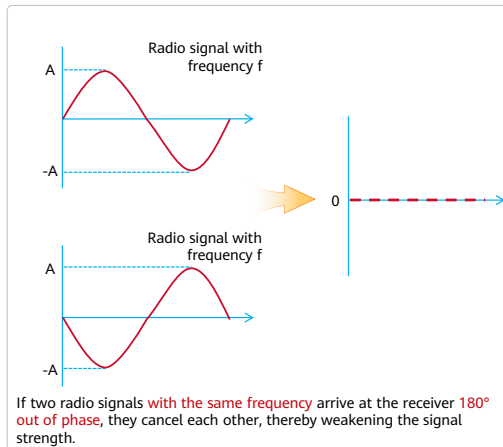


Impact of Phase on Signals

Signal enhanced



Signal weakened



Coding

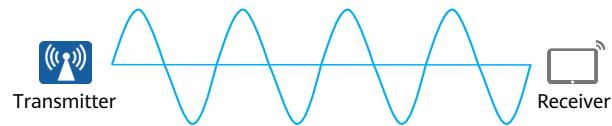
- Coding technologies in wireless communication are classified into source coding and channel coding. They serve different purposes and complement each other.
 - Source coding: **improves efficiency** by representing source information with as few bits as possible, thereby reducing redundancy.
 - Channel coding: **improves reliability** and enhances the anti-interference capability for data transmission over noisy channels.

Source coding	Channel coding
Implementation: - Data compression: removes redundancy from information to the maximum extent possible without causing distortion. - Digitalization: converts analog signals into digital bit streams. Application: Different types of information require different encoding methods. For example, H.264 is a video encoding method.	Implementation: - Redundancy addition: regularly adds check bits to the original data bits to improve the reliability of channel transmission and reduce the bit error rate. - Resistance to noise, attenuation, and interference: copes with unpredictable interference on wireless channels. Application: Channel coding cannot increase the transmission efficiency of valid information; instead, it may decrease the transmission rate, but it increases the success rate of valid information transmission. Selecting an appropriate coding scheme for a communication protocol can achieve the optimal balance between performance and accuracy.

- Channel coding
 - Channel coding restores information at the receiver as much as possible in the presence of noise, reducing the bit error rate in wireless transmission. A widely used channel coding method in WLANs is low-density parity-check (LDPC) coding.
 - Channel coding adds redundant information to the original information, increasing the total bit length. Coding efficiency is the ratio of the number of pre-coding bits (original information) to the number of post-coding bits.

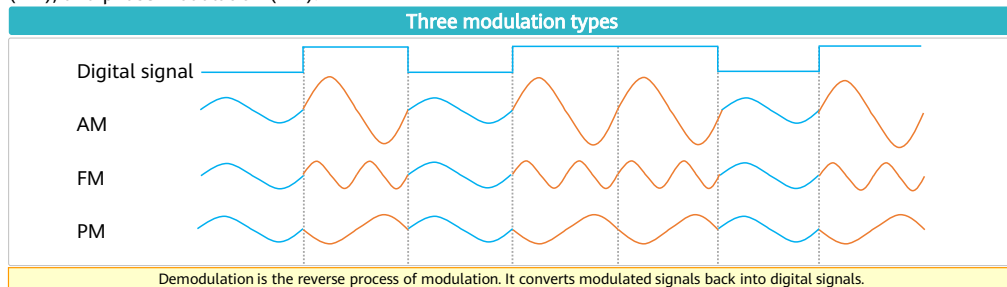
Carrier

- A carrier is an unmodulated radio wave, typically a high-frequency radio wave with a specific frequency. It contains no useful information by itself. Because raw data signals usually occupy low frequencies and are difficult to receive and synchronize at their native frequency, the data is loaded onto the carrier for transmission. The receiver recovers the data by tuning to the carrier frequency.
- In a postal system, for example, the carrier is like a truck that carries goods; this truck is specially designed for long-distance road transportation.
- The carrier is the foundation of wireless communication. The following figure shows a basic carrier. A carrier is characterized by three main parameters: amplitude, frequency, and phase.



Modulation

- Modulation: converts various digital baseband signals into digital modulated signals suitable for channel transmission. In the postal analogy, modulation is the loading process, where a piece of cargo (information) is loaded onto a truck (carrier) for transportation on the road (wireless channel).
- The modulation process is to change the amplitude, frequency, and phase of the carrier according to the variations of the baseband signal. There are three types of modulation: amplitude modulation (AM), frequency modulation (FM), and phase modulation (PM).

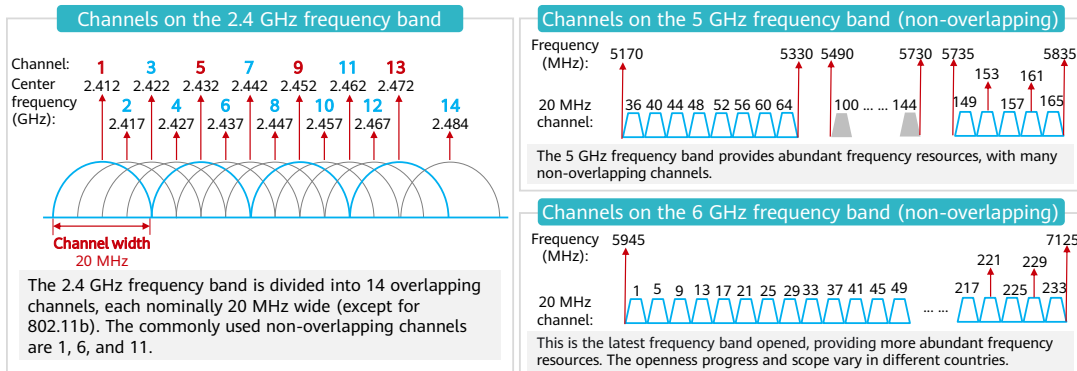


- AM: changes the amplitude of high-frequency carrier signals according to the instantaneous changes of modulation signals. That is, the amplitude of a high-frequency signal is changed by using a modulation signal, so that information of the modulation signal is embedded in the high-frequency signal. The high-frequency signal, which carries information of the modulation signal, is transmitted through an antenna. In this case, the receiver can demodulate the modulation signal, that is, the modulation signal can be obtained by interpreting the amplitude of the high-frequency signal.
- FM: changes the carrier frequency according to the modulation signal. The magnitude of frequency deviation is determined by the modulation signal amplitude, and the deviation period is determined by the modulation signal frequency. The amplitude of the modulated wave remains unchanged. The waveform of an FM wave is like a spring that is compressed unevenly.
- PM: changes the deviation value of the carrier phase relative to the reference phase proportionally according to the instantaneous value of the modulation signal. That is, the initial phase of a carrier changes with the baseband digital signal. For example, digital signal 1 corresponds to phase 180° , and digital signal 0 corresponds to phase 0° .
- In wireless communication, digital signals (binary digits 0 and 1) are transmitted. Modulation is used to map different states of a carrier signal to different binary

sequences.

Channel

- The frequency band is the frequency range a radio can use. Communication protocols define usable bands to avoid interference. WLANs use the 2.4 GHz (2.4–2.48 GHz), 5 GHz (5.17–5.33 GHz; 5.49–5.835 GHz), and 6 GHz (5.945–7.125 GHz) frequency bands. Each frequency band is further divided into frequency ranges called channels.



- WLANs use radio waves generated by the alternating current of an oscillating circuit and transmitted/received through an antenna. The radio wave is also called radio.
- Channel:
 - Overlapping channels: channels whose frequency ranges overlap (for example, channels 1 and 2 on the 2.4 GHz frequency band). Coexistence of overlapping channels in the same space causes interference.
 - Non-overlapping channels: channels whose frequency ranges do not overlap. Traditionally, 2.4 GHz non-overlapping channels are 1, 6, and 11; excluding 802.11b (channel width of 22 MHz, fading away from WLANs) compatibility, channels 1, 5, 9, and 13 can also be treated as non-overlapping. The channels on the 5 GHz and 6 GHz frequency bands are all non-overlapping channels in most cases.
 - Note: Different countries regulate which frequency ranges are available for WLAN use.

Interference

- Interference occurs when unwanted radio signals occupy the frequency band used by a wireless network. These signals superimpose on the valid signals, affecting them and preventing devices from accurately and efficiently parsing the required data.

Interference from non-Wi-Fi devices



Microwave oven



Bluetooth speaker



Wireless headset

Interference from Wi-Fi devices

Authorized transmitter



Broadcasting valid signals



Unauthorized transmitter

Broadcasting signals on the same channel as the valid signals

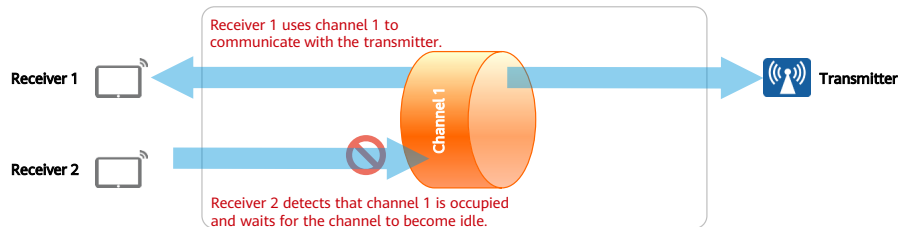


- Unauthorized transmitters and Wi-Fi devices that release radio signals (such as PCs) may cause interference to Wi-Fi in the environment.
- Wi-Fi device interference is the co-channel interference generated when many transmitters operate in the same space, and channel optimization is not performed or there are insufficient non-overlapping channels.

- Non-Wi-Fi device interference occurs when many household devices, such as microwave ovens and Bluetooth headsets, operate on the 2.4 GHz frequency band. In this case, 2.4 GHz channels become heavily occupied, reducing the efficiency in using Wi-Fi transmission channels.

Interference and Channel Utilization

- Channel utilization is also called channel efficiency.
 - For the transmitter, the channel utilization is the ratio of time used to send data to the total sending period.
 - Channel utilization = Time when a channel is busy/Sending period
- WLAN interference increases collisions and backoff. When multiple devices transmit data simultaneously, air interface collisions occur and the receiver cannot parse packets properly. The transmitters then retransmit packets after backoff, which prolongs the idle waiting time and decreases the channel occupancy rate.



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1. WLAN Overview
2. Basic WLAN Concepts
- 3. Key WLAN Technologies**
4. Introduction to 802.11 Standards

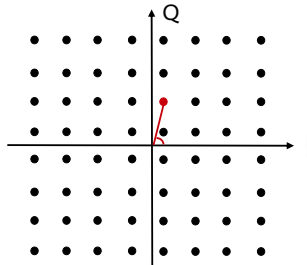
Overview of Key WLAN Technologies

- To avoid data packet collisions during wireless transmission, the 802.11 standard defines a simple protocol—carrier sense multiple access with collision avoidance (CSMA/CA)—at the MAC layer.
- As a wireless communication technology, WLAN needs to be improved in the following four aspects to achieve high-speed wireless transmission.

 Modulation scheme	 Number of carriers	 Channel width	 Number of spatial streams
• Use a better modulation scheme.	• Use orthogonal frequency division multiplexing (OFDM) to increase the number of carriers.	• Use channel bonding technology to increase the channel width.	• Use multiple-input multiple-output (MIMO) technology to increase the number of spatial streams.
Quadrature amplitude modulation (QAM)	OFDM	Channel bonding technology	MIMO

Modulation Technology (QAM)

- WLAN modulation has evolved from simple digital modulation to efficient quadrature amplitude modulation (QAM). QAM simultaneously modulates a carrier's phase and amplitude, using both to transmit information and thereby greatly increasing the data transmission rate with limited bandwidth.
- The constellation diagram is a core tool for understanding and designing QAM. It maps complex signals to a two-dimensional coordinate system for intuitive visualization.



64-QAM constellation diagram

- Constellation point: Each point represents a unique symbol, and its coordinates (I, Q) define the carrier's amplitude and phase.
 - Distance from a point to the origin = Carrier's amplitude
 - Angle between the I axis and the line connecting a point to the origin = Carrier's phase
- For a given system, the number of symbols is 2^n , where n is the number of bits per symbol. For 64-QAM, there are 64 symbols, that is, $n = 6$. Each symbol represents 6 bits, for example, 000000, 000001 and 000010. That is, using 64-QAM modulation, each modulated carrier can carry 6-bit data.
- With the improvement of modulation technology, the data transmission rate increases significantly. Currently, the highest QAM is 4096-QAM, enabling each symbol to carry 12-bit data.

- The core idea of QAM is to use two carriers (in-phase I and quadrature Q) to transmit two independent signals at the same frequency, encoding digital bit information into the carrier's amplitude and phase.

OFDM (1)

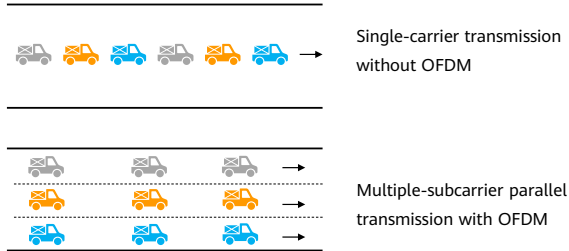
- OFDM is a multi-carrier transmission scheme. Its core idea is to convert a high-speed data stream into parallel low-speed sub-data streams, and modulate each sub-data stream onto each subcarrier for transmission. Each subcarrier is responsible for only low-speed data transmission.
- As shown in the figure, a channel is regarded as a lane. Without OFDM, only one subcarrier is used for transmission at a time, which is inefficient. With OFDM enabled, data transmission vehicles are allocated to different lanes so multiple subcarriers can transmit data simultaneously, greatly improving channel utilization.

Input data: 1 0 0 1 0 1 0 0 1 0 1 0 0 1 0 1 0...

Subcarrier 1: 1 1 0 0 0 1 1 0 0 1 0 1...

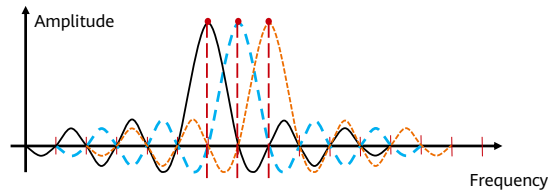
Subcarrier 2: 0 0 0 1 1 0 0 0 1 0 0 0...

Subcarrier 3: 0 1 1 0 0 1 1 1 0 1 1 0...



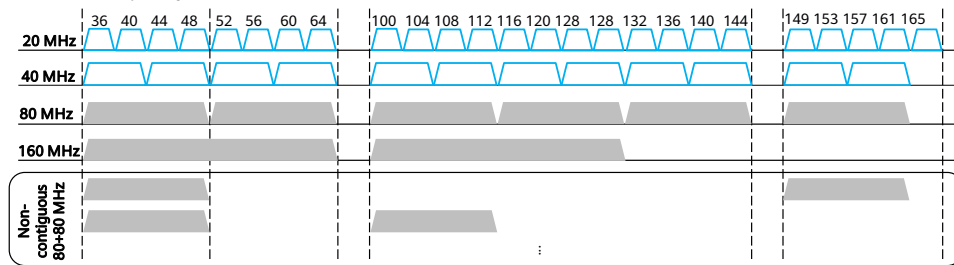
OFDM (2)

- OFDM uses overlapping subcarriers because each selected subcarrier is designed so its waveform is not affected by the others. Orthogonality is the core of OFDM. When one subcarrier reaches its amplitude peak, the amplitudes of the other subcarriers are 0, so they do not interfere with each other.
- As shown in the following figure, a signal is divided into three subcarriers, and the peak (marked in red) of each subcarrier is used for data encoding. OFDM deliberately spaces subcarriers so they are orthogonal. At each peak of a subcarrier, the amplitudes of the other two subcarriers are 0.



Channel Bonding

- The number of valid subcarriers available for transmission depends on the channel width. A larger channel width supports more valid subcarriers. Bonding two or more adjacent non-overlapping channels increases the channel width and the data rate.
- By default, an AP operates on a 20 MHz channel. Two adjacent 20 MHz channels can be bonded into a 40 MHz channel. Similarly, channels can be bonded into 40 MHz+, 40 MHz-, 80 MHz, 80+80 MHz, and 160 MHz channels. On the 6 GHz frequency band, two 160 MHz channels can be bonded into a 320 MHz channel.



Channel distribution on the 5 GHz frequency band

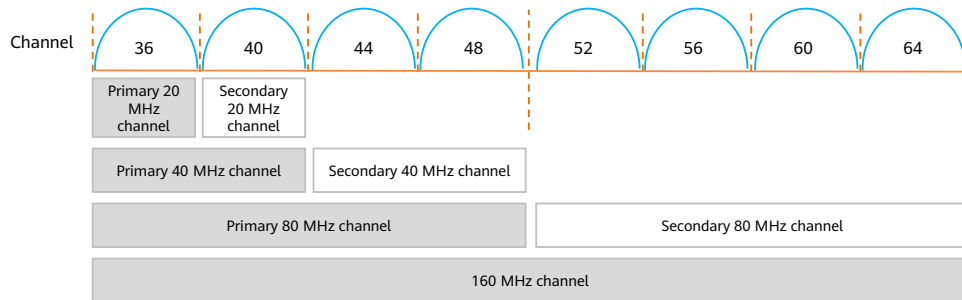
- Channel types:
 - 40 MHz+ and 40 MHz-: Two adjacent 20 MHz channels that do not interfere with each other are bonded into a 40 MHz channel. One of the two 20 MHz channels is the primary channel, and the other is the secondary channel. If the center frequency of the primary channel is higher than that of the secondary channel, the operating channel width is 40 MHz-. Otherwise, the operating channel width is 40 MHz+. For example, 20 MHz channels 36 and 40 are bonded into a 40 MHz channel. If channel 40 is set as the primary channel, the operating channel width is 40 MHz-; if channel 36 is set as the primary channel, the operating channel width is 40 MHz+. In most cases, it is not recommended that the channel width of 40 MHz be configured on the 2.4 GHz frequency band because only one non-overlapping 40 MHz channel can be created on this frequency band. For example, after channels 1 and 5 are bonded into a 40 MHz channel (channel 1 overlaps with channels 2, 3, and 4), bonding the remaining channels must avoid channels 1 to 8 because channel 5 overlaps with channels 6, 7, and 8. Therefore, the remaining channels cannot be bundled into another 40 MHz channel.
 - 80 MHz: Two contiguous 40 MHz channels can be bonded into an 80 MHz channel. Any of the four 20 MHz channels in the 80 MHz channel can be selected as the primary channel. For example, channel 36, 40, 44, and 48

can be bonded into an 80 MHz channel.

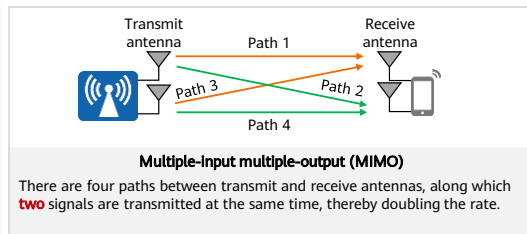
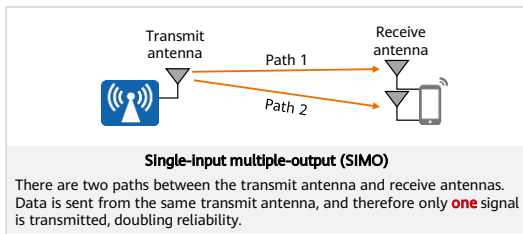
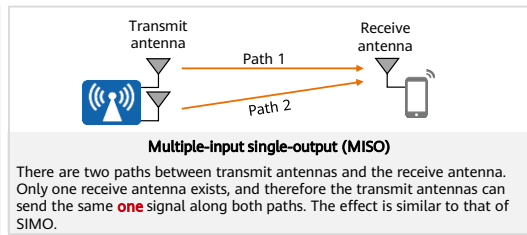
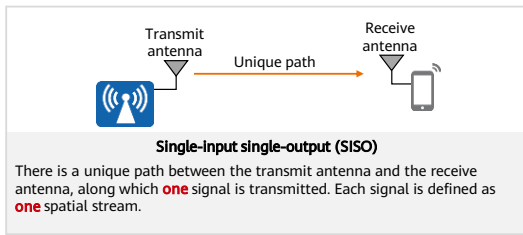
- 80+80 MHz: Two non-contiguous 80 MHz channels can be bonded into an 80+80 MHz channel. For example, channels 36, 40, 44, 48, 100, 104, 108, and 112 can be bonded into an 80+80 MHz channel.
- 160 MHz: Two contiguous 80 MHz channels can be bonded into a 160 MHz channel. Any of the eight 20 MHz channels can be selected as the primary channel. For example, channels 36, 40, 44, 48, 52, 56, 60, and 64 can be bonded into a 160 MHz channel.
- 320 MHz: Two neighboring 160 MHz channels can be bonded into a 320 MHz channel. In a 320 MHz channel, one 20 MHz channel must be configured as the primary channel. The remaining 20 MHz channels in the 160 MHz channel containing the primary channel are called the secondary 20 MHz channels. The 40 MHz, 80 MHz, and 160 MHz channels that do not contain the primary channel are called secondary 40 MHz, secondary 80 MHz, and secondary 160 MHz channels, respectively. The 160 MHz channel that does not contain the primary channel is called the secondary 160 MHz channel. At most one 320 MHz channel (with fixed channel width of 240 MHz available) is supported on the 5 GHz frequency band, and three 320 MHz channels are supported on the 6 GHz frequency band.

Channel Bonding Implementation

- According to the 802.11 standard, two channels are bonded: one is the primary channel and the other is the secondary channel. The key difference between them is that management packets are transmitted on the primary channel, not on the secondary channel. The following figure shows channel bonding for some 5 GHz channels.



SISO, MISO, SIMO, and MIMO



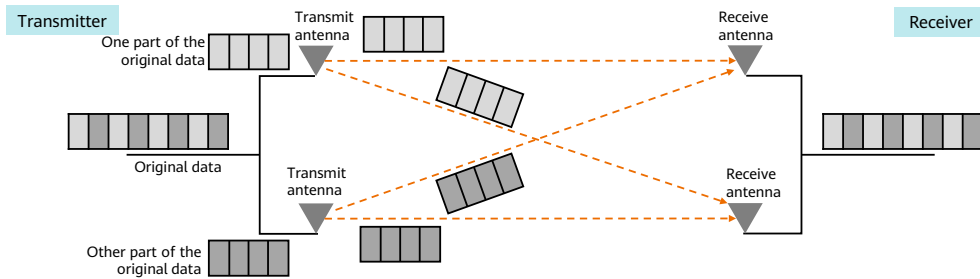
- SISO
 - In SISO, there is a unique path between the transmit antenna and the receive antenna. Apparently, such transmission is unreliable and rate limited. To address this issue, we add more antennas on the receive end (STA) so that two or more signals can be received concurrently, that is, single-input multiple-output (SIMO).
- SIMO
 - There are two paths between the transmit antenna and receive antennas. Data is sent from the same transmit antenna, and therefore only one signal is transmitted, doubling reliability. This mode is also known as receive diversity.
- MISO
 - There are two paths between transmit antennas and the receive antenna. Only one receive antenna exists, and therefore the transmit antennas can send the same one signal along the two paths. The effect is similar to that of SIMO. This mode is also known as transmit diversity.
- MIMO
 - MIMO technology enables multiple antennas to send and receive spatial streams (multiple signals) simultaneously and to differentiate the signals

sent to or received from different spatial orientations. By leveraging technologies such as spatial multiplexing, MIMO boosts system capacity, coverage, and so on, without increasing the occupied bandwidth.

- MIMO technology was officially introduced in 802.11n (Wi-Fi 4) and has been widely used ever since. In this course, we will focus on MIMO technology.

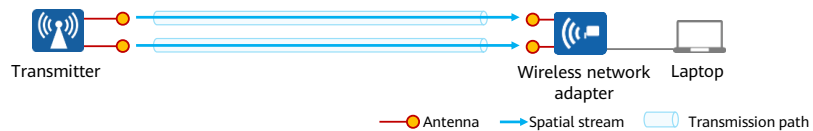
Key MIMO Technology: Spatial Multiplexing

- Spatial multiplexing divides data to be transmitted into multiple data streams, and then encodes and modulates each stream on separate antennas. This increases the transmission rate.
- Working process: Each antenna works independently, with each antenna acting as its own channel. After receiving the signals, the receiver demodulates and decodes them, and then combines the resulting data streams to restore the original signals.



Spatial Stream

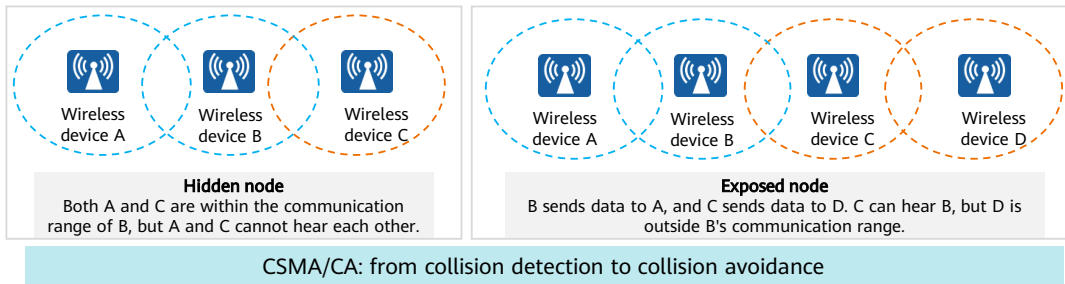
- A radio system can send multiple radio signals simultaneously. Each of these signals is called a spatial stream. Spatial streams are transmitted from the transmitter's antennas and arrive at the receiver through different paths. Generally, one spatial stream can be created between one transmit antenna and one receive antenna.
- A MIMO system is denoted as $M \times N$ MIMO, where M and N indicate the number of transmit antennas and receive antennas, respectively. The number of spatial streams in MIMO is generally less than or equal to the number of transmit or receive antennas. If the number of receive antennas is different from that of transmit antennas, the number of spatial streams is less than or equal to the smaller number of the antennas on the transmitter or receiver end. For example, a 4×4 MIMO system can transmit up to four spatial streams, whereas a 3×2 MIMO system can transmit up to two spatial streams.
- According to 802.11ac and 802.11ax, a radio supports a maximum of eight spatial streams. Therefore, a transmitter with 12 antennas can use at most eight spatial streams.



Background of CSMA/CA

- In early wired Ethernet, carrier sense multiple access with collision detection (CSMA/CD) was used. When networks moved to wireless, CSMA/CD became impractical for the following reasons:

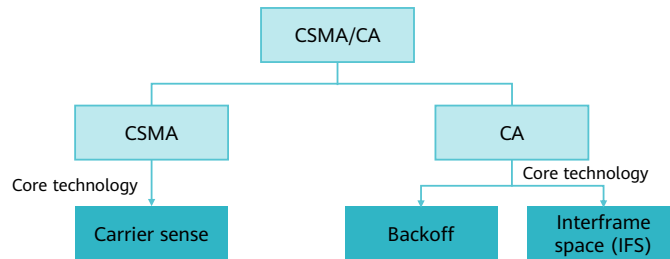
- Collision detection (CD) cannot be implemented. A wireless device typically has a single radio and cannot transmit and listen on the same frequency simultaneously.
- Carrier sense is limited and unreliable. Wireless networks suffer from hidden-node and exposed-node issues.



- Hidden node issue: When wireless device A sends data, it first listens to the channel and finds it idle. Therefore, it starts to send data. At the same time, wireless device C also needs to send data to wireless device B. Because C cannot hear A, it also senses the channel as idle and starts to send data to B. As a result, the signals from A and C collide at B, preventing B from receiving data from either of them.
- Exposed node issue: When C listens to the channel and detects B transmitting to A, C considers the channel as busy and defers its own transmission to D. However, C's transmission to D does not interfere with B's transmission to A, because D is far enough from B that they do not interfere with each other.
- CSMA/CA vs CSMA/CD
 - CSMA/CA: used in wireless networks and uses the collision avoidance mechanism, because reliable collision detection in wireless environments is not as feasible as that in wired environments.
 - CSMA/CD: used in wired networks and uses the collision detection mechanism, allowing a device to stop transmitting immediately after detecting a collision.

Overview of CSMA/CA

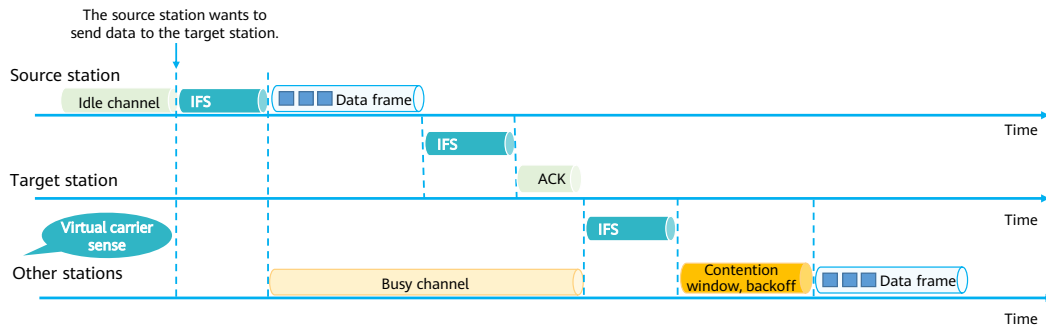
- CSMA/CA is used to determine whether a channel is in use before a wireless device transmits data. If the channel is idle, the device may send data; if the channel is busy, the device waits for a period before attempting to transmit data.
 - CSMA: Before sending data, a wireless device in a broadcast domain listens to the channel to verify it is idle, reducing signal collisions.
 - CA: It aims to minimize the probability of signal collisions.



- Carrier Sense: Before sending data, a device listens to the channel to check whether it is idle. If the channel is busy, the device waits and retries later.
- CA: To avoid collisions, a device waits for a random backoff time before sending data. The backoff time is randomly generated to reduce the possibility of simultaneous transmissions.
- IFS: Before sending data, a device must wait for a specific interval, known as IFS. Frames with different priorities require different waiting times. Frames with a higher priority can gain access to the channel more quickly.

Implementation of CSMA/CA

- On a wireless network, the source station first checks whether the channel is idle. If the channel is idle, the source station waits for the required IFS, transmits the first data frame, and then waits for an acknowledgment. When other stations detect that data frame, they mark the channel as busy and defer their own transmissions.



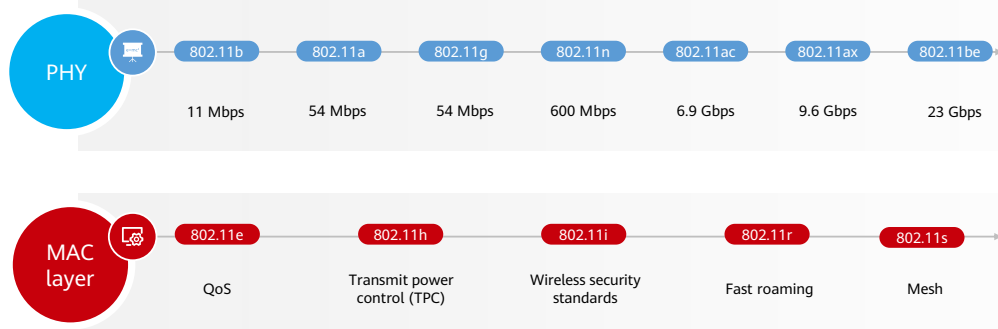
- Acknowledgment (ACK):** After the transmitter sends a data frame, the receiver returns an ACK frame to confirm successful data reception. If the transmitter does not receive the ACK frame, it considers the transmission as failed and retransmits the data.

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IEEE 802.11 Standards

- IEEE 802.11 is a set of international standards developed by the IEEE for WLANs. It defines the MAC-layer and PHY specifications that enable devices to communicate over frequency bands such as 2.4 GHz, 5 GHz, and 6 GHz without using physical cables.



- PHY standards define the operating frequency bands, modulation and coding schemes (MCSs), and the maximum speeds supported by wireless protocols.
- MAC-layer standards define common operations at the MAC layer of wireless networks, such as QoS, security, and roaming. The MAC-layer protocols are not covered in this course. They will be presented in subsequent courses as single-function modules.

IEEE 802.11 Standards and Wi-Fi Generations

Standard		Released In	Frequency Band	Modulation Technology	Number of Spatial Streams	Channel Width (MHz)	Theoretical Rate
-	802.11b	1999	2.4 GHz	-	-	22	11 Mbps
-	802.11a	1999	5 GHz	-	-	20	54 Mbps
-	802.11g	2003	2.4 GHz	64-QAM	-	20	54 Mbps
Wi-Fi 4	802.11n	2009	2.4 GHz or 5 GHz	64-QAM	4	20 and 40	2.4 GHz: 450 Mbps 5 GHz: 600 Mbps
Wi-Fi 5	802.11ac Wave 1	2013	5 GHz	64-QAM	4	20 and 40	3.47 Gbps
	802.11ac Wave 2	2015	5 GHz	256-QAM	8	20, 40, 80, 160, and 80+80	6.9 Gbps
Wi-Fi 6	802.11ax	2019	2.4 GHz or 5 GHz	1024-QAM	8	20, 40, 80, 160, and 80+80	2.4 GHz: 1.15 Gbps 5 GHz: 9.6 Gbps
Wi-Fi 7	802.11be	2024	2.4 GHz, 5 GHz, or 6 GHz	4096-QAM	8	20, 40, 80, 160, 80+80, 160+160, 160+80, and 320	2.4 GHz: 1.37 Gbps 5 GHz: 11.52 Gbps 6 GHz: 23.05 Gbps

Theoretical Rate Calculation

Device rate = Number of spatial streams x Number of encoded bits per subcarrier x Coding rate x Number of valid subcarriers / (Symbol length + Short GI or GI)

Number of Spatial Streams	
802.11ax	802.11be
8	8

Modulation Scheme			
802.11 Standard	Modulation Scheme	Bits/ Symbol	Coding Rate
802.11ax	1024-QAM	10	5/6
802.11be	4096-QAM	12	5/6

Symbol Length and GI		
	802.11ax	802.11be
Symbol length	12.8 μs	12.8 μs
Short GI	/	/
GI	0.8 μs	0.8 μs

Number of Valid Subcarriers		
	802.11ax	802.11be
160 MHz	1960	1960
320 MHz	/	1960 × 2

6 GHz frequency band

Wi-Fi 7: $(8 \times 12 \text{ bits/subcarrier} \times 5/6 \times 1960 \times 2) / (12.8 + 0.8) \mu\text{s} = 23058 \text{ Mbps}$

5 GHz frequency band

Wi-Fi 6: $(8 \times 10 \text{ bits/subcarrier} \times 5/6 \times 1960) / (12.8 + 0.8) \mu\text{s} = 9607 \text{ Mbps}$
Wi-Fi 7: $(8 \times 12 \text{ bits/subcarrier} \times 5/6 \times 1960) / (12.8 + 0.8) \mu\text{s} = 11529 \text{ Mbps}$

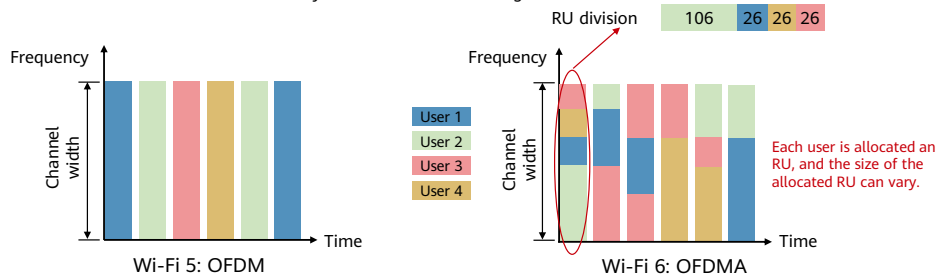
- The parameters in the calculation formula are described as follows:
 - Number of spatial streams: It can be simply understood as the number of antennas of an AP. A larger number of antennas indicate higher throughput for the entire system. Similar to lanes on a highway, an 8-lane expressway carries more traffic than a 4-lane one.
 - Number of bits encoded per subcarrier: The number of bits transmitted per subcarrier varies according to the modulation scheme. For example, in 64-QAM, each subcarrier transmits 6 bits.
 - Coding rate: Theoretical transmission assumes lossless data transmission based on the modulation scheme. During actual transmission, however, transmission requires error correction codes to be added, achieving high reliability through redundancy. The coding rate is the ratio of the actually transmitted data code with the error correction code excluded to the theoretical transmission value.
 - Number of valid subcarriers: A subcarrier is a frequency-domain carrier that carries one symbol. The number of subcarriers varies with channel width. OFDM technology splits a high-speed data stream into multiple low-speed, orthogonal subcarriers for transmission.
 - Symbol and GI: A symbol is a time-domain signal. A guard interval (GI) is used to separate adjacent symbols to avoid inter-symbol interference. This

is similar to truck transportation. Each truck is equivalent to a symbol. There must be a time interval between the two trucks departing from the same station. Otherwise, the two trucks may collide. The GI varies by Wi-Fi standard. Generally, higher transmission speeds require a larger GI. For example, in the truck analogy, the time interval between two 100 km/h trucks running on the same lane is larger than that of two 50 km/h trucks.

- Theoretical rate, achievable rate, and actual access rate:
 - The theoretical rate refers to the highest transmission rate calculable from a protocol standard. For example, the theoretical rate of 802.11ac Wave 2 can reach 6.9 Gbps.
 - The achievable rate refers to the highest rate that can be achieved by a vendor's product based on a protocol standard. For example, a Wi-Fi 7 product may support only four spatial streams, which is less than the specifications defined in the standard. In this case, the achievable rate is less than or equal to the theoretical rate.
 - The actual access rate differs greatly from the achievable rate due to factors such as the transmission distance, interference, and concurrent user access. Physical obstacles (walls, blocking or reflective materials, etc.) and transmission distances can weaken the signal and lower the actual access rate. Therefore, the actual access rate is lower than the achievable rate.

Wi-Fi 6 Technology: OFDMA

- Before Wi-Fi 6, OFDM technology is used for data transmission, and users are distinguished based on time segments. In each time segment, one user occupies all subcarriers to send a complete data packet.
- Wi-Fi 6 introduces orthogonal frequency division multiple access (OFDMA). It allows multiple users to reuse channel resources by allocating subcarriers to different users and adding multiple access to the OFDM system.
- Working mode: OFDMA divides the available bandwidth into multiple resource units (RUs), and users are distinguished by their allocated RUs. In this mode, user data is carried on each RU; therefore, from the perspective of total time-frequency resources, multiple users can transmit data simultaneously within the same time segment.

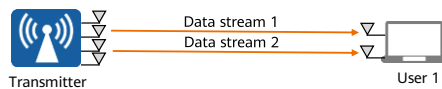


- This means that each time data is sent, one user occupies the entire channel regardless of the user data amount. Let's imagine Wi-Fi communication as an express delivery service, where information represents the goods to be transported to the receiver. In OFDM, a truck delivers one package per trip, regardless of its size. As a consequence, some of the space in the truck is usually wasted. How can we utilize this idle space? OFDMA introduced in Wi-Fi 6 addresses this efficiency issue by dividing a channel into multiple RUs. Different users occupy only their assigned RUs rather than the whole channel, allowing multiple users to transmit data simultaneously.
- RU: In 802.11ax, an RU is a relative concept, meaning that there are many RUs with different sizes. Wi-Fi 6 defines seven types of RUs: 26-tone RU, 52-tone RU, 106-tone RU, 242-tone RU, 484-tone RU, 996-tone RU, and 2×996-tone RU. The 484-tone RU is available only on 40 MHz, 80 MHz, and 160 MHz channels. The 996-tone RU is available only on 80 MHz and 160 MHz channels. The 2×996-tone RU is available only on 160 MHz channels. In terms of *XX*-tone RU, *XX* represents the number of tones included in an RU. For example, a 26-tone RU indicates that the RU includes 26 tones.
- OFDM working mechanism (left figure): Four users occupy channel resources separately in different time segments. In each time segment, a single user occupies all subcarriers for sending data packets.
- OFDMA working mode (right figure): Four users are differentiated by RU. User

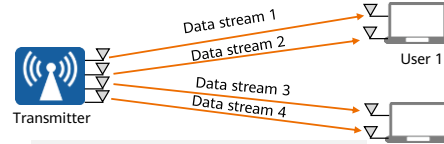
data is carried on each RU. Therefore, multiple users can transmit data simultaneously when the total time-frequency resources remain unchanged. For example, at a given moment, the channel resources can be allocated to users 1 to 4. User 2 uses a 106-tone RU, and users 1, 3, and 4 each use 26-tone RUs.

Wi-Fi 6 Technology: DL/UL MU-MIMO

- MIMO can be classified into single-user MIMO (SU-MIMO) and multi-user MIMO (MU-MIMO) by the number of users.
 - SU-MIMO: Multiple parallel spatial streams that occupy the same time-frequency resources are sent to a single user, increasing that user's throughput.
 - MU-MIMO: Multiple parallel spatial streams that occupy the same time-frequency resources are sent to different users, increasing the throughput for multiple users.
- MU-MIMO was introduced in 802.11ac, but only downlink (DL) 4x4 MU-MIMO is supported. In 802.11ax, the number of MU-MIMO is further increased, and DL 8x8 MU-MIMO is supported.
- Uplink (UL) MU-MIMO is an important feature introduced in 802.11ax. It leverages multi-antenna transmission and reception to transmit data on multiple spatial streams over the same channel, improving multi-user efficiency and greatly reducing application latency.



SU-MIMO
Sends data to a single user at the same time.



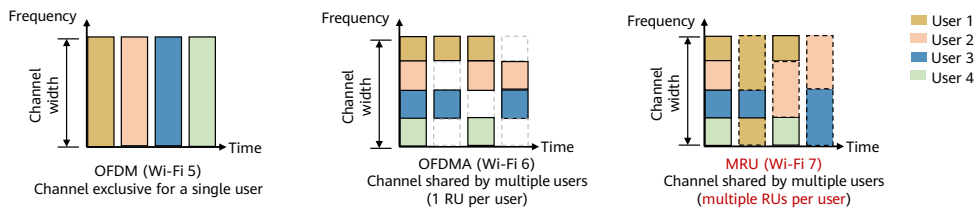
MU-MIMO
Sends data to multiple users at the same time.

- Time-frequency resources combine time domain resources and frequency domain resources.
- MU-MIMO transmits independent data streams with the same bandwidth. Unlike OFDMA, MU-MIMO allows all users to use the entire bandwidth simultaneously, achieving multiplexing gains. As shown in the figure on the left, a transmitter with four antennas (spatial streams) sending data to a receiver with two antennas cannot fully utilize all the four spatial streams because the receiver is limited to two spatial streams. In contrast, MU-MIMO allows a transmitter to send data to multiple users simultaneously. As shown in the figure on the right, the same four-antenna transmitter can send data to two receivers (each with two antennas) simultaneously. This fully utilizes the transmitter's four spatial streams and significantly improves multi-user throughput. Theoretically, the rate can be twice that of SU-MIMO.
- Although 802.11ax allows OFDMA and MU-MIMO to work at the same time, do not confuse OFDMA with MU-MIMO. OFDMA allows multiple users to subdivide channels (subchannels) to improve the concurrency efficiency. MU-MIMO allows multiple users to use different spatial streams to increase the throughput.
- OFDMA:
 - Higher efficiency
 - Reduced latency

- Applicable to low-bandwidth applications
 - Applicable to small-packet transmission
- MU-MIMO
 - Higher capacity
 - Higher per-user rate (in multi-user scenarios)
 - Applicable to high-bandwidth applications
 - Applicable to large-packet transmission

Wi-Fi 7 Technology: MRU

- OFDMA in Wi-Fi 6 allocates different types of RUs to different users, with each user assigned only one RU per transmission interval. When the number of users exceeds the available RUs, some RUs may remain idle. Wi-Fi 7 introduces multi-RU (MRU) technology, which allows a single user to occupy multiple RUs simultaneously and enables combining RUs of different sizes.
- MRU features:
 - Resources allocated to a single user are a combination of multiple RUs, which can be contiguous or non-contiguous.
 - MRU aims to optimize single-user performance, improve anti-interference capability, and increase throughput.

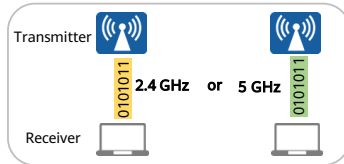


- Features of OFDMA:
 - Each user is allocated one RU.
 - The main objective of OFDMA is to implement multi-user parallelism, reduce latency, and improve efficiency.

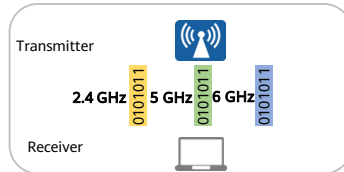
Wi-Fi 7 Technology: Multi-Link

- In Wi-Fi 6 and earlier standards, a transmitter and a receiver can establish a link only on one frequency band at a time even though they both support multiple frequency bands. Wi-Fi 7 introduces multi-link operation (MLO), which allows a transmitter and a receiver to establish multiple links simultaneously on different frequency bands for data communication, further improving throughput.

Wi-Fi 6 and earlier standards
Single-link: 2.4 GHz or 5 GHz



Wi-Fi 7
Multi-link: 2.4 GHz + 5 GHz + 6 GHz



Mode 1: Higher performance



- Multi-link load balancing, improving the peak throughput of a single user

Mode 2: Higher reliability

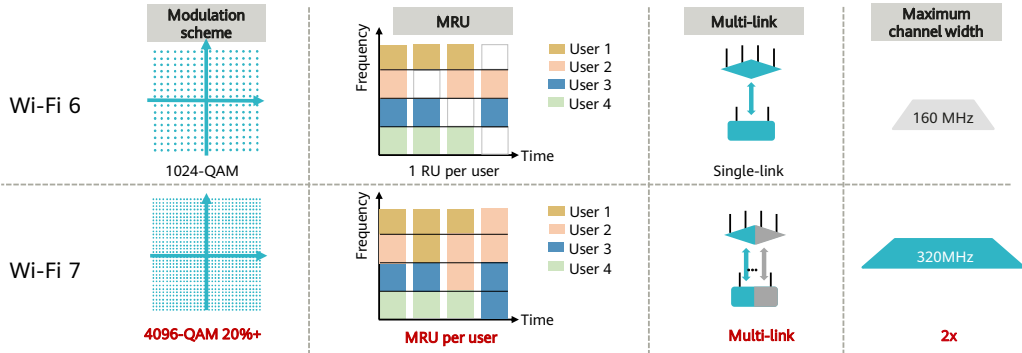


- Multi-fed and selective receiving, improving link reliability

- Multi-link operation (MLO) enables multiple links to be established between the transmitter and receiver for data communication. MLO allows data flows to be sent to multi-link devices (MLDs) on different frequency bands. The following MLO modes are supported:
 - Mode 1: Higher performance. Multiple links are used for load balancing, improving the peak single-user throughput.
 - Mode 2: Higher reliability. Multiple links are used for multi-fed and selective receiving, improving link reliability.

Wi-Fi 7 VS Wi-Fi 6

- Compared with Wi-Fi 6, Wi-Fi 7 has significantly improved performance in the following aspects:
 - Higher-order modulation scheme. Wi-Fi 7 introduces 4096-QAM, so that each modulation symbol carries 12-bit data.
 - Higher spectral efficiency. Technologies such as MRU and MLO are introduced to effectively improve spectrum utilization and reduce interference.
 - More abundant frequency resources. Wi-Fi 7 supports the 6 GHz frequency band, increasing the channel width to 320 MHz and thereby doubling the data transmission capacity.



Quiz

1. (Single-answer question) Which of the following 802.11 protocols supports the 6 GHz frequency band? ()
- A. 802.11a
 - B. 802.11ac
 - C. 802.11ax
 - D. 802.11be

1. D

Summary

- This course covers the basic concepts and key technologies of WLAN, including the modulation scheme, number of carriers, channel width, number of spatial streams, and CSMA/CA mechanism. It also describes the development of 802.11 standards, the PHY technologies used across Wi-Fi generations, and the differences between the 802.11 standards.
- Upon completion of this course, you will have a basic understanding of WLAN fundamentals and be familiar with common WLAN technologies, such as OFDM, MIMO, and CSMA/CA.

Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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WLAN Networking Architecture



Foreword

- Wireless local area networks (WLANs) overcome the disadvantage of poor mobility on wired networks. With the development of Wi-Fi 7 technology, the bandwidth performance gap between wireless and wired networks is narrowing down. WLANs featuring flexible networking can adapt to various complex and changeable application scenarios.
- This course introduces the basic WLAN networking architectures and application scenarios, including WAC + Fit AP, Fat AP, and leader AP networking modes.

Objectives

- On completion of this lesson, you will be able to:
 - Describe basic WLAN concepts.
 - Describe WLAN networking modes.
 - Describe WLAN networking application scenarios.

Contents

- 1. WLAN Networking Overview**
2. WLAN Networking Architecture
3. WLAN Networking Application Scenarios

Wireless Network Identifier: SSID

- When you search for available wireless networks on a phone or laptop, the displayed WLAN names are service set identifiers (SSIDs). An SSID is the most basic identifier of a WLAN and uniquely distinguishes it from others.
- The SSID provides the following functions:
 - Network identification: The Wi-Fi list shown on a STA displays the SSIDs broadcast by nearby WLANs. Users identify and select the WLAN to join by selecting its SSID.
 - Logical isolation: Different SSIDs can be configured on the same group of wireless devices to create logically independent networks.
 - Connection basis: When a user selects an SSID and clicks **Connect**, the STA sends an association request to the specific WLAN.
 - Network management: Each SSID is associated with a comprehensive set of network configurations.



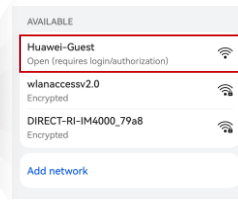
AP



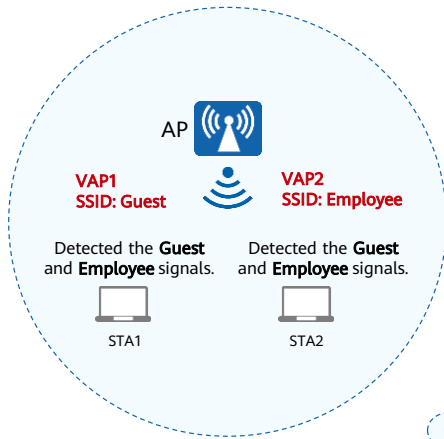
SSID: **Huawei-Guest**



STA



VAP

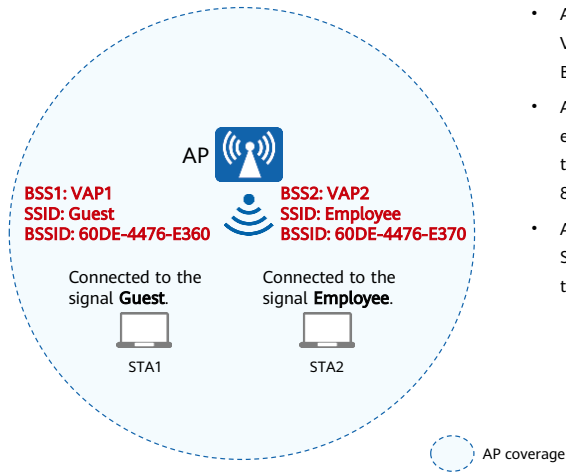


- An AP can host multiple virtual access points (VAPs), with each VAP supporting the configuration of one SSID name. This allows a single AP to provide multiple SSIDs—a capability commonly called multi-SSID.

AP coverage

- With the development of WLAN services, a WLAN provided by a single AP becomes unable to meet the increasing service requirements. However, if multiple WLANs are required in a space, deploying an AP for each WLAN is expensive and requires a lot of channel resources. To address this, SSID management allows you to create multiple VAPs on an AP to provide wireless access services for different user groups.

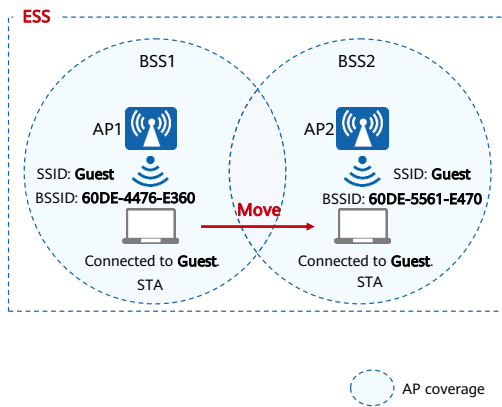
BSS and BSSID



- A basic service set (BSS) is the coverage area provided by a VAP and is also the basic service unit of a WLAN. Typically, a BSS consists of a VAP and STAs connected to the VAP.
- A basic service set identifier (BSSID) is the MAC address of each VAP. Each BSS requires a unique BSSID (represented by the VAP's MAC address), so the AP can correctly forward 802.11 packets.
- As shown in the figure, VAP1 and VAP2 are created on an AP. STA1 associates with the SSID **Guest**, and STA2 associates with the SSID **Employee**.
 - VAP1 and STA1 form a BSS, with the BSSID being **60DE-4476-E360** and the SSID being **Guest**.
 - VAP2 and STA2 form another BSS, with the BSSID being **60DE-4476-E370** and the SSID being **Employee**.

- Multiple VAPs can be created on an AP. Each VAP together with its associated STAs form a BSS, which is identified by a unique BSSID. This means a one-to-one mapping between a VAP and a BSS.
- SSIDs across different BSSs can be identical or different. As such, the SSID does not equal BSSID. Think of the BSSID as the BSS's ID card and the SSID as its name —names may repeat, but the ID card is always unique.

ESS/ESSID



- The effective coverage radius of a BSS is typically 10–15 m, which corresponds to the coverage area of a single AP. To cover a larger area, multiple BSSs can be deployed. To prevent users from perceiving BSS transitions, each BSS can use the same SSID, allowing users to experience a single WLAN as they move.
- This method of extending BSS coverage is called an Extended Service Set (ESS), which comprises multiple BSSs that share the same SSID.
- The same SSID used by all BSSs serves as the ESS identity, known as the extended service set identifier (ESSID), and advertises a continuous WLAN to STAs. In most cases, the SSID is actually the ESSID.

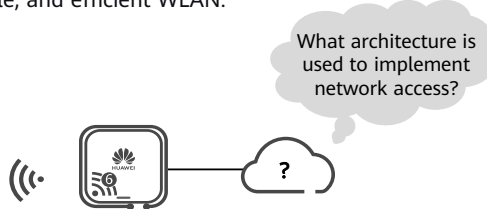
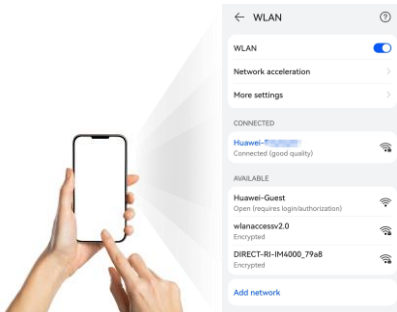
- As shown in the figure, AP1 and the associated STAs form a BSS, with the BSSID **60DE-4476-E360**. AP2 and the associated STAs form another BSS, with the BSSID **60DE-5561-E470**. The two BSSs belong to the same ESS because the SSIDs of the two BSSs are both **Guest**.

Basic WLAN Concepts

Concept	Full Name	Description
SSID	Service set identifier	A unique identifier that identifies a WLAN.
VAP	Virtual access point	A functional entity virtualized on an AP. You can create different VAPs on an AP to provide the wireless access service for different user groups.
BSS	Basic service set	The basic service unit of a WLAN, consisting of an AP and multiple STAs.
BSSID	Basic service set identifier	An identifier that distinguishes VAPs on the same AP at the link layer. It is also used to identify BSSs in an ESS.
ESS	Extended service set	A set of BSSs that share the same SSID, addressing the issue of limited BSS coverage.
ESSID	Extended service set identifier	An identifier of one or a group of WLANs. It is the same as the SSID.

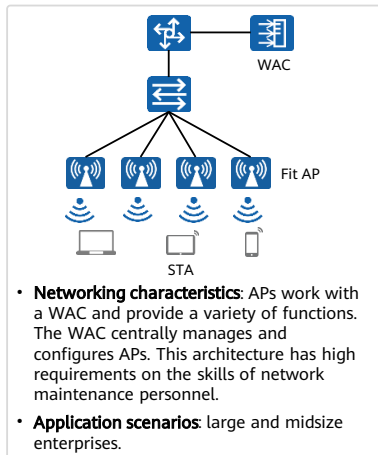
How Does a STA Access the Internet?

- In everyday use, a STA (for example, a smartphone or laptop) scans for available WLANs and selects an SSID on its WLAN interface. After identity authentication (for example, by entering the correct PSK), the STA connects to the WLAN successfully. In modern, complex WLAN deployments, however, an SSID by itself is insufficient to ensure a secure, stable, and efficient WLAN.

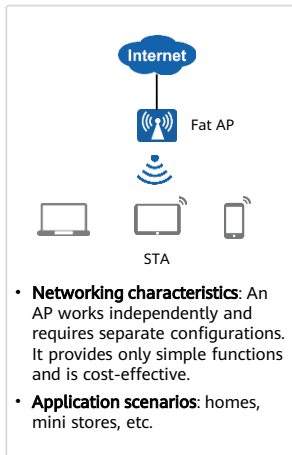


Basic WLAN Networking Architectures

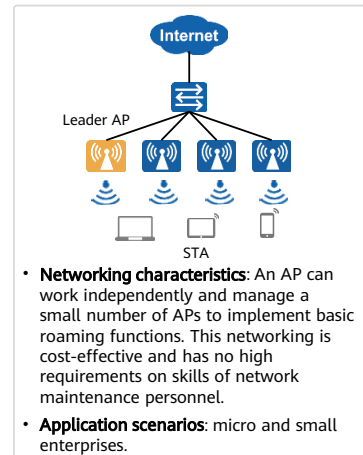
WAC + Fit AP



Fat AP



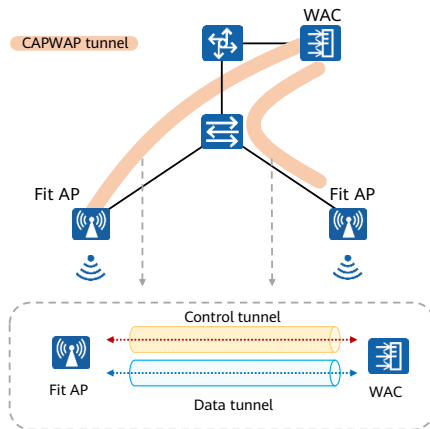
Leader AP



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WAC + Fit AP Networking

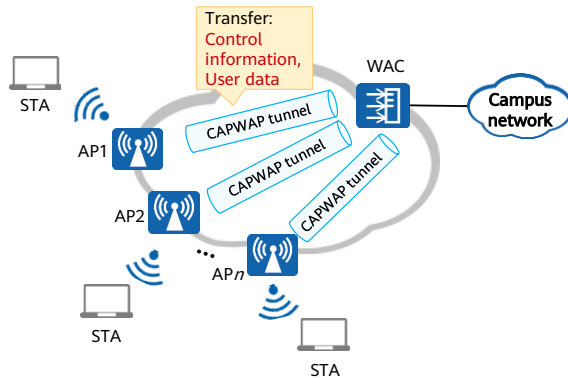


- A WAC provides WLAN access control, data forwarding, AP configuration management, roaming management, and security control.
- Fit APs encrypt and decrypt 802.11 packets, provide 802.11 physical layer (PHY) functions, collect air interface statistics, and are managed by the WAC.
- The WAC communicates with Fit APs using CAPWAP.
- Compared with the Fat AP architecture, the WAC + Fit AP architecture has the following advantages:
 - Easier configuration and deployment in multi-AP scenarios
 - Higher security
 - Easier upgrade and expansion
- This architecture is applicable to large and midsize campus networks and scenarios where enterprises need to centrally manage WLANs.

- The WAC communicates with Fit APs using CAPWAP. The WAC and APs establish CAPWAP tunnels with each other for communication. The CAPWAP tunnels allow for Datagram Transport Layer Security (DTLS) encryption, so that transmitted packets are more secure.
- CAPWAP tunnels include control tunnels and data tunnels.
 - Control tunnels are mainly used to transmit control packets (also called management packets, which are used by the WAC to manage and control APs).
 - Data tunnels are mainly used to transmit data packets (used when the tunnel forwarding mode is used).
- The WAC + Fit AP architecture has the following advantages:
 - Easier configuration and deployment: The WAC centrally configures and manages the wireless network so that each AP does not need to be configured separately. In addition, the channels and power of APs on the entire network are automatically adjusted, eliminating the need for manual adjustment.
 - Security: The WAC provides most security capabilities. Software upgrades and security configurations need to be performed only on the WAC, facilitating quick global security settings.

- Upgrade and extension: The centralized management mode of this architecture enables APs managed by the same WAC to run the same software version. When an upgrade is required, the WAC obtains the new software package or patch and then upgrades the APs in a unified manner.

CAPWAP



What is CAPWAP?

- **Control And Provisioning of Wireless Access Points (CAPWAP)**: defines how to manage and configure services of APs. That is, a WAC manages and controls APs in a centralized manner through CAPWAP tunnels.

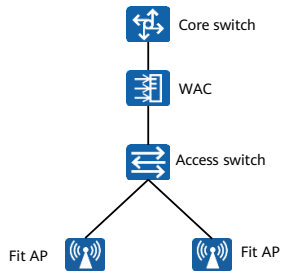
CAPWAP tunnel functions

- Maintaining the connectivity state between the WAC and APs.
- Allowing the WAC to manage APs and deliver configurations to them.
- Forwarding user data encapsulated by APs to the WAC, which then forwards the data to the upper-layer network (This applies when the tunnel forwarding mode is used).

- To meet large-scale networking requirements, multiple APs need to be managed in a unified manner, which is not supported by the traditional WLAN architecture. As such, the Internet Engineering Task Force (IETF) set up the CAPWAP Working Group and developed the CAPWAP protocol.
- CAPWAP is an application-layer protocol based on UDP. CAPWAP packets are classified into control packets (UDP port 5246) that are used to manage APs and data packets (UDP port 5247) that are used to forward user service data.

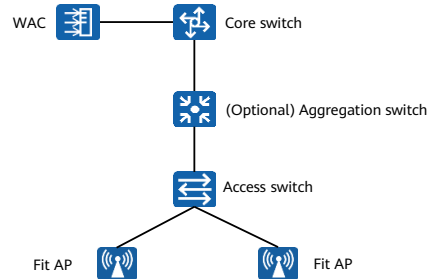
Networking Modes: In-Path and Off-Path Networking

In-path networking



- **Description:** In the in-path networking, the WAC provides functions of both a WAC and an aggregation switch. Data and management packets of APs are all forwarded and processed by the WAC.
- **Application:** new deployments of small and midsize centralized WLANs.

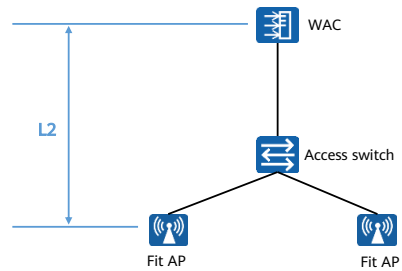
Off-path networking



- **Description:** The WAC is connected to the existing network in off-path mode and processes management services of APs. Data packets of APs can be directly sent to the upper-layer network or centrally forwarded by the WAC.
- **Application:** network reconstruction or deployment of new large and midsize campus networks.

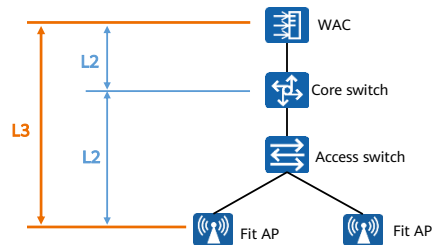
Networking Modes: Layer 2 and Layer 3 Networking

Layer 2 networking



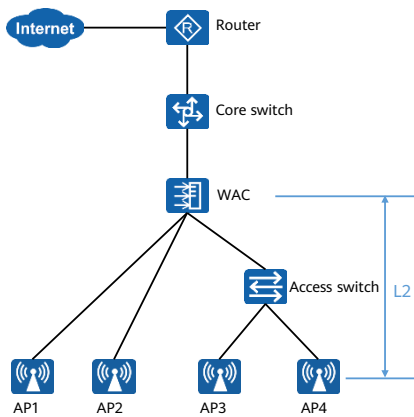
- **Description:** The WAC and Fit APs are located in the same broadcast domain. The APs can discover the WAC by broadcasting packets. The networking, configuration, and management are simple.
- **Application:** small-scale networks, such as small-scale enterprise networks. This networking does not apply to complex and refined WLAN networks of large enterprises.

Layer 3 networking



- **Description:** The WAC and Fit APs are located in different network segments. The intermediate network must ensure that the WAC and Fit APs are reachable to each other. Additional configurations are required to enable the Fit APs to discover the WAC. The networking is flexible and easy to expand.
- **Application:** large and midsize WLANs. For example, on a large campus network, APs are deployed in each office building to provide wireless coverage, and the WAC is deployed in the core equipment room for unified AP management and configuration. Layer 3 networking between the WAC and Fit APs is recommended in this scenario.

Layer 2 In-Path Networking



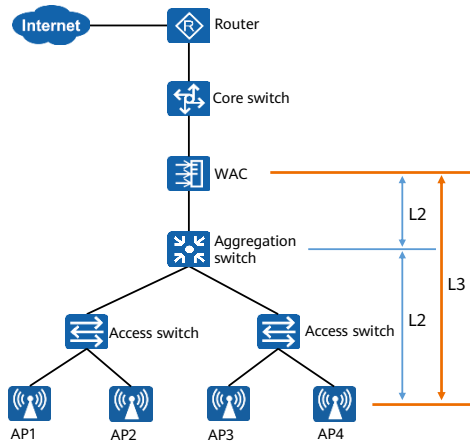
Networking description

- A WAC is directly connected to APs or Layer 2 access switches, and provides functions of both a WAC and an aggregation switch.
- The WAC and APs are in the same broadcast domain, and APs can discover the WAC by broadcasting packets. The network architecture is simple.
- In this networking mode, wireless traffic must pass through the WAC connected along the line. Therefore, the WAC may become a performance bottleneck on the live network.

Application scenarios and networking characteristics

- This networking mode applies to deployment of new small-scale WLANs, such as small enterprise and branch networks.
- The network architecture is simple and easy to maintain.

Layer 3 In-Path Networking



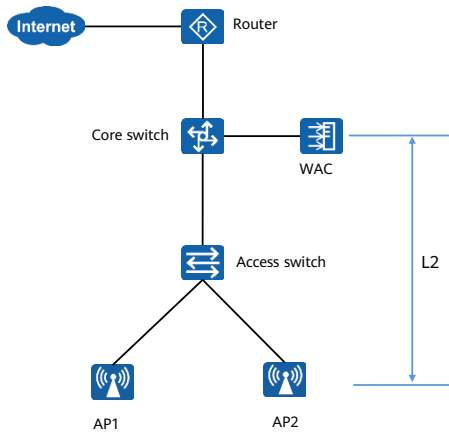
Networking description

- A WAC is connected to the network in in-path mode and located between the aggregation and core switches. Depending on the network scale, the WAC can be connected to multiple aggregation switches.
- The WAC communicates with APs at Layer 3 and the APs can obtain the WAC's IP address through DHCP or other methods.
- Compared with Layer 2 in-path networking, Layer 3 networking is applicable to larger-scale WLANs and facilitates scale-out. However, since all wireless traffic must pass through the WAC, the WAC performance may become a performance bottleneck on the live network.

Application scenarios and networking characteristics

- This networking mode applies to deployment of new small- and medium-scale WLANs, such as office networks of small and midsize enterprises.
- The network architecture is simple and easy to maintain, and facilitates scale-out.

Layer 2 Off-Path Networking



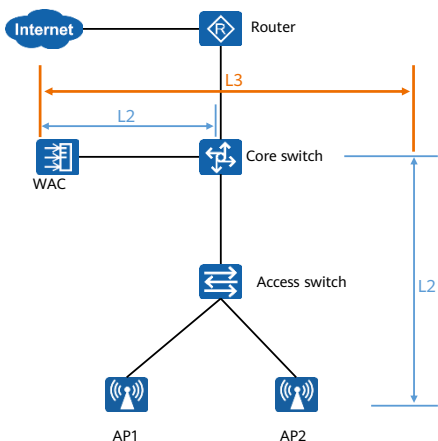
Networking description

- A WAC is connected to the core switch (or another network device) in off-path mode. The WAC and APs are located in the same broadcast domain and communicate with each other at Layer 2.
- APs can discover the WAC by broadcasting packets.
- If the direct forwarding mode is used, wireless traffic does not need to pass through the WAC. If the tunnel forwarding mode is used, all wireless traffic must be forwarded by the WAC.

Application scenarios and networking characteristics

- This networking is applicable to WLAN deployment for small and midsize enterprises and campuses.
- This networking mode is applicable to WLAN construction based on an existing network, and requires only a few changes to the existing network.

Layer 3 Off-Path Networking



Networking description

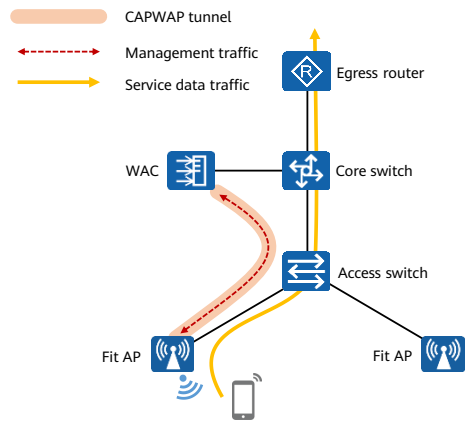
- A WAC is connected to the core switch (or another network device) in off-path mode. The WAC and APs are located in different broadcast domains and communicate with each other at Layer 3.
- Typically, APs obtain the WAC's IP address through DHCP or other methods.
- Compared with Layer 2 off-path networking, Layer 3 networking is applicable to larger-scale WLANs and supports flexible scale-out. The traffic forwarding characteristics are similar to those in the off-path Layer 2 networking, and are not described here.

Application scenarios and networking characteristics

- This networking is applicable to WLAN deployment for large and midsize enterprises and campuses.
- It is also suitable for WLAN reconstruction, and requires only a few changes to the existing network.

VLAN Planning on a WLAN

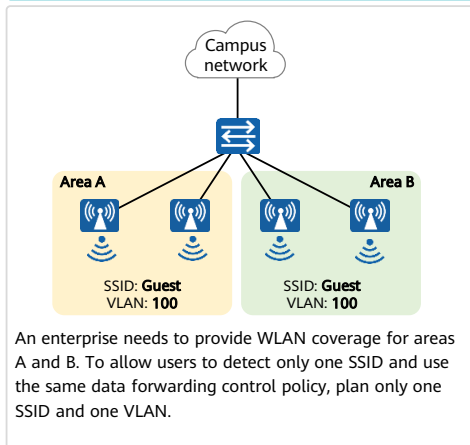
VLAN planning in the WAC + Fit AP networking



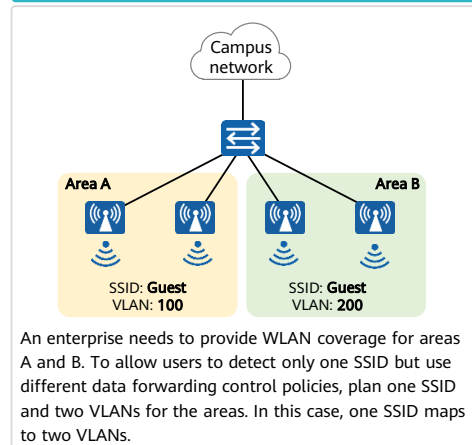
- Two main types of VLANs on a WLAN:
 - **Management VLAN:** transmits packets that are forwarded through CAPWAP tunnels, including management packets and service data packets forwarded through CAPWAP tunnels.
 - **Service VLAN:** transmits service data packets.
- Note the following rules when planning VLANs:
 - It is recommended that the management VLAN and service VLANs be different.
 - Service VLANs need to map to SSIDs based on service requirements.

Mapping Between Service VLANs and SSIDs

SSID:VLAN = 1:1

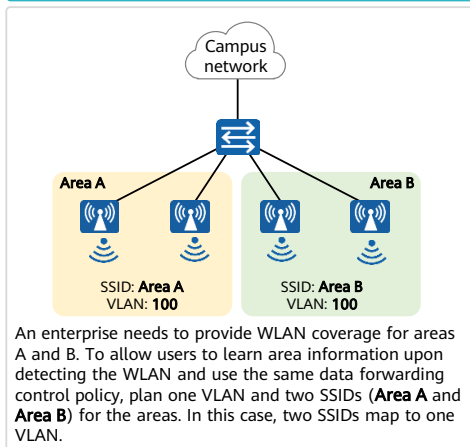


SSID:VLAN = 1:N

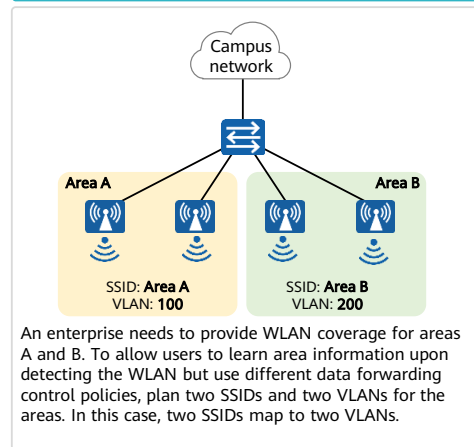


Mapping Between Service VLANs and SSIDs

SSID:VLAN = N:1



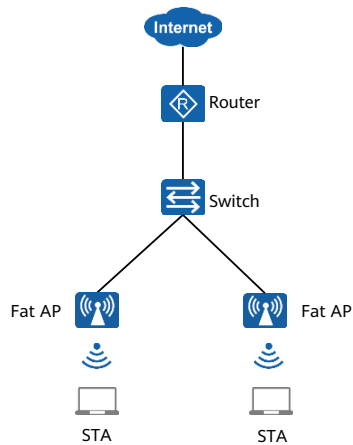
SSID:VLAN = N:M



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Fat AP Networking

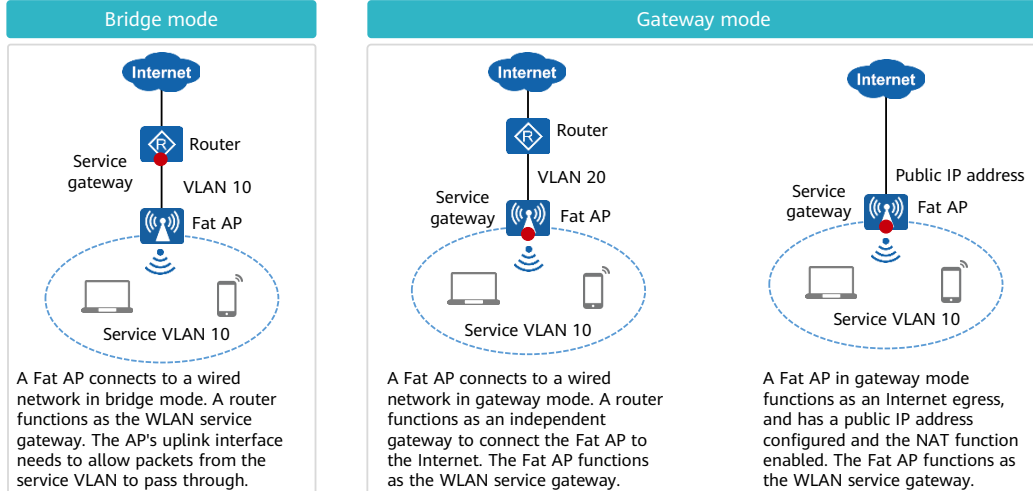


Networking description

- The Fat AP networking architecture is also called autonomous network architecture because it can implement functions such as wireless user access, service data encryption, and service data forwarding, without the need of a dedicated device for centralized control.
- Fat APs work independently and require no additional WAC. Therefore, this networking is easy to deploy and cost-effective, and is mainly applicable to homes and micro stores.
- Since each Fat AP works independently and no centralized control device is used, the Fat APs are difficult to manage and maintain. Therefore, the Fat AP architecture is not recommended for enterprises.

- In the Fat AP networking, APs work in Fat mode.

Common Fat AP Networking Modes

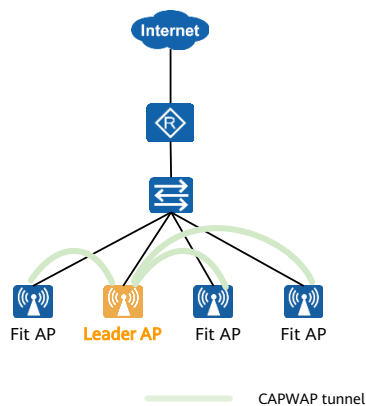


- When wired and wireless users are in the same VLAN and use the same address segment and gateway address, you can configure the Fat AP to work in bridge mode.
- When wired and wireless users are in different VLANs and managed separately, or the Fat AP functions as an Internet egress, you can configure the Fat AP to work in gateway mode.

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Leader AP Networking



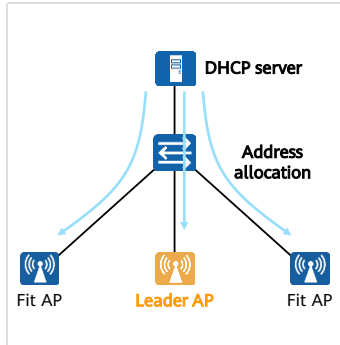
Networking description

- In the leader AP networking, one AP works in Fat mode, and the other APs work in Fit mode. The AP working in Fat mode is also called a leader AP.
- The leader AP and other Fit APs are generally located on the same Layer 2 network.
- Similar to a WAC, the leader AP uses the CAPWAP protocol to centrally manage and configure Fit APs. To enable all APs to provide the same wireless services, you only need to log in to the leader AP to perform configurations.
- This networking requires low network construction costs because no WAC is needed. In addition, this networking has certain scalability and security, and is applicable to small and micro enterprises.
- The wireless roaming function is supported in this networking.

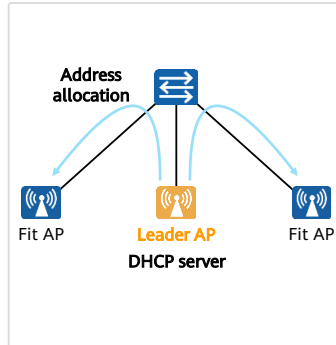
- The leader AP works like a WAC. A Fat AP enabled with the leader AP function can be deployed to build a WLAN together with multiple Fit APs. On this WLAN, the Fat AP manages and configures the Fit APs in a unified manner, enabling STAs to easily roam between APs. The number of Fit APs managed by a leader AP is fewer than that managed by a WAC, but Fit AP management is not under license control.
- WLAN roaming allows STAs to move within the coverage areas of APs belonging to the same extended service set (ESS) with nonstop service transmission.

Address Allocation for APs in the Leader AP Networking

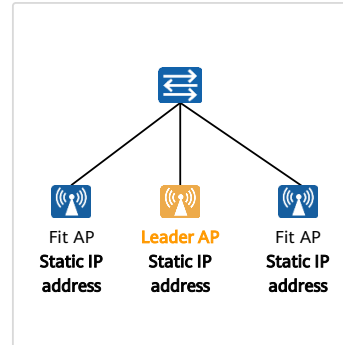
An external DHCP server assigns IP addresses to APs.



The built-in DHCP server assigns IP addresses to APs.

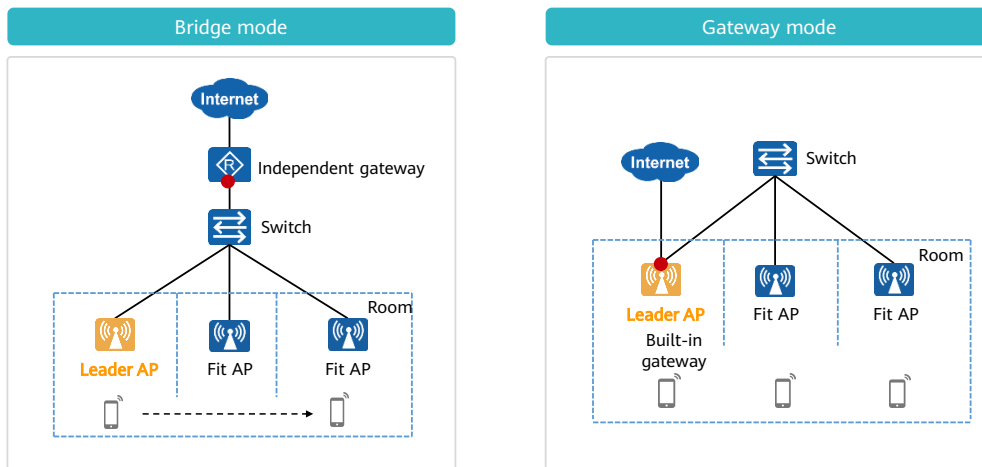


APs use static IP addresses.



- Mode in which Fit APs connect to a leader AP:
 - An external DHCP server can assign IP addresses to all APs. Ensure that the leader AP and Fit APs are in the same VLAN so that the Fit APs can discover the leader AP through CAPWAP broadcast.
 - The leader AP can function as a DHCP server to assign IP addresses to Fit APs. In this case, it is recommended that the leader AP and Fit APs be in the same VLAN.
 - Fit APs can connect to the leader AP using static IP addresses. In this case, it is recommended that the leader AP and Fit APs be in the same VLAN.

Common Leader AP Networking Modes

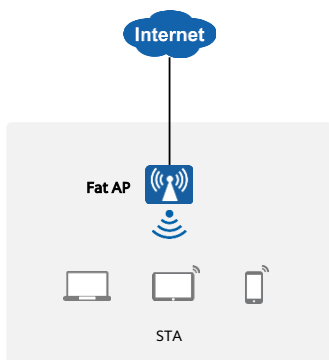


- A leader AP can work in bridge or gateway mode, depending on whether it functions as a gateway.
 - A leader AP in bridge mode functions as a network bridge and works with an independent gateway in the uplink direction. The leader AP and Fit APs communicate with each other on a Layer 2 network. The independent gateway has the DHCP service enabled and assigns IP addresses to STAs and APs. The direct forwarding mode is used, which reduces the load on the leader AP, so that the leader AP can manage more Fit APs. This networking mode is recommended.
 - In gateway mode, a leader AP itself functions as the gateway, and no independent gateway is used. The leader AP and Fit APs communicate with each other on a Layer 2 network. In the uplink direction, the leader AP has NAT enabled and connects to the Internet. In the downlink direction, the leader AP connects to a switch. The leader AP has the DHCP service enabled and assigns IP addresses to Fit APs and STAs. The networking is more simplified than that in bridge mode. In this mode, service traffic is forwarded through CAPWAP tunnels to the leader AP for processing. Therefore, the leader AP is heavily loaded and can manage a limited number of Fit APs. When the service traffic is heavy, the leader AP may become a bottleneck for service forwarding.

Contents

1. WLAN Networking Overview
2. WLAN Networking Architecture
- 3. WLAN Networking Application Scenarios**

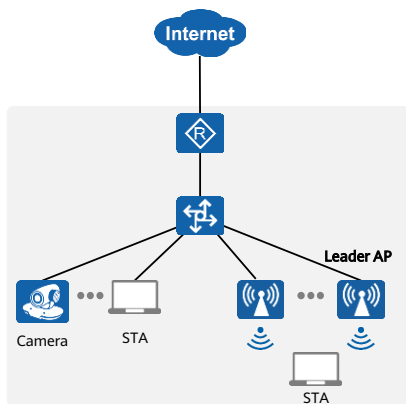
Small and Micro Enterprises and Small Stores



Solution description

- In the single-AP networking, a Fat AP functions as the gateway and Internet egress device for STAs.
- This networking applies to small and micro enterprises and small stores with small areas.
- The number of concurrent online STAs is small.
- Only STA access is required, and roaming is not required.
- STAs have simple Internet access requirements.

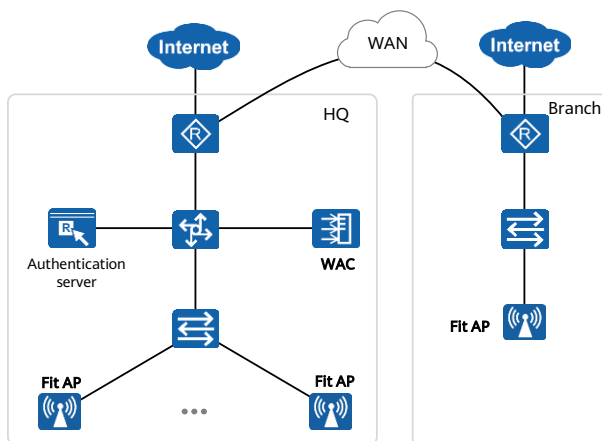
Small Enterprises



Solution description

- Multiple APs are connected to the network through a switch, providing a larger wireless coverage area than a single AP. In addition, the switch can provide network access for wired terminals.
- The leader AP networking is used to implement centralized AP management and configuration and support WLAN roaming.
- This networking applies to small and midsize experience stores and logistics stores, and also applies to small enterprises that require wireless coverage and access of a certain number of wired terminals such as cameras and printers.
- If there are special Internet access requirements, an independent gateway is generally used as the Internet egress device.

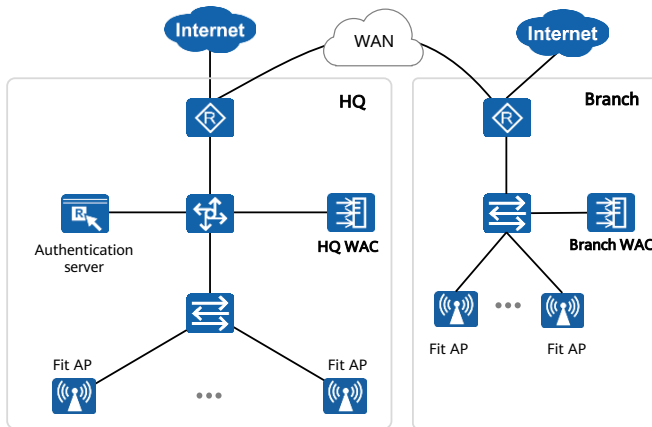
Small and Midsize Enterprises



Solution description

- The HQ and branches of an enterprise are interconnected across a wide area network (WAN) and internal network connectivity is implemented through routing protocols. A WAC and an authentication server are deployed in the HQ. The WAC manages all Fit APs in the HQ and branches.
- Fit APs in the HQ can use the direct or tunnel forwarding mode as required.
- Fit APs in branches usually use the local forwarding mode. Users are assigned IP addresses by branch gateways and directly access Internet resources through the branch egresses.

Large and Midsize Enterprises



Solution description

- The HQ and branches of an enterprise are interconnected across a WAN, and internal network connectivity is implemented through routing protocols.
- WACs are deployed in the HQ and branches to manage Fit APs in the HQ and branches, respectively.
- Generally, the authentication server is deployed only in the HQ.
- This networking applies to large and midsize enterprises and branches.

Quiz

1. (Multiple-answer question) Which of the following are the advantages of the WAC + Fit AP networking compared with the Fat AP networking? ()
- A. Easier configuration and deployment in multi-AP scenarios
 - B. Centralized batch upgrade for APs
 - C. Higher security
 - D. Support for roaming features

1. ABCD

Summary

- This course describes the basic WLAN networking architectures, including Fat AP, leader AP, and WAC + Fit AP networking.
- Upon completing this course, you will be able to understand common WLAN networking architectures and select a suitable one based on site environments and actual requirements.

Thank you.

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每个组织，构建万物互联的智能世界。
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organization for a fully connected,
intelligent world.

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WLAN Fundamentals and Configuration



Foreword

- With the development of wireless networks, more and more places require wireless device deployment. For a large-scale WLAN, the centralized networking architecture (WAC + Fit AP) is typically used. This networking architecture enables a WAC to centrally manage, control, and configure multiple APs, greatly improving WLAN O&M efficiency and reducing O&M costs.
- This course describes the WLAN fundamentals in the centralized networking architecture and explains how to configure basic WLAN services using the CLI.

Objectives

- On completion of this lesson, you will be able to:
 - Describe the AP onboarding process.
 - Describe the STA access process.
 - Understand how to configure WLAN services.

Contents

1. WLAN Fundamentals

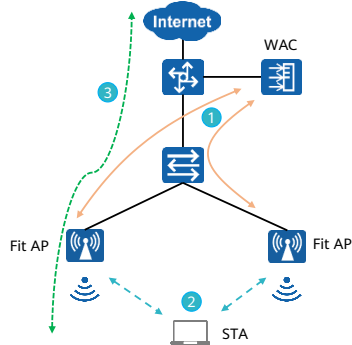
- AP Onboarding Process
 - STA Access Process
 - Data Forwarding for STAs

2. WLAN Service Configuration Procedure

3. WLAN Service Configuration Examples

WLAN Working Process

- In the centralized architecture (WAC + Fit AP), Fit APs need to go online before being managed and controlled by a WAC. After the APs go online, you can configure services on the WAC, so that STAs can access the WLAN.



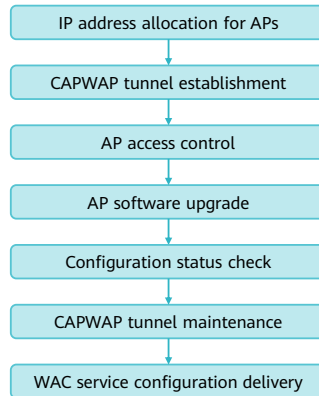
WLAN working process

- 1 **AP onboarding**
 - An AP obtains an IP address, discovers a WAC, and sets up a connection with the WAC.
 - The WAC delivers WLAN service configurations to the AP.
- 2 **STA access**
 - The STA detects the SSID broadcast by the AP.
 - The STA connects to the signal and accesses the network.
- 3 **Data forwarding**
 - The WLAN starts to forward service data.

- This course uses the centralized network architecture—a typical WLAN networking architecture—as an example to describe the WLAN fundamentals. For details about the WLAN fundamentals in other networking architectures, refer to HCIP-WLAN courses.

AP Onboarding Process

- In the centralized network architecture, the WAC can manage and control Fit APs in a centralized manner only after Fit APs go online. The AP onboarding process includes:



IP Address Allocation for APs

IP address
allocation for APs

CAPWAP tunnel
establishment

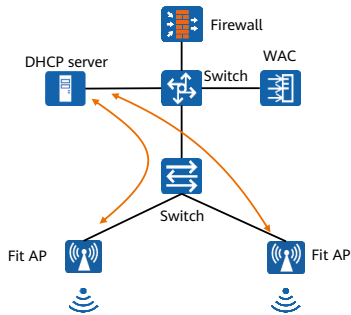
AP access
control

AP software
upgrade

Configuration
status check

CAPWAP tunnel
maintenance

Configuration
delivery



Scenario description:

- An AP must obtain an IP address to communicate with WAC so that the WLAN can work properly.

Mode in which an AP obtains an IP address:

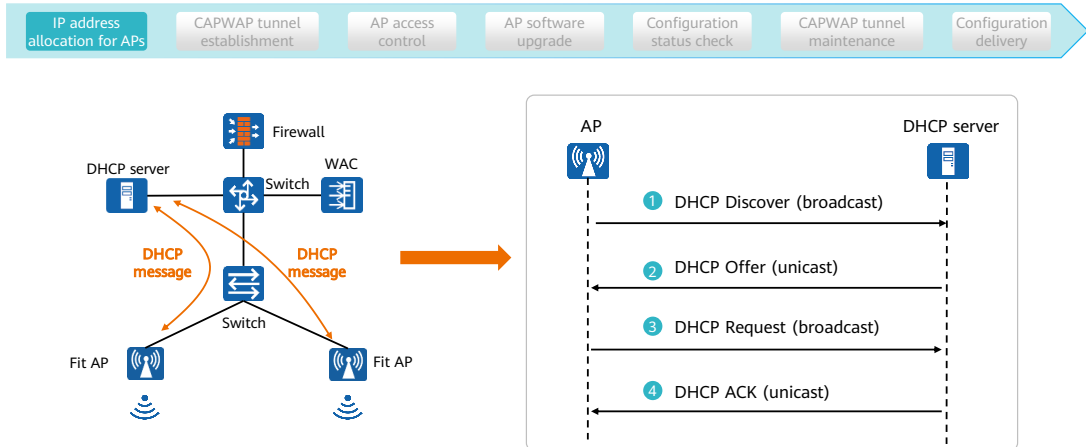
- Static mode: A user logs in to the AP and manually configures its IP address.
- DHCP mode: The AP automatically obtains IP address from the DHCP server.
- Stateless address autoconfiguration (SLAAC) mode: In this mode, an AP can only obtain an IPv6 address.

Typical solutions:

- Deploy a dedicated DHCP server to assign IP addresses to APs.
- Configure the WAC as a DHCP server to assign IP addresses to APs.
- Use a device on the network, such as a core switch, to assign IP addresses to APs.

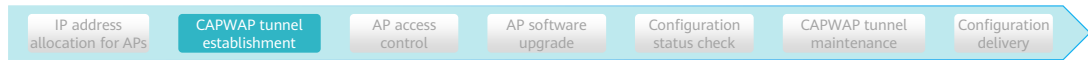
- The DHCP mode is typically used on WLANs to assign IP addresses to APs.

DHCP-based IP Address Allocation for APs

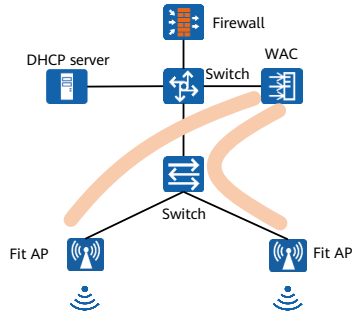


- DHCP four-message exchange:
 - An AP broadcasts a DHCP Discover message to request a response from a DHCP server.
 - After detecting the DHCP Discover message, a DHCP server selects an available IP address from the address pool on the same network segment as the IP address of the interface receiving the DHCP Discover message, and then sends a DHCP Offer message carrying this IP address to the AP.
 - The AP broadcasts a DHCP Request message to notify all DHCP servers of the IP address offered by a specific DHCP server. The unselected IP addresses offered by other DHCP servers are then free to be allocated to other clients.
 - After receiving the DHCP Request message, the DHCP server replies with a DHCP ACK message, which contains the IP address for the AP, lease, gateway information, and DNS server IP address. By now, the lease takes effect and the DHCP four-message exchange is completed.

CAPWAP Tunnel Establishment



CAPWAP tunnel

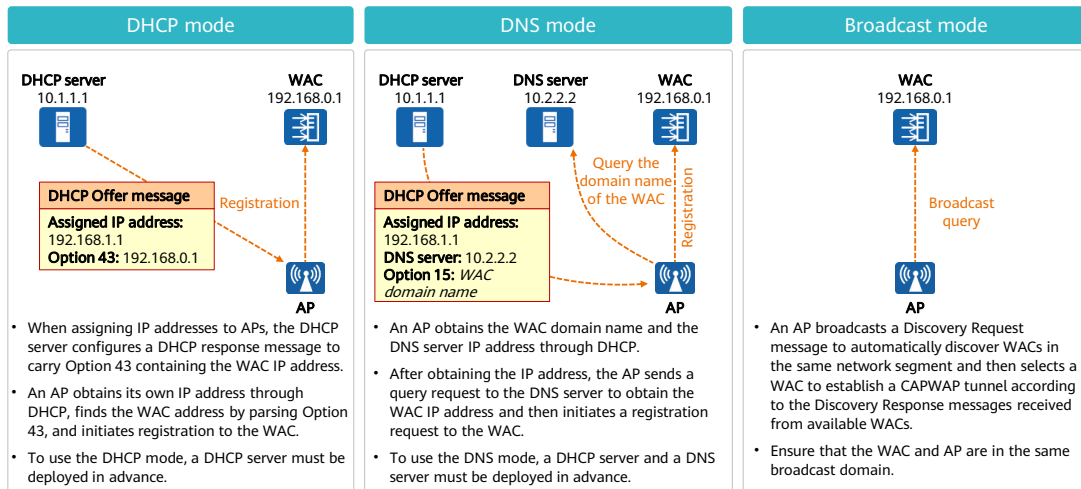


Scenario description:

- The WAC centrally manages and controls APs through CAPWAP tunnels.
- The process of establishing a CAPWAP tunnel between the WAC and an AP is divided into two phases:
 - **Discovery (the AP discovers the WAC)**
 - ✓ Static mode: The IP address list of WACs is preconfigured on APs.
 - ✓ Dynamic mode: An AP can dynamically discover a WAC in DHCP, DNS, or broadcast mode.
 - **CAPWAP tunnel establishment**

- In static mode, an AP discovers a WAC based on preconfigured IP addresses. When the AP goes online, it unicasts a Discovery Request message to each WAC address in its configuration list. The AP then selects a WAC based on the received Discovery Response messages from the WACs and establishes a CAPWAP tunnel with that WAC.
- In most WLAN deployments, WAC discovery is performed in dynamic mode. In this course, we focus on the dynamic mode.

Dynamic WAC Discovery by APs

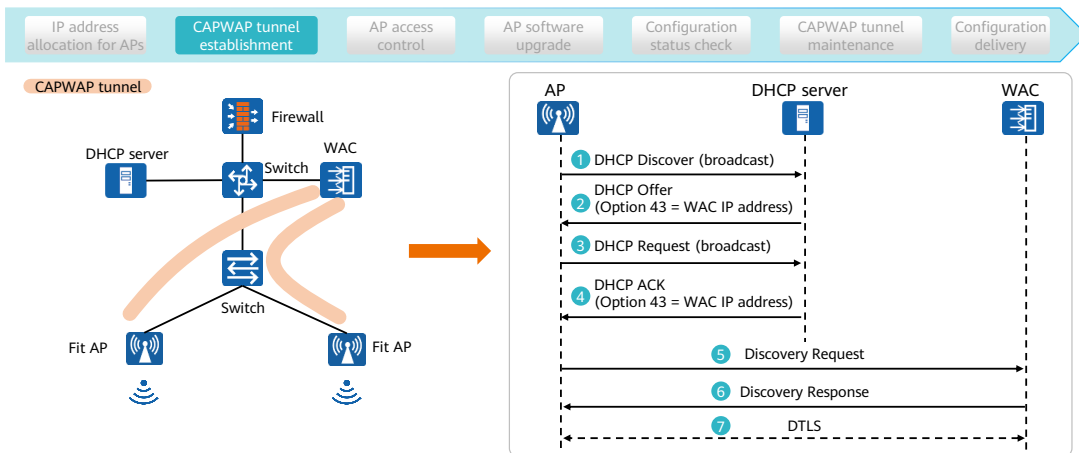


- DHCP mode:
 - An AP obtains the IP address of a WAC through DHCP (by configuring DHCP Offer messages to carry the WAC IPv4 address list in Option 43 or configuring DHCPv6 Advertise messages to carry the WAC IPv6 address list in Option 52 on the DHCP server). The AP then unicasts a Discovery Request message to the WAC. After receiving the Discovery Request message, the WAC responds to the AP with a Discovery Response message.
- DNS mode:
 - An AP obtains the WAC domain name and DNS server IP address through DHCP (by configuring the DHCP server to carry the domain name of the WAC in Option 15 in DHCP Offer messages or Option 24 in DHCPv6 Advertise messages). The AP then sends a request to the DNS server to obtain the IP address corresponding to the WAC domain name. After obtaining the WAC IP address, the AP unicasts a Discovery Request message to the WAC. After receiving the Discovery Request message, the WAC responds to the AP with a Discovery Response message.
 - After obtaining the WAC domain name from Option 15 or 24 in the DHCP Offer message, the AP automatically adds the fixed prefix **huawei-wlan-controller** to this domain name. After sending this "prefix + domain name" to the DNS server, the AP obtains the IP address of the WAC. For example,

after obtaining the WAC domain name **ac.test.com** configured on the DHCP server, the AP adds the prefix **huawei-wlan-controller** to **ac.test.com** and sends **huawei-wlan-controller.ac.test.com** to the DNS server for resolution. The IP address corresponding to **huawei-wlan-controller.ac.test.com** must be configured on the DNS server.

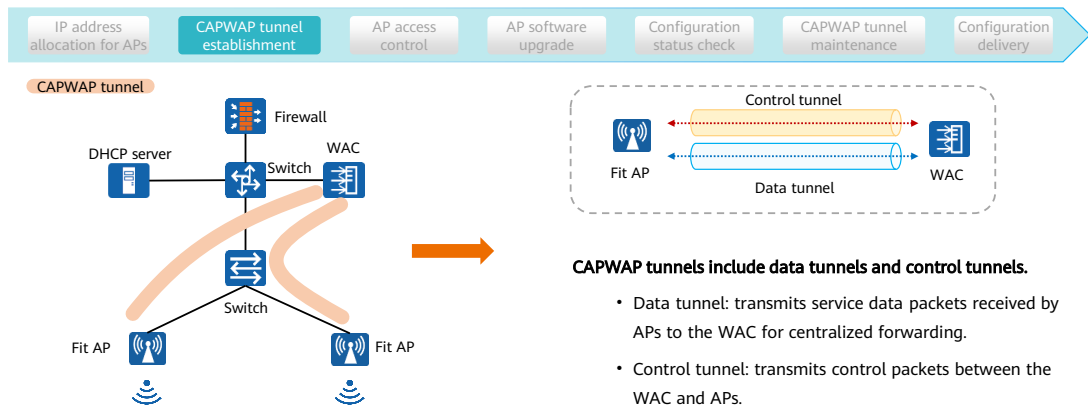
- Broadcast mode:
 - An AP broadcasts a Discovery Request message to discover WACs on the same network segment and then selects one to associate with based on the Discovery Response messages returned by the WACs. The AP and WAC then start to establish a CAPWAP tunnel.
 - Upon a failure to obtain the IP address of any WAC: The AP broadcasts a Discovery Request message. If no WAC responds to the broadcast message twice, the AP determines a failure to discover a WAC. The AP then continues to repeat the preceding operations periodically until it discovers a WAC.
 - Upon successfully obtaining the IP address of a WAC: The AP unicasts a Discovery Request message to the WAC. If no response is received after unicasting three Discovery Request messages, the AP broadcasts the Discovery Request message. If there is still no response from any WAC, the AP repeats the operations of sending unicast and broadcast messages. If no response is received after the process is completed twice, the AP determines a failure to discover a WAC. The AP then continues to repeat the preceding operations periodically until it discovers a WAC.
- After receiving Discovery Response messages from multiple WACs, the AP selects a WAC to set up a link based on WAC information carried in the messages. The WAC selection is determined based on the following priorities in descending order:
 - Maximum number of access APs supported by a WAC. The AP selects the WAC that supports access of the most APs.
 - Maximum number of access STAs supported by a WAC. The AP selects the WAC that supports access of the most STAs.
 - IP address. The AP selects the WAC with the smallest IP address.

CAPWAP Tunnel Establishment Phase 1: WAC Discovery Through DHCP



- During CAPWAP tunnel establishment, Datagram Transport Layer Security (DTLS) encryption can be enabled for CAPWAP tunnels to encrypt and decrypt CAPWAP packets.

CAPWAP Tunnel Establishment Phase 2: Tunnel Establishment



- CAPWAP data tunnels (including DTLS encryption for data tunnels) take effect only when the data forwarding mode is tunnel forwarding.

AP Access Control

IP address
allocation for APs

CAPWAP tunnel
establishment

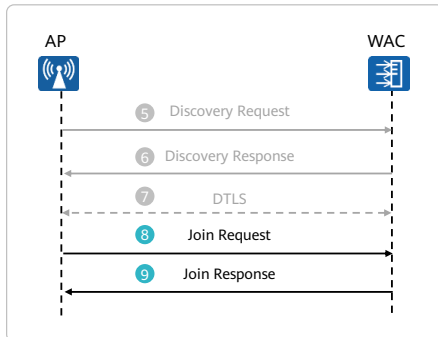
AP access
control

AP software
upgrade

Configuration
status check

CAPWAP tunnel
maintenance

Configuration
delivery



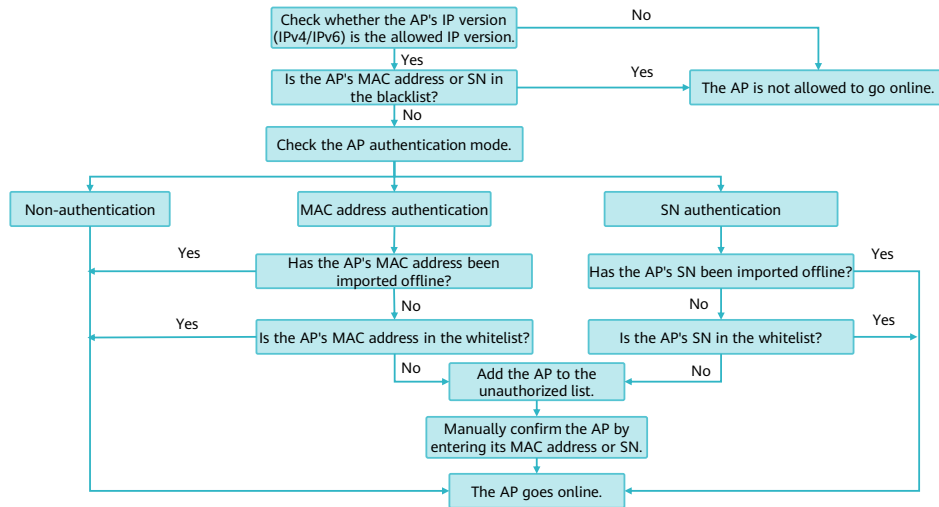
Scenario description:

- The AP sends a Join Request message to the WAC. After receiving the message, the WAC determines whether to allow the AP access and sends a Join Response message to the AP.
- The Join Response message carries the AP software upgrade mode and AP version information configured on the WAC.

Modes in which a WAC authenticates an AP:

- MAC address authentication
- SN authentication
- Non-authentication

AP Access Control Process

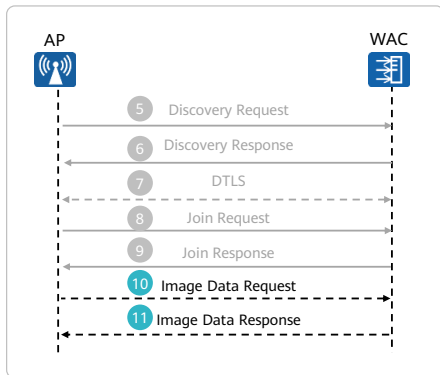
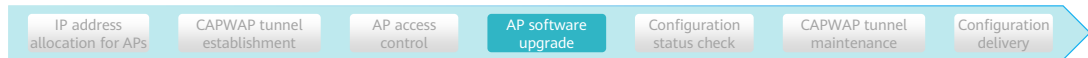


- Process for a WAC to determine whether to allow access of an AP:
 - The WAC first checks whether the AP's IP version (IPv4 or IPv6) is the allowed version for AP onboarding. If not, the AP is denied access.
 - If the IP version is allowed, the WAC checks whether the AP is blacklisted. If so, it is denied access.
 - If the AP is not blacklisted, the WAC checks the AP's authentication mode.
 - If the authentication mode is none, the AP is allowed to access the network.
 - If the authentication mode is MAC address authentication:
 - The WAC checks whether the AP with the corresponding MAC address has been added offline. If so, the AP is allowed to access the network.
 - If the AP has not been added offline, the WAC checks whether the AP's MAC address is in the whitelist. If so, the AP is allowed to access the network. Otherwise, the AP is added to the unauthorized list.
 - If the authentication mode is SN authentication:
 - The WAC checks whether the AP with the corresponding SN has been added offline. If so, the AP is allowed to access the network.
 - If the AP has not been added offline, the WAC checks whether the AP's SN is in the whitelist. If so, the AP is allowed to access the

network. Otherwise, the AP is added to the unauthorized list.

- APs in the unauthorized list can be manually confirmed on the WAC. If an AP in the unauthorized list is not manually confirmed, the AP is denied access.

AP Software Upgrade



Scenario description:

- The AP determines whether its system software version is the same as that specified on the WAC according to parameters in the received Join Response message.
- If the two versions are different, the AP upgrades its software version in AC, FTP, or SFTP mode.

In an AC-mode upgrade:

- The AP sends an Image Data Request packet to the WAC to download the latest software version.
- After receiving the request, the WAC delivers the software version to the AP. After the software version is upgraded, the AP restarts and repeats the preceding three steps.

- AP upgrade modes:
 - AC mode: applies when no file server is deployed on the network. APs download the upgrade file from the WAC during the upgrade.
 - FTP mode: applies to file transfer without high network security requirements. APs download the upgrade file from an FTP server during the upgrade. In this mode, data is transmitted in clear text, which brings security risks.
 - SFTP mode: applies to scenarios that require high network security and provides strict encryption and integrity protection for data transmission. APs download the upgrade file from an SFTP server during an upgrade.

Configuration Status Check

IP address
allocation for APs

CAPWAP tunnel
establishment

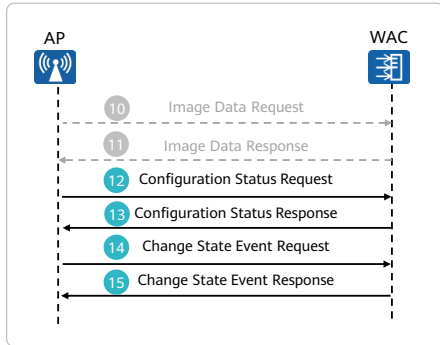
AP access
control

AP software
upgrade

Configuration
status check

CAPWAP tunnel
maintenance

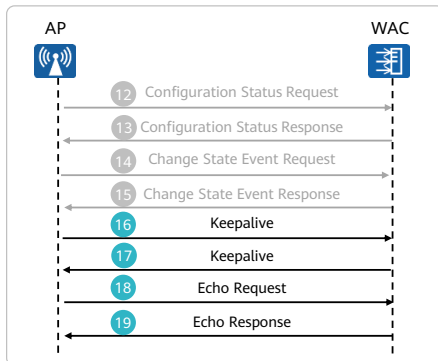
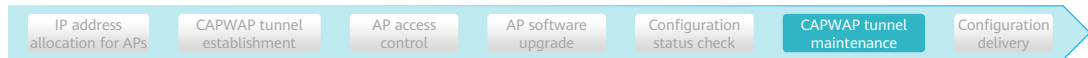
Configuration
delivery



Scenario description:

- The AP sends a Configuration Status Request message to the WAC to request the initial configuration.
- After receiving the Configuration Status Request message, the WAC delivers the initial configuration to the AP.
- After receiving the Configuration Status Response message, the AP performs the initialization configuration based on the message content.
- After the initialization configuration is complete, the AP sends a Change State Event Request message carrying the radio status and configuration execution result to the WAC.
- After receiving the Change State Event Request message, the WAC updates AP information as required.

CAPWAP Tunnel Maintenance

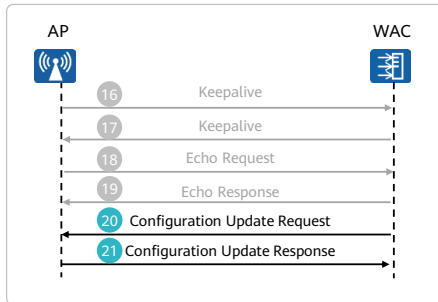
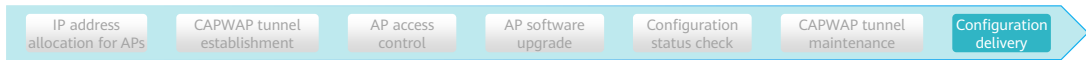


Scenario description:

- The AP and WAC exchange Keepalive packets (through the UDP port 5247) to detect data tunnel connectivity.
- The AP and WAC exchange Echo packets (through UDP port 5246) to detect control tunnel connectivity.

- In CAPWAP heartbeat detection, an AP and a WAC periodically exchange Echo and Keepalive messages to determine whether the control and data tunnels are working properly, respectively. If one end of a tunnel does not receive any response from the other end after sending detection packets (Echo or Keepalive, depending on the tunnel type) for a specified number of times, the device considers that the tunnel is disconnected and needs to be re-established.
 - The CAPWAP heartbeat detection interval refers to the interval for sending Echo or Keepalive messages.
 - The number of CAPWAP heartbeat detection times refers to the number of times Echo or Keepalive messages are sent.
 - By default, the CAPWAP heartbeat detection interval is 25s and the number of CAPWAP heartbeat detections is 6. You can configure the two parameters as required.

Configuration Delivery



Scenario description:

- When the AP configuration needs to be updated, the WAC sends a Configuration Update Request message to the AP.
- After receiving the Configuration Update Request message, an AP sends a Configuration Update Response message to the WAC. After receiving the message, the WAC delivers the service configuration to the AP.

Contents

1. WLAN Fundamentals

- AP Onboarding Process
- STA Access Process
- Data Forwarding for STAs

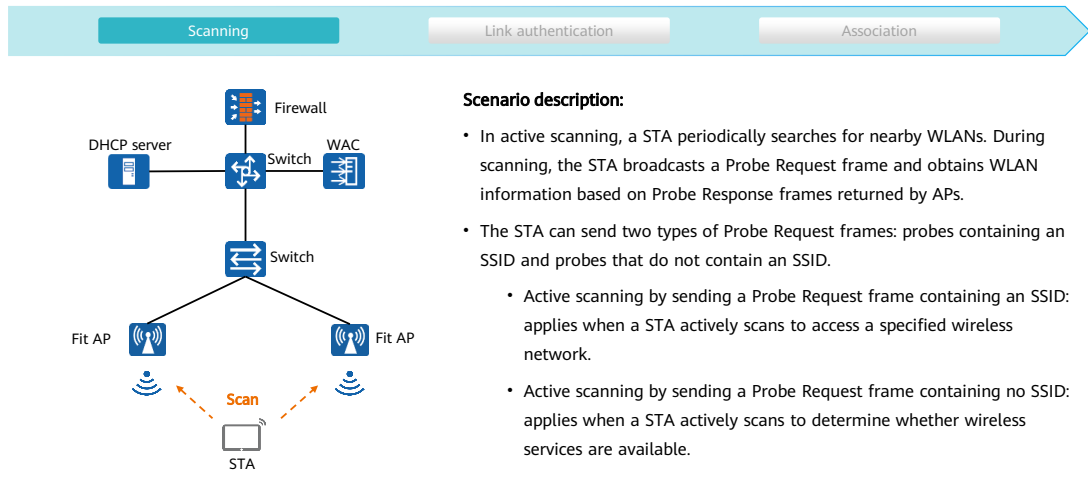
2. WLAN Service Configuration Procedure

3. WLAN Service Configuration Examples

STA Access Process

- After an AP goes online and wireless service configuration is complete, STAs can access the WLAN. STA access involves the following phases:
 - Scanning
 - Link authentication
 - Association
- Whether a STA can access a WLAN depends on the number of access STAs supported by devices and the STA access control function as follows:
 - Maximum number of access STAs supported by a WAC and an AP
 - VAP-based user access control by limiting the number of STAs associated with an SSID on an AP
 - User CAC based on the number of STAs by limiting the number of access STAs connected to a radio of a single AP
 - STA control based on the STA blacklist and whitelist

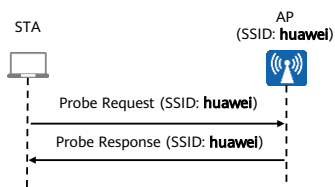
Scanning



- In addition to active scanning, passive scanning can also be used. An AP periodically broadcasts Beacon frames, each of which carries information such as the SSID and supported rates. Therefore, a STA can listen on Beacon frames transmitted on channels in the supported channel list to obtain information about surrounding APs and then connect to the WLAN.
- Passive scanning is typically used by STAs that require power-saving, such as voice over IP (VoIP) terminals.

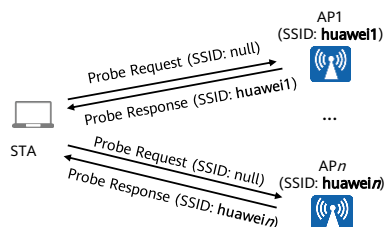
Active Scanning

Active scanning by sending a Probe Request frame containing an SSID



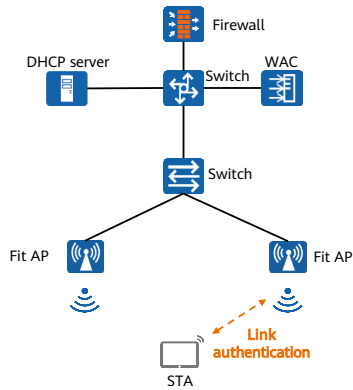
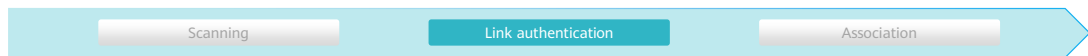
A STA sends a Probe Request frame containing an SSID on each channel to search for the AP with the same SSID. Only the AP with the same SSID will respond to the STA.

Active scanning by sending a Probe Request frame containing no SSID



A STA periodically sends Probe Request frames carrying no SSID information (that is, null SSID) on channels in the supported channel list to scan WLANs. The APs receiving the Probe Request frame will return Probe Response frames to notify the STA of the wireless services they can provide.

Link Authentication

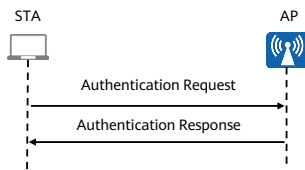


Scenario description:

- To ensure wireless link security, an AP needs to authenticate STAs that attempt to access it. This occurs before STAs associate with the AP.
- 802.11 defines two link authentication mechanisms:
 - Open system authentication: Non-authentication is used, allowing any STA to be successfully authenticated.
 - Shared-key authentication: The same shared key is preconfigured on a STA and an AP. During link authentication, the AP checks whether the STA has the same shared key. If so, the STA is authenticated successfully. If not, STA authentication fails.

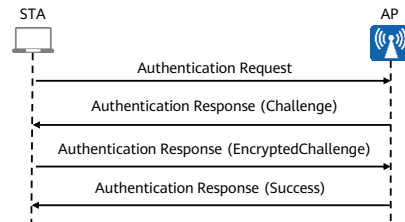
Link Authentication Modes

Open system authentication



- A STA uses its MAC address as proof of identity. The MAC address on a WLAN is unique, the same as that on an Ethernet network.
- In an open system, authentication is not required. Users can access the network as long as the MAC addresses of their STAs are unique.

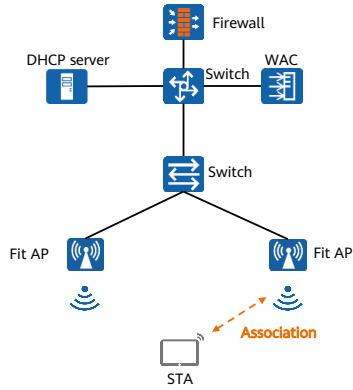
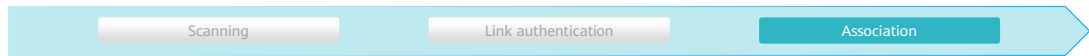
Shared-key authentication



- A STA sends an Authentication Request frame to the AP.
- The AP randomly generates a challenge and sends it to the STA.
- The STA uses the preconfigured key to encrypt the challenge and sends the encrypted challenge to the AP.
- The AP uses the preconfigured key to decrypt the encrypted challenge and compares the decrypted challenge with the challenge sent to the STA earlier. If the two challenges are the same, the STA is authenticated successfully. Otherwise, STA authentication fails.

- Shared-key authentication must use the static Wired Equivalent Privacy (WEP) security policy. Therefore, it can be used only on products that support WEP. In this authentication mode, the shared key must be transmitted to the STA before identity authentication. The theoretical basis of shared key authentication is that a station has a shared key if it can successfully respond to the challenge information sent to it. After the key is successfully exchanged, the STA passes the authentication.

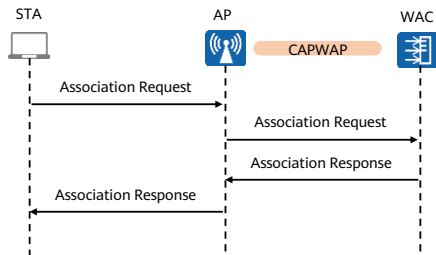
Association



Scenario description:

- The STA association process is essentially a link negotiation process.
- After link authentication is complete, a STA initiates link service negotiation using Association packets.

STA Association Process in the WAC + Fit AP Architecture



- The STA sends an Association Request frame to the AP. This frame carries the STA's own information as well as the parameters selected by the STA as determined by the service configuration, including the supported rates, channels, access authentication algorithm, and encryption algorithm.
- The AP receives the Association Request frame, encapsulates it using CAPWAP, and sends the encapsulated frame to the WAC.
- After receiving the Association Request frame, the WAC determines whether to authenticate the STA and replies with an Association Response frame.
- The AP decapsulates the received Association Response packet using CAPWAP and sends it to the STA.

Contents

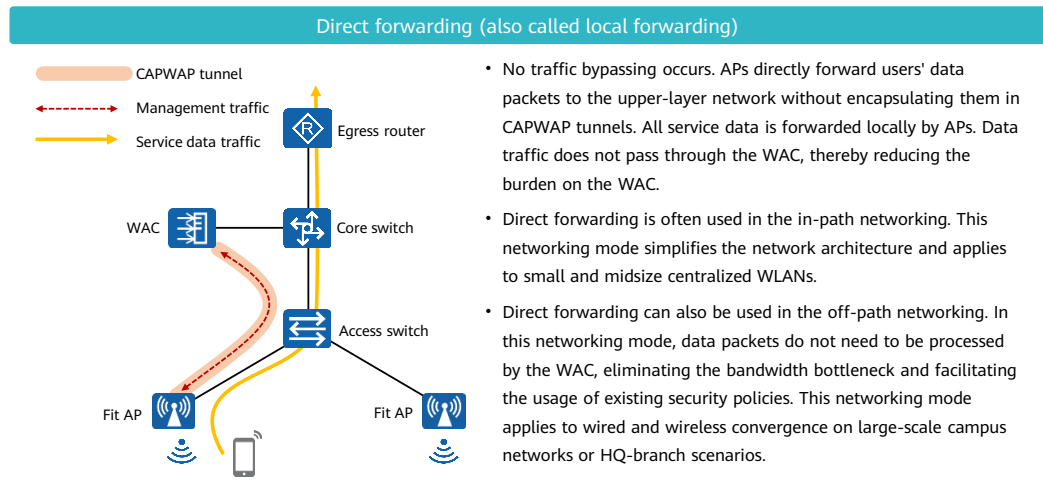
1. WLAN Fundamentals

- AP Onboarding Process
- STA Access Process
- Data Forwarding for STAs

2. WLAN Service Configuration Procedure

3. WLAN Service Configuration Examples

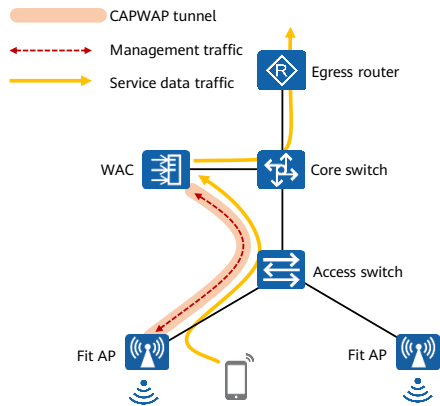
Data Forwarding Mode: Direct Forwarding



- Two types of packets are transmitted on a WLAN: control packets (management packets) and data packets. User data packets can be forwarded in tunnel forwarding (centralized forwarding) or direct forwarding (local forwarding) mode.

Data Forwarding Mode: Tunnel Forwarding

Tunnel forwarding (also called centralized forwarding)



- **Traffic forwarding:** Service data packets are encapsulated by APs and then sent to the WAC through CAPWAP tunnels. The WAC then forwards the packets to the upper-layer network. All data traffic and management traffic pass through the WAC, facilitating security control policy enforcement for wireless users.
- The WAC serves as the control and forwarding center on a WLAN. It manages and configures APs and also forwards service data traffic.
- Tunnel forwarding is typically used in the off-path networking. In this networking mode, the WAC centrally forwards data packets, ensuring high security and facilitating centralized management and control.

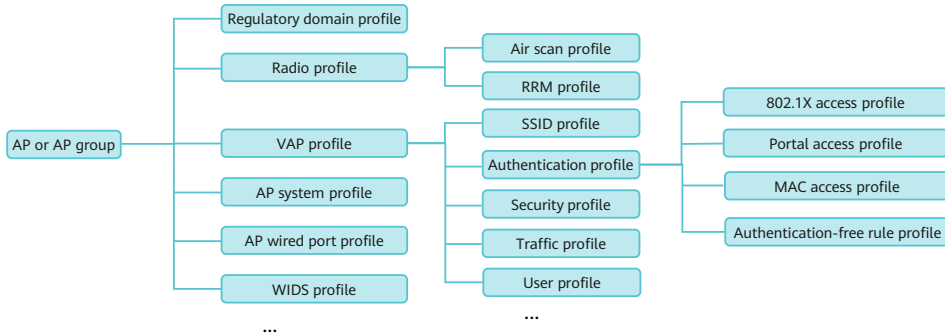
- Service data traffic between the WAC and APs is transmitted over CAPWAP data tunnels, and management traffic is transmitted over CAPWAP control tunnels.

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1. WLAN Fundamentals
- 2. WLAN Service Configuration Procedure**
3. WLAN Service Configuration Examples

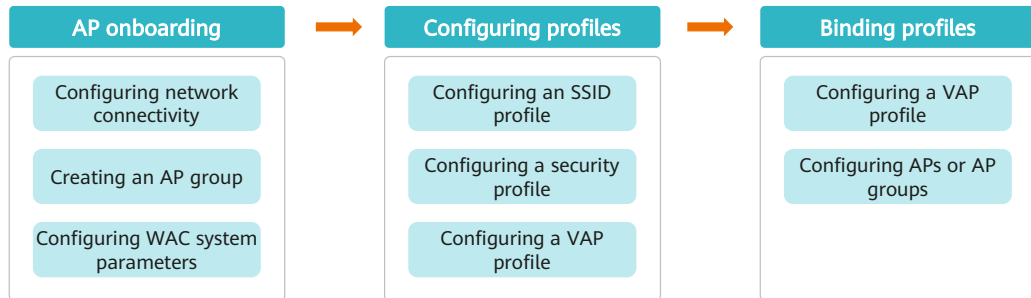
Reference Relationships Between WLAN Profiles

- Various WLAN profiles are designed based on different WLAN service functions.
- WLAN profiles reference each other. Understanding the reference relationships between profiles and the configurable functions in each profile helps you successfully deploy WLAN services.

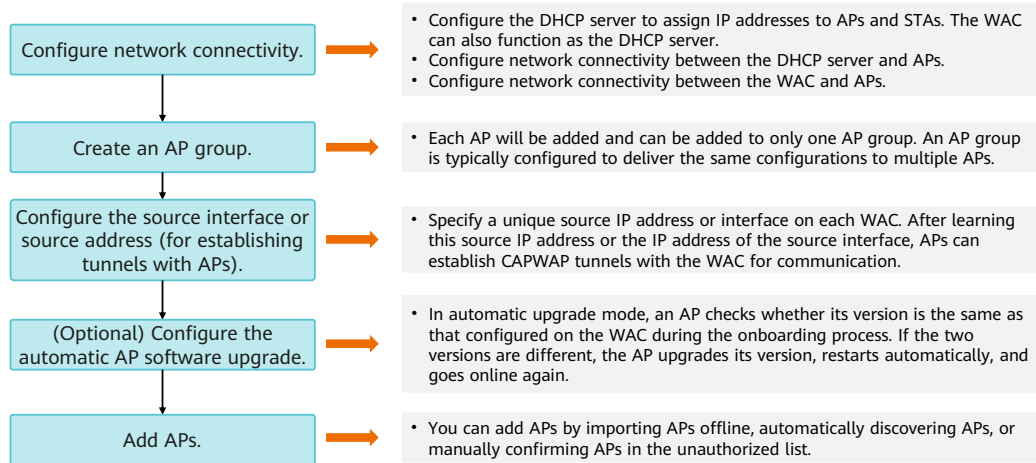


- Multiple profiles can be bound to AP groups and APs, including the regulatory domain profile, AP system profile, AP wired port profile, radio profile, VAP profile. Some of the listed profiles can further reference other profiles, for example, a radio profile can reference an air scan profile and an RRM profile.
- This course describes only some profiles. For details about the profiles and their mapping relationships, see the *Huawei WLAN product documentation*.

Basic WLAN Service Configuration Procedure



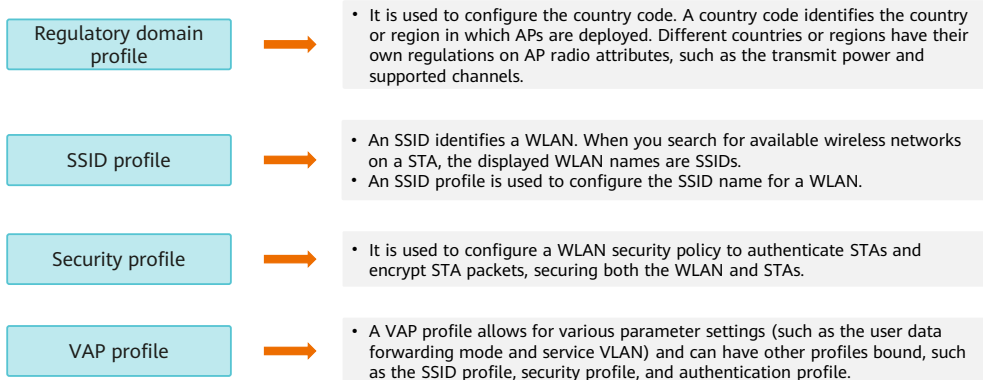
Configuring AP Onboarding



- Configure the WAC source interface or source address.
 - APs can establish CAPWAP tunnels with the WAC only after the WAC source interface or source address is configured.
 - WAC supports a VLANIF or loopback interface as the source interface, and the IP address of the VLANIF or loopback interface as the source address.

Configuring Profiles

- To configure basic WLAN services, you need to configure the following profiles:



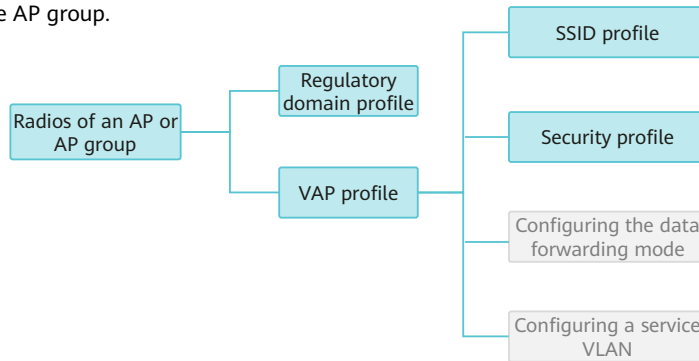
- Regulatory domain profile:
 - A regulatory domain profile provides configurations of the country code, calibration channel set, and calibration bandwidth for APs.
 - The correct country code configuration ensures that AP radio attributes comply with local laws and regulations.
- SSID profile: allows you to configure an SSID name for a WLAN. It also allows for the configuration of other functions, including:
 - SSID hiding: When creating a WLAN, configure APs to hide the SSID name for higher WLAN security. This way, only users who know the SSID can connect their STAs to the WLAN.
 - Maximum number of STAs on a VAP: More access STAs on a VAP indicate fewer network resources that are available to each STA. To ensure Internet access experience, properly configure the maximum number of access STAs on a VAP based on site requirements.
 - SSID hiding when the number of STAs reaches the maximum: With this feature enabled, when the number of access STAs on a WLAN reaches the maximum, the WLAN SSID is hidden to prevent new STAs from discovering it.
- Security profile: allows you to configure a WLAN security policy to authenticate

STAs and encrypt their packets, securing both the WLAN and STAs.

- WLAN security policies support open, WPA, WPA2, WPA3, WPA-WPA2, and WPA2-WPA3. You can configure one of them in a security profile.
- VAP profile: provides wireless access services for STAs. Parameter settings in the VAP profile determine different wireless service capabilities available to STAs.
 - Data forwarding mode in a VAP profile: Data packets can be forwarded in tunnel forwarding or direct forwarding mode.
 - Service VLAN in a VAP profile: Service VLANs are used to transmit service data packets. Layer 2 data packets delivered from a VAP to an AP carry the service VLAN ID.

Binding Profiles

- On a WLAN, there may be a large number of APs, among which many require the same configuration. These APs can be added to an AP group, allowing you to configure them uniformly. If certain APs require unique settings, those configurations can be applied directly to the specific APs instead of through the AP group.



- A VAP profile can have various profiles bound, such as an SSID profile and a security profile. The data forwarding mode and service VLAN are directly configured in the VAP profile.
- The profiles that can be bound to an AP group or a specific AP include the regulatory domain profile, and VAP profile. Some of these profiles can have other profiles bound.
- Each AP must and can join only one AP group when going online on an AC. If an AP receives both AP group configurations and its own specific configurations from the WAC, the AP-specific configurations take priority.

Configuring AP Onboarding (1)

- Configure network connectivity (based on the actual networking).
 - Configure connectivity for management packets. Specifically, configure a proper VLAN or route to ensure that management packets can be properly transmitted between the WAC and APs.
 - Configure connectivity for service packets. In tunnel forwarding mode, you do not need to configure service packet connectivity. In direct forwarding mode, you need to configure service VLAN connectivity between APs and the upper-layer network.
 - Configure STAs to communicate with a DHCP server so that the STAs can obtain IP addresses from the DHCP server.
- Create an AP group.
 - Create an AP group and enter its view.

```
<WAC> system-view  
[WAC] wlan  
[WAC-wlan] ap-group name group-name  
[WAC-wlan-ap-group-group-name]
```

- The **system-view** command displays the system view from the user view.
- The **wlan** command displays the WLAN view from the system view.
- The commands for entering the preceding views are not repeated in subsequent configurations.

Configuring AP Onboarding (2)

- Configure the CAPWAP source interface or source address.
 - Configure the source interface for the WAC to establish a CAPWAP tunnel with APs.
`[WAC] capwap source interface { loopback loopback-number | vlanif vlan-id }`
 - Configure the source IP address of the WAC.
`[WAC] capwap source ip-address ip-address`
- Configure the function of establishing CAPWAP DTLS sessions in non-authentication mode (in deployment scenarios).

```
[WAC] capwap dtls no-auth enable
```

Configuring AP Onboarding (3)

- Add APs.

- Set the AP authentication mode to MAC address, SN authentication, or none authentication. By default, MAC address authentication is used.

```
[WAC-wlan] ap auth-mode { mac-auth | sn-auth | no-auth }
```

- Add an offline AP or enter the AP view, and configure the AP name. (In non-authentication mode, APs do not need to be added offline.)

```
[WAC-wlan] ap-id ap-id [ type-id type-id | ap-type ap-type ] { ap-mac ap-mac | ap-sn ap-sn }  
[WAC-wlan-ap-ap-id] ap-name ap-name
```

- Add APs to an AP group.

```
[WAC-wlan-ap-ap-id] ap-group ap-group
```

- Verify the AP onboarding result.

```
[WAC] display ap { all | ap-group ap-group }
```

- The **ap-id** command adds an offline AP or displays the AP view.
 - *ap-id*: specifies the ID of an AP.
 - **type-id** *type-id*: specifies the ID of an AP type.
 - **ap-type** *ap-type*: specifies the type of an AP.
 - **ap-mac** *ap-mac*: specifies the MAC address of an AP. The value is in the format H-H-H, where H is a 4-digit hexadecimal number.
 - **ap-sn** *ap-sn*: specifies the SN of an AP. The value is a string of 1 to 31 characters, and can contain only letters and digits.

Configuring Profiles (1)

- Configure a regulatory domain profile.
 - Create a regulatory domain profile and configure the country code.

```
[WAC-wlan] regulatory-domain-profile name profile-name  
[WAC-wlan-regulatory-domain-profile-name] country-code country-code
```

- Configure an SSID profile.
 - Create an SSID profile and configure the SSID name.

```
[WAC-wlan] ssid-profile name profile-name  
[WAC-wlan-ssid-prof-profile-name] ssid ssid
```

Configuring Profiles (2)

- Configure a security profile.
 - Create a security profile and configure a security policy (for example, open authentication).

```
[WAC-wlan] security-profile name profile-name  
[WAC-wlan-sec-prof-profile-name] security open
```

- Configure a VAP profile.
 - Create a VAP profile, and configure the data forwarding mode and service VLAN.

```
[WAC-wlan] vap-profile name profile-name  
[WAC-wlan-vap-prof-profile-name] forward-mode { direct-forward | tunnel }  
[WAC-wlan-vap-prof-profile-name] service-vlan { vlan-id vlan-id-value | vlan-pool pool-name }
```

- The security policies that can be configured in a security profile include open, WPA, WPA-WPA2, WPA2, and WPA3.

Binding Profiles

- Configure a VAP profile.
 - Bind the SSID profile and security profile to the VAP profile.

```
[WAC-wlan] vap-profile name profile-name
[WAC-wlan-vap-prof-profile-name] security-profile profile-name
[WAC-wlan-vap-prof-profile-name] ssid-profile profile-name
```

- Configure an AP group.
 - Bind the VAP profile to the AP group so that radios on APs use the VAP profile configuration.

```
[WAC-wlan] ap-group name group-name
[WAC-wlan-ap-group-group-name] vap-profile profile-name wlan wlan-id radio { radio-id | all }
```

- Check VAP information.
 - Display information about service VAPs.

```
[WAC] display vap { ap-group ap-group-name | ap-id ap-id [ radio radio-id ] } [ ssid ssid ]
```

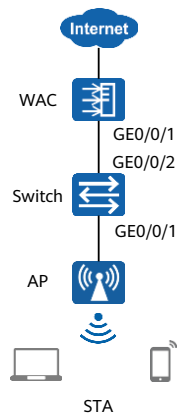
```
[WAC] display vap { all | ssid ssid }
```

- Command: **display vap { ap-group ap-group-name | ap-id ap-id [radio radio-id] } [ssid ssid]**
 - **ap-group ap-group-name**: displays information about all service VAPs in a specified AP group. The value must be the name of an existing AP group.
 - **ap-id ap-id**: displays information about service VAPs on the AP with a specified ID. The value must be the ID of an existing AP.
 - **radio radio-id**: displays information about service VAPs on a specified radio.
 - **ssid ssid**: displays information about service VAPs of a specified SSID. The value must be the name of an existing SSID.
- Command: **display vap { all | ssid ssid }**
 - **all**: displays information about all service VAPs.
 - **ssid ssid**: displays information about service VAPs of a specified SSID. The value must be the name of an existing SSID.

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Topology Design: Layer 2 Tunnel Forwarding Mode in In-Path Networking



Networking requirements

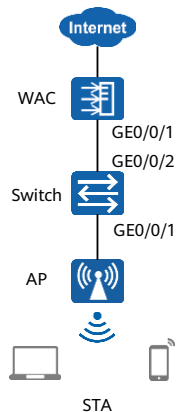
Users access the network through WLANs, meeting the basic mobile office requirements. Additionally, the WLAN is required to ensure continuous user services during STA roaming in the coverage area.

- WAC networking mode: Layer 2 in-path networking
- DHCP deployment mode: WAC functioning as a DHCP server to assign IP addresses to APs and STAs
- Service data forwarding mode: tunnel forwarding

WLAN Data Planning

Configuration Item	Configuration Plan
DHCP server	The WAC functions as a DHCP server to assign IP addresses to APs and STAs, and also serves as the gateway for APs and STAs.
IP address pool for APs	VLAN 100: 10.1.100.2–10.1.100.254/24
IP address pool for STAs	VLAN 101: 10.1.101.2–10.1.101.254/24
IP address of the WAC's source interface	VLANIF 100: 10.1.100.1/24
AP group	Name: ap-group1 Referenced profiles: VAP profile and regulatory domain profile
Regulatory domain profile	Name: domain Country code: CN
SSID profile	Name: employee SSID name: employee
Security profile	Name: employee Security policy: WPA-WPA2+PSK+AES Password: a1234567
VAP profile	Name: employee Forwarding mode: tunnel forwarding Service VLAN: VLAN 101 Referenced profiles: SSID profile employee and security profile employee

Configuring Network Connectivity



- Create VLANs and interfaces on the WAC and switch.
- Configure the WAC as a DHCP server to assign IP addresses to the AP and STAs.

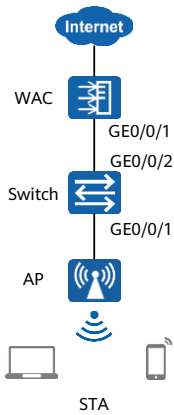
```

[WAC] dhcp enable
[WAC] interface Vlanif 100
[WAC-Vlanif100] ip address 10.1.100.1 24
[WAC-Vlanif100] dhcp select interface
[WAC-Vlanif100] quit
[WAC] interface Vlanif 101
[WAC-Vlanif101] ip address 10.1.101.1 24
[WAC-Vlanif101] dhcp select interface
[WAC-Vlanif101] quit
  
```

- Switch configuration
 - [SW] vlan batch 100
 - [SW] interface gigabitethernet 0/0/1
 - [SW-GigabitEthernet0/0/1] port link-type trunk
 - [SW-GigabitEthernet0/0/1] port trunk pvid 100
 - [SW-GigabitEthernet0/0/1] port trunk allow-pass vlan 100
 - [SW-GigabitEthernet0/0/1] quit
 - [SW] interface gigabitethernet 0/0/2
 - [SW-GigabitEthernet0/0/2] port link-type trunk
 - [SW-GigabitEthernet0/0/2] port trunk allow-pass vlan 100
 - [SW-GigabitEthernet0/0/2] quit
- WAC configuration
 - [WAC] vlan batch 100 101
 - [WAC] interface gigabitethernet 0/0/1
 - [WAC-GigabitEthernet0/0/1] port link-type trunk
 - [WAC-GigabitEthernet0/0/1] port trunk allow-pass vlan 100
 - [WAC-GigabitEthernet0/0/1] quit
- Note: Because the tunnel forwarding mode is used, only the management VLAN needs to be allowed on GE0/0/1 and GE0/0/2 of the switch and GE0/0/1 of the

WAC.

Configuring AP Onboarding (1)



- Configure the CAPWAP source interface.

[WAC] capwap source interface vlanif 100

Set the DTLS PSK(contains 8-32 plain-text characters, or 128 or 148 cipher-text characters that must be a combination of at least two of the following: lowercase letters a to z, uppercase letters A to Z, digits, and special characters):*****

Confirm PSK:*****

Set the user name for FIT APs(The value is a string of 4 to 31 characters, which can contain letters, underscores, and digits, and must start with a letter):**admin**

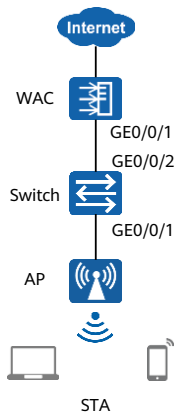
Set the password for FIT APs(plain-text password of 8-128 characters or cipher-text password of 128-268 characters that must be a combination of at least three of the following: lowercase letters a to z, uppercase letters A to Z, digits, and special characters):*****

Confirm password:*****

Set the PSK of the global offline management VAP(plain-text password of 8-63 characters or cipher-text password of 128-188 characters that must be a combination of at least two of the following: lowercase letters a to z, uppercase letters A to Z, digits, and special characters):*****

Confirm PSK:*****

Configuring AP Onboarding (2)



- Create an AP group.

```
[WAC] wlan
[WAC-wlan] ap-group name ap-group1
[WAC-wlan-ap-group-ap-group1] quit
```

- Enable the function of establishing CAPWAP DTLS sessions in non-authentication mode.

```
[WAC] capwap dtls no-auth enable
```

- Import the AP offline on the WAC.

```
[WAC] wlan
[WAC-wlan] ap auth-mode mac-auth
[AC-wlan] ap-id 0 ap-mac 00e0-fc44-4270
[WAC-wlan-ap-0] ap-name ap1
[WAC-wlan-ap-0] ap-group ap-group1
```

Warning: This operation may cause AP to go offline and then online. If the country code changes, it will clear channel, power and antenna gain configurations of the radio, Whether to continue?

```
[Y/N]:y
```

```
[WAC-wlan-ap-0] quit
```

- In this example, the AP authentication mode is MAC address authentication. You need to manually add the AP's MAC address on the WAC.

Verifying the AP Onboarding Configuration

- Power on the AP and run the **display ap all** command to check the AP state. If the **State** field displays **nor**, the AP has gone online.

```
[WAC] display ap all
```

```
Total AP information:
```

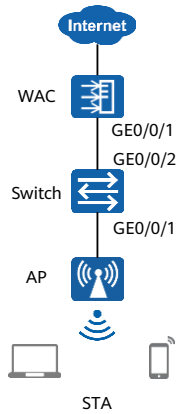
```
nor : normal      [1]
```

```
Extra information: Extra information
```

```
Total: 1
```

ID	MAC	Name	Group	IP	Type	State	STA Uptime	ExtraInfo
0	fa46-0442-0020	ap1	ap-group1	10.1.100.25	AirEngine5773-21	nor	0 10S	-

Configuring WLAN Service Parameters (1)



- Create a regulatory domain profile and configure the country code of the WAC.

```
[WAC-wlan] regulatory-domain-profile name domain
[WAC-wlan-regulate-domain-domain] country-code CN
[WAC-wlan-regulate-domain-domain] quit
```

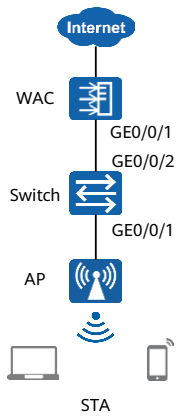
- Create the SSID profile **employee** and set the SSID name to **employee**.

```
[WAC-wlan] ssid-profile name employee
[WAC-wlan-ssid-prof-employee] ssid employee
[WAC-wlan-ssid-prof-employee] quit
```

- Create the security profile **employee** and set the security policy in the profile.

```
[WAC-wlan] security-profile name employee
[WAC-wlan-sec-prof-employee] security wpa-wpa2 psk pass-phrase a1234567 aes
[WAC-wlan-sec-prof-employee] quit
```

Configuring WLAN Service Parameters (2)



- Create a VAP profile, set the data forwarding mode and service VLAN, and bind the security profile and SSID profile to the VAP profile.

```
[WAC-wlan] vap-profile name employee
[WAC-wlan-vap-prof-employee] forward-mode tunnel
[WAC-wlan-vap-prof-employee] service-vlan vlan-id 101
[WAC-wlan-vap-prof-employee] security-profile employee
[WAC-wlan-vap-prof-employee] ssid-profile employee
[WAC-wlan-vap-prof-employee] quit
```

- Bind the VAP profile and regulatory domain profile to the AP group so that configurations in the profiles are applied to all radios on the APs.

```
[WAC-wlan] ap-group name ap-group1
[WAC-wlan-ap-group-ap-group1] vap-profile employee wlan 1 radio all
[WAC-wlan-ap-group-ap-group1] regulatory-domain-profile domain
Warning: This configuration change will clear the channel and power configurations of
radios, and may restart APs. Continue? [Y/N]:y
[WAC-wlan-ap-group-ap-group1] quit
```

Checking VAP Profile Information

- The WLAN service configuration is automatically delivered to the APs. After the service configuration is complete, run the **display vap ssid employee** command to check VAP profile information. If **Status** in the command output is displayed as **ON**, the VAPs have been successfully created for the corresponding AP radios.

```
[WAC-wlan] display vap ssid employee
WID : WLAN ID
Total: 2
```

AP ID	AP name	Rfid	WID	BSSID	Status	Auth type	STA	SSID	Radio Type
0	ap1	0	1	FA46-0442-0020	ON	WPA/WPA2-PSK	0	employee	11be
0	ap1	1	1	FA46-0442-0030	ON	WPA/WPA2-PSK	0	employee	11be

Quiz

1. (True or false) VAP profile can be bound only to an AP group but not to a single AP. ()
 - A. True
 - B. False
2. (Single-answer question) An engineer is configuring basic WLAN services on a Huawei WAC and requires that the wireless signal name be **Employee**. Which of the following configuration is correct? ()
 - A. [WAC-wlan-ssid-prof-employee] ssid employee
 - B. [WAC-wlan-ssid-prof-employee] ssid Employee
 - C. [WAC-wlan-ssid-prof-employee] ssid name employee
 - D. [WAC-wlan-ssid-prof-employee] ssid name Employee

1. B

2. B

Quiz

3. (Multiple-answer question) In the centralized architecture, Fit APs need to go online on the WAC so that the WAC can centrally manage and control them. Which of the following statements are true about the AP onboarding process? ()
- A. APs can obtain IP addresses through static configuration or DHCP.
 - B. The WAC centrally manages and controls APs through CAPWAP tunnel.
 - C. APs can discover WACs in static or dynamic mode.
 - D. The WAC and APs exchange Echo messages to detect the connectivity of both control tunnels and data tunnels.

3. ABC

Summary

- In the centralized networking architecture, the WLAN working process comprises AP onboarding, STA access, and data forwarding. This course describes the AP onboarding and STA access processes to help you understand the WLAN working mechanism and quickly troubleshoot faults if any during WLAN project deployment.
- This course also covers the WLAN service configuration process, including configuring network connectivity, configuring profiles, and binding profiles, helping you master the WLAN service configuration method.

Thank you.

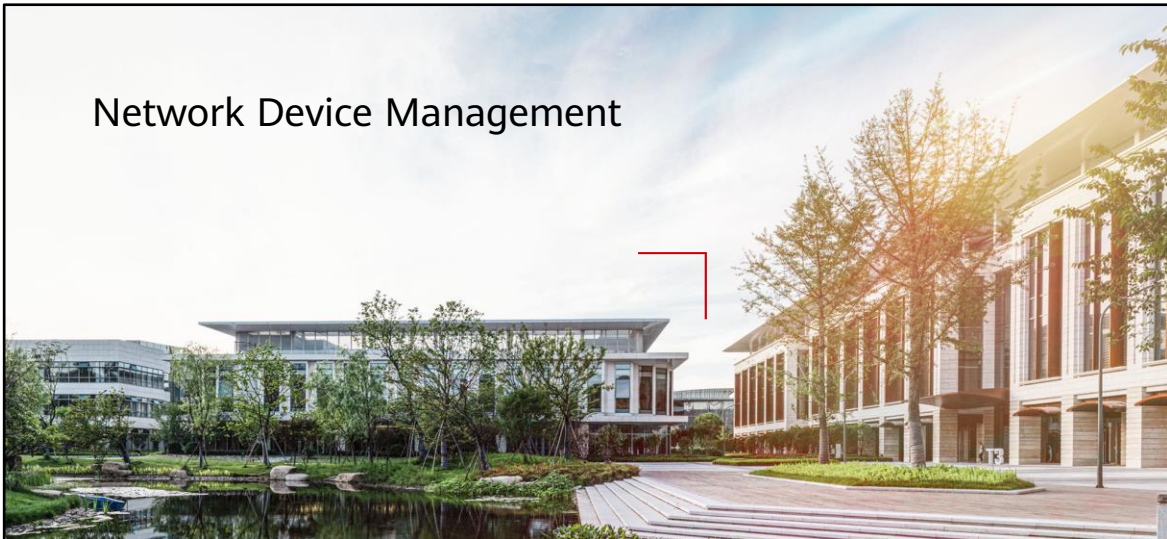
把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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Network Device Management



Foreword

- With the rapid development of new technologies such as artificial intelligence, big data, and cloud computing, various industries will be marching towards digital transformation in the near future. In addition, enterprise services will become diversified along with digital transformation, making network management one of the most challenging tasks for us. We cannot manage the entire network manually. Instead, we are in desperate need of an automated, smart tool to help us manage devices and resources on the network.
- This course starts with concepts related to network management, followed by a detailed description of the protocols used in network management and their respective application scenarios.

Objectives

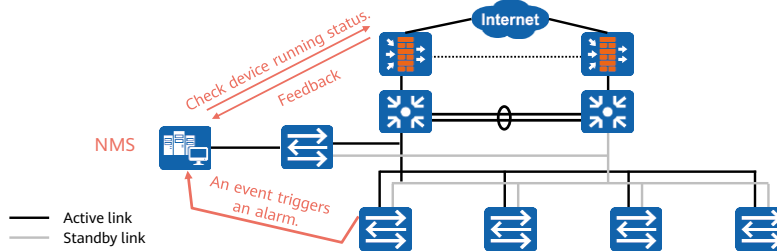
- Upon completion of this course, you will be able to:
 - Understand the functions of network management protocols.
 - Understand the similarities and differences between the implementation mechanisms and application scenarios of different network management protocols.
 - Describe the working principles of SNMP.
 - Describe the differences between NETCONF and RESTCONF.

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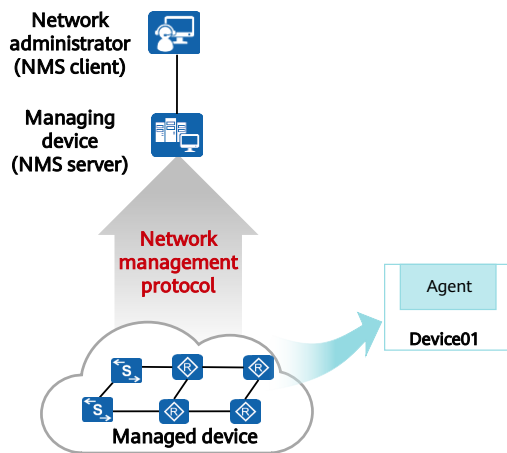
- 1. Basic Concepts of Network Management**
2. SNMP
3. NETCONF
4. RESTCONF
5. Introduction to iMaster NCE

What Is Network Management?

- Network management, simply put, refers to the management of network devices. By managing individual devices and analyzing their running status, a network management station (NMS) can determine the overall status of the network, extending the management scope from single devices to the whole network. Currently, users can manage a network using various protocols, tools, applications, or devices. The managed objects include hardware and software on the network.
- Network management ensures that a network runs properly by means of monitoring, testing, configuring, analyzing, evaluating, and controlling the software and hardware on the network.

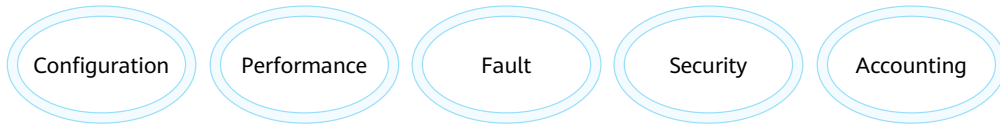


Typical Architecture of a Network Management System



- In most cases, network management systems share a common basic architecture consisting of two key components:
 - Managing device, also called an NMS. It adopts the client/server architecture where both the client and server are implemented as independent software applications. The client software runs on the network administrator's computer, while the server software runs on a physical or virtual server.
 - Managed device, also called an agent device. These are typically various types of network devices, such as switches and routers. A managed device communicates with the NMS through an agent process.
- Managing devices and managed devices communicate with each other using network management protocols.

Five Functions of Network Management

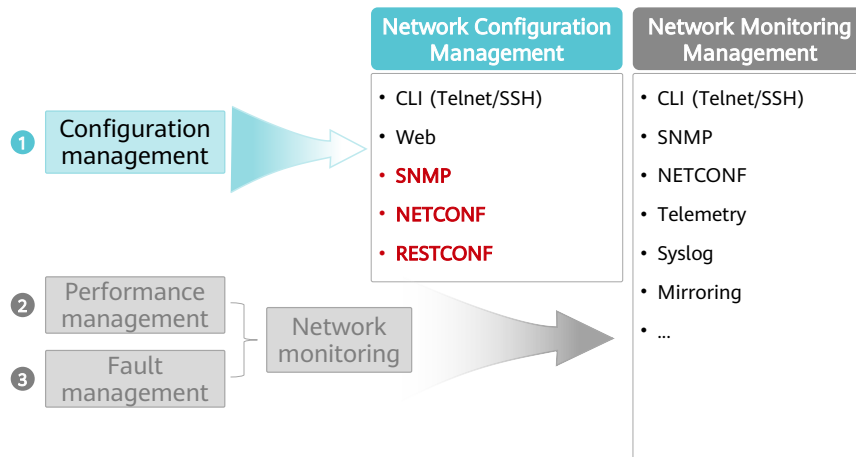


ISO/IEC 7984-4 (Part 4: Management Framework) defines five core functions of network management:

- Configuration management: configures network devices to support network services.
- Performance management: collects and measures network performance metrics to support network optimization.
- Fault management: detects, isolates, and resolves network faults.
- Security management: prevents unauthorized access and protect networks from security attacks.
- Accounting management: collects statistics on network resource usage and determines the cost of resource consumption.

- This course focuses on configuration management. For details about other functions, see related HCIP-Datacom courses.
- For more information, visit <https://www.iso.org/standard/14258.html>.

Network Management Overview



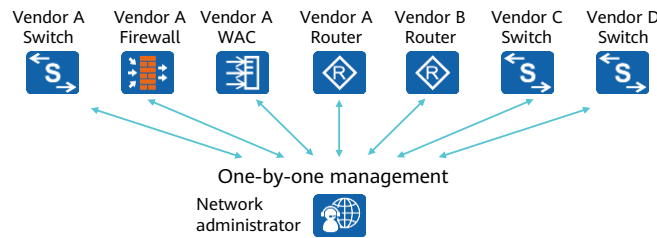
- The command-line interface (CLI) supports both network configuration management and network monitoring management.
- The web system provides a graphical interface for device management through the device's built-in web server. To use the web system to manage and maintain a device, you need to log in to the device through HTTP from a terminal.
- Simple Network Management Protocol (SNMP) supports configuration management through its Set operations and supports monitoring management through Trap notifications.
- Network Configuration Protocol (NETCONF) enables configuration management using Edit operations and supports monitoring management through Get operations.
- Representational State Transfer Configuration Protocol (RESTCONF) is an HTTP-based protocol that provides RESTful programming interfaces for creating, deleting, modifying, and querying data on network devices.
- Telemetry is a high-speed data collection technology that gathers data remotely from physical or virtual devices.
- Syslog is a widely used standardized log transfer protocol for network devices and computer systems. It enables the collection, storage, and management of logs generated by devices.
- This course focuses on protocols used for network configuration management. Topics related to network monitoring management will be covered in HCIP-Datcom courses.

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Web System- and CLI-based Management

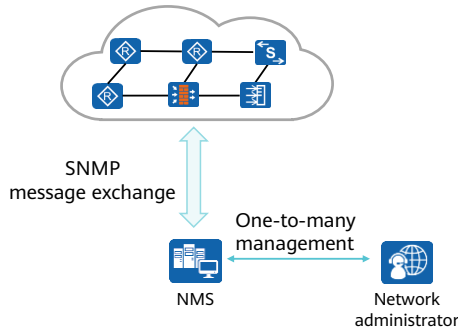
- In small-scale networks, the CLI and web system are commonly used for device management.
 - Network administrators can log in to devices through HTTPS, Telnet, or the console port for management.
 - These methods require no additional software programs or servers, making them cost-effective.
 - Network administrators must have a good master of network knowledge and vendor-specific network configuration commands.
 - These modes are not suitable for large-scale networks with complex network topologies.



- As network technologies rapidly evolve, network administrators face the following problems:
 - The number of network devices grows exponentially and coverage areas continue to expand. Network administrators face mounting challenges in device management, real-time monitoring, and fault locating.
 - Network management is becoming increasingly complex due to the diversity of network devices and varying management interfaces (such as the CLI) provided by different vendors.

SNMP-based Centralized Management

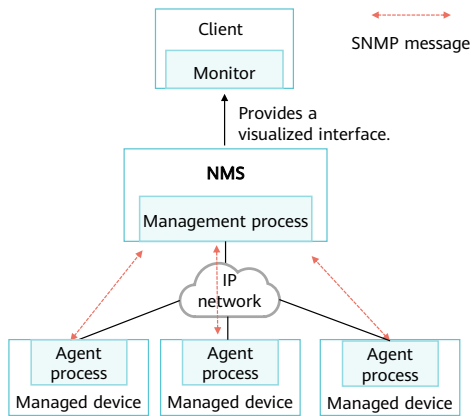
- Simple Network Management Protocol (SNMP) is a standard network management protocol widely used on TCP/IP networks. It provides a method for managing network elements (NEs) through a central computer that runs network management software, that is, an NMS.



- Network administrators can use an NMS to query and modify information and troubleshoot faults on any node on the network, improving work efficiency.
- SNMP enables unified management of network devices of different types and vendors by abstracting differences between them.

- There are three SNMP versions: SNMPv1, SNMPv2c, and SNMPv3.
 - In May 1990, RFC 1157 defined SNMPv1, the first SNMP version. RFC 1157 proposes a systematic method for monitoring and managing networks. SNMPv1 uses community string-based authentication, which has poor security. In addition, it supports only a limited set of error codes.
 - In 1996, the Internet Engineering Task Force (IETF) released RFC 1901 in which SNMPv2c is defined. SNMPv2c introduces enhancements such as a wider range of error codes, support for additional data types such as Counter32 and Counter64, and new operations including GetBulk and Inform.
 - Due to the lack of security protection measures in SNMPv2c, the IETF introduced SNMPv3. SNMPv3 provides user security module (USM)-based encryption and authentication and a view-based access control model (VACM).

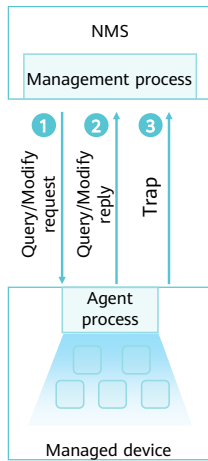
Typical SNMP Architecture



- On an SNMP-managed network, an NMS serves as the centralized management hub and runs a management process. Each managed device needs to run an agent process. The management and agent processes communicate with each other by exchanging SNMP messages.
- An NMS is a system that uses SNMP to manage and monitor network devices. It runs on an NMS server.
- Managed devices refer to devices managed by the NMS.
- An agent process runs on a managed device to maintain device information, respond to NMS requests, and report management data to the requesting NMS.

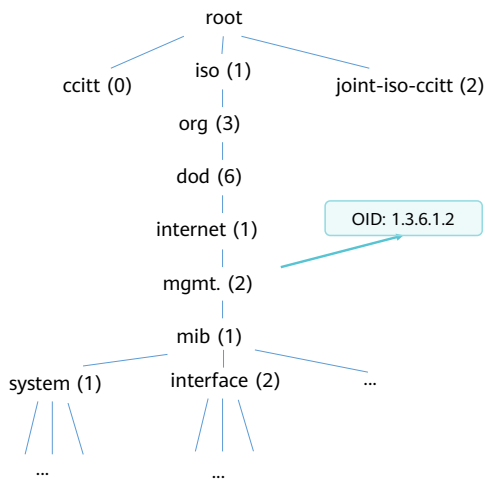
- An NMS is an independent device that runs a network management program. A network management program provides at least one man-machine interface through which the network administrator manages the network. Web-based interaction is a common man-machine interaction mode. That is, a network administrator uses a terminal with a monitor to access the web system provided by the NMS through HTTP or HTTPS.

SNMP Message Exchange



- An NMS and managed devices exchange messages in the following modes:
 - The NMS sends SNMP requests to managed devices to query or modify their configurations. The agent process running on each managed device then responds to these requests.
 - Managed devices proactively report SNMP traps to the NMS so that network administrators can detect faults in a timely manner.
- Managed object: It can be a hardware component or a set of parameters configured on the hardware or software (such as routing protocols) on a device. A managed device may contain multiple managed objects.
- SNMP uses management information bases (MIBs) to describe groups of objects within a manageable entity.

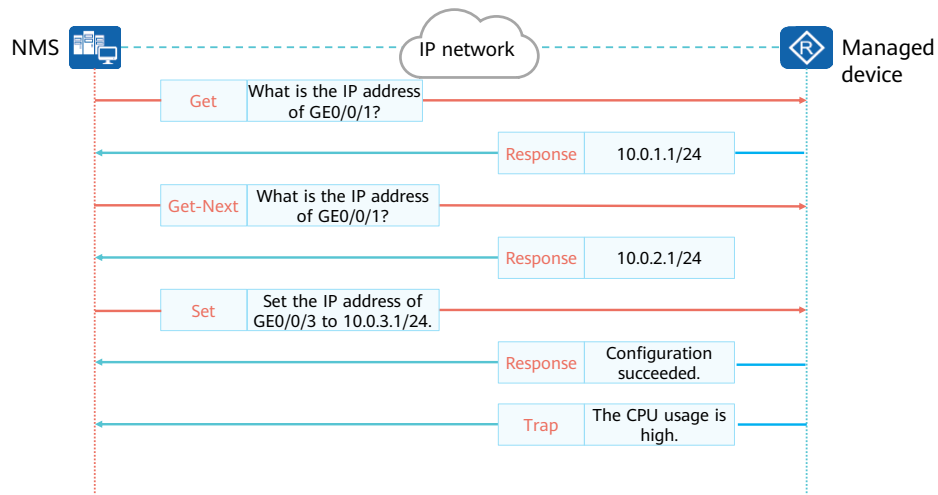
MIB



- A MIB is a database that stores variables maintained by a managed device and can be queried or set by the agent process. The following attributes are defined for managed objects in a MIB:
 - Object identifier (OID)
 - Status
 - Access permission
 - Data type
- A MIB provides a structure that organizes data of all objects that may be managed on the network. Due to its tree-like structure, a MIB is also referred to as an object naming tree.

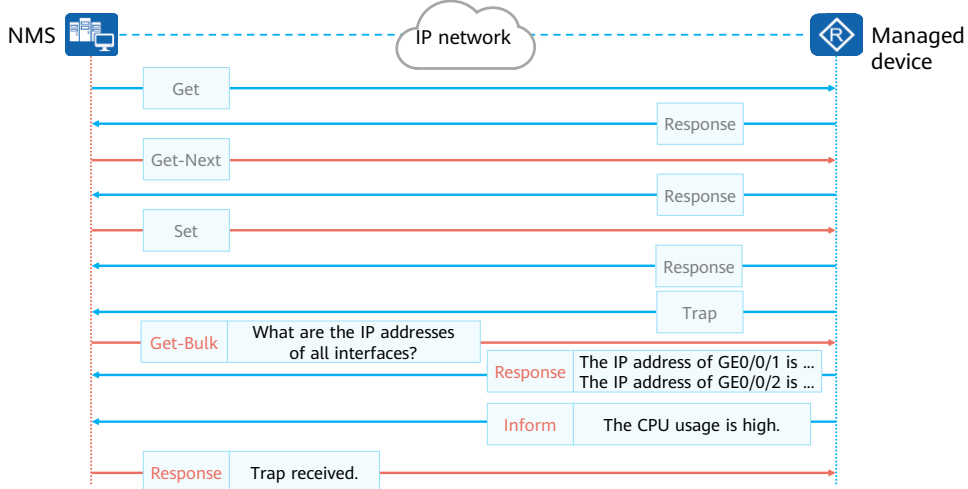
- MIBs are defined independently of network management protocols. Device vendors can integrate SNMP agent software into their products (for example, routers), and must ensure that this software complies with relevant standards after new MIBs are defined. The same network management software can be used to manage devices with different MIB versions, provided that they support MIBs. However, the network management software cannot manage a router that does not support the MIB function.
- MIBs can be classified into the following two types:
 - Public MIBs: defined by RFCs and used for structure design of public protocols and standardization of interfaces. Most vendors need to provide SNMP interfaces in compliance with RFC specifications.
 - Private MIBs: They are necessary supplements to public MIBs. Enterprises use them to enhance SNMP interface management when implementing proprietary protocols or specialized functions. Such MIBs also enable third-party NMS software to manage devices with proprietary protocols or specialized functions. For example, the MIB object of Huawei is 1.3.6.1.4.1.2011.

SNMPv1



- SNMPv1 defines five protocol operations:
 - **Get:** The NMS retrieves one or more parameter values from the MIB of the agent process on a managed device.
 - **Get-Next:** The NMS obtains the next parameter value in lexicographical order from the MIB of the agent process.
 - **Set:** The NMS sets one or more parameter values in the MIB of the agent process.
 - **Response:** The agent process returns one or more parameter values. It is the response to the first three operations.
 - **Trap:** The agent process actively sends messages to the NMS to report critical or major events.

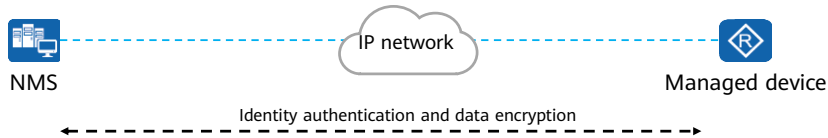
SNMPv2c



- SNMPv2c introduces two additional protocol operations:
 - **Get-Bulk:** consolidates multiple Get-Next operations into a single request. The number of Get-Next operations to be included in one Get-Bulk operation can be set on the NMS as needed.
 - **Inform:** enables a managed device to proactively notify the NMS of traps. Unlike Trap messages, Inform messages require acknowledgment. After sending an Inform request, the managed device expects an Inform response from the NMS. If no response is received, the device stores the trap in the Inform buffer and retransmits it until it receives an acknowledgment message from the NMS or the maximum retransmission limit is reached.

SNMPv3

- SNMPv3 has four functional models:
 - Message processing and control model: responsible for generating and analyzing SNMP messages, and determining whether the messages need to pass through proxy servers during transmission.
 - Local processing model: handles access control, data packaging, and data interruption.
 - User-based security model (USM): provides identity authentication and data encryption. To support these functions, the NMS and agent must share the same keys.
 - View-based access control module (VACM): implements view-based access control for user groups.



- Compared with SNMPv1 and SNMPv2, SNMPv3 implements access control, identity authentication, and data encryption through its local processing model and USM.

- SNMPv3 supports identity authentication and encryption.
 - Identity authentication: refers to the process in which the agent process verifies that a received message originates from an authorized NMS and has not been altered during transmission.
 - Encryption: SNMPv3 messages include header and security parameter fields. For example, when the management process sends an SNMPv3 Get-Request message carrying security parameters such as the username, key, and encryption parameters, the agent process responds with an encrypted message. This mechanism is particularly suited for scenarios where messages between the management process and agent process traverse public networks.

Summary

- SNMP has the following advantages:
 - Simple: As SNMP operates over UDP, it is widely supported across network devices.
 - Powerful: SNMP enables management information exchanges between arbitrary devices on a network. This allows administrators to query status and diagnose faults across the network.
- SNMPv1 is suitable for small-scale networks with minimal security requirements, simple topologies, and stable environments, such as campus or small enterprise networks.
- SNMPv2c is suitable for midsize and large networks where security is not a primary concern or the network environment is secure enough, but higher traffic volumes are expected and traffic congestion may occur.
- SNMPv3 is recommended for networks of all sizes, especially those requiring strong security. It ensures that only authorized administrators can manage devices.

SNMPv3 Configuration (1)

1. Start the SNMP agent service.

```
[HUAWEI] snmp-agent
```

2. Configure the SNMP version.

```
[HUAWEI] snmp-agent sys-info version v3
```

3. Specify the source interface used to receive and respond to requests from the NMS.

```
[HUAWEI] snmp-agent protocol source-interface { protocol-interface-type protocol-interface-number | protocol-interface-name }
```

4. Create or update MIB view information.

```
[HUAWEI] snmp-agent mib-view type view-name oid-tree
```

type indicates the operation type, which can be **excluded** (indicating that the MIB subtree is excluded) or **included** (indicating that the MIB subtree is included).

view-name specifies the name of an SNMP view.

oid-tree specifies an OID MIB subtree of the MIB object. It can be expressed either as the variable's OID string (such as 1.4.5.3.1) or as the variable's name string (such as system).

SNMPv3 Configuration (2)

5. Create an SNMP group and map users in this group to an SNMP view.

```
[HUAWEI] snmp-agent group v3 group-name { authentication | privacy | noauthentication } [ read-view read-view | write-view write-view | notify-view notify-view ] * [ acl { acl-number | acl-name } ]
```

This command creates an SNMP group of the SNMPv3 version, specifies authentication and encryption modes, and specifies one or more of read-only view, read-write view, and notification view for users in this group. It is mandatory on networks that require high security.

6. Add a user to the SNMP group.

```
[HUAWEI] snmp-agent [ remote-engineid engine-id ] usm-user v3 user-name [ group group-name | acl { acl-number | aclName } ] *
```

7. Configure an authentication password for the SNMPv3 user.

```
[HUAWEI] snmp-agent [ remote-engineid engine-id ] usm-user v3 user-name authentication-mode authentication-mode [ localized-configuration ] cipher authKey
```

SNMPv3 Configuration (3)

8. Configure an encryption password for the SNMPv3 user.

```
[HUAWEI] snmp-agent [ remote-engineid engine-id ] usm-user v3 user-name privacy-mode privacy-protocol [ localized-configuration ] cipher privKey
```

9. Configure the target host of trap messages.

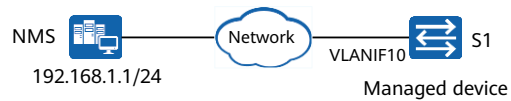
```
[HUAWEI] snmp-agent target-host [ host-name host-name ] trap address udp-domain ip-address params securityname { security-name [ [ v1 | v2c | v3 [ authentication | privacy ] ] }
```

10. Enable all trap functions.

```
[HUAWEI] snmp-agent trap enable
```

This command only enables the device to send traps. It must be used together with the **snmp-agent target-host** command that specifies the target host of the traps.

Example for Configuring SNMP for Device Management



- It is required that SNMPv3 be configured for device management to improve network configuration and maintenance efficiency.

```
[S1] snmp-agent
[S1] snmp-agent sys-info version v3
[S1] snmp-agent protocol source-interface vlanif10
[S1] snmp-agent mib-view included iso iso
[S1] snmp-agent group v3 admin privacy write-view iso notify-view iso read-view iso
[S1] snmp-agent usm-user v3 nms-admin group admin
[S1] snmp-agent usm-user v3 nms-admin authentication-mode sha2-256
Please configure the authentication password (8-255)
Enter Password: HCIA-Datacom
Confirm Password: HCIA-Datacom
[S1] snmp-agent usm-user v3 nms-admin privacy-mode aes128
Please configure the privacy password (8-255)
Enter Password: HCIA-Datacom
Confirm Password: HCIA-Datacom
[S1] snmp-agent target-host trap address udp-domain 192.168.1.1
params securityname nms-admin v3 privacy
[S1] snmp-agent trap enable
```

- For details about the NMS configuration, see the documentation of the corresponding product.

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Background of NETCONF

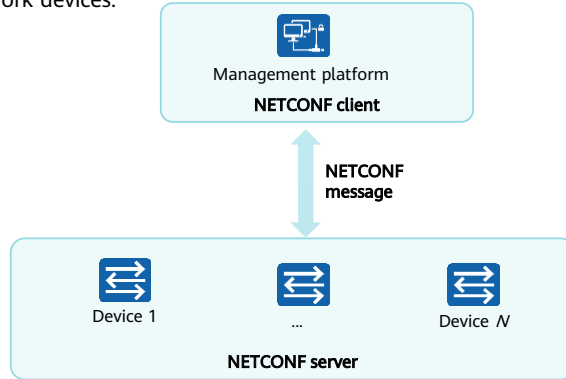
- As networks grow in scale and complexity, SNMP is no longer sufficient, especially for configuration management. This is where NETCONF comes in.

Function	SNMP	NETCONF
Configuration management	SNMP does not provide any locking mechanism when multiple users attempt to perform operations on the same configuration at the same time.	NETCONF provides a locking mechanism to prevent conflicts caused by concurrent operations.
Query	SNMP can be used to query one or more records in a table, but requires multiple interactions to complete a query.	NETCONF allows you to query configuration data of the entire system and supports data filtering.
Scalability	SNMP is poor in scalability.	NETCONF is highly scalable. It adopts a layered protocol model where each layer is independent. Extending one layer has minimal impact on upper layers. In addition, the use of XML encoding enhances both management capabilities and system compatibility.
Security	SNMPv2, released by the Internet Architecture Board (IAB) in 1996, provides only limited security improvements over SNMPv1. SNMPv3 was released in 2002. However, it still suffers from weak transport-layer encryption and no support for security protocol algorithms.	NETCONF leverages existing security protocols to ensure network security, but is not bound to any particular protocol. This makes NETCONF more flexible than SNMP in security protection.

- IAB: Internet Architecture Board.

NETCONF Network Architecture

- NETCONF provides a set of mechanisms for managing network devices. It allows users to add, modify, and delete device configurations, as well as retrieve configuration and status information from network devices.

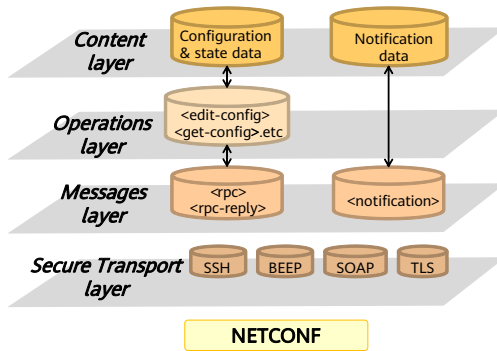


NETCONF involves three objects:

- NETCONF client
- NETCONF message
- NETCONF server

NETCONF Protocol Architecture

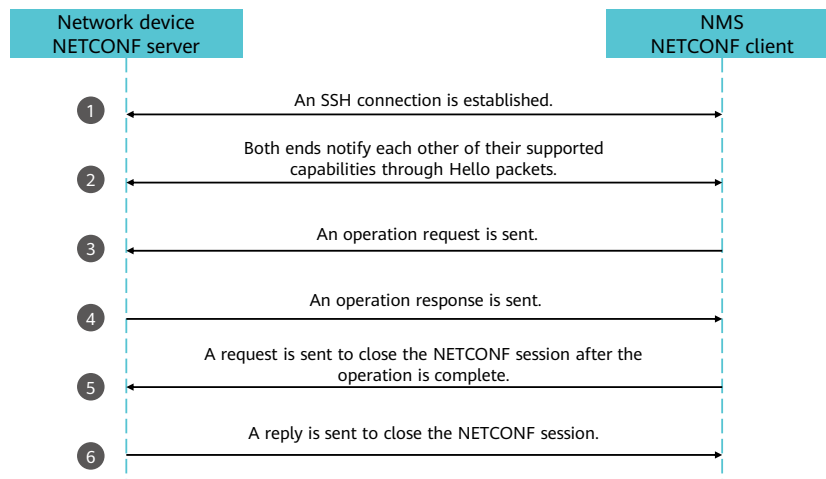
- NETCONF can be conceptually divided into four layers:



- Secure Transport layer: provides a communication path between the client and server. Currently, Huawei devices use Secure Shell (SSH) as the transport protocol for NETCONF.
- Messages layer: provides a simple Remote Procedure Call (RPC) request and response mechanism independent of the underlying transport protocol. The NETCONF client encapsulates an RPC request within an <rpc> element and sends it to the server. After processing the request, the server returns the result in an <rpc-reply> element.
- Operations layer: defines a group of fundamental operations as RPC invoking methods. These operations form the basic capabilities of NETCONF.
- Content layer: describes configuration data involved in network management. This data depends on vendors' devices. Mainstream data models include Schema and Yet Another Next Generation (YANG).

- BEEP: Blocks Extensible Exchange Protocol
- SOAP: Simple Object Access Protocol
- TLS: transport layer security
- Yet Another Next Generation (YANG) is a data modeling language used to define hierarchical data structures and support operations based on network configuration management protocols such as NETCONF. These operations include configuration data, status data, remote procedure calls (RPCs), and notifications. Due to its tight integration with NETCONF and RESTCONF, YANG has become a foundational technology for software-defined networking (SDN) and automated network O&M.
- A schema generally refers to a data scheme definition language, such as XML Schema or JSON Schema. It is used to describe the structure and constraints of documents.

NETCONF Interaction Process



1. An SSH connection is set up between the network device and NMS.
2. The network device and NMS exchange hello packets to advertise their supported capabilities.
3. After a NETCONF session is established between the network device and NMS, the NMS can send operation requests to the device for configuration management.
4. The network device parses and processes each received request and sends a response to the NMS.
5. After the requested operations are complete, the NMS send a request to close the NETCONF session, thereby avoiding unnecessary resource consumption on both the network device and NMS.
6. The network device closes the NETCONF session and sends a response to the NMS.

Datastores Supported by NETCONF

- When the NMS manages a network device through NETCONF, it maintains the device's configuration and state data in three standard datastores.

Datastore	Description	Relationship Among Them
<running/>	Holds the running effective configurations delivered to the device via NETCONF. The configurations in this datastore can be modified individually, and any modification takes effect immediately. It is the only mandatory standard datastore.	- Changes to the configuration in the <candidate/> datastore must be committed to the <running/> datastore to take effect. - At system startup, the configuration in the <startup/> datastore is automatically copied to the <running/> datastore. - The configurations in all these three datastores can be replaced as a whole through <copy-config> operations.
<candidate/>	Stores candidate configurations that have not taken effect. The configurations in this datastore can be modified individually, but any modification does not take effect immediately.	
<startup/>	Stores the configurations that will take effect upon next startup. The configurations in this datastore cannot be modified individually; they can only be replaced as a whole through the <copy-config> operation.	

Operations Supported by NETCONF

Operation	Description
<get-config>	Retrieves all or part of the configuration data from the <running/>, <candidate/>, and <startup/> datastores.
<get>	Retrieves all or part of the running configuration data or device status data from the <running/> datastore.
<edit-config>	Adds, modifies, or deletes the configuration data in the <running/> or <candidate/> datastore.
<copy-config>	Replaces the target datastore with the source datastore. If the target datastore does not exist, it will be created and populated with the configuration data in the source datastore.
<delete-config>	Deletes a datastore. This operation is not applicable to the <running/> datastore.
<lock>	Locks a datastore so that it can be modified exclusively. This operation prevents conflicts caused by concurrent operations from multiple users.
<unlock>	Releases a lock previously set by the same user. Locks set by other users cannot be released.
<close-session>	Closes a NETCONF session.
<kill-session>	Forcibly closes a NETCONF session. Only an administrator has the permission to perform this operation.

- Standard capabilities defined by NETCONF:
 - Writable-Running: indicates that the device supports the <edit-config> and <copy-config> operations on the <running/> datastore.
 - Candidate Configuration: indicates that the device supports the <candidate/> database.
 - Rollback on Error: indicates that rollback can be performed on the device in the case of errors.
 - Distinct Startup: indicates that the device can start independently and supports the <startup/> datastore.
 - Notification: indicates that the device can send traps and event notifications to the NMS, and the NMS manages the device based on the received traps and events.
 - Interleave: indicates that the device supports multiplexing different functions within the same NETCONF session. This allows O&M operations and trap/event management to be performed concurrently within the same session, improving management efficiency.

NETCONF Configuration Commands

1. Enable the NETCONF function.

```
[HUAWEI] snetconf server enable
```

```
[HUAWEI] protocol inbound ssh port 830
```

Both commands can enable NETCONF. If both commands are configured, the client can establish a NETCONF connection to the server using both port 22 and port 830.

2. Enable proactive NETCONF registration so that the device can actively establish a NETCONF connection with an NMS.

```
[HUAWEI] netconf
```

```
[HUAWEI-netconf] callhome callhome-name
```

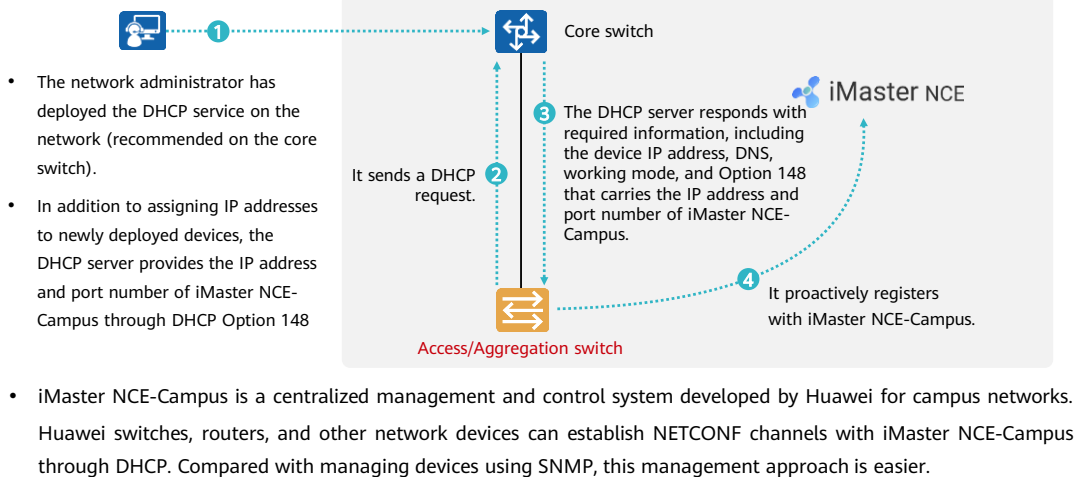
```
[HUAWEI-Netconf-callhome-name] endpoint endpoint-name
```

```
[HUAWEI-Netconf-callhome-name-endpoint-name] peer-ip ip-address port port-number
```

Enter the NETCONF user interface view, create a callhome template and enter its view, create a NETCONF connection instance and enter its view, and configure the IP address and TCP port number of the NMS that the device will connect to using NETCONF.

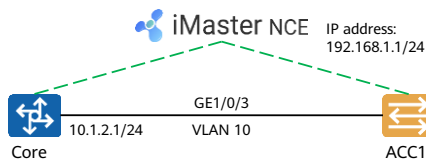
- There are two ways to configure a device to establish a NETCONF connection with an NMS:
 - Enabling proactive NETCONF registration: This method is applicable when the NMS does not support automatic device discovery. With proactive registration enabled, the device initiates a NETCONF connection request as soon as it comes online. For example, when iMaster NCE-Campus acts as the NMS, this mechanism allows the device to automatically establish a NETCONF session with the NMS so that the NMS can manage the device promptly.
 - Configuring a management VLAN: After a management VLAN is created, any device interface that belongs to this VLAN and is connected to a DHCP server can obtain the IP address of the NMS through DHCP. The device then automatically sends a NETCONF connection request to the NMS and establishes an NETCONF session. For the detailed configuration, see the product documentation.

Application Example of NETCONF



1. An administrator deploys the DHCP server function on the core switch or deploys an independent DHCP server on the network, and then configures DHCP Option 148 to carry the NETCONF enabling configuration and iMaster NCE-Campus's IP address/URL and port number.
2. When a switch starts up with factory defaults, it automatically sends a DHCP request to the DHCP server.
3. After receiving the request, the DHCP server replies with a DHCP message that includes Option 148. Based on the information in this option, the switch enables NETCONF and obtains the URL/IP address and port number of iMaster NCE-Campus.
4. Using the obtained URL/IP address and port number of iMaster NCE-Campus, the switch registers and authenticates with iMaster NCE-Campus.

Example for Configuring NETCONF for Device Management (1)



- An enterprise deploys an iMaster NCE-Campus server as an NMS. It is required that network devices be configured and managed on the NMS to improve configuration and maintenance efficiency.
- Procedure:
 - Onboard the core switch (Core) using the CLI.
 - Onboard the access switch (ACC1) using DHCP.

1. Enable the NETCONF function.

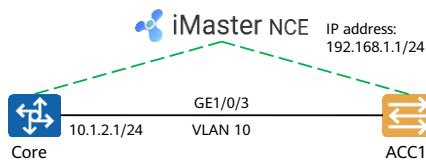
```
[Core] snetconf server enable
```

2. Configure the device to establish a NETCONF connection with the NMS.

```
[Core] netconf
[Core-netconf] callhome NCE-Campus
[Core-netconf-callhome-NCE-Campus] endpoint Campus01
[Core-netconf-callhome-NCE-Campus-endpoint-Campus01] peer-
ip 192.168.1.1 port 10020
[Core-netconf-callhome-NCE-Campus-endpoint-Campus01] quit
[Core-netconf-callhome-NCE-Campus] quit
[Core-netconf] quit
```

- In this example, S1700, S5700, and S6700 switches running V600R024C00 are used.
- In this example, the procedure for configuring an SSH server user and configuring the source interface of the SSH server is not described.
- After the core switch goes online, the onboarding configuration for access switches can be completed and delivered through iMaster NCE-Campus. The detailed operations on iMaster NCE-Campus are described in HCIP-Datacom courses.

Example for Configuring NETCONF for Device Management (2)



- An enterprise deploys an iMaster NCE-Campus server as an NMS. It is required that network devices be configured and managed on the NMS to improve configuration and maintenance efficiency.
- Procedure:
 - Onboard the core switch (Core) using the CLI.
 - Onboard the access switch (ACC1) using DHCP.

3. Configure the core switch as a DHCP server and configure Option 148.

```
[Core] dhcp enable
[Core] ip pool pool1
[Core-ip-pool-pool1] gateway-list 10.1.2.1
[Core-ip-pool-pool1] network 10.1.2.0 mask 255.255.255.0
[Core-ip-pool-pool1] option 148 ascii agilemanage-
mode=ip;agilemanage-domain=192.168.1.1;agilemanage-
port=10020
[Core-ip-pool-pool1] quit
[Core] vlan batch 10
[Core] interface ge 1/0/3
[Core-GE1/0/3] port link-type trunk
[Core-GE1/0/3] port trunk allow-pass vlan 10
[Core-GE1/0/3] quit
[Core] interface vlanif 10
[Core-Vlanif10] ip address 10.1.2.1 24
[Core-Vlanif10] dhcp select global
[Core-Vlanif10] quit
```

- **agilemanage-mode**: indicates the type of the iMaster NCE-Campus address obtained by the device. **agilemanage-mode=domain** indicates that the device obtains the URL, and **agilemanage-domain** must be set to that domain name. **agilemanage-mode=ip** indicates that the device obtains the IP address, and **agilemanage-domain** must be set to that IP address.
- **agilemanage-domain**: indicates the URL or IPv4 address of the active iMaster NCE-Campus system. One or more URLs or IPv4 addresses can be configured, separated by ampersands (&).
- **agilemanage-port**: indicates the port number of the active iMaster NCE-Campus system. One or more port numbers can be configured, separated by ampersands (&).

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Origin of RESTCONF

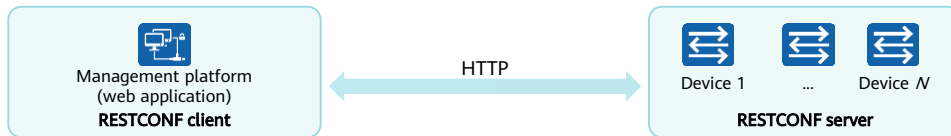
- Representational State Transfer Configuration Protocol (RESTCONF) was developed by combining the concepts of NETCONF with HTTP. It provides the core configuration management functions of NETCONF (create, read, update, and delete operations on network device data) through HTTP methods.

Item	NETCONF	RESTCONF
Transport channel (protocol)	Preferably uses SSH.	Uses HTTP to access device resources. RESTCONF provides programming interfaces that follow the RESTful style commonly used in the IT industry.
Packet format	Uses XML encoding.	Uses XML or JSON encoding.
Operation characteristics	NETCONF operations are relatively complex. NETCONF supports create, delete, modify, and query operations, multiple configuration datastores, and configuration rollback. • NETCONF supports the two-phase commit mechanism where parameter settings are submitted and then committed.	RESTCONF operations are simpler. • RESTCONF supports create, delete, modify, and query operations, and supports only the <running/> datastore. • RESTCONF operations take effect immediately; no two-phase commit is required.

- The increasing network scale and complexity drive the growing demand for automated O&M. NETCONF provides application programming interfaces (APIs) based on the RPC mechanism.
- Currently, network administrators expect devices to provide standardized interfaces that can be accessed and operated through web-based applications. NETCONF, however, cannot meet these requirements.

RESTCONF Network Architecture

- RESTCONF client: uses RESTCONF to manage network devices. It sends a request to the server to create, delete, modify, or query one or more pieces of data.
- RESTCONF server: maintains information about managed devices and responds to requests from clients. After receiving a request from a client, the server parses and processes it, and then returns a response to the client.
- The client must establish a secure, connection-oriented session with the server before communication can occur. Communication between the client and server follows a request-response messaging pattern, with both client requests and server responses encoded in either XML or JSON.



RESTCONF Operation Methods

- Mappings between RESTCONF and NETCONF operations:

RESTCONF	NETCONF
OPTIONS	N/A
HEAD	<get-config>, <get>
GET	<get-config>, <get>
POST	<edit-config> (nc:operation="create")
POST	An RPC operation is invoked.
PUT	<edit-config> (nc:operation="replace")
PATCH	<edit-config> (nc:operation="merge") when the operation object already exists
DELETE	<edit-config> (nc:operation="delete")

- OPTIONS: enables the client to query the server for the RESTCONF operation methods that the server supports.
- GET: retrieves configuration data or state data from the device.
- HEAD: checks whether device configuration or state data exists.
- POST: enables clients to create configuration data or invoke RPCs. It carries operation parameters in the HTTP message body.
- PUT: updates or replaces configuration data.
- PATCH: modifies configuration data.
- DELETE: deletes data and returns the status.

RESTCONF Configuration Commands

1. Enter the HTTP view.

```
[HUAWEI] http
```

2. Create and enter the service-RESTCONF view.

```
[HUAWEI-http] service restconf
```

3. Enable HTTP-based RESTCONF.

```
[HUAWEI-http-restconf] server enable
```

4. Specify the source interface used by the HTTP server to receive and respond to client requests.

```
[HUAWEI-http-restconf] server-source { interface-name | interface-type interface-number }
```

- This slide describes the commands for configuring HTTP-based RESTCONF. To configure HTTPS-based RESTCONF, see the product documentation.

Configuration Example for RESTCONF



- It is required that the network device be managed through HTTP.
- Configure the network device.
 - Configure network connectivity. (The detailed configuration is not provided here.)
 - Configure an AAA user. (The detailed configuration is not provided here.)
 - Enable the HTTP server function.

```
[HUAWEI] http
[HUAWEI-http] service restconf
[HUAWEI-http-service-restconf] server enable
[HUAWEI-http-service-restconf] server-source vlanif20
[HUAWEI-http-service-restconf] quit
[HUAWEI-http] quit
```

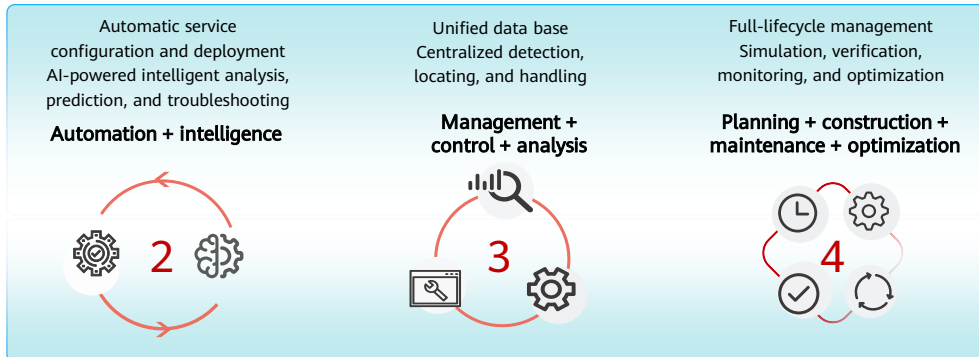
- This slide uses configuration commands on S1700, S5700, and S6700 (V600R024C00) series campus switches as an example.
- You need to configure the HTTP client. The commonly used HTTP client is Postman. To download the Postman installation package, visit <https://www.postman.com/downloads/>.

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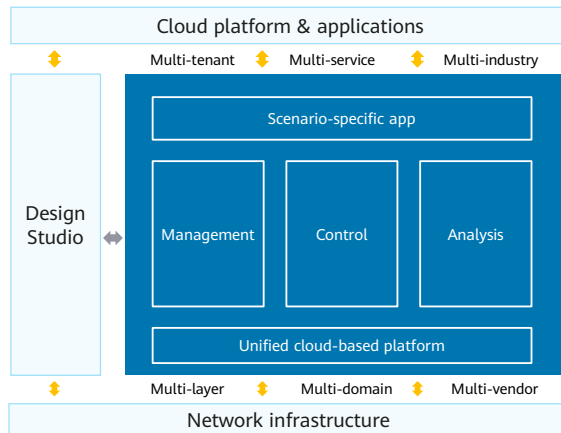
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Getting to Know iMaster NCE

- As the "brain" of cloud-based networks, iMaster NCE integrates network management, service control, and network analysis capabilities to enable resource pooling and intelligent O&M. Its fundamental value lies in centralized management and control, which drives networks toward full autonomy.



iMaster NCE Product Architecture and Customer Benefits



Automated network
Intent-driven, model-based
Multi-domain, multi-layer, and multi-vendor



Intelligent network
Big data- and AI-powered predictive
maintenance



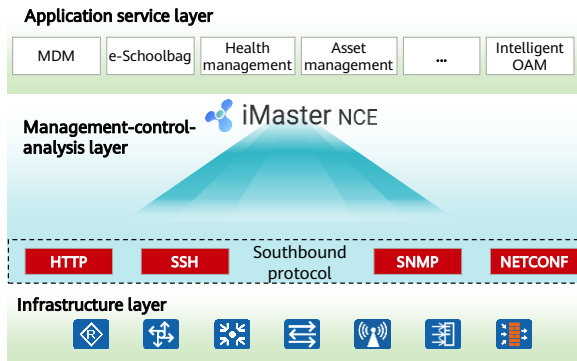
Agile system integration
Programmable design studio
Open REST APIs



Cloud-based elastic expansion
Deployment on both public and private clouds
Elastic expansion & high reliability

Specialized iMaster NCE products are designed for different data communication scenarios.

iMaster NCE for Campus Networks: iMaster NCE-Campus



- iMaster NCE-Campus is the primary configuration and management platform in Huawei's campus network solution.
- It can manage and configure network devices through protocols such as SNMP, NETCONF, HTTP, and SSH.

- Key features of iMaster NCE-Campus:
 - Simplified
 - Simplified network planning: iMaster NCE-Campus uses the cloud-based or offline WLAN Planner to provide building-level engineering views and automated device placement, making network planning simple and efficient.
 - Simplified network deployment: iMaster NCE-Campus provides a plug-and-play solution. In this solution, enterprise network devices can automatically register with iMaster NCE-Campus over the Internet. This implements zero-touch provisioning (ZTP) and end-to-end automated service deployment, making large-scale device deployment faster and easier.
 - Simplified service provisioning: iMaster NCE-Campus supports end-to-end automated network deployment and application-based intelligent traffic steering. It delivers differentiated network services for applications based on key application requirements, and detects link and application quality to preferentially ensure high-quality experience for critical applications through policy-based assurance.
 - Simplified network O&M: iMaster NCE-Campus collects and displays key information such as service traffic statistics, quality, alarms, and

logs across the entire network in real time. It provides an intuitive network topology and geographic information system (GIS) map, giving users full visibility of the network running status and helping them detect and handle problems in a timely manner.

- Elastic

- On-demand network expansion: iMaster NCE-Campus can manage large-scale and geographically distributed device access. Enterprises can expand their networks simply by adding devices and cloud services as needed. iMaster NCE-Campus also supports application-based intelligent traffic steering to deliver differentiated network services for applications based on key application requirements.
- On-demand management expansion: iMaster NCE-Campus supports multiple tenants. Enterprises can select a network management mode based on their capabilities and service requirements. They can either manage and maintain the network themselves or outsource operations to a managed service provider (MSP).

- Open

- Open network data: iMaster NCE-Campus provides enterprise tenants with data such as terminal authentication records and statistical analytics. Enterprises can retrieve and analyze this data to support business optimization.
- Open network platform: iMaster NCE-Campus offers standard northbound APIs, enabling partners and enterprise customers to develop third-party applications and value-added services. This openness supports collaborative building of the cloud ecosystem and continuous enterprise service innovation.

Quiz

1. (Multiple-answer question) Which of the following are basic functions of network management? ()
 - A. Configuration management
 - B. Performance management
 - C. Fault management
 - D. Security management
 - E. Accounting management
2. (True or false) Both NETCONF and RESTCONF allow users to modify, configure, delete, and query the <candidate/> datastore of network devices. ()
 - A. True
 - B. False

1. ABCDE

2. B

Summary

- As digital technologies continue to advance, innovations in big data, AI, and cloud computing, together with business development, are reshaping how networks are managed. These trends are pushing network management toward higher levels of automated management and intelligent O&M.
- Network management is typically structured around four models: organization model, information model, communication model, and functional model.
- SNMP offers three versions: SNMPv1, SNMPv2c, and SNMPv3. Depending on actual requirements, one or several of them can be deployed.
- NETCONF provides a mechanism for managing network devices. It allows users to add, modify, and delete device configurations, as well as retrieve configuration and status information from network devices.
- RESTCONF uses HTTP methods to configure devices.

Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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Network Programming and Automation



Foreword

- Network engineering is evolving rapidly, with new protocols, technologies, delivery modes, and O&M approaches emerging continuously. Traditional networks are under increasing pressure to support new connectivity requirements driven by cloud computing and artificial intelligence. At the same time, enterprises expect their services to be more agile, flexible, and resilient. In this environment, network automation has become ever more critical.
- Network programming and automation help engineers simplify tasks such as configuration, management, monitoring, and troubleshooting. By automating repetitive operations, engineers can deploy services faster and maintain networks more efficiently.
- This course is designed to introduce network engineers to the basics of Python programming and guide them in using Python to implement network automation.

Objectives

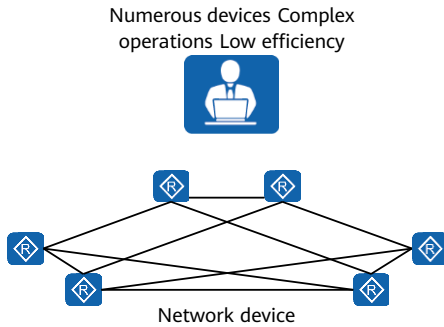
- Upon completion of this course, you will be able to:
 - Understand the challenges faced in traditional network O&M.
 - Describe the major categories of programming languages.
 - Understand Python coding conventions.
 - Use Python paramiko to log in to network devices.

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2. Overview of Programming Languages and Introduction to Python
3. Introduction to Paramiko APIs
4. Example for Network Programming Using Paramiko

Challenges in Traditional Network O&M

- Traditional network O&M requires network engineers to manually log in to network devices, run commands, and verify the configuration. These labor-intensive operations are time-consuming, inefficient, and difficult to audit.

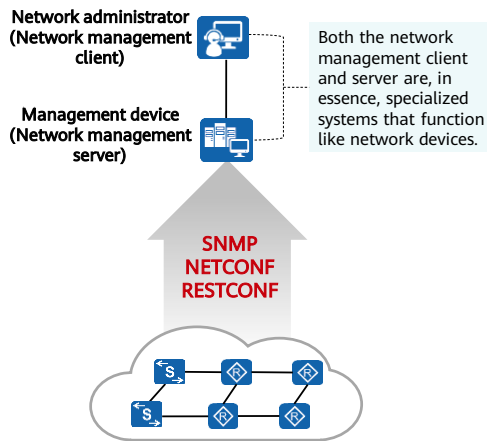


Typical O&M scenario

Have you ever encountered the following scenarios in your work?

1. Device upgrade: Thousands of live-network devices require periodic and batch upgrades.
2. Configuration audit: Enterprises require annual audits to verify device configurations. For example, all devices must have STelnet enabled, and Ethernet switches must implement spanning tree functions. It is essential to quickly identify non-compliant devices.
3. Configuration change: To meet security requirements, device accounts and passwords must be updated every three months. This involves removing existing accounts and creating new ones across thousands of devices.

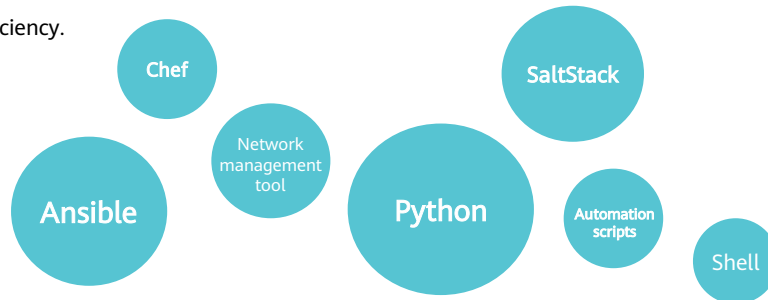
Advantages and Disadvantages of Network Management Software



- Advantages:
 - Network management protocols mask the differences in command line formats across devices from different vendors.
 - Many vendor-specific network management platforms support batch O&M operations, such as bulk software upgrades and configuration changes.
- Disadvantages:
 - Each vendor offers its own network management system with different software features. This often requires separate management systems for devices from different vendors.
 - Because network management software is usually updated only once a year, it can hardly adapt to new service needs quickly, resulting in limited flexibility.
 - Compared with logging in to devices one by one, administrators still need to log in to a management platform to perform operations, which does not fundamentally simplify the overall workflow.

Network Automation

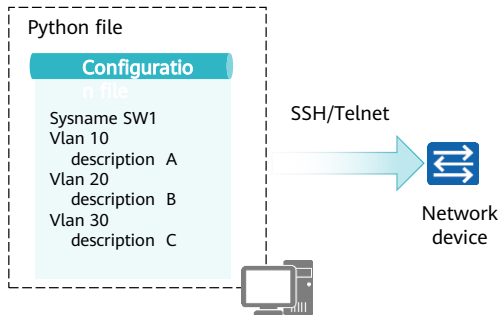
- Network automation refers to automating network deployment, operation, and O&M using tools. It helps minimize human intervention and address challenges in traditional O&M.
- There are many open-source tools for network automation in the industry, such as Ansible, SaltStack, Puppet, and Chef. To build network engineering capabilities, it is recommended that engineers have coding proficiency.



- Network automation is not applicable to all scenarios. For example, occasional or one-off configuration changes may not be worth the considerable amount of time it takes to develop automation scripts.
- Network automation is highly effective in environments with large workloads and numerous repetitive tasks, such as routine device inspections.
- Python is a concise, elegant, cross-platform, and highly extensible high-level interpreted programming language. Designed with the philosophy of making code as readable as English, it has become widely used in areas such as AI and automated O&M.
- Ansible is an open-source automation O&M tool built on Python. It uses SSH to perform batch system configuration, application deployment, and task orchestration.
- SaltStack is an open-source configuration management and remote execution framework written in Python. It adopts the client/server architecture, with the server referred to as the Master and the clients as Minions.
- Puppet is an open-source automation tool written in Ruby. It communicates over HTTPS and requires an agent to be installed on each managed node.
- Chef is an open-source O&M automation and configuration management tool designed to simplify configuration, management, and deployment through an infrastructure-as-code approach.
- A shell script is a collection of shell commands stored in a file, allowing those commands to be executed automatically simply by running the file.

Programming-based Network Automation

- With the rise of network automation technologies in recent years, programming, especially in Python, has become an essential skill for network engineers.
- Automation scripts written in Python are suitable for handling repetitive, time-consuming, and rule-based tasks.



Example: using Python to automate device configurations

- What can network automation do? A fundamental use case of network automation is automatic device configuration. This process involves two steps: compiling configuration files and writing Python code to deploy them to devices.
- Configuration files can be written in command line format, and then uploaded to devices using a Python program. This approach is straightforward and accessible for network engineers new to programming and automation. This course focuses on implementing network automation using this approach.

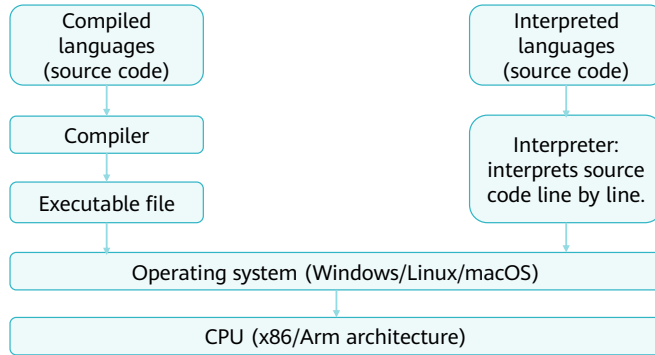
- In addition to device-level automation, mainstream network vendors provide programming interfaces for their network management software as well.
- This course focuses only on device-level automation. For programming against vendor-specific management software, refer to the development guides provided by each vendor.

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Programming Languages

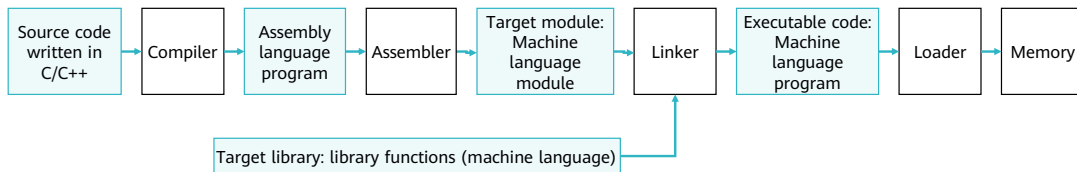
- A programming language is used to write computer programs that define how a computer behaves.
- Programming languages can be categorized by whether they require compilation before execution: compiled languages must be compiled prior to running, while interpreted languages do not.



- Programming languages can also be categorized by their level of abstraction into machine languages, assembly languages, and high-level languages.
 - Machine language is composed of binary instructions—0s and 1s—that computers can interpret and execute directly.
 - Because machine language is difficult for humans to read and understand, these binary instructions were later given symbolic representations (such as MOV and ADD) to make them easier to recognize and remember. This form of language is known as assembly language.
 - The preceding two languages are low-level languages, and all other languages are considered high-level languages, such as C, C++, Java, Python, Pascal, Lisp, Prolog, FoxPro, and Fortran. Programs written in high-level languages cannot be executed directly by computers; they must first be translated into machine language.

Compiled Languages

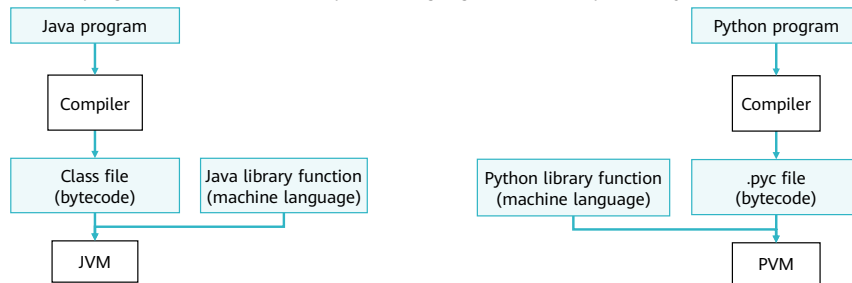
- Programs written in compiled languages require compilation before execution. Source code is translated into a machine language program, which can be executed directly without further translation at runtime. C, C++, and Go are typical compiled languages.
- Process from source code to an executable program: Source code is first translated into machine instructions by a compiler and an assembler. A linker then combines these instructions with functions from libraries to produce a complete machine-language program. Because machine language must match the CPU's instruction set, the resulting program is loaded into memory by a loader at runtime and executed directly by the CPU.



- Code written in compiled languages is translated directly into machine-executable files (for example, .exe files) during compilation. Because compilation and execution are separate steps, the resulting program cannot run across different platforms. For instance, a program compiled for an x86 architecture cannot run on an ARM-based server.

Interpreted Languages

- Programs written in interpreted languages do not require compilation before execution. Instead, they are translated line by line at runtime. Java and Python are typical interpreted languages.
- Process from source code to an executable program: Source code of an interpreted language is first translated into bytecode by a compiler, and then interpreted and executed by a virtual machine, such as a Java Virtual Machine (JVM) or Python virtual machine (PVM). Because virtual machines abstract away differences between CPU instruction sets, programs written in an interpreted language have better portability.



- JVM: Java Virtual Machine.
- PVM: Python Virtual Machine.

What Is Python?

- Python is a fully open-source, high-level programming language, created by Guido Van Rossum.

Advantages:

- Provides elegant and readable syntax, helping engineers focus on program logic rather than low-level syntax details.
- Supports both procedural and object-oriented programming paradigms.
- Offers a rich ecosystem of third-party libraries.
- Provides strong interoperability, allowing Python to interface with code written in other languages (which is why it is often described as a "glue language").

Disadvantages:

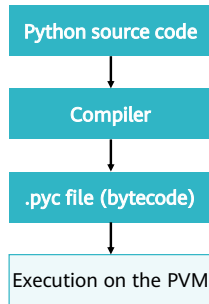
- Slow in execution: Because Python is an interpreted language, there is no need to compile Python code before execution. Instead, Python code is translated into machine instructions that the CPU can understand, line by line at runtime, which can be time-consuming.

Despite this, Python's extensive third-party library ecosystem and its inherent language strengths make it suitable for a wide range of fields, including AI, data science, and O&M script automation.

- Python is also a dynamically typed language. A dynamically typed language determines the type of an object automatically at runtime, so there is no need to declare variable types.

Python Code Execution Process

Python program compilation and execution



Operation procedure

1. Install Python on the OS.
2. Write source code in Python.
3. The compiler translates source code into bytecode (.pyc file).
4. The PVM interprets the bytecode and converts it into machine instructions.
5. The hardware executes the machine instructions.

- Python source code does not need to be compiled into binary code. A program can be executed directly from source code. During Python code execution, the Python interpreter first translates source code into bytecode, which is then executed by the PVM.
- The PVM is not a separate program and does not need to be installed independently.

Getting Started with Python Coding: Code Execution in Interactive Mode

- Python programs can be executed in two ways: interactive mode and script mode.
- Python programming with interactive execution does not require creating script files. Code can be written and executed directly in the Python interpreter's interactive prompt.

```
C:\Users>python
Python X.X.X (default, Aug 9 XXXX, 18:34:13) [MSC v.1915 64 bit (AMD64)] ::
Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
1. Input  -- >>> print ("hello world")
2. Output -- hello world
3. Input  -- >>> a = 1
4. Input  -- >>> b = 2
5. Input  -- >>> print ( a + b )
6. Output -- 3
>>>
```

Getting Started with Python Coding: Code Execution in Script Mode

- In script mode, Python code can be executed using various Python interpreters or integrated development environments (IDEs), such as IDLE (bundled with Python), Atom, Visual Studio, PyCharm, and Anaconda.

demo.py

```
print ("hello world")  
a = 1  
b = 2  
print ( a + b )
```

```
1. Input  -- C:\Users\Richard>python demo.py  
2. Output -- hello world  
3. Output -- 3
```

1 Compile a Python script file.

2 Execute the script file.

Python Coding Conventions

- Coding conventions define the rules that should be observed when you write Python code, including naming practices, indentation, and how code and statements are structured. Following these conventions improves code readability and makes code easier to maintain and modify.
- For example, recommended practices for using semicolons, parentheses, blank lines, and spaces are as follows:

Semicolon

- Python allows semicolons at the end of a line, but using them to separate statements is not recommended.
- You are advised to write one statement per line.

Blank line

- Blank lines can be used to separate functions or code blocks. This helps distinguish sections of code and improves overall readability.

Parentheses

- Parentheses can be used to continue a long statement over multiple lines. Unnecessary parentheses should be avoided.

Space

- Spaces are not recommended inside parentheses.
- Whether to add spaces around operators is a matter of personal preference.

Identifier Naming

- Python identifiers are used to represent the names of constants, variables, functions, and other objects.
- Identifiers typically consist of letters, digits, and underscores (_), but must not start with a digit. They are case-sensitive and must be unique. If an identifier does not comply with these rules, the compiler will raise a **SyntaxError** when executing the code.

What is the output?

1. Numeric assignment	--	User_ID = 10	print (User_ID)
2. Numeric assignment	--	user_id = 20	print (user_id)
3. String assignment	--	User_Name = 'Richard'	print (User_Name)
4. Numeric assignment	--	Count = 1 + 1	print (Count)
5. Invalid identifier	--	4_passwd = 'Huawei'	SyntaxError

print() is a built-in Python function used to output the value inside the parentheses.

- The fundamental data types in Python include Boolean values (True/False), integers, floating-point numbers, and strings. In Python, all data—whether Boolean values, integers, floating-point numbers, strings, as well as complex data structures, functions, and even programs—exist in the form of objects. This object-based design gives Python a high level of consistency.
- In the example above, the output values are 10, 20, Richard, and 2. The identifier 4_passwd is invalid because it begins with a digit, and Python will raise a syntax error when encountering it.
- This document does not provide a full introduction to Python syntax. For more details, see the official documentation at <https://www.python.org/>.

Code Indentation

- In Python, indentation defines the scope of a code block. If a code block contains two or more statements, all of them must use the same level of indentation. Indentation is a fundamental part of Python's syntax. Together with colons (:), it indicates the code structure and hierarchy.
- When writing Python code, you are advised to use four spaces for each level of indentation. If incorrect indentation is used, Python will raise an **IndentationError** at runtime.

Correct indentation	--	if True:
		print ("Hello")
Correct indentation	--	else:
		print (0)
Incorrect indentation	--	a = "Python"
		print (a)

- The **if...else...** structure forms a complete code block, and all statements within it must use the same indentation level.
- The statement **print(a)** uses **a** as its argument and belongs to the same code block as the **if...else...** statement, so it must be indented at the same level.

Comments

- Comments are explanatory notes added to code, helping improve code readability. In Python, a comment may appear on a single line or span multiple lines.
- A single-line comment starts with a number sign (#) and continue to the end of the line.
- A multi-line comment spans several lines and is enclosed within a pair of triple quotes (''' ... ''' or """ ... """).

Single-line comment	--	<pre>#Assign a character string to variable a. a = "Python" print (a)</pre>
Multi-line comment	--	<pre>""" The output is Python. """</pre>

Source File Structure

- A complete Python source file consists of interpreter and encoding declarations, docstring, module import statements, and executable code.
- When a program needs to use classes from the standard library or third-party libraries, the corresponding modules must be imported using import or from ... import statement. Import statements always appear at the top of the file, following module comments or the docstring.

Interpreter declaration	--	#!/usr/bin/env python
Encoding format declaration	--	#-*- coding:utf-8 -*-
Module comments or docstring	--	"""Document description (docstring)
		This document is used to...
		"""
Importing the time module	--	import time
Executable code	--	...

- The interpreter declaration specifies the path of the compiler that runs the file. This is useful when the compiler is installed in a non-default location or when multiple Python compilers are installed. On Windows, this declaration is unnecessary.
- The encoding declaration specifies the character encoding used in the source file, ensuring that the code is read correctly. Python 3 uses UTF-8 as the default encoding.
- The docstring provides a general description of the program's purpose and functionality.
- The time module is a built-in Python module that provides time-related functions.

Functions and Modules

- A function is an organized, reusable block of code that helps improve program modularity and promotes code reuse. Functions are defined using the keyword **def**.
- A module is a saved Python file that can contain functions or classes. The only difference between a module and a regular Python program lies in their intended use: a module is designed to be imported and invoked by other programs. For this reason, a module typically does not include a main function.

demo.py file

```
def sit(): #Define a function.  
    print ('A dog is now sitting')  
  
sit() #Call the function.
```

Execution result:

A dog is now sitting.

- 1 Compile a Python file (.py).

test.py file

```
import demo #Import a module.  
  
demo.sit() #Call the function.
```

Execution result:

A dog is now sitting.
A dog is now sitting.

Why are two lines output?

- 2 Invoke a module.

- The test.py file contains two lines of code. The first line imports the code from the **demo.py** file. When the code is imported, a line of output is automatically printed. The second line calls the sit() function, which generates another line of output.

Classes and Methods

- A class is a blueprint that defines a group of objects sharing the same attributes and behaviors. Classes are defined using the keyword **class**.
- Functions defined inside a class are called methods when the class is instantiated. Methods inside a class typically include the keyword **self**, which represents the instance of the class itself.

demo.py file

```
class Dog(): #Define a class.
    def sit(self): #Define a method.
        print("A dog is now sitting.")
Richard = Dog() #Instantiate the class.
Richard.sit()
print (type(Richard.sit)) #The type of the instantiated class is method.
print (type(Dog.sit)) #The type is function.
```

Execution result:

```
A dog is now sitting.
<class 'method'>
<class 'function'>
```

1 Compile a Python file (.py).

test.py file

```
import demo
demo.Dog.sit
```

Execution result:

```
A dog is now sitting.
<class 'method'>
<class 'function'>
```

2 Invoke a module.

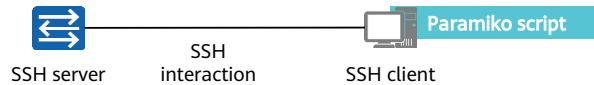
- According to the Python official website, functions and methods are defined as follows:
 - Function: A series of statements which returns some value to a caller. It can also be passed zero or more arguments which may be used in the execution of the body.
 - Method: A function which is defined inside a class body. If called as an attribute of an instance of that class, the method will get the instance object as its first argument (which is usually called self).
- type() is a built-in Python function used to obtain the type of an object and to dynamically create classes.
- For more information, visit <https://docs.python.org/3/tutorial/classes.html>.

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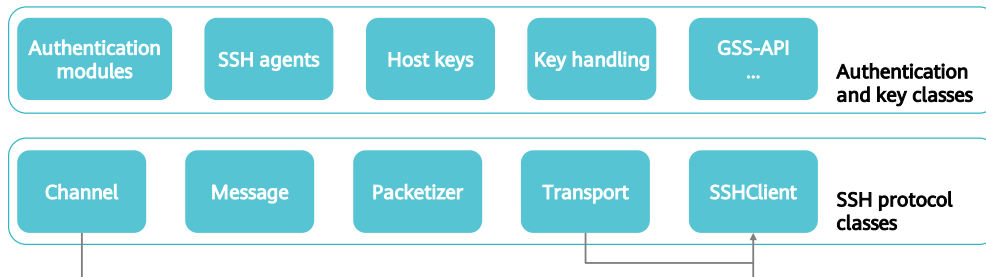
Introduction to Paramiko

- Paramiko is a Python module that implements SSHv2. It supports both password and public key authentication, enabling functions such as secure remote command execution and file transfer.
- Engineers can use paramiko to write Python code to implement SSH functionalities.



Introduction to Paramiko APIs

- Paramiko provides several key classes, primarily categorized into the following:
 - Authentication and key classes: used to generate, load, and manage SSH keys.
 - SSH protocol classes: used to establish and manage SSH sessions and channels.
- The SSHClient class provides a high-level interface for interacting with an SSH server. It encapsulates Transport, Channel, and SFTPClient to handle authentication and channel setup.



- An application programming interface (API) is a collection of predefined functions, such as the paramiko classes shown on this page, that allow different software systems to communicate with one another. With APIs, you do not need to understand the internal implementation of other systems. You simply send requests in the defined format to retrieve data or trigger specific functions.
- In most cases, when working with paramiko, you only need to create an SSHClient instance. This course focuses on practical examples and commonly used methods of the SSHClient class.
- Other classes and methods are not described in this course. For more information, visit <https://docs.paramiko.org/en/stable/#authentication-keys>.

Typical Use Cases and Methods of the SSHClient Class

- Typical use case of the SSHClient class:

```
client=paramiko.client.SSHClient() #Create an SSHClient instance.  
client.connect (hostname='192.168.56.100', port=22, username='client', password='123456') #Set up an SSH connection.  
stdin,stdout,stderr=client.exec_command ('dis cu') #Execute a command.
```

Common Method	Function
connect()	Establishes a connection and authenticates with a remote server.
set_missing_host_key_policy()	Defines the policy for connecting to servers without a known host key.
exec_command()	Executes Linux commands on a remote server.
invoke_shell()	Starts an interactive shell session on a remote server.
close()	Closes the connection.

Introduction to Methods: connect()

- Purpose: establishes a connection and authenticates with a remote server.
- Sample:

```
client.connect(hostname='192.168.56.100',port=22,username='client',key_filename='id_rsa')
```

```
client.connect(hostname='192.168.56.100',port=22,username='client',password='123456')
```

Parameter	Description
hostname	Target host to be connected. This is the only mandatory parameter.
port	Port number. The default value is 22.
username	User name for authentication. This parameter is left blank by default.
password	User password for authentication. This parameter is left blank by default.
key_filename	File name or file list that specifies the private key file. This parameter is left blank by default.
pkey	Private key used for identity authentication.
...	...

Introduction to Methods: set_missing_host_key_policy

- Purpose: defines the policy for connecting to servers without a known host key.
- Sample:

```
client.set_missing_host_key_policy(paramiko.client.AutoAddPolicy())
```

Parameter	Description
AutoAddPolicy	Automatically adds the host name and host key to the local HostKeys object, regardless of the load_system_host_key configuration. No confirmation (yes or no) is required when establishing a new SSH connection.
WarningPolicy	Accepts the connection if the host key is unknown and logs a Python warning. It functions similarly to AutoAddPolicy , but prompts a new connection.
RejectPolicy	Automatically rejects the connection if the host name or key is unknown. This is the default policy.

Introduction to Methods: exec_command

- Purpose: executes commands on a remote server.
- Sample:

```
stdin,stdout,stderr = client.exec_command('dis cu') #Execute the dis cu command.  
print(stdout.read().decode()) #Print the output.
```

- stdin: sends input data. If the command requires additional input (such as passwords or confirmation prompts), data can be written using stdin.write().
- stdout: reads the normal output of the command. The stdout.read() or stdout.readlines() method can be used to retrieve the output.
- stderr: reads error information generated during command execution. If the command fails to be executed or an error is reported, error information is displayed here.

Only one command can be executed at a time.

Introduction to Methods: invoke_shell

- Purpose: starts an interactive shell session over an SSH session.

- Sample:

```
cli = client.invoke_shell()
cli.send('screen-length 0 temporary\n')    #Issue the first command.
cli.send('display current-configuration\n') #Issue the second command.
print (cli.recv(65535).decode('utf-8'))    #Print the output.
```

Multiple commands can be sent.

- `cli.recv(65535).decode('utf-8')`
 - `cli.recv(65535)`: The `recv` method is used to receive data through the SSH channel. The argument 65535 defines the size of the receive buffer, meaning that it can buffer up to 65,535 bytes of data. This value can be adjusted as needed, but it is typically set to a large number to ensure that the complete command output can be received.
 - `.decode('utf-8')`: The received data is presented in bytes. The `decode('utf-8')` method decodes the bytes into character strings, making them easier to process and display.

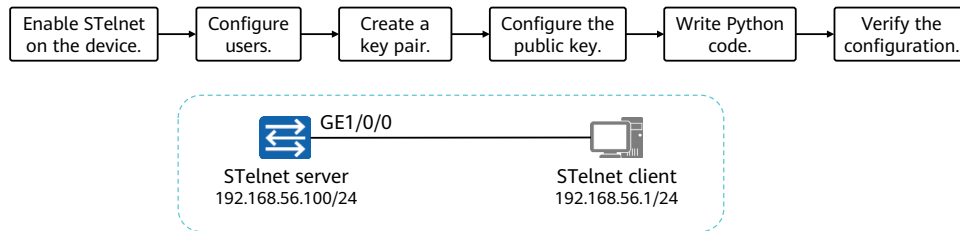
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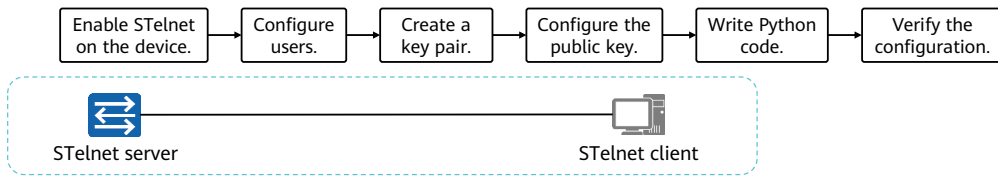
Example: Logging In to a Device Using SSH

- Case description:

- As shown in the figure, after the STelnet server function is enabled on the switch functioning as the SSH server, an SSH client can log in to the SSH server using various authentication methods, such as password or RSA authentication.
- This example demonstrates how an SSH client uses Python paramiko to log in to an SSH server using RSA authentication.



Configuration Roadmap



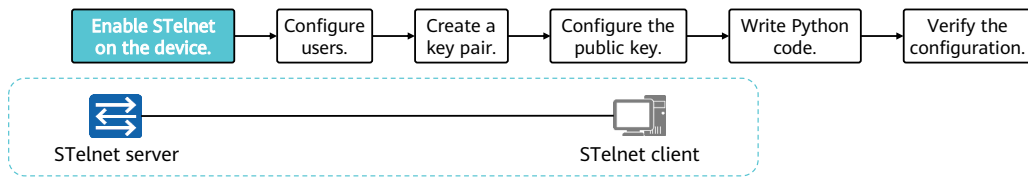
- Server configuration:

- Enabling STelnet: Configure a management IP address, enable STelnet, and configure user interfaces.
- Configuring users: Create local and SSH users, and configure the service type and authentication mode for these users.
- Configuring the public key: Add the public key generated by the client and allocate it to the users.

- Client configuration:

- Creating a key pair: Generate a local pair of public and private keys.
- Write Python code.
- Verify the configuration.

Example: Configuring STelnet on the Server



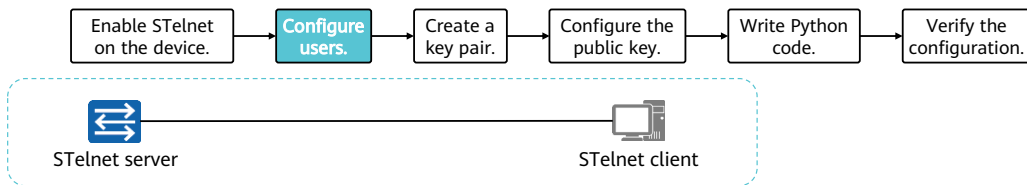
1. Configure an IP address for the management interface on the server.

```
<HUAWEI> system-view immediately
[HUAWEI] sysname SSH Server
[SSH Server] interface GE 1/0/0
[SSH Server-GE1/0/0] ip add 192.168.56.100 24
[SSH Server-GE1/0/0] quit
```

2. Enable STelnet on the server and configure VTY user interfaces.

```
[SSH Server] stelnet server enable
[SSH Server] ssh server-source all-interface
[SSH Server] user-interface vty 0 4
[SSH Server-ui-vty0-4] authentication-mode aaa
[SSH Server-ui-vty0-4] protocol inbound ssh
[SSH Server-ui-vty0-4] quit
```

Example: Configuring Users on the Server



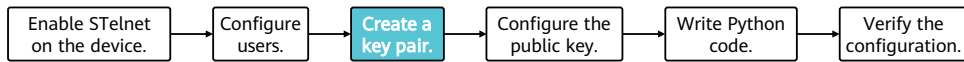
3. Create a local user and configure a service type for the user.

```
[SSH Server] aaa
[SSH Server-aaa] local-user client password irreversible-cipher Huawei@123
[SSH Server-aaa] local-user client privilege level 3
[SSH Server-aaa] local-user client service-type ssh
[SSH Server-aaa] quit
```

4. Create an SSH user, and configure an authentication mode and service type for the user.

```
[SSH Server] ssh user client
[SSH Server] ssh user client authentication-type rsa
[SSH Server] ssh user client service-type stelnet
```

Example: Creating an RSA Key Pair on the Client



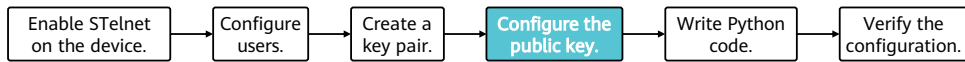
5. Create an RSA key pair (private key **id_rsa** and public key **id_rsa.pub**) on the client and view the public key.

```
Generate an RSA public-private key pair. -- C:\Users\example01> ssh-keygen -t rsa -b 4096
Specify the path for saving the key pair. -- Generating public/private rsa key pair.
(Press Enter to use the default path.) -- Enter file in which to save the key (C:\Users\example01\.ssh\id_rsa):
Enter the passphrase. -- Enter passphrase (empty for no passphrase):
(Press Enter to use the default passphrase.) -- Enter same passphrase again:
Enter the passphrase again. -- Your identification has been saved in C:\Users\example01\.ssh\id_rsa
(Press Enter to use the default passphrase.) -- Your public key has been saved in C:\Users\example01\.ssh\id_rsa.pub
Path of the private key file on the client --
Path of the public key file on the client --
.....

Check the public key. -- C:\Users\example01> type .ssh\id_rsa.pub
ssh-rsa AAAAB3NzaC1y.....
```

- Windows 11 is used as an example to describe how to generate an RSA key pair and view the public key.
 - `ssh-keygen -t rsa -b 4096`: `-t` specifies that the key type is RSA, and `-b` specifies that the key is 4096 bits long.
- Note: Only part of the output is provided on this page.

Example: Configuring the Public Key on the Server

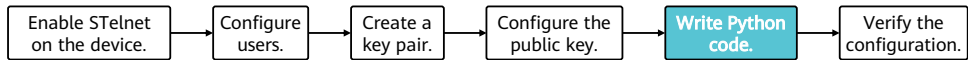


6. Add the public key generated by the client to the server and allocate the public key to the user.

```
[SSH Server] rsa peer-public-key rsa01 encoding-type openssh
[SSH Server-rsa-public-key] public-key-code begin
[SSH Server-rsa-public-key-rsa-key-code] ssh-rsa AAAAB3NzaC1y.....
[SSH Server-key-code] peer-public-key end
[SSH Server] ssh user client assign rsa-key rsa01
```

Public key generated by the client

Example: Writing Python Code on the Client



7. Write Python code for logging in to the server through SSH.

Import modules.

Instantiate an SSH object.

Allow unknown hosts to be connected.

Establish an SSH session.

Open an interactive session.

Issue the command for canceling screen splitting.

Issue the command for displaying the current configuration.

Stop the program for 3 seconds before continuing.

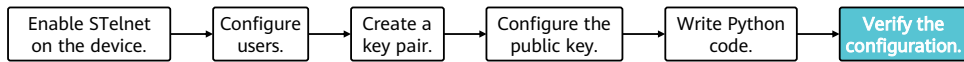
Instantiate the received data.

Print the output.

Close the SSH connection.

```
-- import paramiko
-- import time
-- ssh = paramiko.SSHClient()
-- ssh.set_missing_host_key_policy(paramiko.AutoAddPolicy())
-- ssh.connect(hostname='192.168.56.100',port=22,username='client',
-- key_filename=r'C:\Users\exampleuser\.ssh\id_rsa')
-- cli = ssh.invoke_shell()
-- cli.send('screen-length 0 temporary\n')
-- cli.send('display cu\n')
-- time.sleep(3)
-- dis_cu = cli.recv(999999).decode()
-- print(dis_cu)
-- ssh.close()
```

Example: Verifying the Configuration on the Client



8. Execute the code and check whether the current configuration of the SSH server is displayed.

```
Info: The max number of VTY users is 5, the number of current VTY users online is 1,
and total number of terminal users online is 2.
<SSH Server>screen-length 0 temporary
Info: The configuration takes effect on the current user terminal interface only.
<SSH Server>display cu
...
```

Quiz

1. (True or false) Python is a compiled language. ()
 - A. True
 - B. False
2. (Single-answer question) Which of the following is not a policy of `set_missing_host_key_policy` in paramiko? ()
 - A. `AutoAddPolicy`
 - B. `WarningPolicy`
 - C. `RejectPolicy`
 - D. `AutoDelPolicy`

1. B

2. D

Summary

- Network automation refers to the use of tools to automate network deployment and O&M, minimizing manual intervention.
- Network automation can be implemented through programming languages or specialized automation tools.
- Python is a fully open-source, high-level programming language known for its simple syntax and ease of learning. With its extensive standard library and rich ecosystem of third-party libraries, Python is well-suited for network engineering.
- Paramiko is a Python module that implements the SSHv2 protocol, enabling secure remote command execution, file transfer, and other SSH-related operations.
- This course demonstrates only how to log in to a network device via SSH and execute commands. Other features, such as SFTP, are not covered and are left for you to explore on your own.

Thank you.

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每个组织，构建万物互联的智能世界。
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Network Troubleshooting



Foreword

- An enterprise data communication network is composed of various network devices, which are interconnected through different types of physical links. To ensure accurate forwarding of data packets, these network devices rely on the coordinated operation of multiple network protocols. In practice, faults may occur on network devices, physical cables, or protocol configurations, affecting network services.
- As a network engineer, you need to be proficient in basic commands such as ping and tracert, and be able to locate common device and link-layer faults. Additionally, you need to master skills such as fault detection, quick response, and service recovery.
- This course focuses on the methods for network troubleshooting, introduces common network faults, and uses case-based scenarios to strengthen the fault handling capabilities essential for network engineers.

Objectives

- Upon completion of this lesson, you will be able to:
 - Describe general methods for network troubleshooting.
 - Master troubleshooting methods for terminal IP address acquisition failures.
 - Master troubleshooting methods for intra-VLAN communication failures.
 - Master troubleshooting methods for ping failures.
 - Master troubleshooting methods for access point (AP) onboarding failures.
 - Master troubleshooting methods for station (STA) access failures.

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1. Overview of Network Troubleshooting

- General Troubleshooting Procedure
- Common System Maintenance Commands

2. Common Network Troubleshooting Cases

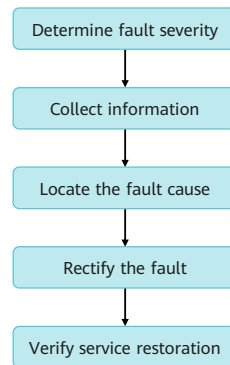
What Is a Network Fault?

- A network fault occurs when the network fails to deliver essential functions due to specific causes, affecting service delivery.
- From the perspective of users, any phenomenon that affects services can be regarded as a fault.
- Common fault symptoms and categories are as follows.

Symptom Category	Alarm	Loop	Service Unreachable	Service Interruption	Transient Service Interruption	Packet Loss	Protocol Anomaly	Protocol Flapping	Routing Anomaly
Hardware	√			√		√			
Configuration		√	√				√		√
Network		√	√	√	√	√	√	√	√
Performance	√				√	√		√	√
Software							√		√
Interconnection		√	√				√		
Others	√		√	√	√	√			

- The mapping between the preceding fault categories and symptoms varies according to scenarios.

General Troubleshooting Procedure



Determining Fault Severity

- After a fault occurs, the first step is to analyze the symptoms and confirm the scope of impact, which helps determine the next steps.
 - If all users are affected, check whether other devices or users connected to the same upper-layer device also experience the fault.
 - If only some users are affected, proceed as follows:
 - Identify the type of service involved, and confirm whether other services are also impacted.
 - Check whether the affected users and normal users are connected to the same device port.
- Faults are classified into four levels based on their severity.

Critical	Major	Minor	Warning
Faults causing large-scale service outage across the entire area	Faults occurring at medium-scale sites or affecting certain VIP sites	Faults occurring at small-scale sites	Faults occurring at scattered sites or resulting in degraded coverage quality

Collecting Information

- Effective information collection during the initial stages of troubleshooting helps narrow down the scope of investigation and improves the accuracy of fault locating. Collect the following information before handling a fault:

No.	Item	Collection Method
1	Networking information	Draw a network diagram, showing the upstream and downstream devices and interconnection interfaces.
2	Operating status	Record the system operating status before and after a fault, along with any generated log information.
3	Symptom	Record the fault occurrence time, observed symptoms, and any operations that may have triggered the fault.
4	Hardware status	Record the model, software version, and operating status of the device.
5	System information	Run the display current-configuration command in the system view to collect configuration information.

Locating the Fault Cause

- Locating a fault quickly and accurately is critical for troubleshooting because it:
 - Improves troubleshooting efficiency.
 - Effectively prevents fault escalation and other man-made incidents caused by blind troubleshooting.
 - Provides valuable guidance and reference for further troubleshooting.
- Preliminary fault locating

Symptom	Possible Cause
Service failure on individual devices	User terminal or line faults, switch port faults, power supply issues, etc.
Service failure on multiple devices	Upper-layer device faults, device configuration errors, etc.
Service failure on all devices	Device or line faults, network attacks, etc.

Common Methods for Locating Faults

- **Observation:** Check device alarms, indicator lights, and other visible signs to identify potential issues.
- **Elimination:**
 - **Software:** Disable a suspicious function or feature to determine its impact. For example, if the fault is rectified after encrypted authentication is disabled, the fault is caused by this function. If the fault persists, the fault is not related to this function.
 - **Hardware:** If a card is suspected to be faulty, replace it directly to check whether the fault is rectified.
- **Comparison:** Compare the faulty component or observed symptom with a functional component or normal condition to pinpoint the issue.
- **Swap:** Replace a suspected faulty component with a known functional one, and then compare the operating status before and after the swap to determine the scope or location of the fault.

Rectifying the Fault

- Fault rectification involves taking appropriate measures to eliminate faults and restore system functionality. After determining the cause of the fault, proceed as follows:
 - Isolate the fault to prevent it from spreading to other areas and to minimize its impact on services.
 - Rectify the fault by checking and repairing lines, replacing components, or modifying configuration data.
 - Hardware faults: Reset or replace the faulty component.
 - Configuration faults: Modify configuration data or upgrade software.

Verifying Services

- After rectifying a fault, conduct verification tests on the affected services, based on the scope of the fault, to ensure the affected services have been fully restored.
 - Verify that the fault has been eliminated and no new issues have arisen.
 - Compile a troubleshooting report and summarize the case promptly for future reference.

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Common System Maintenance Commands: display (1)

- When a network fault occurs, you can run appropriate display commands to check the current device status and collect relevant network information.

Item	Command	Function
Basic information	display diagnostic-information	Displays basic system information by integrating the outputs of multiple commonly used display commands, such as display version and display current-configuration .
Device information	display device	Displays the type and status of components on a device.
Interface information	display interface	Displays interface information that can be used to identify interconnection faults and check packet loss statistics.
Version information	display version	Displays device version information and the capacity of various storage media.
Patch information	display patch-information	Displays current patch information, such as the patch package version and patch package name.
Electronic label information	display elabel	Displays the electronic label information of a device, which identifies its hardware details.
Device status information	display health	Displays information about the temperature, power modules, fan modules, power consumption, CPU usage, memory usage, and storage medium usage of a device.
CPU usage	display cpu-usage	Displays the CPU usage of a device.

- The commands may vary according to the device model. For details, see the product documentation for the specific device model in use.

Common System Maintenance Commands: display (2)

Item	Command	Function
Memory usage	display memory-usage	Displays the memory usage of a device.
Current system configuration	display current-configuration	Displays all current configurations of a device. You can use a regular expression to filter and obtain the required configurations.
Saved system configuration	display saved-configuration	Displays the startup configuration of a device. • To check the system configuration saved last time, run the display saved-configuration last command. • To check the last time when the system configuration is saved, run the display saved-configuration time command.
Time information	display clock	Displays the current date and time of the system.
User log information	display logfile buffer	Displays user logs saved in the log buffer. This command is run in the diagnostic view.
Diagnostic log information	display diag-logfile buffer	Displays diagnostic logs saved in the log buffer. This command is run in the diagnostic view.
Alarm Information	display trapbuffer	Displays information recorded in the trap buffer of the information center.
AP operating information	display ap run-info	Displays the operating information of an AP.
AP status	display ap all	Displays the AP status.
Access user information	display access-user/display station	Displays information about access users.

Common System Maintenance Commands: Ping & Tracert

- During routine system maintenance, you can run the **ping** command to check network connectivity. If the ping operation fails, run the **tracert** command to pinpoint the fault location, providing a basis for further diagnosis.

- The **ping** command checks network connectivity and device reachability.

```
ping [-a source-ip-address] [-c count] [-f] [-s packetsize] [-t timeout] host
```

- The **tracert** command traces the path of data packets from the source to the destination, helping you check network reachability and identify where faults occur along the path.

```
tracert [-a source-ip-address] [-f first-ttl] [-m max-ttl] [-q nqueries] [-w timeout] host
```

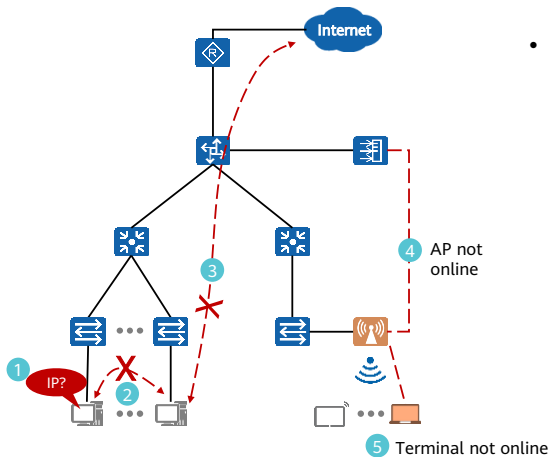
- Parameter description of the **ping** command
 - a**: specifies the source IP address of ICMP Echo Request packets. If this parameter is not specified, the IP address of the outbound interface is used as the source address.
 - c**: specifies the number of ICMP Echo Request packets to send. The default value is 5. In poor network conditions, you can increase the value and use the packet loss rate to evaluate network quality.
 - f**: indicates that the sent packets will not be fragmented. If the size of a packet is greater than the MTU, the packet is discarded.
 - s**: specifies the length of an ICMP Echo Request packet, excluding the IP and ICMP headers.
 - t**: specifies the timeout period for ICMP Echo Reply packets. You can set a larger timeout period if the network conditions are poor. The default value is 2 seconds. If no reply is received within 2 seconds, the destination is considered unreachable.
 - host**: specifies the domain name or IP address of the destination host.
- Parameter description of the **tracert** command
 - a**: specifies the source address for sending tracert packets. If this parameter is not specified, the IP address of the outbound interface is used as the source address.
 - f**: specifies the initial TTL. If the number of hops is less than the initial TTL, the TTL field remains greater than 0, and the intermediate nodes will not return ICMP Time Exceeded packets to the source host.

- **-m:** specifies the maximum TTL, which is usually set to the number of hops a packet passes through.
- **-q:** specifies the number of UDP probe packets to be sent each time. In poor network conditions, you can increase the value to ensure a higher probability of probe packets reaching the destination.
- **-w:** specifies the timeout period for response packets. If a tracert packet times out at a gateway, an asterisk (*) is displayed. You are advised to increase the timeout period if network quality is poor or transmission is slow.
- **host:** specifies the domain name or IP address of the destination host.

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 - Intra-VLAN Communication Failures
 - Ping Failures
 - AP Onboarding Failures
 - STA Access Failures

Common Campus Network Faults



- Campus network faults can be classified into the following types based on terminal access types:
 - Typical wired network faults:
 1. Terminals fail to obtain IP addresses.
 2. Terminals in the same VLAN cannot communicate with each other.
 3. Terminals cannot ping an external network.
 - Typical wireless network faults:
 1. APs fail to go online.
 2. STAs fail to access the network.

- Wired terminal access:
 - The first step for any terminal to access the network is to obtain an IP address. Therefore, when troubleshooting a network fault, first check whether the terminal has obtained an IP address.
 - After verifying that a terminal has obtained an IP address, check whether it can communicate with other terminals on the same network segment to determine if a Layer 2 fault exists (for example, incorrect VLAN configuration).
 - If a terminal cannot access the external network, first check whether it can ping an external IP address.
- Wireless terminal access:
 - In WAC + Fit AP networking, the prerequisite for STA access is that APs are managed by the WAC.
 - After a Fit AP goes online on the WAC, if STAs cannot go online, verify the WLAN service configuration.
- The network faults listed in this course do not cover all scenarios. This course focuses on typical cases to help you understand the general troubleshooting roadmap.
- For more troubleshooting cases, refer to the troubleshooting guide of the related

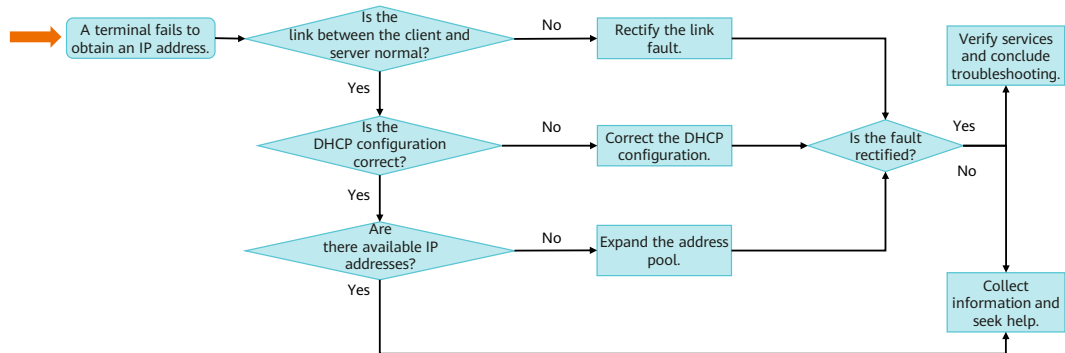
product, for example, the Huawei S Series Campus Switches Troubleshooting Guide at

<https://support.huawei.com/hedex/hdx.do?docid=EDOC1100313768&tocURL=resources%2Fhedex-homepage.html>.

Common Causes of Terminal IP Address Acquisition Failures

- A DHCP server processes requests for address allocation, renewal, and release from DHCP clients or DHCP relay agents, and allocates IP addresses and other network configuration parameters to DHCP clients.
- When a device functions as a DHCP server to allocate IP addresses to terminals, possible reasons why terminals fail to obtain an IP address include the following:
 - The link between the client and DHCP server is abnormal.
 - The DHCP configuration is incorrect.
 - The address pool has no available IP addresses.

Troubleshooting Flowchart for Terminal IP Address Acquisition Failures



- This course covers only the basic troubleshooting process. For more detailed information, refer to the troubleshooting guide of the corresponding product.

Checking Whether the Link Between the Client and Server Is Normal

- Check the physical interface and VLANIF interface. If the physical interface is Up but the number of error packets keeps increasing, replace the Ethernet cable. If the VLANIF interface is Down, create the corresponding VLAN on the device, and configure interfaces to allow packets from this VLAN to pass through.

```
[HUAWEI] display interface brief | include up
PHY: Physical
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
(b): BFD down
(e): ETHOAM down
...
InUti/OutUti: input utility rate/output utility rate
Interface      PHY      Protocol  InUti  OutUti  inErrors  outErrors
10GE1/0/1      up       up        0%     0%     0         0
10GE1/0/2      up       up        0%     0%     0         0
Vlanif10       up       up        0%     0%     0         0
```

Checking Whether the DHCP Configuration Is Correct

- A device functioning as a DHCP server allocates IP addresses to clients using either of the following methods:
 - Allocates IP addresses from a global address pool.
 - Allocates IP addresses from an interface address pool.
- Check whether the configuration of the interface address pool is correct.

```
<HUAWEI> system-view
[HUAWEI] interface vlanif 10
[HUAWEI-Vlanif10] display this
#
interface Vlanif10
ip address 192.168.4.1 255.255.255.0
dhcp select interface
dhcp server gateway-list 192.168.4.1
#
return
```

Checking Whether the DHCP Address Pool Has Available IP Addresses

- Run the **display ip pool** command to check whether the value of **Idle** in the output is 0.

```
<HUAWEI> display ip pool interface vlanif10
Pool-name : Vlanif10
Pool-No : 0
...
-----
Network section
Start      End          Total   Used   Idle(Expired)  Conflict  Disabled
-----
192.168.4.1 192.168.4.254 254     0      254(0)         0         0
-----
```

- If the value of **Idle** is not 0, the server's address pool has available IP addresses. In this case, proceed to the next step.
- If the value of **Idle** is 0, the server's address pool has no available IP addresses. See the table below for solutions.

Solution	Interface Address Pool	Global Address Pool
Increase the number of IP addresses.	- Reduce the address mask length. - Assign clients to additional VLANs to increase the number of interface address pools. - Increase the number of DHCP servers.	
Reduce the IP address lease.	Run the dhcp server lease lease command in the interface view.	Run the lease lease command in the global address pool view.

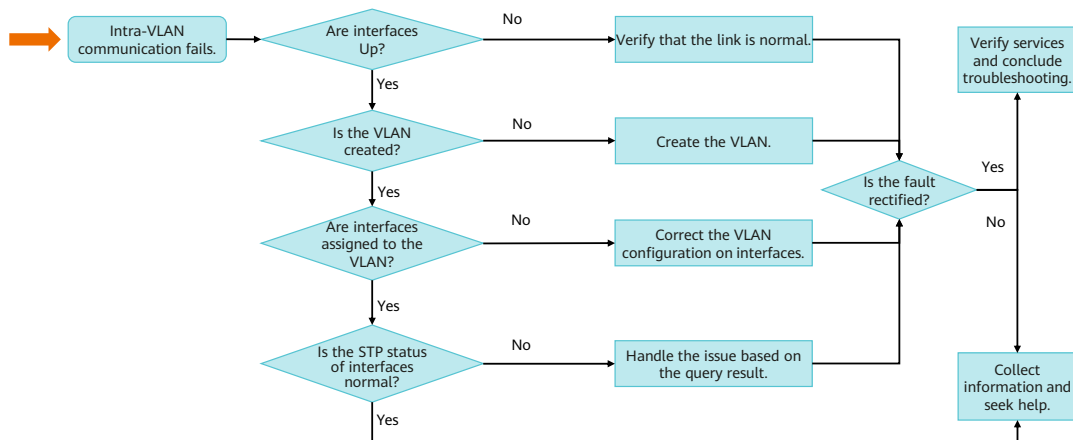
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 - Ping Failures
 - AP Onboarding Failures
 - STA Access Failures

Common Causes of Intra-VLAN Communication Failures

- If users in the same VLAN cannot communicate with each other, the possible causes are as follows:
 - A link fails.
 - An interface is shut down manually or a physical interface is damaged.
 - The device learns incorrect MAC addresses.
 - Layer 2 port isolation is configured on the device.
 - Incorrect static ARP entries are configured on hosts.
 - Incorrect mappings between interfaces and MAC addresses are configured on the device.

Troubleshooting Flowchart for Intra-VLAN Communication Failures



- This course covers only the basic troubleshooting process. For more detailed information, refer to the troubleshooting guide of the corresponding product.

Checking Whether Interfaces Are Up

- Check whether the inbound and outbound interfaces of packets are Up. If an interface is not Up, check whether the link is normal.
- For example, if packets are received on 10GE1/0/1 and sent out through 10GE1/0/2, check whether both 10GE1/0/1 and 10GE1/0/2 are Up.

```
[HUAWEI] display interface brief | include up
PHY: Physical
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
(b): BFD down
(e): ETHOAM down
...
InUti/OutUti: input utility rate/output utility rate
Interface      PHY      Protocol InUti  OutUti  inErrors  outErrors
10GE1/0/1      up       up       0%    0%      0         0
10GE1/0/2      up       up       0%    0%      0         0
NULL0          up       up(s)    0%    0%      0         0
```

Checking Whether a VLAN Is Created

- Check whether the VLAN for forwarding packets has been created. For example, when packets are forwarded within VLAN 10, run the **display vlan 10** command to check whether VLAN 10 has been created.

```
<HUAWEI> display vlan 10
```

```
-----  
U: Up;           D: Down;           TG: Tagged;       UT: Untagged;  
MP: Vlan-mapping; ST: Vlan-stacking;  
#: ProtocolTransparent-vlan; *: Management-vlan;  
MAC-LRN: MAC-address learning; STAT: Statistic;  
BC: Broadcast;   MC: Multicast;   UC: Unknown-unicast;  
FWD: Forward;   DSD: Discard;  
-----
```

```
VID      Ports
```

```
10  UT:10GE1/0/1(U) 10GE1/0/2(U)
```

```
VID  Type   Status  Property  MAC-LRN  STAT  BC   MC   UC   Description
```

```
10   common enable default  enable   disable FWD  FWD  FWD  VLAN 0010
```

- If the VLAN has not been created, create it in the system view.
- <HUAWEI> system-view
- [HUAWEI] vlan 10 //Create VLAN 10.
- [HUAWEI-vlan10]

Checking Whether Interfaces Are Assigned to the VLAN

- Check whether the inbound and outbound interfaces of packets belong to the same VLAN and whether their interface types are correct. If the VLANs are different or the interface types do not match, correct the configurations in the interface view.
- For example, verify that the inbound interface 10GE1/0/1 and the outbound interface 10GE1/0/2 are both assigned to VLAN 10 and in Up state.

```
<HUAWEI> display vlan 10
-----
U: Up;          D: Down;      TG: Tagged;     UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;
MAC-LRN: MAC-address learning; STAT: Statistic;
BC: Broadcast;  MC: Multicast; UC: Unknown-unicast;
FWD: Forward;   DSD: Discard;
-----
VID      Ports
-----
10  UT:10GE1/0/1 (U) 10GE1/0/2(U)
-----
VID  Type   Status Property  MAC-LRN  STAT  BC  MC  UC  Description
-----
10   common enable  default  enable   disable FWD FWD FWD  VLAN 0010
```

Checking Whether the STP Status of Interfaces Is Normal

- Check whether the STP status of interfaces is normal. If STP is enabled on the device, the STP status of the inbound and outbound interfaces must be **FORWARDING**. Otherwise, packets will be discarded.

<HUAWEI> display stp brief						
MSTID	Port	Role	STP State	Protection	Cost	Edged
0	10GE1/0/1	DESI	FORWARDING	none	20000	disable
0	10GE1/0/2	DESI	FORWARDING	none	20000	disable

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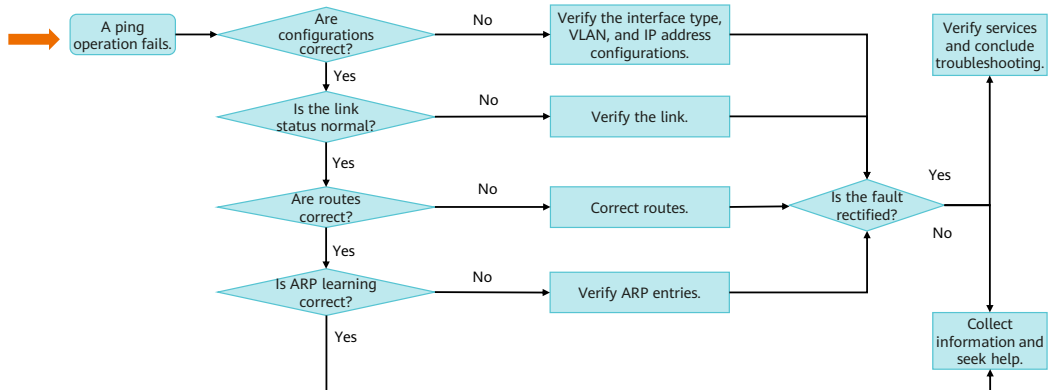
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Possible Causes of Ping Failures

- A ping failure occurs when all ICMP Echo Reply messages cannot reach the destination. The possible causes include:
 - Improper ping command parameters
 - Incorrect configuration
 - Abnormal physical link status
 - Routing issues
 - Failed or incorrect ARP learning
 - Peer IP address included in a blacklist policy
 - Abnormal packet transmission or reception

- This section focuses on troubleshooting ping failures between directly connected devices—for example, when a PC cannot ping its directly connected gateway or when two adjacent network devices fail to ping each other.

Troubleshooting Flowchart for Ping Failures



- This course covers only the basic troubleshooting process. For more detailed information, refer to the troubleshooting guide of the corresponding product.

Checking Whether Configurations Are Correct

- If a PC is directly connected to the device, verify that the PC's IP address is on the same network segment as the IP address of the VLANIF interface corresponding to the PC's VLAN.
- If the device is directly connected to another network device, verify that the interface link types and VLAN configurations are consistent on both ends, and that the VLANIF addresses on both ends reside on the same network segment.
 - Run the **display port vlan** command to check the interface link type and VLAN configuration. The **Link Type** field specifies the interface link type, and the **Trunk VLAN List** field specifies the VLAN IDs to which an interface is statically and dynamically assigned.

```
<HUAWEI> display port vlan
Port      Link Type  PVID    Trunk VLAN List  Port Description
-----
10GE1/0/1  trunk      1        10-30             --
10GE1/0/2  access     30        2-3               --
...
```

- Run the **display ip interface brief** command to check the IP address of the VLANIF interface. The **IP Address/Mask** field specifies the IP address and mask of the interface.

```
<HUAWEI> display ip interface brief
...
Interface  IP Address/Mask  Physical  Protocol  VPN
Vlanif10   192.168.1.10/24  up        up        -
...
```

Checking Whether Physical Links Are Normal (1)

- Verify that the physical link media is functioning properly and that the optical fiber or Ethernet cable is connected to the correct port. If any issue is found, try using another port or replacing the cable.
- Verify that the VLANIF interface is Up. This is a prerequisite for a successful ping. If the VLANIF interface is Down, no member interface in the VLAN is Up. Check the cable condition and connection to ensure that the interface is Up.

```
<HUAWEI> display ip interface brief
```

```
...
Interface  IP Address/Mask  Physical  Protocol
Vlanif10   192.168.10.1/24    down     down
Vlanif30   unassigned         *down    down
...
```

```
<HUAWEI> display interface brief
```

```
...
Interface  PHY      Protocol  InUti   OutUti   inErrors  outErrors
Eth-Trunk0  down     down      0%      0%      0         0
10GE1/0/1   down     down      0%      0%      0         0
10GE1/0/2   down     down      0%      0%      0         0
```

Checking Whether Physical Links Are Normal (2)

- If both the VLANIF and physical interfaces are Up, check whether Layer 2 protocols such as Spanning Tree Protocol (STP) or Smart Link are running on the device and whether the physical interface transmitting ping packets is blocked. If the interface is blocked, modify the configuration to restore normal packet forwarding.
 - Run the **display stp brief** command to check the STP status. In the command output, if **STP State** is **FORWARDING**, packets are forwarded normally. If **STP State** is **DISCARDING**, the corresponding interface is blocked.

<HUAWEI> display stp brief						
MSTID	Port	Role	STP State	Protection	Cost	Edged
0	10GE1/0/1	DESI	FORWARDING	none	20000	disable
0	10GE1/0/2	ALTE	DISCARDING	none	20000	disable

Checking Whether Routes Are Correct

- Check whether there are direct routes.
 - If the connected device is a terminal, check whether the correct gateway address is configured on the terminal.
 - If the connected device is a switch or router, check whether the device has a correct return route.
 - Run the **display ip routing-table** *ip-address* command on the source device to check whether a route to the peer device exists. If a route exists, information similar to the following is displayed, where the value **Direct** in the **Proto** field indicates a direct route.

```
<HUAWEI> display ip routing-table 192.168.1.11
Route Flags: R - relay, D - download to fib
-----
Routing Table : Public
Summary Count : 1
Destination/Mask    Proto   Pre  Cost  Flags  NextHop  Interface
192.168.1.0/24      Direct  0    0     D      192.168.1.10  Vlanif10
```

- If no route exists, no information will be displayed. In this case, check whether the routing protocol configuration is correct.

Checking Whether ARP Is Functioning Properly

- Check whether ARP learning for directly connected addresses is successful.

```
<HUAWEI> display arp | include 192.168.1.11
IP ADDRESS   MAC ADDRESS   EXPIRE(M) TYPE/VLAN   INTERFACE   VPN-INSTANCE
-----
192.168.1.11  4c1f-cc2f-3634  19      D/10      10GE1/0/1
.....
```

- If ARP learning is successful, run the **display mac-address** *mac-address* command to check whether the outbound interface of the MAC address entry is the same as the physical outbound interface of the ARP entry. If they are different, check whether a loop or MAC address conflict occurs.

```
<HUAWEI> display mac-address 4c1f-cc2f-3634
Flags: * - Backup
# - forwarding logical interface, operations cannot be performed based on the interface.
BD : bridge-domain Age : dynamic MAC learned time in seconds
-----
MAC Address   VLAN/VSI/BD   Learned-From   Type   Age
-----
4c1f-cc2f-3634  1/-/-      10GE1/0/1      dynamic  182
-----
Total items: 1
```

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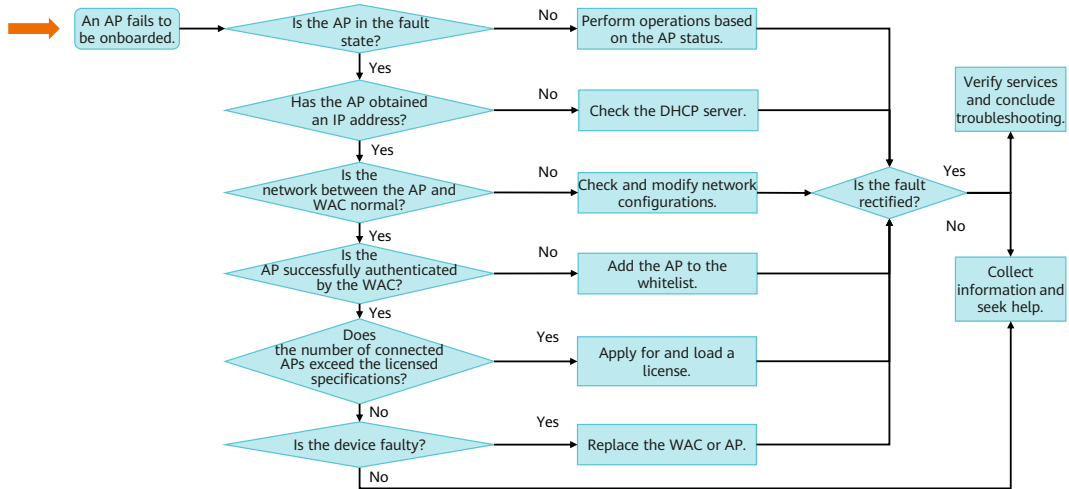
2. Common Network Troubleshooting Cases

- Terminal IP Address Acquisition Failures
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- STA Access Failures

Possible Causes of AP Onboarding Failures

- In the WAC + Fit AP architecture, a Fit AP may fail to go online on the WAC for the following reasons:
 - The AP has not obtained an IP address, or its static IP address is incorrectly configured.
 - The network between the AP and WAC is disconnected due to incorrect configurations.
 - The AP fails MAC address or SN authentication.
 - The number of APs connected to the WAC reaches the license limit.
 - The AP is faulty.

Troubleshooting Flowchart for AP Onboarding Failures



- AP troubleshooting is carried out based on the AP onboarding process. Therefore, WLAN engineers must be thoroughly familiar with the CAPWAP tunnel establishment process to perform troubleshooting effectively.
- This course covers only the basic troubleshooting process. For more detailed information, refer to the troubleshooting guide of the corresponding product.

Checking the AP Status

- Log in to the WAC, and run the **display ap all** command to check the AP status.

```
<WAC> display ap all
```

```
Total AP information:
```

```
idle : idle      [1]
```

```
nor  : normal    [1]
```

ID	MAC	Name	Group	IP	Type	State	STA	Uptime
0	60de-4476-e360	L1_003	default	192.168.109.254	AP7060DN	nor	0	2D:5H:48M:44S
1	dcd2-fc04-b500	dcd2-fc04-b500	default	-	AP7060DN	idle	0	-

```
Total: 2
```

Common states include the following:

- normal**: The AP is operating properly.
- standby**: It is the AP status on the standby WAC.
- idle**: It is the initial state of the AP before it establishes a link with the WAC.
- download**: The AP is upgrading.
- fault**: The AP fails to go online.
- unauth**: The AP fails authentication.

- Common AP states include the following:
 - normal**: The AP has successfully gone online on the WAC.
 - standby**: It is the AP status on the standby WAC.
 - idle**: The AP is in idle state after it is added offline. If an AP is in this state, verify that the AP is properly connected to the network.
 - download**: The AP is loading system software during an upgrade. Wait until the upgrade completes, and then check the AP state again.
 - fault**: The AP fails to register with the WAC. If an AP is in this state, proceed to the next step.
 - committing**: The WAC is delivering services to the AP.
 - config-failed**: The AP's initial configuration has failed. Verify that the network connection is normal.
 - name-conflicted**: The AP name conflicts with another. Change the AP name.
 - ver-mismatch**: The AP software version does not match the WAC software version. Verify that the two versions are compatible.

Querying the Cause of an AP Onboarding Failure

- Log in to the WAC, and run the **display ap online-fail-record** command to check the reason why the AP fails to be onboarded.

```
<WAC> display ap online-fail-record mac 00e0-fcb1-56a0
Total APs: 1 Total records: 1
-----
MAC          Last fail time      Reason
-----
00e0-fcb1-56a0  20xx-xx-xx/15:xx:xx  The AP is added to the AP blacklist.
```

- The following lists the common causes of AP onboarding failures in the command output. For detailed information, refer to the product documentation of the corresponding version. Troubleshoot the issue based on the specific cause of the AP onboarding failure.
 - Insufficient license resources.
 - The AP is not in the SN whitelist.
 - The AP is not in the MAC whitelist.
 - The AP is added to the AP blacklist.
 - ...

Checking Whether the AP Has Obtained an IP Address

- On the DHCP server, run commands to check whether the AP has obtained an IP address.
- When using the WAC as a DHCP server, run the **display ip pool** command to check the allocated IP addresses. Compare the MAC address to check whether the AP has obtained an IP address. If so, perform a ping test to that address.

```
<WAC> display ip pool interface Vlanif1219 used | include dcd2-fc22-d880
Pool-name      : Vlanif1219
Pool-No       : 4
Lease         : 1 Days 0 Hours 0 Minutes
...
Network section :
-----
Index      IP          MAC          Lease  Status
-----
4090    10.1.15.251  dcd2-fc22-d880  9368  Used
-----
```

- When using another device as a DHCP server, check whether the AP has obtained an IP address from it. If not, check whether the DHCP server and related device interfaces are correctly configured.

Checking Whether the Number of Connected APs Has Reached the Upper Limit

- Run the **display license resource usage** command to check the current service capability set specified by **ResourceUsage**. If the number of normal APs has reached the licensed capability, new APs cannot go online. In this case, purchase a new license.

```
<WAC> display license resource usage
Activated License: flash:/LIC92680232*****_*****5396810CB000006.dat
FeatureName      | ConfigureItemName      | ResourceUsage
CRFEA1           | LH85WLANAP01           | 0/256
```

- The number of APs that can connect to a WAC is limited by the license resource item. The total number of connected APs cannot exceed the licensed upper limit.

Checking Whether the AP Is Successfully Authenticated by the WAC

- Run the **display ap global configuration** command on the WAC to check the AP authentication mode.

```
<WAC> display ap global configuration
-----
AP auth-mode      : MAC-auth
...
```

- If MAC address or SN authentication is used, run the **display ap unauthorized record** command to check for unauthenticated APs.

```
<WAC> display ap unauthorized record
Unauthorized AP record:
Total number: 1
-----
AP type: AirEngine8760-X1-PRO
AP SN: xxxxxxxxxxxxxxxxx
AP MAC address: dcd2-fc22-xxxx
AP IP address: 192.168.109.252
-----
```

- For APs that fail authentication, run the **ap-confirm** command to confirm the unauthenticated APs so that they can go online. Alternatively, run the **ap whitelist** command to add the specified APs to the AP whitelist.

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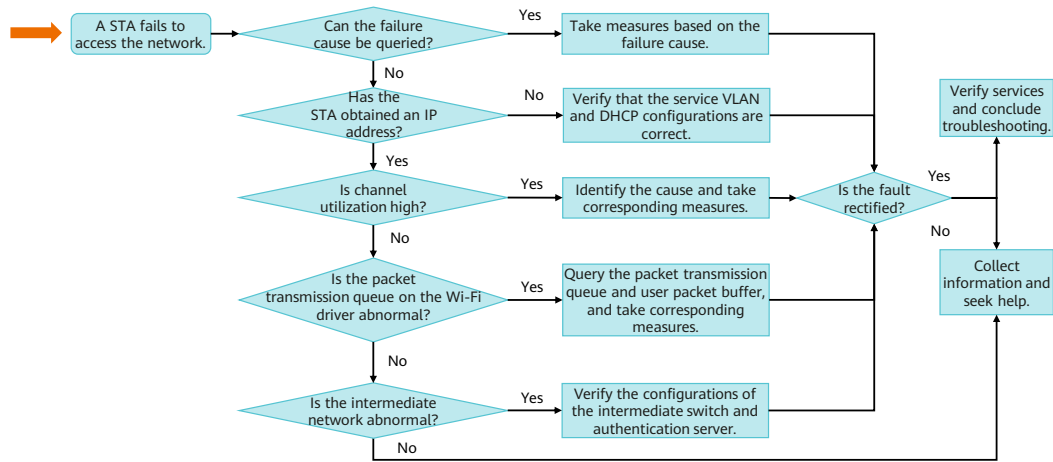
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- Terminal IP Address Acquisition Failures
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Possible Causes of STA Access Failures

- The possible causes of STA access failures include:
 - No service VLAN is configured in the VAP profile, or the service VLAN is incorrect.
 - A VLAN pool is configured in the VAP profile, but some VLANs in the pool are not enabled.
 - The STA blacklist or whitelist function is configured.
 - Channel utilization on the AP radio is excessively high, causing the air interface to be busy.

Troubleshooting Flowchart for STA Access Failures



- This course covers only the basic troubleshooting process. For more detailed information, refer to the troubleshooting guide of the corresponding product.

Querying the Cause of a STA Access Failure

- Run the **display station online-fail-record** command to check the reason why a STA fails to access the network.

```
[WAC-wlan] display station online-fail-record all
Reason distribution
-----
Reason Count Percent
-----
The STA is in the VAP's blacklist. 2 100.00%
-----
Total Count: 2
Recent records
Rf/WLAN: Radio ID/WLAN ID
-----
STA MAC      Device type  Device ID  Device name  Rf/WLAN  Last record time  Reason
-----
14cf-9202-13dc AP          1         AP-area_1    0/1      20xx-xx-xx/15:xx:xx  The STA is in the VAP's blacklist.
14cf-9202-13dd AP          2         AP-area_2    1/1      20xx-xx-xx/15:xx:xx  The STA is in the VAP's blacklist.
-----
Total stations: 2 Total records: 2
```

- Rectify the fault based on the specific cause.

- The common causes of STA access failures identified through the above command are as follows:
 - STA authentication times out.
 - The STA is not authenticated.
 - The number of STAs exceeds the physical specifications allowed by the AP.
 - The number of associated STAs exceeds the maximum allowed by the AC.
 - The STA is in the global blacklist.
 - The STA is not in the global whitelist.
 - ...

Checking Whether the STA Has Obtained an IP Address

- On the DHCP server, run commands to check whether the STA has obtained an IP address. When using the WAC as a DHCP server, run the **display ip pool** command to check the allocated IP addresses. Compare the MAC address to check whether the STA has obtained an IP address.

```
<WAC> display ip pool interface Vlanif100 used | include dcd2-fc9a-c800
Pool-name      : Vlanif100
...
Network section :
-----
Index      IP      MAC      Lease  Status
-----
4085      10.1.15.246      dcd2-fc9a-c800      7375  Used
-----
```

- Check whether the service VLAN is created and configured correctly. Ensure that the service VLAN is different from the management VLAN.
 - Direct forwarding mode: Check whether the service VLAN is created on the devices deployed between the AP and the STA gateway and whether the devices allow packets from the service VLAN to pass through. If not, service VLAN packets cannot be forwarded.
 - Tunnel forwarding mode: If the DHCP server that allocates IP addresses to STAs is the WAC, check whether the service VLAN is created on the WAC. If not, service VLAN packets cannot be forwarded.

Checking Whether the AP Channel Utilization Is Too High

- Check whether the AP's channel utilization is within a normal range. Typically, channel utilization should remain below 80%. If it exceeds 85%, packet transmission over the air interface will be affected, and STAs may fail to access the network. In this case, check the air interface interference, traffic load on the AP's Ethernet interfaces, and the number of broadcast and multicast packets. Identify the root cause of high channel utilization, and optimize the network accordingly.

[WAC-wlan] **display radio ap-id 0**

CH/BW:Channel/Bandwidth

CE:Current EIRP (dBm)

ME:Max EIRP (dBm)

CU:Channel utilization

ST:Status

WM:Working Mode (normal/monitor/monitor dual-band-scan/monitor proxy dual-band-scan)

AP ID	Name	RfID	Band	Type	ST	CH/BW	CE/ME	STA	CU	WM
0	60de-4474-9640	0	2.4G	bgn	on	6/20M	24/24	0	55%	normal
0	60de-4474-9640	1	5G	an	on	56/20M	25/25	0	13%	normal

Total:2

Quiz

1. (Single-answer question) Which of the following commands is used on Huawei WACs to check the causes of STA access failures? ()
- A. display ap online-fail-record
 - B. display vap create-fail-record
 - C. display station offline-record
 - D. display station online-fail-record

1. D

Summary

- This course covers troubleshooting of common network faults, including terminal IP address acquisition failures, intra-VLAN communication failures, ping failures, AP onboarding failures, and STA access failures.
- Upon completion of this section, you are expected to have mastered universal network troubleshooting methods and typical troubleshooting workflows.

Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。
Bring digital to every person, home, and
organization for a fully connected,
intelligent world.

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Typical Campus Networking Solution



Foreword

- When you are studying at school, working at your office, or shopping in a mall, you may notice that networks are available in all of these places. Through the networks, you can access internal resources of the school, access printers in the company to print documents, or access the Internet to read news.
- These networks are all campus networks, which are generally built by enterprises or organizations themselves. A campus network can not only improve the operational efficiency of enterprises, but also provide network access services for external users.
- In this course, we will explore the basic architecture of campus networks and learn how to build a campus network.

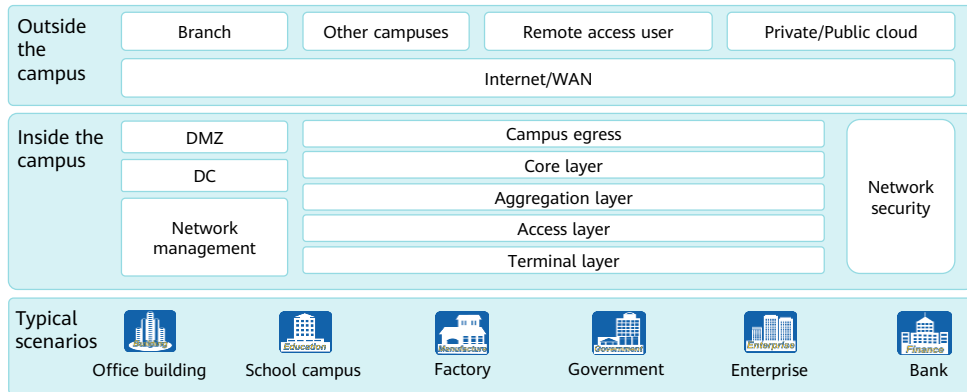
Objectives

- Upon completion of this course, you will be able to:
 - Grasp the definition of campus networks.
 - Understand the typical networking architecture of a campus network.
 - Learn the planning and design methods of a small campus network.
 - Master the deployment and implementation methods of a small campus network.
 - Comprehend the concepts of O&M and optimization of a small campus network.
 - Independently set up a campus network.

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- 1. Basic Concepts of Campus Networks**
2. Practices in Campus Network Projects

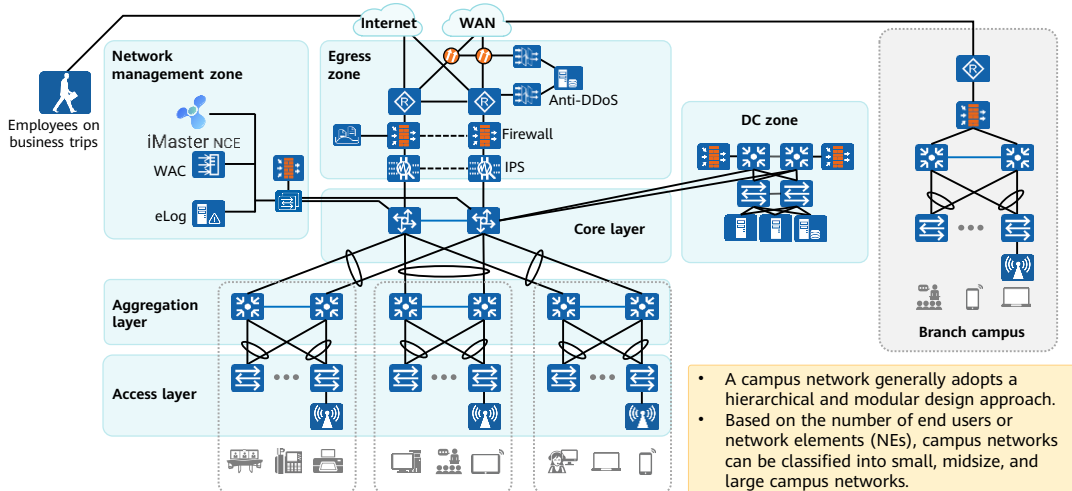
What Is a Campus Network?



A campus network is a local area network (LAN) that connects people and things within a specified area. It is usually managed by only one entity. If there are multiple management entities, each part managed by an entity is regarded as a campus network.

- The scale of a campus network can vary greatly, ranging from Small Office Home Office (SOHO) to large environments such as school campuses, enterprise campuses, or shopping centers. The scale of a campus network is limited. Generally, a large campus network, such as a university campus or industrial campus, covers an area of several square kilometers. In this area, LAN technologies can be used to build a network. A network that exceeds this range is usually considered as a metropolitan area network (MAN), and MAN or WAN technologies are required.
- Typical LAN technologies used on campus networks include IEEE 802.3-compliant Ethernet (wired) and IEEE 802.11-compliant Wi-Fi (wireless).
- DMZ: Demilitarized Zone

Typical Architecture of a Campus Network

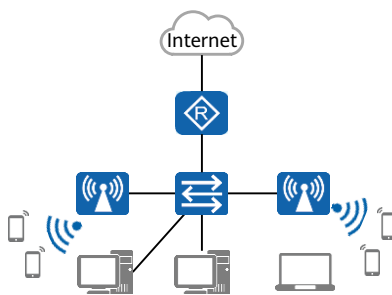


- A campus network generally adopts a hierarchical and modular design approach.
 - Hierarchical design principles:
 - Simplified management and fault locating: Each layer has a clear and unique function, making the network structure clear. When a fault occurs, it can be quickly pinpointed at the specific layer, greatly improving troubleshooting efficiency.
 - Scalability and predictability: As the network scale and traffic increase, each layer can be expanded independently (for example, adding access switches and upgrading core links) without affecting other layers. This ensures that the capacity expansion solution is predictable and the cost is controllable.
 - Traffic pattern optimization: Traffic is forwarded along the following path: access -> aggregation -> core -> destination, which facilitates unified traffic control and policy management at the aggregation and core layers.
 - Modular design principles:
 - Function isolation and security: Different functional areas (such as the data center zone and the egress zone) are divided based on modules. An issue in one module (for example, an attack on the data center) can be confined within the module, preventing the issue from spreading across the entire campus.
 - Convenient policy implementation: Security policies, access control

lists (ACLs), and routing policies can be deployed at module boundaries (such as firewalls in the egress zone and front end of the data center) in a unified manner, enabling more centralized and efficient management.

- Independent evolution: Different modules can use different technologies or be upgraded at different paces.
- Typical layers and zones of a campus network:
 - Access layer: As the edge of a campus network, this layer connects end users to the campus network.
 - Aggregation layer: This layer is a middle layer of a campus network, and completes data aggregation or switching. Some key network functions, such as routing and security, are also provided at this layer.
 - Core layer: This layer is the backbone of the campus network and the core of data switching. It connects various parts of the campus network, such as the data center zone, network management zone, and campus egress.
 - Egress zone: This zone is the border between the campus internal network and the external network and enables internal users to access the public network and external users to access the internal network. In this zone, a large number of network security devices, such as intrusion prevention systems (IPSs), anti-DDoS devices, and firewalls, are deployed to defend against attacks from external networks.
 - Data center zone: In this zone, servers and application systems are deployed to provide data and application services for internal and external users of the enterprise.
 - Network management zone: In this zone, network management systems, including the SDN controller, wireless access controller, and eLog (log server), are deployed to manage and monitor the entire campus network.

Typical Architecture of a Small Campus Network

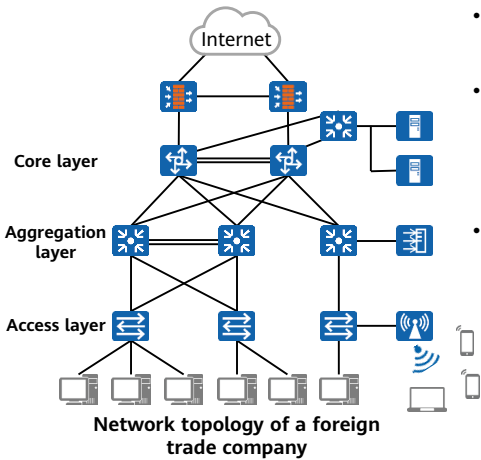


Network topology of a chain coffee shop

- A small campus network is designed for environments with a limited number of access users, generally accommodating several to several dozen users. The network covers only one site and does not have a hierarchical structure. The network is built to facilitate communication among internal resources.
- A small campus network has the following characteristics:
 - Small number of users
 - Single site
 - Simple network architecture
 - Simple network requirements

Number of end users	< 200
Number of NEs	< 25

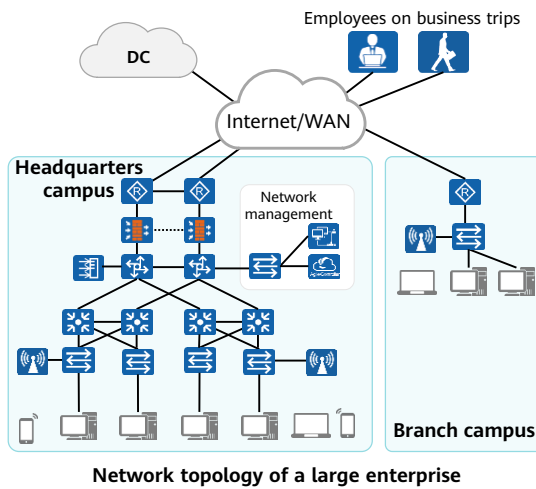
Typical Architecture of a Midsize Campus Network



- A midsize campus network can accommodate hundreds to thousands of users.
- A midsize network adopts a modular design approach. It is generally organized by functional areas, such as service requirements. Typically, the number of functional modules in a midsize network is small.
- A midsize campus network has the following characteristics:
 - Medium scale
 - Most widely used
 - Design by function
 - Typically, a three-layer network architecture is used: core layer, aggregation layer, and access layer.

Number of end users	200 to 2000
Number of NEs	25 to 100

Typical Architecture of a Large Campus Network



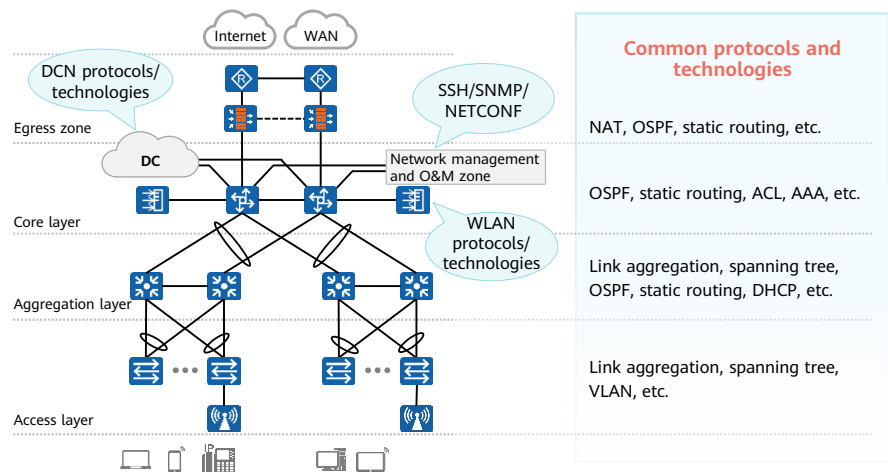
- A large campus network may be a network that covers multiple buildings or a network that connects multiple campuses in a city through the WAN. Generally, access services are provided to allow employees on business trips to access the internal network through technologies such as virtual private network (VPN).

- A large campus network has the following characteristics:

Number of end users	> 2000
Number of NEs	> 100

- Wide coverage
- Large number of users
- Complex network requirements
- Comprehensive functional modules
- Abundant network layers

Main Protocols and Technologies for Campus Networks



Contents

1. Basic Concepts of Campus Networks
- 2. Practices in Campus Network Projects**

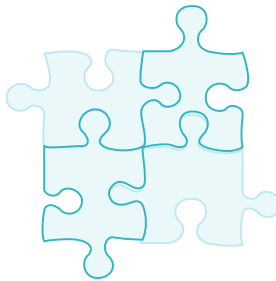
Network Requirements

- A company with about 200 employees plans to build a new campus network to meet its service development requirements. The requirements for the new network are as follows:
 - The network can meet the company's current service requirements.
 - The network topology is simple and easy to maintain.
 - Wired access is provided for employees, and Wi-Fi services are provided for guests.
 - Simple network traffic management is supported.
 - The network ensures a certain degree of security.

Campus Network Project Lifecycle

1 Planning and design

- Device model selection
- Physical topology
- Logical topology
- Technologies and protocols



2 Deployment and implementation

- Device installation
- Device commissioning
- Joint commissioning
- Cutover

3 Network O&M

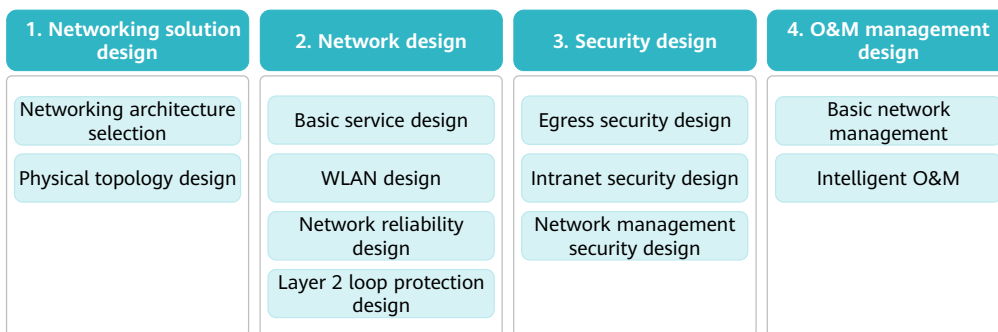
- Routine maintenance
- Centralized network management and monitoring
- Troubleshooting

4 Network optimization

- Network security enhancement
- Software and configuration backup
- Software upgrade
- User experience improvement

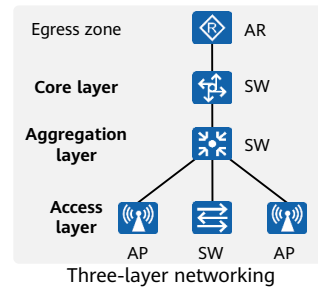
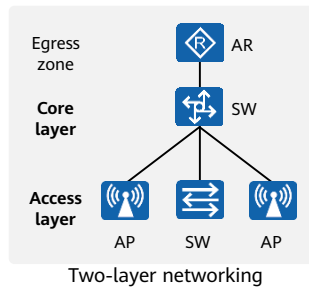
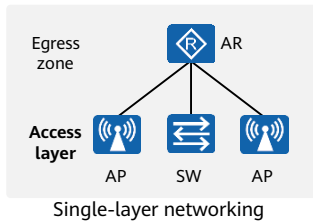
- **Planning and design:** Network planning and design is the starting point of a project. A well-planned project provides a solid foundation for subsequent project operations.
- **Deployment and implementation:** Project implementation is the specific operation phase of project delivery by engineers. Systematic management and efficient processes are the basic elements for ensuring successful project implementation.
- **Network O&M:** To ensure that network functions run properly and support the smooth development of user services, routine maintenance and troubleshooting need to be performed on the network.
- **Network optimization:** As services develop, users' requirements on network functions also keep changing. If the current network cannot meet service requirements, or potential problems are found while the network is running, the network needs to be optimized.

Small Campus Network Design



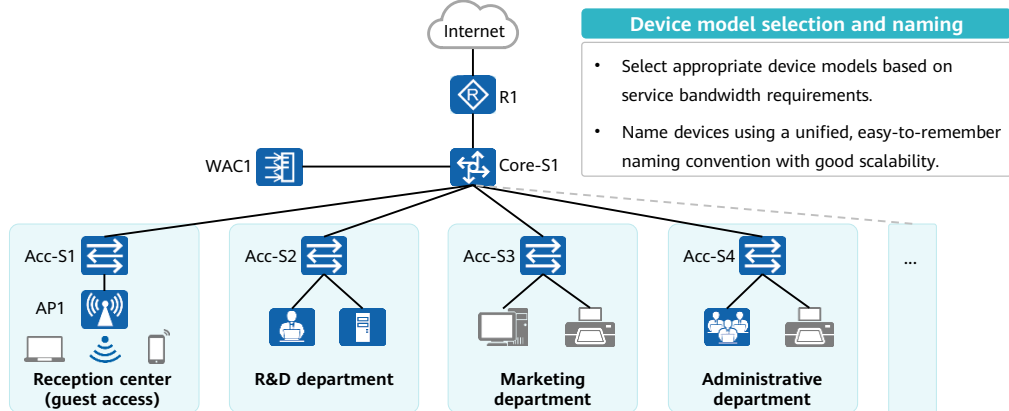
Networking Architecture Selection

- The networking architecture of a small or midsize campus network can be selected based on the network scale.
- Generally, small and midsize campus networks have a relatively small scale. For a single campus or branch, the single-layer architecture (access layer) or two-layer architecture (core layer + access layer) is used. For a large or midsize campus, the three-layer architecture (core layer + aggregation layer + access layer) can be used.



Physical Topology Design

- Considering factors such as the budget and service requirements, the company uses a two-layer architecture. The following figure shows the physical topology.

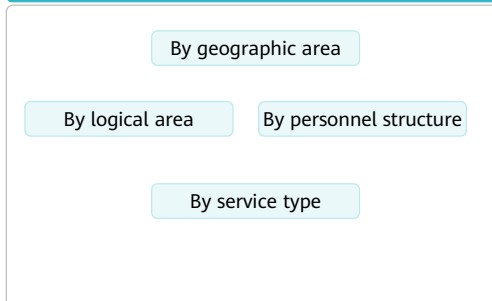


- The network uses the egress zone and two-layer architecture.
 - A NetEngine AR series router is used as the egress device of the entire campus network. It connects to external networks and implements security management and control for internal and external traffic.
 - The core layer uses a CloudEngine S series switch as the gateway for wired and wireless services.
 - The access layer uses CloudEngine S series switches to provide 100 Mbit/s wired and wireless network access for employees' PCs/laptops, printers, and guest terminals.
 - The wireless network uses the WAC + Fit AP networking architecture and AirEngine series wireless devices to provide wireless access for the campus network.
- Note: Acc is short for access, and is used to represent access layer devices.

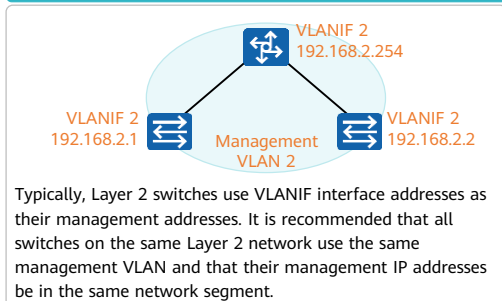
Basic Service Design: VLAN Design

- You are advised to assign consecutive VLAN IDs to ensure proper use of VLAN resources.
- VLANs are classified into management VLANs, interconnection VLANs, and service VLANs.
- Interface-based VLAN assignment is the most commonly used method.

Service VLAN design



Management VLAN design



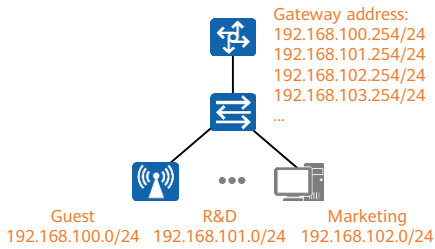
VLAN Planning

- **Management VLAN:** The management VLAN of Layer 2 devices and the VLAN required for establishing CAPWAP tunnels between Fit APs and WACs are reserved.
- **Interconnection VLAN:** Interconnection VLANs are reserved for Layer 3 switches to communicate with routers and WACs through VLANIF interfaces.
- **Service VLAN:** Service VLANs are assigned to different departments, including the guest VLAN, R&D VLAN, marketing VLAN, and administrative VLAN.

VLAN Type	VLAN Resource	VLAN ID	VLAN Description
Management VLAN	VLAN 2 to VLAN 9	2	Layer 2 device management VLAN
		3	WLAN management VLAN
Interconnection VLAN	VLAN 10 to VLAN 99	10	VLAN for interconnection between Core-S1 and R1
		11	VLAN for interconnection between Core-S1 and WAC1
Service VLAN	VLAN 100 to VLAN 199	100	Service VLAN for guests (WLAN service VLAN)
		101	Service VLAN for the R&D department
		102	Service VLAN for the marketing department
		103	Service VLAN for the administrative department

Basic Service Design: IP Address Design

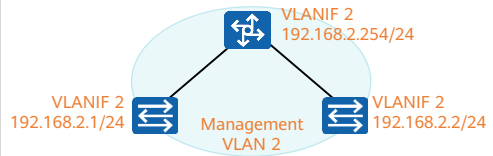
Service IP address



A service IP address is the IP address of a server, host, or gateway.

- It is recommended that gateway IP addresses use the same last digits, for example, .254 or .1.
- The IP address segment of each service must be clearly distinguished, and the IP addresses of each type of service terminals must be contiguous and can be aggregated.
- An IP address segment with a 24-bit mask is recommended.

Management IP address



A Layer 2 device uses the VLANIF interface's IP address as the management IP address. It is recommended that all Layer 2 switches connected to a gateway be on the same network segment.

Interconnection IP address

It is recommended that network devices use an interconnection IP address with a 30-bit mask and the core device use a smaller host IP address.

IP Address Planning (1)

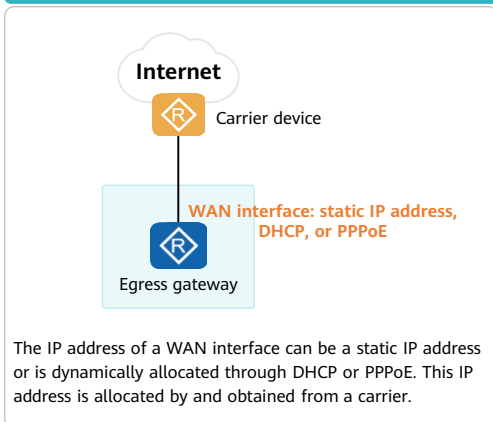
- Management IP address: Plan a network segment for management IP addresses, which are used to manage Layer 2 devices and wireless APs and also used as the source address of the CAPWAP tunnel.
- Interconnection IP address: Plan a network segment for interconnection IP addresses, which are used for interconnection between Layer 3 devices.
- Service IP address: Plan network segments and gateway addresses for each type of service based on the numbers of access clients and IP addresses to be reserved.

IP Address Planning (2)

IP Address Type	IP Address Segment/Mask	Gateway IP Address	Network Segment Description
Management IP address	192.168.2.0/24	192.168.2.254	Management network segment for Layer 2 devices, with Core-S1 serving as the gateway
	192.168.3.0/24	192.168.3.254	Management network segment for the WLAN, with Core-S1 serving as the gateway
Interconnection IP address	192.168.10.0/30	/	Interconnection network segment between Core-S1 and R1
	192.168.11.0/30	/	Interconnection network segment between Core-S1 and WAC1. The interconnection IP address of WAC1 is also used as the source IP address of the CAPWAP tunnel.
	1.2.3.1/30	1.2.3.2/30	Public IP address assigned by the carrier to R1 for connecting to the external network
Service IP address	192.168.100.0/24	192.168.100.254	Network segment for guests, with Core-S1 serving as the gateway
	192.168.101.0/24	192.168.101.254	Network segment for the R&D department, with Core-S1 serving as the gateway
	192.168.102.0/24	192.168.102.254	Network segment for the marketing department, with Core-S1 serving as the gateway
	192.168.103.0/24	192.168.103.254	Network segment for the administrative department, with Core-S1 serving as the gateway

Basic Service Design: IP Address Assignment Design

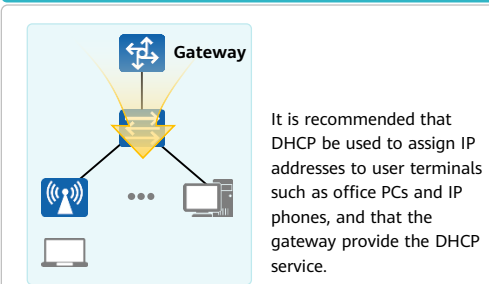
Egress gateway



Devices such as servers and printers

Static IP addresses are recommended for servers and special terminals, such as punch-card machines, printing servers, and IP video security devices.

End user



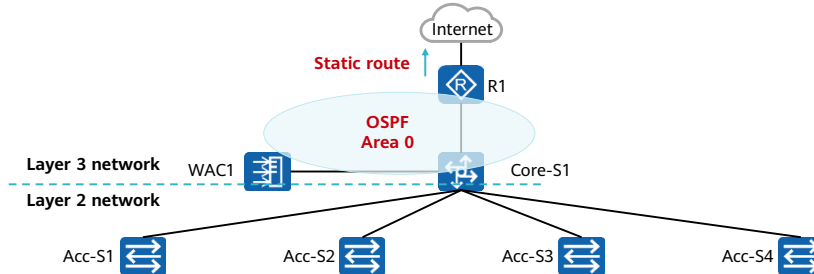
IP Address Assignment Planning

- Egress gateway: The IP address for the egress gateway to connect to the WAN is manually configured.
- Network devices: The IP addresses of all network devices (except Fit APs) are manually configured.
- Fit APs and terminals: IP addresses are dynamically obtained through DHCP. Fixed IP addresses are assigned to servers and printers through DHCP.

IP Address Type	IP Address Segment/Mask	Assignment Mode	Network Segment Description
Management IP address	192.168.2.0/24	Static	Device management IP address, which is statically configured.
	192.168.3.0/24	DHCP	The IP address of the Fit AP is dynamically allocated by the gateway Core-S1 that also functions as the DHCP server.
Interconnection IP address	192.168.10.0/30	Static	Interconnection IP address, which is statically configured.
	192.168.11.0/30	Static	Interconnection IP address and source IP address of the CAPWAP tunnel, which is statically configured.
	1.2.3.1/30	Static	Public IP address assigned by the carrier to R1 for connecting to the external network.
Service IP address	192.168.100.0/24	DHCP	The IP addresses are dynamically allocated by the gateway Core-S1 that also functions as the DHCP server. Fixed IP addresses are allocated to fixed devices such as servers and printers.
	192.168.101.0/24	DHCP	
	192.168.102.0/24	DHCP	
	192.168.103.0/24	DHCP	

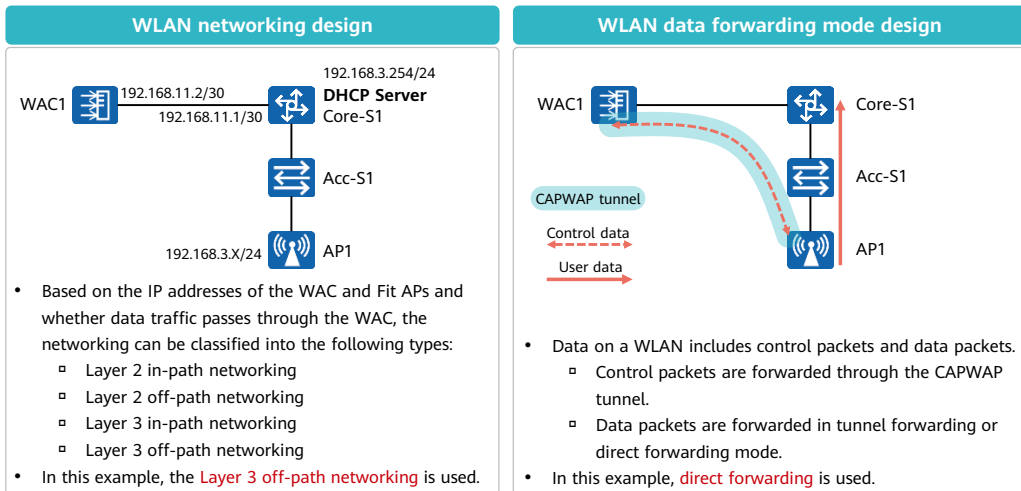
Basic Service Design: Routing Design

- Internal routing design of the campus network:
 - On the same network segment: After IP addresses are allocated through DHCP, a default route is generated, with Core-S1 serving as the Layer 3 gateway.
 - Between different network segments: Open Shortest Path First (OSPF) is deployed on all devices that need to forward Layer 3 data to implement Layer 3 interworking.
- Routing design at the campus egress: Static default routes need to be configured.



- The routing design of small and midsize campus networks includes design of internal routes and the routes between the campus egress and the Internet or WAN devices.
 - The internal routing design of the campus network must meet the communication requirements of devices and terminals on the campus network and enable interaction with external routes. A small or midsize campus network has a simple architecture. Layer 3 connectivity can be enabled by configuring a single OSPF area or static routes.
 - The egress route design must be able to support Internet and WAN access of intranet users. When the egress device is connected to the Internet or WAN, you are advised to configure static routes on the egress device.

WLAN Design



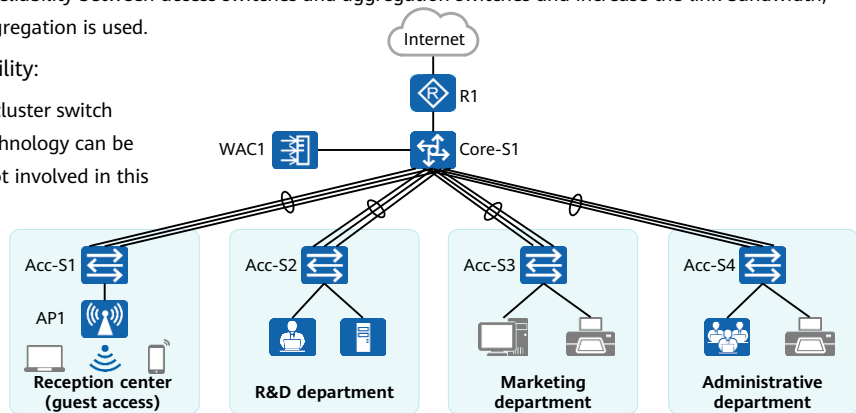
- In addition to networking and data forwarding mode planning, you also need to perform the following tasks:
 - Network coverage design: Design and plan the wireless network coverage area to ensure that the signal strength in the coverage area meets user requirements and solve the co-channel interference problem between neighboring APs.
 - Network capacity design: Design the number of APs required based on the bandwidth requirements and number of STAs, concurrency rate, and per-AP performance. This ensures that WLAN performance can meet the Internet access requirements of all STAs.
 - AP deployment design: Based on the network coverage design, adjust and confirm the actual deployment positions, deployment modes, and power supply and cabling principles of APs.
 - In addition to the preceding tasks, you also need to perform other tasks such as WLAN security design and roaming design, which will not be detailed in this course.

WLAN Data Planning

Configuration Item	Setting
Management VLAN/network segment for APs	VLAN 3, 192.168.3.0/24 (gateway: 192.168.3.254)
Service VLAN/network segment for STAs	VLAN 100, 192.168.100.0/24 (gateway: 192.168.100.254)
DHCP server	Core-S1 functions as the DHCP server to assign IP addresses to APs and STAs.
WAC's source interface/IP address	VLANIF 11, 192.168.11.2/24
Regulatory domain profile	Name: default Country code: CN
SSID profile	Name: WLAN-Guest SSID: WLAN-Guest
Security profile	Name: WLAN-Guest Security policy: WPA-WPA2+PSK+AES Password: WLAN@Guest123
VAP profile	Name: WLAN-Guest Forwarding mode: direct forwarding Service VLAN: VLAN 100 Referenced profiles: SSID profile WLAN-Guest and security profile WLAN-Guest
AP group	Name: ap-group1 Referenced profiles: VAP profile WLAN-Guest and regulatory domain profile default

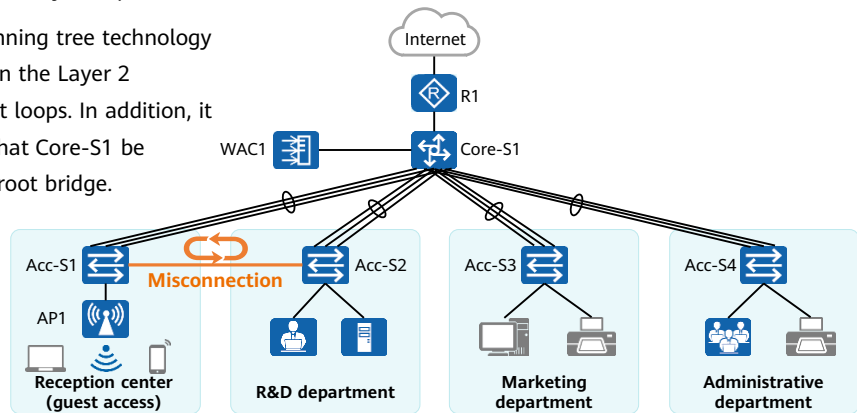
Network Reliability Design

- Link-level reliability:
 - To improve the reliability between access switches and aggregation switches and increase the link bandwidth, Ethernet link aggregation is used.
- Device-level reliability:
 - The stacking or cluster switch system (CSS) technology can be used, which is not involved in this networking.



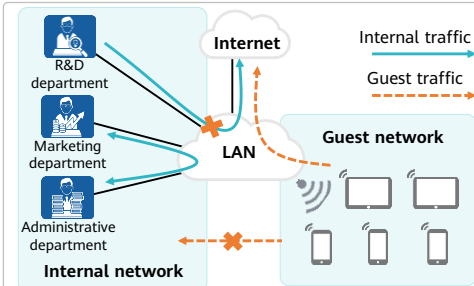
Layer 2 Loop Protection Design

- Question: Currently, the network segment does not have redundant links. How can we prevent Layer 2 network loops caused by misoperations?
- Suggestion: A spanning tree technology can be deployed on the Layer 2 network to prevent loops. In addition, it is recommended that Core-S1 be configured as the root bridge.



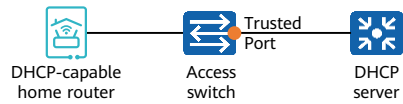
Security Design

Egress security design: traffic control



- Different departments can communicate with each other but are not allowed to access the Internet.
- Guests can access the Internet but are not allowed to access the internal campus network.
- Traffic policy technologies can be used to isolate the internal network from the Internet, and NAT can be used to control the access from the internal network to the Internet.

Intranet security design: DHCP security



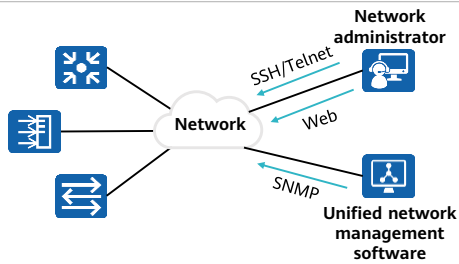
- On a campus network, employees may connect unauthorized DHCP-enabled wireless routers to the network, causing internal IP address conflicts and Internet access failures.
- To prevent this, DHCP snooping can be enabled on access switches.

Network management security design

- When a device is managed through Telnet or web, ACLs can be used to allow only authorized users to log in to the device for management.
- For centralized network management, SNMPv3 provides identity authentication and encryption, which greatly improves network management security.

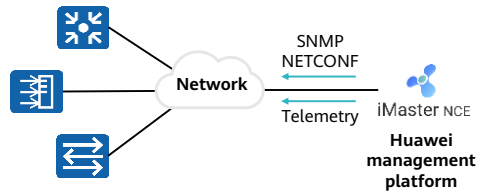
O&M Management Design

Basic network management



- If the network administrator can reach the device IP address, the administrator can manage the device through Telnet, SSH, or web.
- When there are a large number of devices on the network, SNMP-based unified network management software can be deployed to perform network O&M and management.

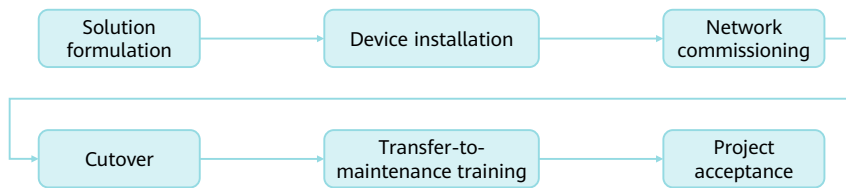
Intelligent O&M



- In addition to the traditional SNMP-based network management software, Huawei iMaster NCE can be used to manage and maintain networks, implementing autonomous driving networks.

Small Campus Network Deployment and Implementation

- The deployment and implementation of a campus network project need to follow a defined process, including:



- The specific process needs to be determined based on the actual project conditions.

Configuration Solution for Small Campus Networks

- After devices are installed and connected through physical cables, you need to configure and deploy services on the small campus network according to the service plan.

Configuration Item	Sub-configuration Item	Description
Network Reliability	Eth-Trunk configuration	If redundant links are deployed between devices, configure link aggregation and add interface description.
Layer 2 loop protection	STP configuration	Enable STP on all switches and configure the core switch as the root bridge.
Basic services	VLAN configuration	Assign VLANs based on interfaces.
	IP address configuration	Terminals and APs dynamically obtain IP addresses through DHCP, and devices are configured with static IP addresses.
	Route configuration	OSPF is deployed on Layer 3 devices (Core-S1, WAC1, and R1) to implement interworking.
Wireless services	WLAN configuration	Perform the configuration according to the WLAN plan.
Network security	Traffic monitoring	Configure an advanced ACL to deny traffic from network segment <i>X</i> to network segment <i>Y</i> , and permit other traffic. Configure a traffic policy that references the ACL and apply the traffic policy to an interface to filter traffic.
	Egress NAT	Configure a NAT policy to permit only the traffic from network segment <i>X</i> to the Internet, and enable the NAT function on the outbound interface.
	DHCP security	Enable DHCP snooping on all access switches and configure uplink interfaces as trusted interfaces.
	Network management security	Configure a basic ACL to permit only the traffic from the administrator's IP address.
O&M management	O&M management	Configure STelnet-based remote management and set the authentication mode to AAA.

Small Campus Network Commissioning

- After campus network services are deployed, you need to commission the campus network.

1. Connectivity test

Basic link interconnection

Layer 2 connectivity

Layer 3 connectivity

2. Reliability test

Loop protection

Path switchover

Hot standby

3. Service performance test

Service traffic

Access control

Small Campus Network O&M

- After a campus network project is deployed and put into operation, network engineers need to perform routine maintenance to detect and eliminate potential risks of devices, thereby ensuring stable running of devices. Such work is generally referred to as network O&M.
- Common tasks in the network O&M phase include but are not limited to the following:
 - Routine maintenance: Periodically check devices (such as the running environment, basic information, and performance status), monitor service traffic and quality (such as the packet error rate and packet loss rate) in real time, and handle various alarms in a timely manner to ensure that the network running is visible and controllable.
 - Configuration management: Standardize network change operations (such as new user access and policy adjustment), periodically back up and archive configurations, and upgrade the software version based on service and function requirements to ensure the accuracy and consistency of network configurations.
 - Troubleshooting: Establish a closed-loop process, including quick fault detection, accurate fault locating, and effective fault rectification. In addition, develop preventive measures through root cause analysis to reduce the probability of fault recurrence.
- As the scale and complexity of campus networks increase, traditional manual O&M methods face challenges. Using professional network management platforms or introducing automation scripts and tools has become an important development direction for improving O&M efficiency and ensuring O&M quality.

Small Campus Network Optimization

- In addition to network O&M, network engineers need to continuously analyze the network to improve performance and eliminate bottlenecks, thereby enhancing network performance, efficiency, and user experience. This phase is generally referred to as network optimization.
- Common tasks in the network optimization phase include but are not limited to the following:
 - Network performance optimization: Periodically evaluate network quality (such as bandwidth utilization, delay, jitter, and packet loss rate), proactively identify network health status, network faults, and wireless network performance, and dynamically adjust parameters and policies to improve network performance.
 - Network resource adjustment: Monitor network traffic characteristics, adjust network resource configurations (such as bandwidth expansion and link load balancing), periodically evaluate the network capacity and bearing capability, and implement service quality policies based on service priorities to ensure key service experience and proper resource allocation.
 - User experience improvement: Evaluate user experience of key applications through proactive detection or user feedback. When a fault occurs, quickly locate the root cause and rectify the fault. In addition, preferentially ensure VIP user experience.
- As the digitalization of services accelerates and user experience requirements are becoming more stringent, traditional optimization methods through static configurations are facing challenges. Introducing an intelligent analysis platform or deploying automatic optimization and AI prediction tools has become a key approach to achieving dynamic network optimization and ensuring service experience.

Quiz

1. (Short-answer question) What is the lifecycle of a campus network?
2. (Short-answer question) What is the function of a management IP address?

1. Planning and design, deployment and implementation, network O&M, and network optimization
2. This IP address is used by the network administrator to manage devices.

Summary

- This course introduces the concept, types, and common technologies of campus networks.
- This course covers the lifecycle of campus networks:
 - Planning and design
 - Deployment and implementation
 - Network O&M
 - Network optimization
- On the basis of the previous courses, this course focuses on the planning and design, as well as deployment and implementation of campus networks, and demonstrates how to build a small campus network.

Thank you.

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intelligent world.

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